

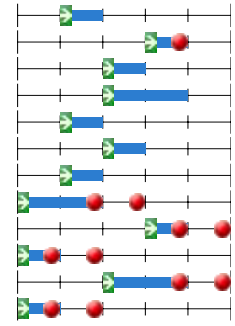
Program overview

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Year 2024/2025
 Organization Aerospace Engineering
 Education Master Aerospace Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master LR 2024							
Master Aerospace Engineering 2024							
Track 1: Aerodynamics and Wind Energy							
Profile I: Aerodynamics							
AE4115-23	Experimental Simulations	4					
AE4120	Viscous Flows	3					
AE4130-22	Aircraft Aerodynamics	4					
AE4135	Rotor / wake Aerodynamics	4					
AE4136-24	CFD 2: Discretisation Techniques	5					
AE4140	Gas Dynamics	3					
AE4180	Flow Measurement Techniques	3					
AE4202	CFD for Aerospace Engineers	3					
AE5051	Internship	15					
AE5122	Thesis Aerodynamics & Wind Energy	45					
WI3150TU	Partial Differential Equations A	3					
WM0324LR	Ethics and Engineering for Aerospace Engineering	3					
Preferred Electives Profile Aerodynamics							
AE4006E	Sustainability in Engineering	5					
AE4007	AI for Mindfulness in Engineering	5					
AE4010	Research Methodologies	2					
AE4117	Fluid-Structure Interaction	4					
AE4138-18	CFD 4: Uncertainty Quantification	3					
AE4139	CFD 3: Large Eddy Simulation	3					
AE4143	Hypersonic Aerodynamics	3					
AE4185	Fluid Flow Data Processing and Visualization	3					
AE4240	Advanced Aircraft Design I	4					
AE4260A	Fundamental of Aeroacoustics	2					
AE4260B	Experimental Applications of Aeroacoustics	2					
AE4262	Combustion for propulsion and power technologies	4					
AE4ASM506	Fundamentals of Aeroelasticity	3					
AE4ASM514TU	Continuum Mechanics	4					
AE4T40	Airborne Wind Energy	3					
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4					
AE4W21-14	Wind Turbine Aeroelasticity	2					
AE4W30	Wind Resource and Wind Farm Yield	4					
AE4W31	Floating Offshore Wind Energy	3					
ME45001	Advanced Heat Transfer	4					
ME45030	Turbulence	5					
ME46060	Engineering Optimization: Concepts and Applications	3					
WI3151TU	Partial Differential Equations B	3					
WI4014TU	Numerical Analysis	6					
WI4201	Scientific Computing	6					
WI4212	Advanced Numerical Methods	6					
Profile II: Wind Energy							
AE4120	Viscous Flows	3					
AE4135	Rotor / wake Aerodynamics	4					
AE4202	CFD for Aerospace Engineers	3					
AE4ASM506	Fundamentals of Aeroelasticity	3					
AE4T40	Airborne Wind Energy	3					
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4					
AE4W09	Wind Turbine Design	5					
AE4W13	Site Conditions for Wind Turbine Design	3					
AE4W21-14	Wind Turbine Aeroelasticity	2					
AE5051	Internship	15					
AE5122	Thesis Aerodynamics & Wind Energy	45					
WM0324LR	Ethics and Engineering for Aerospace Engineering	3					
Preferred Electives Profile Wind Energy							
AE4006E	Sustainability in Engineering	5					
AE4007	AI for Mindfulness in Engineering	5					
AE4010	Research Methodologies	2					
AE4115-23	Experimental Simulations	4					
AE4117	Fluid-Structure Interaction	4					
AE4130-22	Aircraft Aerodynamics	4					

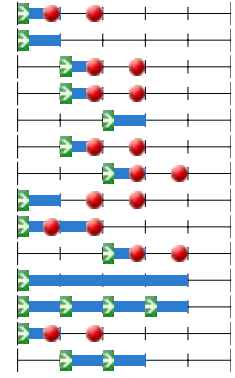
AE4136-24	CFD 2: Discretisation Techniques	
AE4138-18	CFD 4: Uncertainty Quantification	3
AE4139	CFD 3: Large Eddy Simulation	3
AE4180	Flow Measurement Techniques	3
AE4260A	Fundamental of Aeroacoustics	2
AE4260B	Experimental Applications of Aeroacoustics	2
AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4
AE4W30	Wind Resource and Wind Farm Yield	4
AE4W31	Floating Offshore Wind Energy	3
ME45001	Advanced Heat Transfer	4
ME45030	Turbulence	5
WI3150TU	Partial Differential Equations A	3



Track 2: Control and Operations

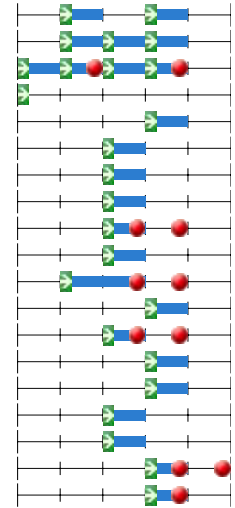
Profile I: Control and Simulation

AE4301	Automatic Flight Control System Design	3
AE4301P	Exercise Automatic Flight Control System Design	1
AE4302	Avionics and Operations	3
AE4304	Stochastic Aerospace Systems	3
AE4304P	Stochastic Aerospace Systems Practical	1
AE4316	Aerospace Human-Machine Systems	4
AE4322-24	Piloted Flight Simulation	3
AE4353	Artificial Intelligence for Aerospace Control and Operations	4
AE4441-16	Operations optimisation	4
AE4462-17	Aircraft Emissions and Climate Effects	4
AE5051	Internship	15
AE5322	Thesis Control and Operations	45
WI2056LR	Systems Theory	4
WM0324LR	Ethics and Engineering for Aerospace Engineering	3



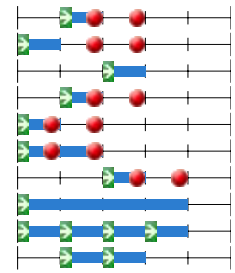
Preferred Electives Profile Control and Simulation

AE4006E	Sustainability in Engineering	5
AE4007	AI for Mindfulness in Engineering	5
AE4010	Research Methodologies	2
AE4270	Control and Operations Project	4
AE4311	Nonlinear and Adaptive Flight Control	4
AE4313-20	Spacecraft Attitude Dynamics and Control	4
AE4314-21	Helicopter Performance, Stability and Control	4
AE4316P	Advanced Topics in Aerospace Human-Machine Systems	3
AE4317	Autonomous Flight of Micro Air Vehicles	4
AE4320	System Identification of Aerospace Vehicles	4
AE4321-15	Air Traffic Management	4
AE4323	Real-time Distributed Flight and Space Simulation	3
AE4324	Physical Interaction for Aerial and Space Robots	4
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
AE4351	Robust Flight Control	4
AE4352	Mathematical and human-inspired decision making	3
AE4422-20	Agent-based Modelling and Simulation in Air Transport	4
RO47019	Intelligent Control Systems	4
SC42035	Integration Project Systems and Control	5



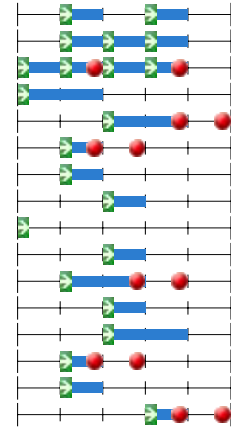
Profile II: Sustainable Air Transport

AE4302	Avionics and Operations	3
AE4353	Artificial Intelligence for Aerospace Control and Operations	4
AE4422-20	Agent-based Modelling and Simulation in Air Transport	4
AE4423-20	Airline Planning and Optimisation	4
AE4431-23	Aircraft noise	3
AE4441-16	Operations optimisation	4
AE4462-17	Aircraft Emissions and Climate Effects	4
AE5051	Internship	15
AE5322	Thesis Control and Operations	45
WM0324LR	Ethics and Engineering for Aerospace Engineering	3



Preferred Electives Profile Sustainable Air Transport

AE4006E	Sustainability in Engineering	5
AE4007	AI for Mindfulness in Engineering	5
AE4010	Research Methodologies	2
AE4130-22	Aircraft Aerodynamics	4
AE4135	Rotor / wake Aerodynamics	4
AE4238	Aero Engine Technology	4
AE4260A	Fundamental of Aeroacoustics	2
AE4260B	Experimental Applications of Aeroacoustics	2
AE4270	Control and Operations Project	4
AE4320	System Identification of Aerospace Vehicles	4
AE4321-15	Air Traffic Management	4
AE4352	Mathematical and human-inspired decision making	3
AE4439	Sustainable Air Transport Modelling Project	2
AE4446	Airport and Cargo Operations	4
AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4
AE4465	Maintenance Modeling & Analysis	4



WI4062TU	Transport, Routing and Scheduling	3	
Track 3: Space			
Profile I: Space Engineering			
AE4874 I	Fundamentals of Astrodynamics	4	
AE4890-11	Planetary Exploration	4	
AE4896	Space instrumentation	4	
AE4S07	Micropropulsion	4	
AE4S10	Microsat Engineering	4	
AE4S12-23	Space systems engineering	4	
AE4S15	Space Embedded Systems	3	
AE5051	Internship	15	
AE5822	Thesis Space	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Choose 1 course			
AE4897	Space Engineering Practical	8	
AE4S52-21	Collaborative Space System Design Project	5	
Preferred Electives Space			
AE4006E	Sustainability in Engineering	5	
AE4007	AI for Mindfulness in Engineering	5	
AE4010	Research Methodologies	2	
AE4143	Hypersonic Aerodynamics	3	
AE4202	CFD for Aerospace Engineers	3	
AE4204	Knowledge Based Engineering	4	
AE4205	MDO for Aerospace Applications	4	
AE4238	Aero Engine Technology	4	
AE4302	Avionics and Operations	3	
AE4313-20	Spacecraft Attitude Dynamics and Control	4	
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3	
AE4866-1	Propagation and Optimisation in Astrodynamics	5	
AE4889	Special Topics in Astrodynamics	2	
AE4895-24	Measurement Strategies for Planetary Science Missions	4	
AE4897	Space Engineering Practical	8	
AE4898	Space debris tracking and mitigation	4	
AE4ASM003	Linear Modeling (incl. F.E.M)	3	
AE4ASM524	Spacecraft Structures Development	3	
AE4ASM525	Materials for Space	3	
AE4ASM526	Spacecraft Thermal Design	3	
AE4S52-21	Collaborative Space System Design Project	5	
AESM2001	Physical Principles of Earth System Observation	5,5	
AP3141	Environmental Physics	6	
AP3181	Applied Multiphase Flows	6	
AP3222	Nanotechnology	6	
AP3701	Submm and terahertz physics and applications	3	
BM41155	3D Printing	4	
CS4090	Quantum Communication and Cryptography	5	
CS4205	Evolutionary Algorithms	5	
EE4685	Machine Learning, a Bayesian Perspective	5	
EE4C06	Networking	5	
EE4C12	Machine Learning for Electrical Engineering	5	
ET3604LR	Electronic Circuits	3	
ET4378	Photovoltaic Systems	4	
ET4386	Estimation and Detection	5	
GEO1016	Photogrammetry and 3D Computer Vision	5	
ME44312	Machine Learning for Transport and Multi-Machine Systems	3	
ME45001	Advanced Heat Transfer	4	
ME45165	Equipment for Heat & Mass Transfer	5	
ME45170	Turbomachinery	4	
ME45203	Electrolysers, Fuel Cells and Batteries	4	
ME46006	Physics for Mechanical Engineers	4	
ME46060	Engineering Optimization: Concepts and Applications	3	
ME46085-23	Mechatronic System Design	4	
ME46095	Multiphysics Modelling using COMSOL	4	
ME46300	Optics	4	
ME46310	Opto-Mechatronics	4	
MS43310	Materials at High Temperature	4	
QIST4310	Fundamentals of Quantum Information	4	
RO47001	Robot Dynamics & Control	5	
RO47002	Machine Learning for Robotics	5	
RO47007	Multidisciplinary Project	5	
RO47015	Applied Experimental Methods	5	
RO47019	Intelligent Control Systems	4	
SC42030	Control for High Resolution Imaging	3	
SC42065	Adaptive Optics Design Project	3	
SET3085	Hydrogen Technology	4	
WI4019-SP	Non-linear Differential Equations	6	
WI4062TU	Transport, Routing and Scheduling	3	

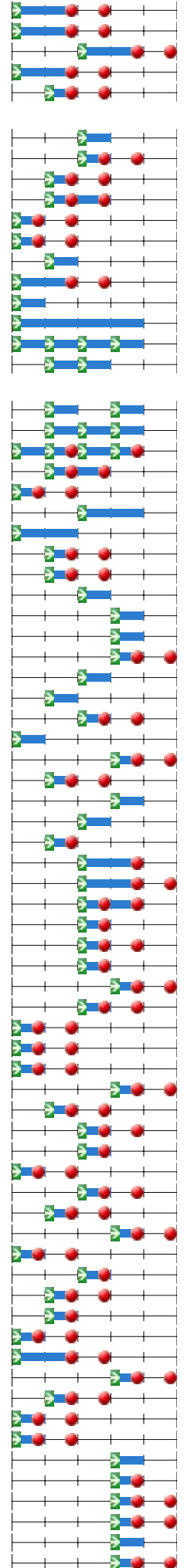
WI4156(TU)	Game theory	6
WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
WI4630	Statistical Learning	6
WI4635	Linear Algebra and Optimization for Machine Learning	6
WI4771TU	Object Oriented Scientific Programming with C++	3

Profile II: Spaceflight Dynamics

AE4868-1	Numerical Astrodynamics	4
AE4870A	Rocket Motion	3
AE4870B	Re-Entry Systems	3
AE4872	Satellite Orbit Determination	6
AE4874 I	Fundamentals of Astrodynamics	4
AE4890-11	Planetary Exploration	4
AE4894	Practical Astrodynamics	3
AE4896	Space instrumentation	4
AE4S12-23	Space systems engineering	4
AE5051	Internship	15
AE5822	Thesis Space	45
WM0324LR	Ethics and Engineering for Aerospace Engineering	3

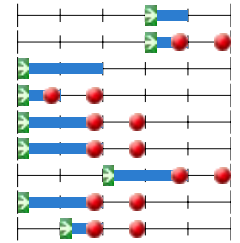
Preferred Electives Space

AE4006E	Sustainability in Engineering	5
AE4007	AI for Mindfulness in Engineering	5
AE4010	Research Methodologies	2
AE4143	Hypersonic Aerodynamics	3
AE4202	CFD for Aerospace Engineers	3
AE4204	Knowledge Based Engineering	4
AE4205	MDO for Aerospace Applications	4
AE4238	Aero Engine Technology	4
AE4302	Avionics and Operations	3
AE4313-20	Spacecraft Attitude Dynamics and Control	4
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
AE4866-1	Propagation and Optimisation in Astrodynamics	5
AE4889	Special Topics in Astrodynamics	2
AE4895-24	Measurement Strategies for Planetary Science Missions	4
AE4897	Space Engineering Practical	8
AE4898	Space debris tracking and mitigation	4
AE4ASM003	Linear Modeling (incl. F.E.M)	3
AE4ASM524	Spacecraft Structures Development	3
AE4ASM525	Materials for Space	3
AE4ASM526	Spacecraft Thermal Design	3
AE4S52-21	Collaborative Space System Design Project	5
AESM2001	Physical Principles of Earth System Observation	5,5
AP3141	Environmental Physics	6
AP3181	Applied Multiphase Flows	6
AP3222	Nanotechnology	6
AP3701	Submm and terahertz physics and applications	3
BM41155	3D Printing	4
CS4090	Quantum Communication and Cryptography	5
CS4205	Evolutionary Algorithms	5
EE4685	Machine Learning, a Bayesian Perspective	5
EE4C06	Networking	5
EE4C12	Machine Learning for Electrical Engineering	5
ET3604LR	Electronic Circuits	3
ET4378	Photovoltaic Systems	4
ET4386	Estimation and Detection	5
GEO1016	Photogrammetry and 3D Computer Vision	5
ME44312	Machine Learning for Transport and Multi-Machine Systems	3
ME45001	Advanced Heat Transfer	4
ME45165	Equipment for Heat & Mass Transfer	5
ME45170	Turbomachinery	4
ME45203	Electrolysers, Fuel Cells and Batteries	4
ME46006	Physics for Mechanical Engineers	4
ME46060	Engineering Optimization: Concepts and Applications	3
ME46085-23	Mechatronic System Design	4
ME46095	Multiphysics Modelling using COMSOL	4
ME46300	Optics	4
ME46310	Opto-Mechatronics	4
MS43310	Materials at High Temperature	4
QIST4310	Fundamentals of Quantum Information	4
RO47001	Robot Dynamics & Control	5
RO47002	Machine Learning for Robotics	5
RO47007	Multidisciplinary Project	5
RO47015	Applied Experimental Methods	5
RO47019	Intelligent Control Systems	4
SC42030	Control for High Resolution Imaging	3
SC42065	Adaptive Optics Design Project	3
SET3085	Hydrogen Technology	4



WI4019-SP	Non-linear Differential Equations	6	
WI4062TU	Transport, Routing and Scheduling	3	
WI4156(TU)	Game theory	6	
WI4525TU	Monte Carlo Simulation and Stochastic Processes	5	
WI4630	Statistical Learning	6	
WI4635	Linear Algebra and Optimization for Machine Learning	6	
WI4771TU	Object Oriented Scientific Programming with C++	3	
Profile III: Planetary Exploration			
AE4868-1	Numerical Astrodynamics	4	
AE4872	Satellite Orbit Determination	6	
AE4874 I	Fundamentals of Astrodynamics	4	
AE4876-24	Planetary Missions	5	
AE4890-11	Planetary Exploration	4	
AE4893-24	Physics of Planetary Interiors	5	
AE4895-24	Measurement Strategies for Planetary Science Missions	4	
AE4896	Space instrumentation	4	
AE4S12-23	Space systems engineering	4	
AE5051	Internship	15	
AE5822	Thesis Space	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Space			
AE4006E	Sustainability in Engineering	5	
AE4007	AI for Mindfulness in Engineering	5	
AE4010	Research Methodologies	2	
AE4143	Hypersonic Aerodynamics	3	
AE4202	CFD for Aerospace Engineers	3	
AE4204	Knowledge Based Engineering	4	
AE4205	MDO for Aerospace Applications	4	
AE4238	Aero Engine Technology	4	
AE4302	Avionics and Operations	3	
AE4313-20	Spacecraft Attitude Dynamics and Control	4	
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3	
AE4866-1	Propagation and Optimisation in Astrodynamics	5	
AE4889	Special Topics in Astrodynamics	2	
AE4895-24	Measurement Strategies for Planetary Science Missions	4	
AE4897	Space Engineering Practical	8	
AE4898	Space debris tracking and mitigation	4	
AE4ASM003	Linear Modeling (incl. F.E.M)	3	
AE4ASM524	Spacecraft Structures Development	3	
AE4ASM525	Materials for Space	3	
AE4ASM526	Spacecraft Thermal Design	3	
AE4S52-21	Collaborative Space System Design Project	5	
AESM2001	Physical Principles of Earth System Observation	5,5	
AP3141	Environmental Physics	6	
AP3181	Applied Multiphase Flows	6	
AP3222	Nanotechnology	6	
AP3701	Submm and terahertz physics and applications	3	
BM41155	3D Printing	4	
CS4090	Quantum Communication and Cryptography	5	
CS4205	Evolutionary Algorithms	5	
EE4685	Machine Learning, a Bayesian Perspective	5	
EE4C06	Networking	5	
EE4C12	Machine Learning for Electrical Engineering	5	
ET3604LR	Electronic Circuits	3	
ET4378	Photovoltaic Systems	4	
ET4386	Estimation and Detection	5	
GEO1016	Photogrammetry and 3D Computer Vision	5	
ME44312	Machine Learning for Transport and Multi-Machine Systems	3	
ME45001	Advanced Heat Transfer	4	
ME45165	Equipment for Heat & Mass Transfer	5	
ME45170	Turbomachinery	4	
ME45203	Electrolysers, Fuel Cells and Batteries	4	
ME46006	Physics for Mechanical Engineers	4	
ME46060	Engineering Optimization: Concepts and Applications	3	
ME46085-23	Mechatronic System Design	4	
ME46095	Multiphysics Modelling using COMSOL	4	
ME46300	Optics	4	
ME46310	Opto-Mechatronics	4	
MS43310	Materials at High Temperature	4	
QIST4310	Fundamentals of Quantum Information	4	
RO47001	Robot Dynamics & Control	5	
RO47002	Machine Learning for Robotics	5	
RO47007	Multidisciplinary Project	5	
RO47015	Applied Experimental Methods	5	
RO47019	Intelligent Control Systems	4	
SC42030	Control for High Resolution Imaging	3	

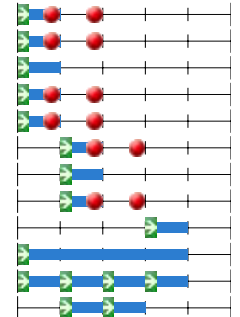
SC42065	Adaptive Optics Design Project	3
SET3085	Hydrogen Technology	4
WI4019-SP	Non-linear Differential Equations	6
WI4062TU	Transport, Routing and Scheduling	3
WI4156(TU)	Game theory	6
WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
WI4630	Statistical Learning	6
WI4635	Linear Algebra and Optimization for Machine Learning	6
WI4771TU	Object Oriented Scientific Programming with C++	3



Track 4: Aerospace Structures and Materials

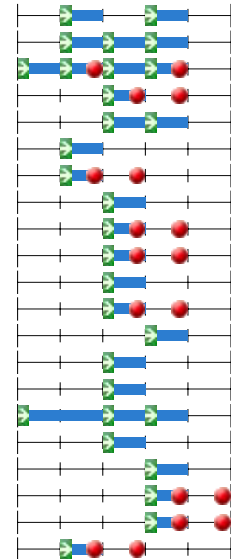
Profile I: Materials and Manufacturing

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
AE4ASM002	Designing Materials with Aerospace Specific Properties	3
AE4ASM003	Linear Modeling (incl. F.E.M)	3
AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
AE4ASM005	Fatigue of Structures & Materials	3
AE4ASM101TU-22	Polymer Science	4
AE4ASM110-22	Polymer Composite Manufacturing	4
AE4ASM515-22	Characterization of materials and components	4
AE4ASM521	Additive Manufacturing	3
AE5051	Internship	15
AE5722	Thesis Aerospace Structures & Materials	45
WM0324LR	Ethics and Engineering for Aerospace Engineering	3



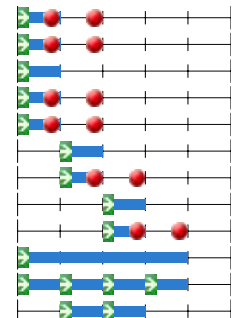
Preferred Electives Profile Materials and Manufacturing

AE4006E	Sustainability in Engineering	5
AE4007	AI for Mindfulness in Engineering	5
AE4010	Research Methodologies	2
AE4ASM103	Functional Coatings	3
AE4ASM105	Trinity Exercise	4
AE4ASM106	Stability & Analysis of Structures I	3
AE4ASM108	Experimental Techniques & NDT	3
AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
AE4ASM503	Sheet Metal Forming	3
AE4ASM504	Structural Integrity and Maintenance	3
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
AE4ASM506	Fundamentals of Aeroelasticity	3
AE4ASM510	Design & Analysis of Composite Structures II	3
AE4ASM511	Stability & Analysis of Structures II	3
AE4ASM514TU	Continuum Mechanics	4
AE4ASM517	Aircraft Manufacturing Laboratory	6
AE4ASM520	Industrial Composite Manufacturing	3
AE4ASM521	Additive Manufacturing	3
AE4ASM522	Applied Aircraft Aeroelasticity	3
AE4ASM524	Spacecraft Structures Development	3
AE4ASM525	Materials for Space	3



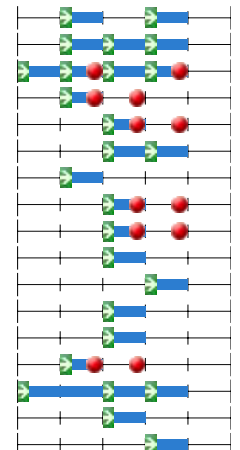
Profile II: Design and Safety of Structures

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
AE4ASM002	Designing Materials with Aerospace Specific Properties	3
AE4ASM003	Linear Modeling (incl. F.E.M)	3
AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
AE4ASM005	Fatigue of Structures & Materials	3
AE4ASM106	Stability & Analysis of Structures I	3
AE4ASM108	Experimental Techniques & NDT	3
AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
AE4ASM506	Fundamentals of Aeroelasticity	3
AE5051	Internship	15
AE5722	Thesis Aerospace Structures & Materials	45
WM0324LR	Ethics and Engineering for Aerospace Engineering	3



Preferred Electives Profile Design and Safety of Structures

AE4006E	Sustainability in Engineering	5
AE4007	AI for Mindfulness in Engineering	5
AE4010	Research Methodologies	2
AE4ASM101TU-22	Polymer Science	4
AE4ASM103	Functional Coatings	3
AE4ASM105	Trinity Exercise	4
AE4ASM110-22	Polymer Composite Manufacturing	4
AE4ASM503	Sheet Metal Forming	3
AE4ASM504	Structural Integrity and Maintenance	3
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
AE4ASM510	Design & Analysis of Composite Structures II	3
AE4ASM511	Stability & Analysis of Structures II	3
AE4ASM514TU	Continuum Mechanics	4
AE4ASM515-22	Characterization of materials and components	4
AE4ASM517	Aircraft Manufacturing Laboratory	6
AE4ASM520	Industrial Composite Manufacturing	3
AE4ASM521	Additive Manufacturing	3



AE4ASM522	Applied Aircraft Aeroelasticity	3	
AE4ASM524	Spacecraft Structures Development	3	
AE4ASM525	Materials for Space	3	
Profile III: Structures for Space			
AE4870A	Rocket Motion	3	
AE4870B	Re-Entry Systems	3	
AE4ASM001	Design of lightweight structures I: Composites & Metals	3	
AE4ASM002	Designing Materials with Aerospace Specific Properties	3	
AE4ASM003	Linear Modeling (incl. F.E.M)	3	
AE4ASM004	Manufacturing of Aerospace Structures & Materials	3	
AE4ASM005	Fatigue of Structures & Materials	3	
AE4ASM106	Stability & Analysis of Structures I	3	
AE4ASM524	Spacecraft Structures Development	3	
AE4ASM525	Materials for Space	3	
AE5051	Internship	15	
AE5722	Thesis Aerospace Structures & Materials	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Structures for Space			
AE4006E	Sustainability in Engineering	5	
AE4007	AI for Mindfulness in Engineering	5	
AE4010	Research Methodologies	2	
AE4ASM101TU-22	Polymer Science	4	
AE4ASM103	Functional Coatings	3	
AE4ASM105	Trinity Exercise	4	
AE4ASM108	Experimental Techniques & NDT	3	
AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5	
AE4ASM110-22	Polymer Composite Manufacturing	4	
AE4ASM503	Sheet Metal Forming	3	
AE4ASM504	Structural Integrity and Maintenance	3	
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4ASM510	Design & Analysis of Composite Structures II	3	
AE4ASM511	Stability & Analysis of Structures II	3	
AE4ASM514TU	Continuum Mechanics	4	
AE4ASM515-22	Characterization of materials and components	4	
AE4ASM517	Aircraft Manufacturing Laboratory	6	
AE4ASM520	Industrial Composite Manufacturing	3	
AE4ASM521	Additive Manufacturing	3	
AE4ASM522	Applied Aircraft Aeroelasticity	3	
AE4ASM526	Spacecraft Thermal Design	3	
AE4S10	Microsat Engineering	4	
Track 5: Flight Performance and Propulsion			
Profile I: Flight Performance			
AE4115-23	Experimental Simulations	4	
AE4130-22	Aircraft Aerodynamics	4	
AE4202	CFD for Aerospace Engineers	3	
AE4204	Knowledge Based Engineering	4	
AE4205	MDO for Aerospace Applications	4	
AE4238	Aero Engine Technology	4	
AE4240	Advanced Aircraft Design I	4	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE5051	Internship	15	
AE5222	Thesis Flight Performance & Propulsion	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Flight Performance			
AE4006E	Sustainability in Engineering	5	
AE4007	AI for Mindfulness in Engineering	5	
AE4010	Research Methodologies	2	
AE4120	Viscous Flows	3	
AE4135	Rotor / wake Aerodynamics	4	
AE4136-24	CFD 2: Discretisation Techniques	5	
AE4138-18	CFD 4: Uncertainty Quantification	3	
AE4139	CFD 3: Large Eddy Simulation	3	
AE4140	Gas Dynamics	3	
AE4180	Flow Measurement Techniques	3	
AE4206-24	Turbomachinery	4	
AE4260A	Fundamental of Aeroacoustics	2	
AE4260B	Experimental Applications of Aeroacoustics	2	
AE4261	Internal Flows	3	
AE4262	Combustion for propulsion and power technologies	4	
AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5	
AE4301	Automatic Flight Control System Design	3	
AE4301P	Exercise Automatic Flight Control System Design	1	
AE4314-21	Helicopter Performance, Stability and Control	4	
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3	
AE4431-23	Aircraft noise	3	

AE4462-17	Aircraft Emissions and Climate Effects	4	
AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4	
AE4ASM003	Linear Modeling (incl. F.E.M)	3	
AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5	
AE4ASM517	Aircraft Manufacturing Laboratory	6	
ET4117	Electrical Machines and Drives	4	
ME45001	Advanced Heat Transfer	4	
ME46060	Engineering Optimization: Concepts and Applications	3	
SET3085	Hydrogen Technology	4	
Profile II: Power and Propulsion			
AE4130-22	Aircraft Aerodynamics	4	
AE4202	CFD for Aerospace Engineers	3	
AE4205	MDO for Aerospace Applications	4	
AE4206-24	Turbomachinery	4	
AE4238	Aero Engine Technology	4	
AE4261	Internal Flows	3	
AE4262	Combustion for propulsion and power technologies	4	
AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5	
AE5051	Internship	15	
AE5222	Thesis Flight Performance & Propulsion	45	
ME45001	Advanced Heat Transfer	4	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Power and Propulsion			
AE4006E	Sustainability in Engineering	5	
AE4007	AI for Mindfulness in Engineering	5	
AE4010	Research Methodologies	2	
AE4115-23	Experimental Simulations	4	
AE4120	Viscous Flows	3	
AE4136-24	CFD 2: Discretisation Techniques	5	
AE4138-18	CFD 4: Uncertainty Quantification	3	
AE4139	CFD 3: Large Eddy Simulation	3	
AE4140	Gas Dynamics	3	
AE4180	Flow Measurement Techniques	3	
AE4204	Knowledge Based Engineering	4	
AE4240	Advanced Aircraft Design I	4	
AE4260A	Fundamental of Aeroacoustics	2	
AE4260B	Experimental Applications of Aeroacoustics	2	
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3	
AE4431-23	Aircraft noise	3	
AE4462-17	Aircraft Emissions and Climate Effects	4	
AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4	
AE4ASM003	Linear Modeling (incl. F.E.M)	3	
AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4ASM517	Aircraft Manufacturing Laboratory	6	
ET4117	Electrical Machines and Drives	4	
ME46060	Engineering Optimization: Concepts and Applications	3	
MS43310	Materials at High Temperature	4	
SET3085	Hydrogen Technology	4	
Track 6: European Wind Energy Master Track Rotor Design			
Profile I: Aerodynamics			
AE4135	Rotor / wake Aerodynamics	4	
AE4180	Flow Measurement Techniques	3	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4W09	Wind Turbine Design	5	
AE4W13	Site Conditions for Wind Turbine Design	3	
AE4W21-14	Wind Turbine Aeroelasticity	2	
AE5912	Thesis Wind Energy Rotor Design	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Aerodynamics			
AE4010	Research Methodologies	2	
AE4117	Fluid-Structure Interaction	4	
AE4W31	Floating Offshore Wind Energy	3	
AE5055	Professional Training	6	
Profile II: Structures and Composites			
AE4ASM105	Trinity Exercise	4	
AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5	
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4W09	Wind Turbine Design	5	
AE4W21-14	Wind Turbine Aeroelasticity	2	
AE5912	Thesis Wind Energy Rotor Design	45	
WM0324LR	Ethics and Engineering for Aerospace Engineering	3	
Preferred Electives Profile Structures and Composites			
AE4010	Research Methodologies	2	

AE4117	Fluid-Structure Interaction	4	
AE4ASM510	Design & Analysis of Composite Structures II	3	
AE4ASM511	Stability & Analysis of Structures II	3	
AE4W31	Floating Offshore Wind Energy	3	
AE5055	Professional Training	6	
Master LR Keuzevakken			
AE4006E	Sustainability in Engineering	5	
AE4010	Research Methodologies	2	
AE4115-23	Experimental Simulations	4	
AE4117	Fluid-Structure Interaction	4	
AE4120	Viscous Flows	3	
AE4130-22	Aircraft Aerodynamics	4	
AE4135	Rotor / wake Aerodynamics	4	
AE4136-24	CFD 2: Discretisation Techniques	5	
AE4138-18	CFD 4: Uncertainty Quantification	3	
AE4139	CFD 3: Large Eddy Simulation	3	
AE4140	Gas Dynamics	3	
AE4143	Hypersonic Aerodynamics	3	
AE4180	Flow Measurement Techniques	3	
AE4185	Fluid Flow Data Processing and Visualization	3	
AE4202	CFD for Aerospace Engineers	3	
AE4204	Knowledge Based Engineering	4	
AE4206-24	Turbomachinery	4	
AE4238	Aero Engine Technology	4	
AE4240	Advanced Aircraft Design I	4	
AE4260A	Fundamental of Aeroacoustics	2	
AE4260B	Experimental Applications of Aeroacoustics	2	
AE4261	Internal Flows	3	
AE4262	Combustion for propulsion and power technologies	4	
AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5	
AE4270	Control and Operations Project	4	
AE4301	Automatic Flight Control System Design	3	
AE4301P	Exercise Automatic Flight Control System Design	1	
AE4311	Nonlinear and Adaptive Flight Control	4	
AE4313-20	Spacecraft Attitude Dynamics and Control	4	
AE4314-21	Helicopter Performance, Stability and Control	4	
AE4316P	Advanced Topics in Aerospace Human-Machine Systems	3	
AE4317	Autonomous Flight of Micro Air Vehicles	4	
AE4320	System Identification of Aerospace Vehicles	4	
AE4321-15	Air Traffic Management	4	
AE4323	Real-time Distributed Flight and Space Simulation	3	
AE4324	Physical Interaction for Aerial and Space Robots	4	
AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3	
AE4351	Robust Flight Control	4	
AE4352	Mathematical and human-inspired decision making	3	
AE4422-20	Agent-based Modelling and Simulation in Air Transport	4	
AE4431-23	Aircraft noise	3	
AE4439	Sustainable Air Transport Modelling Project	2	
AE4446	Airport and Cargo Operations	4	
AE4462-17	Aircraft Emissions and Climate Effects	4	
AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4	
AE4465	Maintenance Modeling & Analysis	4	
AE4889	Special Topics in Astrodynamics	2	
AE4897	Space Engineering Practical	8	
AE4898	Space debris tracking and mitigation	4	
AE4ASM003	Linear Modeling (incl. F.E.M)	3	
AE4ASM101TU-22	Polymer Science	4	
AE4ASM103	Functional Coatings	3	
AE4ASM105	Trinity Exercise	4	
AE4ASM106	Stability & Analysis of Structures I	3	
AE4ASM108	Experimental Techniques & NDT	3	
AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5	
AE4ASM110-22	Polymer Composite Manufacturing	4	
AE4ASM503	Sheet Metal Forming	3	
AE4ASM504	Structural Integrity and Maintenance	3	
AE4ASM505	Non-Linear Modeling (using F.E.M.)	3	
AE4ASM506	Fundamentals of Aeroelasticity	3	
AE4ASM510	Design & Analysis of Composite Structures II	3	
AE4ASM511	Stability & Analysis of Structures II	3	
AE4ASM514TU	Continuum Mechanics	4	
AE4ASM515-22	Characterization of materials and components	4	
AE4ASM517	Aircraft Manufacturing Laboratory	6	
AE4ASM520	Industrial Composite Manufacturing	3	
AE4ASM521	Additive Manufacturing	3	
AE4ASM522	Applied Aircraft Aeroelasticity	3	
AE4ASM524	Spacecraft Structures Development	3	

AE4ASM525	Materials for Space		
AE4ASM526	Spacecraft Thermal Design	3	
AE4S10	Microsat Engineering	4	
AE4S52-21	Collaborative Space System Design Project	5	
AE4T40	Airborne Wind Energy	3	
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4	
AE4W21-14	Wind Turbine Aeroelasticity	2	
AE4W30	Wind Resource and Wind Farm Yield	4	
AE4W31	Floating Offshore Wind Energy	3	
AE5055	Professional Training	6	
AESM2001	Physical Principles of Earth System Observation	5,5	
ET3604LR	Electronic Circuits	3	
ET4117	Electrical Machines and Drives	4	
ME45001	Advanced Heat Transfer	4	
ME45030	Turbulence	5	
ME46060	Engineering Optimization: Concepts and Applications	3	
MS43310	Materials at High Temperature	4	
RO47019	Intelligent Control Systems	4	
SC42035	Integration Project Systems and Control	5	
SET3085	Hydrogen Technology	4	
WI3150TU	Partial Differential Equations A	3	
WI3151TU	Partial Differential Equations B	3	
WI4014TU	Numerical Analysis	6	
WI4019-SP	Non-linear Differential Equations	6	
WI4062TU	Transport, Routing and Scheduling	3	
WI4201	Scientific Computing	6	
WI4212	Advanced Numerical Methods	6	
WI4525TU	Monte Carlo Simulation and Stochastic Processes	5	
WI4771TU	Object Oriented Scientific Programming with C++	3	
WM1115TU	Dutch Elementary 1	3	
Master AE PPD Courses			
Master AE PPD Courses (Personal and Professional Development)			
AE4006E	Sustainability in Engineering	5	
AE4007	AI for Mindfulness in Engineering	5	
AE4010	Research Methodologies	2	
AR1A066	Delft Lectures on Architectural History and Theory	5	
BM51007	Professional Development Journey	1	
CESE4020	Effective and Responsible Engineering	5	
CME4200	Intercultural Relations and Diversity Competence	2	
CME4300	Engineering Asset Management Systems	5	
CME4400	Entrepreneurial Engineering	5	
CME4601	Leadership and Strategic Management	5	
CS4225	Educational Technologies	5	
CT3101	Project Management Basics	5	
EPA114A	Oral Presentation	0	
EPA142A	Ethics of Large System Interventions	5	
EPA143A	Macro-economics for Policy Analysis	5	
EPA223A	Academic Writing	0	
IDEM102	Build Your Start-up 1: Idea Validation	5	
IDEM103	Build Your Start-up 2: Prototyping	5	
MOT9611	Project Entrepreneurship Thesis related	5	
MS3320	The Biography of Art Objects: Understanding the Material Life Cycle of Artwork through Physical and	5	
SEN1131	Institutional Economics for Designing in Socio-technical Systems	3	
SEN1141	Managing Multi-actor Decision-making	5	
SEN1211	Agent-based Modelling	5	
SEN124A	Design in Networked Systems	5	
SEN1522	Electricity and Gas: Market Design and Policy Issues	5	
SEN1541	Sociotechnology of Future Energy Systems	5	
SEN1611	I&C Architecture Design	5	
SEN1622	I&C Service Design	5	
SEN171A	Advanced Evaluation Methods for Transport Policy Decision-making	5	
SEN1721	Travel Behaviour Research	5	
SEN1741	Innovations in Transport and Logistics	5	
SEN9110	Simulation Masterclass	5	
SEN9235	Game Design Project	5	
SEN9720	Logistics and Supply Chain Innovations	5	
SEN9725	Supply Chain Gaming	5	
TPM001B	Sociotechnology of Future Energy Systems	5	
TPM007A	Talk Like TED	3	
TPM010A	Cyber Crime Science	5	
TPM018A	English Grammar for the University	2	
TPM021A	Economics and Finance	6	
TPM022A	Intermediate Economics	5	
TPM025B	User-Centred Security	5	
TPM027A	Cyber Risk Management	5	
TPM028B	Decision Making for Logistics and Transport Systems	5	

TPM029A	Responsible Innovation	3	
TPM030A	Introduction to Cloud as Infrastructure: The effects of the new business of computing on practice	5	
TPM033A	Sustainable Innovation and Transitions	3	
TPM041A	Diversity and inclusion in Science, Technology, Engineering and Mathematics	3	
TPM043A	Resilience of socio-technical-environmental systems	5	
TPM0637SET	Economic Policy for Sustainable Energy	4	
TPM0931SET	Sustainable Energy Innovation and Transitions	4	
TPM301C	Spoken English for International Communication (for Master students)	2	
TPM302B	Spoken English for Academic Purposes Advanced	2	
TPM305A	Writing a Masters Thesis in English	2	
TPM306A	Writing Academic English using AI	2	
TPM308A	Higher Advanced English for the University (for Master students)	3	
TPM401A	Technology Entrepreneurship and Innovation	5	
TPM402A	Technology Entrepreneurship and Health	5	
TPM403SET	Technology Entrepreneurship and Sustainability	4	
TPM404B	Technology Entrepreneurship and Internationalization	5	
TPM405A	Patent Law and Patent Policy	5	
TPM406A	Corporate Entrepreneurship and Start-ups	6	
TPM407B	Technology based marketing and sales	5	
TPM408A	Financing Technology Ventures	5	
TPM409A	Digital technology entrepreneurship and management	5	
TPM411A	Idea to Start-up IT & AI	5	
TPM412A	Idea to Start-up Health & Life Sciences	5	
TPM413A	Idea to Start-up Energy & Sustainability	5	
TPM414A	Idea to Start-up Deep Tech	5	
TPM415B	Idea to Start-up - Climate Action	5	
TPM416A	Turning Technology into Business	6	
TPM420A	Ready to Start-up	6	
TPM423A	Technology Venture Development	15	
TPM425B	Start-up Challenge	3	
TPM426B	Start-up Challenge - Advanced	5	
TPM959A	Technology Battles	3	
WM-ITAV-4010	Scientific Writing	2	
WM0201TU	Technical Writing	2	
WM0203TU	Oral Presentations	2	
WM0204TU	Meetings and Negotiations	2	
WM0312CIE	Philosophy, Technology Assessment and Ethics for CIE	4	
WM0318TN	Argumentation and Philosophy of Science	3	
WM0320TU	Ethics and Engineering	3	
WM0329TU	Ethics and Engineering	6	
WM0348TU	Christian Philosophy: Recent Developments	3	
WM0349WB	Philosophy of Engineering Science and Design	3	
WM0353TU	Climate Ethics	3	
WM0361TU	Filosofie I: techniek- en wetenschapsfilosofie	3	
WM0371TU	Todays Philosophy of Technology and MSc Ethics	3	
WM0375TU	Ethics of Technological Risks	3	
WM0376TU	Ethics of Technological Risks	5	
WM0801TU	Introduction to Safety Science	3	
WM1028AM	Ethics for Applied Mathematics	3	
WM1301TU	Ethics of Transportation	3	
WM1302TU	Ethics of Transportation	5	
WM1401TU	Ethics of Healthcare Technologies	3	
WM1402TU	Ethics of Healthcare Technologies	5	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Master LR 2024

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 1: Aerodynamics and Wind Energy

MSc Coordinator	Dr.ir. A.H. van Zuijlen
Contact for students	Programme coordinator Aerodynamics: A.H. van Zuijlen Programme coordinator for Wind Energy: F. Avallone
Introduction 1	The master Track Aerodynamics and Wind Energy combines fundamental and applied research disciplines of aerospace and wind-power systems focusing on development and optimization. The track is of interest to students who want to acquire expertise with advanced aerodynamic tools (experimental and numerical modeling techniques) or desire experience with the applications of aerodynamics in design and sustainable energy conversion. The Track offers two Profiles: Aerodynamics and Wind Energy. The first is supported by the Aerodynamics research group, while the second is supported by the Wind Energy research groups.
Program Goals	Learning objectives of the MSc Track: The objective of this track is to provide students with the opportunity to become a specialist with specific knowledge in analysis of aerodynamic systems, and the methods used for their application in design. The students will obtain a thorough fundamental basis in aerodynamics as well as in modern techniques to investigate such systems.
Program Structure 1	In the first year of the MSc, the student follows core courses from the Aerodynamics and Wind Energy program, which is designed to provide a fundamental background in fluid mechanics and its applications. Aerodynamics analysis by means of experimental, theoretical and computational methods is also taught in the first year. Concurrently, the student follows three or four elective courses associated with a selected profile. After completing the courses and an internship, the student joins one of the research groups for a literature study and final thesis. The literature study and final thesis are performed under the supervision of a scientific staff member, working on a project of current relevance to the field. In some cases, the thesis work may be carried out in conjunction with a company or a research institute. The supervisor and project should be chosen early in the MSc to aid the selection of pertinent courses, nevertheless they can be defined at latest during the second part of the course period.

Profile I: Aerodynamics	
Program Coordinator	Dr.ir. A.H. van Zuijlen
Introduction 1	The Aerodynamics Profile is concerned with the understanding and control of flows associated with aerospace vehicles. On the fundamental side, it considers the design of measurement and computational methods for the provision of detailed flow descriptions. On the applied side, it considers the design of flow control devices and techniques to maximise vehicle performance. The course program of the profile focuses on the knowledge which must be acquired to understand modern flow analysis. This includes courses in gasdynamics and viscous flows, computational fluid dynamics, flow measurement techniques and experimental simulations. It also includes practical courses which establish the context of this knowledge, including aircraft and rotor aerodynamics. The final thesis, which combines the literature study and thesis project, can be performed within the Aerodynamics research group, our partner research centres, or industry. The topics cover a large area, from the development of new computational methods for difficult flow problems, to the design of high-speed flow measurement systems, to the application of such methods and systems in the design of aerospace vehicles. A key advantage of the profile is its focus on fundamental knowledge. As flows are ubiquitous in engineering systems, much of the experience gained in the profile is easily transferable to other fields. Graduates of the Aerodynamics profile can therefore expect a wide range of employment opportunities.

AE4115-23	Experimental Simulations	4
Responsible Instructor	Dr.ir. T. Sinnige	
Instructor	Dr.ir. W.J. Baars	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling; fractional analysis; dimensional analysis (Buckingham-Pi theorem); rigorous methods of obtaining dimensionless parameters; 3 Fundamentals of aerodynamic similarity: fundamentals of aerodynamics necessary for the interpretation of similarity between aerodynamic test results and full-scale; 4 Mach number and Reynolds number effects: practical considerations with respect to implications of dissimilarities between model-scale testing and full-scale applications; 5 Wind tunnel facilities: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 6 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 7 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 8 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 9 Design of experiments: factorial designs, response-surface models; 10 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report; 11 Engine integration: engine simulators, performance of those and their similarity with full-scale engines; 12 Acoustic testing: noise generation mechanisms and their appropriate scaling; definition of sound pressure levels; measurement requirements and devices; far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report;	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stromungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports (Q2 + Q3) and oral exam (wk 3.10 or 4.1).	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter.	

AE4120	Viscous Flows	3
Responsible Instructor	M. Kotsonis	
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Instructor	Dr.ir. T. Michelis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Prof.dr.ir. L.L.M. Veldhuis	
Instructor	L.T. Lima Pereira	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130-22, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during the first 3 weeks of Q1).	
Assessment	The assessment is done through 5 assignments that are performed on a personal basis. The assignments are to be combined in 1 report that should be delivered after the finalization of the lecture at the end of Q2. Additional information on the timing and contents of the assignments will be presented on BrightSpace.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Responsible Instructor	W. Yu	
Gast Instructor	Dr.ir. T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic aerodynamics, level BSc Aerospace Engineering.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade elementmomentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsens Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4136-24	CFD 2: Discretisation Techniques	5
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results. After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignement and oral examination.	
Enrolment / Application	Brightspace	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4140	Gas Dynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 none	
Course Language	English	
Expected prior knowledge	AE2130-III Compressible aerodynamics (chapter 7 - 12 of Fundamentals of Aerodynamics by J.D. Anderson) WI3150TU Partial differential equations	
Course Contents	Knowledge of computer programming (Python, Matlab or similar) in order to successfully complete the tasks - How to apply the basic laws of mechanics and thermodynamics to describe compressible flows. - How can we properly describe a compressible flow field using the Euler equation + jump relations. - Understand the concept of characteristics and invariants in the context of linear and non-linear flows. - How to apply characteristics for non-linear flows for the computation of isentropic unsteady flows. - Understand the relation between shock waves and characteristics. - Application of Hugoniot and Poisson curves to solve a Riemann problem. - Definition of characteristics for 2D steady flows, and similarity with 1D unsteady flows (time-like and space-like). - How can use the method of characteristics and method of waves to compute a 2D compressible flow field. - Investigate the effect of viscosity and heat transfer in a 1D flow (Fanno and Rayleigh flow).	
Study Goals	At the end of this course, the student will be able to: - Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. - Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow and discuss the terms. - Derive the jump relations for the Euler equations and describe their relation to shock waves. - Discuss the role of entropy in combination with the jump relation for a correct description of a flow field. - Explain the concept of characteristics and invariants for 1D unsteady and 2D steady flows and how to use them for flow field computations (MOC). - Understand the role of characteristics in shock wave formation, elaborate on the theory of simple waves. - Derive the equations governing 1D flows through channels and nozzles in presence of viscosity and heat transfer. Explain the physical phenomena and processes that occur.	
Education Method	Lectures and self study	
Literature and Study Materials	Reader (can be downloaded from Brightspace) Further recommended literature M.J. Zucrow, J.D.Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. John D. Anderson jr., Modern Compressible Flow with Historical Perspective 2nd edition MacGraw-Hill 1990; H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001	
Prerequisites	Compressible aerodynamics Fundamentals of Aerodynamics by J.D. Anderson ch 7 - 10	
Assessment	Take home tasks + oral examination	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	M. Li	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Introduction to measurement techniques 2. Flow visualization (classroom assignment) 3. Pressure measurements 4. Hot Wire Anemometry (lab exercise) 5. Laser Doppler Velocimetry 6. Particle Image Velocimetry (classroom assignment+lab exercise) 7. InfraRed Thermography 8. Schlieren, Shadowgraphy and Background Oriented Schlieren 9. Fundamentals of data processing	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written exam and report on final assignment.	
Permitted Materials during Tests	none	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5122	Thesis Aerodynamics & Wind Energy	45
Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal). If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go' (fail). Please refer to Brightspace for more information.	

WI3150TU		Partial Differential Equations A	3
Responsible Instructor	T. Akkaya		
Contact Hours / Week x/x/x/x	4/0/0/0		
Education Period	1		
Start Education	1		
Exam Period	1 2		
Course Language	English		
Expected prior knowledge	Calculus, ordinary differential equations		
Course Contents	I: (WI3150TU) Introduction. Types of second order equations. Initial and initial boundary value problems. Fourier series. Quasi-linear, first order partial differential equations. Waves and reflections of waves. Separation of variables. Sturm-Liouville problems. Parabolic, elliptic and hyperbolic equations. Maximum principle. Diffusion and heat transport problems. Lectures (3 ECTS). II: (WI3151TU) Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).		
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.		
Education Method	Lectures		
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972.		
Assessment	Written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. The resit consists of a written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		
Co-Instructor	Dr.ir. W.T. van Horssen		
Co-Instructor	Dr. H.M. Schuttelaars		

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Aerodynamics

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub., 1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Responsible Instructor	Dr. S.J. Hulshoff	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4143	Hypersonic Aerodynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Instructor	Dr.ir. L.M.G.F.M. Walpot	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* AE2130-III Aerodynamics 2 (Compressible Aerodynamics) or equivalent. In case you did not have such a course, please study chapters 7 to 12 of Fundamentals of Aerodynamics by J.D. Anderson * It is very highly recommended to also follow AE4120 (Viscous flows) * AE4140 Gasdynamics is nice to follow as well but not absolutely necessary	
Parts	Lecture and study material 1. Introduction: characteristics of hypersonic flow; applications. 2. Inviscid flow: 2D shocks and expansions; Newtonian theory. 3. Tangent- wedge/tangent cone-, shock-expansion theories; similarity parameters. 4. Euler equations; small perturbation theory; hypersonic equivalence principle; blast-wave theory. 5. Thin shock layer theory; computational results. 6. Viscous flow: boundary layer equations; laminar boundary layers. 7. Transition; turbulent boundary layers. 8. Viscous interactions; flat plate application. 9. Heat transfer: convective heating; results for flat plate, cone, blunt body. 10. Laminar and turbulent stagnation region heating; radiation: effects on surface heating; radiation from the shock layer; nocturnal radiation. 11. High temperature effects; chemical reactions; inviscid equilibrium flow; difference between equilibrium flow and frozen flow. 12. Non-equilibrium flow; high temperature effects in flow through normal and oblique shocks. 13. High temperature effects in flow through nozzles. 14. Chemical reacting viscous flow	
Course Contents	Hypersonic Flow: Inviscid flow, compressions and expansions, surface inclination methods, similarity rules, hypersonic small perturbations, Euler equations. Viscous flow, boundary layers, viscous interactions, heat transfer, radiation effects. High temperature effects, inviscid and viscous equilibrium and nonequilibrium flow. Rarefied gasdynamics.	
Study Goals	This course is designed to provide the student with fundamental knowledge of hypersonic aerothermodynamics. The course is set up to deliver engineering tools for conceptual design problems together with a physical understanding of various aspects of the broad gamma of hypersonic flow.	
Education Method	Lecture	
Literature and Study Materials	J.D. Anderson, High temperature gas dynamics, , - ISBN 0070016712 Recommended literature J.J. Bertin, Hypersonic aerothermodynamics, , - ISBN 1563470365	
Assessment	Oral exam	

AE4185	Fluid Flow Data Processing and Visualization	3
Responsible Instructor	Dr. D. Modesti	
Responsible Instructor	Dr. N.A.K. Doan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is a general introduction to data processing techniques used in fluid mechanics to analyse unsteady flows. Specifically, techniques related to the flow visualization and the postprocessing of numerical data will be presented. The overall content of the course is: 1. Review of fundamentals of turbulent flows (energy cascade, Reynolds-averaged equations, Reynolds stress, turbulent kinetic energy, vorticity equation, turbulence spectrum, energy budget, ...) 2. Characteristics of flow datasets (Nyquist criterion, sampling and resolution, first principle of visualization, vortex/Q criteria, ...) 3. Statistical analysis in flows - Deterministic/random data - Averaging (ensemble, condition, phase) - Probability: PDFs, statistical moments, conditional PDFs - Correlation function - Ergodic hypothesis - Statistical convergence and uncertainty 4. Temporal and spectral data processing - Fourier transform - Physical/spectral space and multi-dimensional FFTs - Auto/cross spectra - Integral length/time scale - Convolution (filtering) and transfer function 5. Modal analysis and coherent structure eduction - Proper Orthogonal Decomposition - Dynamic Mode Decomposition - Basics of reduced-order modelling	
Study Goals	At the end of this course, the student should be able to: - Visualize fluid mechanics datasets using relevant quantities - Analyse fluid mechanics datasets using statistical methods - Apply temporal and spectral analysis methods to analyse flow datasets - Postprocess fluid mechanics datasets using modal-based methods	
Education Method	The course will be a combination of lectures, scheduled tutorials and homework assignments.	
Assessment	The course will be assessed with a final assignment (60% of the final grade) and an oral exam (40% of the final grade). To get a passing grade (6.0 or higher), the student must get at least a 5.0 on both the final assignment and the oral exam.	

AE4240	Advanced Aircraft Design I	4
Responsible Instructor	Dr.ir. R. Vos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The design of high-subsonic transport aircraft is the topic of this course. The main goal is to demonstrate how the geometry of the aircraft can be shaped to ensure favorable flow conditions that satisfy requirements on performance, stability and controllability. The course is divided in the following lectures: 1. Design objectives 2. Boundary-layer aerodynamics and subsonic cruise drag. 3. Relation between supersonic velocity and pressure coefficient. Relation between geometry and pressure distribution. 4. Interference drag and area ruling. 5. Pressure distribution on airfoil sections at high and low speed. Development of the transonic airfoil. Design of supercritical and sonic airfoils. 6. Reynolds number effects. 7. Low-speed stalling characteristics of airfoil sections. Reynolds number and Mach number effects on maximum lift coefficient. High-speed stall (buffet) on airfoil sections. 8. Airfoils equipped with high lift devices: mutual interaction, types of flow separation. 9. Development of the swept-wing concept: historical background, principle of wing sweep, transonic flow characteristics, effect on boundary layer, effect on maximum lift coefficient, fences. 10. First generation of swept wing aircraft: tip-stall, aeroelastic deformation. 11. Pressure distribution on finite swept wings: spanwise drag distribution, forward sweep, fuselage interference, improving velocity distribution at root and tip, minimizing pitching moment. Design considerations for pressure distributions on finite swept wings. 12. Stalling characteristics of full scale aircraft: stall requirements, complexity vs. maximum lift, spanwise stall onset, deep stall. 13. Horizontal tail surface design: functions and requirements, planform shape, aeroelastic effects, elevator lock. 14. The buffet onset boundary: requirements, effect on flight envelope. Flight characteristics between MMO and MD: requirements, Mach tuck, Mach trim compensation. 15. Vertical tail surface design: functions and requirements, planform shape, effect of dorsal fin. Control surface design: functions and requirements, linearity, aeroelasticity, stall. 16. Wing control surfaces: spoilers and ailerons 17. Propeller slipstream effects: effect on spanwise lift distribution, effect on longitudinal balance, effect on directional balance in OEI condition. 18. Engine intake design: lip design for take-off and cruise conditions, drag in OEI condition. 19. Exhaust design: effect of ambient pressure, boat tail drag. Thrust reversers: types, interference with empennage. 20. Examples of modern wing design	
Course Contents	In this course it is demonstrated how the exterior design of high-subsonic transport aircraft can be manipulated in order to meet specified performance objectives and satisfy relevant certification constraints. It is shown what geometric characteristics of the airplane allow for a high operating Mach number while keeping fuel burn at acceptable levels. In particular, it is demonstrated how the exterior of all major airplane components can be shaped such as to adequately handle low-subsonic and transonic flow. It is demonstrated how the aerodynamic design and the operation at low speed and high speed affects the airplane in terms of balance, stability, aeroelastic behavior, weight, controllability, cruise performance, and field performance. Through many historic and contemporary examples the student learns to relate the functionality of high-subsonic transport airplanes to their external design.	
Study Goals	Students are able to: 1. Recognize the aerodynamic phenomena occurring around external shapes at a variety of conditions such as Mach number, Reynolds numbers or angles of attack. 2. Derive how these phenomena can be affected by the detailed external shape and interposition of various airplane components such as wing, fuselage, tail, nacelle, pylon, and fuselage. 3. Show how wing and tail movables can be applied to locally affect the flow at low subsonic and/or high-subsonic conditions. 4. Explain the effect of the aerodynamic characteristics on the airplanes cruise performance, field performance, stability, balance, and controllability. 5. Identify the advantages and disadvantages of various external design features on the airplanes aerodynamic performance and cross-disciplinary characteristics such as aeroelasticity, weight, or practical limitations resulting from the operational use of the aircraft.	
Education Method	Lectures	
Literature and Study Materials	Mandatory textbooks: - Aerodynamic Design of Transport Aircraft, Obert, E, IOS Press, 2009 - Introduction to Transonic Aerodynamics, Vos, R. and Farokhi, S, Springer, 2015	
Prerequisites	Aerospace Design and Systems Engineering Elements I Aerospace Design and Systems Engineering Elements II Aerodynamics I (subsonic aerodynamics) Aerodynamics II (supersonic aerodynamics) Systems Engineering and Aerospace Design Flight Dynamics and Simulation	
Assessment	Written exam	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor	Dr. D. Ragni	
Instructor	Prof.dr.ir. M. Snellen	
Co-responsible for assignments	Dr. R. Zamponi	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources.	
Study Goals	The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4262	Combustion for propulsion and power technologies	4
Responsible Instructor	Dr. I. Langella	
Responsible Instructor	F. De Domenico	
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	G. Kelesidis	
Contact Hours / Week x/x/x/x	0/0/0/4+2lab	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology	
Summary	This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.	
Course Contents	1. Thermodynamics of combustion 2. Chemical kinetics 3. Transport processes 4. Premixed and diffusion flames 5. Laminar and turbulent reactive flows 6. Combustion experiments and diagnostics 7. Combustion modelling 8. Laboratory Tutorial & Exercise 9. Practical combustion systems for aerospace applications 10. Pollutants and emissions 11. Hydrogen combustion physics and technology	
Study Goals	Write chemical equations and perform stoichiometry calculations. Compute heating values of fuels, enthalpies of formation and adiabatic flame temperature. Apply principles of thermodynamics to analyse reacting mixtures at chemical equilibrium. Describe the fundamentals of the physics involved in premixed- and diffusion flames, laminar and turbulent. Discuss the physics related to turbulent reacting flows and chemistry interaction. Explain experimental measurement techniques and their role in understanding combustion. Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems. Provide phenomenological descriptions of pollutant formation of conventional combustion systems. Elaborate on the potential and challenges related to the use of alternative fuels in combustion systems.	
Education Method	The course consists of weekly lectures and exercise sessions (practicals). In addition, the students are required to complete a project assignment and perform a laboratory exercise. The project assignment is relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with the investigation of methane or hydrogen laminar flames. The students will gain hands-on experience in designing the experimental matrix, acquiring data, performing data analysis and writing a technical report. Both laboratory exercise and assignment are mandatory to be admitted to the oral exam.	
Course Relations	Aero Engine Technology Hydrogen Technology	
Literature and Study Materials	Slides, lecture notes and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION;	
Books	J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"	
Assessment	To pass the course, the students must Complete laboratory exercise and project assignment; and Pass the oral exam with a minimum grade of 5.5 out of 10 The project assignment consists of several parts, including analysis on the data collected during the laboratory exercise and further theoretical and numerical analyses of combustion phenomena. Note attendance to the laboratory exercise and submission of the assignment are both compulsory for the admission to the oral exam. The work on the assignment should be summarized in a report, to be written by a single student or in a group of maximum three. The final grade is the result of the oral examination (60%) and the grade obtained in the assignment (40%).	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Adisson-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE4ASM514TU	Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.	
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.	
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.	
Education Method	Lectures.	
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.	
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.	

AE4T40	Airborne Wind Energy	3
Responsible Instructor	Dr.ing. R. Schmehl	
Instructor	Dr. D. Ragni	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	BSc. completed	
Parts	Lectures include latest material on the developments in the field of airborne wind energy, with a thematical focus on kite power generation.	
Course Contents	Lecture 1: Introduction to airborne wind energy Lecture 2: History of kite applications Lecture 3: Implemented AWE concepts Lecture 4: Physics of tethered flight Lecture 5: Theory of tethered flight in three dimensions Lecture 6: Flight navigation and maneuvering Lecture 7: Operation of AWE systems Lecture 8: Practical use of AWE systems Lecture 9: AWE in the future energy system Lecture 10: Design and flight dynamics of kites Lecture 11: Aerodynamics of kites Lecture 12: Aeroelasticity of kites Practical: AWE workshop with Jupyter Notebook	
Study Goals	Give an overview of developments in airborne wind energy, developng mathematical models for kites and designing kite power systems for industrial scale generation of energy.	
Education Method	Lecture & workshop	
Prerequisites	Any MSc student can take this course, given a good understanding of vector analysis and basic trigonometry.	
Assessment	Written exam	
Set-up	12 lectures of two lecture hours each in the first semester followed by a written examination.	

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr.ir. M.B. Zaayer	
Instructor	Prof.dr. D.A. von Terzi	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: - describe the components and configurations of wind turbines - describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine - describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design - explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. - identify the drivers of dynamic behaviour and calculate their typical values - describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure - describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Remarks	This course is an elective course for students from various faculties (AE, AS, ME, EEMCS, CEG, TPM). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	W. Yu	
Instructor	Dr.ir. C.J. Simao Ferreira	
Gast Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Preferred: AE4ASM506 (Aeroelasticity). It's preferred that the student has completed or is attending this or an equivalent course.	
Course Contents	The course will cover the main structural and aerodynamic phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. What are the key characterized parameters for an aeroelastic system? 2. How to construct a structural model of a wind turbine blade? 3. How to define if resonance occurs to a system and understand its relevance to wind turbines and how to mitigate it? 4. How to define if instability occurs to a system and its relevance to wind turbines and understand the role of aerodynamic damping? 5. Dynamic inflow and dynamic stall. 6. How to construct an aeroelastic model by coupling the structural and aerodynamic effects?	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand the dynamics and loads of a wind turbine due to the strong coupling of a) external loads(aerodynamics) b) internal structural loads c) inertia loads; 2. solve the dynamic structural response based on the equations of motions; 3. Calculate the unsteady aerodynamic loads of a wind turbine under dynamic inflow and dynamic stall; 4. apply the knowledge to program their own aeroelastic models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE4W30	Wind Resource and Wind Farm Yield	4
Responsible Instructor	Prof.dr. S.J. Watson	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	This course will cover how the output of wind farm arrays is predicted based on modelling of the wind resource and combined wake interactions. The first part of the course will cover the different wind resource models which are used ranging from simple mass consistent models, to linearised solvers, computational fluid dynamics models and mesoscale models. The use of models in both onshore and offshore environments will be covered including the importance of atmospheric stability. The second part of the course will consider the application of models to assess the yield from wind farms considering wakes effects. Wake models will be reviewed including simple semi-empirical models to kinematic models and the modelling of wakes in CFD models. Once again, the importance of atmospheric stratification will be considered and the potential impact on wind farms of certain mesoscale phenomena, e.g. gravity waves offshore, will be assessed and quantified. Large wind farm phenomena will be studied including the so-called deep array effect and wind farm blockage.	
Study Goals	Course Aim: to give students a firm foundation in how to predict wind farm yield from prevailing wind conditions. Course Objectives: 1) To understand what tools are available to predict wind resource over a wide area 2) To apply to such models to potential wind farm sites onshore and offshore 3) To quantify the importance of terrain effects and atmospheric stability on the wind resource 4) To understand what tools are available to predict wake effects in a large wind farm 5) To apply wake models in a realistic wind farm situation 6) To appreciate the importance to wind farm yield of phenomena such as terrain complexity (onshore), proximity to the coast (offshore), atmospheric stability, mesoscale effects (e.g. gravity waves), wind farm layout and wind farm scale	
Education Method	Students will learn through a combination of guided reading, group work, online resources (e.g. lectures, videos, quizzes), face-to-face lectures, coursework and exams.	
Assessment	The students will undertake two course assignments (resource assessment and wind farm yield prediction) worth 30% each, spread out during the course and an oral exam worth 40% at the end of the course. Assignment 1: application of a model to assess and optimise wind resource and energy yield for an onshore site with a report to be assessed. Assignment 2: application of a model to assess and optimise energy yield for an offshore site incorporating different wake models with a report to be assessed. Exam: oral test to assess understanding of the principles of wind resource assessment and wind farm yield prediction.	

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Prof.dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Instructor	Dr. F. Taruffi	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'). The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Instruction if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Process & Energy	

ME45030	Turbulence	5
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Prof.dr.ir. J. Westerweel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Advanced Fluid Dynamics (or equivalent)	
Course Contents	The course provides an introduction to the physical aspects of turbulent flows.	
Study Goals	(1) basic equations that describe turbulence; (2) transition to turbulence; (3) turbulence closure models; (4) energy cascade; (5) turbulent dispersion; (6) selected topics in turbulence	
Education Method	lectures and instructions	
Literature and Study Materials	Turbulence (2016) by Nieuwstadt, Boersma & Westerweel	
Assessment	Written exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Fluid Mechanics Mechanics Numeric Methods Physics	
Department	ME Department Process & Energy	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Prof.dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Dr. O. Nejadseyfi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be 5.8 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest decimal. The maximum bonus score is 0.8. The bonus only applies to reports with grade ≥ 5.5 and submitted before the first report deadline.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	ME Department Precision & Microsystems Engineering	
Contact	Prof.dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

WI3151TU	Partial Differential Equations B	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	WI3150TU	
Course Contents	Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures and a feedback assignment.	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	The feedback assignment and a written exam. The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The resit consists of a written exam (100%). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Judgement	The feedback assignment is mandatory, and as long as the assignment is not completed no final grading will be given.	
Co-Instructor	Dr. H.M. Schuttelaars	

WI4014TU	Numerical Analysis	6
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	Classification of partial differential equations and boundary and initial conditions. Maximum principle, uniqueness and non-negativity of the solution, elements of variational calculus. Finite- Difference, Finite-Volume and Finite-Element discretization methods. Solution methods for time-dependent and nonlinear problems. Direct and iterative linear solvers.	
Study Goals	Students will learn: the principles of finite-difference, finite-volume, and finite-element discretizations; the accuracy, consistency, and convergence of numerical solutions; the basic ideas of numerical linear algebra. They will be able to implement a simple PDE solver and to work with existing sophisticated numerical software.	
Education Method	Lectures (possibly, online), homework assignments, programming assignments, written report.	
Literature and Study Materials	[1] J.van Kan, A.Segal, and F.Vermolen, Numerical methods in scientific computing. VSSD, Delft, 2005, improved 2008, ISBN-13 978-90-71301-50-6 [2] Python programming: - Python Help: https://www.python.org/about/help/ - Numpy for Matlab users: https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html [3] FENICS Project: https://fenicsproject.org/ - Installation: https://fenicsproject.org/download/ - Tutorial: https://fenicsproject.org/tutorial/	
Prerequisites	Calculus of functions of several variables, vector calculus, introductory course on numerical analysis (numerical integration, error analysis), introductory course on partial differential equations, some experience with Python or Matlab.	
Assessment	Theoretical and computational assignments, graded reports.	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). Mathematics Modelling Numeric Methods	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The final grade of the course consists of the following components: - homework exercises deadline start of Q2 leads to grade G1, - take home exam deadline half of January grade G2 - written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework exercises: resubmit deliverable - take home exam: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	mr.ir.drs. V.N.S.R. Dwarka	

WI4212	Advanced Numerical Methods	6
Responsible Instructor	Dr.ir. S. Jain	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Introductory numerical analysis, Introductory partial differential equations, Introductory continuum mechanics.	
Course Contents	This course is an introduction to hyperbolic partial differential equations and a powerful class of numerical methods for approximating their solution, including both linear problems and nonlinear conservation laws. These equations describe a wide range of wave propagation and transport phenomena arising in nearly every scientific and engineering discipline. Several applications are described in a self-contained manner, along with much of the mathematical theory of hyperbolic problems. High-resolution versions of Godunov's method are developed, in which Riemann problems are solved to determine the local wave structure and limiters are then applied to eliminate numerical oscillations. These methods were originally designed to capture shock waves accurately, but are also useful tools for studying linear wave-propagation problems, particularly in heterogeneous material.	
Study Goals	apply numerical methods to hyperbolic systems, study convergence, stability, monotonicity, know methods for 2 dimensional hyperbolic systems	
Education Method	Lectures	
Literature and Study Materials	Finite volume methods for hyperbolic problems R.J. LeVeque Cambridge, UK: Cambridge University Press, 2002. # ISBN-10: 0521009243 # ISBN-13: 978-0521009249	
Assessment	The final grade of the course consists of the following components: - Homework assignments - Group assignment - oral exam Final grade calculation: Provided that G_3 is larger than or equal to 6, the final grade is computed by the formula: $(G_1 + G_2 + 2 \cdot G_3)/4$. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework assignments: resubmit deliverable - Group assignment: resubmit deliverable - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. C.A. Urzúa Torres	

Profile II: Wind Energy	
Program Coordinator	Dr.ir. D.A.M. De Tavernier
Introduction 1	Profile II Wind Energy The Wind Energy profile focuses on methods and systems of energy extraction from wind. Both wind turbine and kite power systems are addressed. The profile offers courses on atmospheric wind conditions, rotor aerodynamics, wind turbine design, design of rotor blades and kite systems and of offshore wind farms. Graduation projects range from numerical wind field modeling and wind turbine aerodynamics to design aspects of large offshore wind farms, kites and other non conventional wind energy systems. Projects related to energy extraction from other moving media such as water currents and waves are also possible. After successful completion of the MSc programme you: - Will have in-depth knowledge of low speed aerodynamics and its application to energy extracting devices such as wind turbines and kites; - Will have thorough system knowledge regarding these energy extraction devices; - Will be able to apply advanced analysis and design techniques in at least one of the following areas: wind turbines, wind farms or kite power systems; - Will have insight in related disciplines such as boundary layer meteorology, structural design and dynamics of structures, control and operations. - Will have knowledge of the noise generation mechanisms in wind-turbines, of sound propagation in realistic terrain and atmospheric conditions, and the implications of noise reduction on wind-farm economy.

AE4120	Viscous Flows	3
Responsible Instructor	M. Kotsonis	
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Instructor	Dr.ir. T. Michelis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Responsible Instructor	W. Yu	
Gast Instructor	Dr.ir. T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic aerodynamics, level BSc Aerospace Engineering.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade elementmomentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsens Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE4T40	Airborne Wind Energy	3
Responsible Instructor	Dr.ing. R. Schmehl	
Instructor	Dr. D. Ragni	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	BSc. completed	
Parts	Lectures include latest material on the developments in the field of airborne wind energy, with a thematic focus on kite power generation.	
Course Contents	Lecture 1: Introduction to airborne wind energy Lecture 2: History of kite applications Lecture 3: Implemented AWE concepts Lecture 4: Physics of tethered flight Lecture 5: Theory of tethered flight in three dimensions Lecture 6: Flight navigation and maneuvering Lecture 7: Operation of AWE systems Lecture 8: Practical use of AWE systems Lecture 9: AWE in the future energy system Lecture 10: Design and flight dynamics of kites Lecture 11: Aerodynamics of kites Lecture 12: Aeroelasticity of kites Practical: AWE workshop with Jupyter Notebook	
Study Goals	Give an overview of developments in airborne wind energy, developng mathematical models for kites and designing kite power systems for industrial scale generation of energy.	
Education Method	Lecture & workshop	
Prerequisites	Any MSc student can take this course, given a good understanding of vector analysis and basic trigonometry.	
Assessment	Written exam	
Set-up	12 lectures of two lecture hours each in the first semester followed by a written examination.	

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr.ir. M.B. Zaayer	
Instructor	Prof.dr. D.A. von Terzi	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: - describe the components and configurations of wind turbines - describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine - describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design - explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. - identify the drivers of dynamic behaviour and calculate their typical values - describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure - describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Remarks	This course is an elective course for students from various faculties (AE, AS, ME, EEMCS, CEG, TPM). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Prof.dr. D.A. von Terzi	
Instructor	Dr.ir. M.B. Zaayer	
Instructor	Dr.ir. D.A.M. De Tavernier	
Instructor	Dr. D. Zappalá	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control The use of standards for load calculations Other optional material (tbd)	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): - AE4W13 Site conditions for wind turbine design - AE4135 Rotor / wake aerodynamics - AE4W21-14 Wind turbine aeroelasticity - AE4W31 Floating offshore wind energy - ET4117 Electrical machines and drives - OE44135 Offshore wind support structures	

AE4W13	Site Conditions for Wind Turbine Design	3
Responsible Instructor	Dr.ir. W.A.A.M. Bierbooms	
Instructor	Dr.ing. R. Schmehl	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 Different, to be announced	
Course Language	English	
Expected prior knowledge	It is assumed that the student has a basic understanding of fluid mechanics and stochastics (on Brightspace self tests are available). A basic skill of Matlab is assumed: you should be able to read from and write to a text file, write and/or modify simple functions (m-files). Furthermore, it is recommended that you already have a basic knowledge of wind turbines; e.g. AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Course Contents	Introduction Meteorology Atmospheric Boundary Layer Fluid Dynamics Turbulence Wind power meteorology Measurement instruments Statistics Methods in turbulence Offshore wind climate External conditions in standards Wave and current climate and loading Example analysis site data Turbulence spectra Satellite remote sensing techniques Simulation of turbulent wind fields and gusts	
Study Goals	After successfully completing the course, students will be able to: Discuss the relevance of particular site data for wind turbine design, operation and siting. Analyse simple problems related to wind power meteorology Interpret scientific dissemination by experts on main points Apply analysis methods to convert basic site data to wind turbine design parameters	
Education Method	Lecture + assignments + practical	
Course Relations	Follow up courses AE4W09, AE4135	
Literature and Study Materials	www.eos.ubc.ca/books/Practical_Meteorology Chapters 12 to 14; Pijush K. Kundu, Ira M. Cohen, Fluid Mechanics, Elsevier. Lecture notes	
Prerequisites	AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Assessment	Written / digital exam (scheduled in the exam period after Q4). The resit is an written exam on basis of pass/fail (i.e. in case of a pass the grade 6 will be put in Osiris).	
Set-up	The course combines lectures and student assignments/practical. For several topics guest speakers will be invited.	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	W. Yu	
Instructor	Dr.ir. C.J. Simao Ferreira	
Gast Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Preferred: AE4ASM506 (Aeroelasticity). It's preferred that the student has completed or is attending this or an equivalent course.	
Course Contents	The course will cover the main structural and aerodynamic phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. What are the key characterized parameters for an aeroelastic system? 2. How to construct a structural model of a wind turbine blade? 3. How to define if resonance occurs to a system and understand its relevance to wind turbines and how to mitigate it? 4. How to define if instability occurs to a system and its relevance to wind turbines and understand the role of aerodynamic damping? 5. Dynamic inflow and dynamic stall. 6. How to construct an aeroelastic model by coupling the structural and aerodynamic effects?	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand the dynamics and loads of a wind turbine due to the strong coupling of a) external loads(aerodynamics) b) internal structural loads c) inertia loads; 2. solve the dynamic structural response based on the equations of motions; 3. Calculate the unsteady aerodynamic loads of a wind turbine under dynamic inflow and dynamic stall; 4. apply the knowledge to program their own aeroelastic models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project.

The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5122	Thesis Aerodynamics & Wind Energy	45
Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal). If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go' (fail). Please refer to Brightspace for more information.	

WM0324LR		Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch		
Responsible for assignments	Dr.ir. U. Pesch		
Co-responsible for assignments	A.R. Gammon		
Contact Hours / Week x/x/x/x	0/4/4/0		
Education Period	2 3		
Start Education	2 3		
Exam Period	none		
Course Language	English		
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering		
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations		
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.		
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).		
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.		
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace		
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.		
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.		
Remarks	Q3: priority for EWEM students		
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/		
	MSc level		

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Wind Energy

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4115-23	Experimental Simulations	4
Responsible Instructor	Dr.ir. T. Sinnige	
Instructor	Dr.ir. W.J. Baars	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling; fractional analysis; dimensional analysis (Buckingham-Pi theorem); rigorous methods of obtaining dimensionless parameters; 3 Fundamentals of aerodynamic similarity: fundamentals of aerodynamics necessary for the interpretation of similarity between aerodynamic test results and full-scale; 4 Mach number and Reynolds number effects: practical considerations with respect to implications of dissimilarities between model-scale testing and full-scale applications; 5 Wind tunnel facilities: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 6 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 7 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 8 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 9 Design of experiments: factorial designs, response-surface models; 10 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report; 11 Engine integration: engine simulators, performance of those and their similarity with full-scale engines; 12 Acoustic testing: noise generation mechanisms and their appropriate scaling; definition of sound pressure levels; measurement requirements and devices; far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report;	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stroemungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports (Q2 + Q3) and oral exam (wk 3.10 or 4.1).	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub., 1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Prof.dr.ir. L.L.M. Veldhuis	
Instructor	L.T. Lima Pereira	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130-22, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during the first 3 weeks of Q1).	
Assessment	The assessment is done through 5 assignments that are performed on a personal basis. The assignments are to be combined in 1 report that should be delivered after the finalization of the lecture at the end of Q2. Additional information on the timing and contents of the assignments will be presented on BrightSpace.	

AE4136-24	CFD 2: Discretisation Techniques	5
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results. After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignement and oral examination.	
Enrolment / Application	Brightspace	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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6	Variational inverses	[Inversion]	Finalize []	--																																						
7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.																																						

	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Responsible Instructor	Dr. S.J. Hulshoff	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	M. Li	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Introduction to measurement techniques 2. Flow visualization (classroom assignment) 3. Pressure measurements 4. Hot Wire Anemometry (lab exercise) 5. Laser Doppler Velocimetry 6. Particle Image Velocimetry (classroom assignment+lab exercise) 7. InfraRed Thermography 8. Schlieren, Shadowgraphy and Background Oriented Schlieren 9. Fundamentals of data processing	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor	Dr. D. Ragni	
Instructor	Prof.dr.ir. M. Snellen	
Co-responsible for assignments	Dr. R. Zamponi	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources.	
Study Goals	The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course AE4463P-23 aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) must be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and also applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B, C and F are required. A choice can be made between D and E.	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4W30	Wind Resource and Wind Farm Yield	4
Responsible Instructor	Prof.dr. S.J. Watson	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	This course will cover how the output of wind farm arrays is predicted based on modelling of the wind resource and combined wake interactions. The first part of the course will cover the different wind resource models which are used ranging from simple mass consistent models, to linearised solvers, computational fluid dynamics models and mesoscale models. The use of models in both onshore and offshore environments will be covered including the importance of atmospheric stability. The second part of the course will consider the application of models to assess the yield from wind farms considering wakes effects. Wake models will be reviewed including simple semi-empirical models to kinematic models and the modelling of wakes in CFD models. Once again, the importance of atmospheric stratification will be considered and the potential impact on wind farms of certain mesoscale phenomena, e.g. gravity waves offshore, will be assessed and quantified. Large wind farm phenomena will be studied including the so-called deep array effect and wind farm blockage.	
Study Goals	Course Aim: to give students a firm foundation in how to predict wind farm yield from prevailing wind conditions. Course Objectives: 1) To understand what tools are available to predict wind resource over a wide area 2) To apply to such models to potential wind farm sites onshore and offshore 3) To quantify the importance of terrain effects and atmospheric stability on the wind resource 4) To understand what tools are available to predict wake effects in a large wind farm 5) To apply wake models in a realistic wind farm situation 6) To appreciate the importance to wind farm yield of phenomena such as terrain complexity (onshore), proximity to the coast (offshore), atmospheric stability, mesoscale effects (e.g. gravity waves), wind farm layout and wind farm scale	
Education Method	Students will learn through a combination of guided reading, group work, online resources (e.g. lectures, videos, quizzes), face-to-face lectures, coursework and exams.	
Assessment	The students will undertake two course assignments (resource assessment and wind farm yield prediction) worth 30% each, spread out during the course and an oral exam worth 40% at the end of the course. Assignment 1: application of a model to assess and optimise wind resource and energy yield for an onshore site with a report to be assessed. Assignment 2: application of a model to assess and optimise energy yield for an offshore site incorporating different wake models with a report to be assessed. Exam: oral test to assess understanding of the principles of wind resource assessment and wind farm yield prediction.	

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Prof.dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Instructor	Dr. F. Taruffi	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Instruction if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Process & Energy	

ME45030	Turbulence	5
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Prof.dr.ir. J. Westerweel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Advanced Fluid Dynamics (or equivalent)	
Course Contents	The course provides an introduction to the physical aspects of turbulent flows.	
Study Goals	(1) basic equations that describe turbulence; (2) transition to turbulence; (3) turbulence closure models; (4) energy cascade; (5) turbulent dispersion; (6) selected topics in turbulence	
Education Method	lectures and instructions	
Literature and Study Materials	Turbulence (2016) by Nieuwstadt, Boersma & Westerweel	
Assessment	Written exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Tags	Fluid Mechanics Mechanics Numeric Methods Physics	
Department	ME Department Process & Energy	

WI3150TU	Partial Differential Equations A	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, ordinary differential equations	
Course Contents	I: (WI3150TU) Introduction. Types of second order equations. Initial and initial boundary value problems. Fourier series. Quasi-linear, first order partial differential equations. Waves and reflections of waves. Separation of variables. Sturm-Liouville problems. Parabolic, elliptic and hyperbolic equations. Maximum principle. Diffusion and heat transport problems. Lectures (3 ECTS). II: (WI3151TU) Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972.	
Assessment	Written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. The resit consists of a written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. W.T. van Horssen	
Co-Instructor	Dr. H.M. Schuttelaars	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 2: Control and Operations

MSc Coordinator	Dr.ir. C.C. de Visser
Introduction 1	Control and Operations is focused on the through-life operations associated with aerospace industry. It studies the avionics systems of individual aircraft, flight control and flight deck design, airline operations and support (maintenance), air traffic control and air traffic management, airports and air transport operations, as well as the impact on the environment, including noise, climate and air pollution effects. By selecting one of the two C&O profiles, the student chooses an educational programme that contains a particular balance of these elements. Theory and practice are combined in exciting and challenging fields of study for talented and ambitious students. Our graduates easily find their way in finding a good job in engineering, research, consulting, and management. The C&O profiles incorporate a research project that builds on the learning gained from the taught modules. C&O research projects focus on academic quality and the contribution to the body of knowledge in the field. There is also a strong recognition of the importance of understanding problems coming from industry, through industrial collaboration, and experimental validation of the novel operational concepts in real-life scenarios. The Track offers two Profiles: I. Control and Simulation II. Sustainable Air Transport
Program Goals	Learning objectives The focus in C&O lies on first establishing a sound theoretical and framework and then to sharing our department's expertise in applying this knowledge in practical problems.
Program Structure 1	The Track consists of two Profiles that share a common core of 18 ECTS. Each profile programme consists of a 'Profile core' of mandatory courses, ranging between 15 and 17 ECTS, supplemented by Electives (13 to 20 ECTS), and finally, up to 12 ECTS on Personal Development (PPD) electives that can be adjusted to either the subject of the final thesis work, or to the student's own taste. The MSc thesis project is an individual research assignment, conducted 'in house' or at an appropriate external organization partner.

Profile I: Control and Simulation	
Program Coordinator	E.J.J. Smeur
Introduction 1	The C&S profile focuses on design, analysis and testing of automatic and manual guidance, navigation, and control (GNC) systems. Our aim is to enhance safety, sustainability and performance of aerospace vehicles by creating better interfaces, procedures and automation. The profile builds on systems and control theory, stochastic dynamic systems and signals, flight simulation, and modelling human cognition and manual control. For their electives and thesis, students can focus on autonomous guidance and control, micro aerial vehicles, the air traffic management system, or human-in-the-loop control systems. Students can work with state-of-the-art facilities to close the loop of theory and application, using the laboratory aircraft, the SIMONA flight simulator, the ATC lab, or the CyberZoo indoor flight-test area. For autonomous systems, students can focus on the design of highly-automated, intelligent and (semi-)autonomous aerial vehicles. They can learn advanced control theory, global optimisation approaches, state and parameter estimation techniques, and modern dynamic modelling techniques. Aeronautical research examples are fault tolerant and re-configurable flight control, flight envelope clearance and protection, vehicle dynamics model identification, and advanced flight test instrumentation systems. Space examples are rendezvous/docking vehicle control, GNC for formation flying spacecraft, modelling of spacecraft with flexible structures and liquid sloshing, and optimal terminal area energy management of re-entry vehicles. The Micro Aerial Vehicle (MAV) research and education is a multi-disciplinary field, combining flight dynamics and control of innovative flying vehicles, obstacle avoidance with computer vision and machine learning, interaction with the environment and swarming. Students are encouraged to apply theory to real flying vehicles and perform flight test experiments. Example research projects are drone racing through machine learning, spiking neural networks for obstacle avoidance and control, flight control of hybrid VTOL drones, and drones with a sense of touch. With Air Traffic Management (ATM), students will learn about both the conventional air traffic management system, as well as challenges and approaches for new airspace users such as drones and electric vehicles (Urban Traffic Management and mixed operations). Research examples are design and evaluation of new (emission-reducing) procedures for conventional aviation, development of conflict detection and resolution algorithms (both AI and analytical), airspace design, complexity analysis, data science in ATM, and analysis of communication, navigation, and surveillance applications. For human-in-the-loop systems, students will learn to understand and model human perception and skill-based control behaviour, using modern identification techniques and classical control. Research examples are the design of haptic force-feedback systems for aerospace (and other) vehicles, modelling human visual-vestibular perception and control, neuromuscular system dynamics and biodynamic coupling effects, the optimal tuning of flight simulator visual and motion cues, aircraft handling qualities and advanced fly-by-wire systems, 3D flight guidance displays, and the tele-operation of unmanned aerial vehicles. Learning objectives: 1. Knowledge on modern methods for analysis of control problems and synthesis of control systems 2. Knowledge of modelling human perception and control behaviour, including modern methods for identification 3. Skills in the use of modern control and simulation programs 4. Skills in the programming, setup and execution of experiments in flight simulators or in flight tests 5. Insight in related disciplines such as artificial intelligence, statistics, sensor fusion, distributed computing, fast-time and real-time simulation

AE4301	Automatic Flight Control System Design	3
Responsible Instructor	Dr. A. Jamshidnejad	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4303	
Expected prior knowledge	The following prior knowledge is required: AE3302 AE2204 (until 2012/2013) AE2235-I (from 2013/2014)	
Parts	1. Introduction & recapitulation of systems and control theory . Laplace transformation . Elementary closed loop systems . Transfer functions . Poles and zeros . First order systems . Second order systems . Pole placement for simple systems . Root locus method . Characteristic equation . Angle and magnitude conditions . State space formulation . Controllability, observability . Basic controllers: P,PI,PD,PID . Frequency response (Bode plots) 2. Introduction: Course use and arrangement a. Why automatic flight control systems? b. Function of the flight control system in civil aviation c. Recapitulation of theory on flight dynamics d. Review of the different frames of reference: wind, stability, and body. e. Non-linear equations of motion. f. Trim and linearization of the non-linear equations of motion. g. Linearized longitudinal dynamics using a statespace representation and the equivalent frequency domain form. 3. Performance and handling qualities a. Handling quality criteria b. The Control Anticipation Parameter (CAP) c. Gibsons Phase rate and Frequency criterion 4. Dynamic stability augmentation a. yaw dampers b. pitch dampers c. phugoid dampers 5. Static stability augmentation a. angle of attack feedback to improve static margin b. load factor feedback to improve manoeuvre margin c. sideslip feedback to improve directional static stability 6. Basic longitudinal autopilot modes a. pitch attitude hold mode b. altitude hold mode c. airspeed hold mode (using autothrottle) d. vertical speed 7. Basic lateral autopilot modes a. roll angle hold mode b. coordinated roll angle hold mode c. turn rate at constant altitude and speed d. heading angle hold mode 8. Longitudinal and lateral guidance modes a. glideslope hold mode b. automatic flare mode c. localizer hold mode d. VOR hold mode	
Course Contents	Classical control is still predominantly used in aerospace industry for the design and analysis of automatic flight control systems. Various existing control systems such as Stability Augmentation Systems (SAS), Control Augmentation Systems (CAS) and fly-bywire systems are reviewed in detail. The emphasis of the course lies in demonstrating, through application of classical frequency domain and state space techniques, how to design systems that fulfill the requirements imposed by the aviation authorities, with emphasis on understanding the benefits and limitations of such systems.	
Study Goals	After this course the student should be able to: - Understand dynamics of linear time invariant systems and design controllers for such system for the desired performance	

	- substantiate the function of a Flight Control System(FCS) in civil aviation. - apply the theory of flight dynamics and control to FCS design for civil aviation. - verify if a given FCS satisfies the handling qualities for civil aviation criteria. - design static and dynamic stability augmentation systems. - design all longitudinal and lateral autopilot modes for civil aviation.
Education Method	Lectures
Literature and Study Materials	Course material to support the exercises will be posted on the Brightspace. Recommended literature - Any standard textbook on systems and control theory (e.g., Modern Control Engineering by Ogata) - M.V. Cook, Principles in flight dynamics, Edward Arnold, London, 1997 ISBN 0-340-63200-3. - B.L. Stevens, F.L.Lewis, Aircraft control and simulation, Wiley, New York, 1992 ISBN 0471613975. - J. Roskam, Airplane flight dynamics and control Part II, , ISBN 1-8845885-18-7.
Assessment	Written closed-book examination
Set-up	At the end of each lecture, a simple take home assignment is given in order to gain experience in working with the course material. There will be a written examination at the end of the course. In the related practical AE4301P a control system must be designed that satisfies certain desired requirements.

AE4301P	Exercise Automatic Flight Control System Design	1
Responsible Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301 Automatic Flight Control System Design AE3212-I Flight Dynamics AE2235-I Systems and Control Theory	
Course Contents	The goal of this exercise is to design a flight control system using classical control theory, using a simulation model in Simulink. The exercise makes you in a practical way familiar with the implementation of inner control loop design, handling qualities satisfying CAP/Gibson/MIL-specifications, and the design of a higher level autopilot mode such as glideslope tracking or terrain following.	
Study Goals	To become familiar with classical flight controllers and their design, and to gain insight in handling qualities of open-loop and controlled aircraft.	
Education Method	General Practice Exercise.	
Literature and Study Materials	Course material to support the exercise will be posted on Brightspace.	
Assessment	Report	
Remarks	A short report about the design of this controller using classical control theory needs to be written. This report must be handed in during the second quarter. There is a resit-opportunity in the 3rd or 4th quarter.	
Set-up	The exercise is a homework assignment to be solved using Matlab and Simulink.	

AE4302	Avionics and Operations	3
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Ir. T.J. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4316 AE4316P AE4321-15 AE4322	
Expected prior knowledge	Flight Dynamics System and Control Theory	
Parts	Week arrangement Lecture and study material 1. Introduction to avionics systems. -. Air Data systems (home study). 2. Gyroscopes, attitude reference systems. 3. Compasses, heading reference systems. 5. Flight deck instruments and integrated systems. 6. Navigation equations. 7. The Flight Management System (FMS). 8. Inertial Navigation Systems. 9. Radio navigation systems (ADF, VOR, DME). 10. Landing guidance systems (ILS, GPS). 11. Communication, Navigation, Surveillance (CNS). 12. Satellite navigation systems (GPS). 13. Air Traffic Management (ATM). 14. The Future Air Navigation System (FANS).	
Course Contents	(see week arrangement) This course provides a comprehensive, unified coverage of the principles of modern navigation equipment and systems, both in the aircraft and on the ground, including the aircraft instrumentation and flight-deck systems, with a special emphasis on the important trends in the global air navigation and air traffic management system.	
Study Goals	1. The student can describe in detail the working principles of the avionics systems treated in the course. 2. The student can demonstrate the avionics systems' functionalities, identify their strong points and weaknesses, and make comparisons between the avionics systems. 3. The student can evaluate, criticize, and appraise their usage in the current and future operational context.	
Education Method	Lecture	
Literature and Study Materials	Lecture slides and other lecture material are put on the BrightSpace The books mentioned below (Pallett & Coombs, Kayton & Fried) can be hired at the Control & Simulation division. Recommended literature: E.H.J. Pallett, L.F.E.Coombs, Aircraft Instruments and Integrated Navigation Systems, Longman, 1997 ISBN 0-582-08627-2. M. Kayton, W.R.Fried, Avionics Navigation Systems, Wiley, 1997 ISBN 0471547956.	
Assessment	Digital, closed-book exam on campus	

AE4304	Stochastic Aerospace Systems	3
Responsible Instructor	Prof.dr.ir. M. Mulder	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4306 AE4307	
Expected prior knowledge	AE2235-I Aerospace System and Control Theory AE2235-II Signal Analysis and Telecommunication Principles AE4301 Automatic Flight Control Systems Design	
Parts	Week arrangement See Detailed Description of Course 1. Introduction 2. Chapter 2 of the lecture notes. 3. Chapter 3. 4. Chapter 3. 5. Chapter 4. 6. Chapter 4. 7. Chapter 5.	
Course Contents	The course AE4304 covers ONLY the first five chapters of the lecture notes, the practical assignment AE4304P covers chapters six to eight. Chapter 9 (Etkin's 4 point model) serves as background reading. So, the lecture AE4304 (and its exam) covers: 1. Introduction (aircraft do respond to atmospheric turbulence, effects on flight control system design). 2. Scalar stochastic processes (probability theory, joint probability density functions, covariance and correlation functions, stochastic processes, ergodic processes). 3. Spectral analysis of stochastic processes in continuous time (Fourier analysis, power spectral densities, analysis of dynamic linear system responses in frequency domain). 4. Spectral analysis of stochastic processes in discrete time (discrete time Fourier transform, Fast Fourier Transform, spectral estimates-smoothing). 5. Multivariable stochastic processes (covariance function matrix and spectral density matrix, multi-variable system responses in the frequency and in the time domain). The practical assignment AE4304P (Matlab or Python) covers: 6. Description of atmospheric turbulence (physical mechanisms, stochastic models of atmospheric turbulence, the two fundamental correlation functions, von Karman en Dryden spectra, models in the time domain). 7. Symmetric aircraft response to atmospheric turbulence (symmetrical aerodynamic forces and moments due to turbulence, gust derivatives, equations of motion of aircraft flying in symmetrical atmospheric turbulence). 8. Asymmetric aircraft response to atmospheric turbulence (elementary two-dimensional fields of turbulence, asymmetrical aerodynamic forces and moments, asymmetrical gust derivatives, equations of motion).	
Study Goals	Introduction to stochastic processes, spectral analysis, understanding the physics of aircraft responses to atmospheric turbulence, derivation of equations of motion of symmetrical and asymmetrical responses to atmospheric turbulence.	
Education Method	Lecture	
Literature and Study Materials	J.A. Mulder, J.C. van der Vaart, W.H.J.J. van Staveren, M. Mulder: Aircraft responses to atmospheric turbulence, Lecture notes AE4304, November 2020. All lecture slides will be made available electronically via the Brightspace.	
Assessment	Written (with formula page)	
Remarks	The written examination (on probability and stochastic processes, chapters 1-5 of the lecture notes) is followed by the assignment ae4304P (related to the theories of chapters 6-8 of the lecture notes).	
Set-up	Lectures, MATLAB/PYTHON demonstrations.	

AE4304P	Stochastic Aerospace Systems Practical	1
Responsible Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE3212-I Flight Dynamics AE2235-I Systems and Control Theory AE2235-II Signal Analysis and Telecommunication Principles AE4301 Automatic Flight Control System Design	
Course Contents	Application of MATLAB or Python software to aircraft specific turbulence responses: 1. Calculation of aircraft time-histories due to both symmetrical and asymmetrical, longitudinal, lateral and vertical turbulence components. 2. Calculation of analytical transfer functions, frequency response functions, and auto- and cross Power Spectral Density (PSD) functions of state- and output variables (e.g. acceleration levels). 3. Numerical calculation of frequency response functions, and auto- and cross Power Spectral Density (PSD) functions of state- and output-variables. 4. Calculation of (co)variance- and correlation-functions of aircraft state-and output-variables. 5. The effects of Automatic Flight Control Systems on the aircrafts responses to atmospheric turbulence.	
Study Goals	Introduction to both time- and frequency-domain identification and simulation techniques using MATLAB or Python. The techniques are applied to example aircraft (amongst others Cessna Citation 500).	
Education Method	General Practice Exercise	
Literature and Study Materials	Lecture Notes AE4304, by J.A. Mulder, J.C. van der Vaart, W.H.J.J. van Staveren, Q.P. Chu & M. Mulder: Aircraft Responses to Atmospheric Turbulence, November 2020. Recommended literature: D.E Newland, Random Vibrations and Spectral Analysis, Longman, New York, 1984 3rd. ed ISBN 0582215846.	
Assessment	Take-home assignment.	
Set-up	Take-home assignment. At the end of the second lecture period, the assignment will be available.	

AE4316	Aerospace Human-Machine Systems	4
Responsible Instructor	Dr.ir. M.M. van Paassen	
Instructor	Prof.dr.ir. M. Mulder	
Instructor	Dr.ir. C. Borst	
Instructor	Dr.ir. D.M. Pool	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Follow-up course (elective) AE4316P. That course focuses on advanced topics in aerospace human-machine systems by having small groups of students participate in an experiment (in either a manual or supervisory control setting), collect data and analyze their own generated data. In other words, experience the theory behind human-machine systems in practice!	
Expected prior knowledge	Control theory (stability and Bode plots); signals AE2235 Aerospace Signals, Systems & Control, AE4301 Automatic Flight Control System Design Stochastic signals; AE4304 Stochastic Aerospace Systems Flight instruments and procedures; AE4302 Avionics and Operations	
Course Contents	This course focuses on the various aspects of actual and future flight deck and air traffic control human-machine interfaces. It provides extensive theoretical as well as practical knowledge on the specific characteristics of human behavior, such as human control behavior (i.e., cybernetics), human perception (visual & haptic), human mental processing, cognitive factors, and human-automation interaction in manual and supervisory control tasks.	
Study Goals	Overall, the student will have a working knowledge of human operator (pilot) characteristics that are relevant for the design and evaluation of human-machine systems. Specific study goals: The student: * Is able to classify different types of human behaviour according to Rasmussen's rule-skill and knowledge taxonomy * Is able to predict performance and human behaviour in manual control tasks, using McRuer's cross-over theory, and is able to reason on the effect of task variables (motion, haptics, etc.) * Knows the physiology and characteristics of human sensory systems and actuation processes (visual, vestibular, proprioceptive senses and neuromuscular system) as relevant for human behaviour, and is able to predict the implications of these properties for human perception and behaviour, and relate this to design choices for displays and manipulators. * can describe the pitfalls of automation (i.e., ironies of automation) and is familiar with various taxonomies on levels & stages of automation. * can analyse accident and incident reports, find latent and active errors and classify these with Rasmussen's SRK taxonomy and identify Reason's error shaping factor. The student understands the wider context of human error (Dekker's "new view"). * is familiar with workload and situation awareness, and knows which methods are used to measure these properties * understands the nature of human cognition, can distinguish between different views and models of cognition and knows when these are applicable * can describe the differences and similarities between interface design approaches that aim to support both system and human (control & cognitive) performances * is familiar with workload theory, knows different metrics for workload, and when these are applicable	
Education Method	Lectures with activating elements. Week arrangement: 1. Introduction + SRK 2. Crossover model I 3. Crossover model II 4. Crossover Model II 5. Visual systems (the eye and visual motion perception) 6. Vestibular system 7. Neuromuscular system 8. Human-Automation Interaction 9. Human Error 10. Cognition 11. Ecological Interface Design 12. Display Design Principles 13. Applications of HMS theory 14. Workload (self study module, no lecture) The course also includes a demonstration session in the human machine laboratory, in which students will encounter experimentation settings and experience manual control tasks.	
Assessment	Written exam. The exam typically consists of 6 questions, covering a selection of the course's topics.	

AE4322-24	Piloted Flight Simulation	3
Responsible Instructor	Ir. O. Stroosma	
Gast Instructor	Dr.ir. M.M. van Paassen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	AE4323 (Real-time Distributed Flight and Space Simulation)	
Expected prior knowledge	AE2235 (Aerospace Signals, Systems & Control, or equivalent) AE3212 (Aerospace Flight Dynamics, or equivalent) AE4302 (Avionics and Operations, or equivalent) AE4316 (Aerospace Human-Machine Systems, or equivalent)	
Course Contents	Lecture topics, not necessarily in chronological order 1) Introduction to piloted flight simulation, including fidelity considerations 2) Use of piloted flight simulation, including training and qualification 3) Simulator sub-systems 4) Modelling of vehicle dynamics 5) Real-time software engineering 6) Distributed simulation 7) Motion bases, including cueing 8) Visual systems, including image generation 9) Control loading	
Study Goals	1) You can explain the use cases of piloted flight simulators and the resulting (sub-)system requirements. 2) You can explain the working principles of piloted flight simulator subsystems (vehicle model, real-time and distributed software, motion system, visual system, and control loading). 3) You can identify the design options for a flight simulator's functions, distinguish strong and weak points, as well as relate these to the simulator's intended use. 4) You can identify, analyze, and evaluate piloted flight simulator systems in an operational context, e.g. in a flight training centre or a simulator manufacturing plant. 5) You can identify and explain the different ways in which the fidelity of a simulator (system) can be evaluated (physical, perceptual, behavioural). 6) You can use common methodologies used for assessing simulator fidelity in industry (QTG, OMCT, numerical stability).	
Education Method	Lectures, simulator demonstrations, and excursions to industry.	
Literature and Study Materials	Lecture slides and other materials are put on Brightspace. Recommended but not required literature: D. Allerton, Principles of Flight Simulation, Wiley, 2009, ISBN 978-0-470-75436-8	
Assessment	Written examination and excursion report.	

AE4353	Artificial Intelligence for Aerospace Control and Operations	4
Responsible Instructor	Dr. G.C.H.E. de Croon	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Programming skills in Python are required, as obtained, e.g., with the course AE1205.	
Course Contents	This course forms an introduction to artificial intelligence for use in aerospace control and operations. The emphasis is on both the theory and application of machine learning methods, with a large emphasis on deep neural networks.	
Study Goals	The main study goals are: 1. Students can independently apply AI methods to aerospace problems for regression and classification tasks. 2. Students can explain the theoretical properties, learning methods, and functioning of AI methods, with a focus on machine learning (ML) and deep neural networks (DNNs). Topics covered in the course include: Supervised, unsupervised and self-supervised learning Artificial neural networks (perceptrons, multi-layer perceptrons, deep neural networks) Learning methods and advanced structures for deep nets Uncertainty estimation with ANNs Recurrency mechanisms Explainable AI / analysis of trained neural networks Hyperparameter tuning AI tools	
Education Method	The course will consist of 7 lectures and 7 practical sessions, in which the students apply the learned theory to practical examples from aerospace control and operations practice (e.g., using AI to interpret acoustic data, aircraft maintenance data or drone data).	
Assessment	The course assessment consists of a digital exam that requires both the answering of questions on the theory and programming to solve machine learning problems. A bonus point for the exam can be obtained by submitting a valid entry to a machine learning competition that is presented at the start of the course.	

AE4441-16	Operations optimisation	4
Responsible Instructor	Ir. P.C. Roling	
Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	4/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course aims at providing the students with knowledge and experience to set-up and analyze linear and nonlinear optimization problems. The course covers the following topics: 1. Introduction to Operations Research, examples of OR models for air transport. 2. Linear programming (LP) models and the simplex Method. 3. Sensitivity analysis and Duality. 4. Transportation and assignment problems. 5. Network optimization and dynamic programming. 6. Mixed Integer Linear programming. 7. Nonlinear programming.	
Study Goals	At the end of the course, the students will be able to: 1. Understand the theory behind basic linear and non-linear optimization problems. 2. Model a problem as a linear program, a (mixed) integer program or a network optimization problem; 3. Verify the model using self created test data set. 4. Create and apply a sensitivity analysis.	
Education Method	The course consists of two studio classroom sessions (2 times 2hrs every week), where the theory is practiced and help is available. Students need to watch lecture video's and/or read up on the subject using the book before the studio classroom sessions.	
Books	Introduction to Operations Research Frederick S. Hillier and Gerald L. Lieberman, Tenth Edition Mc Graw-Hill International Edition	
Assessment	The final grade consists of two parts: 50% individual exam (computer-based open book); 50% an assignment (groups of 2-3 students). Both partial grades must be 6.0 or higher.	

AE4462-17	Aircraft Emissions and Climate Effects	4
Responsible Instructor	Prof.dr. V. Grewe	
Instructor	Dr. I.C. Dedoussi	
Responsible for assignments	V.R. Meijer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on calculus, fluid mechanics and first year physics.	
Course Contents	The course introduces the students in the state-of-the-art capabilities, research issues and challenges in climate effects of aviation. Topics include: - Combustion and emissions - Overview of all environmental effects of aviation - Air pollution relevant atmospheric chemistry and physics - Local air pollution modelling - Aviation environmental policy, new technologies, and other mitigation options - General circulation of the atmosphere - Atmospheric chemistry and contrail formation - Climate change caused by air traffic emissions - Modelling of climate impact from aviation - Application and show cases for the assessment of mitigation options supporting decision making, e.g. for climate optimized trajectories	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the climate impact of air traffic. These elements are required, for example, for the assessment of the air quality and climate impact of aviation and mitigation measures, including climate optimised trajectories/routing, low climate impact routing and consequences for aircraft design and supersonic transport.	
Education Method	Lectures and exercises	
Literature and Study Materials	Lecture slides and hand-outs	
Assessment	Written exam (Resits may be oral, depending on the situation) Assignments Both the written exam and the assignments need to be passed with a 6.0 or higher.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project.

The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5322	Thesis Control and Operations	45
Responsible Instructor	Dr.ir. C.C. de Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic	
Assessment	Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved. When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal). If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go' (fail). Please refer to Brightspace for more information.	

WI2056LR	Systems Theory	4
Responsible Instructor	Dr.ir. J.G. Maks	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Linear Algebra. Laplace transforms.	
Course Contents	During this course the following topics will be covered: State-space representation of input-output system (both for continuous-time and discrete-time case). Linearization of a system. Solution of a linear system (both for continuous-time and discrete-time case). Impulse response and step response of a linear system (both for continuous-time and discrete-time case). Asymptotic stability, BIBO stability (both for continuous-time and discrete-time case). Controllability and observability (both for continuous-time and discrete-time case). Kalman decomposition. State feedback (both for continuous-time and discrete-time case). State reconstruction by observer (both for continuous-time and discrete-time case). System description in frequency domain. Composition of systems in frequency domain. Realization of transfer function.	
Study Goals	After a successful completion of the course you will be able to model an input-output system by a state space model (both for continuous-time and discrete-time case). linearize a system around a given solution. determine whether an equilibrium point of a linear system is asymptotically stable, weakly stable or unstable (both for continuous-time and discrete-time case). compute the solution of a linear time-invariant system (both for continuous-time and discrete-time case). compute the impulse response and the step response of a linear time-invariant system (both for continuous-time and discrete-time case). determine whether or not a linear system is controllable (both for continuous-time and discrete-time case). determine whether or not a linear system is observable (both for continuous-time and discrete-time case). construct a Kalman decomposition of a linear system. design a feedback control (if it exists) which makes an unstable system stable or one which reduces the effect of disturbing signals (both for continuous-time and discrete-time case). design an observer (if it exists) which produces an approximation of the state of the system such that the error converges to zero (both for continuous-time and discrete-time case). represent a linear system in the frequency domain. construct various realizations of a given transfer function.	
Education Method	Lectures	
Literature and Study Materials	Mathematical Systems Theory, 4th edition, VSSD, Delft, 2011, by G.J. Olsder, J.W. van der Woude, J.G. Maks and D. Jeltsema (open access).	
Assessment	Written exam of 3 hours.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Control and Simulation

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4270	Control and Operations Project	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. J. Ellerbroek	
Instructor	Dr. G.C.H.E. de Croon	
Instructor	E.J.J. Smeur	
Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	///	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This course code is a placeholder for miscellaneous student assignments within the Control and Operations track. For example, a lecturer would like to have some data processed or have a nice simulation model that can be used for demonstration purposes in the lectures. Through this course code, we now have a way to compensate students for their efforts by considering it as an elective of 4 ECTS. If there are such assignments available, they will be announced.	
Study Goals	n/a, depends on the nature of the assignment	
Education Method	n/a	
Assessment	A (short) report needs to be written about the assignment. The content needs to be discussed with the supervisor of the assignment.	
Elective	Yes	

AE4311	Nonlinear and Adaptive Flight Control	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE2235-I AE3212-I AE4301 AE4301P AE4312/AE4320	
Course Contents	Safety and survivability of modern civil and military aircraft when airframe and structural failures have occurred in flight have become a key issue in the aerospace industry nowadays. This course introduces nonlinear and adaptive control techniques to enhance reconfigurabilities of aircraft flight control systems. Advanced nonlinear and adaptive control techniques such as Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion (INDI), Adaptive Backstepping (ABS) and Incremental Backstepping are described together with applications to fault tolerant and reconfigurable aircraft flight control system designs. Unconventional yet available control effectors are used through control allocation algorithms to achieve required performance.	
Study Goals	Students will be able to acquire the skills in designing fault tolerant and reconfigurable flight control systems using Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion, Adaptive Backstepping (ABS) and Incremental Backstepping control techniques.	
Education Method	Lectures altered with assisted computer exercises	
Literature and Study Materials	Handouts of lecture slides (available on Brightspace)	
	Recommended literature: J.-J.E. Slotine & Weiping Li, Applied non-linear control, Englewood Cliffs : Prentice-Hall, 1991, ISBN 0-13-040890-5 M. Krstic, et al, Nonlinear and adaptive control design, John Wiley & Sons, Inc., ISBN 0-471-12732-9	
Assessment	Individual Take-home assignment	
Remarks	Hand-in exercise and a report of the design of an advanced controller. The report must be handed in at the request date given by the course instructor.	
Set-up	Course contents Introduction to advanced flight control Nonlinear aircraft flight control based on Nonlinear Dynamic Inversion Adaptive Nonlinear Dynamic Inversion Incremental Nonlinear Dynamic Inversion Aircraft fault tolerant flight control based on Adaptive and Incremental Nonlinear Dynamic Inversion Nonlinear control based on backstepping Nonlinear aircraft flight control based on backstepping Adaptive Backstepping Incremental Backstepping Aircraft fault tolerant flight control based on Adaptive and Incremental Backstepping Control allocation	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical singularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments. The assignments will be handed out near the end of Q3 and the hand-in deadline will be near the end of Q4. Exact dates will be announced during the lectures.	

AE4314-21	Helicopter Performance, Stability and Control	4
Responsible Instructor	Dr. M.D. Pavel	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Aircraft performance Aerodynamics Flight dynamics	
Parts	Week Arrangement Lecture and study material 1. Lay-out of the helicopter. Qualitative description of the rotor and control systems. Definition of the tip plane, control plane, hub plane and several co-ordinate systems. Blade element analysis of hover performance. 2. Performance calculations of vertical flight. Turbulent Wake State. Autorotation. Performance calculations in forward flight. Blade element analysis. Glauerts hypothesis. Comparison with performance of fixed wing aircraft. 3. Flapping dynamics of centrally hinged rotors. Equations of motion of helicopter with only pitch degree of freedom. Stability and control. 4. Flapping dynamics in forward flight. Symmetric equations of motion. Stability and control in forward flight. 5. Trim conditions, linearised equations of motions. Representation in the complex plane and comparison with fixed wing aircraft. 6. Influence of semi-rigid rotorsystems on the equations of motion and on the characteristic handling characteristics. 7. Flap/Torsion blade dynamics. Divergence. Classical blade flutter, Structural lay-up to avoid flutter. Introduction to ground resonance instability.	
Course Contents	General lay-out of helicopters, performance calculations during hover, vertical climb and descent and forward flight. Helicopter blade dynamics. Introduction to helicopter control characteristics: symmetrical equations of motion handling- and control properties. Introduction to aeroelasticity of rotorcraft.	
Study Goals	1. An example of the synthesis of a large diversity of subject areas, aimed at the analysis of a complicated technical object. 2. Introduction to aerodynamics, flight mechanics and aeroelasticity of helicopters, which enables the student to independently study more advanced literature.	
Education Method	A combination of Online and Classical lecturing courses. Articles Readings and discussions. Questions and Discussions fora	
Literature and Study Materials	"Helicopter performance, Stability and Control", reader AE4-214, 2002, Faculty of Aerospace Engineering "Basic Helicopter Aerodynamics" by John Seddon& Simon Newman, Blackwell Science, 2002 "Principle of Helicopter Aerodynamics", Gordon Leishman, Cambridge Aerospace Series, 2000	
Assessment	Multiple-choice tests during the course (optional). Presentation of a helicopter-related subject in a small Symposium (obligatory) One final take-home assignment at the end of the course (obligatory) +Oral Exam (obligatory)	
Special Information	A helicopter flight in SIMONA simulator may be given to the students following this course depending on the availability of SIMONA simulator.	
Set-up	Online education and Classical lecturing	

AE4316P	Advanced Topics in Aerospace Human-Machine Systems	3
Responsible Instructor	Dr.ir. D.M. Pool	
Instructor	Dr.ir. M.M. van Paassen	
Instructor	Dr.ir. C. Borst	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	This course is an elective extension to the AE4316 (core) course for the C&S master profile. Prior knowledge from AE4316 is expected and mandatory.	
Course Contents	This course further explores the various aspects of supervisory control and manual control (cybernetics) and their applications in aerospace as introduced in the AE4316 lecture series. Through a number of lectures in the first weeks, this course provides detailed practical information on the current advanced state-of-the art methods used for measuring, analyzing, and modeling human performance and behavior in supervisory and manual control tasks and how such methods are applied in applications relevant to aerospace. Furthermore, through extensive practical work performed in small groups, students will gain practical experience with the advanced analysis of human-machine system performance through an experiment they need to plan and perform themselves. In this experiment, which will be performed in one of the simulator setups available at Control & Simulation (i.e., the Air Traffic Management Laboratory, the Human-Machine Laboratory or the SIMONA Research Simulator), students will collect data for an experiment of their choice, analyze the collected data, and reflect on the conclusions drawn from their experimental measurements.	
Study Goals	1. Formulate a good research question and hypotheses for a supervisory or manual control experiment. 2. Demonstrate understanding of the direct link between an experiments independent, dependent and control variables and the data analysis that can be performed on its data. 3. Analyze data from a supervisory or manual control experiment using state-of-the-art data analysis techniques and statistical tests. 4. Compare experimental findings with theoretical constructs and elementary models of human performance or control behavior. 5. Reflect on the scientific conclusions drawn from a set of human subject experiment data and their reliability. 6. Reflect on the consequences of experiment design choices on the scope and validity of human subject experiment results.	
Education Method	Classroom lectures, twice a week in the first three lecture weeks (3.1-3.3). After that, students will work in groups on the design and execution of an experiment they will perform in one of the simulation facilities at Control & Simulation.	
Literature and Study Materials	The presentation slides of all lectures will be made available through Brightspace Recommended reading for this course: 1. Jagacinski, Richard J., and John Flach. Control theory for humans: Quantitative approaches to modeling performance. Taylor & Francis, 2011. 2. Wickens, Christopher D., Lee, John D., Liu, Yili, and Gordon Becker, Sallie E. An Introduction to Human Factors Engineering, Second Edition, Pearson Education Inc., 2004. 3. Field, Andy. Discovering statistics using SPSS. Sage publications, 2009.	
Assessment	No written exam. Instead, grading for AE4316P will be based on the quality of two group deliverables: an Experiment plan (deadline end week 3.4, 20% of final grade) and a final Scientific Article (deadline end of Q3, 80% of final grade).	
Elective	Yes	
Set-up	Classroom lectures will be provided in the first three weeks. The following topics will be addressed in the lectures: 1. Introduction and experiment descriptions. 2. Human subject experiment design. 3. Experimental data analysis and statistics. 4. Cognitive Work Analysis. 5. Multi-channel manual control behavior identification. In addition to the lectures, students work in small groups on a practical assignment. This effort includes: developing an experimental plan, collecting and analyzing data, and writing a scientific article.	

AE4317	Autonomous Flight of Micro Air Vehicles	4
Responsible Instructor	Dr.ir. C. de Wagter	
Instructor	Dr. G.C.H.E. de Croon	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Programming experience (any language) If the student has no previous background in C / C++, we expect the student to do an online tutorial before/at the beginning of the course. Desired prior knowledge: Programming experience with C / C++ AE4393 (Avionics and Operations)	
Course Contents	This course covers the challenges and existing state-of-the-art methods for enabling autonomous flight of Micro Air Vehicles (MAVs), ranging from 20-gram flapping wings to 1 kg quad rotors. The emphasis is on computationally efficient, bio-inspired approaches to MAV autonomous flight.	
Study Goals	The theoretical knowledge will be applied by the students in the practical assignment, in which student groups program quad rotors in order to avoid obstacles in TU Delft's Cyberzoo. (1) Students can explain the differences between MAVs and larger aircraft in terms of control, design, aerodynamics, and electronics. (2) Students can explain how the above-mentioned differences affect the task of achieving autonomous flight with MAVs. (3) Students can list and explain Artificial Intelligence (AI) and control techniques useful for autonomous flight of Micro Air Vehicles (MAVs). a) Bio-inspired robotics b) Vision-based navigation (4) Students can apply the learned AI and control techniques to, and create new algorithms for an autonomous flight task. (5) Students can reproduce the current state of regulations for MAV operations. In addition, they can assess and discuss the safety and ethical aspects of MAV operations.	
Education Method	The course will consist of theory (7 lectures) and a practical assignment (6 practical sessions and a competition session). Brief description of practical assignment: Student teams will develop an efficient, purely vision-based approach to navigate through an obstacle field autonomously. The algorithm must run onboard the Parrot Bebop Drone, a commercially available quadrotor with a range of sensors. Importantly for the project, it has a single frontal camera. Hence, the developed algorithm concerns a monocular vision solution to obstacle avoidance.	
Assessment	The assessment of the course will consist of 50% of the theoretical exam and 50% of the practical assignment. Assessment of the practical assignment is based on the performance of the drone in the given task, the approach as explained in a presentation, and the code produced by the team.	

AE4320	System Identification of Aerospace Vehicles	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. D.M. Pool	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301.	
Parts	Week arrangement 1. Introduction. A global overview of the complete system identification cycle applied to aerospace vehicles, from sensor data acquisition and state estimation to aircraft aerodynamic model identification and validation. Introduction to sensor data acquisition and management, state and parameter estimation, and the two-step method for aerodynamic model identification. Finally an overview is give of advanced modelling techniques. 2. State estimation. Introduction to the essential role of the Kalman Filter (KF) in aircraft state estimation. The extension of the standard KF to nonlinear systems will be introduced in the form of the Extended Kalman Filter/Smotherer (EKF/EKS), and iterated extended Kalman filter (IEKF). Applications of the Kalman filters in aerospace navigation, attitude determination, on-board sensor calibration, sensor integration, and data fusion will be discussed. 3. Parameter estimation. Introduction to the process of parameter estimation and its role in creating accurate aerodynamic models of aircraft. Introduction to four of the most widely used parameter estimation techniques; Least squares (LS), Weighted Least squares (WLS), Total Least Squares (TLS), and Maximum Likelihood (ML) parameter estimation. Additionally, recursive parameter estimation methods like Recursive Least Squares (RLS) will be introduced. 4. Advanced optimization techniques. Some problems cannot be solved adequately with standard linear optimization techniques. In that case, a nonlinear optimization method must be used. In this part of the course, a powerful global nonlinear optimization technique based on interval analysis will be introduced. 5. Advanced modelling methods: Neural Networks. Neural Networks are a class of black-box function approximators that can be used to model nonlinear systems like Micro Air Vehicles and high performance aircraft. The application of neural networks to aircraft system identification is discussed, and it will be shown how neural networks can be used inside a framework for parameter estimation. 6. Advanced modelling methods: Multivariate Splines. Multivariate Simplex B-Splines are a powerful type of white-box function approximators that can be used to create accurate models of aircraft with highly nonlinear aerodynamics such as MAVs and high performance aircraft. A number of unique methods for assessing the quality of the created models are discussed, and pointers towards future research are given. 7. Demonstration: Aircraft system identification with multivariate splines. An in-depth demonstration will be given of the process of Aircraft system identification with multivariate splines using real flight test data from the Cessna Citation II laboratory aircraft. The goal is not only to create an accurate flight dynamics model of the Citation II, but also to apply various model validation techniques to assess the quality of the created model.	
Course Contents	Accurate aerodynamic models play a crucial role in the design and operation of flight simulators and flight control systems. The creation of accurate aerodynamic models from CFD, wind tunnel, and flight test data has historically been a highly challenging task. This is a direct result of the nonlinear nature of aircraft (aero)dynamics and the fact that not all aircraft states can be measured directly. As a consequence, the aerospace vehicle parameter identification problem constitutes a joint parameter-state estimation problem. It is the aim of this course to provide the student with a complete overview of the system identification cycle as it is currently applied to aerospace systems, and to introduce the student to the current state of the art in the field of aerodynamic model identification. The course consists of 7 parts covering the entire system identification cycle. In the first part of the course, the process of data acquisition using on-board sensors including accelerometers, gyroscopes, GPS and various air data sensors is discussed. It will be demonstrated that measurements made by real-world sensors are contaminated with noise, and are sometimes biased. Additionally, some aircraft states, like the true angle of attack, cannot be measured directly and must be reconstructed by combining sensor measurements. The second part of the course introduces the concept of state estimation in which prior physical knowledge of the system is used in a Kalman filter (KF) to estimate the true aircraft states from the measured states. Next to the ordinary Kalman filter, the Extended Kalman filter (EKF) and the Iterated extended Kalman filter (IEKF) will be introduced. In the third part of the course various methods for the estimation of model parameters will be discussed. It will be shown how the results from the state estimation are used in combination with a parameter estimator. This part not only focuses on offline batch parameter estimation methods like ordinary least squares (OLS), weighted least squares (WLS), total least squares (TLS), and maximum likelihood (ML) estimators, but also on recursive parameter estimation methods like recursive least squares (RLS) that can be used online. Special attention will be paid on the process of choosing a parameter estimator that is right for the job. The fourth part of the course introduces an advanced global nonlinear optimization method based on interval analysis. In aerospace system identification, so-called non-convex and nonlinear optimization problems are often encountered. Such problems can be solved with global nonlinear optimization methods like interval analysis. The fifth and sixth part of the course focuses on two advanced model structures that can be used in combination with the earlier introduced parameter estimation methods. In the fifth part of the course the neural network black-box function approximator is introduced. It is shown how neural networks are used to approximate scattered multidimensional data. In the sixth part of the course a new method for aerodynamic model identification based on multivariate simplex B-splines is introduced. This method was recently developed at the Faculty of Aerospace Engineering of the TU-Delft, and has a number of advantages over existing methods. For example, the simplex B-splines have a transparent model structure, are general in any number of dimensions, and can be computed efficiently in real-time.	

Study Goals	In the final part of the course, all theory introduced in the first six parts is used in a demonstration that uses real flight data from a real-life system like the Cessna Citation II laboratory aircraft, or a Micro Air Vehicle to identify a multivariate spline based aerodynamic model. The created model is validated using various techniques. Finally, pointers are given towards further research in the field of aerodynamic model validation. After completing the course, students will have a thorough understanding of the theoretical elements of aircraft system identification. Students will be able to apply these elements to go from real CFD, wind tunnel, and flight test data to an accurate aerodynamic model that can be used in simulator or flight control system applications.
Education Method	Lectures.
Literature and Study Materials	Q.P. Chu, Modern flight test technologies and system identification, lecture notes, Faculty of Aerospace Engineering, Delft University of technology. C.C. de Visser, Global Nonlinear Model Identification with Multivariate Splines, PhD thesis, Faculty of Aerospace Engineering, Delft University of technology. Recommended literature V. Klein and E. A. Morelli, Aircraft System Identification, AIAA, 2006. J.A. Mulder, J.K.Sridhar, J.Breeman, Nonlinear Analysis and Manoeuvre Design, Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 2, , May 1994 R.E. Maine, K.W.Illiff, AGARD Flight Test Techniques Series on Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 1, , December 1986 . Eric Walter and Luc Pronzato, Identification of parametric models, Springer, ISBN 3-540-76119-5.
Assessment	Individual Take-home assignments

AE4321-15	Air Traffic Management	4
Responsible Instructor	Prof.dr.ir. J.M. Hoekstra	
Instructor	Dr.ir. J. Ellerbroek	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Course Contents	The course gives an Introduction in Air Traffic Management and ATM research topics. 1. Introduction and history of Air Traffic Management, organisational & regulatory aspects 2. Airspace set-up, flight planning and navigation 3. Communication: Information services, Voice R/T, Data Link 4. Navigation: Great Circle calculations, WGS'84, approximations, beacons/navaids, landing aids 5. Surveillance: Radar basics, primary-secondary radar, radar types, ADS-B 6. Separation: Separation Minima, Near Misses, Closest Point of Approach Calculations, wind vortex categories 7. New concepts: Airborne separation assurance, 4D contract TBO, paired approaches, etc. 8. ATC Simulations: fast-time, human in the loop 9. ATC Research organisations and large scale programs (SESAR, NextGen) 10. ATM research topics: interactively discussed in class based on presentations by students While the first part of the course is mainly lectures, the second part of the course is much more interactive. Here pairs of students get assigned a paper, project or case to study and form an opinion on. They should explain and review it in both a brief essay and during a short presentation in class, followed by a discussion with the class in which they need to defend their view on this study. In this way the students are familiarized with different aspects and opinions on the 'hot topics' in ATM research currently.	
Study Goals	This course is designed to give the students an appreciation of the current ATM systems, throughout the world, with emphasis on the core area of Europe, in preparing the existing ATC systems for the predicted increase of traffic. Emerging technologies will be described, with the associated implementation aspects like Air Traffic Controllers' systems, the history and future of ATC, airline perspective, European and US new ATM programmes SESAR and NEXTGEN. Emphasis is on both the current technology as well as future concepts and technology which are currently being researched.	
Education Method	Lectures will be given on-line via livestreaming. Assignments will be given to the students for the second period. Students need to present their understanding of and opinion on a case, project or paper on-line. Discussions on this with the other students are part of the interactive method used in the second period.	
Reader	Hand-outs of the presentations are available on Brightspace.	
Assessment	Written, on site exam (50%) and the assignment (essay+presentation, also 50%). Both parts need at least a 5.0.	

AE4323	Real-time Distributed Flight and Space Simulation	3
Responsible Instructor	Ir. O. Stroosma	
Instructor	Ing. E.H.H. Thung	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE1205 (Programming & Scientific Computing in Python, or equivalent introduction to programming) AE4322 (Piloted Flight Simulation)	
Course Contents	Lecture topics, not necessarily in chronological order 1) C++ introduction 2) Real-time software engineering 3) Delft University Environment for Communication and Activation (DUECA) 4) Modelling tools 5) Verification and Validation 6) Software project management Programming assignments, individually and in a team. The course is completed with a practical project in which students program, demonstrate and document a distributed simulation with an aerospace theme. Attendance in the project meetings is mandatory.	
Study Goals	1) The student can identify the software issues related to real-time, distributed simulation. 2) The student can actively apply the theory of rigid body dynamics in real-time simulation. 3) The student can combine the modelling of dynamical systems with numerical integration routines. 4) The student can use quaternions to represent rigid body attitudes. 5) The student can implement simulation models using different software languages and tools (C++, Simulink). 6) The student can create 3D models for computer visualisation and optimise these for real-time simulation. 7) The student can (in a team) implement a real-time simulation on a distributed platform, using advanced software tools.	
Education Method	Lectures, programming instruction, and programming assignments.	
Literature and Study Materials	Lecture slides and other materials are put on Brightspace.	
Assessment	Individual assignment during 2nd/3rd week and team project report. Attendance at the team project meetings in week 3 and later is mandatory.	

AE4324	Physical Interaction for Aerial and Space Robots	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	S. Hamaza	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course covers the fundamentals of Robotics and delves deeper into the mechanics of physical interaction for aerial and free-flying systems. Topics span from modelling, design and motion planning of fixed robots, to the latest developments in control of floating-base robots physically interacting with objects. Many of these concepts will then be reproduced on a real robotic system, as part of the assignment. The course is structured in the format of lectures, in-class exercises and home assignment. The final examination will be a written exam. TOPICS: Introduction & Applications Robot Morphologies Robot Kinematics The Jacobian Matrix Robot Dynamics Trajectory Planning Interaction Control for Fixed Robots Interaction Control for Floating-base Robots Contact-based Navigation ASSIGNMENT: The goal is to code the kinematics, workspace, trajectory planning and an arbitrary interaction task of a real robotic arm, e.g. grasping/contour following/writing/. The grade includes the simulated experiments in ROS/MATLAB/OPENAIGYM; the real-world demo; and the report. The grade of the assignment should be higher than 5.	
Study Goals	After completing this course, the students are able to: -Identify different robot morphologies and associate the correct modelling rules -Derive the kinematics of simple and complex morphologies -Implement contact-less control laws for robot arms -Implement contact-based control laws for robot arms -Explain and implement the different methods in interaction control theory -Implement a simulation model in Python or MATLAB -Integrate software & hardware, familiarise with robotic hardware and learn how to debug it.	
Education Method	-Lectures & in-class exercises (4 hours per week) -Home assignment	
Assessment	Individual assignment (weight 40%) - Simulated results of robot kinematics and planning task - Real-world implementation on a 5-DoFs arm - Report & Code Written Exam (weight 60%) To pass, the exam grade must be sufficient (≥ 6) and the assignment grade cannot be lower than 5.	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Physical intelligence for interaction - Insect intelligence and Swarming - Spiking neural networks	
Study Goals		After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4351	Robust Flight Control	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	S. Theodoulis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE2235-I AE3212-I WI2056LR AE4301 AE4301P	
Course Contents	The vast majority of aerospace vehicles, spanning from UAVs and rotorcrafts to military or commercial aircraft and all the way up to space launchers, need to fly in a secure, predictable and repeatable way. To achieve this, the on-board flight control system needs to be designed to withstand (i.e. to be robust against) a whole array of uncertainties, disturbances, noise, rapidly-varying flight conditions and imperfect components. Furthermore, the designer needs to provide with solid guarantees and verifiable certificates that the flight control system must meet. This course is aimed at offering to the students hands-on theoretical and practical knowledge on the modelling, design and analysis of robust flight control laws, paying special attention to the industrial applicability of the tools albeit with a research twist.	
Study Goals	The learning objectives of this course are the following: LO1: Calculate multivariable mathematical models of aerospace systems that include system uncertainty LO2: Evaluate feedback loop robust stability and performance via the Structured Singular Value (SSV) and compare against more traditional methods LO3: Breakdown the fundamental control performance requirements and limitations intrinsic to the aerospace system models LO4: Identify and fix robustness deficiencies of linear-quadratic methods (LQR, LQG, H₂) for aerospace control system design (LQG/LTR) LO5: Compute and simulate robust control laws using modern H₂/H_∞ theory (mixed sensitivity, loop shaping) using the latest tools in MATLAB/Simulink To illustrate the above concepts the short period dynamics of a highly manoeuvrable aerospace vehicle are considered and various control and stability augmentation systems are designed. The results and simulations are done in MATLAB/Simulink.	
Education Method	Lectures together with assisted computer exercises in MATLAB/Simulink	
Books	Some books that may be used from the students for reference are: - Multivariable Feedback Control: Analysis and Design, Ian Postlethwaite and Sigurd Skogestad - Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems, Brian L. Stevens, Eric N. Johnson, and Frank L. Lewis - Robust Multivariable Control of Aerospace Systems, Declan Bates and Ian Postlethwaite	
Assessment	The assessment is a take-home assignment which involves the step-by-step design and comparison of various robust control laws taught in the course. The system used is the same one considered during the course, namely the longitudinal short period dynamics of a highly agile aerospace vehicle.	

AE4352	Mathematical and human-inspired decision making	3
Responsible Instructor	Dr. A. Jamshidnejad	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course you will learn about model predictive control (MPC) as well as human-inspired decision making (particularly, how fuzzy logic is used for decision making)	
Study Goals	After finishing this course successfully, the student should be able to: - Formulate model predictive control (MPC) problem for mathematical decision making - Develop fuzzy logic controllers for human-inspired decision making - Implement MPC controllers and fuzzy logic controllers for case study problems	
Education Method	Lectures	
Assessment	Group project (report + presentation of results)	

AE4422-20	Agent-based Modelling and Simulation in Air Transport	4
Responsible Instructor	Dr. O.A. Sharpans'kykh	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction. Agents and Multiagent systems. Agent-based modelling architectures. Examples from air transportation. Emergence in Multiagent systems. Agent-based simulation. Agent-based modeling and simulation tools. Agent-based coordination, planning, and scheduling in air transportation. Nature-inspired approaches to solve optimization problems. Swarm intelligence. Adaptive behavior and learning in agent-based systems. Collaborative decision making in air transportation. Negotiation, auctions, game-theoretic approaches. Agent-based model analysis: sensitivity, uncertainty, robustness. Validation of agent-based models.	
Study Goals	The student has to be able: - to formulate a practical air transportation problem as an agent or a multiagent system model; - to identify appropriate agent-based methods and to apply them; - to implement agent-based models; - to perform agent-based simulation, interpret and analyze simulation results; - to be able to apply agent-based optimization techniques.	
Education Method	Lectures and practical sessions	
Assessment	65% of the final grade for the course is the grade for a group assignment; 35% is the grade for the individual defense.	
Tags	Adventurous Algorithmics Analysis Artificial intelligence Group work Intensive Modelling Programming Small groups Transport & Logistics Transport phenomena	

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory	
Course Contents	Fundamentals of Coding in Python and MatLab Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Symbolic regression * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control We will aim for a balance of practical intuition and theoretical understanding: we will discuss examples of applications and go deeper into the theory when feasible.	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures (recorded and in class) and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: (i) practical assignment and (ii) written exam, open book. You must score at least 5.5 on both assignments to get a pass. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity to resit if they fail the written examination. * If your practical assignment is graded between 5 and 5.75, you will have two weeks to integrate the assessors feedback. If this is done satisfactorily, you can be rewarded with a mark that is not higher than 6/10. The exact instructions for the addition will be announced after the grading of the PA is concluded.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding course registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) exam registration is obligatory. More information regarding exam registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Artificial intelligence Complex Mathematics Intensive Linear Algebra Mathematics Modelling Programming Project Proofs	
Department	ME Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42035	Integration Project Systems and Control	5
Responsible Instructor	D. Boskos	
Instructor	Ir. S.C. Bregman	
Instructor	Dr. L. Laurenti	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	SC42015 Control Theory SC42155 Modeling of Dynamical Systems SC42095 Control Engineering SC42025 Filtering and Identification	
Course Contents	The course is based on practical laboratory sessions, in which students will gain hands-on experience with the application of control theory to real-world systems. Matlab and Simulink are used as the primary software environment for the design, analysis, and real-time implementation of the algorithms. Students work in groups of two in the lab, with a setup of their choice. The setups include for instance the rotary pendulum, helicopter model, inverted pendulum on a cart, rotational pendulum, reaction wheel pendulum, and magnetic levitation system.	
Study Goals	The goal of this course is to integrate and apply the theoretical knowledge gained in the compulsory courses of the Systems and Control curriculum. These courses cover the subjects of Control Theory, Physical Modeling, Filtering and Identification. The concepts and tools to be used include mechanistic modeling (based on principles like mass balances, Lagrange equations, etc.), filtering and estimation (e.g., Kalman filtering), linear control design and performance analysis, and system identification in open and closed loop. It is assumed that the students already know these concepts or are able to look them up in the literature. At the end of the course, the students should be able to: - Develop a mathematical model of a real-world dynamical system - Build a simulation model of the system in Matlab/Simulink - Identify the model parameters - Design a state estimator to recover unmeasured system states - Design controllers that regulate the derived system model - Integrate their acquired control engineering knowledge in laboratory setups	
Education Method	Project, demonstration	
Assessment	There is no written exam. The final grade is determined on the basis of a written report (50% of the grade), the final discussion of the results with the lecturers (50%) and the performance during the entire course period (rounding off the grade).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. Due to the lab work that's needed to complete the project, it is required that you are enrolled on the Brightspace page and take part in the group/setup assignment procedure during the first week of the period. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC52035	
Department	ME Department Delft Center for Systems and Control	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Sustainable Air Transport	
Program Coordinator	A. Bombelli
Introduction 1	In operating airlines and airports, a large number of (competing) requirements are encountered. These comprise minimizing environmental effects (noise, air quality and climate effects) and costs, and maximizing resilience and, of course, safety. To deal with all requirements two elements are important. First, you need to be able to quantify all these so-called Key Performance Indicators (value, environmental impact, cost, capacity, efficiency and safety). You will learn to deal with questions like How can we define how efficient operations are? or What is the contribution of aviation to greenhouse gasses in the atmosphere?. Simulations, using appropriate modelling techniques, are an essential element in this. Three SAT profile core courses aim to provide you with the relevant theories. - The SAT profile course Agent-based modelling and simulation in Air Transport introduces you to the important and widely used concept of agent-based modelling of complex distributed dynamic systems. - Through the course Aircraft Noise you will be able understand the relationship between aviation and the resulting noise levels, especially around airports. - Finally, the Stochastic Processes and Simulation course brings in the concept of randomness, accounting for the unpredictability of daily life. The second element draws on this ability to quantify the Performance Indicators and aims to actually optimize airline and airport operations given the requirements agreed upon by the relevant stakeholders. The SAT profile course Airline Planning & Optimization expands upon the track course on operations optimization, now putting a special focus on airlines. Transfer of the developed tools and insights to daily life should be realized, for example through the use of decision support systems or monitoring strategies. The increasing availability of (big) data, ranging from airline network data and aircraft maintenance monitoring to noise and pollution measurements near airports, together with innovative AI and machine learning methods provide new insights and opportunities in the field of air transportation. Both modelling and such data-driven approaches are covered throughout the courses in this profile. Although all courses apply theories to the case of aircraft, airports and airlines, the theories are general and transferable to other fields.
Program Goals	The SAT profile learning objectives are: 1. Understanding of the key operational challenges faced now and in the future by airlines and airports. 2. Understanding the theories needed to quantify the performance indicators related to the operation of airports and airlines (environmental impact, cost, capacity, efficiency, and safety). 3. Apply these theories for modelling, measurements, and analysis in applications related to the operation of existing and future air transport systems. 4. Developing the ability to transfer the methods to society both through technological advancements (decision support systems) and communication.

AE4302	Avionics and Operations	3
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Ir. T.J. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4316 AE4316P AE4321-15 AE4322	
Expected prior knowledge	Flight Dynamics System and Control Theory	
Parts	Week arrangement Lecture and study material 1. Introduction to avionics systems. -. Air Data systems (home study). 2. Gyroscopes, attitude reference systems. 3. Compasses, heading reference systems. 5. Flight deck instruments and integrated systems. 6. Navigation equations. 7. The Flight Management System (FMS). 8. Inertial Navigation Systems. 9. Radio navigation systems (ADF, VOR, DME). 10. Landing guidance systems (ILS, GPS). 11. Communication, Navigation, Surveillance (CNS). 12. Satellite navigation systems (GPS). 13. Air Traffic Management (ATM). 14. The Future Air Navigation System (FANS).	
Course Contents	(see week arrangement) This course provides a comprehensive, unified coverage of the principles of modern navigation equipment and systems, both in the aircraft and on the ground, including the aircraft instrumentation and flight-deck systems, with a special emphasis on the important trends in the global air navigation and air traffic management system.	
Study Goals	1. The student can describe in detail the working principles of the avionics systems treated in the course. 2. The student can demonstrate the avionics systems' functionalities, identify their strong points and weaknesses, and make comparisons between the avionics systems. 3. The student can evaluate, criticize, and appraise their usage in the current and future operational context.	
Education Method	Lecture	
Literature and Study Materials	Lecture slides and other lecture material are put on the BrightSpace The books mentioned below (Pallett & Coombs, Kayton & Fried) can be hired at the Control & Simulation division. Recommended literature: E.H.J. Pallett, L.F.E.Coombs, Aircraft Instruments and Integrated Navigation Systems, Longman, 1997 ISBN 0-582-08627-2. M. Kayton, W.R.Fried, Avionics Navigation Systems, Wiley, 1997 ISBN 0471547956.	
Assessment	Digital, closed-book exam on campus	

AE4353	Artificial Intelligence for Aerospace Control and Operations	4
Responsible Instructor	Dr. G.C.H.E. de Croon	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Programming skills in Python are required, as obtained, e.g., with the course AE1205.	
Course Contents	This course forms an introduction to artificial intelligence for use in aerospace control and operations. The emphasis is on both the theory and application of machine learning methods, with a large emphasis on deep neural networks.	
Study Goals	The main study goals are: 1. Students can independently apply AI methods to aerospace problems for regression and classification tasks. 2. Students can explain the theoretical properties, learning methods, and functioning of AI methods, with a focus on machine learning (ML) and deep neural networks (DNNs). Topics covered in the course include: Supervised, unsupervised and self-supervised learning Artificial neural networks (perceptrons, multi-layer perceptrons, deep neural networks) Learning methods and advanced structures for deep nets Uncertainty estimation with ANNs Recurrency mechanisms Explainable AI / analysis of trained neural networks Hyperparameter tuning AI tools	
Education Method	The course will consist of 7 lectures and 7 practical sessions, in which the students apply the learned theory to practical examples from aerospace control and operations practice (e.g., using AI to interpret acoustic data, aircraft maintenance data or drone data).	
Assessment	The course assessment consists of a digital exam that requires both the answering of questions on the theory and programming to solve machine learning problems. A bonus point for the exam can be obtained by submitting a valid entry to a machine learning competition that is presented at the start of the course.	

AE4422-20	Agent-based Modelling and Simulation in Air Transport	4
Responsible Instructor	Dr. O.A. Sharpan's'kykh	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction. Agents and Multiagent systems. Agent-based modelling architectures. Examples from air transportation. Emergence in Multiagent systems. Agent-based simulation. Agent-based modeling and simulation tools. Agent-based coordination, planning, and scheduling in air transportation. Nature-inspired approaches to solve optimization problems. Swarm intelligence. Adaptive behavior and learning in agent-based systems. Collaborative decision making in air transportation. Negotiation, auctions, game-theoretic approaches. Agent-based model analysis: sensitivity, uncertainty, robustness. Validation of agent-based models.	
Study Goals	The student has to be able: - to formulate a practical air transportation problem as an agent or a multiagent system model; - to identify appropriate agent-based methods and to apply them; - to implement agent-based models; - to perform agent-based simulation, interpret and analyze simulation results; - to be able to apply agent-based optimization techniques.	
Education Method	Lectures and practical sessions	
Assessment	65% of the final grade for the course is the grade for a group assignment; 35% is the grade for the individual defense.	
Tags	Adventurous Algorithmics Analysis Artificial intelligence Group work Intensive Modelling Programming Small groups Transport & Logistics Transport phenomena	

AE4423-20	Airline Planning and Optimisation	4
Responsible Instructor	M.J. Ribeiro	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	It is assumed that students are familiar with modelling and optimization techniques. It is assumed that students have experience in code development in Python or MatLab (or any other programming language).	
Course Contents	This course provides students with knowledge to analyse planning problems related to airline operations and to develop modeling approaches to solve these problems. The focus is on the relationship between planning models and their operational implications. It starts with a general overview of the airline demand analysis, followed by the study of the planning framework which airlines operate in. This planning framework includes strategic decisions, namely fleet planning and network development, tactical decisions, such as scheduling and maintenance planning.	
Study Goals	At the end of the course, the students should be able to: (1) Define the main strategic, tactical, and operational problems of an airline. (2) Discuss multiple modelling and solution techniques. (3) Formulate an airline strategic and tactical planning problem in a way that it can be solved. (4) Evaluate the implications of airline planning decisions.	
Education Method	Weekly meetings will be held to discuss the theory. These sessions should be supplemented with the available online videos. Additionally, there will be tutorial lectures to discuss the assignment(s).	
Literature and Study Materials	The students are assessed according to the grades of a group assignment (60%) and an individual online exam (40%). References literature: - Belobaba, P., Odoni, A., Barnhart, C. (2009), "The Global Airline Industry", John Wiley & Sons Ltd, West Sussex, UK. [ISBN: 978-0-470-74077-4] - Lectures notes and additional material posted on Brightspace during the course. Additional Literature: - Doganis, R. (2010), "Flying Off Course: Airline economics and Marketing", 4th Edition, Routhledge: London, UK. [ISBN: 0-415-44737-2] - Bazargan, M. (2010), "Airline Operations and Scheduling", 2nd Edition, Ashgate Publishing Company, Surrey, UK. [ISBN: 978-0-7546-7900-4] - Clark, P. (2007), "Buying the Big Jets: Fleet Planning for Airlines", Ashgate Publishing Company, Hampshire, UK. [ISBN: 978-0-7546-7090-2]	
Prerequisites	The students should have completed the AE4441 course, or the ME44206 course, or an equivalent operations research course. The students should also be familiar with programming. For python courses, students can follow TIL6010, AE1205, or TI3105TU.	
Assessment	Group assignments - divided into two parts (60%) Final exam (40%) The students need to have a grade equal to or higher than 5.5 in both examination elements. The group assignments are done by groups of (maximum) 3 students.	
Permitted Materials during Tests	None	

AE4431-23	Aircraft noise	3
Responsible Instructor	Prof.dr. D.G. Simons	
Gast Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is required for AE4463P-23 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics (dBA meter) - Dutch aircraft noise model NRM or airframe noise model (ANOPP method) - digital signal analysis and acoustic imaging	
Expected prior knowledge	Basic knowledge on calculus and first year physics.	
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and terts band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phone and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. Acoustic imaging: Principle of beamforming. Imaging aircraft noise data. (Note: this topic is only touched upon in this course, the details (including application) are treated in AE4463P-23) Aircraft noise sources: Mechanism of propeller noise and blade passage frequency. Mechanism of turbo engine noise. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise. ANOPP modelling method. Programming assignment about modelling airframe noise (landing gear, flaps, slats, etc.) using the ANOPP method.	
Study Goals	Understand the relationship between aviation and the resulting noise levels, especially around airports.	
Education Method	Lectures, exercises, programming assignment	
Literature and Study Materials	Lecture slides, exercises with worked-out solutions, programming assignment Reader: Course AE4431-23 Aircraft Noise Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2023	
Assessment	- Written exam (max 8 points) (a formula sheet will be provided) - programming assignment (max 2 points)	
Set-up	Lectures, exercises, programming assignment	

AE4441-16	Operations optimisation	4
Responsible Instructor	Ir. P.C. Roling	
Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	4/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course aims at providing the students with knowledge and experience to set-up and analyze linear and nonlinear optimization problems. The course covers the following topics: 1. Introduction to Operations Research, examples of OR models for air transport. 2. Linear programming (LP) models and the simplex Method. 3. Sensitivity analysis and Duality. 4. Transportation and assignment problems. 5. Network optimization and dynamic programming. 6. Mixed Integer Linear programming. 7. Nonlinear programming.	
Study Goals	At the end of the course, the students will be able to: 1. Understand the theory behind basic linear and non-linear optimization problems. 2. Model a problem as a linear program, a (mixed) integer program or a network optimization problem; 3. Verify the model using self created test data set. 4. Create and apply a sensitivity analysis.	
Education Method	The course consists of two studio classroom sessions (2 times 2hrs every week), where the theory is practiced and help is available. Students need to watch lecture video's and/or read up on the subject using the book before the studio classroom sessions.	
Books	Introduction to Operations Research Frederick S. Hillier and Gerald L. Lieberman, Tenth Edition Mc Graw-Hill International Edition	
Assessment	The final grade consists of two parts: 50% individual exam (computer-based open book); 50% an assignment (groups of 2-3 students). Both partial grades must be 6.0 or higher.	

AE4462-17	Aircraft Emissions and Climate Effects	4
Responsible Instructor	Prof.dr. V. Grewe	
Instructor	Dr. I.C. Dedoussi	
Responsible for assignments	V.R. Meijer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on calculus, fluid mechanics and first year physics.	
Course Contents	The course introduces the students in the state-of-the-art capabilities, research issues and challenges in climate effects of aviation. Topics include: - Combustion and emissions - Overview of all environmental effects of aviation - Air pollution relevant atmospheric chemistry and physics - Local air pollution modelling - Aviation environmental policy, new technologies, and other mitigation options - General circulation of the atmosphere - Atmospheric chemistry and contrail formation - Climate change caused by air traffic emissions - Modelling of climate impact from aviation - Application and show cases for the assessment of mitigation options supporting decision making, e.g. for climate optimized trajectories	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the climate impact of air traffic. These elements are required, for example, for the assessment of the air quality and climate impact of aviation and mitigation measures, including climate optimised trajectories/routing, low climate impact routing and consequences for aircraft design and supersonic transport.	
Education Method	Lectures and exercises	
Literature and Study Materials	Lecture slides and hand-outs	
Assessment	Written exam (Resits may be oral, depending on the situation) Assignments Both the written exam and the assignments need to be passed with a 6.0 or higher.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5322	Thesis Control and Operations	45
Responsible Instructor	Dr.ir. C.C. de Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal). If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go' (fail). Please refer to Brightspace for more information.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Sustainable Air Transport

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2	
Start Education	4	
Exam Period	2	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics.	
	The following topics will be covered:	
	- Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to:	
	- Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases.	
Enrolment / Application	The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course.	
	The selection procedure for this course will be as follows:	
	- The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59.	
	Applications submitted outside of these slots will not be considered.	
	- In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course?	
	- The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period.	
	- The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Prof.dr.ir. L.L.M. Veldhuis	
Instructor	L.T. Lima Pereira	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130-22, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during the first 3 weeks of Q1).	
Assessment	The assessment is done through 5 assignments that are performed on a personal basis. The assignments are to be combined in 1 report that should be delivered after the finalization of the lecture at the end of Q2. Additional information on the timing and contents of the assignments will be presented on BrightSpace.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Responsible Instructor	W. Yu	
Gast Instructor	Dr.ir. T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic aerodynamics, level BSc Aerospace Engineering.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade elementmomentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsens Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4238	Aero Engine Technology	4
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	Dr. F. Yin	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery", "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Inlets and Exhaust - o Turbomachinery - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls - o New Engine /propulsion Concepts	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an flipped course in which the lectures are pre-recorded and the students are supposed to watch the lectures before coming to the classroom. The classroom interaction is mainly to have discussions and for tutorials. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13. Evaluate the calorific values for various fuels 14. Understand the basics of combustion and emission formation in Aero engines 15. Understand the working of gas turbine combustor 16. Understand the basic principles of gas turbine controls and evaluate different strategies for control 17. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a flipped teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

	1. Gas Turbine Theory by H.I.H.Saravanamuttoo, G.F.C. Rogers, and H. Cohen 2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons
Reader	A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Prerequisites	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Assessment	The assessment is based on 2-3 assignments given during the course and a final exam. All assignments together will form 30% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
Permitted Materials during Tests	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor	Dr. D. Ragni	
Instructor	Prof.dr.ir. M. Snellen	
Co-responsible for assignments	Dr. R. Zamponi	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources.	
Study Goals	The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4270		Control and Operations Project	4
Responsible Instructor	Dr.ir. C.C. de Visser		
Instructor	Dr.ir. J. Ellerbroek		
Instructor	Dr. G.C.H.E. de Croon		
Instructor	E.J.J. Smeur		
Instructor	A. Bombelli		
Contact Hours / Week x/x/x/x	///		
Education Period	Different, to be announced		
Start Education	1		
Exam Period	none		
Course Language	English		
Course Contents	This course code is a placeholder for miscellaneous student assignments within the Control and Operations track. For example, a lecturer would like to have some data processed or have a nice simulation model that can be used for demonstration purposes in the lectures. Through this course code, we now have a way to compensate students for their efforts by considering it as an elective of 4 ECTS. If there are such assignments available, they will be announced.		
Study Goals	n/a, depends on the nature of the assignment		
Education Method	n/a		
Assessment	A (short) report needs to be written about the assignment. The content needs to be discussed with the supervisor of the assignment.		
Elective	Yes		

AE4320	System Identification of Aerospace Vehicles	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. D.M. Pool	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301.	
Parts	Week arrangement 1. Introduction. A global overview of the complete system identification cycle applied to aerospace vehicles, from sensor data acquisition and state estimation to aircraft aerodynamic model identification and validation. Introduction to sensor data acquisition and management, state and parameter estimation, and the two-step method for aerodynamic model identification. Finally an overview is give of advanced modelling techniques. 2. State estimation. Introduction to the essential role of the Kalman Filter (KF) in aircraft state estimation. The extension of the standard KF to nonlinear systems will be introduced in the form of the Extended Kalman Filter/Smoothen (EKF/EKS), and iterated extended Kalman filter (IEKF). Applications of the Kalman filters in aerospace navigation, attitude determination, on-board sensor calibration, sensor integration, and data fusion will be discussed. 3. Parameter estimation. Introduction to the process of parameter estimation and its role in creating accurate aerodynamic models of aircraft. Introduction to four of the most widely used parameter estimation techniques; Least squares (LS), Weighted Least squares (WLS), Total Least Squares (TLS), and Maximum Likelihood (ML) parameter estimation. Additionally, recursive parameter estimation methods like Recursive Least Squares (RLS) will be introduced. 4. Advanced optimization techniques. Some problems cannot be solved adequately with standard linear optimization techniques. In that case, a nonlinear optimization method must be used. In this part of the course, a powerful global nonlinear optimization technique based on interval analysis will be introduced. 5. Advanced modelling methods: Neural Networks. Neural Networks are a class of black-box function approximators that can be used to model nonlinear systems like Micro Air Vehicles and high performance aircraft. The application of neural networks to aircraft system identification is discussed, and it will be shown how neural networks can be used inside a framework for parameter estimation. 6. Advanced modelling methods: Multivariate Splines. Multivariate Simplex B-Splines are a powerful type of white-box function approximators that can be used to create accurate models of aircraft with highly nonlinear aerodynamics such as MAVs and high performance aircraft. A number of unique methods for assessing the quality of the created models are discussed, and pointers towards future research are given. 7. Demonstration: Aircraft system identification with multivariate splines. An in-depth demonstration will be given of the process of Aircraft system identification with multivariate splines using real flight test data from the Cessna Citation II laboratory aircraft. The goal is not only to create an accurate flight dynamics model of the Citation II, but also to apply various model validation techniques to assess the quality of the created model.	
Course Contents	Accurate aerodynamic models play a crucial role in the design and operation of flight simulators and flight control systems. The creation of accurate aerodynamic models from CFD, wind tunnel, and flight test data has historically been a highly challenging task. This is a direct result of the nonlinear nature of aircraft (aero)dynamics and the fact that not all aircraft states can be measured directly. As a consequence, the aerospace vehicle parameter identification problem constitutes a joint parameter-state estimation problem. It is the aim of this course to provide the student with a complete overview of the system identification cycle as it is currently applied to aerospace systems, and to introduce the student to the current state of the art in the field of aerodynamic model identification. The course consists of 7 parts covering the entire system identification cycle. In the first part of the course, the process of data acquisition using on-board sensors including accelerometers, gyroscopes, GPS and various air data sensors is discussed. It will be demonstrated that measurements made by real-world sensors are contaminated with noise, and are sometimes biased. Additionally, some aircraft states, like the true angle of attack, cannot be measured directly and must be reconstructed by combining sensor measurements. The second part of the course introduces the concept of state estimation in which prior physical knowledge of the system is used in a Kalman filter (KF) to estimate the true aircraft states from the measured states. Next to the ordinary Kalman filter, the Extended Kalman filter (EKF) and the Iterated extended Kalman filter (IEKF) will be introduced. In the third part of the course various methods for the estimation of model parameters will be discussed. It will be shown how the results from the state estimation are used in combination with a parameter estimator. This part not only focuses on offline batch parameter estimation methods like ordinary least squares (OLS), weighted least squares (WLS), total least squares (TLS), and maximum likelihood (ML) estimators, but also on recursive parameter estimation methods like recursive least squares (RLS) that can be used online. Special attention will be paid on the process of choosing a parameter estimator that is right for the job. The fourth part of the course introduces an advanced global nonlinear optimization method based on interval analysis. In aerospace system identification, so-called non-convex and nonlinear optimization problems are often encountered. Such problems can be solved with global nonlinear optimization methods like interval analysis. The fifth and sixth part of the course focuses on two advanced model structures that can be used in combination with the earlier introduced parameter estimation methods. In the fifth part of the course the neural network black-box function approximator is introduced. It is shown how neural networks are used to approximate scattered multidimensional data. In the sixth part of the course a new method for aerodynamic model identification based on multivariate simplex B-splines is introduced. This method was recently developed at the Faculty of Aerospace Engineering of the TU-Delft, and has a number of advantages over existing methods. For example, the simplex B-splines have a transparent model structure, are general in any number of dimensions, and can be computed efficiently in real-time.	

Study Goals	In the final part of the course, all theory introduced in the first six parts is used in a demonstration that uses real flight data from a real-life system like the Cessna Citation II laboratory aircraft, or a Micro Air Vehicle to identify a multivariate spline based aerodynamic model. The created model is validated using various techniques. Finally, pointers are given towards further research in the field of aerodynamic model validation. After completing the course, students will have a thorough understanding of the theoretical elements of aircraft system identification. Students will be able to apply these elements to go from real CFD, wind tunnel, and flight test data to an accurate aerodynamic model that can be used in simulator or flight control system applications.
Education Method	Lectures.
Literature and Study Materials	Q.P. Chu, Modern flight test technologies and system identification, lecture notes, Faculty of Aerospace Engineering, Delft University of technology. C.C. de Visser, Global Nonlinear Model Identification with Multivariate Splines, PhD thesis, Faculty of Aerospace Engineering, Delft University of technology. Recommended literature V. Klein and E. A. Morelli, Aircraft System Identification, AIAA, 2006. J.A. Mulder, J.K.Sridhar, J.Breeman, Nonlinear Analysis and Manoeuvre Design, Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 2, , May 1994 R.E. Maine, K.W.Illiff, AGARD Flight Test Techniques Series on Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 1, , December 1986 . Eric Walter and Luc Pronzato, Identification of parametric models, Springer, ISBN 3-540-76119-5.
Assessment	Individual Take-home assignments

AE4321-15	Air Traffic Management	4
Responsible Instructor	Prof.dr.ir. J.M. Hoekstra	
Instructor	Dr.ir. J. Ellerbroek	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Course Contents	The course gives an Introduction in Air Traffic Management and ATM research topics. 1. Introduction and history of Air Traffic Management, organisational & regulatory aspects 2. Airspace set-up, flight planning and navigation 3. Communication: Information services, Voice R/T, Data Link 4. Navigation: Great Circle calculations, WGS'84, approximations, beacons/navaids, landing aids 5. Surveillance: Radar basics, primary-secondary radar, radar types, ADS-B 6. Separation: Separation Minima, Near Misses, Closest Point of Approach Calculations, wind vortex categories 7. New concepts: Airborne separation assurance, 4D contract TBO, paired approaches, etc. 8. ATC Simulations: fast-time, human in the loop 9. ATC Research organisations and large scale programs (SESAR, NextGen) 10. ATM research topics: interactively discussed in class based on presentations by students While the first part of the course is mainly lectures, the second part of the course is much more interactive. Here pairs of students get assigned a paper, project or case to study and form an opinion on. They should explain and review it in both a brief essay and during a short presentation in class, followed by a discussion with the class in which they need to defend their view on this study. In this way the students are familiarized with different aspects and opinions on the 'hot topics' in ATM research currently.	
Study Goals	This course is designed to give the students an appreciation of the current ATM systems, throughout the world, with emphasis on the core area of Europe, in preparing the existing ATC systems for the predicted increase of traffic. Emerging technologies will be described, with the associated implementation aspects like Air Traffic Controllers' systems, the history and future of ATC, airline perspective, European and US new ATM programmes SESAR and NEXTGEN. Emphasis is on both the current technology as well as future concepts and technology which are currently being researched.	
Education Method	Lectures will be given on-line via livestreaming. Assignments will be given to the students for the second period. Students need to present their understanding of and opinion on a case, project or paper on-line. Discussions on this with the other students are part of the interactive method used in the second period.	
Reader	Hand-outs of the presentations are available on Brightspace.	
Assessment	Written, on site exam (50%) and the assignment (essay+presentation, also 50%). Both parts need at least a 5.0.	

AE4352	Mathematical and human-inspired decision making	3
Responsible Instructor	Dr. A. Jamshidnejad	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course you will learn about model predictive control (MPC) as well as human-inspired decision making (particularly, how fuzzy logic is used for decision making)	
Study Goals	After finishing this course successfully, the student should be able to: - Formulate model predictive control (MPC) problem for mathematical decision making - Develop fuzzy logic controllers for human-inspired decision making - Implement MPC controllers and fuzzy logic controllers for case study problems	
Education Method	Lectures	
Assessment	Group project (report + presentation of results)	

AE4439	Sustainable Air Transport Modelling Project	2
Responsible Instructor	A. Bombelli	
Instructor	Ir. P.C. Roling	
Instructor	M.J. Ribeiro	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	All mandatory SAT courses.	
Course Contents	The course has a project format. It is meant to deepen knowledge and skills in mathematical and computational modelling in application to specific sustainable air transport problems. The course can be done under supervision of any SAT lecturer, who is willing to supervise such a project. Specific project themes can be very diverse, but should be in line with the research areas of the SAT profile. A specific theme can be defined in consultation with a supervisor. It is advisable to work on a project in Q3 or Q4, when the basic modelling techniques have been learned in Q1 and Q2.	
Study Goals	n/a, depend on the project	
Education Method	The course has a project format.	
Assessment	A report needs to be written about the project. The content needs to be discussed with the supervisor.	

AE4446	Airport and Cargo Operations	4
Responsible Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of the following lectures: Lecture 1 Introduction Lecture 2 Landside operations Lecture 3 Airside operations Lecture 4 The Gate assignment problem Lecture 5 The Aircraft taxiing problem Lecture 6 Air cargo supply chain Lecture 7 Bin packing problems in air cargo operations Lecture 8 Introduction to graph theory and scheduling problems Lecture 9 Introduction to stochastic programming Lecture 10-14 Guest lectures A short quiz is given after lectures 2 to 9. Bonus points for the test can be earned by participating in these quizzes. Note that the sequence of the lectures might change pending the availability of the invited speakers. In addition, if not enough invited speakers are found, such lectures will be replaced with lectures covering additional topics (e.g., uncertainty in airport operations) or a Q&A related to the assignment. Knowledge of linear optimization and familiarity with Mixed Integer Linear Programming (MILP) formulations is a strongly suggested pre-requisite because many models presented in the lecture and the assignment are MILPs that the students must understand and (potentially) reproduce. Additional resources, examples, and tutorials will be provided to help students.	
Study Goals	At the end of the course, students will be able to: - Understand and critically assess key airport characteristics - Understand the air cargo supply chain in terms of stakeholders and operations - Understand, reproduce, and modify mathematical models describing different operations occurring inside or around airports	
Education Method	The course consists of about 8/9 theoretical lectures and 5/6 guest lectures (based on availability of guests). Students work in a group of up to 4 students to assess and reproduce an airport/cargo related optimization model (MILP model) and write a report. As such, some knowledge about linear programming is a strongly suggested pre-requisite for the course (e.g., courses such as Operations Optimization or Quantitative Methods for Logistics are recommended courses). A closed-book computer test is used to test and grade the individual knowledge of the students.	
Literature and Study Materials	Slides and other information and supplementary material are posted on Brightspace. Some lectures and the assignment are based on academic papers describing MILP models. Such papers are also uploaded on Brightspace.	
Books	Airport Systems: Planning, Design, and Management, Richard de Neufville and Amadeo Odoni, McGraw-Hill	
Assessment	40% Individual computer test (must be 5.0 or higher) 50% Group report (must be 5.0 or higher) 10% Homework quizzes	
Permitted Materials during Tests	Non graphical calculator	

AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course AE4463P-23 aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) must be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and also applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B, C and F are required. A choice can be made between D and E.	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4465	Maintenance Modeling & Analysis	4
Responsible Instructor	A. Bombelli	
Instructor	Dr. I.I. de Pater	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Maintenance is a critical operational activity in aviation, and is crucial to retain the aircrafts airworthiness. With the current developments in AI, the field of maintenance is quickly evolving as well, and new maintenance strategies involve methods from machine learning and data science. However, also classical reliability methods, probability and statistics remain very important in the field of maintenance. This course will cover all these topics. Specifically, the course consists of three parts where you will learn to: I Introduction to maintenance: 1. Describe the concepts of maintenance and reliability. 2. Apply the basic theories and models of probability and statistics to a variety of problems (recap). II Preventive maintenance (i.e., classical reliability methods): 1. Describe the concepts of preventive maintenance. 2. Fit an appropriate distribution to historically observed lifetimes of the component 3. Optimize a preventive maintenance strategy based on the usage of the component, using stochastic processes. If there is time left, you will also learn the basics of optimizing the inventory (spare parts) management. III Predictive maintenance: 1. Describe the concepts from predictive maintenance. 2. Estimate the current and future degradation and predict the failure time (remaining useful life) of aircraft components with various machine learning/data science techniques.	
Study Goals	By the end of the course, you should be able to 1. Optimize a preventive maintenance strategy for a component based on the usage of this component, including a. Fit an appropriate distribution to historically observed lifetimes of the component b. Identify the optimal maintenance time of the component based on a distribution of the historical lifetime, with various problem characteristics c. Explain the stochastic processes underlying this optimization procedure. 2. Apply the techniques from predictive maintenance to predict the future health of a component based on the sensor measurements of this component, including a. Estimate the current and future degradation in a component b. Predict the failure time of a component using various machine learning/data science techniques.	
Education Method	The content of the course will be given via on-campus lectures and tutorials. The students will also have access to the lecture slides and exercises on Brightspace. There will be assignment(s) to be solved either in groups or individually. The assignment(s) may involve coding problems, open questions, or mathematical exercises. The primary bibliography for this course will be: The presentation slides of all lectures will be made available through Brightspace. Non-mandatory reference literature: - Kumar, U. D., Crocker, J., Knezevic, J., & El-Haram, M. (2000). Reliability, Maintainability and Logistic Support - A Lifecycle Approach. Boston: Kluwer Academic Publishers. - Arts, J. J. (2017). Maintenance modeling and optimization. (BETA publicatie : working papers; Vol. 526). Technische Universiteit Eindhoven. - Axsäter, S. (2015). Inventory control (Vol. 225). Springer.	
Assessment	The course's final grade will comprise the individual exam grade (60%) and individual or group assignment(s) (40%). It is required that students have a grade equivalent or higher than 5.75 in the individual exam and an average grade equivalent or higher than 5.75 in the assignments to pass the course.	
Elective	Yes	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	After this course you are able to: - recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - solve the shortest path problem and the transportation problem as well as some small flow shop problems. - solve the travelling salesman problem by the Branch and Bound algorithm. - apply several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - reproduce basic theorems concerning the mentioned problems and is able to prove some of these theorems. - explain how to solve large scale problems, especially shortest path and vehicle routing problems. - define scheduling problems and complexity theory. - implement the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam and a written resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Elective	Yes	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr.ir. J.T. van Essen	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 3: Spacef

MSc Coordinator	Ir. K.J. Cowan
Introduction 1	The Space MSc Track (formerly: Spaceflight MSc Track) is offered to students who seek an academic program that prepares them well for a career in the international space sector. The MSc curriculum covers the full life cycle of space missions, from conceptual design to implementation and utilization. In line with this lifecycle, this Track focuses on spaceflight dynamics, space engineering and instrumentation, and planetary exploration. In astronautics, each space mission is uniquely designed to perform specific tasks in order to achieve its operational or scientific objectives. This requires an end-to-end approach where the objectives drive the design of the mission, and data processing is an integral part of the mission. In this Track you can participate in ongoing engineering and scientific projects. You will develop skills to independently and individually carry out a scientific engineering research project. This culminates in a highly focused MSc thesis project under the supervision of a scientific staff member. In addition, you will acquire broad knowledge in the field of space and its applications. Pursuing these learning objectives will make you a well-rounded aerospace engineer with widely applicable engineering and research skills and with a clear area of specialization. The Track consists of three Profiles: Spaceflight Dynamics, Space Engineering, and Planetary Exploration. Each Profile has a different focus, yet they are also interrelated. On the one hand, you develop as an all-round space professional with a focused area of specialization, and on the other hand you will acquire broader skills that enable you to pursue a career in a wide spectrum of industrial and research environments.
Program Goals	Learning objectives of the MSc Track You will develop skills to independently and individually carry out a scientific engineering research project. This culminates in a highly focused MSc thesis project under close supervision. In addition, you will acquire broad knowledge in the field of space and its applications. Pursuing these learning objectives will make you a well-rounded aerospace engineer with widely applicable engineering and research skills and with a clear area of specialization. Experience has shown that all students who selected this Track have gone on to successful careers in industry, knowledge institutes, agencies, and academia, both within as well as outside the aerospace world.
Program Structure 1	The Track consists of three Profiles that share a common core of 19 ECTS. The Track core contains four space courses covering spacecraft trajectories, space systems engineering and instrumentation, and planetary sciences, plus a compulsory course on ethics. These courses provide a strong basis for any space engineer. Profile courses (with a mandatory minimum number of ECTS) are intended to broaden this basis and depend on the student's interest in the dynamics, engineering, or planetary exploration aspects of space missions. Profile SD (Spaceflight Dynamics) focuses on flight mechanics, simulation, optimization, data processing, and applications. For this Profile, the student selects a minimum of 13 ECTS of coursework from a total of 19 ECTS. Profile SE (Space Engineering) goes deeper into the propulsion, instrumentation, thermal characteristics, and design of spacecraft and rockets. For this Profile, the student selects a minimum of 13 ECTS of coursework from a total of 24 ECTS. Note also: the student may only choose either ae4s52-21 (Collaborative Space System Design Project) or ae4897 (Space Engineering Practical) - and thus not both of these courses. Profile PE (Planetary Exploration) utilizes instrumentation, flight mechanics, simulation, data processing, and applications, with a particular focus on the study of natural objects in the Solar System and beyond. For this Profile, the student selects a minimum of 14 ECTS of coursework from a total of 24 ECTS. In addition, students will also select elective courses, some of which will be courses for Personal and Professional Development (PPD). The program is completed with a mandatory internship (or equivalent) of 15 ECTS and the thesis project (45 ECTS). The choice of the elective courses is done in consultation with the potential supervisor of the thesis project, the Profile Coordinator of the Profile of interest, and/or the Track Coordinator. The electives are preferably tailored to the student's area of interest, already with an eye on a particular thesis theme (or even specific project). The MSc Space Track provides a spectrum of recommended electives, but it is also possible, in consultation with the potential thesis supervisor, and after approval by the Profile/Track Coordinator, to choose elective courses from other Tracks within or outside of the AE faculty, or even outside of TU Delft at other (preferably Dutch) universities.

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile I: Space Engineering	
Program Coordinator	Dr.ir. J. Bouwmeester
Introduction 1	Profile I "Space Engineering" Short description: this profile combines multidisciplinary engineering fields to realise high-performance space systems and system components. Areas of interest include the engineering of space missions, space vehicles and instruments, sensors, actuators, mechanisms, propulsive means, vehicle control, distributed space systems, and Systems Engineering. In Space Engineering, applying state-of-the-art technologies is daily business. The Space Engineering profile aims to educate future academic engineers in end-to-end space engineering. You will learn how to engineer complex space vehicles: launch vehicles, satellites and deep space probes, as well as their subsystems, instruments and components using a systematic approach. You will gain expertise on every phase of a spacecraft system development including design, development, integration, verification/testing and operation. You will also learn about management aspects of engineering including scheduling and planning, and about working in multi-disciplinary design teams to formulate spacecraft system solutions. This will enable you to apply your gained knowledge also to other complex technical systems outside of aerospace engineering. NOTE: For the academic year 24/25 you can select as a profile course either the course "Space Engineering Practical" or the "Collaborative Space Design Project". You cannot select both as for both we have a limited number of places.

AE4874 I	Fundamentals of Astrodynamics	4
Responsible Instructor	Ir. K.J. Cowan	
Instructor	Dr. S. Gehly	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	ae4866 ae4868 ae4870 ae4872 ae4889 ae4894	
Summary	Thorough treatment of the fundamentals of astrodynamics; topics are treated in a mathematically cohesive way. You will acquire fundamental insights into the theory and methods of astrodynamics as well as working knowledge to compute and analyze spacecraft trajectories.	
Course Contents	Introduction to astrodynamics; two-body and many-body problems; relative motion; coordinates, reference frames, orbital elements, and time; geocentric, cislunar, and interplanetary flight; introduction to orbital perturbations.	
Study Goals	You will be able to: 1. Provide full and accurate descriptions of the fundamental concepts and phenomena of astrodynamics; 2. Explain the relationships between astrodynamics concepts & phenomena and the mathematics & physics theory that describes them; 3. Apply the relevant relationships, models, and methods for astrodynamics analysis; 4. Derive and manipulate the equations of motion for a spacecraft subject to gravitational and other forces; 5. Apply the equations of motion, their solutions, and related equations to analyze the absolute and relative positions and velocities of spacecraft as a function of time. Additional detail will be available in the course syllabus.	
Education Method	Lectures, workshops, self study	
Literature and Study Materials	Fundamentals of Astrodynamics, Karel Wakker, 2015. ISBN: 978-946-186-4192 available for download from the TU Delft Library / Institutional Repository also available to order as university reader in hardcopy Astrodynamics course website Orbital Mechanics for Engineering Students, Howard Curtis, 2013, 3rd edition. ISBN: 978-008-097-7478 (print) 978-008-097-7485 (ebook) available for download from the TU Delft Library also available to order from booksellers as hardcopy	
Assessment	Written exam	

AE4890-11	Planetary Exploration	4
Responsible Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	Dr.ir. W. van der Wal	
Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J.M. Potin	
Instructor	Dr. S. Paardekooper	
Instructor	Dr. S.J. de Vet	
Responsible for assignments	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4876 Planetary Sciences II	
Parts	Planetary Sciences: - Origin and Evolution of our Solar System - Planetary Surfaces and Interiors - Planetary Atmospheres - Planetary Magnetism - Planetary Ring Systems - Asteroids and Comets - Extrasolar Planets - Recent Topics in Planetary Science Current topics in planetary science with a focus on a specific topic each time the course is offered, such as Mars, Venus, Galilean satellites, comets, recent discoveries, spacecraft observations, or other developments.	
Course Contents	Planetary science is a major interdisciplinary field, combining aspects of astrophysics with geology, geophysics, meteorology, atmospheric and space science. This close relationship to geophysics, atmospheric and space sciences implies that the study of the planetary bodies in our Solar System and beyond offers the unique opportunity for comparison available to Earth scientists. This course teaches the concepts in the planetary physical sciences and solar system properties. The learning process is greatly enhanced by involving students in solving related problems.	
Study Goals	After the course the student should be able to: 1. Describe the primary physical processes in solar and planetary systems and the properties involved 2. Apply modelling techniques that are commonly used to describe the primary physical processes in solar and planetary systems 3. Demonstrate the use of physical principles to derive knowledge on the state and evolution of (extra-)solar system bodies from observations 4. Assess the quality of evidence of current knowledge of (extra-)solar system bodies 5. Assess the habitability of a (extra-)solar system body based on inferences of its properties and orbit	
Education Method	Classical oral lectures and classroom/take home assignments.	
Literature and Study Materials	Jack J. Lissauer and Imke de Pater, Fundamental Planetary Science, Cambridge University Press, ISBN: 9780521618557, 2013. PowerPoint/PDF files of the lectures and lecture notes.	
Books	Jack J. Lissauer & Imke de Pater, Fundamental Planetary Science, Cambridge Univ. Press, 2013.	
Assessment	The final grade for the planetary sciences course is determined from: - mark of a written exam - mark for mandatory assignments The mark for the written exam needs to be 5.0 or higher in all cases, no exceptions. If the mark for the exam is 5.0 or higher, grading for this course follows from averaging two parts with equal (so 0.5) weight: the mark for the classroom/take home exercises and the mark for the written exam. No opportunity will be given to redo the classroom/take home exercises within the same academic year. The grade for the classroom/homework assignments will remain valid for the next consecutive academic year, and so will annul the year thereafter	
Exam Hours	3 hours written exam	
Permitted Materials during Tests	Closed-Book exam, Non-programmable calculator is allowed.	
Set-up	Oral lectures and classroom/take home assignments.	

AE4896	Space instrumentation	4
Responsible Instructor	Dr. J.J.D. Loicq	
Contact Hours / Week x/x/x/x	4/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	The aim of this course is to provide students with an understanding of instrumentation and its role in observational strategies for missions. The course will explore the main observables of the universe and how they are measured in space. For each of the observables, the course will provide: i) The physical context of the observables and how they are generated, ii) The physics of the measure, iii) A review and explanation of the existing and under development space instruments, iv) The missions related to these observables. The course will be divided into three main parts: - The Fundamentals of Instrumentation - Remote sensing space Instrumentation - In-situ space Instrumentation.	
Study Goals	This course aims to familiarize students with the wide range of space instrumentation and observables. By the end of the course, students will be able to: - Understand and describe the relationships between instruments, missions, and observation strategies - Identify and explain the physical observables that space instruments measure, including light, ions, magnetic and electric fields, etc and related phenomena - Distinguish between the different types of space instruments, and explain how they are designed and constructed - Derive instrument requirements from scientific questions - Perform preliminary analysis and use the key driver parameters to size space instruments. Overall, this course will equip students with a comprehensive understanding of space instrumentation, enabling them to dive into space observables diversity and the complex processes involved in their measurement.	
Education Method	Lectures	
Assessment	The exam will test all theories provided in the lectures and skills practised in the provided exercises. It will be a closed-book written exam (3h). Formula sheet will be provided. The resit will have the same settings.	

AE4S07	Micropropulsion	4
Responsible Instructor	A. Cervone	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	2 3 4	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	IMPORTANT: starting from Academic Year 2024-25, taking this course will be allowed only to students enrolled in the Spaceflight MSc track. No mandatory requirements on previously taken courses. Advised: Propulsion and Power (BSc), Thermal Rocket Propulsion (MSc).	
Parts	The first part of the course (1 ECTS) will consist of online study material (available on Brightspace) covering the basics of micro-propulsion for small satellites. At the end of the first part, the student will be asked to take an individual test (1 hour, multiple-choice questions) to check their understanding of the material studied in Brightspace. Students who pass the test with a sufficient grade will be admitted to an individual project (3 ECTS), chosen by the student from a list of options regularly updated by the responsible instructor and aligned with the current micro-propulsion research at the SpE department.	
Course Contents	(1) Fundamental theory and state-of-art of micro-propulsion systems for small satellites; (2) Down-scaling and manufacturing of miniaturized propulsion systems and components; (3) Individual project (variable topic and goals, see below).	
Course Contents Continuation		
Study Goals	Understand the basics of micro-propulsion systems for small satellites and their fundamental differences to larger scale space propulsion. Apply the basic theory to identify the most important requirements for a micro-propulsion system, starting from the relevant top-level mission and satellite requirements. Understand and apply down-scaling rules for the miniaturization of propulsion and fluidic systems. Know, compare and apply (if required by the specific individual project taken by the student) the main available manufacturing techniques for micro-propulsion systems and components, their range of applicability and their advantages/drawbacks. Critically analyze the available state-of-art micro-propulsion options, identify their peculiar characteristics and ranges of applicability, discuss and justify the most suitable one(s) based on given mission and satellite requirements. Actively contribute to solve the challenges and achieve the goals of the current micro-propulsion research in the SpE department, by performing the tasks of a specific project, chosen among the ones proposed by the responsible instructor. Acquire hands-on skills on one or more of the following (depending on the specific individual project selected by the student): micro-propulsion laboratory and testing activities; design of micro-propulsion systems and components; micro-propulsion verification/optimization by means of analytical or numerical tools; or combinations of the above.	
Education Method	Self study (using the study material made available by the responsible instructor on Brightspace). Individual project (supervised by the responsible instructor or other members of the micro-propulsion research team).	
Literature and Study Materials	All study material (preparation material for the exam, project descriptions, lists of reference documents, etc.) will be made available on Brightspace.	
Assessment	1) Interim test (after taking the online part of the course): 1 hour, multiple choice questions, taken individually in front of the responsible instructor (in a date/time agreed with him). With a grade of 6.0 or higher, the student will be admitted to the individual project. With a grade lower than 6.0, the student will have to wait no less than 4 weeks before re-taking the interim test. With a grade higher than 6.0, the student is allowed to ask for a re-sit to improve the grade; this re-sit shall however be scheduled no less than 4 weeks after the first test. In line with the current general assessment rules of the MSc program in the AE faculty, only two test opportunities will be allowed per student per Academic Year (first exam + one re-sit). 2) Individual project: chosen from a list prepared and regularly updated by the responsible instructor. For all projects, a description will be provided to the students including: general background, tasks, research question(s) to be answered, expected deliverables and grading criteria. The project is designed for a total individual effort of 3 ECTS (= 84 hours) and will be considered successful when graded 6.0 or higher. With a grade lower than 6.0, the project will be considered a fail and the student will have to re-take the full course (thus, both interim exam and a new project), not earlier than 3 calendar months after the previous project has been failed. Kick-Off date for the project will be considered as the date when the student is provided by the responsible instructor with the required input material and general instructions on the selected project, as indicated in the project description. After this Kick-Off date, it is NOT allowed to the student, under any circumstance, to change the project choice. The only constraint on the project duration is that the final deliverables have to be provided to the responsible instructor not later than 3 calendar months after the Kick-Off date. A student who has not provided yet the deliverables will receive a reminder about it approximately two calendar weeks before the deadline. A student who doesn't meet the 3-months deadline will be given an additional extension of one calendar month to complete the project, but in this case the final course grade of that student will be reduced by 0.5 points. If a student doesn't provide the deliverables within this additional one month extension, the project will be considered a fail and the following will apply: -) The student will have to start a completely new project, not earlier than 3 calendar months after the previous one has been failed; -) The final course grade of that student will be reduced by 1.0 points. No exceptions to the above rules for schedule and deadlines will be granted, independently on the reason for the exception request. Final course grade: 25% interim test grade, 75% project grade.	
Permitted Materials during Tests	The only allowed materials during the interim test are a calculator and (when applicable) a formula sheet provided by the responsible instructor. No other materials are allowed, including lecture notes or internet-connected devices.	
Enrolment / Application	IMPORTANT: starting from Academic Year 2024-25, taking this course will be allowed only to students enrolled in the Spaceflight MSc track. Students have to enroll via the Brightspace page of the course, study the online material by following the instructions given in the same page, and inform the responsible instructor as soon as they are ready to take the interim test, deciding with him the date and time of the test. The test date shall be within 15 working days from the time when the responsible instructor is contacted by the student, except specific cases when this is not possible (e.g. holidays or business trips of the responsible instructor during the 15 working days period).	
Set-up	The course will continuously be available to students, who can decide when to start it based on their own study and schedule needs. The list of individual projects will regularly be updated in order to be always aligned with the ongoing micro-propulsion research activities in the SpE department. Various types of projects will be offered to best meet the preferences and	

characteristics of different students, ranging from lab/testing activities to design tasks, analytical/numerical work, or combinations of them. No specific limitations will be given on the duration of the individual projects, provided that the expected 3 ECTS (84 hours) effort is put by the student in it and that all project tasks are completed within a maximum of 3 calendar months after the project Kick-Off date.

After the student provides to the responsible instructor the deliverable items at the end of the project, the responsible instructor will check these deliverables and send his comments and remarks to the student. The student will then be allowed to revise the deliverables based on the comments received, and this revised version will be graded. No further iterations on the deliverables after this one will be allowed. The revised version of the deliverables shall be delivered by the student not later than 1 calendar month after receiving the comments from the responsible instructor; if this deadline is not met, the responsible instructor will grade the original version of the deliverables as is, without any modifications. The end date of the project will be considered the day when the student has sent the first draft of the required deliverables, provided that these deliverables are integral, logically formulated and complete.

The final project grade will take into account both the initial version of the deliverables and the improvements/adjustments made in the final version, after receiving the comments and remarks from the responsible instructor.

AE4S10	Microsat Engineering	4
Responsible Instructor	Dr. J. Guo	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Lectures from experts on various aspects of small satellites and group/individual projects for practicing.	
Course Contents	The course covers the engineering of small satellites. The concept and definition of micro-, nano-, pico-, and femto-satellites is introduced along with the past, present, and future of small satellites in general. The course comprises seven to eight lectures, partly from professionals in the space industry. The lectures are intended to familiarise the students with current industry trends and state-of-the-art research in small satellite engineering. The lectures vary each year and potential lecture topics include communications, AOCS, onboard computer, distributed space systems, miniaturised payloads, small satellite missions and applications, etc.	
Study Goals	The course will enable students to design and engineer small satellite missions. The high-level learning objectives of the course are: The students shall be able to explain key aspects of small satellite missions explain and describe elements in the architecture of a spacecraft mission discuss in detail at least one element in the architecture of a spacecraft mission present and defend their design of an element in the architecture of a spacecraft mission show familiarity with current trends in space industry and small satellite research	
Education Method	Lectures (Provided partly by professionals from the space industry). Group work (Students work on assignment in groups; Coaching will be provided to guide the students). Individual work (Students work on small individual assignments).	
Literature and Study Materials	Optional: Wertz J.R., Larson W.J. (eds.); Space Mission Analysis and Design; 3rd edition Kluwer Academic Publishers (2003). Mandatory: Lecture Notes AE4-S10 MicroSatellite Engineering.	
Prerequisites	General mathematical, physical and engineering knowledge from completed BSc study. In particular, AE1101 + AE1102 + AE3201, or similar	
Assessment	There will be two assessment components A Group Assignment and an individual assignment. Group assignment The final demonstration or report will be graded on the following aspects: Planning Presentation, argumentation and defence of the design Functionality and Usability of the deliverable Context of Small satellite (How well has the student team incorporated current small satellite research and industry trends/needs in the design) Independence Creativity Individual assignment This assignment will be graded on the following aspects: Understanding and summary Critical review of articles in a specified technical field of small satellites Please note: there is no written exam and, therefore, also no re-sit exam for this course.	
Permitted Materials during Tests	N/A	
Enrolment / Application	Students have to enroll via the BrightSpace site of the course. Due to the limited capacity, this course is open to Space Engineering profile students only.	
Set-up	The course comprises lectures, guest lectures, team sessions, group work and individual work.	

AE4S12-23	Space systems engineering	4
Responsible Instructor	Prof.dr. E.K.A. Gill	
Contact Hours / Week x/x/x/x	4/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	General engineering knowledge at completed BSc level. AE1110, AE1111, AE2111, AE3211, or similar; AE3524 (for students, not having AE BSc)	
Parts	1. Introducing Space Systems Engineering 2. Designing Systems 3. Integrating Systems 4. Managing the Systems Engineering Process 5. Managing the technical effort 6. Managing interfaces and configuration; Conclusions	
Course Contents	The course covers Space System Engineering exemplified by the design of a spacecraft subsystem for a satellite mission of high societal impact. It introduces a process which allows the creation of successful systems, such as space applications, space missions and satellite subsystems. Methods and tools are presented and exercised which will improve the depth and breadth of Space Systems Engineering graduate level education at the TU Delft by emphasizing the need for the end-to-end approach and life cycle of space systems, including cost and risk handling and using state-of-the-art methods, such as elements of Model-based Systems Engineering. Upon completion, the student will have a firm understanding of advanced Systems Engineering and be able to apply adequate methods and tools which help to create successful systems.	
Study Goals	The course mission is to enable students to realize a successful space system in an end-to-end Systems Engineering approach. There are three high-level learning objectives: 1. Participants shall be able to explain the objectives and tasks of Systems Engineering for realizing successful systems together with their needs, potentials, benefits and limitations in a context which comprises Business Engineering and Management. 2. Participants shall be able to design an end-to-end Systems Engineering process demonstrating a smart balance of risks of cost, schedule, and performance. 3. Participants shall be able to use Systems Engineering methods and tools to design a high-level satellite subsystem, such as the Attitude and Orbit Control System (AOCS).	
Education Method	Lectures and tutorial Individual homeworks from lectures to practice material Group Assignment including peer-reviews to design a satellite sub-system, done by students in small groups The graded group report and graded individual homeworks comprise the formal course close-out.	
Literature and Study Materials	Required: Lecture Slides AE4-S12 Space Systems Engineering. Optional: Larson W. J.; Kirkpatrick D.; Sellers J. J.; Thomas L.D.; Verma D.: Applied Space Systems Engineering. McGraw Hill (2009). Optional: Space Mission Analysis and Design, 3rd edition (Space Technology Library, Vol. 8) Eds.: Wiley J. Larson, James R. Wertz (1999)	
Assessment	4 graded individual homeworks + 1 Group Assignment. Acceptance of 6 homeworks is required to pass the course. Passing both assessment elements is required to pass course.	
Enrolment / Application	Students have to enroll via Brightspace. Professionals are welcome to the course. Conditions for their participation and registration are available with the Management Assistants of the Department of Space Engineering.	
Set-up	The course comprises lectures, guest lectures, tutorial, group work, peer-review, and individual homeworks.	

AE4S15	Space Embedded Systems	3
Responsible Instructor	Dr. A. Menicucci	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basics of programming and electronics	
Course Contents	The first part of the course will consist of lectures covering the following topics: 1. Introduction to embedded systems 2. Hardware and software elements 3. Space environment and effects on embedded systems 4. Fault tolerant techniques in space application 5. Radiation testing The second part of the course will be dedicated to a Group Assignment: students, in groups of 2-4, will build and program an embedded system for a possible space application which will feature fault tolerant techniques	
Study Goals	The students will learn to: Explain the basic elements of embedded systems and their characteristics. Explain the different radiation environments encountered in space and their effects on electronics components. Design and implement (hardware and software) an embedded system based on microcontrollers. Apply fault tolerance techniques both on software and hardware parts. Define the system requirements and select the components Evaluate the performances of the selected components to ensure they satisfy the system requirements.	
Education Method	Lectures, partly given by professionals from space area Group assignment, allowing actual hands-on application of the concepts learned. The group assignment will be carried out in period 4 but mostly off-campus. The assignment hand-in deadline will be agreed with the students based on their internships constraints.	
Assessment	Written exam + 1 group assignment	
Permitted Materials during Tests	Lectures notes and books. Internet connection with electronic devices is NOT allowed.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project.

The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5822	Thesis Space	45
Responsible Instructor	Prof.dr. E.K.A. Gill	
Responsible Instructor	Prof.dr. L.L.A. Vermeersen	
Responsible Instructor	Prof.dr.ir. P.N.A.M. Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	e MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal).	

If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go' (fail). Please refer to Brightspace for more information.

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Choose 1 course	
Program Coordinator	Ir. B.T.C. Zandbergen
Program Coordinator	Dr.ir. J. Bouwmeester

AE4897	Space Engineering Practical	8
Responsible Instructor	I. Akay	
Responsible Instructor	Ir. B.T.C. Zandbergen	
Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr. J. Guo	
Instructor	Dr.ir. R. Saathof	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	M.S. Uludag	
Instructor	Dr. B.V.S. Jyoti	
Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you will be taught on and experience the basic steps needed to successfully conduct experiments/simulations to verify/characterize a product/system as part of a real space engineering research project brought forward by industry and/or academia. Everyone who ever designed an experiment or conducted simulations to verify product/system requirements has experienced issues with improperly formulated requirement on one hand and the preparation and execution of simulations and experiments on the other. Typical problems that occur are inadequate planning (not planning in advance, lack of time), system/object not being in agreement with the design specifications, improper test set-up, instrument failures, missing data, convergence issues, bugs, extremely long calculation times, and improper requirements verification. Learning about the basic steps will allow the students to conduct such projects successfully in the future. The course is made up of two complementary parts, which are described in more detail below. Part 1 provides in-class instructions on how to perform specific tasks needed for conducting a proper test/verification project. The subjects dealt with in class (lectures/workshops) include (not necessarily in the order indicated): 1. Why this course, course organization, project phases, timeline, deliverables 2. Generating requirements including also test requirements, like test ranges and test tolerances, 3. Hypothesis formulation and assessment, 4. Design of Experiments, Experimental Measurements, and Error Propagation, 5. Test planning and the development of the test set-up and the accompanying Assembly, Test and Safety Procedures. 6. Data gathering/acquisition (both qualitative and quantitative) & automation 7. Results analysis 8. Other, like Safety, Types of Testing, Use of Standards Part 2 focuses on teams stepping through the project cycle of a experimentally oriented project assigned to them. The steps include: - Orientation (on project assignment, object/system to be researched/tested/verified under what conditions, goals) - Estimation of the performances of the test object by design analysis/simulations - Development of test hypothesis, and test requirements and criteria - Generation of test specification with success criteria - Generating a test plan - Assembly/build of test object (may be a mock-up in case of use of hazardous materials) - Assembly of (mock-up) test set up. early testing (including calibrations) and data elaboration to develop/check procedures - Test readiness review (if review is passed go on to next step, if not revise test plan) - Test execution - Data elaboration (observations and experimental data) - Discussion of results (data review) - Communicating the results/outcomes + project close-out For each project, project supervisors may arrange for more specific training/instructions related to the specific test set-up to be used. Project supervisors are also responsible for students conducting safe testing.	
Study Goals	Upon completion of the project the student shall be able to: 1. Describe and motivate the project cycle of an experimental project. 2. Formulate a test specification and test plan. 3. Set up and assemble/develop an appropriate test set-up and test procedures. 4. Perform data acquisition (both qualitative and quantitative data) with some degree of automation. 5. Execute measurements in a reproducible way and determine measurement uncertainties. 6. Analyze experimental data taking into account measurement uncertainties. 7. Discuss the outcomes of the project including the statistical validity of the data. 8. Report on the outcomes of an experimental project.	
Education Method	Lectures/workshops and Project The course is a practical lab project combined with lectures/workshops. The lectures/workshops are given by a variety of instructors. All instructors are expected to link the material they are teaching to the various projects. Instructors shall also strive to let students experience the work through a number of workshops/assignments that are geared to the projects the various teams are conducting. During the projects, instructors can be approached in case teams need help on elements instructed upon. Most of the instructions are scheduled in Q2, but to allow for proper alignment with the projects, some may move to Q3. The size of this part of the course is 2 EC (roughly 56 working hours for each participating student). Each project on offer is of size 6 EC and is conducted by a team of 5-6 students. All teams are assigned a staff member, with expertise in the field, as project supervisor. The role of the project supervisors is to provide intellectual guidance and offer advice on the planning and progress of the work. Project supervisors also are to check and review the progress made by the team and determine if a team can move on to the next phase in the project, thereby ensuring project progress and that test preparation activities and testing are conducted in a safe way. To streamline this, two formative assessments will be performed, which help students identify their strengths and weaknesses and target areas that need work. The first such assessment relates to the project	

	specification and the second one is prior to getting approval to conduct full scale testing. Additionally, during their project, students have an open invitation to consult the lecturers/instructors of the lecture/workshops part of the course for advise/feedback on issues taught in class or exercised upon in the workshops that are relevant to their project. Projects can be conducted in the Q2 to Q4 time frame of the academic year. The exact time frame is subject to negotiation between the team and the respective project supervisor. For instance, teams can propose running the project in a single quarter or spread over multiple quarters.
Literature and Study Materials	Appropriate literature and study materials will be provided throughout the course. For the lectures/workshops, material is provided for by the instructors relating to the topic/subject they instruct upon. For the specific projects, a project assignment will be made available that describes the test object/system to a great detail, including the relevant characteristics of the object/system. The project assignment will also describe the test goals as well as the type of testing to be performed (functional testing, life testing, reliability testing, failure testing, etc.) and the test conditions under which the testing is to be performed.
Assessment	The result of this course will be entirely based upon the results of the experimentally oriented project and the associated deliverables. These deliverables include: - Test specification - Test plan - Results Summary document, test data files, and a visual demonstration (video/photos/poster) of the testing conducted and the equipment used. For each deliverable a template/content description (or example) will be provided. Assessment will take place according to the detailed rubrics available on Brightspace. Assessment is by the project supervisor of the project (40% of the final grade) and the course instructors involved in the instructions (60% of the final grade). Assessment is coordinated by the course responsible instructor. A minimum grade of 6 is needed after rounding to the nearest half to pass the project. Late submission of the work results in a maximum penalty of 1.5 point (on a 10-scale). Due dates for submission will be communicated in due time via Brightspace. In case of an insufficient grade, teams are allowed one opportunity to resubmit an improved version of their report.
Enrolment / Application	This course is only open for enrollment for students enrolled in the Space Engineering profile of the track Space. A maximum of 45 students will be accommodated on a first-come, first-served basis. Students will have the opportunity to choose from a list of available projects that encompass different knowledge fields in the field of Space Engineering. Some past projects include: - Battery project: Novel capacitor-based spacecraft energy storage system - Propulsion project: Solid rocket motor assembly and testing (in cooperation with DARE) - Attitude control project: Assembly and testing of a PocketQube attitude determination and control system - Payload controller project: Assembly and testing of a small sat onboard data handling system

they can apply for a project. Students can then apply and only if they are accepted, their enrollment will be made definite. All non-accepted students will be dis-enrolled.

For the year 24/25, the maximum number of participants for this course has been set to 20. Priority will be given to students enrolled in the Space Engineering profile of the Space track. If places remain, also students from the other profiles in the Track Space can participate.

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Space

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4143	Hypersonic Aerodynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Instructor	Dr.ir. L.M.G.F.M. Walpot	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* AE2130-III Aerodynamics 2 (Compressible Aerodynamics) or equivalent. In case you did not have such a course, please study chapters 7 to 12 of Fundamentals of Aerodynamics by J.D. Anderson * It is very highly recommended to also follow AE4120 (Viscous flows) * AE4140 Gasdynamics is nice to follow as well but not absolutely necessary	
Parts	Lecture and study material 1. Introduction: characteristics of hypersonic flow; applications. 2. Inviscid flow: 2D shocks and expansions; Newtonian theory. 3. Tangent- wedge/tangent cone-, shock-expansion theories; similarity parameters. 4. Euler equations; small perturbation theory; hypersonic equivalence principle; blast-wave theory. 5. Thin shock layer theory; computational results. 6. Viscous flow: boundary layer equations; laminar boundary layers. 7. Transition; turbulent boundary layers. 8. Viscous interactions; flat plate application. 9. Heat transfer: convective heating; results for flat plate, cone, blunt body. 10. Laminar and turbulent stagnation region heating; radiation: effects on surface heating; radiation from the shock layer; nocturnal radiation. 11. High temperature effects; chemical reactions; inviscid equilibrium flow; difference between equilibrium flow and frozen flow. 12. Non-equilibrium flow; high temperature effects in flow through normal and oblique shocks. 13. High temperature effects in flow through nozzles. 14. Chemical reacting viscous flow	
Course Contents	Hypersonic Flow: Inviscid flow, compressions and expansions, surface inclination methods, similarity rules, hypersonic small perturbations, Euler equations. Viscous flow, boundary layers, viscous interactions, heat transfer, radiation effects. High temperature effects, inviscid and viscous equilibrium and nonequilibrium flow. Rarefied gasdynamics.	
Study Goals	This course is designed to provide the student with fundamental knowledge of hypersonic aerothermodynamics. The course is set up to deliver engineering tools for conceptual design problems together with a physical understanding of various aspects of the broad gamma of hypersonic flow.	
Education Method	Lecture	
Literature and Study Materials	J.D. Anderson, High temperature gas dynamics, , - ISBN 0070016712 Recommended literature J.J. Bertin, Hypersonic aerothermodynamics, , - ISBN 1563470365	
Assessment	Oral exam	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4204	Knowledge Based Engineering	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ing. A. Heidebrecht	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some familiarity with Object Oriented modelling and (Python) programming	
Course Contents	The course will provide a theoretical and practical introduction to Knowledge Based Engineering (KBE) technology, as a means to address complex engineering problems through design automation/digitalization principles. Other addressed topics, strictly related to KBE include - UML/SysML modelling language (class and activity diagrams) - object oriented modelling paradigm	
Study Goals	The main goals are the following: - understanding of Knowledge Based Engineering theoretical foundations and programming principles and their application to support complex engineering design problems - hands-on experience in developing simple but representative KBE applications by means of a provided KBE system - ability to model product and process knowledge by means of UML/SysML	
Education Method	The course is organized into a set of interactive lectures and tutorials to be performed part in class and part at independently at home. During the tutorials, students are introduced to the KBE programming approach. After a quick refresh of the python language, the main functionalities of a commercial KBE system will be illustrated, through interactive programming sessions. A significant amount of time is left for self-study, where students will have to develop a KBE application in teams of two, to address an engineering design problem of their choice. Attendance to lecture and tutorials is not compulsory but very highly recommended. Although the course is spread over two periods, the second one (Q4) is reserved for self-study (i.e. the development of the KBE application) and the oral examinations (See assessment section)	
Computer Use	The use of a personal computer is required, with the Windows OS installed A selected KBE system will be used for the practical sessions and assignment (temporary licenses for this course will be provided)	
Course Relations	This course has a logical connection to the MDO course AE4205, being KBE an enabling technology to multidisciplinary design, analysis and optimization. However, the MDO course is not propaedeutic to this course, thus students who did not followed MDO can take the KBE course without any problem	
Literature and Study Materials	- lecture sheets and some lecture notes (only available digitally on Brightspace) - a few scientific papers - several programming tutorials and code samples	
Books	Suggested (not compulsory) book: Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 Please note that only Chapter 9 is required for this course (single chapters can be purchased on line at Wiley). The rest of the book is suggested literature for the MDO course AE4205 The book is available as e-book through the TU Delft library	
Assessment	There is no conventional written exam for this course. The assessment is based on the quality of a KBE app to be developed at home by teams of two students and demonstrated live during an oral examination session. Two examination sessions are planned per academic year. The KBE app is supposed to address a relevant engineering design problem, which will be proposed by the student teams in the early phase of the course. The proposed KBE assignment descriptions will require lecturer's approval. If during the oral examination the quality of the developed KBE application and its demonstration are evaluated to be unsatisfactory, the team will have to use the retake session. For the retake session, two options are offered: 1) submit and present an improved version of the app proposed for the nominal session. One penalty point applies. 2) submit and present a totally different app than proposed for the nominal session. No penalty points. The new KBE assignment proposal will have to be delivered after the nominal session and will require lecture's approval. The oral examinations (nominal and retake sessions) are taking place around the beginning and the end of the second period of the course (i.e. 4th quarter of the academic year), not necessarily during the standard exam weeks. Delivery dates and examinations days are announced at the beginning of the course	
Special Information	Temporary student licenses of ParaPy, the KBE system used in this course, will be provided only to the students actively participating to the course. Thus, not to guests or other participants enrolled via Brightspace. The licenses shall be used exclusively for the course assignment.	
Tags	Artificial intelligence Computer Engineering Group work Integrated Modelling Programming Programming concepts Programming Software Software Engineering	

AE4205	MDO for Aerospace Applications	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ir. M.F.M. Hoogreef	
Instructor	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Familiarity with MATLAB and basic aircraft design knowledge. MATLAB is used for tutorials and home assignment, thus students without MATLAB experience must gain this through self-study Students with no experience in aircraft design should consult textbooks like Torenbeek, Raymer and Roskam to familiarize with the fundamentals of wing sizing process and general concepts such as wing loading, aerodynamic coefficients, fuel fraction method, drag and its components, etc.	
Course Contents	The course will introduce students to multidisciplinary design optimization (MDO), as a modern methodology to support complex product development. Main topics: - What and why MDO - Parameterization techniques - Numerical optimisation methods (focus on gradient based methods) - Decomposition and coordination of complex computational systems - the eXtended Design Structure Matrix (XDSM) - MDO architectures - Intro to surrogate modeling, discrete&mixed integer problems, genetic algorithms and Multi Objective Optimisation	
Study Goals	The main goals are the following: - understanding of Multidisciplinary Design Optimization (MDO) methods and architectures - hands-on experience in setting up and solving MDO problems	
Education Method	The course consists of classical lectures and computer tutorials using own laptop. During the tutorials, students are introduced to some of the tools discussed during lecture and can make hands-on experience during interactive sessions with the teachers. In particular, students will be introduced to the MATLAB optimization function fmincon and its use for gradient based optimization. A significant amount of time is left for self-study, where students will have to work in pairs on one take-home assignment.	
Computer Use	The use of a personal computer is required MATLAB (including its optimization toolbox) will be used for the MDO practical sessions and assignment. Student licenses are provided by the TUD	
Literature and Study Materials	- lecture sheets and some notes (only available digitally on Brightspace) - scientific papers - some software modules - Suggested (not compulsory) book: - Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszczanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 The book is available in e-format from the TUD library (Chapter 9 is used in the course AE4204 on Knowledge Based Engineering) - Engineering Design Optimization, by J.R.R.A. Martins and A. Ning (free on line)	
Assessment	There is no conventional exam for this course. The assessment is based on the quality of one take-home assignment, which will be executed in teams of two students. The assignment is about setting up and solving an MDO problem using Matlab. The deliverables are the Matlab code and a report. Two different take-home assignments are defined for the two yearly examination sessions.	
Tags	Algorithmics Analysis Matlab Modelling Optimalisation Programming Project	

AE4238	Aero Engine Technology	4
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	Dr. F. Yin	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery", "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Inlets and Exhaust - o Turbomachinery - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls - o New Engine /propulsion Concepts	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an flipped course in which the lectures are pre-recorded and the students are supposed to watch the lectures before coming to the classroom. The classroom interaction is mainly to have discussions and for tutorials. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13. Evaluate the calorific values for various fuels 14. Understand the basics of combustion and emission formation in Aero engines 15. Understand the working of gas turbine combustor 16. Understand the basic principles of gas turbine controls and evaluate different strategies for control 17. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a flipped teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

Reader	1. Gas Turbine Theory by H.I.H.Saravanamuttoo, G.F.C. Rogers, and H. Cohen 2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons
Prerequisites	A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Assessment	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Permitted Materials during Tests	The assessment is based on 2-3 assignments given during the course and a final exam. All assignments together will form 30% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4302	Avionics and Operations	3
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Ir. T.J. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4316 AE4316P AE4321-15 AE4322	
Expected prior knowledge	Flight Dynamics System and Control Theory	
Parts	Week arrangement Lecture and study material 1. Introduction to avionics systems. -. Air Data systems (home study). 2. Gyroscopes, attitude reference systems. 3. Compasses, heading reference systems. 5. Flight deck instruments and integrated systems. 6. Navigation equations. 7. The Flight Management System (FMS). 8. Inertial Navigation Systems. 9. Radio navigation systems (ADF, VOR, DME). 10. Landing guidance systems (ILS, GPS). 11. Communication, Navigation, Surveillance (CNS). 12. Satellite navigation systems (GPS). 13. Air Traffic Management (ATM). 14. The Future Air Navigation System (FANS). (see week arrangement)	
Course Contents	This course provides a comprehensive, unified coverage of the principles of modern navigation equipment and systems, both in the aircraft and on the ground, including the aircraft instrumentation and flight-deck systems, with a special emphasis on the important trends in the global air navigation and air traffic management system.	
Study Goals	1. The student can describe in detail the working principles of the avionics systems treated in the course. 2. The student can demonstrate the avionics systems' functionalities, identify their strong points and weaknesses, and make comparisons between the avionics systems. 3. The student can evaluate, criticize, and appraise their usage in the current and future operational context.	
Education Method	Lecture	
Literature and Study Materials	Lecture slides and other lecture material are put on the BrightSpace The books mentioned below (Pallett & Coombs, Kayton & Fried) can be hired at the Control & Simulation division. Recommended literature: E.H.J. Pallett, L.F.E.Coombs, Aircraft Instruments and Integrated Navigation Systems, Longman, 1997 ISBN 0-582-08627-2. M. Kayton, W.R.Fried, Avionics Navigation Systems, Wiley, 1997 ISBN 0471547956.	
Assessment	Digital, closed-book exam on campus	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical singularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments. The assignments will be handed out near the end of Q3 and the hand-in deadline will be near the end of Q4. Exact dates will be announced during the lectures.	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Physical intelligence for interaction - Insect intelligence and Swarming - Spiking neural networks	
Study Goals	After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.	
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4866-1	Propagation and Optimisation in Astrodynamics	5
Responsible Instructor	D. Dirks	
Instructor	Dr. S. Gehly	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4874 I (Astrodynamics I) is required; AE4868-1 (Numerical Astrodynamics) is required; Basic programming knowledge is expected.	
Course Contents	1. Introduction. Explanation of course assignment. Introduction to software. 2. Numerical propagation. Formulations of equations of motion. Selection of propagation method. 3. Overview of formulation of acceleration models. Relation between acceleration models and environment models. Influence of acceleration model selection on simulation fidelity/runtime. 4. Introduction to local optimization methods. Constraints in optimization problems. Robustness of optimal solutions. Optimal control. Examples of local optimization in astrodynamics. 5. Global optimization methods (genetic algorithms, particle swarm, etc.). Multi-objective optimization. Examples of global optimization in astrodynamics 6. Pre/post-processing of results. Analysis of numerical propagation/optimization results. Use of numerical tools (e.g. interpolation).	
Study Goals	1. Select and define the settings of a numerical astrodynamics simulation problem, based on top-level description and requirements. 2. Describe the relation between models of the physical environment and the propagated dynamical behaviour. 3. Define and set up local and global optimization problems for typical astrodynamics problems 4. Assess the quality of numerical orbit propagation for typical astrodynamics problems 5. Assess the quality of optimization results for typical astrodynamics problems	
Education Method	The course uses the following education methods: Weekly live lectures covering specifics of the assignments, examples, topic introduction, etc. The bulk of the material is covered as pre-recorded lectures on propagators, environment, acceleration models, design space exploration, optimization Working Lectures are held twice per week, in weeks 1-10 of the quarter. The course requires significant self-study by the students, both to master the material, and complete the assignments. However, ample support is provided through Brightspace, forum, and the working lecture. Successfully doing the course within a reasonable time requires students to make good use of these options.	
Computer Use	Laptop computer with minimum 4 GB RAM	
Course Relations	AE4870A (Rocket Motion); AE4870B (Re-Entry Systems); AE4872 (Satellite Orbit Determination); AE4890-11 (Planetary Sciences I); AE4894 (Practical Astrodynamics)	
Literature and Study Materials	Selected articles; lecture notes; lecture slides (details to be announced and distributed during the course).	
Assessment	The assessment is a combination of two individual and one group take-home assignments. The individual assignments are thematically related to the group assignments, and are used as a starting point for the group work . Students will use Python for generating results during the course, and the language/tool of their choice for visualizing results. It is assumed that students are sufficiently proficient in Python (or are willing to independently make themselves so) to finish and run partially finished Python scripts. It is assumed that students are proficient in a data visualization tool (e.g. Matlab, Python's matplotlib, etc.) For the group work, individual contributions to the group work must be clearly identifiable. The total grade is composed of: Individual assignments (individual grade): 50% Project (individual grade): 25% Project (group grade): 25% Each of these three grades must be ≥ 5.0 to pass the course. The final grade for the course must be ≥ 6 to pass the course. In the assignments, students will apply the material covered in the course to the analysis of a real astrodynamics problem. Weekly practical sessions will be held to discuss questions and issues with homework, the practical assignment and the numerical implementation. The resit of the course will be done during the summer period and/or Q1 of the following academic year. During the resit, there will be no opportunities to ask questions of any kind (installation, coding, assignment, theory, interpretation, etc.). It is not possible to resit single assignments, only a resit of the full course is offered.	
Remarks	**NOTE:** This course will have limited capacity (maximum ~25-30 students). The procedure on student admittance (if the maximum is reached) will be communicated to the students at the beginning of the academic year through Brightspace. In case of oversubscription, preference will be given to students from the Spaceflight track. To take part in the course, students must submit their preferences for an assignment topic. These must be submitted through Brightspace on the first Wednesday of the course's first lecture week, at the latest. Only enrolling for the course on Brightspace is NOT sufficient to indicate you wish to take part in the course. Instructions for submitting a topic preference will be posted on Brightspace one week before the start of the course, at the latest. Not submitting your topic preference in a timely manner means that you will NOT be able to take part in the course. For this course, it is expected that students are on campus for the full duration of this course (weeks 1-8 of the quarter in which it is taught). Being away for one week or more may result in exclusion from the course. In case of doubts or concerns, please contact the course manager. Working lectures are held during week 1-8 of the period, where students are expected to be present. It is required that students have followed the course AE4868 (Numerical Astrodynamics), and have submitted all assignments, and can still reasonably pass that course, before taking part in this course.	

AE4889	Special Topics in Astrodynamics	2
Responsible Instructor	Dr.ir. M.J. Heiligers	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	A modular and research-focussed course on advanced topics in astrodynamics. You will extend and deepen your insights and analytical abilities based on Fundamentals of Astrodynamics and related coursework, devoting particular attention to their application to a selection of advanced astrodynamics topics (see below).	
Course Contents	The course treats a selected set of advanced topics in astrodynamics; the exact topics change from year to year depending on the latest research developments and the instructors involved. Strictly as examples, the following topics were covered in previous years: - Circular restricted three-body problem (equilibrium points, their stability and manifolds, and periodic orbits around the equilibrium points, their stability and manifolds) - Solar-sail orbital dynamics (acceleration models and artificial equilibrium points in the circular restricted three-body problem) - Machine learning applied to astrodynamics	
Study Goals	Within the context of the selected astrodynamics topics: - Extend your insight and analytical abilities beyond the Fundamentals of Astrodynamics and related coursework; - Actively apply your knowledge and skills accumulated through Fundamentals of Astrodynamics and related coursework; - Tackle "open-ended assignments with a hands-on, scientific approach as a first foray into astrodynamics research.	
Education Method	Workshops, lectures, self-study	
Computer Use	Yes	
Prerequisites	Fundamentals of Astrodynamics (AE4874 I)	
Assessment	Assignments	
	Additional detail will be available on the course website (Brightspace)	

AE4895-24	Measurement Strategies for Planetary Science Missions	4
Responsible Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	For planetary science the research question has a profound impact on the measurement strategy and instrument to be used. Studying planetary objects and materials in the Solar System thus hinges on observing and obtaining measurements from either nearby (in contact) and from afar (remote sensing). In this course you will follow lectures, learn more about planetary analogues and gain hands-on experience as you explore different measurement techniques in the lab. By applying measurement techniques to a specific rocky or ice sample, and to a planetary analogue material, you will try to find out how the data products and the scale at which it is applied influences the research questions that a science mission can address.	
Study Goals	After completion of the course, the student will have obtained practical experience and gained a deeper understanding of how measurement techniques contribute to characterizing a physical-chemical property of a planetary object. After this course the student will be able to: LO1: Explain key scientific questions that space exploration seeks to answer for rocky (Mars) and icy (icy moons) planetary bodies. LO2: Understand the underlying physical principles of a measurement technique. LO3: Demonstrate to have working knowledge of measurement techniques by participating in a structured laboratory exercise. LO4: Evaluate how measurement techniques result in an improved understanding of materials and processes in planetary sciences.	
Education Method	Lectures; structured laboratory exercises focussing on either a rocky or an icy planetary analogue material.	
Assessment	Laboratory activities and maintaining a lab journal (20%) Presentation lab results and planetary integration (20%) Final report, in form of a short communication in paper format (60%)	
Enrolment / Application	The course has a limited capacity of 20 students, which will be allocated to students from the profile planetary exploration first, and then on the basis of first come, first serve (potentially filled up with students from other profiles and/or programs). This limit of 20 could be higher depending on the actual capacity of the laboratory facilities	

AE4897	Space Engineering Practical	8
Responsible Instructor	I. Akay	
Responsible Instructor	Ir. B.T.C. Zandbergen	
Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr. J. Guo	
Instructor	Dr.ir. R. Saathof	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	M.S. Uludag	
Instructor	Dr. B.V.S. Jyoti	
Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you will be taught on and experience the basic steps needed to successfully conduct experiments/simulations to verify/characterize a product/system as part of a real space engineering research project brought forward by industry and/or academia. Everyone who ever designed an experiment or conducted simulations to verify product/system requirements has experienced issues with improperly formulated requirement on one hand and the preparation and execution of simulations and experiments on the other. Typical problems that occur are inadequate planning (not planning in advance, lack of time), system/object not being in agreement with the design specifications, improper test set-up, instrument failures, missing data, convergence issues, bugs, extremely long calculation times, and improper requirements verification. Learning about the basic steps will allow the students to conduct such projects successfully in the future. The course is made up of two complementary parts, which are described in more detail below. Part 1 provides in-class instructions on how to perform specific tasks needed for conducting a proper test/verification project. The subjects dealt with in class (lectures/workshops) include (not necessarily in the order indicated): 1. Why this course, course organization, project phases, timeline, deliverables 2. Generating requirements including also test requirements, like test ranges and test tolerances, 3. Hypothesis formulation and assessment, 4. Design of Experiments, Experimental Measurements, and Error Propagation, 5. Test planning and the development of the test set-up and the accompanying Assembly, Test and Safety Procedures. 6. Data gathering/acquisition (both qualitative and quantitative) & automation 7. Results analysis 8. Other, like Safety, Types of Testing, Use of Standards Part 2 focuses on teams stepping through the project cycle of a experimentally oriented project assigned to them. The steps include: - Orientation (on project assignment, object/system to be researched/tested/verified under what conditions, goals) - Estimation of the performances of the test object by design analysis/simulations - Development of test hypothesis, and test requirements and criteria - Generation of test specification with success criteria - Generating a test plan - Assembly/build of test object (may be a mock-up in case of use of hazardous materials) - Assembly of (mock-up) test set up. early testing (including calibrations) and data elaboration to develop/check procedures - Test readiness review (if review is passed go on to next step, if not revise test plan) - Test execution - Data elaboration (observations and experimental data) - Discussion of results (data review) - Communicating the results/outcomes + project close-out For each project, project supervisors may arrange for more specific training/instructions related to the specific test set-up to be used. Project supervisors are also responsible for students conducting safe testing.	
Study Goals	Upon completion of the project the student shall be able to: 1. Describe and motivate the project cycle of an experimental project. 2. Formulate a test specification and test plan. 3. Set up and assemble/develop an appropriate test set-up and test procedures. 4. Perform data acquisition (both qualitative and quantitative data) with some degree of automation. 5. Execute measurements in a reproducible way and determine measurement uncertainties. 6. Analyze experimental data taking into account measurement uncertainties. 7. Discuss the outcomes of the project including the statistical validity of the data. 8. Report on the outcomes of an experimental project.	
Education Method	Lectures/workshops and Project The course is a practical lab project combined with lectures/workshops. The lectures/workshops are given by a variety of instructors. All instructors are expected to link the material they are teaching to the various projects. Instructors shall also strive to let students experience the work through a number of workshops/assignments that are geared to the projects the various teams are conducting. During the projects, instructors can be approached in case teams need help on elements instructed upon. Most of the instructions are scheduled in Q2, but to allow for proper alignment with the projects, some may move to Q3. The size of this part of the course is 2 EC (roughly 56 working hours for each participating student). Each project on offer is of size 6 EC and is conducted by a team of 5-6 students. All teams are assigned a staff member, with expertise in the field, as project supervisor. The role of the project supervisors is to provide intellectual guidance and offer advice on the planning and progress of the work. Project supervisors also are to check and review the progress made by the team and determine if a team can move on to the next phase in the project, thereby ensuring project progress and that test preparation activities and testing are conducted in a safe way. To streamline this, two formative assessments will be performed, which help students identify their strengths and weaknesses and target areas that need work. The first such assessment relates to the project	

specification and the second one is prior to getting approval to conduct full scale testing. Additionally, during their project, students have an open invitation to consult the lecturers/instructors of the lecture/workshops part of the course for advise/feedback on issues taught in class or exercised upon in the workshops that are relevant to their project. Projects can be conducted in the Q2 to Q4 time frame of the academic year. The exact time frame is subject to negotiation between the team and the respective project supervisor. For instance, teams can propose running the project in a single quarter or spread over multiple quarters.

Literature and Study Materials

Appropriate literature and study materials will be provided throughout the course.

For the lectures/workshops, material is provided for by the instructors relating to the topic/subject they instruct upon.

For the specific projects, a project assignment will be made available that describes the test object/system to a great detail, including the relevant characteristics of the object/system. The project assignment will also describe the test goals as well as the type of testing to be performed (functional testing, life testing, reliability testing, failure testing, etc.) and the test conditions under which the testing is to be performed.

Assessment

The result of this course will be entirely based upon the results of the experimentally oriented project and the associated deliverables. These deliverables include:

- Test specification
- Test plan
- Results Summary document, test data files, and a visual demonstration (video/photos/poster) of the testing conducted and the equipment used.

For each deliverable a template/content description (or example) will be provided.

Assessment will take place according to the detailed rubrics available on Brightspace. Assessment is by the project supervisor of the project (40% of the final grade) and the course instructors involved in the instructions (60% of the final grade). Assessment is coordinated by the course responsible instructor. A minimum grade of 6 is needed after rounding to the nearest half to pass the project.

Late submission of the work results in a maximum penalty of 1.5 point (on a 10-scale). Due dates for submission will be communicated in due time via Brightspace.

In case of an insufficient grade, teams are allowed one opportunity to resubmit an improved version of their report.

Enrolment / Application

This course is only open for enrollment for students enrolled in the Space Engineering profile of the track Space.

A maximum of 45 students will be accommodated on a first-come, first-served basis. Students will have the opportunity to choose from a list of available projects that encompass different knowledge fields in the field of Space Engineering. Some past projects include:

- Battery project: Novel capacitor-based spacecraft energy storage system
- Propulsion project: Solid rocket motor assembly and testing (in cooperation with DARE)
- Attitude control project: Assembly and testing of a PocketQube attitude determination and control system
- Payload controller project: Assembly and testing of a small sat onboard data handling system

AE4898	Space debris tracking and mitigation	4
Responsible Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	AE4874 I (Fundamentals of Astrodynamics) is required. AE4868-1 (Numerical Astrodynamics), AE4894 (Practical Astrodynamics) and AE4870B (Re-Entry Systems) are recommended.	
Course Contents	The course will introduce the problem of space debris and associated aspects, such as how to detect it, how to avoid it, and potentially how to remove it. The contents of the course will be presented in a hybrid format, i.e., a combination of classical lectures and working lectures, where the students can practise with the introduced material, tools and techniques. The course content is roughly split into two parts. The first half of the course will introduce the problem of space debris: how much is out there, how large and heavy is it, where does it come from, and what kind of problems does it pose. Having identified the problems for the safe operation of space, the focus shifts to how we can predict potential collisions and how we can avoid those. Finally, how can the larger chunks of space debris be removed (Active Debris Removal), and if the debris enters the atmosphere and does not fully burn up: where will it impact on earth. The second part focuses on the observation and estimation of space objects in general, and space debris in particular. The students will learn about different sensors that can be used, either ground-based or space-based, the accuracy of measurements, and how this will affect the estimation process. Linear and nonlinear estimation techniques will be discussed, with application to determine translational state vectors, characterise physical parameters, and detect manoeuvres of space objects. The course will further consider methods of sensor management and data fusion.	
Study Goals	After the course the student shall: 1. Assess the risks posed by orbital debris on current and future space missions, including the probability and consequences of on-orbit collisions 2. Analyze the mitigation of space debris hazards through the use of post-mission disposal and active debris removal 3. Estimate the orbits and physical characteristics of resident space objects using simulated space object tracking data, including measurements from optical and/or radar sensors 4. Design a collision detection and avoidance strategy for use with a simulated space object catalog	
Education Method	Per week, two lectures are scheduled. One is a regular ("classical") lecture where topics related to the field are introduced, clarified with examples. The second lecture is a working lecture, where tools are introduced by means of tutorials and where students can begin on their assignments/lab work.	
Computer Use	Personal laptop that can run Tudat. The choice of programming language is free (Python knowledge is required to work with Tudat). No coding support is provided.	
Literature and Study Materials	The main course material consists of the lecture slides. Recommended papers and books will be suggested during the course.	
Assessment	The assessment is a combination of one individual and two group take-home assignments. A tutorial assignment (not graded) will be handed out at the beginning of the course. For the group work, individual contributions to the group work must be clearly identifiable. The total grade is composed of: Individual assignments (individual grade): 30% Project (individual grade): 30% Project (group grade): 40% Each of these three grades must be ≥ 5.0 to pass the course (final grade is the weighted average, which should be 5.75 or higher). The resit of the course will be done based on individual assignments. Support during this will be limited. It is not possible to resit single assignments, only a resit of the full course is offered.	
Enrolment / Application	The course is open to students of the Spaceflight Dynamics profile only; the maximum number of students that can participate is limited to a maximum of 30. To enroll, the student should write a motivation (limited to 5 lines) why this course is important/relevant to them, and send this by email to both instructors: e.mooij@tudelft.nl (Dr Erwin Mooij) and s.gehly@tudelft.nl (Dr. Steve Gehly). Enrollment closes 2 weeks before the start of the course, and students will be notified about the status of their enrollment before the course starts.	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods.	
Study Goals	Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Education Method	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Assessment	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments (only assignment 3 and 4 since they have the highest weight; assignment 1 and 2 cannot be redone). Results for individual assignments are only valid for one academic year: when re-taking the course all assignments have to be done in that academic year.	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	P.P. Sundaramoorthy	
Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of lectures, working sessions, design review sessions and a final symposium. The course structure will be explained in detail in the first weekly module. The time and date of the design review(s) and final symposium will also be communicated in the first weekly module. Students are requested to reserve all scheduled timeslots in their agendas. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. Note that, the course will require group work. Groups will be formed in the fourth week of the course. Only students who pass the individual test are allowed to participate in a group. Students who are not part of a group cannot participate in the remainder of the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. During each module it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Individual test on course content 2. Written feedback on design specification sheet of another group in the course Students with a failing grade for the individual test, i.e. a grade below 5.8, will have a resit opportunity of the individual test in week 4 of the course. This resit will be in the form of a short oral exam. This resit is only available for students who did not pass the individual test. On passing the oral resit exam, students will receive a 5.8 as grade for the individual test. Students who fail the resit opportunity will not be able to participate in the group work, and therefore cannot participate in the remainder of the course. It is mandatory that students are present at the design review(s) and the final symposium presentations of their own group and those of other groups. The individual grade of students who fail to be present will be withheld. These students will not receive a final grade for the course as a result. Students of whom the individual grade which has been withheld, may take a resit. The resit will be in the form of a short oral exam. It will be possible to apply for a resit up to 15 working days after the publication of the grades in Osiris, but no later. A resit exam will then be scheduled by the principle lecturer of the course in consultation with the student in question. Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, Introduction to materials science" or books of similar nature. Be sure to check the entry level quiz before making up your mind about this course.	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, ISS, Hubble telescope, James Webb telescope, Venus orbiters, Launchers) will be presented as a carrier for material selection analysis. A holistic view of "materials science - manufacturing - structural analysis - system engineering for missions" will be presented. While students come mostly from different backgrounds, the goal of this class is to help them identify the perspective they use in the holistic view and to learn about the materials science perspective. Fundamentals of materials classes and their typical properties will be introduced. Current material choice will be analyzed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. In sessions of "ESA forum + student poster seminar", students will be given opportunities to interact with ESA experts and present their own findings from analyzing given missions, by translating desired properties into material structures and microstructures. In class discussion is an important part of learning. The structure of the lectures will be tailored to maximize the student involvement. The course contains the following topics: week 1: Holistic view and materials science perspective; Typical materials - metals, ceramics and polymers week 2: Typical materials - metals, ceramics and polymers week 3: Material degradation in space - vacuum; thermal cycling; week 4: Material degradation in space - AO; charging week 5: Material degradation in space - space radiation week 6: Material degradation in space - meteoroids and orbital debris; week 7: "ESA forum + student poster seminar" Materials testing for space applications; Extreme missions; Materials challenge in future space age (ESA guest speakers); Student poster seminar (Jan 6, 2025) week 8: Study period (Q and A) week 9: exam week (Jan 20. 2025)	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 12 lectures of 2h per week for the first 6 weeks. The last week is a whole-morning session of "ESA guest speaker forum + student poster presentation".	
Literature and Study Materials	A compilation of book chapters, lecture notes and review papers.	
Assessment	Poster presentation (40%) + final exam (60%). The students are to work in groups and do research on prescribed question/topics and deliver a final poster presentation (15% individual score, 25% group score) in week 7 to show their discoveries throughout the course. The final grade will be the weighted average of poster presentation and final exam. Example questions/topics will be made available on Brightspace.	
Enrolment / Application	For students not from "structures for space" profile, there will be a cap of 20 students maximum. Please enroll in brightspace (enrollment period Oct 28 to Nov 10, 2024) as soon as you decide to take the course. Once the maximal number is reached, enrollment will be closed.	

AE4ASM526	Spacecraft Thermal Design	3
Responsible Instructor	I. Uriol Balbin	
Instructor	C. Falsetti	
Instructor	N. van der Pas	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of orbital mechanics Understanding of general design spacecraft systems	
Course Contents	General knowledge of space thermal control hardware, system design and analysis processes are presented. Using the principles of heat transfer, and more specific of radiation and conduction, the methods and tools for thermal design of a spacecraft are presented. Numerical analysis will also be introduced via the software ESATAN-TMS.	
Study Goals	The student is able to generate a thermal design of a satellite using heat transfer analysis and thermal engineering techniques. In more detail the student will be able to: - Understand the spacecraft thermal environment - Define the thermal subsystem and compare the different hardware options. - Understand the design cycle of the thermal subsystem. - Apply the concepts of heat transfer (radiation, conduction), heat balance and heat flows - Generate and use analytical thermal analysis models - Generate and use numerical thermal analysis models	
Education Method	Lectures + homework Students are given access to a discussion board provided with the course in Brightspace. The discussion board will be used as the main communication tool to ask questions on the material taught or in preparation of an assignment. Answers to the questions may be given by fellow students and/or the lecturer(s).	
Computer Use	Software licenses are available during the lecture period Software can be used in the computer room or in the personal computer (Windows/Linux)	
Literature and Study Materials	Lecture notes available on Brightspace.	
Assessment	The final grade of this course is based on 2 homework assignments (60%) and one final exam (40%). A minimum grade of 5 is required in each part.	
Enrolment / Application	This class has a limit of 80 students, due to the limit on the software licenses. Please enroll in Brightspace before 4th of March 2025 if you intend to take this class.	

AE4S52-21	Collaborative Space System Design Project	5
Responsible Instructor	Ir. B.T.C. Zandbergen	
Instructor	Dr. S.M. Cazaux	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The Collaborative Space (System) Design Project (CSDP) is a TU Delft/Faculty of Aerospace Engineering (AE) Master course in engineering design. The course focuses on the conceptualization and preliminary design phase of a space mission, spacecraft, or space instrument and starts on a design challenge co-created by the CSDP organization team and a company, research group or other entity. All projects on offer are multi-disciplinary projects, i.e. projects that require different types of design work and knowledge to be combined to provide a design solution. All projects are carried out by a team of students that have to organize and manage themselves. Each project is to be organized in 3 phases similar to NASA's System Design Process as defined in a.o. "NASA's System Engineering Handbook". Some adaptations have been made though to make it fit in the limited course time. The phases are: 1) Exploration phase; In this phase the team is to explore the problem, the needs, the competition and past missions and to develop a proposal for the next phase of the project. This includes the identification of a range of high-potential concepts for study in phase 2, the work distribution, etc. 2) Concepts design studies phase; In phase 2, the high-potential concepts defined in phase 1 are analyzed in detail for feasibility, traded and a best concept is selected. Additionally the plan for the next phase is to be generated. 3) Detailed design phase. In this final phase of the project the single concept selected in phase 2 is worked out in detail and a plan is developed for the further development of the design. Each project phase ends with a review (feedback moment) wherein other teams and expert staff reflect on the outcomes generated and the engineering design methods used. During the quarter, workshops/instructions will be held to provide knowledge and training on selected management and engineering design topics, including agile management, and the use of integrated design modelling, i.e. the integration of all the geometry, configuration, analysis, and requirements verification into a generative, parametric, unified computational model where data is shared seamlessly between the different disciplines. The course does not focus on teaching the required disciplinary knowledge and experience, but rather focuses on decision making, the collaborative integration of the knowledge and experience available in the team, and the iterative design method using different levels of model fidelity to create a feasible design solution in answer to the problem identified by the "customer". As projects vary from year to year and may encompass knowledge that is not available in the team, this may require that participants actively acquire the knowledge required.	
Study Goals	The course aims to develop student skills in multi-disciplinary team projects from a challenge-driven perspective. In more detail, students will advance their ability in ...: - ... disciplinary design including modelling, simulation, visualization, quantitative analysis of alternatives, design tool verification, calibration and validation, and design refinement. - ... the process of engineering design (ABET definition), including the steps in design, the (iterative) nature of the process, development of a Straw Man design, and development of process models. - ... multidisciplinary design, thereby taking into account differences between the different disciplines involved in terms of a.o. differences in fidelity level of disciplinary models, and dissimilar assumptions; - ... systems engineering, including the design phasing, work breakdown and work distribution, modeling and interfaces, and the role of specialty engineering (e.g. cost-, RAMS-, and mass modelling and configuration design). - ... concurrent engineering, as opposed to the more classical sequential engineering (the waterfall method). - ... project management and teamwork (assigning roles and responsibilities, setting goals and objectives, coordination and management of team process, decision making, handling conflicts, creativity, empowerment and motivation, communication, and reflection on own work and work of others).	
Education Method	The course consists of: - Group project (4 EC or 112 working hours) - Workshops/instructions (1 EC or 28 working hours) For the group work, students work in teams of 6-12 students. Prior to the start of the course, students will be informed on the available projects as to allow them to state their preference(s), and project teams are formed. During the project, each team is guided by a design coach, who mainly focuses on the problem solving/engineering design process. Additionally, each team is free to organize experts (from industry, academia or other) for the different disciplines involved that can advise the team during the design. During the quarter workshops/instructions are given to train teams' project management and design integration skills.	
Assessment	The assessment of each team consists of an assessment by the workshop instructors and project coaches (80% of the final grade), team of experts not involved in instruction and workshops (10%) and by the other student teams (10%). The assessment of the workshop instructors and project coaches is per design phase: - Exploration phase (10%) - Concepts design studies phase (20%) - Final design phase (70%) The assessment of the instructors and coaches is based on a design report, a (design) management log (record keeping) and one or more design presentations and reviews. Elements that will be assessed relate to design competencies (problem formulation, requirements generation, design option identification, design analysis, design trades and design evaluation), workplace competencies (e.g. teamwork, creative thinking, decision making, planning and organizing), academic and personal effectiveness competencies (e.g. communications, modelling/experimenting and critical and analytical thinking). Assessment by the team of experts and by peer teams will be based on the performance of the teams at a project presentation session at the end of quarter 3.	
Enrolment / Application	Only students enrolled in the Space track can participate. There is no free enrollment for this course. Still at the start of Q1, the students can preliminary enroll for the course as to learn more about the CSDP, and past projects conducted (no solutions on offer). In Q1, an information session will be held informing interested students of the track Space on the projects available and on how	

they can apply for a project. Students can then apply and only if they are accepted, their enrollment will be made definite. All non-accepted students will be dis-enrolled.

For the year 24/25, the maximum number of participants for this course has been set to 20. Priority will be given to students enrolled in the Space Engineering profile of the Space track. If places remain, also students from the other profiles in the Track Space can participate.

AESM2001	Physical Principles of Earth System Observation	5.5
Responsible Instructor	Prof.dr.ir. F. Lopez Dekker	
Module Manager	Prof.dr.ir. F. Lopez Dekker	
Instructor	Dr. P.G. Ditmar	
Instructor	Prof.dr.ir. S.C. Steele-Dunne	
Instructor	Prof.dr.ir. F. Lopez Dekker	
Instructor	K. L��er	
Contact Hours / Week x/x/x/x	4 hours lectures or practicals	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Measuring is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Electromagnetic, seismic and gravity potential-field observations inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. The measurements can be acquired from spaceborne, airborne, surface and sub-surface-based sensors. This programme core module aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain geophysical variables. For example, students will learn how electromagnetic theory allows to use the intensity of radar echoes to yield information about rain rate, soil moisture, ocean roughness, or the layering of the subsurface. Similarly, they will learn how potential field theory can be applied to quantify mass changes of, e.g., the ice sheets. Students will be able to weigh the advantages and disadvantages of, for example, microwave versus visible and near-infrared observations for monitoring the Earth surface, and seismic and electromagnetic imaging in mapping the subsurface. Review basics of electromagnetic (EM) waves (propagation, polarization, spectra, Doppler, etc.) and propagation effects (attenuation, refraction, dispersion, polarimetric effects) Optical-infrared scattering and propagation (different reflection types, albedo, effect of wavelength, BRDF, Radiative transfer modelling of optical EM waves). Physical principles and limitations of VNIR/SWIR sensing systems (photographic systems, LiDAR, spectral systems including TIR) in terms of spatial and spectral resolution, sensitivity, atmospheric disturbances and corrections. Microwave scattering and propagation (Rayleigh, Mie scattering, rough surface, volume scattering, clear air, radiative transfer modelling of microwave scattering in the atmosphere and at the land surface). Radiometry (Radiometric Quantities and Units, Blackbody Radiation, emissivity at different frequencies). Radiative transfer modelling in passive microwave remote sensing of the Earths atmosphere and surface. Physical principles and limitations of microwave sensing systems (radiometer, non-imaging radar, imaging radar, ground-penetrating radar) in terms of spatial and spectral resolution, sensitivity, atmospheric propagation effects and corrections, radar range equation Seismic (including acoustic) waves (Newtons second law, stress tensor, equation of motion, elasto-dynamic wave equation, Plane P- and S-waves, spatial and temporal resolution) Potential fields (gravity, magnetic and electric fields) and their relationship to physical properties of a medium (Gauss law and its applications, Newtonian/Magnetic Potential, Laplace equation and harmonic functions, representation of potential fields, spherical harmonics) Choose an approach for the observation and measurement of a geophysical variable.	
Study Goals	After completing this module, students will be able to: Characterize electromagnetic waves and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and & subsurface). [ILO A] Describe the interaction of optical-infrared waves with the Earths atmosphere and surface and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of optical/infrared remote sensing in determining geophysical variables. [ILO A, D] Describe microwave scattering in the Earths atmosphere and at the Earths surface, and analyse the fundamental sensitivity to geophysical variables of interest. [ILO A] Describe the emission of thermal waves and microwaves from the Earths atmosphere and surface, and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of microwave remote sensing in determining geophysical variables. [ILO A, D] Explain mathematically how the elastodynamic wave equation is derived, characterize elastic waves, and explain their sensitivity to contrast parameters. [ILO A] Establish the linearized relationship between an observable and a given representation of the potential field, and synthesize the potential from functionals given on the surface of a sphere. [ILO A] Critically discuss the potential value and limitations of different types of measurements (I.e., different bands of the EM spectrum and EMR/seismic/potential field) and propose and defend a type of measurement to observe certain geophysical variables. [ILO F, G]	
Education Method	Lectures and group practicals/exercises. Feedback via quizzes (formative) and intermediate assignment results (formative).	
Assessment	Module is summatively assessed with a project, using 1 rubric (similar as thesis project) at the end of the period based on: Assignment report (may comprise different parts, i.e., based on a several assignments, report includes reflection on LOs) Presentation Oral discussion (may include questions on LOs not directly covered in the assignment work).	

Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Some background in electromagnetic theory and fundamentals of wave propagation. Solving linear partial and ordinary differential equations.
Academic Skills	Earth Observation, data interpretation.
Literature & Study Materials	Provided course material, published online lectures and material, book/e-books
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N.A.
Collegerama	No

AP3141	Environmental Physics	6
Responsible Instructor	Dr. S.R. de Roode	
Instructor	Prof.dr.ir. J.L. Kloosterman	
Instructor	Dr.ir. D.J. Verschuur	
Instructor	Ir. R.A. Verzijlbergh	
Instructor	Prof.dr.ir. B.J.H. van de Wiel	
Instructor	Dr.ir. S.J.A. van der Linden	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Physics at BSc level, including basic Transport Phenomena and Fluid Dynamics, thermodynamics, calculus and differential equations; use of Python and/or Matlab.	
Course Contents	*) Fundamentals of the environmental physics, world climate and the greenhouse-model, durable energy-resources (solar-, wind-, wave-, nuclear-, biomass-energy), transport of pollution in ecosystems, Air Quality; *) Fundamentals of harnessing sustainable nuclear energy and physical models for nuclear reactors to do so; *) Land-atmosphere interactions and turbulent transport processes. Fundamentals of dimensional analysis & generalization in environmental physics; *) Radiative transfer & Solar cell physics, PV system design & Renewable energy integration; *) Environmental noise, basic theory, noise calculation schemes	
Study Goals	To provide physicists with the knowledge and tools so that they can contribute to a more sustainable society from their own specific field of expertise: 1. Acquiring advanced knowledge of the basic physics underlying environmental issues such as: the earth energy budget; air-, water- and ground pollution; solar, wind, tidal, wave, biomass and nuclear energy technologies and environmental noise. 2. Being able to translate problems in the above areas into appropriate physical models and their relevant equations. 3. Being able to translate physical models and model equations into (e.g. Matlab/Python) computer models. 4. Being able to find (quantitative) solutions to model equations, using either theoretical analytical methods, or numerical methods. 5. Being able to quantitatively and physically evaluate, analyse and critically acclaim proposed solutions and strategies for environmental issues. 6. Being able to contribute to solutions and strategies for environmental issues.	
Education Method	Lectures and mandatory homework assignments.	
Computer Use	Homework requires computer programming in Matlab or Python	
Literature and Study Materials	Egbert Boeker and Rienk van Grondelle, Environmental Physics, Third edition, Wiley, 2011, ISBN 978-0-470-66676-0 + lecture notes and handouts	
Assessment	There is a series of mandatory (homework) Assignments. Please note that the Assignments have to be done in the prescribed periods, and that grades cannot be improved later. Lecturers can invite students for an oral assessment of their Assignments. The grade for the Assignments is the mean over all of them. At the end of the course there will be a written Exam. This Exam will be of an open-book type (see below). The final grade of the course is the average of the Exam grade and the Assignment grade. A passing grade can only be obtained if a score of at least 5.0 is obtained for both the Exam and for the Assignments. The Assignments and Exam must be done in the same academic year. The validity of result of the Assignments is restricted to one academic year. Resit exam can be a written exam (if enough candidates) or via an oral assessment (by appointment). In case an on-campus exam cannot be provided, there will an online open-book exam followed by an oral fraud-assessment of a random subset of students. See announcements on Brightspace.	
Permitted Materials during Tests	The exam will be done in a computer room. The computer will contain the PDFs of the book and PDFs of all handouts of the lectures, without access to internet. Besides this, you can take one A4 double-sides printed or hand-written with anything you prefer. And a simple calculator (not programmable) for possible calculations is allowed.	
Studyload/Week	2 lecture hours per week (14 weeks), 5 series of homework of 20 hours each, 40 hours of exam preparation.	

AP3181	Applied Multiphase Flows	6
Responsible Instructor	Dr. L. Portela	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course assumes some elementary knowledge of Fluid Mechanics and/or Transport Phenomena. The level of prior knowledge required is typically acquired in a BSc course in Engineering or Applied Physics. Prior knowledge of multiphase flow is not required.	
Course Contents	1. Introduction. Overview of multiphase flow. Examples of industrial and environmental flows. Examples of flows with heat transfer and phase change. Classification of multiphase flows: separated vs. dispersed. Brief introduction to flow patterns. 2. Balance equations. Physical mechanisms involved and material behavior. Constitutive equations and mechanistic models of behavior. Interfacial interactions and forces. Analogies and differences with respect to single-phase flow. Flow parameters, non-dimensional numbers and scaling. 3. Introduction to turbulence, turbulence modeling and turbulence effects. Analogies and differences with respect to single-phase flow. 4. Separated flows and interfacial phenomena. Stability associated with interfacial flows. Interfacial waves. 5. Dispersed flows. Interaction between the dispersed and continuous phases. Momentum, heat and mass transfer. Differences and similarities between solid particles, droplets and bubbles. 6. Dynamics of single particles: solid particles, droplets and bubbles. Bubble and droplet dynamics. Bubble growth and collapse. Introduction to cavitation. Droplet dynamics and breakup. 7. Inter-particle interactions and collisions. Droplet and bubble coalescence and breakup. Collision kernel. Population balance models and simulations. 8. Quasi-1D flows, both internal and external (channel, pipe, jet, etc.). Balance equations and simplifying assumptions. Simple mechanistic models. Similarities and differences between: gas-liquid, liquid-gas, liquid-liquid, solid-liquid, and solid-gas. Flow classification and flow patterns, both separated and dispersed. Physical explanation and underlying mechanisms in the construction of a flow map. Steady and transient phenomena. 9. Stirred flows, both internal and external (bubble columns, fluidized beds, stirred tanks, ocean and atmospheric turbulence, etc.). Similarities and differences with respect to quasi-1D flows. Motion-driven, pressure-driven and gravity-driven flows. Dilute and dense flows. Flows dominated by inter-particle interactions (dense fluidized beds, bubble columns, etc.). 10. Multiphase flows with complex fluids (granular flow, slurries, agglomerates, emulsions, foams, etc.). Internal mesoscale structures. Thermodynamic and physicochemical interactions. Complex particles (with a complex structure and/or a complex particle-dynamics). 11. Heat transfer and phase change. Boiling and condensation. Radiation. 12. Industrial flow examples. Complex flow in pipelines in the oil-gas industry. Pneumatic conveying and solid-gas flows. Process equipment. Microfluidics applications. 13. Environmental flow examples. Sediment transport in rivers and the ocean. Aerosols and particulates dispersion in the atmosphere. Droplets in clouds.	
Study Goals	To give a general unified perspective on multiphase flow, with an emphasis on: (i) the understanding of the several essential physical mechanisms involved; and (ii) the formulation of complex engineering problems in terms of simple physically-based models. The students will develop the ability to: (i) formulate the problems in a structured way, starting from first principles; (ii) obtain (approximate) solutions using standard analytical and numerical techniques; (iii) interpret and critically analyse the solutions obtained; (iv) report the formulation of the problems and its solution in a clear well-structured form; and (v) sustain a critical discussion on the formulation of the problems and its solution. Also, during the course the students (i) will acquire a broad knowledge of multiphase flow; and (ii) will develop the ability to read and understand scientific literature on multiphase flow.	
Education Method	The education method is based on the combination of regular lectures, homework and interaction with the instructor outside the lectures (regular and impromptu office hours). The Learning by Doing is a key component of the education method: the homework involves both standard problems and open-ended problems, with the initiative and creativity of the students being strongly encouraged. The interaction with the instructor outside the lectures plays a key role, and it involves an iterative process, based on the Learning by Struggling philosophy.	
Literature and Study Materials	The study material consists mostly of lecture notes. Also, during the course, sections of books, on specific topics, and articles, from the scientific literature, are suggested for reading. Possible auxiliary general reference-books on Multiphase Flow are: 1. Brennen, C.E. (2005). Fundamentals of Multiphase Flow. Cambridge University Press. 2. Ghiaasiaan, S. M. (2007). Two-Phase Flow, Boiling and Condensation: in Conventional and Miniature Systems. Cambridge University Press. 3. Kleinstreuer, C. (2003). Two-Phase Flow: Theory and Applications. Taylor & Francis. 4. Crowe, C.T., Schwarzkopf, J.D., Sommerfeld, M. and Tsuji, Y. (2012). Multiphase Flows with Droplets and Particles. Second Edition. CRC Press. 5. Fan, L-S and Zhu, C. (1998). Principles of Gas-Solid Flows. Cambridge University Press. 6. Peker, S.M. and Helvacı, S.S. (2008). Solid-Liquid Two Phase Flow. Elsevier. Possible auxiliary general reference-books on Fluid Mechanics and Transport Phenomena are: 1. Kundu, P.K. and Cohen, I.M. (2008). Fluid Mechanics. Fourth Edition. Academic Press. 2. Ghiaasiaan, S.M. (2011). Convective Heat and Mass Transfer. Cambridge University Press.	
Assessment	The assessment is based on the homework proposed during the course (30%) and on the final exam (70%). The final exam consists of a written part followed by an oral discussion with the instructor. The written part is open book: the students can consult any reference they wish (including the internet) and use any tool they wish (including a laptop). The oral discussion will be arranged individually, shortly after the written exam, in consultation between the student and the instructor. It is possible to have a resit of the final exam; the format of the resit is the same as the format of the original final exam, consisting of a written exam followed by an oral discussion with the instructor.	
Permitted Materials during Tests	Any material (including laptop).	
Elective	Yes	
Tags	Fluid Mechanics Physics	

AP3222	Nanotechnology	6
Responsible Instructor	S. Goswami	
Instructor	R.A. Norte	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic physics and mathematics are required; solid state physics, materials science and physics of microelectronics are recommended. Good understanding of the typical physics approach to problems is helpful.	
Course Contents	The synthesis and study of objects with nanometer-size functional elements require dedicated fabrication methods. This course will provide the technological and scientific background needed to master the methods of nanofabrication. Subjects include: material properties, growth and etching methods; pattern definition and transfer (=lithography); self-assembly; molecular manipulation; methods of inspection, analysis and characterization. The subjects will be taught with a focus on practical applications.	
Study Goals	The student can evaluate the merits of the fabrication technologies for nanotechnological applications. The student can design the processing steps for a nanotechnological application. The student can find, understand and apply the relevant scientific literature. The student can evaluate, model and understand the functioning of nano-objects. The student can execute a simple but complete nano fabrication task.	
Education Method	Lectures, expert lectures, student presentation & discussion sessions, and lab work.	
Literature and Study Materials	1. Sami Franssila, Introduction to Microfabrication, 2nd edition, Wiley 2. Copies of the lectures 3. Scientific papers for the student presentations and discussions 4. Lab-work manual	
Assessment	Written exam (35%), home work (15%), lab work (30%), and student presentation (20%). The scores for the written exam and the lab work should be both at least 5.0.	
Permitted Materials during Tests	Ruler, pocket calculator	
Enrolment / Application	Because of limited space and support in the cleanroom, a maximum of 30 students can do the course. Enrollment will start on Monday 6 January 2025 at 8:00am via MyTUDelft. Enrollment requests done before January 6 at 8:00am will not be considered. If you encounter any problems with this registration procedure, please contact Arno Haket, A.J.W.Haket@tudelft.nl, from 6 January. Students who do not have a Physics background are asked to submit a motivation (on prior knowledge, current activities and interest in nanotechnology) by e-mail to Srijit Goswami S.Goswami@tudelft.nl. Students who do not attend the first lecture (without prior notice) will have lower priority than those present. Students who do not have their bachelor degree at the start of the second semester (10 February 2025), will have lower priority than those with the degree.	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/education/admission-and-application/elective-course#c1323630	
Elective	Yes	
Tags	Lab Research Physics	

AP3701	Submm and terahertz physics and applications	3
Responsible Instructor	Dr. J.R. Gao	
Responsible Instructor	Dr. A.J.L. Adam	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	The course is intended for MSc students from different disciplines, including those from Applied Physics, Electrical Engineering, and Space Engineering, but it is also open to interested PhD students.	
Expected prior knowledge	Electromagnetism, Any introduction of Optics, and Introduction of solid-state Physics. If you miss some of them, we will also add necessary background knowledge during the course	
Course Contents	The submm and terahertz region of the electromagnetic spectrum, which spans from 100 GHz to 10 THz, has attracted a lot of attention in recent years due to the developments in new detection and generation techniques and due to the applications for astronomic observations in space and the potential applications as an imaging technique for security, communications and biomedical applications. The Submm&THz frequency range is rapidly emerging which allows unearthing new science and technology. We wish to provide a course in which students can get an introduction and learn the basic physics & principle, and will also be updated with the latest developments. The course will focus on the topics like Photonics, Time-Domain Spectroscopy, Antenna/Waveguide, terahertz detectors and sources, THz Space application. Here the titles of all the different lectures given: 1. Introduction to THz. Light Interaction with Matter 2. THz photonics (THz fibers, Photonic crystals & Metamaterials. High THz fields. Nonlinear interaction of THz waves with matter) 3. THz Imaging (THz Time Domain spectroscopy, Transmission, Reflection, Time of Flight Tomography, Near field, Tomography) 4. THz Sensing (Application to Chemistry, Security, Biology & Medicine; Pump-probe experiment) 5. Submm & THz Coupling (Waveguide, Si Lens, Antennas) 6. Sources 7. Nano-detectors and cameras 8. Submm & THz Sensing instruments from space (as a few examples) Additionally: A case study and design of an space instrument by a subgroup of students (Adam/Gao) (To be announced)	
Study Goals	Understand submm & terahertz electromagnetic radiation and propagation fundamentals, and the basics of typical terahertz detectors and sources. Distinguish different terahertz antennas with regard to their advantages and disadvantages for terahertz radiations. Explain the working principle of THz Time-Domain Spectroscopy. Build practical experience on a THz imaging setup and a heterodyne receiver system. Highlight recent advances of THz research and development from the academic and astronomic sectors. Summarize state-of-the-art THz applications, including space applications.	
Education Method	The course will be given as following: 1) one 2x45 min lecture per week 2) Home work every week on the topics following the lecture. The homework counts 30% of the grade. 3) Design an instrument concept and address a specific topics with groups and present 4) a written exam on all topics discussed during the lectures and homework We based our lectures on current research and science on THz technologies	
Literature and Study Materials	The lectures will be based on a few selected chapters of different books. For Gao's lectures: Chapter 27 "FIR heterodyne receivers by J.R. Gao" of a new book "Observing Photons in Space" and the book of "Terahertz Astronomy" by Christopher K. Walker. For Aurele's lectures: Terahertz Technology, Volume 77, A. Rostami, H. Rasooli, and H. Baghban, An Overview of the Technological and Scientific Achievements of the Terahertz, pp. 189. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011. (http://link.springer.com/chapter/10.1007/978-3-642-15793-6_1) After each lecture, the teachers will provide also the powerpoint slides used during the lectures.	
Assessment	20% from the homework 30 % from case/project study report/presentation; 50 % from the written exam	
Permitted Materials during Tests		
Elective	Yes	
Tags	Building & Spatial Development Physics Projects Technology	

BM41155	3D Printing	4
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Responsible Instructor	Dr. J. Zhou	
Instructor	Ing. M.A. Leeftang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	in Track I (Musculoskeletal Biomechanics)and Track II (Medical Devices and Bioelectronics) of MSc Biomedical Engineering BME) and in the specialization of Bio-Inspired Technology (ME-BMD-BITE)	
Expected prior knowledge	BSc in mechanical engineering, materials engineering, industrial design engineering and aerospace engineering with a basic knowledge of materials (e.g. WB2330)	
Parts	- Overview of AM technologies, their principles and applications - Materials selection (polymers, ceramics, metals and composites) - Powder-bed fusion processes - Powdered materials and powder fabrication - Powder characteristics in relation to AM - Structural integration during melting and sintering - Extrusion-based systems - Design for functionality and 3D printability Software issues and AM limitations - Other powder-based manufacturing technologies as alternatives to AM - Post-processing of AM products - Mechanical testing of AM products - Case studies AM for engineering and medical applications - Assignment and presentations	
Course Contents	Additive manufacturing (AM), also known as 3D printing the process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to conventional subtractive manufacturing technologies, is set to revolutionize manufacturing across a wide range of industries. The course 3D printing - AM technologies is designed to equip students with the fundamental knowledge of AM and basic hands-on experience in AM. The topics include various AM processes, their principles, capabilities and limitations, materials used for AM processes, powdered material fabrication and characterization, CAD designs for AM, alternative powder-based manufacturing technologies, current and future AM applications, as well as the mechanical evaluation of AM parts. The primary focus is placed on powder-based AM for metallic parts. In addition to a series of lectures, project groups, each composed of 3-5 students, will create their own designs of the objects made of polymer PLA, formulate requirements in terms of geometry, dimensions, tolerances, mass, surface finish, functionality and mechanical properties of these objects, redesign them by making use of SolidWorks, print them, test and evaluate 3D printed products against the requirements, and document the projects in the form of reports and presentations. The new design is expected to reflect the distinctive features and freedom that AM has to offer (thin walls, lattice structures, form-free channels, integrated parts/functions, and customization), but within the capacity of the printing machines (bounding box size, minimum feature size or wall thickness). The report and presentation are expected to contain: reason for the selection of the object, weakness of the existing design, new design features, actual testing results and recommendations.	
Study Goals	Students will be familiar with the concept of AM, various AM processes, AM applications, materials for AM, product design for AM, post-processing techniques and evaluation methods. In addition, students will gain basic hands-on experience in product development via AM. Once successfully completing the course, the students will: - Have a fundamental knowledge of 3D printing/AM technologies for a wide range of applications - Be able to explain the benefits, advantages, challenges and limitations of individual AM technologies as well as alternative powder-based manufacturing technologies - Have an insight into selective laser sintering and melting (SLS/SLM) processes regarding process control in relation to the geometry and mechanical properties of the products - Become familiar with the common (metallic) materials for 3D printing and their characterization techniques - Understand printable geometry design rules and the common methods of digital file creation - Become familiar with the common techniques for post-processing - Become familiar with the common methods for the mechanical evaluation of 3D printed materials and functional parts.	
Education Method	Lectures and student projects	
Computer Use	Attending lectures and performing assignment are both compulsory	
Course Relations	- Object design with Software (SolidWorks Student Engineering Kit)	
Literature and Study Materials	Biomaterials, Tissue biomechanics and other engineering courses in MSc biomedical engineering, materials science and engineering, mechanical engineering, marine technology, aerospace engineering, design for interaction, integrated design and strategic design - Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21) - Lecture materials	
Books	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21)	
Reader	- Lecture materials	
Assessment	Written exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information) or CSE3130 (Introduction to Quantum Computer Science)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution.	
Study Goals	The student will acquire: - A good understanding of the fundamental concepts of quantum communication - The ability to understand, design and analyze quantum communication protocols - Insight into the differences between classical and quantum communication and cryptography	
Education Method	- Flipped classroom (Online videos and materials, prepared for flip classroom teaching) - Weekly Quiz and Discussion Session (In person) - Weekly Tutorials on Homework Problems (In person)	
Literature and Study Materials	Book on tools and quantum cryptography: Introduction to Quantum Cryptography, Cambridge University Press, 2023 Lecture Notes Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: - Homework (weighting 60%) - Final Project (weighting 30%) - Weekly Quiz (weighting 10%) Final grade calculation: $0.6 * \text{Homework} + 0.3 * \text{Final Project} + 0.1 * \text{Quiz}$ A passing final grade for the course can only be earned when for the homework and final project at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: - Homework: re-submit deliverable - Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4205	Evolutionary Algorithms	5
Responsible Instructor	Dr.ir. P.A. Bouter	
Responsible Instructor	Prof.dr. P.A.N. Bosman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks.	
	This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, (GPU) parallelization, and real-world applications.	
Study Goals	At the end of this course, the student is able to: 1. Explain the key concepts underlying the main streams in Evolutionary Algorithm (EA) research, with in particular genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2. Explain key ingredients underlying the rationale of when these algorithms work and when they do not work. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3. Explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4. Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5. Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	8-10 lectures 4 smaller (individual) projects 1 larger group project	
Literature and Study Materials	Various scientific papers, course-specific texts, and slides	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment (weighting 30%) - Individual Assignments: weekly assignments (weighting 10%) Final grade calculation = 0.6 * Written Exam + 0.3 * Group Assignments + 0.1 * Individual Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? - Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework. On the slides, references to articles (i.e. numbered paragraphs) or to pages in the book "Graph Spectra for Complex Networks" (see Literature and Study Material) are made. The information in the book thus complements the slides.	
Literature and Study Materials	Graph Spectra For Complex Networks, second edition (2023) Cambridge University Press ISBN: 978-009-36680-9 See: https://www.nas.ewi.tudelft.nl/people/Piet/bookGraphSpectraCN.html	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Course Relations	This course is mutual exclusive with the course SET3125. Students can not take this course if they have taken the course SET3125.	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (70%), project (20%), homework (10%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET3604LR	Electronic Circuits	3
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	Major topics: Electronics in information processing (sensor readout) Design principles and trajectory Signal processing functions Electronic design analysis Negative feedback amplifiers Operational amplifiers Components and networks Semiconductor diodes Power supplies and voltage regulators Semiconductor transistors and IC technology (basic) Oscillators DC motor basics	
Course Contents	This course is intended to provide gain insight, in a structured way, in the function of electronics as information-processing technology. The way of modelling will be discussed in relation to the physical characteristics of electronic components, circuits and systems. The course will provide the students with the skills to interpret circuit diagrams and measurement results and formulate specifications for electronic circuits that enable expert electronics designers to take proper design decisions when requested to design a high-performance application specific circuit. The students will learn the terminology that enables them to understand the explanation for these decisions by the experts (and potentially critically review them with respect to system level decisions). During the course many lecture demonstrations will be given, where the students both can observe the practical behavior of the test object and see what type of measurement equipment is used and how it is operated.	
Study Goals	After this course, the student -Is able to interpret electronic circuit diagrams and draw them according to standards -Is able to recognize and select electronic components -Is able to interpret and create Bode plots that describe the frequency behavior of an electronic circuit -Is able to analyze and design basic frequency filters -Is able to explain the basic behavior of semiconductor devices, like bipolar transistors, field effect transistors, diodes and special devices like thyristors. -Is able to determine the proper readout conditions for a sensor -Is able to determine the proper drive conditions for an actuator -Is able to interpret basic diode circuits like rectifiers -Is able to explain the basics of negative feedback systems and able to calculate and interpret the essential parameters -Is able to design a negative feedback amplifier using an Operational Amplifier (OpAmp) as active circuit that optimally meets the requirements -Is able to explain the basics of power circuits -Is able to name the different oscillator classes and explain the working principle the oscillators in each class -Is able to determine the class of a given oscillator -Is able to name the specific good and bad properties of each oscillator class -Is able to communicate with experts in electronic design and translate system specifications to specifications for the electronics in such a way that the expert can create an optimal design	
Education Method	Lecture. There is a mandatory practical with its own course code: ET3604LRP	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276 (older versions can very well be used for this course)	
Assessment	The exam is multiple-choice. During the lecture period, there will be one bonus test on oscillators. A pass grade gives a bonus point for the exam. The bonus point remains valid for the re-sit exam. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Dr. M.R. Vogt	
Instructor	Prof.dr.ir. O. Isabella	
Instructor	Dr.ir. H. Ziar	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	By the end of this course, you will be able to: 1. Explain the purpose of the components in different PV Systems using trends in their physical quantities. 2. Examine different power tracking methods and the importance of bypass diodes. 3. Calculate the energy yield of a PV system in conjunction with the following steps: 3a. Employ the calculation for optimal PV panel orientation from geographical survey. 3b. Determine the advantages and disadvantages of different PV systems topologies. 4. Design PV systems for different applications based on your own cost estimation and technical evaluation justified in a report	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	For students of the 2024 cohort: Final project report (33% of the final mark) and written test (67% of the final mark). For students of previous cohorts: Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr. R.T. Rajan	
Instructor	Dr.ir. R.C. Hendriks	
Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Tags	Algebra Algorithmics Artificial intelligence Linear Algebra Optimisation Signals Software Stochastics	

GEO1016	Photogrammetry and 3D Computer Vision	5
Course Coordinator	Dr. L. Nan	
Contact Hours / Week x/x/x/x	6 hours/week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Photogrammetry and 3D computer vision aim at recovering the structure of real-world objects/scenes from images. This course is about the theories, methodologies, and techniques of 3D computer vision for the built environment. In the term of this course, students will learn the basic knowledge and algorithms in 3D computer vision through a series of lectures, reading materials, lab exercises, and group assignments. The topics cover the whole pipeline of reconstructing 3D models from images: - Cameras models: how a point from the real world gets projected onto the image plane and how to recover the camera parameters from a set of observations; - Epipolar geometry: the geometric relations between 3D points and their images points; the constraints between the image points; - Image matching: define and match image features (SIFT) to establish correspondences between images; - Structure from motion: recover/refine geometry and camera parameters from a set of images; - Multi-view stereo and learning-based approaches for recovering dense geometry (e.g., point clouds) from images; - Surface reconstruction: obtain 3D surface models of real-world objects from point clouds.	
Study Goals	After finishing this course, the students will be able to: - apply linear algebra knowledge to implement basic 3D computer vision algorithms; - explain the main concepts in 3D computer vision (i.e., camera models, epipolar constraints, fundamental matrix, image matching, triangulation, structure from motion, bundle adjustment, multi-view stereo, and surface reconstruction); - explain the principles of the state-of-the-art 3D computer vision pipelines for 3D dense reconstruction from images; - evaluate methods for reconstructing smooth surfaces and piecewise planar objects, and choose applicable methods to solve specific reconstruction problems; - propose and implement solutions for reconstructing real-world buildings from images.	
Education Method	Lectures, reading materials, group assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is related to all fields of Geomatics for the Built Environment, from data acquisition, data management, data visualization, data analysis, 3D modeling, to applications.	
Literature and Study Materials	In addition to the lectures and handouts (distributed after each lecture), you will find relevant topics discussed in great detail in the following materials. [1] Y. Ma, et al. An Invitation to 3-D Vision: From Images to Models. 2001 [2] R. Hartley and A. Zisserman. Multiple View Geometry in Computer Vision. Second Edition. Cambridge University Press, 2004. [3] D. Forsyth and J. Ponce. Computer Vision: A Modern Approach. Second Edition. Pearson, 2012. [4] R. Hartley and P. Strum. Triangulation. Computer vision and image understanding. 1997. [5] Berthold K.P. Horn. Recovering baseline and orientation from essential matrix. 1990. [6] Y. Furukawa and J. Ponce. Accurate, Dense, and Robust Multiview Stereopsis. PAMI, 2010 [7] H. Hoppe, et al. Surface reconstruction from unorganized points. SIGGRAPH, 1992 [8] M. Kazhdan, M. Bolitho, H. Hoppe. Poisson surface reconstruction. SPG, 2006, 61-70 [9] L. Nan and P. Wonka. PolyFit: Polygonal Surface Reconstruction from Point Clouds. ICCV 2017.	
Practical Guide	In the assignments and group work sessions, students will experiment with the 3D computer vision techniques introduced in the lectures. - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; - Discuss the effect of parameters (if possible); - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Basic linear algebra and calculus; - Basic knowledge and skills in C++ programming. Students who have no experience in the above topics can still participate but should be willing to make efforts and invest some more time.	
Assessment	The assessment of this course consists of three group assignments and the final written exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded lower than 5.75 is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.	
Period of Education	Quarter 3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ME44312	Machine Learning for Transport and Multi-Machine Systems	3
Responsible Instructor	F. Schulte	
Instructor	Dr. B. Atasoy	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic programming	
Course Contents	The course gives an introduction to the basic concepts of machine learning (ML) (supervised, unsupervised, reinforcement learning) and puts them into relation to artificial intelligence and data analytics. Important fundamental machine learning algorithms including regression, classification, clustering, and neural networks are introduced and implemented in Python. The course also gives a brief overview of more advanced ML techniques that are used in the transportation and multi-machine domains such as reinforcement learning, graph-neural networks, and recommender systems. On this basis, machine learning methods are applied in group projects to transport and multi-machine systems using different types of data from different domains (e.g., AIS, GPS, New York Taxi, or machine failure data).	
Study Goals	-Understand the basic concepts of machine learning -Learn to implement widely-used machine learning algorithms -Apply machine learning methods to transport and multi-machine systems -Evaluate different machine learning methods for a problem setting -Conduct small ML projects	
Education Method	-Lectures -Group Work -Individual exercises for each machine learning algorithm	
Computer Use	Python on laptop	
Assessment	Group report	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Abstract Algorithmics Artificial intelligence Broad Business Design Group work Industry Modelling Optimalisation Programming Research Methods Stochastics Sustainability Technology Transport & Logistics Transport phenomena	
Percentage of Design	40%	
Design Content	Designing the right machine learning methodologies for different problem settings in the context of transport and multi-machine systems	
Department	ME Department Maritime & Transport Technology	
Contact	f.schulte@tudelft.nl b.atasoy@tudelft.nl	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Instruction if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Process & Energy	

ME45165	Equipment for Heat & Mass Transfer	5
Responsible Instructor	Prof.dr. K. Hooman	
Instructor	Dr. H.B. Eral	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Advanced reaction & separation systems Process plant design	
Expected prior knowledge	ME4500 (Advanced heat transfer)	
Summary	The course covers basic principles and methods required for the design and selection of heat and mass transfer equipment (for instance heat exchangers and thermal separation processes). This allows for the integration of heat & mass transfer principles to design process equipment.	
Course Contents	Rating and sizing of heat and mass transfer equipment such as heat exchangers, distillation, absorption, tripping, crystallization, adsorption, extraction and membrane separations will be covered. In particular: classification of various separation and heat transfer processes, the separation efficiency in single and multi-stage processes (and the influence of the process conditions and contacting modes on the separation). design procedures of different separation processes based on limitations of the thermodynamic equilibria. quantification of the influence of process condition on the design procedures of different separation processes based on limitations of the thermodynamic equilibria. characteristics of equipment for various separation processes the role of mass transfer in the sizing and design of the separation equipment. methods for the design and dimensioning of separation equipment. the use of computer tools for the design of heat exchangers and separation processes	
Study Goals	At the end of the course, the student will be able to rate and size heat and mass transfer equipment. In particular the student: 1. Can calculate overall heat transfer coefficient for heat exchangers; 2. Can select heat transfer equipment for given application; 3. Can apply effectiveness and Number of Transfer Units method to design heat exchangers; 4. Can apply pinch technology and heat integration for heat exchanger network. 5. Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes, 6. Comprehends the role of mass transfer in the dimensioning of equipment for separation processes 7. Can determine the separation efficiency for single and multi-stage separations and can quantify the influence of the process conditions and the contacting modes on the separation. 8. Can use computer based process simulation tools to analyze and design separation equipment	
Education Method	Lectures, assignments, self study, team work, tutorials	
Computer Use	Simulation of performance of distillation trays Standard engineering packages including ASPEN	
Literature and Study Materials	Course material: 1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738 2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Wiley & Sons, 2011. 3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer. 4: Course sheets / relevant papers 5: R K Sinnott, Gavin P Towler: Chemical engineering design, Coulson & Richardson's Chemical Engineering Series; Butterworth-Heinemann, an imprint of Elsevier: Kidlington, Oxford, United Kingdom, 2020. 6: Kays, W. M.; London, A. L. Compact Heat Exchangers, 3rd ed.; McGraw-Hill: New York, 1984. 7: Kroger Detlev G. Air-Cooled Heat Exchangers and Cooling Towers; Penwell Corp: Tulsa, Okl., 2004. Further reading Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son Geoffrey F. Hewitt, "Heat Exchanger Design Handbook", ISBN Online: 978-1-56700-422-9, Imperial College, England, UK	
Assessment	Formative: In-class exercises/quizzes Summative: Group project report & presentation (50%) Individual exam (50%)	
Permitted Materials during Tests	Course books and lecture notes	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Percentage of Design	50	
Design Content	Thermohydraulic design of heat exchangers; heat exchanger network, dimensioning pieces of equipment used in the	

	abovementioned processes
Department	ME Department Process & Energy

ME45170	Turbomachinery	4
Responsible Instructor	Prof.dr.ir. S.A. Klein	
Responsible Instructor	Prof.dr. R. Pecnik	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Thermodynamics, Fluid dynamics	
Course Contents	The course consists of the following parts: - Introduction and recap of the required thermodynamics. This class will give an overall overview of the relevance of turbomachinery, the general basics and the history. Furthermore the basic thermodynamics will be repeated (1 class) - Turbomachinery: basic theory and the application to the design of turbomachinery (Conservation equations and velocity triangles, turbines, compressors and pumps, 3D flow, radial machinery) (5 classes) - Application of compressor, pump and turbine maps. Limitations in the operation (stall, surge, NPSH etc) (1 class) - Gas turbine: components (turbine, compressor, combustor), gas turbine cycles, performance calculation,	
Study Goals	The student is able to describe the fundamental aspects of the working principle of turbomachinery. More specifically, the student must be able to: 1. reproduce and apply thermodynamic principles of turbomachinery 2. describe the basic fluid dynamics of turbomachinery and the way work is transferred between the working fluid and the turbomachinery, including the impact on performance and limitations of operations 3. apply compressor, pump and turbine curves in practical applications 4. describe different gas turbine cycles 5. describe the influence of different losses on the performance (power and efficiency) of gas turbines 6. synthesize different gas turbine cycles with respect to their application. 7. describe the principles of combustion as applied in gas turbines with respect to functionality and emissions 8. describe the part load characteristics of gas turbines and its relation to the functional characteristics of gas turbine components	
Education Method	Lectures and homework assignments	
Assessment	Closed book exam (60%) and home work assignments (40%). All homework assignments must be submitted. The grade for the homework assignments will only be included in the final grade if the exam grade is above 5.5/10	
Enrolment / Application	Registration: Standard enrollment via Brightspace	
Tags	Energy Energy & Industry Fluid Mechanics Thermodynamics	
Department	ME Department Process & Energy	

ME45203	Electrolysers, Fuel Cells and Batteries	4
Responsible Instructor	J.W. Haverkort	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Familiarity with transport phenomena (e.g. a course on heat transport), basic fluid dynamics, elementary differential equations, and introductory electrochemistry. Ideal preparation include the courses ME45201, CH3513, or SET3110. A previous course on fuel cells, batteries, or similar, or additional self-study may also be sufficient.	
Course Contents	Electrochemistry plays a pivotal role in the transition towards a sustainable energy future. Renewable electrical energy can be stored in batteries or used to synthesise chemicals or fuels in electrochemical reactors, like an electrolyser producing hydrogen. This course will introduce to a student, the knowledge and skills required to understand, model and optimize such electrochemical devices. Content overview: Porous electrode theory, multiphase flow, and transport models will be used to derive analytical current-voltage relations of various electrochemical devices including fuel cells, batteries, redox flow batteries, and electrolysers. Some of the topics that will be discussed in this course include: - The Nernst-Planck equations for ionic transport in binary electrolytes. - Porous electrode theory using the concept of an electrochemical Thiele modulus. - The flooded agglomerate and relative permeability models for gas diffusion electrodes. - The single particle and Newman's P2D battery models. - A vertical current and gas distribution model for electrolysers.	
Study Goals	After successful completion of this course, the student is able to independently: 1. explain and apply porous electrode theory. 2. derive 0D, 1D, and 2D mathematical models to describe the current-voltage relationship of various electrochemical cells including flow, multiple phases, and porous electrodes. 3. optimize an electrochemical cell design with respect to, for example, energy efficiency.	
Education Method	Two lecture hours + two exercise hours per week.	
Course Relations	This course is a natural sequel to ME45201, CH3513, and SET3110 for students interested in analytical modelling of electrochemical systems.	
Books	Electrolysers, Fuel Cells and Batteries: Analytical Modelling. (J.W. Haverkort) https://doi.org/10.59490/tb.93 (open access)	
Assessment	Written exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	

ME46006	Physics for Mechanical Engineers	4
Responsible Instructor	Dr. G.J. Verbiest	
Instructor	Prof.dr. P.G. Steeneken	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have Physics knowledge at the level of the final exam of the physics exam at Dutch highschools (VWO) and to have Mathematics knowledge at the level of the BSc education in Mechanical Engineering (e.g. WBMT2048). If they have not, they can still do the course but it will require more self-study from the book of Tipler.	
Summary	The course ME46006 Physics for Mechanical Engineers, aims to provide mechanical engineering students with a broad background in physics. The course offers basic understanding and skills in the field of physics, that will provide a background that can be useful in many engineering situations. It can serve as a starting point for more advanced physics studies and can also help mechanical engineers to interact with electrical engineers and physicists in their future career. Since the course aims for a broad coverage of the field of physics, the course material includes many different subjects at a basic to intermediate level. In contrast to other courses, this course favours to cover a relatively large number of different topics in order to provide a broad background, instead of covering a small number of topics in-depth. This means that the amount of material that needs to be studied is larger than for other courses. A bachelor degree in mechanical, electrical, aerospace, industrial engineering or applied physics is considered prerequisite.	
Course Contents	- Waves - Electrostatics and electricity - Magnetism - Electrodynamics - Optics - Interference and diffraction - Outlook into quantum mechanics	
Study Goals	At the end of the course, students should have knowledge, and should be able to solve problems on the following topics: Waves (Chapters 15&16 of Tipler): - The wave equation - Propagation, reflection and transmission of waves - The Doppler effect Electrostatics (Chapters 21-24): - Coulomb's law and electric forces - Electric fields and Gauss' law - Voltage and capacitors Electricity (Chapters 25 & 29): - Ohm's law - DC circuits Magnetism (Chapters 27-28, 30): - Gauss' law for magnetism - Lorentz', Ampere's, Lenz' and Faraday's law - Electromagnetic motors and inductors - Maxwell's equations Optics (Chapter 31-34): - Propagation and properties of light - Light-matter interaction - Imaging systems with light - Interference and diffraction - Limits of optical systems and quantum mechanics	
Education Method	Learning methods include power point presentations, class demonstrations, and recommended homework. Lecture notes will be made available on the Brightspace. Recommended homework will be on the concepts discussed in the class and solutions will be made available on the Brightspace. The course ends with a written exam.	
Literature and Study Materials	The course will closely follow selected chapters of the book of Tipler. In addition lecture slides will be provided on Brightspace.	
Books	Physics for Engineers and Scientists, 6th edition by Tipler and Mosca.	
Assessment	Written exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Precision & Microsystems Engineering	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Prof.dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Dr. O. Nejadseyfi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be 5.8 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest decimal. The maximum bonus score is 0.8. The bonus only applies to reports with grade ≥ 5.5 and submitted before the first report deadline.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	ME Department Precision & Microsystems Engineering	
Contact	Prof.dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

ME46085-23	Mechatronic System Design	4
Responsible Instructor	Dr. S.H. Hossein Nia Kani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The following are strictly required to be able to follow the course: - Basics of Multi-body Dynamics, including the ability to form equations of motion. - Basics of Electromagnetism, particularly Maxwell's equations. - Fundamentals of signals and systems (Laplace transforms, transfer function, bode plot, Nyquist stability). It is expected that students are able to use MATLAB or are interested in spending time to learn it while passing the course. This is required to complete the assignments.	
Course Contents	Course Contents Mechatronic system design deals with the design of controlled motion systems by the integration of functional elements from a multitude of disciplines. It starts with thinking how the required function can be realised by the combination of different subsystems according to a Systems Engineering approach. It should be noted that the control principles used in this course place a strong emphasis on frequency domain methods by linearising the system at its working point with the use of Bode- and Nyquist plots. The main reason for this emphasis is the strong focus in other control related courses on (non-linear) time domain related methods while linearised frequency domain related methods are still dominantly applied in the industry. A mechatronic engineer should be able to work with both methods and use them where appropriate. Some supporting disciplines, like power-electronics and electromechanics, are not part of the BSc program of mechanical engineers. For this reason this course introduces these disciplines in connection with mechanical dynamics and PID-motion control principles to realise an optimally designed motion system. The target application for the lectures are motion systems that combine high speed movements with extreme precision. The course covers the following three main subjects: 1: Dynamics of motion systems in the time and frequency domain, including analytical frequency transfer functions that are represented in Bode and Nyquist plots. 2: Electromechanical actuators, mainly based on the electromagnetic Lorentz principle. Reluctance force and piezoelectric actuators will be shortly presented to complete the overview. 3: Motion control in the frequency domain with PID and model-based feedforward control-principles that effectively deal with the mechanical dynamic anomalies (resonances and eigenmodes) of the plant. The other relevant discipline, electronics and position measurement systems is dealt with in another course: ME46005, Physics and measurement. The most important educational element that will be addressed is the necessary knowledge of the physical phenomena that act on motion systems, to be able to critically judge results obtained with simulation software. The lectures challenge the capability of students to match simulation models with reality, to translate a real system into a sufficiently simplified dynamic model and use the derived dynamic properties to design a suitable, practically realisable controller. This course increases the understanding what a position control system does in reality in terms of virtual mechanical properties like stiffness and damping that are added to the mechanical plant by a closed loop feedback controller. It is shown how a motion system can be analysed and modelled top-down with approximating (scalar and linearised) calculations by hand, giving a sufficient feel of the problem to make valuable concept design decisions in an early stage. With this method students learn to work more efficiently by starting their design with a quick and dirty global analysis to prove feasibility or direct further detailed modelling in specific problem areas.	
Study Goals	Can analyse and derive improvements to the dynamic behaviour of an actuator-driven mechanical structure with maximum 6th order plant dynamics (incl actuator and amplifier) by means of Bode and Nyquist plots. Can select and calculate a single-axis functional electromagnetic actuator for a given specification, working according to the Lorentz and reluctance force generation principle. Can select a proper amplifier to drive the above mentioned electromagnetic actuator. Can understand the basic concept of piezoelectric actuators and their application in mechatronic systems Can identify and apply feedforward and feedback P-, PD- or PID-motion controller settings for a given plant, consisting of a dynamically realistic power amplifier, electromagnetic actuator and mechanical structure with an ideal sensor, to achieve a stable system targeting a specified maximum bandwidth or disturbance rejection.	
Education Method	Lectures are given in 12*2 lecture hours, with presentations on the theory and practice of active-controlled motion systems. In addition, students will work on assignments to implement the theory learned in the course.	
Literature and Study Materials	Only the book is allowed at the examination. Therefore, no printed copy of the first edition, no computers, e-readers, smartphones, or other items are permitted. However, small notes written on the pages of the book are allowed.	
Books	The Design of High Performance Mechatronics - 3rd Revised Edition by R. Munnig Schmidt, G. Schitter, A. Rankers and Jan Van Eijk	
Assessment	The Final Assignment will account for 50% of the final grade. The remaining 50% of the score will be based on an open-book written examination. The examination will consist of questions covering the above-mentioned study goals.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	

Contact	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations. s.h.hosseinniakani@tudelft.nl
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ME46095	Multiphysics Modelling using COMSOL	4
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Contact Hours / Week x/x/x/x	2 hour lectures on a weekly basis	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In applied mechanics one is often confronted with a multi-physics or coupled problem: A problem that requires the (simultaneous) solution of more than one type of physical process or phenomenon in order to accurately describe the problem. Examples of multiphysics problems are fluid-structure interaction, thermal-structure interaction and electro-thermal-structure interaction, possibly combined with a control problem. Particularly in the field of Mechatronic design and MEMS multiphysics problems are frequently encountered. COMSOL MultiPhysics is a finite element code, which can be used both as a MATLAB toolbox and as a standalone program, which is particularly suited for the simulation of multi-physics systems. In this course the student will learn to recognize different types of multi-physics coupling and methods for their efficient numerical solution using COMSOL. Short homework assignments are used to practise the use of COMSOL on different types of problems and in a final assignment the student is asked to study a multi-physics problem using COMSOL.	
Study Goals	The student must be able to: 1. recognize multiphysics coupling in complex problems 2. distinguish between different types of coupling one-directional vs. bi- or multi-directional interface vs. field strong vs. weak 3. describe numerical solution techniques applicable to coupled problems 4. use COMSOL MultiPhysics on coupled problems	
Education Method	(Online) Lectures (2 hours per week) / Self study	
Assessment	Report	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	
Remarks	*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations. Old course code: WB2454-07	
Department	ME Department Precision & Microsystems Engineering	

ME46300	Optics	4
Responsible Instructor	Dr.ir. S.F. Pereira	
Instructor	Dr. N. Bhattacharya	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to optics: light propagation, the vectorial nature of light, interference, diffraction, optical spectra, geometrical optics	
Study Goals	At the completion of this course the student will be able to - have a good understanding of the basics of optics (i.e. making new connections between learned concepts and applying what has been learned to new situations) - explain the basic laws of polarization, birefringence, interference and diffraction - apply the theory to solve simple problems - put the theory in practice by performing experiments.	
Education Method	Lectures, exercise sections and practicals	
Books	Textbook that will be used in this course: Introduction to Modern Optics, G.T. Fowles, second edition, ISBN 100486659577 plus lecture notes and slides (available on Brightspace)	
Assessment	Individual written exam (90% of the grade). If it is required that the exam will be online, we will have an oral check after the exam and similarity check. The grade of the practicals (10% of the final grade) will be the average of the written group reports.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Precision & Microsystems Engineering	

ME46310	Opto-Mechatronics	4
Responsible Instructor	Dr. N. Bhattacharya	
Instructor	Dr.ir. S. Iskander-Rizk	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	Optical sensors and actuators are being integrated at an accelerated rate into mechatronics systems making these systems more precise, autonomous and intelligent. In this course the principles of optics, optomechanics and optical detectors are taught and discussed from the perspective of integration in optomechatronics systems for increased performance and improvement/addition in functionality. Topics included in this course are: 1) The different models of optics 2) Optical measurement methods. 3) Tolerances and sensitivities of mechanical positioning which influence optical performance of a machine. 4) Total system design where optical requirements can be translated to mechanical and mechatronical requirements. 5) Sources of light and optical detectors which can influence system performance (SNR, reliability, accuracy).	
Study Goals	At the end of this course the student can LO 1: Understand the practical applications and limitations of the different models of optics (geometric optics, diffraction optics, electromagnetic waves). LO 2: Understand the basic principles of imaging, interferometry, and spectrometry. LO 3: Understand how aberrations are related to mechanical tolerances and imaging quality. LO 4: Evaluate the function and performance of an optical system including the electronic readout module LO 5: Design an optical system for given environmental and application requirements (stress, thermal, kinematic effects) LO 6: Justify the mechatronic system components choices which can improve the performance of a given optomechanical system LO 7: Understand the influence of material choices and types of mounting optics	
Education Method	Lectures and Quizzes	
Assessment	Written exam + Quiz grade + 2 Guest lecture summary.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Broad Challenging Industry Mechanics Mechatronics Physics Signals	
Department	ME Department Precision & Microsystems Engineering	

MS43310	Materials at High Temperature	4
Responsible Instructor	Dr. V.A. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on Brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Materials Science & Engineering

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. L. di DiCarlo	
Responsible Instructor	Dr. M. Blaauboer	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	The final grade for the final exam is composed of the homework assignment grades (total of 30%), online quiz grades (total of 10%) and the grade obtained for the final exam (60%). A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course. The final grade for the resit exam consists of the resit exam grade only (100%). In other words grades for homework and quizzes do not count towards the final grade of the resit exam.	

RO47001	Robot Dynamics & Control	5
Responsible Instructor	L. Peternel	
Instructor	B. Shyrokau	
Instructor	J.M. Prendergast	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides the basics for deriving kinematic/dynamical models of a robot (such as a factory robotic arm, mobile robot, etc.) and for designing corresponding robot controllers. Students will learn how to derive the equations of motion and basic control equations themselves, and they will also learn how to use existing frameworks to do the same. THEORY SUBJECTS Introduction & Fundamentals -Course layout, instructors, grade scheme, etc. -Importance of the course and the big picture -Connection to real-world -Basic concepts/terms in robotics -Methods to describe robot pose -Numerical integration Recap of Control Theory -State-space model -Root locus -Bode plot -PID control -Concept of LQR and MPC control -Reference to online materials Endpoint Control for Robotic Manipulators -Big picture + motivation -Difference between position and torque-controlled robots -Endpoint position control (with a PID controller) -Endpoint impedance control (for position tracking tasks) -Endpoint force control (for force tracking tasks) Robotic Arm Kinematics -Deriving forward kinematics and Jacobian (for a 2-DoF planar arm) -Transformations between endpoint and joint space using Jacobian -Kinematic singularities and their solution -Kinematic redundancy and task priority control (for a 4-DoF planar arm) -Kinematics of parallel robots and differences compared to serial robots Robotic Arm Dynamics -Rigid-body dynamics model -Deriving dynamics model with Lagrange method for 2-DoF planar arm -Using the dynamics model in the control -Dynamic task priority control 3D Robotic Arm -Deriving 3D kinematics using the Denavit-Hartenberg convention -3D trajectory generation Planning & Motion Control -Orientation control -Stability analysis of the controller Kinematics & Dynamics of Mobile Robot / Automated Vehicle -Forward kinematic models -Wheel kinematic constraints -Robot kinematic constraints -Non-holonomic system -Derivation of linear bicycle model -Steady-state analysis Dynamics & Control of Mobile Robot / Automated Vehicle -Transient analysis -Frequency response -Path-following control PRACTICAL ASSIGNMENTS (individual) Assignment 1: The goal of this assignment is to learn how to control a 2-DoF robotic arm in Python to follow a prescribed trajectory using kinematic and dynamic equations. Assignment 2: The goal of this assignment is to learn how to control a multi-DoF robotic manipulator (3D kinematics, 3D trajectory, orientation control). Assignment 3: The goal of this assignment is to learn how to implement a linear bicycle model, and calculate and discuss primary handling characteristics, poles, stability analysis, and feedback control (PID).	
Study Goals	After completing this course, the students are able to:	

Education Method	-Derive the equations of motion for different types of robots -Create joint-level and endpoint-level controllers (including trajectory following) -Implement a simulation model in Python or Matlab -Analyse the designed control systems and reflect on the design choices -Lectures & interactive sessions (4 hours per week) -Practical assignments (homework)
Literature and Study Materials	Note: While attending lectures & interactive sessions is not mandatory, it is highly recommended. Assignments are not strictly mandatory, however, if you do not submit them you will lose the corresponding percentage of the final grade. That might mean that even if you pass the exam, you might not have a high enough final grade to pass the course. Study materials -Lecture slides -Papers, books, videos, etc. (links provided in Brightspace) Robotics textbooks (either one) -Lynch, K. M., & Park, F. C. (2017). Modern robotics. Cambridge University Press. -Craig, J. J. (2017). Introduction to robotics: mechanics and control. Pearson Education. -Siciliano, B., Sciavicco, L., Villani, L., & Oriolo, G. (2010). Robotics: modelling, planning and control. Springer Science & Business Media. Control theory textbooks (either one) -Ogata, K. (2010). Modern control engineering. Prentice hall. -Dorf, R. C., & Bishop, R. H. (2001). Modern Control Systems: Solutions Manual (Vol. 12). Prentice Hall. Useful toolboxes -Corke, Peter. Robotics Toolbox for Python. -Foote, Tully. "tf: The transform library." 2013 IEEE Conference on Technologies for Practical Robot Applications (TePRA). IEEE, 2013.
Prerequisites	-Linear algebra (including matrix and vector operations) -Calculus -Linear system theory at the BSc level (e.g., simple differential equations, Laplace transforms) -Basic programming experience at the BSc level (Python/Matlab)
Assessment	Number of assessments: -Four (3 assignments, 1 exam), all based on individual work. -The exam will be a computer/digital exam in ANS on campus. Grade calculation: -If and only if the exam grade is 5.5 or higher, the grade will consist of 30% of the average score of the assignments and 70% of the exam grade. The final grade must be 6.0 or higher to pass the course. Assignment submission lateness criteria: <1 hour: -1 point <1 day: -2 points <2 days: -4 points <3 days: -6 points >3 days: no points given
Enrolment / Application	Note: Assignments and exam grades are only valid for the academic year they were obtained in. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department Contact	ME Department Cognitive Robotics For general course inquiries: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 1 and related content: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 2 and related content: Micah Prendergast (J.M.Prendergast@tudelft.nl) For Assignment 3 and related content: Barys Shyrokau (B.Shyrokau@tudelft.nl)

RO47002	Machine Learning for Robotics	5
Responsible Instructor	Dr.ir. J. Kober	
Instructor	Dr. H. Caesar	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides a broad overview of machine learning techniques and their practical application in robotics. Intro to Machine Learning and python - Machine Learning in Robotics - Understanding the goal of machine learning, fundamental problems, high-level overview. Visualizing your data - Supervised vs Unsupervised vs Reinforcement learning - Model-driven vs Data-driven: White box, Gray box, Black box - Prior knowledge vs Unstructured. Feature extraction, linear regression + Deep Learning. Interpretability Hands-on Machine Learning - Example machine learning project in Robotics - Binary classification, decision boundaries, using logistic regression/SVM (one parameter) - How to perform a classification experiment. Dataset splitting, learning curves, metrics, comparing results Regression & Data collection - Regression and Data Collection in Robotics - Regression methods, least squares fitting - Overfitting, cross validation, regularization - Collecting (noisy) data, labelling data, outliers - High-dimensional data, data augmentation - Hyper-parameter optimization (grid search vs random search) Classification - Classification in Robotics - Parametric vs Non-parametric classifiers - Logistic regression - Decision tree, forest - Bayesian classification: Bayes' rule, naive Bayes, Gaussian Mixture Model - k-nearest neighbour - SVM, kernel-SVM, dual problem - Multi-class classification, metrics (confusion matrix), class imbalance Unsupervised Learning - Unsupervised Learning in Robotics - Clustering: K-means, Gaussian Mixture Model, DBSCAN - Dimensionality reduction: PCA, Local Linear Embedding (LLE) Neural Networks - Neural Networks in Robotics - Multi-Layer Perceptron, gradient descent - Neural Networks, and deep learning - Vanishing gradient problem, DropOut, Optimizers Advanced machine learning - Outlook: Recurrent neural networks, convolutional neural networks - Outlook: Data augmentation, AutoEncoder - Outlook: Reinforcement learning	
Study Goals	Knowledge, insight, judgement and skills: After completing this course, students will be able to - explain the role of ML in robotics, and describe possible use cases, and when not to use ML - explain the main concepts in ML (overfitting, training and validation, classification vs regression, cross validation, regularization, parametrics vs. non-parametrics, supervised/unsupervised/reinforcement learning) - explain the principles of well-established ML methods (e.g. Logistic regression, SVM, RFs, Neural Networks) - collect and preprocess data (normalization, feature selection, augmentation, splitting) for ML - analyze a ML problem, consider available data and computational resources, and select the appropriate ML methods to apply - perform experiments, evaluate the results, and draw sound conclusions - present the findings of a ML experiment clearly in a report using graphs and scientific English - write python code to visualize data and implement ML methods using popular python frameworks, such as pandas, numpy, matplotlib, jupyter notebook, sklearn, keras Transferable and interpersonal skills: After completing this course, students will be able to - collaborate and coordinate with a lab partner on coding and reporting tasks	
Education Method	Contact hours - Flipped classroom: pre-recorded lectures (2 hours per week) + classroom discussion (2 hours per week) - Q&A on practica (2 hours per week) Formative feedback - Quizzes -- Individual -- On material of previous lecture(s) -- Via BrightSpace at set time slots - Practica -- Pairwise collaboration -- Submission via BrightSpace -- Counts for knock-out criteria -- No late submission -- No resit Deadlines - For each practicum, the deadline is approx. one week after the practicum is published.	
Literature and Study Materials	Study materials - Slides (released shortly before lectures) - Book: "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent	

	Systems" 3rd edition, Aurélien Geron
Prerequisites	Suggested literature - "Pattern Recognition and Machine Learning", C. Bishop - "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Hastie, Tibshirani, Friedman TU Delft BSc Mechanical Engineering curriculum, or equivalent. Especially familiarity with - Programming in python - Linear Algebra - Probability theory
Assessment	Final assignment (30%) - Pairwise collaboration - Late submission: -1 grade point per day - No resit - Deadline: Will be released after the 6th practicum, giving the students approx. two weeks to read the assignment, collect the necessary data, perform their experiments, and write their report Exam (70%) - Individual - Written closed-book exam - Exam in Q1, resit in Q2 Knock-out criteria - At least 4 practicum grades need to be a pass (of current or previous academic year) - Minimum grade for final assignment is a 5 (of current or previous academic year) - Minimum grade for the exam is a 5 - Final grade minimum is a 6
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics

RO47007	Multidisciplinary Project	5
Responsible Instructor	Dr. C. Pek	
Instructor	Dr.ir. G.N. Saunders	
Instructor	Prof.dr.ir. M. Wisse	
Instructor	C. Hernandez Corbato	
Instructor	Ir. C.C. Claij	
Contact Hours / Week x/x/x/x	0.0.0.2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In this capstone project of the first year of the MSc Robotics, you will be challenged to use all the knowledge and skills you have gathered so far. Within a multidisciplinary team of up to 5 people, you will develop and test software for use in a real robot whilst working together for a real client in an industrial setting. We expect all your robotics skills (including your programming skills) to be up to date! The estimated workload of this course is 18 hours per week over 8 weeks. In this project, you will not just produce a software solution for a robot. You will also implement and test your software in a real robot and report and present your results as well as document your code, mirroring industry practice. Just like you would in industry, you will likely spend only 50% of your time coding and testing and the other 50% on meetings, systems design, report writing, (preparing) presentations, and documenting your code. These are no longer afterthoughts. In this project, you work closely with an industry client, realistically a possible future employer. That means there are expectations regarding your commitment and presence during the project. We expect you to behave cordially and respectfully towards the client and be on time for scheduled sessions. You will be given the option to express a preference for a client when you enroll if multiple clients take part in the course. Each of you will sign up with a specialist role in mind within the project, in which you are responsible for a specific part of the software design and implementation. It is no longer possible for one person to do all the programming. The specializations for which you can express a preference when you enroll are: - human-robot interaction, - navigation, planning, control - perception, and relate to the different disciplines within robotics. Within your team, you are expected to make a translation of societal issues from different perspectives (human, sustainability, safety, ethics, economic, etc. into intelligent robot solutions. Practically, each student team develops a complete, integrated software package for a complex robot system. You develop and test in a simulated environment, and then implement it in a real robot during the course before presenting your results in the final presentation to your client at the end. This course consists of 5 iterative phases with continuous reporting varying in lengths: - Phase 0: Introduction, meet your team, and get to know your robot. - Phase 1: Initiation and Planning - Getting organized, defining and planning your project. - Phase 2: Design & Execution - Software architecture design, initial (simulation) implementations and demonstration of concept to the client. - Phase 3: Implementation - Further design and implementation of your solution on your robot. - Phase 4: Testing & Completion - Finalizing your project and final reporting and code documentation, with a presentation of your implementation for the client in week 4.8. A detailed schedule will be available on Brightspace after the introduction lecture, likely in week 3.8.	
Study Goals	The learning objectives for this course are divided over two domains: 1) Knowledge, insight, judgment and skills By the end of the course, you should be able to: LO 1: Define a robotics use case with its functional and non-functional requirements. LO 2: Design robotic solutions by integrating knowledge of system engineering, current opportunities for robots, trends in robotics, and societal aspects. LO 3: Produce a robotic solution that meets the design specifications and considers sustainable, safety, ethical, economical and other relevant societal aspects. LO 4: Use functional architecture for planning and communicating robot software. LO 5: Communicate the multidisciplinary robot solution orally, in writing, and in code documentation according to robot software development guidelines. 2) Transferable and interpersonal skills. By the end of the course, you should be able to: LO 1: Formulate learning goals for your own personal development. LO 2: Reflect on your competencies (e.g., Teamwork, Leadership, Entrepreneurial thinking, Strategic multidisciplinary problem-solving), their development, and your personal learning goals. LO 3: Apply structured feedback to improve your performance or the performance of others. LO 4: Apply structured multidisciplinary software project management, with the use of different team roles and responsibilities.	
Education Method	Project work in teams supported by (video) lectures and workshops.	
Computer Use	During the project, we will be using different types of software and collaboration platforms. To save time and frustration, please ensure you have a working laptop that has the following software installed and have gained access to the following repository before the project starts: ROS: We expect you to have a working ROS environment on your laptop. Please use your knowledge and experience from RO47003 Robot Software Practicals and make sure that you align with the Ubuntu and ROS versions used in your clients robot. We will be using native installations of Ubuntu and ROS, and do not provide support for any other installation. Instructions on this will be published on Brightspace. Gitlab: All code and documentation for this project will be stored on GitLab. To be added to your groups repository on the GitLab environment for this course, go to: https://gitlab.tudelft.nl and log in using your NetID. Please do this as soon as you have been admitted to the course and have been assigned a group. Otherwise, you will not be able to access the required code for your project.	
Literature and Study Materials	During the project we expect you to use and refer to the following books. Please avail yourself prior to the course starting of either a hardcopy or an electronic copy of the book. [1] A Gentle Introduction to ROS Jason OKane, E-Book: Accessible via the authors website: https://jokane.net/agitr/ [2] Practical Model-Based Systems Engineering Jose Luis Fernandez & Carlos Hernandez Corbato, E-book: Accessible with the TU Delft library web extension In addition, you may also want to refer to these books. These are, however, not mandatory for the course: [3] Beginning Software Engineering Rod Stephens, Hardcopy: Available via Leeghwater, E-book: available through the TU Delft library extension. [4] Information Technology Project Management K. Schwalbe, Hardcopy: any online bookshop, E-Book: no free legal website available	
Prerequisites	Starting the project is only allowed when a student has successfully completed the course RO47003 "Robot Software Practicals" and has completed RO47000 Vision & Reflection. Without these preliminaries, the student will not be allowed to participate. Exemptions require written approval of the teachers of RO47003 and RO47000 that the student has obtained equivalent	

Assessment	qualifications. It is expected that you can code when you start this course. At the end of the course, your project needs to comply with a few requirements before we will grade it. These will be clearly listed in the course manual on Brightspace. Moreover, you must attend all mandatory sessions of the course. Excused absences are only permitted by receiving prior approval of a study advisor and the course staff. Absences due to medical reasons require a confirmation, e.g., by a study advisor. For this project, an individual grade is given. The grade for this project is a combination of the group grades for your final project including your final presentation, the Sprint assignments, and an individual grade based on your personal reflection. An individual bonus-malus of max. 1 grade point can be applied by staff in rare cases of excellence or underperformance. For more details on grading, see the course manual on Brightspace.
Enrolment / Application	Enrollment for the course will open at the beginning of Q3. Further details on how to enroll will be announced on Brightspace. All students must have registered before the announced registration deadline. It is the students responsibility to acquire the required registration information and deadline. After the registration deadline, late participation in the project is not possible anymore. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Special Information	Note: The mandatory introduction lecture and the ROS software & robot programming sessions will take place at the end of Q3, likely in week 3.8.
Remarks	This is a very intensive course which requires you to work 18 hours per week over a period of 8 weeks regardless of Public Holidays. The course requires intensive teamwork and good time management. On average there is 1 mandatory session per week you must be available for which can vary over the weeks depending on the availability of robots and industry clients. Past experience has shown that combining this course with too many electives and/or other intensive activities during weekdays is incompatible. Be prepared you will likely spend 40h per week on this project and any electives combined. It is also not recommended to start with your literature research during MDP.
Department	ME Department Cognitive Robotics

RO47015	Applied Experimental Methods	5
Responsible Instructor	Dr. Y. Vardar	
Instructor	Prof.dr.ir. D.A. Abbink	
Contact Hours / Week x/x/x/x	0.0.0.x	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Study Goals	After completing this course, students will be able to: - o independently design, execute and analyse a human factors experiment - o formulate a clear hypothesis in the field of human factors that can be tested experimentally, in a realistic amount of time with a realistic amount of means - o set up a methodologically sound experiment to test this hypothesis - o select, adapt or build an experimental setup - o gather reliable experimental data on test subjects - o choose appropriate metrics - o apply correct statistical techniques - o choose the most relevant results, and construct the most clear figures to show them - o critically reflect on your own experimental findings - o communicate clearly through presentations and writing	
Education Method	Students will perform a mini-graduation project in six weeks, in small groups (2-3 students). They will independently design, perform, analyze and present a human factors experiment, guided by four supervisors.	
Literature and Study Materials	Scientific papers	
Assessment	The final grade is composed of the following sub grades 10% Presentation 1: Hypothesis 10% Presentation 2: Methods 10% Presentation 3: Analyzing Results (two main results figures, explain) 20% First Draft for Paper 50% Final Paper In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment can take place in a similar way.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Broad Design Group Dynamics/Project Organisation Group work Integrated Intensive Involved Lab Research Project planning / management Small groups	
Department	ME Department Cognitive Robotics	

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory	
Course Contents	Fundamentals of Coding in Python and MatLab Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Symbolic regression * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control We will aim for a balance of practical intuition and theoretical understanding: we will discuss examples of applications and go deeper into the theory when feasible.	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures (recorded and in class) and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: (i) practical assignment and (ii) written exam, open book. You must score at least 5.5 on both assignments to get a pass. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity to resit if they fail the written examination. * If your practical assignment is graded between 5 and 5.75, you will have two weeks to integrate the assessors feedback. If this is done satisfactorily, you can be rewarded with a mark that is not higher than 6/10. The exact instructions for the addition will be announced after the grading of the PA is concluded.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding course registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) exam registration is obligatory. More information regarding exam registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Artificial intelligence Complex Mathematics Intensive Linear Algebra Mathematics Modelling Programming Project Proofs	
Department	ME Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42030	Control for High Resolution Imaging	3
Responsible Instructor	Dr.ing. C.S. Smith	
Responsible Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	sc42065	
Expected prior knowledge	Basic knowledge of physics and control theory is expected. The first lectures recap in a condensed form the necessary knowledge in optics. Understanding of signal processing (in particular, the Fourier analysis) is expected for successful work on the homework assignments. The course involves some basic numeric simulations, so practical knowledge of Matlab/python/... is required.	
Course Contents	High-resolution imaging is crucial in scientific breakthroughs, such as discovering exoplanets in the habitable zone or understanding the origin and progress of diseases at a molecular level. For that purpose, special optical instruments like Extreme Large Telescopes or STED microscopy are developed. Optical aberrations are the key obstacle that hampers a clear vision and invites control engineers to step in. These are the disturbances induced by the medium, like turbulence in the case of astronomy, or by the specimen under investigation, like the change in refraction index due to inhomogeneities in the biological tissue. Adaptive Optics is a modern way to remove the effect of aberrations. This fascinating and expanding field in science is providing an excellent challenge to control engineers to improve image quality significantly by active control. This course will review the hardware necessary to control light waves in contemporary optical instruments, their modelling from a control engineering perspective, and discuss model-based control methodologies to do disturbance rejection.	
Study Goals	Understand the propagation of light, imaging and aberrations in the imaging process. Understand the operation principle of pupil-plane and focal-plane sensors to estimate the wavefront aberrations. Understand the design principles of opto-mechatronic wavefront corrector devices to correct wavefront aberrations. Develop spatial and temporal models of complete imaging systems and use these models in the design of model-based controllers for aberration correction.	
Education Method	Oral Presentations (seven lectures), homework assignments (4 in total), and self-study in the form of reading and presenting a scientific paper. Seven seminars (interactive discussions related to the homework assignments) are interleaved with the lectures, using the same time slots. Some homework assignments are allowed to be done in small groups; paper presentations are done in groups.	
Literature and Study Materials	Course Notes; selected chapters from published books and relevant scientific papers are provided as complementary reading material.	
Assessment	SC42030: (homework and paper presentation) 80% + oral examination 20%.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC4045	
Elective	Yes	
Tags	Broad Mathematics Optimalisation Physics Signals and Systems	
Department	ME Department Delft Center for Systems and Control	

SC42065	Adaptive Optics Design Project	3
Responsible Instructor	Dr.ing. C.S. Smith	
Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of signal/Fourier analysis, linear algebra, and optimisation is required. Basic skills in Python are required.	
Course Contents	The course consists of the realization of laboratory experiments to design and operate Adaptive Optics equipment to realize high-resolution imaging systems. Crucial in the design is the alignment of active optics systems and in the operation the development of algorithms for acquiring accurate information about the wavefront aberrations from intensity-based imaging components (like Shack-Hartmann sensors or CCD cameras) and using this wavefront information in the tuning of multivariable dynamic controllers to compensate in real-time the wavefront aberrations. The design is conducted under close supervision by world-leading experts in the field and is performed in groups of students. The size of the groups depends on the number of participants in this course. The course requires hands-on experiments, Python coding, and presenting the results in a report and/or a joint final presentation.	
Study Goals	Building insights about the key components in Adaptive Optics such as the wavefront reconstruction and the deformable mirror. As well as building the controller methodology to obtain a smart optics system for high-resolution imaging.	
Education Method	Project Based The work is subdivided into sessions. Each session results in a deliverable (e.g. some measurements or a piece of code). The work is performed in groups, which should be well balanced in, for instance, writing, coding, and theoretical skills.	
Computer Use	Lab computers are available, own laptops can be used	
Course Relations	Participation in SC42030 is required	
Practical Guide	A practical guide subdivided into sessions is provided.	
Assessment	Oral Presentation and evaluation of the written report in combination with the course sc42030	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Remarks	Old course code: SC4115	
Department	ME Department Delft Center for Systems and Control	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. R. Kortlever	

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced, analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	The final grade of the course consists of the following components: - Take home exam (50%) - oral exam (50%) Final grade calculation: $(0.5 * \text{take home exam}) + (0.5 * \text{oral exam})$ The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 take home exam: resubmit deliverable oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. H.M. Schuttelaars	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	After this course you are able to: - recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - solve the shortest path problem and the transportation problem as well as some small flow shop problems. - solve the travelling salesman problem by the Branch and Bound algorithm. - apply several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - reproduce basic theorems concerning the mentioned problems and is able to prove some of these theorems. - explain how to solve large scale problems, especially shortest path and vehicle routing problems. - define scheduling problems and complexity theory. - implement the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam and a written resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Elective	Yes	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr.ir. J.T. van Essen	

WI4156(TU)	Game theory	6
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Open to all MSc students with a strong command of mathematics. Ideally suited for math students with an interest in finance.	
Expected prior knowledge	Standard BSc math courses: calculus, linear algebra, probability.	
Course Contents	This is a course in "mathematical" game theory, which is the counterpart of "algorithmic" game theory (agent based modelling, this part of game theory is taught in the computer science department). We cover all subjects from combinatorial games to cooperative games.	
Study Goals	Understanding the mathematical theory of rational behaviour/conflicts/negotiations/social choice. Game theory is one of the most prominent economic theories: three recent Nobel prizes have been awarded to game theorists. We will cover a very wide area, ranging from pure mathematics to economics, with an emphasis on the mathematical aspects. This is a course in the Finance Track of the Applied Maths programme, but it is open to all MSc students (math and non-math).	
Education Method	Lectures	
Literature and Study Materials	Game Theory - Thomas Ferguson - UCLA: find it on the net.	
Books	T.S. Ferguson, A course in game theory, World Scientific, 2020	
Assessment	The final grade of the course consists of the following components: written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students are expected to have: superior knowledge and skill in elementary probability (details below); experience working with Python, Matlab, or R; and a mathematical inclination. If this does not apply to you and you still want to enroll: contact the instructor. Good skills in elementary probability are indispensable as they are the foundation for the more advanced material of the course. We build on the concepts, tools, and methods of elementary probability, and students taking this course must be (or rapidly become) black-belt masters in these foundations, specifically: Chapters 2-5, 7, 9, and 10 of Dekking et al; prior knowledge of Chapters 13 and 14 is helpful, though we will revisit these subjects during the course. Testing yourself and doing a thorough review in the first few weeks is required; we will also spend some time on this in class to help you do this efficiently; after this a diagnostic test (carrying part of the course grade) will assess your mastership; sub par performance means you cannot follow the course.	
Course Contents	This course on Monte Carlo simulation and stochastic processes consists of a mixture of these components: modelling and models (and some of their pluses and minuses), learning a number of stochastic processes and some of their properties, doing Monte Carlo simulations, analyzing the generated data and drawing conclusions from them. Contents and format of this course are intended to combine the following: 1. real (possibly stylized) problems as starting points; 2. enough about modelling to make participants aware of the impact of modelling decisions; 3. identification of a number of stochastic processes present in/related to models and problems; 4. (limited) treatment of underlying theory, guided by: (a) what is necessary to ensure that participants know how to do things right and (b) what enables them to do things smartly; 5. facts, properties, and theorems: (a) sincere attempt to understand where possible; (b) focus on practical implications and uses; (c) illustrations rather than proofs or theory; 6. useful tools and techniques from statistics, probability, and stochastic processes; 7. simulation as aid to: obtaining answers, intuitive understanding, illustration of theory; 8. providing as much opportunity as feasible for participants to work on/with problems from their own field. Tentative topics/contents list: Review basic probability: most to be done as self-study, some in class, plus a mandatory test (see Assessment). Models, modelling. Simulation: Convert a model to a program and run it. How to expose and locate errors. Data, empirical distribution; estimation. Accuracy, efficiency. Sensitivity. Rare event simulation. Experimental design. Conditional distributions and expectations; conditioning as a tool. Multivariate distributions: general (some examples); multivariate normal. Stochastic processes: Random walk. Poisson process; renewal processes. Markov chains: discrete time; continuous time; long-term behaviour; embedded Markov chain; birth and death process; Brownian motion. A snippet of diffusions, (solutions to) stochastic differential equations (SDEs).	
Study Goals	Learning objectives: after following the course, participants will be able to 1. in an iid setting, carry out Monte Carlo simulations; and 2. extract and collect, summarize, and interpret the data thus generated; and 3. assess accuracy/uncertainty in estimates, and solve questions like How many runs for 1% accuracy?; 4. the above (three) also in some non-iid settings, e.g., for a stationary process, using the batch-means method; 5. for small-sized practical problems: build a suitable model and simulate it; 6. extract and collect required model parameters from a problem description or background; 7. recognize where modelling choices are made; 8. make elementary modelling decisions themselves; 9. recognize and describe in mathematical terms (the properties of) a number of stochastic processes, among others: Markov chains, random walks, Brownian motion; 10. recognize/identify the covered stochastic processes in practical settings/problems; 11. use the tools and concepts from probability theory to describe, discuss, and manipulate (the components of) stochastic models; 12. where applicable, solve problems by means other than Monte Carlo, i.e., analytically or numerically, procuring exact answers, by identifying a stochastic process and using its known properties; for example, concerning long-term system-behaviour.	
Education Method	Weekly class meeting of two hours. About two thirds spent on lecture; about one third on questions and exercises. A number of so-called "Hands-on" assignments: a collection of exercises that combine the analytical content with computer simulation; to be done individually or with a partner. The participants are expected to perform a significant amount of simulation in Python, Matlab, or R (or other suitable language). Spread over the semester, five homework assignments. I strive to make a classroom available during the (probably) Friday lunch break, in which the participants can work on their own and where the TA (if there is one) is available for questions. In addition, I will hold a weekly drop-in office hour, most likely at lunch time (details on Brightspace).	
Literature and Study Materials	Unfortunately, for this course there is no really appropriate textbook, one that combines modelling, Monte Carlo simulation, and stochastic processes. For the review of basic probability and statistics, you are referred to Dekking et al, "A modern introduction to probability and statistics," Springer, 2005; the text is very well suited for individual review and practice, as it contains many worked examples, (quick) exercises, answers and full solutions. To be successful in the course, participants must have developed superior skill wrt the tools and techniques in the first half of the book (see "Expected prior knowledge"). Most of the material on stochastic processes can be found in "Probability Models" by Sheldon M. Ross; the book contains many worked examples and exercises; its heuristic and intuitive approach fits that of the course; references will be to the 10th Edition. A slightly more mathematical, but still very accessible reference covering the same area is "An introduction to stochastic modeling" by Howard M. Taylor and Samuel Karlin. Lecture slides, notes, and additional references will be posted on Brightspace.	

	A laptop is required and it is recommended that you bring it to class. Many languages and/or software packages are suitable: I prefer you use the one you are most familiar with. I am migrating to Python, but in the material presented in class and distributed via Brightspace, there still is some Matlab code. You may use a language/tool other than those three, but talk to me first (because you should be able to use your software on the exam and it should be possible to arrange this). In all cases: you should have the software installed and be familiar with its use.
Assessment	1. Probability test on knowledge and skills needed in/for the course (15%); passing score must be obtained to take the course. 2. Assignments (five; each counts for 4%), which may be done with a (=one) partner. 3. Final exam (on-campus): closed-book computer exam (with cheatsheet allowed, specs on Brightspace); counts for 65%. To pass the course, the probability test must be passed and the final exam grade must be 6 or higher; course grade is weighted average.
Permitted Materials during Tests	To the final exam you may bring a self-compiled formula sheet: one A4 page with normal-sized text on about 25 lines (or any reasonable equivalent of this), made by you, of things you would find helpful.
Elective	Yes
Tags	Adventurous Diverse Mathematics Modelling Practicals Stochastics
Co-Instructor	Dr.ir. G.N.J.C. Bierkens

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: * Bachelor level knowledge and ability in calculus, probability, statistics, optimization, linear algebra. * Programming skills in Python (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Recommended: * Linear Algebra and Optimization for Machine Learning (WI4635) * Statistical Inference (WI4455) * Knowledge of measure theory (as, for example, in AM2090)	
Course Contents	Statistical learning provides a probabilistic and statistical understanding of topics in machine learning. Note: this year the course will be part of the MasterMath programme. This may lead to some last-minute deviations from the information in the study guide. The overarching goal of the course is to develop methods for estimating (or 'learning') an unknown function from data or making predictions for unseen function outputs. The course aims to empower the student to make a justified decision in adopting machine learning approaches for practical problems and even design their own machine learning methodology using sound mathematical principles. This is achieved by studying key ideas, models, algorithms and theories related to the subject of statistical learning. In the first half of the course a collection of essential models, algorithms and techniques is introduced. In the second half of the course we adopt a Bayesian perspective on machine learning and study associated methods for analysing models and developing computational methodology.	
Study Goals	By the end of the course, you are able to: 1. Explain the challenges of statistical learning (e.g., curse of dimensionality, overfitting, bias-variance trade-off). 2. Implement machine learning algorithms and computational methods on artificial and real-world data sets (e.g., linear regression, logistic regression, sparse regression, neural networks, decision trees). 3. Compare the theoretical properties and practical aspects of machine learning approaches. 4. Validate machine learning approaches on arbitrary data sets. 5. Design tailor-made computational approaches for new learning problems by adapting, combining and/or extending methods and/or by employing a Bayesian approach.	
Education Method	Lectures, exercises, practical assignments, practice exam	
Literature and Study Materials	Lecture notes and slides will be made available. You may wish to consult additional references for background reading. Some suggestions of publicly available textbooks are (listed in order of suitability for this course): * Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/ (very explicit in terms of derivations; focusses on a Bayesian approach) * Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ (more advanced reference; requires some mathematical and statistical maturity since it shows relatively few derivations; focusses on a non-Bayesian approach) * Murphy, Kevin P (2022). Probabilistic machine learning: an introduction. MIT press, https://probml.github.io/pml-book/book1.html (very recent textbook on machine learning which discusses most topics; however not a very well thought out as textbook)	
Assessment	The final grade of the course consists of the following components: * Practical assignments during the course (30%) * Final written exam (70%) Final grade calculation: $(0.3 * \text{practical assignments} + 0.7 * \text{final written exam})$ Note: For both components, a sufficient grade (5.8) is required. Repair possibilities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5. * Assignments: resubmit deliverable * Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course material and literature are mostly self-contained, but prior knowledge equivalent to contents of the courses AM2010 Linear Algebra 2 and AM2020 Optimization from the Bachelor of Applied Mathematics programme is strongly recommended. Programming experience in Python or Matlab is expected.	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will discuss algorithmic foundations from the areas of linear algebra and optimization. This also includes aspects of their theory and efficient implementation. Among others, we will study + matrix decomposition and factorization techniques, + regression and classification problems and regularization, + iterative and continuous optimization methods, + hyper parameter optimization, + graph-based algorithms and clustering, + basic introduction to neural networks and algorithmic aspects neural networks (e.g. computational graphs and back-propagation)	
Study Goals	Upon completion of the course, you should be able to: + Categorize typical data sets based on their structure + Derive suitable mathematical formulations for common supervised and unsupervised machine learning tasks + Choose suitable algorithmic building blocks for solving mathematical formulations of common machine learning tasks + Analyze machine learning algorithms with respect to their properties, such as efficiency and robustness + Implement machine learning algorithms in a computationally efficient way	
Education Method	The course comprises 14 lectures and 14 exercise classes.	
Literature and Study Materials	Lecture notes, slides, and the main literature: + Charu Aggarwal. Linear algebra and optimization for machine learning. Springer, 2020. + Gilbert Strang. Linear algebra and learning from data. Wellesley-Cambridge Press Cambridge, 2019. + Jeremy Watt, Reza Borhani, and Aggelos K Katsaggelos. Machine learning refined: Foundations, algorithms, and applications. Cambridge University Press, 2020.	
Assessment	The final grade of the course consists of the following components: - Two group projects (0.4) - Written exam (0.6) Final grade calculation: $(0.4 * \text{group projects} + 0.6 * \text{written exam})$ Note: For the written exam, a score of at least 5.0 must be obtained. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group projects: resubmit deliverable - Written exam: There will be a resit for the written exam at a later point in the same academic year Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. D. de Laat	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	The final grade of the course consists of the following components: - Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). - Group programming project (group work) starting mid Q2 with submission deadline end Q2 (2/3 of final grade). Final grade calculation: $(1/3 * \text{homework assignments}) + (2/3 * \text{group programming project})$ but it is only valid when all homework programming assignments have been completed with sufficient results. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework programming assignments: submit new set of homework - Group programming project: submit improved version Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. J. Thies	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Spaceflight Dynamics	
Program Coordinator	Prof.dr.ir. P.N.A.M. Visser
Introduction 1	Profile II "Spaceflight Dynamics" This profile addresses many aspects of space missions: from concept to application, and from launch to end-of-life. It aims to deliver skilled space mission engineers and scientists. You will learn about techniques and methods to track, predict, optimize, and very precisely compute spacecraft trajectories, both for vehicles around and beyond Earth and for launch and (re-)entry vehicles. Space debris is another main research element. You will also learn about methods and tools to employ spacecraft for a wide range of scientific and societally relevant issues and applications, including monitoring all kinds of aspects that are related to Earths climate such as sea level and the ice caps. This profile is generally theoretical in nature.

AE4868-1	Numerical Astrodynamics	4
Responsible Instructor	D. Dirks	
Instructor	Ir. K.J. Cowan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4874 I (Astrodynamics I) is required. Basic programming knowledge is expected.	
Course Contents	Basic orbit propagation/integration methods and options; simulation options in Tudat. Defining input for orbit propagation Linking analytical and numerical design methods in astrodynamics Details of numerical integration methods: multi-stage, multi-step and extrapolation methods (fixed and variable step size) Selection of numerical integration methods; quantification of integration errors In principle, the course will only use Tudatpy (the Python interface for the Tudat C++ software). Depending on student interest, part of the C++ layer of Tudat may be covered.	
Study Goals	Set up and run a numerical orbit propagation using the Tudat software suite from a given set of simulation settings Critically evaluate and analyze numerical orbit propagation results Interpret numerical propagation results, and relate numerical results to theoretical predictions Select numerical integration scheme for a given astrodynamics problem	
Education Method	The course uses the following education methods: Weekly live lectures covering specifics of the assignments, examples, topic introduction, etc. The bulk of the material is covered as pre-recorded Lectures on Tudat, Dynamical Modelling, Analytical vs. Numerical Mission Design and Numerical integration Working Lectures are held twice per week, in weeks 1-8 of the quarter. The course requires significant self-study by the students, both to master the material, and complete the assignments. However, ample support is provided through Brightspace, forum, and the working lecture. Successfully doing the course within a reasonable time requires students to make good use of these options.	
Computer Use	Laptop computer with at least 4 GB of RAM	
Course Relations	AE4866 (Propagation and Optimization in Astrodynamics); AE4870A (Rocket Motion); AE4872 (Satellite Orbit Determination); AE4890-11 (Planetary Sciences I); AE4894 (Practical Astrodynamics)	
Assessment	Individual take-home assignments. Assignments will consist of setting up, running, analyzing, and documenting astrodynamics simulations in Tudat. Partially completed Python code will be provided for the assignments, which is to be completed by the students, and used to answer questions in the assignments. Students will use Python for generating results during the course, and the language/tool of their choice for visualizing results. It is assumed that students are sufficiently proficient in Python (or are willing to independently make themselves so) to finish and run partially finished Python scripts. It is assumed that students are proficient in a data visualization tool (e.g. Matlab, Python's matplotlib, etc.) The resit of the course will be done during the summer period. During the resit, there will be no opportunities to ask questions of any kind (installation, coding, assignment, theory, interpretation, etc.). It is not possible to resit single assignments, only a resit of the full course is offered.	
Remarks	In case of capacity issues, access to the course may be limited for students not enrolled in the Spaceflight track. In case you want to take this course, but are following a different track, please contact the course manager.	

AE4870A	Rocket Motion	3
Responsible Instructor	Ir. M.C. Naeije	
Instructor	Ir. W.J.F. Simons	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	Course Overview	
Study Goals	1. Introduction and recap previous courses on orbital mechanics 2. Fundamentals of rocket motion 3. Launch trajectories in a homogeneous gravity field, in vacuum and in an atmosphere (2D); vertical flight, constant pitch angle, gravity, and sounding rockets. 4. Theory of the multi-stage rocket; optimal mass distribution. 5. Ballistic flight over the Earth; 2D, and 3D over spherical (non-rotating and rotating Earth). 6. Launch systems design, unconventional launchers like air launch 7. Summary and question hour (workout exam question example) This course introduces the fundamentals of rocket motion by studying the motion of rockets under different circumstances. Emphasising on analytical solutions of the equations of motion, it will provide insight into the qualitative and quantitative aspects of launch trajectories of rockets in a homogeneous gravitational field, the performance of single and multi-stage rockets, and the exoatmospheric ballistic flight over the Earth. It discusses launcher design considerations and unconventional launch systems like air launch. At the end of the course you should be able to LO1: Evaluate the performance of existing space launchers that make use of single and multi-stage rockets. LO2: Analytically derive equations of rocket motion for specific launch and rocket applications in a homogeneous gravitational field. LO3: Simulate a range of rocket trajectory scenarios (e.g. in Matlab or python).	
Education Method	Lectures	
Literature and Study Materials	To be studied: Lecture slides AE4870A Rocket Motion Lecture notes AE4-870A Rocket Motion (selected chapters) Further reading: Turner, Martin J.L., Rocket and Spacecraft propulsion: principles, practice and new developments. 3rd Edition. ISBN 978-3-540-69202-7 (selected parts)	
Assessment	Closed-book written exam	

AE4870B	Re-Entry Systems	3
Responsible Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	IMPORTANT NOTICE: DUE TO THE LARGE INFLUX OF STUDENTS, THE COURSE IS ONLY OPEN TO STUDENTS FROM THE FOLLOWING PROFILES: SPACEFLIGHT DYNAMICS STRUCTURES FOR SPACE THIS CAP MAY BE LIFTED IN CASE CAPACITY IS AVAILABLE. ===== The course aims at giving a complete overview of re-entry systems, mainly from a mission point of view, but also from the system point of view. To begin with the mission, the fundamental principles of mechanics (Newton's Laws) are applied to derive the equations of motion for translational and rotational motion. Subsequently, simplifications are applied to give a description of planar ballistic, gliding and skipping flight, where the focus will be on analytical solutions for quantifying maximum mechanical and thermal loads. Next, typical mission applications, such as planetary entry and descent with parachutes and/or propulsion systems are discussed, as well as terminal-area flight of winged entry vehicles. Special topics (not covered each year): To extend the mission applications, attention is also paid to the atmospheric flight aspects during an aero-gravity assist, aerocapture and aerobraking. As many re-entry missions rely on the application of guidance, navigation and control systems, the fundamentals of each of these three sub-systems will be discussed and applied in simple examples, such that after the course the student has a starting point for further study. The relation between vehicle configurations (capsule, winged, lifting body, etc.) and mission aspects, such as flight time and (maximum) system loads is a typical system aspect that will pass review. Finally, all presented theory will be combined in a so-called development plan for building a simulator that can be used for performance analysis of both controlled and uncontrolled re-entry vehicles. How to use such a simulator efficiently is depending on what kind of simulation and analysis technique will be used. Therefore, the course is concluded with an overview of such techniques like Monte-Carlo Analysis, and how to extract the required information from the simulated data.	
Course Contents Continuation	Lecture Topics (the actual topics covered may vary per year): #01 Introduction #02 Entry Environment and Aeroheating #03 Fundamentals of Motion #04 Ballistic entry #05 Gliding Entry #06 Skip Entry #07 Guidance, Navigation and Control #08 Planetary Entry and Descent #09 Advanced Descent and Landing Systems #10 Terminal Area Energy Management	
Study Goals	At the end of this course, the student will be able to (depending on topics covered): 1. Identify the influence of the planetary environment on the motion of and loads on an entry vehicle 2. Derive the equations of planar motion, as well as key performance indicators 3. Understand the three unpowered entry mechanisms (ballistic, gliding and skip entry), and derive the corresponding equations of motion 4. Apply a sizing methodology to design an entry descent system (parachute and/or propulsion) 5. Understand the fundamental functions of a Guidance, Navigation and Control System and design a simple system 6. Name the characteristic phases of the Terminal Area flight, and derive the equations for maximum range and maximum dive (steady-state approximation) 7. Solve typical re-entry problems through a combination of physical insight, analytical skills, and numerical evaluation NOTE: particular emphasis is put on deriving the relevant equations, as this provides insight in the physics and underlying assumptions. These derivations are an integral part of the exam	
Education Method	Classical lectures (2 x 2 hrs per week). The course is structured in a number of blocks, which each deal with an aspect of re-entry missions and systems. Many real-life examples will be used to highlight certain theoretical aspects, and to show the applicability of the presented theory. Typical exam questions will be used to prepare the student for the exam by testing their insight in the presented material.	
Course Relations	AE3212-I - Aerospace Flight Dynamics and Simulation AE3212-II - Simulation, Verification and Simulation AE4870A - Rocket Motion	
Literature and Study Materials	1. Lecture Notes and Lecture Slides (Mandatory) The lecture notes are in the process of being published, so they may be replaced by the published book. To be announced through Brightspace and in class. Selected references for additional study: 1. Anderson, J. (1989). Hypersonic and High Temperature Gas Dynamics. McGraw-Hill. 2. Bertin, J. (1994). Hypersonic Aerothermodynamics. AIAA. 3. Cornelisse, J., Schoeyer, H., and Wakker, K.F. (1979). Rocket propulsion and spaceflight dynamics. Pitman, London. 4. Hankey, W. (1988). Re-Entry Aerodynamics. AIAA. 5. Mulder, J., van Staveren, W., van der Vaart, J., de Weerd, E., In 't Veld, A. and Mooij, E. (2013). Aerospace Flight Dynamics and Simulation. Delft University of Technology, Lecture Notes AE3-212-I. 6. Regan, F. and Anandakrishnan, S. (1993). Dynamics of Atmospheric Re-Entry. AIAA. 7. Vinh, N. (1981). Optimal trajectories in atmospheric flight. Elsevier. 8. Vinh, N., Busemann, A., and Culp, R. (1980). Hypersonic and planetary entry flight mechanics. The University of Michigan.	
Assessment	Closed-book, written examination.	

Note: in previous years the grade was composed of partial grades for homework and exam. Exam grades from previous years are not carried over to the current year

AE4872	Satellite Orbit Determination	6
Responsible Instructor	Dr.ir. E.J.O. Schrama	
Instructor	Prof.dr.ir. P.N.A.M. Visser	
Instructor	Dr.ir. J.A.A. van den IJssel	
Instructor	Dr.ir. B.C. Root	
Instructor	C. Siemes	
Responsible for assignments	C. Siemes	
Contact Hours / Week x/x/x/x	0/4/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	We assume that the participants are familiar with mechanics, analysis, linear algebra, and differential equations. We make use of MATLAB and the algebraic manipulation tool MAPLE, there are excellent primers for both tools. Some of the exercises assigned as student projects we expect that MATLAB and MAPLE will be used. If participants make use of other programming environments than no support can be guaranteed.	
Course Contents	Course contents 1 Dynamics - Introduction to dynamics - o Planetary Gravity field - o Tides and the three-body problem - o Hill radius and Roche limit - o Relation to planetary sciences and astrodynamics - Solving Equations of motion - o reformulate orbit problem as a system of ordinary differential equations - o efficiency and accuracy of numerical integration methods - o implementation of numerical integration methods 2 Observations techniques and reference systems - Observation techniques - o Laser, Doppler, optical and VLBI observations - o Refraction, Electromagnetism, radio- and optical technology - o Tropospheric and ionospheric refraction - o Relativity and the definition of time, - o Classification of time systems (UTC, TAI, etc) - o Light-time effect - o Quality of clocks (Allan Variance behaviour of clocks) - Reference systems - o Local and global coordinate systems - o Definition of geoid and reference ellipsoid, height systems - o Precession and nutation, polar motion, polar wander. - o Newton or Einstein, consequences for reference systems 3 Statistics - Random variables, probability density functions, moments, hypothesis testing - Least squares minimisation - o unconstrained linear parameter estimation, - o data weighting - o nonlinear parameter estimation. - Rank deficient equation systems - o Singular value decomposition and Empirical Orthogonal functions - o Penrose Moore inverse - o compatibility conditions - o general and homogeneous solutions - o constrained linear parameter estimation - Mechanisation of parameter estimation algorithms - o Choice of algorithms 4 Orbit determination - Perturbation analysis and variational problems - o state transition matrix for initial state vector problems - o partial derivatives for dynamical parameters - Parameter estimation - o Identification of parameters - o batch least squares - o Kalman filter, theory and implementation 5 Applications - Global Navigation Satellite Systems: - o Technology and terminology, - o various data processing strategies and available software - o Modelling deformation of the solid Earth, - o the Earths gravity field and thermospheric density - Satellite laser ranging and Doppler tracking via DORIS - o technology and terminology, results and applications - Observing changes in the cryosphere with satellites - Hydrology and Oceanography observed with satellites 6 Individual assignments in Q2 - An exercise related to dynamics, observation systems or reference systems - An exercise related to GNSS applied to orbit determination - An exercise related to Kalman filtering	

	7 Numerical Practical in Q3 Exercises with the Ghost software, typical examples are to solve: - o initial value problems (state vector estimation) - o problems with parameters in a dynamic model (drag parameter estimation) - o problems with time bias parameters
Study Goals	The candidate should be able to: 1) Explain the physical and mathematical aspects of orbit determination (OD), the topics are 1.1) Solar system dynamics 1.2) Equations of motion and variational equations 1.3) Parameter estimation 2) Construct transformations between various coordinate and time systems that play a role in OD 3) Examine error sources in satellite tracking data and implement error mitigation strategies 4) Make use of parameter estimation methods in the context of tracking data for OD 5) Apply relevant statistical techniques within the framework of OD 6) Discuss details relevant to scientific applications of satellite missions that depend on precise OD 6.1) GNSS techniques to model the deformation of the Earth 6.2) Satellite gravimetry to model the gravity field of the Earth 6.3) Satellite altimetry to model the ocean topography 6.4) Satellite altimeter missions to measure variations of land and sea ice 6.5) Combinations of 6.1 to 6.4 7) Apply OD with state-of-the-art software
Education Method	We provide lectures and classroom instructions during quarter 2. In quarter 3 there is a numerical practical to introduce the topics mentioned under the course contents on an MSc level for aerospace engineering students. In quarter 2 all lecture material will be made available, instruction hours will be organised to complement the material presented during the lectures, also, the instruction hours will be used to introduce individual assignments which are part of this course. In quarter 3 we run a numerical practical. During this practical software tools will be used to compute precise satellite orbits and which comes with the estimation of parameters with a state-of-the-art software package (e.g. GHOST or TUDAT). In addition a quality assessment of the results will be part of the practical. There are two variants of the numerical practical, before we start with Q3 participants will be able to make a choice between the variants.
Computer Use	Use will be made of Python or MATLAB (preferred) during the assignments, for the numerical practical use will be made of Linux and Fortran, or Python and Jupyter notebooks
Literature and Study Materials	Lecture notes will be provided on BrightSpace and they containing most of the material used in this course. Optional reference material: Statistical Orbit Determination, Schutz, Tapley and Born, Academic Press, ISBN: 9780126836301 Montenbruck and Gill, Satellite Orbits, Springer Verlag, doi: doi: 10.1007/978-3-642-58351-3 Planetary Sciences, Imke de Pater and Jack Lissauer, Cambridge University Press Additional information could be provided via BrightSpace
Assessment	The final grade will come together from: 1) Written examination resulting in a grade 2) Nominally two individual assignments resulting in a weighted average grade 3) Numerical practical resulting in "pass" or "fail". The rules are as follows: 1) The numerical practical result should be: "pass" 2) The written examination result should be at least a 5,0 3) In order to participate in the numerical practical you must have passed the individual assignments with at least a weighted average of 5,0 4) No claims can be made with regard to individual assignment results or numerical practical results obtained before the previous academic year. For ae4-872 in 2024-2025 it means that results from 2023-2024 are still accepted, however, we do not accept results from 2022-2023 or earlier academic years. (Academic years run from September to August)
Exam Hours	Conform the regulations there are two written exam opportunities within an academic year. These are three hour written exams at the end of quarter 2 and 3.
Permitted Materials during Tests	We provide a closed book examination, required information for the exam (like physical constants or a selected number of equations) can be made available on the examination form.
Tags	Analysis Elektricity & Magnetism Lineair Algebra Matlab Numeric Methods Physics Signals and Systems Technology

AE4874 I	Fundamentals of Astrodynamics	4
Responsible Instructor	Ir. K.J. Cowan	
Instructor	Dr. S. Gehly	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	ae4866 ae4868 ae4870 ae4872 ae4889 ae4894	
Summary	Thorough treatment of the fundamentals of astrodynamics; topics are treated in a mathematically cohesive way. You will acquire fundamental insights into the theory and methods of astrodynamics as well as working knowledge to compute and analyze spacecraft trajectories.	
Course Contents	Introduction to astrodynamics; two-body and many-body problems; relative motion; coordinates, reference frames, orbital elements, and time; geocentric, cislunar, and interplanetary flight; introduction to orbital perturbations.	
Study Goals	You will be able to: 1. Provide full and accurate descriptions of the fundamental concepts and phenomena of astrodynamics; 2. Explain the relationships between astrodynamics concepts & phenomena and the mathematics & physics theory that describes them; 3. Apply the relevant relationships, models, and methods for astrodynamics analysis; 4. Derive and manipulate the equations of motion for a spacecraft subject to gravitational and other forces; 5. Apply the equations of motion, their solutions, and related equations to analyze the absolute and relative positions and velocities of spacecraft as a function of time. Additional detail will be available in the course syllabus.	
Education Method	Lectures, workshops, self study	
Literature and Study Materials	Fundamentals of Astrodynamics, Karel Wakker, 2015. ISBN: 978-946-186-4192 available for download from the TU Delft Library / Institutional Repository also available to order as university reader in hardcopy Astrodynamics course website Orbital Mechanics for Engineering Students, Howard Curtis, 2013, 3rd edition. ISBN: 978-008-097-7478 (print) 978-008-097-7485 (ebook) available for download from the TU Delft Library also available to order from booksellers as hardcopy	
Assessment	Written exam	

AE4890-11	Planetary Exploration	4
Responsible Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	Dr.ir. W. van der Wal	
Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J.M. Potin	
Instructor	Dr. S. Paardekooper	
Instructor	Dr. S.J. de Vet	
Responsible for assignments	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4876 Planetary Sciences II	
Parts	Planetary Sciences: - Origin and Evolution of our Solar System - Planetary Surfaces and Interiors - Planetary Atmospheres - Planetary Magnetism - Planetary Ring Systems - Asteroids and Comets - Extrasolar Planets - Recent Topics in Planetary Science Current topics in planetary science with a focus on a specific topic each time the course is offered, such as Mars, Venus, Galilean satellites, comets, recent discoveries, spacecraft observations, or other developments.	
Course Contents	Planetary science is a major interdisciplinary field, combining aspects of astrophysics with geology, geophysics, meteorology, atmospheric and space science. This close relationship to geophysics, atmospheric and space sciences implies that the study of the planetary bodies in our Solar System and beyond offers the unique opportunity for comparison available to Earth scientists. This course teaches the concepts in the planetary physical sciences and solar system properties. The learning process is greatly enhanced by involving students in solving related problems.	
Study Goals	After the course the student should be able to: 1. Describe the primary physical processes in solar and planetary systems and the properties involved 2. Apply modelling techniques that are commonly used to describe the primary physical processes in solar and planetary systems 3. Demonstrate the use of physical principles to derive knowledge on the state and evolution of (extra-)solar system bodies from observations 4. Assess the quality of evidence of current knowledge of (extra-)solar system bodies 5. Assess the habitability of a (extra-)solar system body based on inferences of its properties and orbit	
Education Method	Classical oral lectures and classroom/take home assignments.	
Literature and Study Materials	Jack J. Lissauer and Imke de Pater, Fundamental Planetary Science, Cambridge University Press, ISBN: 9780521618557, 2013. PowerPoint/PDF files of the lectures and lecture notes.	
Books	Jack J. Lissauer & Imke de Pater, Fundamental Planetary Science, Cambridge Univ. Press, 2013.	
Assessment	The final grade for the planetary sciences course is determined from: - mark of a written exam - mark for mandatory assignments The mark for the written exam needs to be 5.0 or higher in all cases, no exceptions. If the mark for the exam is 5.0 or higher, grading for this course follows from averaging two parts with equal (so 0.5) weight: the mark for the classroom/take home exercises and the mark for the written exam. No opportunity will be given to redo the classroom/take home exercises within the same academic year. The grade for the classroom/homework assignments will remain valid for the next consecutive academic year, and so will annul the year thereafter	
Exam Hours	3 hours written exam	
Permitted Materials during Tests	Closed-Book exam, Non-programmable calculator is allowed.	
Set-up	Oral lectures and classroom/take home assignments.	

AE4894	Practical Astrodynamics	3
Responsible Instructor	J.G. De Teixeira da Encarnacao	
Instructor	D. Dirkx	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The topics covered in this course are chosen to expose you to the practical side of Astrodynamics. These topics are based on elementary algorithms, straightforward to implement numerically, make use of publicly available data, and are addressed by the students through weekly practical assignments. Any implementation is to be done from scratch, i.e., without resorting to existing black-box proprietary libraries or toolboxes. Each academic year (refer to Brightspace), a selection of topics will be made from: 1. Open satellite orbit data: Two-Line Elements (TLEs) and JPL Ephemerides 2. Applications and transformation of reference frames and element types 3. Numerical integration of trajectories 4. Orbit perturbations and incorporation into numerical orbit integration 5. Basics of numerical optimisation 6. Orbit Control	
Study Goals	By the end of this course, you will be able to: 1. Explain different algorithms that model astrodynamics. 2. Develop numerical algorithms that solve the practical assignments by applying the acquired knowledge without resorting to existing libraries or toolboxes. 3. Analyse the results obtained by the computer code you implemented. 4. Validate your results against alternative approaches. 5. Evaluate the strengths and weaknesses of the algorithms used.	
Education Method	During this course, you will a. recall knowledge and theory gained in this and other Astrodynamics-related courses, b. extend their understanding to a higher level of detail, c. acquire new skills through the hands-on application of this knowledge and theory to the development algorithms and the interpretation of their results, d. gain a different perspective on the applicability, usefulness, and scope of selected astrodynamics concepts. The learning material will be distributed through Brightspace. During class, we will largely focus on the theoretical foundations to prepare and guide you to handle the practical exercises effectively. Support will be given during lectures and open office hours to solve practical assignments at the conceptual level. It is up to you to handle the technical implementation details needed to solve each assignment. You are encouraged to solve the exercises using Python, but no specific programming language is enforced. You are free to choose the programming language you prefer. You must use git through TU Delfts GitLab server, where an account will be created for you. There is no other way to submit your assignments than using git. This course is focused on solving practical assignments individually from a blank slate. The libraries and toolboxes allowed to solve the practical exercises must not be related in any way to any of the topics discussed in the lectures. The course total study load is 3 ECTS x 28 hours/ECTS = 84 hours. With weekly, two-hour lectures for 6 weeks (12 hours total) and 1 hour for preparing for each lecture (6 hours total), you can expect to need 11 hours to solve each of the weekly assignments. This course relies heavily on your programming skills. The time it takes to solve the assignments will be much longer if you intend to develop your programming skills at the same time. You are not encouraged to follow more than 2 courses in the same quarter that are assessed with periodic assignments.	
Computer Use	You need to provide (or have access to) a personal computer (desktop or laptop).	
Books	- Python for Engineers: https://teachbooks.github.io/learn-python - Wakker, K.F. 2015. Fundamentals of Astrodynamics. TU Delft Library. http://resolver.tudelft.nl/uuid:3fc91471-8e47-4215-af43-718740e6694e - Iaco Veris, Alessandro de. 2018. Practical Astrodynamics. Springer Aerospace Technology. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-62220-0 - Vallado, David A, and Wayne D McClain. 2020. Fundamentals of Astrodynamics and Applications. (\$90: https://astrobooks.com/fundamentals-of-astrodynamics-and-applications-4th-edition-vallado-2013-softcover.aspx)	
Reader	For lecture notes AE4984, see the course content on Brightspace.	
Prerequisites	Fundamentals of Astrodynamics (AE4874-I)	
Assessment	There is no written exam. The final grade is the average of the weekly individual reports. The rubrics are available in Brightspace.	
Enrolment / Application	Enrollment in this course is exclusively through Brightspace and is limited to the first 50 students. Only students from the Spaceflight Dynamics profile may enrol. Please note: enrolled students are expected to submit all assignments. If this is not done, students will receive a penalty of 0.5 final grade points per missing assignment if they enrol in this course in the future.	

AE4896	Space instrumentation	4
Responsible Instructor	Dr. J.J.D. Loicq	
Contact Hours / Week x/x/x/x	4/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	The aim of this course is to provide students with an understanding of instrumentation and its role in observational strategies for missions. The course will explore the main observables of the universe and how they are measured in space. For each of the observables, the course will provide: i) The physical context of the observables and how they are generated, ii) The physics of the measure, iii) A review and explanation of the existing and under development space instruments, iv) The missions related to these observables. The course will be divided into three main parts: - The Fundamentals of Instrumentation - Remote sensing space Instrumentation - In-situ space Instrumentation.	
Study Goals	This course aims to familiarize students with the wide range of space instrumentation and observables. By the end of the course, students will be able to: - Understand and describe the relationships between instruments, missions, and observation strategies - Identify and explain the physical observables that space instruments measure, including light, ions, magnetic and electric fields, etc and related phenomena - Distinguish between the different types of space instruments, and explain how they are designed and constructed - Derive instrument requirements from scientific questions - Perform preliminary analysis and use the key driver parameters to size space instruments. Overall, this course will equip students with a comprehensive understanding of space instrumentation, enabling them to dive into space observables diversity and the complex processes involved in their measurement.	
Education Method	Lectures	
Assessment	The exam will test all theories provided in the lectures and skills practised in the provided exercises. It will be a closed-book written exam (3h). Formula sheet will be provided. The resit will have the same settings.	

AE4S12-23	Space systems engineering	4
Responsible Instructor	Prof.dr. E.K.A. Gill	
Contact Hours / Week x/x/x/x	4/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	General engineering knowledge at completed BSc level. AE1110, AE1111, AE2111, AE3211, or similar; AE3524 (for students, not having AE BSc)	
Parts	1. Introducing Space Systems Engineering 2. Designing Systems 3. Integrating Systems 4. Managing the Systems Engineering Process 5. Managing the technical effort 6. Managing interfaces and configuration; Conclusions	
Course Contents	The course covers Space System Engineering exemplified by the design of a spacecraft subsystem for a satellite mission of high societal impact. It introduces a process which allows the creation of successful systems, such as space applications, space missions and satellite subsystems. Methods and tools are presented and exercised which will improve the depth and breadth of Space Systems Engineering graduate level education at the TU Delft by emphasizing the need for the end-to-end approach and life cycle of space systems, including cost and risk handling and using state-of-the-art methods, such as elements of Model-based Systems Engineering. Upon completion, the student will have a firm understanding of advanced Systems Engineering and be able to apply adequate methods and tools which help to create successful systems.	
Study Goals	The course mission is to enable students to realize a successful space system in an end-to-end Systems Engineering approach. There are three high-level learning objectives: 1. Participants shall be able to explain the objectives and tasks of Systems Engineering for realizing successful systems together with their needs, potentials, benefits and limitations in a context which comprises Business Engineering and Management. 2. Participants shall be able to design an end-to-end Systems Engineering process demonstrating a smart balance of risks of cost, schedule, and performance. 3. Participants shall be able to use Systems Engineering methods and tools to design a high-level satellite subsystem, such as the Attitude and Orbit Control System (AOCS).	
Education Method	Lectures and tutorial Individual homeworks from lectures to practice material Group Assignment including peer-reviews to design a satellite sub-system, done by students in small groups The graded group report and graded individual homeworks comprise the formal course close-out.	
Literature and Study Materials	Required: Lecture Slides AE4-S12 Space Systems Engineering. Optional: Larson W. J.; Kirkpatrick D.; Sellers J. J.; Thomas L.D.; Verma D.: Applied Space Systems Engineering. McGraw Hill (2009). Optional: Space Mission Analysis and Design, 3rd edition (Space Technology Library, Vol. 8) Eds.: Wiley J. Larson, James R. Wertz (1999)	
Assessment	4 graded individual homeworks + 1 Group Assignment. Acceptance of 6 homeworks is required to pass the course. Passing both assessment elements is required to pass course.	
Enrolment / Application	Students have to enroll via Brightspace. Professionals are welcome to the course. Conditions for their participation and registration are available with the Management Assistants of the Department of Space Engineering.	
Set-up	The course comprises lectures, guest lectures, tutorial, group work, peer-review, and individual homeworks.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5822	Thesis Space	45
Responsible Instructor	Prof.dr. E.K.A. Gill	
Responsible Instructor	Prof.dr. L.L.A. Vermeersen	
Responsible Instructor	Prof.dr.ir. P.N.A.M. Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	e MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal).	

If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go' (fail). Please refer to Brightspace for more information.

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
	MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Space

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4143	Hypersonic Aerodynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Instructor	Dr.ir. L.M.G.F.M. Walpot	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* AE2130-III Aerodynamics 2 (Compressible Aerodynamics) or equivalent. In case you did not have such a course, please study chapters 7 to 12 of Fundamentals of Aerodynamics by J.D. Anderson * It is very highly recommended to also follow AE4120 (Viscous flows) * AE4140 Gasdynamics is nice to follow as well but not absolutely necessary	
Parts	Lecture and study material 1. Introduction: characteristics of hypersonic flow; applications. 2. Inviscid flow: 2D shocks and expansions; Newtonian theory. 3. Tangent- wedge/tangent cone-, shock-expansion theories; similarity parameters. 4. Euler equations; small perturbation theory; hypersonic equivalence principle; blast-wave theory. 5. Thin shock layer theory; computational results. 6. Viscous flow: boundary layer equations; laminar boundary layers. 7. Transition; turbulent boundary layers. 8. Viscous interactions; flat plate application. 9. Heat transfer: convective heating; results for flat plate, cone, blunt body. 10. Laminar and turbulent stagnation region heating; radiation: effects on surface heating; radiation from the shock layer; nocturnal radiation. 11. High temperature effects; chemical reactions; inviscid equilibrium flow; difference between equilibrium flow and frozen flow. 12. Non-equilibrium flow; high temperature effects in flow through normal and oblique shocks. 13. High temperature effects in flow through nozzles. 14. Chemical reacting viscous flow	
Course Contents	Hypersonic Flow: Inviscid flow, compressions and expansions, surface inclination methods, similarity rules, hypersonic small perturbations, Euler equations. Viscous flow, boundary layers, viscous interactions, heat transfer, radiation effects. High temperature effects, inviscid and viscous equilibrium and nonequilibrium flow. Rarefied gasdynamics.	
Study Goals	This course is designed to provide the student with fundamental knowledge of hypersonic aerothermodynamics. The course is set up to deliver engineering tools for conceptual design problems together with a physical understanding of various aspects of the broad gamma of hypersonic flow.	
Education Method	Lecture	
Literature and Study Materials	J.D. Anderson, High temperature gas dynamics, , - ISBN 0070016712 Recommended literature J.J. Bertin, Hypersonic aerothermodynamics, , - ISBN 1563470365	
Assessment	Oral exam	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4204	Knowledge Based Engineering	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ing. A. Heidebrecht	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some familiarity with Object Oriented modelling and (Python) programming	
Course Contents	The course will provide a theoretical and practical introduction to Knowledge Based Engineering (KBE) technology, as a means to address complex engineering problems through design automation/digitalization principles. Other addressed topics, strictly related to KBE include - UML/SysML modelling language (class and activity diagrams) - object oriented modelling paradigm	
Study Goals	The main goals are the following: - understanding of Knowledge Based Engineering theoretical foundations and programming principles and their application to support complex engineering design problems - hands-on experience in developing simple but representative KBE applications by means of a provided KBE system - ability to model product and process knowledge by means of UML/SysML	
Education Method	The course is organized into a set of interactive lectures and tutorials to be performed part in class and part at independently at home. During the tutorials, students are introduced to the KBE programming approach. After a quick refresh of the python language, the main functionalities of a commercial KBE system will be illustrated, through interactive programming sessions. A significant amount of time is left for self-study, where students will have to develop a KBE application in teams of two, to address an engineering design problem of their choice. Attendance to lecture and tutorials is not compulsory but very highly recommended. Although the course is spread over two periods, the second one (Q4) is reserved for self-study (i.e. the development of the KBE application) and the oral examinations (See assessment section)	
Computer Use	The use of a personal computer is required, with the Windows OS installed A selected KBE system will be used for the practical sessions and assignment (temporary licenses for this course will be provided)	
Course Relations	This course has a logical connection to the MDO course AE4205, being KBE an enabling technology to multidisciplinary design, analysis and optimization. However, the MDO course is not propaedeutic to this course, thus students who did not followed MDO can take the KBE course without any problem	
Literature and Study Materials	- lecture sheets and some lecture notes (only available digitally on Brightspace) - a few scientific papers - several programming tutorials and code samples	
Books	Suggested (not compulsory) book: Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 Please note that only Chapter 9 is required for this course (single chapters can be purchased on line at Wiley). The rest of the book is suggested literature for the MDO course AE4205 The book is available as e-book through the TU Delft library	
Assessment	There is no conventional written exam for this course. The assessment is based on the quality of a KBE app to be developed at home by teams of two students and demonstrated live during an oral examination session. Two examination sessions are planned per academic year. The KBE app is supposed to address a relevant engineering design problem, which will be proposed by the student teams in the early phase of the course. The proposed KBE assignment descriptions will require lecturer's approval. If during the oral examination the quality of the developed KBE application and its demonstration are evaluated to be unsatisfactory, the team will have to use the retake session. For the retake session, two options are offered: 1) submit and present an improved version of the app proposed for the nominal session. One penalty point applies. 2) submit and present a totally different app than proposed for the nominal session. No penalty points. The new KBE assignment proposal will have to be delivered after the nominal session and will require lecture's approval. The oral examinations (nominal and retake sessions) are taking place around the beginning and the end of the second period of the course (i.e. 4th quarter of the academic year), not necessarily during the standard exam weeks. Delivery dates and examinations days are announced at the beginning of the course	
Special Information	Temporary student licenses of ParaPy, the KBE system used in this course, will be provided only to the students actively participating to the course. Thus, not to guests or other participants enrolled via Brightspace. The licenses shall be used exclusively for the course assignment.	
Tags	Artificial intelligence Computer Engineering Group work Integrated Modelling Programming Programming concepts Programming Software Software Engineering	

AE4205	MDO for Aerospace Applications	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ir. M.F.M. Hoogreef	
Instructor	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Familiarity with MATLAB and basic aircraft design knowledge. MATLAB is used for tutorials and home assignment, thus students without MATLAB experience must gain this through self-study Students with no experience in aircraft design should consult textbooks like Torenbeek, Raymer and Roskam to familiarize with the fundamentals of wing sizing process and general concepts such as wing loading, aerodynamic coefficients, fuel fraction method, drag and its components, etc.	
Course Contents	The course will introduce students to multidisciplinary design optimization (MDO), as a modern methodology to support complex product development. Main topics: - What and why MDO - Parameterization techniques - Numerical optimisation methods (focus on gradient based methods) - Decomposition and coordination of complex computational systems - the eXtended Design Structure Matrix (XDSM) - MDO architectures - Intro to surrogate modeling, discrete&mixed integer problems, genetic algorithms and Multi Objective Optimisation	
Study Goals	The main goals are the following: - understanding of Multidisciplinary Design Optimization (MDO) methods and architectures - hands-on experience in setting up and solving MDO problems	
Education Method	The course consists of classical lectures and computer tutorials using own laptop. During the tutorials, students are introduced to some of the tools discussed during lecture and can make hands-on experience during interactive sessions with the teachers. In particular, students will be introduced to the MATLAB optimization function fmincon and its use for gradient based optimization. A significant amount of time is left for self-study, where students will have to work in pairs on one take-home assignment.	
Computer Use	The use of a personal computer is required MATLAB (including its optimization toolbox) will be used for the MDO practical sessions and assignment. Student licenses are provided by the TUD	
Literature and Study Materials	- lecture sheets and some notes (only available digitally on Brightspace) - scientific papers - some software modules - Suggested (not compulsory) book: - Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 The book is available in e-format from the TUD library (Chapter 9 is used in the course AE4204 on Knowledge Based Engineering) - Engineering Design Optimization, by J.R.R.A. Martins and A. Ning (free on line)	
Assessment	There is no conventional exam for this course. The assessment is based on the quality of one take-home assignment, which will be executed in teams of two students. The assignment is about setting up and solving an MDO problem using Matlab. The deliverables are the Matlab code and a report. Two different take-home assignments are defined for the two yearly examination sessions.	
Tags	Algorithmics Analysis Matlab Modelling Optimalisation Programming Project	

AE4238	Aero Engine Technology	4
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	Dr. F. Yin	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery", "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Inlets and Exhaust - o Turbomachinery - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls - o New Engine /propulsion Concepts	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an flipped course in which the lectures are pre-recorded and the students are supposed to watch the lectures before coming to the classroom. The classroom interaction is mainly to have discussions and for tutorials. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13. Evaluate the calorific values for various fuels 14. Understand the basics of combustion and emission formation in Aero engines 15. Understand the working of gas turbine combustor 16. Understand the basic principles of gas turbine controls and evaluate different strategies for control 17. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a flipped teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

Reader	1. Gas Turbine Theory by H.I.H.Saravanamuttoo, G.F.C. Rogers, and H. Cohen 2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons
Prerequisites	A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Assessment	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Permitted Materials during Tests	The assessment is based on 2-3 assignments given during the course and a final exam. All assignments together will form 30% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4302	Avionics and Operations	3
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Ir. T.J. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4316 AE4316P AE4321-15 AE4322	
Expected prior knowledge	Flight Dynamics System and Control Theory	
Parts	Week arrangement Lecture and study material 1. Introduction to avionics systems. -. Air Data systems (home study). 2. Gyroscopes, attitude reference systems. 3. Compasses, heading reference systems. 5. Flight deck instruments and integrated systems. 6. Navigation equations. 7. The Flight Management System (FMS). 8. Inertial Navigation Systems. 9. Radio navigation systems (ADF, VOR, DME). 10. Landing guidance systems (ILS, GPS). 11. Communication, Navigation, Surveillance (CNS). 12. Satellite navigation systems (GPS). 13. Air Traffic Management (ATM). 14. The Future Air Navigation System (FANS). (see week arrangement)	
Course Contents	This course provides a comprehensive, unified coverage of the principles of modern navigation equipment and systems, both in the aircraft and on the ground, including the aircraft instrumentation and flight-deck systems, with a special emphasis on the important trends in the global air navigation and air traffic management system.	
Study Goals	1. The student can describe in detail the working principles of the avionics systems treated in the course. 2. The student can demonstrate the avionics systems' functionalities, identify their strong points and weaknesses, and make comparisons between the avionics systems. 3. The student can evaluate, criticize, and appraise their usage in the current and future operational context.	
Education Method	Lecture	
Literature and Study Materials	Lecture slides and other lecture material are put on the BrightSpace The books mentioned below (Pallett & Coombs, Kayton & Fried) can be hired at the Control & Simulation division. Recommended literature: E.H.J. Pallett, L.F.E.Coombs, Aircraft Instruments and Integrated Navigation Systems, Longman, 1997 ISBN 0-582-08627-2. M. Kayton, W.R.Fried, Avionics Navigation Systems, Wiley, 1997 ISBN 0471547956.	
Assessment	Digital, closed-book exam on campus	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical singularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments. The assignments will be handed out near the end of Q3 and the hand-in deadline will be near the end of Q4. Exact dates will be announced during the lectures.	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Physical intelligence for interaction - Insect intelligence and Swarming - Spiking neural networks	
Study Goals	After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.	
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4866-1	Propagation and Optimisation in Astrodynamics	5
Responsible Instructor	D. Dirksen	
Instructor	Dr. S. Gehly	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4874 I (Astrodynamics I) is required; AE4868-1 (Numerical Astrodynamics) is required; Basic programming knowledge is expected.	
Course Contents	1. Introduction. Explanation of course assignment. Introduction to software. 2. Numerical propagation. Formulations of equations of motion. Selection of propagation method. 3. Overview of formulation of acceleration models. Relation between acceleration models and environment models. Influence of acceleration model selection on simulation fidelity/runtime. 4. Introduction to local optimization methods. Constraints in optimization problems. Robustness of optimal solutions. Optimal control. Examples of local optimization in astrodynamics. 5. Global optimization methods (genetic algorithms, particle swarm, etc.). Multi-objective optimization. Examples of global optimization in astrodynamics 6. Pre/post-processing of results. Analysis of numerical propagation/optimization results. Use of numerical tools (e.g. interpolation).	
Study Goals	1. Select and define the settings of a numerical astrodynamics simulation problem, based on top-level description and requirements. 2. Describe the relation between models of the physical environment and the propagated dynamical behaviour. 3. Define and set up local and global optimization problems for typical astrodynamics problems 4. Assess the quality of numerical orbit propagation for typical astrodynamics problems 5. Assess the quality of optimization results for typical astrodynamics problems	
Education Method	The course uses the following education methods: Weekly live lectures covering specifics of the assignments, examples, topic introduction, etc. The bulk of the material is covered as pre-recorded lectures on propagators, environment, acceleration models, design space exploration, optimization Working Lectures are held twice per week, in weeks 1-10 of the quarter. The course requires significant self-study by the students, both to master the material, and complete the assignments. However, ample support is provided through Brightspace, forum, and the working lecture. Successfully doing the course within a reasonable time requires students to make good use of these options.	
Computer Use	Laptop computer with minimum 4 GB RAM	
Course Relations	AE4870A (Rocket Motion); AE4870B (Re-Entry Systems); AE4872 (Satellite Orbit Determination); AE4890-11 (Planetary Sciences I); AE4894 (Practical Astrodynamics)	
Literature and Study Materials	Selected articles; lecture notes; lecture slides (details to be announced and distributed during the course).	
Assessment	The assessment is a combination of two individual and one group take-home assignments. The individual assignments are thematically related to the group assignments, and are used as a starting point for the group work . Students will use Python for generating results during the course, and the language/tool of their choice for visualizing results. It is assumed that students are sufficiently proficient in Python (or are willing to independently make themselves so) to finish and run partially finished Python scripts. It is assumed that students are proficient in a data visualization tool (e.g. Matlab, Python's matplotlib, etc.) For the group work, individual contributions to the group work must be clearly identifiable. The total grade is composed of: Individual assignments (individual grade): 50% Project (individual grade): 25% Project (group grade): 25% Each of these three grades must be ≥ 5.0 to pass the course. The final grade for the course must be ≥ 6 to pass the course. In the assignments, students will apply the material covered in the course to the analysis of a real astrodynamics problem. Weekly practical sessions will be held to discuss questions and issues with homework, the practical assignment and the numerical implementation. The resit of the course will be done during the summer period and/or Q1 of the following academic year. During the resit, there will be no opportunities to ask questions of any kind (installation, coding, assignment, theory, interpretation, etc.). It is not possible to resit single assignments, only a resit of the full course is offered.	
Remarks	**NOTE:** This course will have limited capacity (maximum ~25-30 students). The procedure on student admittance (if the maximum is reached) will be communicated to the students at the beginning of the academic year through Brightspace. In case of oversubscription, preference will be given to students from the Spaceflight track. To take part in the course, students must submit their preferences for an assignment topic. These must be submitted through Brightspace on the first Wednesday of the course's first lecture week, at the latest. Only enrolling for the course on Brightspace is NOT sufficient to indicate you wish to take part in the course. Instructions for submitting a topic preference will be posted on Brightspace one week before the start of the course, at the latest. Not submitting your topic preference in a timely manner means that you will NOT be able to take part in the course. For this course, it is expected that students are on campus for the full duration of this course (weeks 1-8 of the quarter in which it is taught). Being away for one week or more may result in exclusion from the course. In case of doubts or concerns, please contact the course manager. Working lectures are held during week 1-8 of the period, where students are expected to be present. It is required that students have followed the course AE4868 (Numerical Astrodynamics), and have submitted all assignments, and can still reasonably pass that course, before taking part in this course.	

AE4889	Special Topics in Astrodynamics	2
Responsible Instructor	Dr.ir. M.J. Heiligers	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	A modular and research-focussed course on advanced topics in astrodynamics. You will extend and deepen your insights and analytical abilities based on Fundamentals of Astrodynamics and related coursework, devoting particular attention to their application to a selection of advanced astrodynamics topics (see below).	
Course Contents	The course treats a selected set of advanced topics in astrodynamics; the exact topics change from year to year depending on the latest research developments and the instructors involved. Strictly as examples, the following topics were covered in previous years: - Circular restricted three-body problem (equilibrium points, their stability and manifolds, and periodic orbits around the equilibrium points, their stability and manifolds) - Solar-sail orbital dynamics (acceleration models and artificial equilibrium points in the circular restricted three-body problem) - Machine learning applied to astrodynamics	
Study Goals	Within the context of the selected astrodynamics topics: - Extend your insight and analytical abilities beyond the Fundamentals of Astrodynamics and related coursework; - Actively apply your knowledge and skills accumulated through Fundamentals of Astrodynamics and related coursework; - Tackle "open-ended assignments with a hands-on, scientific approach as a first foray into astrodynamics research.	
Education Method	Workshops, lectures, self-study	
Computer Use	Yes	
Prerequisites	Fundamentals of Astrodynamics (AE4874 I)	
Assessment	Assignments	
	Additional detail will be available on the course website (Brightspace)	

AE4895-24	Measurement Strategies for Planetary Science Missions	4
Responsible Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	For planetary science the research question has a profound impact on the measurement strategy and instrument to be used. Studying planetary objects and materials in the Solar System thus hinges on observing and obtaining measurements from either nearby (in contact) and from afar (remote sensing). In this course you will follow lectures, learn more about planetary analogues and gain hands-on experience as you explore different measurement techniques in the lab. By applying measurement techniques to a specific rocky or ice sample, and to a planetary analogue material, you will try to find out how the data products and the scale at which it is applied influences the research questions that a science mission can address.	
Study Goals	After completion of the course, the student will have obtained practical experience and gained a deeper understanding of how measurement techniques contribute to characterizing a physical-chemical property of a planetary object. After this course the student will be able to: LO1: Explain key scientific questions that space exploration seeks to answer for rocky (Mars) and icy (icy moons) planetary bodies. LO2: Understand the underlying physical principles of a measurement technique. LO3: Demonstrate to have working knowledge of measurement techniques by participating in a structured laboratory exercise. LO4: Evaluate how measurement techniques result in an improved understanding of materials and processes in planetary sciences.	
Education Method	Lectures; structured laboratory exercises focussing on either a rocky or an icy planetary analogue material.	
Assessment	Laboratory activities and maintaining a lab journal (20%) Presentation lab results and planetary integration (20%) Final report, in form of a short communication in paper format (60%)	
Enrolment / Application	The course has a limited capacity of 20 students, which will be allocated to students from the profile planetary exploration first, and then on the basis of first come, first serve (potentially filled up with students from other profiles and/or programs). This limit of 20 could be higher depending on the actual capacity of the laboratory facilities	

AE4897	Space Engineering Practical	8
Responsible Instructor	I. Akay	
Responsible Instructor	Ir. B.T.C. Zandbergen	
Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr. J. Guo	
Instructor	Dr.ir. R. Saathof	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	M.S. Uludag	
Instructor	Dr. B.V.S. Jyoti	
Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you will be taught on and experience the basic steps needed to successfully conduct experiments/simulations to verify/characterize a product/system as part of a real space engineering research project brought forward by industry and/or academia. Everyone who ever designed an experiment or conducted simulations to verify product/system requirements has experienced issues with improperly formulated requirement on one hand and the preparation and execution of simulations and experiments on the other. Typical problems that occur are inadequate planning (not planning in advance, lack of time), system/object not being in agreement with the design specifications, improper test set-up, instrument failures, missing data, convergence issues, bugs, extremely long calculation times, and improper requirements verification. Learning about the basic steps will allow the students to conduct such projects successfully in the future. The course is made up of two complementary parts, which are described in more detail below. Part 1 provides in-class instructions on how to perform specific tasks needed for conducting a proper test/verification project. The subjects dealt with in class (lectures/workshops) include (not necessarily in the order indicated): 1. Why this course, course organization, project phases, timeline, deliverables 2. Generating requirements including also test requirements, like test ranges and test tolerances, 3. Hypothesis formulation and assessment, 4. Design of Experiments, Experimental Measurements, and Error Propagation, 5. Test planning and the development of the test set-up and the accompanying Assembly, Test and Safety Procedures. 6. Data gathering/acquisition (both qualitative and quantitative) & automation 7. Results analysis 8. Other, like Safety, Types of Testing, Use of Standards Part 2 focuses on teams stepping through the project cycle of a experimentally oriented project assigned to them. The steps include: - Orientation (on project assignment, object/system to be researched/tested/verified under what conditions, goals) - Estimation of the performances of the test object by design analysis/simulations - Development of test hypothesis, and test requirements and criteria - Generation of test specification with success criteria - Generating a test plan - Assembly/build of test object (may be a mock-up in case of use of hazardous materials) - Assembly of (mock-up) test set up. early testing (including calibrations) and data elaboration to develop/check procedures - Test readiness review (if review is passed go on to next step, if not revise test plan) - Test execution - Data elaboration (observations and experimental data) - Discussion of results (data review) - Communicating the results/outcomes + project close-out For each project, project supervisors may arrange for more specific training/instructions related to the specific test set-up to be used. Project supervisors are also responsible for students conducting safe testing.	
Study Goals	Upon completion of the project the student shall be able to: 1. Describe and motivate the project cycle of an experimental project. 2. Formulate a test specification and test plan. 3. Set up and assemble/develop an appropriate test set-up and test procedures. 4. Perform data acquisition (both qualitative and quantitative data) with some degree of automation. 5. Execute measurements in a reproducible way and determine measurement uncertainties. 6. Analyze experimental data taking into account measurement uncertainties. 7. Discuss the outcomes of the project including the statistical validity of the data. 8. Report on the outcomes of an experimental project.	
Education Method	Lectures/workshops and Project The course is a practical lab project combined with lectures/workshops. The lectures/workshops are given by a variety of instructors. All instructors are expected to link the material they are teaching to the various projects. Instructors shall also strive to let students experience the work through a number of workshops/assignments that are geared to the projects the various teams are conducting. During the projects, instructors can be approached in case teams need help on elements instructed upon. Most of the instructions are scheduled in Q2, but to allow for proper alignment with the projects, some may move to Q3. The size of this part of the course is 2 EC (roughly 56 working hours for each participating student). Each project on offer is of size 6 EC and is conducted by a team of 5-6 students. All teams are assigned a staff member, with expertise in the field, as project supervisor. The role of the project supervisors is to provide intellectual guidance and offer advice on the planning and progress of the work. Project supervisors also are to check and review the progress made by the team and determine if a team can move on to the next phase in the project, thereby ensuring project progress and that test preparation activities and testing are conducted in a safe way. To streamline this, two formative assessments will be performed, which help students identify their strengths and weaknesses and target areas that need work. The first such assessment relates to the project	

specification and the second one is prior to getting approval to conduct full scale testing. Additionally, during their project, students have an open invitation to consult the lecturers/instructors of the lecture/workshops part of the course for advise/feedback on issues taught in class or exercised upon in the workshops that are relevant to their project. Projects can be conducted in the Q2 to Q4 time frame of the academic year. The exact time frame is subject to negotiation between the team and the respective project supervisor. For instance, teams can propose running the project in a single quarter or spread over multiple quarters.

Literature and Study Materials

Appropriate literature and study materials will be provided throughout the course.

For the lectures/workshops, material is provided for by the instructors relating to the topic/subject they instruct upon.

For the specific projects, a project assignment will be made available that describes the test object/system to a great detail, including the relevant characteristics of the object/system. The project assignment will also describe the test goals as well as the type of testing to be performed (functional testing, life testing, reliability testing, failure testing, etc.) and the test conditions under which the testing is to be performed.

Assessment

The result of this course will be entirely based upon the results of the experimentally oriented project and the associated deliverables. These deliverables include:

- Test specification
- Test plan
- Results Summary document, test data files, and a visual demonstration (video/photos/poster) of the testing conducted and the equipment used.

For each deliverable a template/content description (or example) will be provided.

Assessment will take place according to the detailed rubrics available on Brightspace. Assessment is by the project supervisor of the project (40% of the final grade) and the course instructors involved in the instructions (60% of the final grade). Assessment is coordinated by the course responsible instructor. A minimum grade of 6 is needed after rounding to the nearest half to pass the project.

Late submission of the work results in a maximum penalty of 1.5 point (on a 10-scale). Due dates for submission will be communicated in due time via Brightspace.

In case of an insufficient grade, teams are allowed one opportunity to resubmit an improved version of their report.

Enrolment / Application

This course is only open for enrollment for students enrolled in the Space Engineering profile of the track Space.

A maximum of 45 students will be accommodated on a first-come, first-served basis. Students will have the opportunity to choose from a list of available projects that encompass different knowledge fields in the field of Space Engineering. Some past projects include:

- Battery project: Novel capacitor-based spacecraft energy storage system
- Propulsion project: Solid rocket motor assembly and testing (in cooperation with DARE)
- Attitude control project: Assembly and testing of a PocketQube attitude determination and control system
- Payload controller project: Assembly and testing of a small sat onboard data handling system

AE4898	Space debris tracking and mitigation	4
Responsible Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	AE4874 I (Fundamentals of Astrodynamics) is required. AE4868-1 (Numerical Astrodynamics), AE4894 (Practical Astrodynamics) and AE4870B (Re-Entry Systems) are recommended.	
Course Contents	The course will introduce the problem of space debris and associated aspects, such as how to detect it, how to avoid it, and potentially how to remove it. The contents of the course will be presented in a hybrid format, i.e., a combination of classical lectures and working lectures, where the students can practise with the introduced material, tools and techniques. The course content is roughly split into two parts. The first half of the course will introduce the problem of space debris: how much is out there, how large and heavy is it, where does it come from, and what kind of problems does it pose. Having identified the problems for the safe operation of space, the focus shifts to how we can predict potential collisions and how we can avoid those. Finally, how can the larger chunks of space debris be removed (Active Debris Removal), and if the debris enters the atmosphere and does not fully burn up: where will it impact on earth. The second part focuses on the observation and estimation of space objects in general, and space debris in particular. The students will learn about different sensors that can be used, either ground-based or space-based, the accuracy of measurements, and how this will affect the estimation process. Linear and nonlinear estimation techniques will be discussed, with application to determine translational state vectors, characterise physical parameters, and detect manoeuvres of space objects. The course will further consider methods of sensor management and data fusion.	
Study Goals	After the course the student shall: 1. Assess the risks posed by orbital debris on current and future space missions, including the probability and consequences of on-orbit collisions 2. Analyze the mitigation of space debris hazards through the use of post-mission disposal and active debris removal 3. Estimate the orbits and physical characteristics of resident space objects using simulated space object tracking data, including measurements from optical and/or radar sensors 4. Design a collision detection and avoidance strategy for use with a simulated space object catalog	
Education Method	Per week, two lectures are scheduled. One is a regular ("classical") lecture where topics related to the field are introduced, clarified with examples. The second lecture is a working lecture, where tools are introduced by means of tutorials and where students can begin on their assignments/lab work.	
Computer Use	Personal laptop that can run Tudat. The choice of programming language is free (Python knowledge is required to work with Tudat). No coding support is provided.	
Literature and Study Materials	The main course material consists of the lecture slides. Recommended papers and books will be suggested during the course.	
Assessment	The assessment is a combination of one individual and two group take-home assignments. A tutorial assignment (not graded) will be handed out at the beginning of the course. For the group work, individual contributions to the group work must be clearly identifiable. The total grade is composed of: Individual assignments (individual grade): 30% Project (individual grade): 30% Project (group grade): 40% Each of these three grades must be ≥ 5.0 to pass the course (final grade is the weighted average, which should be 5.75 or higher). The resit of the course will be done based on individual assignments. Support during this will be limited. It is not possible to resit single assignments, only a resit of the full course is offered.	
Enrolment / Application	The course is open to students of the Spaceflight Dynamics profile only; the maximum number of students that can participate is limited to a maximum of 30. To enroll, the student should write a motivation (limited to 5 lines) why this course is important/relevant to them, and send this by email to both instructors: e.mooij@tudelft.nl (Dr Erwin Mooij) and s.gehly@tudelft.nl (Dr. Steve Gehly). Enrollment closes 2 weeks before the start of the course, and students will be notified about the status of their enrollment before the course starts.	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods.	
Study Goals	Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Education Method	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Assessment	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments (only assignment 3 and 4 since they have the highest weight; assignment 1 and 2 cannot be redone). Results for individual assignments are only valid for one academic year: when re-taking the course all assignments have to be done in that academic year.	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	P.P. Sundaramoorthy	
Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of lectures, working sessions, design review sessions and a final symposium. The course structure will be explained in detail in the first weekly module. The time and date of the design review(s) and final symposium will also be communicated in the first weekly module. Students are requested to reserve all scheduled timeslots in their agendas. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. Note that, the course will require group work. Groups will be formed in the fourth week of the course. Only students who pass the individual test are allowed to participate in a group. Students who are not part of a group cannot participate in the remainder of the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. During each module it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Individual test on course content 2. Written feedback on design specification sheet of another group in the course Students with a failing grade for the individual test, i.e. a grade below 5.8, will have a resit opportunity of the individual test in week 4 of the course. This resit will be in the form of a short oral exam. This resit is only available for students who did not pass the individual test. On passing the oral resit exam, students will receive a 5.8 as grade for the individual test. Students who fail the resit opportunity will not be able to participate in the group work, and therefore cannot participate in the remainder of the course. It is mandatory that students are present at the design review(s) and the final symposium presentations of their own group and those of other groups. The individual grade of students who fail to be present will be withheld. These students will not receive a final grade for the course as a result. Students of whom the individual grade which has been withheld, may take a resit. The resit will be in the form of a short oral exam. It will be possible to apply for a resit up to 15 working days after the publication of the grades in Osiris, but no later. A resit exam will then be scheduled by the principle lecturer of the course in consultation with the student in question. Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, Introduction to materials science" or books of similar nature. Be sure to check the entry level quiz before making up your mind about this course.	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, ISS, Hubble telescope, James Webb telescope, Venus orbiters, Launchers) will be presented as a carrier for material selection analysis. A holistic view of "materials science - manufacturing - structural analysis - system engineering for missions" will be presented. While students come mostly from different backgrounds, the goal of this class is to help them identify the perspective they use in the holistic view and to learn about the materials science perspective. Fundamentals of materials classes and their typical properties will be introduced. Current material choice will be analyzed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. In sessions of "ESA forum + student poster seminar", students will be given opportunities to interact with ESA experts and present their own findings from analyzing given missions, by translating desired properties into material structures and microstructures. In class discussion is an important part of learning. The structure of the lectures will be tailored to maximize the student involvement. The course contains the following topics: week 1: Holistic view and materials science perspective; Typical materials - metals, ceramics and polymers week 2: Typical materials - metals, ceramics and polymers week 3: Material degradation in space - vacuum; thermal cycling; week 4: Material degradation in space - AO; charging week 5: Material degradation in space - space radiation week 6: Material degradation in space - meteoroids and orbital debris; week 7: "ESA forum + student poster seminar" Materials testing for space applications; Extreme missions; Materials challenge in future space age (ESA guest speakers); Student poster seminar (Jan 6, 2025) week 8: Study period (Q and A) week 9: exam week (Jan 20, 2025)	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 12 lectures of 2h per week for the first 6 weeks. The last week is a whole-morning session of "ESA guest speaker forum + student poster presentation".	
Literature and Study Materials	A compilation of book chapters, lecture notes and review papers.	
Assessment	Poster presentation (40%) + final exam (60%). The students are to work in groups and do research on prescribed question/topics and deliver a final poster presentation (15% individual score, 25% group score) in week 7 to show their discoveries throughout the course. The final grade will be the weighted average of poster presentation and final exam. Example questions/topics will be made available on Brightspace.	
Enrolment / Application	For students not from "structures for space" profile, there will be a cap of 20 students maximum. Please enroll in brightspace (enrollment period Oct 28 to Nov 10, 2024) as soon as you decide to take the course. Once the maximal number is reached, enrollment will be closed.	

AE4ASM526	Spacecraft Thermal Design	3
Responsible Instructor	I. Uriol Balbin	
Instructor	C. Falsetti	
Instructor	N. van der Pas	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of orbital mechanics Understanding of general design spacecraft systems	
Course Contents	General knowledge of space thermal control hardware, system design and analysis processes are presented. Using the principles of heat transfer, and more specific of radiation and conduction, the methods and tools for thermal design of a spacecraft are presented. Numerical analysis will also be introduced via the software ESATAN-TMS.	
Study Goals	The student is able to generate a thermal design of a satellite using heat transfer analysis and thermal engineering techniques. In more detail the student will be able to: - Understand the spacecraft thermal environment - Define the thermal subsystem and compare the different hardware options. - Understand the design cycle of the thermal subsystem. - Apply the concepts of heat transfer (radiation, conduction), heat balance and heat flows - Generate and use analytical thermal analysis models - Generate and use numerical thermal analysis models	
Education Method	Lectures + homework Students are given access to a discussion board provided with the course in Brightspace. The discussion board will be used as the main communication tool to ask questions on the material taught or in preparation of an assignment. Answers to the questions may be given by fellow students and/or the lecturer(s).	
Computer Use	Software licenses are available during the lecture period Software can be used in the computer room or in the personal computer (Windows/Linux)	
Literature and Study Materials	Lecture notes available on Brightspace.	
Assessment	The final grade of this course is based on 2 homework assignments (60%) and one final exam (40%). A minimum grade of 5 is required in each part.	
Enrolment / Application	This class has a limit of 80 students, due to the limit on the software licenses. Please enroll in Brightspace before 4th of March 2025 if you intend to take this class.	

AE4S52-21	Collaborative Space System Design Project	5
Responsible Instructor	Ir. B.T.C. Zandbergen	
Instructor	Dr. S.M. Cazaux	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The Collaborative Space (System) Design Project (CSDP) is a TU Delft/Faculty of Aerospace Engineering (AE) Master course in engineering design. The course focuses on the conceptualization and preliminary design phase of a space mission, spacecraft, or space instrument and starts on a design challenge co-created by the CSDP organization team and a company, research group or other entity. All projects on offer are multi-disciplinary projects, i.e. projects that require different types of design work and knowledge to be combined to provide a design solution. All projects are carried out by a team of students that have to organize and manage themselves. Each project is to be organized in 3 phases similar to NASA's System Design Process as defined in a.o. "NASA's System Engineering Handbook". Some adaptations have been made though to make it fit in the limited course time. The phases are: 1) Exploration phase; In this phase the team is to explore the problem, the needs, the competition and past missions and to develop a proposal for the next phase of the project. This includes the identification of a range of high-potential concepts for study in phase 2, the work distribution, etc. 2) Concepts design studies phase; In phase 2, the high-potential concepts defined in phase 1 are analyzed in detail for feasibility, traded and a best concept is selected. Additionally the plan for the next phase is to be generated. 3) Detailed design phase. In this final phase of the project the single concept selected in phase 2 is worked out in detail and a plan is developed for the further development of the design. Each project phase ends with a review (feedback moment) wherein other teams and expert staff reflect on the outcomes generated and the engineering design methods used. During the quarter, workshops/instructions will be held to provide knowledge and training on selected management and engineering design topics, including agile management, and the use of integrated design modelling, i.e. the integration of all the geometry, configuration, analysis, and requirements verification into a generative, parametric, unified computational model where data is shared seamlessly between the different disciplines. The course does not focus on teaching the required disciplinary knowledge and experience, but rather focuses on decision making, the collaborative integration of the knowledge and experience available in the team, and the iterative design method using different levels of model fidelity to create a feasible design solution in answer to the problem identified by the "customer". As projects vary from year to year and may encompass knowledge that is not available in the team, this may require that participants actively acquire the knowledge required.	
Study Goals	The course aims to develop student skills in multi-disciplinary team projects from a challenge-driven perspective. In more detail, students will advance their ability in ...: - ... disciplinary design including modelling, simulation, visualization, quantitative analysis of alternatives, design tool verification, calibration and validation, and design refinement. - ... the process of engineering design (ABET definition), including the steps in design, the (iterative) nature of the process, development of a Straw Man design, and development of process models. - ... multidisciplinary design, thereby taking into account differences between the different disciplines involved in terms of a.o. differences in fidelity level of disciplinary models, and dissimilar assumptions; - ... systems engineering, including the design phasing, work breakdown and work distribution, modeling and interfaces, and the role of specialty engineering (e.g. cost-, RAMS-, and mass modelling and configuration design). - ... concurrent engineering, as opposed to the more classical sequential engineering (the waterfall method). - ... project management and teamwork (assigning roles and responsibilities, setting goals and objectives, coordination and management of team process, decision making, handling conflicts, creativity, empowerment and motivation, communication, and reflection on own work and work of others).	
Education Method	The course consists of: - Group project (4 EC or 112 working hours) - Workshops/instructions (1 EC or 28 working hours) For the group work, students work in teams of 6-12 students. Prior to the start of the course, students will be informed on the available projects as to allow them to state their preference(s), and project teams are formed. During the project, each team is guided by a design coach, who mainly focuses on the problem solving/engineering design process. Additionally, each team is free to organize experts (from industry, academia or other) for the different disciplines involved that can advise the team during the design. During the quarter workshops/instructions are given to train teams' project management and design integration skills.	
Assessment	The assessment of each team consists of an assessment by the workshop instructors and project coaches (80% of the final grade), team of experts not involved in instruction and workshops (10%) and by the other student teams (10%). The assessment of the workshop instructors and project coaches is per design phase: - Exploration phase (10%) - Concepts design studies phase (20%) - Final design phase (70%) The assessment of the instructors and coaches is based on a design report, a (design) management log (record keeping) and one or more design presentations and reviews. Elements that will be assessed relate to design competencies (problem formulation, requirements generation, design option identification, design analysis, design trades and design evaluation), workplace competencies (e.g. teamwork, creative thinking, decision making, planning and organizing), academic and personal effectiveness competencies (e.g. communications, modelling/experimenting and critical and analytical thinking). Assessment by the team of experts and by peer teams will be based on the performance of the teams at a project presentation session at the end of quarter 3.	
Enrolment / Application	Only students enrolled in the Space track can participate. There is no free enrollment for this course. Still at the start of Q1, the students can preliminary enroll for the course as to learn more about the CSDP, and past projects conducted (no solutions on offer). In Q1, an information session will be held informing interested students of the track Space on the projects available and on how	

they can apply for a project. Students can then apply and only if they are accepted, their enrollment will be made definite. All non-accepted students will be dis-enrolled.

For the year 24/25, the maximum number of participants for this course has been set to 20. Priority will be given to students enrolled in the Space Engineering profile of the Space track. If places remain, also students from the other profiles in the Track Space can participate.

AESM2001	Physical Principles of Earth System Observation	5.5
Responsible Instructor	Prof.dr.ir. F. Lopez Dekker	
Module Manager	Prof.dr.ir. F. Lopez Dekker	
Instructor	Dr. P.G. Ditmar	
Instructor	Prof.dr.ir. S.C. Steele-Dunne	
Instructor	Prof.dr.ir. F. Lopez Dekker	
Instructor	K. Löer	
Contact Hours / Week x/x/x/x	4 hours lectures or practicals	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Measuring is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Electromagnetic, seismic and gravity potential-field observations inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. The measurements can be acquired from spaceborne, airborne, surface and sub-surface-based sensors. This programme core module aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain geophysical variables. For example, students will learn how electromagnetic theory allows to use the intensity of radar echoes to yield information about rain rate, soil moisture, ocean roughness, or the layering of the subsurface. Similarly, they will learn how potential field theory can be applied to quantify mass changes of, e.g., the ice sheets. Students will be able to weigh the advantages and disadvantages of, for example, microwave versus visible and near-infrared observations for monitoring the Earth surface, and seismic and electromagnetic imaging in mapping the subsurface. Review basics of electromagnetic (EM) waves (propagation, polarization, spectra, Doppler, etc.) and propagation effects (attenuation, refraction, dispersion, polarimetric effects) Optical-infrared scattering and propagation (different reflection types, albedo, effect of wavelength, BRDF, Radiative transfer modelling of optical EM waves). Physical principles and limitations of VNIR/SWIR sensing systems (photographic systems, LiDAR, spectral systems including TIR) in terms of spatial and spectral resolution, sensitivity, atmospheric disturbances and corrections. Microwave scattering and propagation (Rayleigh, Mie scattering, rough surface, volume scattering, clear air, radiative transfer modelling of microwave scattering in the atmosphere and at the land surface). Radiometry (Radiometric Quantities and Units, Blackbody Radiation, emissivity at different frequencies). Radiative transfer modelling in passive microwave remote sensing of the Earths atmosphere and surface. Physical principles and limitations of microwave sensing systems (radiometer, non-imaging radar, imaging radar, ground-penetrating radar) in terms of spatial and spectral resolution, sensitivity, atmospheric propagation effects and corrections, radar range equation Seismic (including acoustic) waves (Newtons second law, stress tensor, equation of motion, elasto-dynamic wave equation, Plane P- and S-waves, spatial and temporal resolution) Potential fields (gravity, magnetic and electric fields) and their relationship to physical properties of a medium (Gauss law and its applications, Newtonian/Magnetic Potential, Laplace equation and harmonic functions, representation of potential fields, spherical harmonics) Choose an approach for the observation and measurement of a geophysical variable.	
Study Goals	After completing this module, students will be able to: Characterize electromagnetic waves and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and & subsurface). [ILO A] Describe the interaction of optical-infrared waves with the Earths atmosphere and surface and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of optical/infrared remote sensing in determining geophysical variables. [ILO A, D] Describe microwave scattering in the Earths atmosphere and at the Earths surface, and analyse the fundamental sensitivity to geophysical variables of interest. [ILO A] Describe the emission of thermal waves and microwaves from the Earths atmosphere and surface, and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of microwave remote sensing in determining geophysical variables. [ILO A, D] Explain mathematically how the elastodynamic wave equation is derived, characterize elastic waves, and explain their sensitivity to contrast parameters. [ILO A] Establish the linearized relationship between an observable and a given representation of the potential field, and synthesize the potential from functionals given on the surface of a sphere. [ILO A] Critically discuss the potential value and limitations of different types of measurements (I.e., different bands of the EM spectrum and EMR/seismic/potential field) and propose and defend a type of measurement to observe certain geophysical variables. [ILO F, G]	
Education Method	Lectures and group practicals/exercises. Feedback via quizzes (formative) and intermediate assignment results (formative).	
Assessment	Module is summatively assessed with a project, using 1 rubric (similar as thesis project) at the end of the period based on: Assignment report (may comprise different parts, i.e., based on a several assignments, report includes reflection on LOs) Presentation Oral discussion (may include questions on LOs not directly covered in the assignment work).	

Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Some background in electromagnetic theory and fundamentals of wave propagation. Solving linear partial and ordinary differential equations.
Academic Skills	Earth Observation, data interpretation.
Literature & Study Materials	Provided course material, published online lectures and material, book/e-books
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N.A.
Collegerama	No

AP3141	Environmental Physics	6
Responsible Instructor	Dr. S.R. de Roode	
Instructor	Prof.dr.ir. J.L. Kloosterman	
Instructor	Dr.ir. D.J. Verschuur	
Instructor	Ir. R.A. Verzijlbergh	
Instructor	Prof.dr.ir. B.J.H. van de Wiel	
Instructor	Dr.ir. S.J.A. van der Linden	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Physics at BSc level, including basic Transport Phenomena and Fluid Dynamics, thermodynamics, calculus and differential equations; use of Python and/or Matlab.	
Course Contents	*) Fundamentals of the environmental physics, world climate and the greenhouse-model, durable energy-resources (solar-, wind-, wave-, nuclear-, biomass-energy), transport of pollution in ecosystems, Air Quality; *) Fundamentals of harnessing sustainable nuclear energy and physical models for nuclear reactors to do so; *) Land-atmosphere interactions and turbulent transport processes. Fundamentals of dimensional analysis & generalization in environmental physics; *) Radiative transfer & Solar cell physics, PV system design & Renewable energy integration; *) Environmental noise, basic theory, noise calculation schemes	
Study Goals	To provide physicists with the knowledge and tools so that they can contribute to a more sustainable society from their own specific field of expertise: 1. Acquiring advanced knowledge of the basic physics underlying environmental issues such as: the earth energy budget; air-, water- and ground pollution; solar, wind, tidal, wave, biomass and nuclear energy technologies and environmental noise. 2. Being able to translate problems in the above areas into appropriate physical models and their relevant equations. 3. Being able to translate physical models and model equations into (e.g. Matlab/Python) computer models. 4. Being able to find (quantitative) solutions to model equations, using either theoretical analytical methods, or numerical methods. 5. Being able to quantitatively and physically evaluate, analyse and critically acclaim proposed solutions and strategies for environmental issues. 6. Being able to contribute to solutions and strategies for environmental issues.	
Education Method	Lectures and mandatory homework assignments.	
Computer Use	Homework requires computer programming in Matlab or Python	
Literature and Study Materials	Egbert Boeker and Rienk van Grondelle, Environmental Physics, Third edition, Wiley, 2011, ISBN 978-0-470-66676-0 + lecture notes and handouts	
Assessment	There is a series of mandatory (homework) Assignments. Please note that the Assignments have to be done in the prescribed periods, and that grades cannot be improved later. Lecturers can invite students for an oral assessment of their Assignments. The grade for the Assignments is the mean over all of them. At the end of the course there will be a written Exam. This Exam will be of an open-book type (see below). The final grade of the course is the average of the Exam grade and the Assignment grade. A passing grade can only be obtained if a score of at least 5.0 is obtained for both the Exam and for the Assignments. The Assignments and Exam must be done in the same academic year. The validity of result of the Assignments is restricted to one academic year. Resit exam can be a written exam (if enough candidates) or via an oral assessment (by appointment). In case an on-campus exam cannot be provided, there will be an online open-book exam followed by an oral fraud-assessment of a random subset of students. See announcements on Brightspace.	
Permitted Materials during Tests	The exam will be done in a computer room. The computer will contain the PDFs of the book and PDFs of all handouts of the lectures, without access to internet. Besides this, you can take one A4 double-sides printed or hand-written with anything you prefer. And a simple calculator (not programmable) for possible calculations is allowed.	
Studyload/Week	2 lecture hours per week (14 weeks), 5 series of homework of 20 hours each, 40 hours of exam preparation.	

AP3181	Applied Multiphase Flows	6
Responsible Instructor	Dr. L. Portela	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course assumes some elementary knowledge of Fluid Mechanics and/or Transport Phenomena. The level of prior knowledge required is typically acquired in a BSc course in Engineering or Applied Physics. Prior knowledge of multiphase flow is not required.	
Course Contents	1. Introduction. Overview of multiphase flow. Examples of industrial and environmental flows. Examples of flows with heat transfer and phase change. Classification of multiphase flows: separated vs. dispersed. Brief introduction to flow patterns. 2. Balance equations. Physical mechanisms involved and material behavior. Constitutive equations and mechanistic models of behavior. Interfacial interactions and forces. Analogies and differences with respect to single-phase flow. Flow parameters, non-dimensional numbers and scaling. 3. Introduction to turbulence, turbulence modeling and turbulence effects. Analogies and differences with respect to single-phase flow. 4. Separated flows and interfacial phenomena. Stability associated with interfacial flows. Interfacial waves. 5. Dispersed flows. Interaction between the dispersed and continuous phases. Momentum, heat and mass transfer. Differences and similarities between solid particles, droplets and bubbles. 6. Dynamics of single particles: solid particles, droplets and bubbles. Bubble and droplet dynamics. Bubble growth and collapse. Introduction to cavitation. Droplet dynamics and breakup. 7. Inter-particle interactions and collisions. Droplet and bubble coalescence and breakup. Collision kernel. Population balance models and simulations. 8. Quasi-1D flows, both internal and external (channel, pipe, jet, etc.). Balance equations and simplifying assumptions. Simple mechanistic models. Similarities and differences between: gas-liquid, liquid-gas, liquid-liquid, solid-liquid, and solid-gas. Flow classification and flow patterns, both separated and dispersed. Physical explanation and underlying mechanisms in the construction of a flow map. Steady and transient phenomena. 9. Stirred flows, both internal and external (bubble columns, fluidized beds, stirred tanks, ocean and atmospheric turbulence, etc.). Similarities and differences with respect to quasi-1D flows. Motion-driven, pressure-driven and gravity-driven flows. Dilute and dense flows. Flows dominated by inter-particle interactions (dense fluidized beds, bubble columns, etc.). 10. Multiphase flows with complex fluids (granular flow, slurries, agglomerates, emulsions, foams, etc.). Internal mesoscale structures. Thermodynamic and physicochemical interactions. Complex particles (with a complex structure and/or a complex particle-dynamics). 11. Heat transfer and phase change. Boiling and condensation. Radiation. 12. Industrial flow examples. Complex flow in pipelines in the oil-gas industry. Pneumatic conveying and solid-gas flows. Process equipment. Microfluidics applications. 13. Environmental flow examples. Sediment transport in rivers and the ocean. Aerosols and particulates dispersion in the atmosphere. Droplets in clouds.	
Study Goals	To give a general unified perspective on multiphase flow, with an emphasis on: (i) the understanding of the several essential physical mechanisms involved; and (ii) the formulation of complex engineering problems in terms of simple physically-based models. The students will develop the ability to: (i) formulate the problems in a structured way, starting from first principles; (ii) obtain (approximate) solutions using standard analytical and numerical techniques; (iii) interpret and critically analyse the solutions obtained; (iv) report the formulation of the problems and its solution in a clear well-structured form; and (v) sustain a critical discussion on the formulation of the problems and its solution. Also, during the course the students (i) will acquire a broad knowledge of multiphase flow; and (ii) will develop the ability to read and understand scientific literature on multiphase flow.	
Education Method	The education method is based on the combination of regular lectures, homework and interaction with the instructor outside the lectures (regular and impromptu office hours). The Learning by Doing is a key component of the education method: the homework involves both standard problems and open-ended problems, with the initiative and creativity of the students being strongly encouraged. The interaction with the instructor outside the lectures plays a key role, and it involves an iterative process, based on the Learning by Struggling philosophy.	
Literature and Study Materials	The study material consists mostly of lecture notes. Also, during the course, sections of books, on specific topics, and articles, from the scientific literature, are suggested for reading. Possible auxiliary general reference-books on Multiphase Flow are: 1. Brennen, C.E. (2005). Fundamentals of Multiphase Flow. Cambridge University Press. 2. Ghiaasiaan, S. M. (2007). Two-Phase Flow, Boiling and Condensation: in Conventional and Miniature Systems. Cambridge University Press. 3. Kleinstreuer, C. (2003). Two-Phase Flow: Theory and Applications. Taylor & Francis. 4. Crowe, C.T., Schwarzkopf, J.D., Sommerfeld, M. and Tsuji, Y. (2012). Multiphase Flows with Droplets and Particles. Second Edition. CRC Press. 5. Fan, L-S and Zhu, C. (1998). Principles of Gas-Solid Flows. Cambridge University Press. 6. Peker, S.M. and Helvacı, S.S. (2008). Solid-Liquid Two Phase Flow. Elsevier. Possible auxiliary general reference-books on Fluid Mechanics and Transport Phenomena are: 1. Kundu, P.K. and Cohen, I.M. (2008). Fluid Mechanics. Fourth Edition. Academic Press. 2. Ghiaasiaan, S.M. (2011). Convective Heat and Mass Transfer. Cambridge University Press.	
Assessment	The assessment is based on the homework proposed during the course (30%) and on the final exam (70%). The final exam consists of a written part followed by an oral discussion with the instructor. The written part is open book: the students can consult any reference they wish (including the internet) and use any tool they wish (including a laptop). The oral discussion will be arranged individually, shortly after the written exam, in consultation between the student and the instructor. It is possible to have a resit of the final exam; the format of the resit is the same as the format of the original final exam, consisting of a written exam followed by an oral discussion with the instructor.	
Permitted Materials during Tests	Any material (including laptop).	
Elective	Yes	
Tags	Fluid Mechanics Physics	

AP3222	Nanotechnology	6
Responsible Instructor	S. Goswami	
Instructor	R.A. Norte	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic physics and mathematics are required; solid state physics, materials science and physics of microelectronics are recommended. Good understanding of the typical physics approach to problems is helpful.	
Course Contents	The synthesis and study of objects with nanometer-size functional elements require dedicated fabrication methods. This course will provide the technological and scientific background needed to master the methods of nanofabrication. Subjects include: material properties, growth and etching methods; pattern definition and transfer (=lithography); self-assembly; molecular manipulation; methods of inspection, analysis and characterization. The subjects will be taught with a focus on practical applications.	
Study Goals	The student can evaluate the merits of the fabrication technologies for nanotechnological applications. The student can design the processing steps for a nanotechnological application. The student can find, understand and apply the relevant scientific literature. The student can evaluate, model and understand the functioning of nano-objects. The student can execute a simple but complete nano fabrication task.	
Education Method	Lectures, expert lectures, student presentation & discussion sessions, and lab work.	
Literature and Study Materials	1. Sami Franssila, Introduction to Microfabrication, 2nd edition, Wiley 2. Copies of the lectures 3. Scientific papers for the student presentations and discussions 4. Lab-work manual	
Assessment	Written exam (35%), home work (15%), lab work (30%), and student presentation (20%). The scores for the written exam and the lab work should be both at least 5.0.	
Permitted Materials during Tests	Ruler, pocket calculator	
Enrolment / Application	Because of limited space and support in the cleanroom, a maximum of 30 students can do the course. Enrollment will start on Monday 6 January 2025 at 8:00am via MyTUDelft. Enrollment requests done before January 6 at 8:00am will not be considered. If you encounter any problems with this registration procedure, please contact Arno Haket, A.J.W.Haket@tudelft.nl, from 6 January. Students who do not have a Physics background are asked to submit a motivation (on prior knowledge, current activities and interest in nanotechnology) by e-mail to Srijit Goswami S.Goswami@tudelft.nl. Students who do not attend the first lecture (without prior notice) will have lower priority than those present. Students who do not have their bachelor degree at the start of the second semester (10 February 2025), will have lower priority than those with the degree.	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/education/admission-and-application/elective-course#c1323630	
Elective	Yes	
Tags	Lab Research Physics	

AP3701	Submm and terahertz physics and applications	3
Responsible Instructor	Dr. J.R. Gao	
Responsible Instructor	Dr. A.J.L. Adam	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	The course is intended for MSc students from different disciplines, including those from Applied Physics, Electrical Engineering, and Space Engineering, but it is also open to interested PhD students.	
Expected prior knowledge	Electromagnetism, Any introduction of Optics, and Introduction of solid-state Physics. If you miss some of them, we will also add necessary background knowledge during the course	
Course Contents	The submm and terahertz region of the electromagnetic spectrum, which spans from 100 GHz to 10 THz, has attracted a lot of attention in recent years due to the developments in new detection and generation techniques and due to the applications for astronomic observations in space and the potential applications as an imaging technique for security, communications and biomedical applications. The Submm&THz frequency range is rapidly emerging which allows unearthing new science and technology. We wish to provide a course in which students can get an introduction and learn the basic physics & principle, and will also be updated with the latest developments. The course will focus on the topics like Photonics, Time-Domain Spectroscopy, Antenna/Waveguide, terahertz detectors and sources, THz Space application. Here the titles of all the different lectures given: 1. Introduction to THz. Light Interaction with Matter 2. THz photonics (THz fibers, Photonic crystals & Metamaterials. High THz fields. Nonlinear interaction of THz waves with matter) 3. THz Imaging (THz Time Domain spectroscopy, Transmission, Reflection, Time of Flight Tomography, Near field, Tomography) 4. THz Sensing (Application to Chemistry, Security, Biology & Medicine; Pump-probe experiment) 5. Submm & THz Coupling (Waveguide, Si Lens, Antennas) 6. Sources 7. Nano-detectors and cameras 8. Submm & THz Sensing instruments from space (as a few examples) Additionally: A case study and design of an space instrument by a subgroup of students (Adam/Gao) (To be announced)	
Study Goals	Understand submm & terahertz electromagnetic radiation and propagation fundamentals, and the basics of typical terahertz detectors and sources. Distinguish different terahertz antennas with regard to their advantages and disadvantages for terahertz radiations. Explain the working principle of THz Time-Domain Spectroscopy. Build practical experience on a THz imaging setup and a heterodyne receiver system. Highlight recent advances of THz research and development from the academic and astronomic sectors. Summarize state-of-the-art THz applications, including space applications.	
Education Method	The course will be given as following: 1) one 2x45 min lecture per week 2) Home work every week on the topics following the lecture. The homework counts 30% of the grade. 3) Design an instrument concept and address a specific topics with groups and present 4) a written exam on all topics discussed during the lectures and homework We based our lectures on current research and science on THz technologies	
Literature and Study Materials	The lectures will be based on a few selected chapters of different books. For Gao's lectures: Chapter 27 "FIR heterodyne receivers by J.R. Gao" of a new book "Observing Photons in Space" and the book of "Terahertz Astronomy" by Christopher K. Walker. For Aurele's lectures: Terahertz Technology, Volume 77, A. Rostami, H. Rasooli, and H. Baghban, An Overview of the Technological and Scientific Achievements of the Terahertz, pp. 189. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011. (http://link.springer.com/chapter/10.1007/978-3-642-15793-6_1) After each lecture, the teachers will provide also the powerpoint slides used during the lectures.	
Assessment	20% from the homework 30 % from case/project study report/presentation; 50 % from the written exam	
Permitted Materials during Tests		
Elective	Yes	
Tags	Building & Spatial Development Physics Projects Technology	

BM41155	3D Printing	4
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Responsible Instructor	Dr. J. Zhou	
Instructor	Ing. M.A. Leeftang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	in Track I (Musculoskeletal Biomechanics)and Track II (Medical Devices and Bioelectronics) of MSc Biomedical Engineering BME) and in the specialization of Bio-Inspired Technology (ME-BMD-BITE)	
Expected prior knowledge	BSc in mechanical engineering, materials engineering, industrial design engineering and aerospace engineering with a basic knowledge of materials (e.g. WB2330)	
Parts	- Overview of AM technologies, their principles and applications - Materials selection (polymers, ceramics, metals and composites) - Powder-bed fusion processes - Powdered materials and powder fabrication - Powder characteristics in relation to AM - Structural integration during melting and sintering - Extrusion-based systems - Design for functionality and 3D printability Software issues and AM limitations - Other powder-based manufacturing technologies as alternatives to AM - Post-processing of AM products - Mechanical testing of AM products - Case studies AM for engineering and medical applications - Assignment and presentations	
Course Contents	Additive manufacturing (AM), also known as 3D printing the process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to conventional subtractive manufacturing technologies, is set to revolutionize manufacturing across a wide range of industries. The course 3D printing - AM technologies is designed to equip students with the fundamental knowledge of AM and basic hands-on experience in AM. The topics include various AM processes, their principles, capabilities and limitations, materials used for AM processes, powdered material fabrication and characterization, CAD designs for AM, alternative powder-based manufacturing technologies, current and future AM applications, as well as the mechanical evaluation of AM parts. The primary focus is placed on powder-based AM for metallic parts. In addition to a series of lectures, project groups, each composed of 3-5 students, will create their own designs of the objects made of polymer PLA, formulate requirements in terms of geometry, dimensions, tolerances, mass, surface finish, functionality and mechanical properties of these objects, redesign them by making use of SolidWorks, print them, test and evaluate 3D printed products against the requirements, and document the projects in the form of reports and presentations. The new design is expected to reflect the distinctive features and freedom that AM has to offer (thin walls, lattice structures, form-free channels, integrated parts/functions, and customization), but within the capacity of the printing machines (bounding box size, minimum feature size or wall thickness). The report and presentation are expected to contain: reason for the selection of the object, weakness of the existing design, new design features, actual testing results and recommendations.	
Study Goals	Students will be familiar with the concept of AM, various AM processes, AM applications, materials for AM, product design for AM, post-processing techniques and evaluation methods. In addition, students will gain basic hands-on experience in product development via AM. Once successfully completing the course, the students will: - Have a fundamental knowledge of 3D printing/AM technologies for a wide range of applications - Be able to explain the benefits, advantages, challenges and limitations of individual AM technologies as well as alternative powder-based manufacturing technologies - Have an insight into selective laser sintering and melting (SLS/SLM) processes regarding process control in relation to the geometry and mechanical properties of the products - Become familiar with the common (metallic) materials for 3D printing and their characterization techniques - Understand printable geometry design rules and the common methods of digital file creation - Become familiar with the common techniques for post-processing - Become familiar with the common methods for the mechanical evaluation of 3D printed materials and functional parts.	
Education Method	Lectures and student projects	
Computer Use	Attending lectures and performing assignment are both compulsory	
Course Relations	- Object design with Software (SolidWorks Student Engineering Kit)	
Literature and Study Materials	Biomaterials, Tissue biomechanics and other engineering courses in MSc biomedical engineering, materials science and engineering, mechanical engineering, marine technology, aerospace engineering, design for interaction, integrated design and strategic design - Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21) - Lecture materials	
Books	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21) - Lecture materials	
Reader	- Lecture materials	
Assessment	Written exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information) or CSE3130 (Introduction to Quantum Computer Science)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution.	
Study Goals	The student will acquire: - A good understanding of the fundamental concepts of quantum communication - The ability to understand, design and analyze quantum communication protocols - Insight into the differences between classical and quantum communication and cryptography	
Education Method	- Flipped classroom (Online videos and materials, prepared for flip classroom teaching) - Weekly Quiz and Discussion Session (In person) - Weekly Tutorials on Homework Problems (In person)	
Literature and Study Materials	Book on tools and quantum cryptography: Introduction to Quantum Cryptography, Cambridge University Press, 2023 Lecture Notes Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: - Homework (weighting 60%) - Final Project (weighting 30%) - Weekly Quiz (weighting 10%) Final grade calculation: $0.6 * \text{Homework} + 0.3 * \text{Final Project} + 0.1 * \text{Quiz}$ A passing final grade for the course can only be earned when for the homework and final project at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: - Homework: re-submit deliverable - Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4205	Evolutionary Algorithms	5
Responsible Instructor	Dr.ir. P.A. Bouter	
Responsible Instructor	Prof.dr. P.A.N. Bosman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks.	
	This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, (GPU) parallelization, and real-world applications.	
Study Goals	At the end of this course, the student is able to: 1. Explain the key concepts underlying the main streams in Evolutionary Algorithm (EA) research, with in particular genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2. Explain key ingredients underlying the rationale of when these algorithms work and when they do not work. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3. Explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4. Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5. Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	8-10 lectures 4 smaller (individual) projects 1 larger group project	
Literature and Study Materials	Various scientific papers, course-specific texts, and slides	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment (weighting 30%) - Individual Assignments: weekly assignments (weighting 10%) Final grade calculation = 0.6 * Written Exam + 0.3 * Group Assignments + 0.1 * Individual Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? - Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework. On the slides, references to articles (i.e. numbered paragraphs) or to pages in the book "Graph Spectra for Complex Networks" (see Literature and Study Material) are made. The information in the book thus complements the slides.	
Literature and Study Materials	Graph Spectra For Complex Networks, second edition (2023) Cambridge University Press ISBN: 978-009-36680-9 See: https://www.nas.ewi.tudelft.nl/people/Piet/bookGraphSpectraCN.html	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Course Relations	This course is mutual exclusive with the course SET3125. Students can not take this course if they have taken the course SET3125.	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (70%), project (20%), homework (10%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET3604LR	Electronic Circuits	3
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	Major topics: Electronics in information processing (sensor readout) Design principles and trajectory Signal processing functions Electronic design analysis Negative feedback amplifiers Operational amplifiers Components and networks Semiconductor diodes Power supplies and voltage regulators Semiconductor transistors and IC technology (basic) Oscillators DC motor basics	
Course Contents	This course is intended to provide gain insight, in a structured way, in the function of electronics as information-processing technology. The way of modelling will be discussed in relation to the physical characteristics of electronic components, circuits and systems. The course will provide the students with the skills to interpret circuit diagrams and measurement results and formulate specifications for electronic circuits that enable expert electronics designers to take proper design decisions when requested to design a high-performance application specific circuit. The students will learn the terminology that enables them to understand the explanation for these decisions by the experts (and potentially critically review them with respect to system level decisions). During the course many lecture demonstrations will be given, where the students both can observe the practical behavior of the test object and see what type of measurement equipment is used and how it is operated.	
Study Goals	After this course, the student -Is able to interpret electronic circuit diagrams and draw them according to standards -Is able to recognize and select electronic components -Is able to interpret and create Bode plots that describe the frequency behavior of an electronic circuit -Is able to analyze and design basic frequency filters -Is able to explain the basic behavior of semiconductor devices, like bipolar transistors, field effect transistors, diodes and special devices like thyristors. -Is able to determine the proper readout conditions for a sensor -Is able to determine the proper drive conditions for an actuator -Is able to interpret basic diode circuits like rectifiers -Is able to explain the basics of negative feedback systems and able to calculate and interpret the essential parameters -Is able to design a negative feedback amplifier using an Operational Amplifier (OpAmp) as active circuit that optimally meets the requirements -Is able to explain the basics of power circuits -Is able to name the different oscillator classes and explain the working principle the oscillators in each class -Is able to determine the class of a given oscillator -Is able to name the specific good and bad properties of each oscillator class -Is able to communicate with experts in electronic design and translate system specifications to specifications for the electronics in such a way that the expert can create an optimal design	
Education Method	Lecture. There is a mandatory practical with its own course code: ET3604LRP	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276 (older versions can very well be used for this course)	
Assessment	The exam is multiple-choice. During the lecture period, there will be one bonus test on oscillators. A pass grade gives a bonus point for the exam. The bonus point remains valid for the re-sit exam. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Dr. M.R. Vogt	
Instructor	Prof.dr.ir. O. Isabella	
Instructor	Dr.ir. H. Ziar	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	By the end of this course, you will be able to: 1. Explain the purpose of the components in different PV Systems using trends in their physical quantities. 2. Examine different power tracking methods and the importance of bypass diodes. 3. Calculate the energy yield of a PV system in conjunction with the following steps: 3a. Employ the calculation for optimal PV panel orientation from geographical survey. 3b. Determine the advantages and disadvantages of different PV systems topologies. 4. Design PV systems for different applications based on your own cost estimation and technical evaluation justified in a report	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	For students of the 2024 cohort: Final project report (33% of the final mark) and written test (67% of the final mark). For students of previous cohorts: Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr. R.T. Rajan	
Instructor	Dr.ir. R.C. Hendriks	
Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Tags	Algebra Algorithmics Artificial intelligence Linear Algebra Optimisation Signals Software Stochastics	

GEO1016	Photogrammetry and 3D Computer Vision	5
Course Coordinator	Dr. L. Nan	
Contact Hours / Week x/x/x/x	6 hours/week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Photogrammetry and 3D computer vision aim at recovering the structure of real-world objects/scenes from images. This course is about the theories, methodologies, and techniques of 3D computer vision for the built environment. In the term of this course, students will learn the basic knowledge and algorithms in 3D computer vision through a series of lectures, reading materials, lab exercises, and group assignments. The topics cover the whole pipeline of reconstructing 3D models from images: - Cameras models: how a point from the real world gets projected onto the image plane and how to recover the camera parameters from a set of observations; - Epipolar geometry: the geometric relations between 3D points and their images points; the constraints between the image points; - Image matching: define and match image features (SIFT) to establish correspondences between images; - Structure from motion: recover/refine geometry and camera parameters from a set of images; - Multi-view stereo and learning-based approaches for recovering dense geometry (e.g., point clouds) from images; - Surface reconstruction: obtain 3D surface models of real-world objects from point clouds.	
Study Goals	After finishing this course, the students will be able to: - apply linear algebra knowledge to implement basic 3D computer vision algorithms; - explain the main concepts in 3D computer vision (i.e., camera models, epipolar constraints, fundamental matrix, image matching, triangulation, structure from motion, bundle adjustment, multi-view stereo, and surface reconstruction); - explain the principles of the state-of-the-art 3D computer vision pipelines for 3D dense reconstruction from images; - evaluate methods for reconstructing smooth surfaces and piecewise planar objects, and choose applicable methods to solve specific reconstruction problems; - propose and implement solutions for reconstructing real-world buildings from images.	
Education Method	Lectures, reading materials, group assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is related to all fields of Geomatics for the Built Environment, from data acquisition, data management, data visualization, data analysis, 3D modeling, to applications.	
Literature and Study Materials	In addition to the lectures and handouts (distributed after each lecture), you will find relevant topics discussed in great detail in the following materials. [1] Y. Ma, et al. An Invitation to 3-D Vision: From Images to Models. 2001 [2] R. Hartley and A. Zisserman. Multiple View Geometry in Computer Vision. Second Edition. Cambridge University Press, 2004. [3] D. Forsyth and J. Ponce. Computer Vision: A Modern Approach. Second Edition. Pearson, 2012. [4] R. Hartley and P. Strum. Triangulation. Computer vision and image understanding. 1997. [5] Berthold K.P. Horn. Recovering baseline and orientation from essential matrix. 1990. [6] Y. Furukawa and J. Ponce. Accurate, Dense, and Robust Multiview Stereopsis. PAMI, 2010 [7] H. Hoppe, et al. Surface reconstruction from unorganized points. SIGGRAPH, 1992 [8] M. Kazhdan, M. Bolitho, H. Hoppe. Poisson surface reconstruction. SPG, 2006, 61-70 [9] L. Nan and P. Wonka. PolyFit: Polygonal Surface Reconstruction from Point Clouds. ICCV 2017.	
Practical Guide	In the assignments and group work sessions, students will experiment with the 3D computer vision techniques introduced in the lectures. - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; - Discuss the effect of parameters (if possible); - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Basic linear algebra and calculus; - Basic knowledge and skills in C++ programming. Students who have no experience in the above topics can still participate but should be willing to make efforts and invest some more time.	
Assessment	The assessment of this course consists of three group assignments and the final written exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded lower than 5.75 is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.	
Period of Education	Quarter 3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ME44312	Machine Learning for Transport and Multi-Machine Systems	3
Responsible Instructor	F. Schulte	
Instructor	Dr. B. Atasoy	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic programming	
Course Contents	The course gives an introduction to the basic concepts of machine learning (ML) (supervised, unsupervised, reinforcement learning) and puts them into relation to artificial intelligence and data analytics. Important fundamental machine learning algorithms including regression, classification, clustering, and neural networks are introduced and implemented in Python. The course also gives a brief overview of more advanced ML techniques that are used in the transportation and multi-machine domains such as reinforcement learning, graph-neural networks, and recommender systems. On this basis, machine learning methods are applied in group projects to transport and multi-machine systems using different types of data from different domains (e.g., AIS, GPS, New York Taxi, or machine failure data).	
Study Goals	-Understand the basic concepts of machine learning -Learn to implement widely-used machine learning algorithms -Apply machine learning methods to transport and multi-machine systems -Evaluate different machine learning methods for a problem setting -Conduct small ML projects	
Education Method	-Lectures -Group Work -Individual exercises for each machine learning algorithm	
Computer Use	Python on laptop	
Assessment	Group report	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Abstract Algorithmics Artificial intelligence Broad Business Design Group work Industry Modelling Optimalisation Programming Research Methods Stochastics Sustainability Technology Transport & Logistics Transport phenomena	
Percentage of Design	40%	
Design Content	Designing the right machine learning methodologies for different problem settings in the context of transport and multi-machine systems	
Department	ME Department Maritime & Transport Technology	
Contact	f.schulte@tudelft.nl b.atasoy@tudelft.nl	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Instruction if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Process & Energy	

ME45165	Equipment for Heat & Mass Transfer	5
Responsible Instructor	Prof.dr. K. Hooman	
Instructor	Dr. H.B. Eral	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Advanced reaction & separation systems Process plant design	
Expected prior knowledge	ME4500 (Advanced heat transfer)	
Summary	The course covers basic principles and methods required for the design and selection of heat and mass transfer equipment (for instance heat exchangers and thermal separation processes). This allows for the integration of heat & mass transfer principles to design process equipment.	
Course Contents	Rating and sizing of heat and mass transfer equipment such as heat exchangers, distillation, absorption, tripping, crystallization, adsorption, extraction and membrane separations will be covered. In particular: classification of various separation and heat transfer processes, the separation efficiency in single and multi-stage processes (and the influence of the process conditions and contacting modes on the separation). design procedures of different separation processes based on limitations of the thermodynamic equilibria. quantification of the influence of process condition on the design procedures of different separation processes based on limitations of the thermodynamic equilibria. characteristics of equipment for various separation processes the role of mass transfer in the sizing and design of the separation equipment. methods for the design and dimensioning of separation equipment. the use of computer tools for the design of heat exchangers and separation processes	
Study Goals	At the end of the course, the student will be able to rate and size heat and mass transfer equipment. In particular the student: 1. Can calculate overall heat transfer coefficient for heat exchangers; 2. Can select heat transfer equipment for given application; 3. Can apply effectiveness and Number of Transfer Units method to design heat exchangers; 4. Can apply pinch technology and heat integration for heat exchanger network. 5. Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes, 6. Comprehends the role of mass transfer in the dimensioning of equipment for separation processes 7. Can determine the separation efficiency for single and multi-stage separations and can quantify the influence of the process conditions and the contacting modes on the separation. 8. Can use computer based process simulation tools to analyze and design separation equipment	
Education Method	Lectures, assignments, self study, team work, tutorials	
Computer Use	Simulation of performance of distillation trays Standard engineering packages including ASPEN	
Literature and Study Materials	Course material: 1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738 2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Wiley & Sons, 2011. 3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer. 4: Course sheets / relevant papers 5: R K Sinnott, Gavin P Towler: Chemical engineering design, Coulson & Richardson's Chemical Engineering Series; Butterworth-Heinemann, an imprint of Elsevier: Kidlington, Oxford, United Kingdom, 2020. 6: Kays, W. M.; London, A. L. Compact Heat Exchangers, 3rd ed.; McGraw-Hill: New York, 1984. 7: Kroger Detlev G. Air-Cooled Heat Exchangers and Cooling Towers; Penwell Corp: Tulsa, Okl., 2004. Further reading Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son Geoffrey F. Hewitt, "Heat Exchanger Design Handbook", ISBN Online: 978-1-56700-422-9, Imperial College, England, UK	
Assessment	Formative: In-class exercises/quizzes Summative: Group project report & presentation (50%) Individual exam (50%)	
Permitted Materials during Tests	Course books and lecture notes	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Percentage of Design	50	
Design Content	Thermohydraulic design of heat exchangers; heat exchanger network, dimensioning pieces of equipment used in the	

	abovementioned processes
Department	ME Department Process & Energy

ME45170	Turbomachinery	4
Responsible Instructor	Prof.dr.ir. S.A. Klein	
Responsible Instructor	Prof.dr. R. Pecnik	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Thermodynamics, Fluid dynamics	
Course Contents	The course consists of the following parts: - Introduction and recap of the required thermodynamics. This class will give an overall overview of the relevance of turbomachinery, the general basics and the history. Furthermore the basic thermodynamics will be repeated (1 class) - Turbomachinery: basic theory and the application to the design of turbomachinery (Conservation equations and velocity triangles, turbines, compressors and pumps, 3D flow, radial machinery) (5 classes) - Application of compressor, pump and turbine maps. Limitations in the operation (stall, surge, NPSH etc) (1 class) - Gas turbine: components (turbine, compressor, combustor), gas turbine cycles, performance calculation,	
Study Goals	The student is able to describe the fundamental aspects of the working principle of turbomachinery. More specifically, the student must be able to: 1. reproduce and apply thermodynamic principles of turbomachinery 2. describe the basic fluid dynamics of turbomachinery and the way work is transferred between the working fluid and the turbomachinery, including the impact on performance and limitations of operations 3. apply compressor, pump and turbine curves in practical applications 4. describe different gas turbine cycles 5. describe the influence of different losses on the performance (power and efficiency) of gas turbines 6. synthesize different gas turbine cycles with respect to their application. 7. describe the principles of combustion as applied in gas turbines with respect to functionality and emissions 8. describe the part load characteristics of gas turbines and its relation to the functional characteristics of gas turbine components	
Education Method	Lectures and homework assignments	
Assessment	Closed book exam (60%) and home work assignments (40%). All homework assignments must be submitted. The grade for the homework assignments will only be included in the final grade if the exam grade is above 5.5/10	
Enrolment / Application	Registration: Standard enrollment via Brightspace	
Tags	Energy Energy & Industry Fluid Mechanics Thermodynamics	
Department	ME Department Process & Energy	

ME45203	Electrolysers, Fuel Cells and Batteries	4
Responsible Instructor	J.W. Haverkort	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Familiarity with transport phenomena (e.g. a course on heat transport), basic fluid dynamics, elementary differential equations, and introductory electrochemistry. Ideal preparation include the courses ME45201, CH3513, or SET3110. A previous course on fuel cells, batteries, or similar, or additional self-study may also be sufficient.	
Course Contents	Electrochemistry plays a pivotal role in the transition towards a sustainable energy future. Renewable electrical energy can be stored in batteries or used to synthesise chemicals or fuels in electrochemical reactors, like an electrolyser producing hydrogen. This course will introduce to a student, the knowledge and skills required to understand, model and optimize such electrochemical devices. Content overview: Porous electrode theory, multiphase flow, and transport models will be used to derive analytical current-voltage relations of various electrochemical devices including fuel cells, batteries, redox flow batteries, and electrolysers. Some of the topics that will be discussed in this course include: - The Nernst-Planck equations for ionic transport in binary electrolytes. - Porous electrode theory using the concept of an electrochemical Thiele modulus. - The flooded agglomerate and relative permeability models for gas diffusion electrodes. - The single particle and Newman's P2D battery models. - A vertical current and gas distribution model for electrolysers.	
Study Goals	After successful completion of this course, the student is able to independently: 1. explain and apply porous electrode theory. 2. derive 0D, 1D, and 2D mathematical models to describe the current-voltage relationship of various electrochemical cells including flow, multiple phases, and porous electrodes. 3. optimize an electrochemical cell design with respect to, for example, energy efficiency.	
Education Method	Two lecture hours + two exercise hours per week.	
Course Relations	This course is a natural sequel to ME45201, CH3513, and SET3110 for students interested in analytical modelling of electrochemical systems.	
Books	Electrolysers, Fuel Cells and Batteries: Analytical Modelling. (J.W. Haverkort) https://doi.org/10.59490/tb.93 (open access)	
Assessment	Written exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	

ME46006	Physics for Mechanical Engineers	4
Responsible Instructor	Dr. G.J. Verbiest	
Instructor	Prof.dr. P.G. Steeneken	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have Physics knowledge at the level of the final exam of the physics exam at Dutch highschools (VWO) and to have Mathematics knowledge at the level of the BSc education in Mechanical Engineering (e.g. WBMT2048). If they have not, they can still do the course but it will require more self-study from the book of Tipler.	
Summary	The course ME46006 Physics for Mechanical Engineers, aims to provide mechanical engineering students with a broad background in physics. The course offers basic understanding and skills in the field of physics, that will provide a background that can be useful in many engineering situations. It can serve as a starting point for more advanced physics studies and can also help mechanical engineers to interact with electrical engineers and physicists in their future career. Since the course aims for a broad coverage of the field of physics, the course material includes many different subjects at a basic to intermediate level. In contrast to other courses, this course favours to cover a relatively large number of different topics in order to provide a broad background, instead of covering a small number of topics in-depth. This means that the amount of material that needs to be studied is larger than for other courses. A bachelor degree in mechanical, electrical, aerospace, industrial engineering or applied physics is considered prerequisite.	
Course Contents	- Waves - Electrostatics and electricity - Magnetism - Electrodynamics - Optics - Interference and diffraction - Outlook into quantum mechanics	
Study Goals	At the end of the course, students should have knowledge, and should be able to solve problems on the following topics: Waves (Chapters 15&16 of Tipler): - The wave equation - Propagation, reflection and transmission of waves - The Doppler effect Electrostatics (Chapters 21-24): - Coulomb's law and electric forces - Electric fields and Gauss' law - Voltage and capacitors Electricity (Chapters 25 & 29): - Ohm's law - DC circuits Magnetism (Chapters 27-28, 30): - Gauss' law for magnetism - Lorentz', Ampere's, Lenz' and Faraday's law - Electromagnetic motors and inductors - Maxwell's equations Optics (Chapter 31-34): - Propagation and properties of light - Light-matter interaction - Imaging systems with light - Interference and diffraction - Limits of optical systems and quantum mechanics	
Education Method	Learning methods include power point presentations, class demonstrations, and recommended homework. Lecture notes will be made available on the Brightspace. Recommended homework will be on the concepts discussed in the class and solutions will be made available on the Brightspace. The course ends with a written exam.	
Literature and Study Materials	The course will closely follow selected chapters of the book of Tipler. In addition lecture slides will be provided on Brightspace.	
Books	Physics for Engineers and Scientists, 6th edition by Tipler and Mosca.	
Assessment	Written exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Precision & Microsystems Engineering	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Prof.dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Dr. O. Nejadseyfi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be 5.8 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest decimal. The maximum bonus score is 0.8. The bonus only applies to reports with grade ≥ 5.5 and submitted before the first report deadline.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	ME Department Precision & Microsystems Engineering	
Contact	Prof.dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

ME46085-23	Mechatronic System Design	4
Responsible Instructor	Dr. S.H. Hossein Nia Kani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The following are strictly required to be able to follow the course: - Basics of Multi-body Dynamics, including the ability to form equations of motion. - Basics of Electromagnetism, particularly Maxwell's equations. - Fundamentals of signals and systems (Laplace transforms, transfer function, bode plot, Nyquist stability). It is expected that students are able to use MATLAB or are interested in spending time to learn it while passing the course. This is required to complete the assignments.	
Course Contents	Course Contents Mechatronic system design deals with the design of controlled motion systems by the integration of functional elements from a multitude of disciplines. It starts with thinking how the required function can be realised by the combination of different subsystems according to a Systems Engineering approach. It should be noted that the control principles used in this course place a strong emphasis on frequency domain methods by linearising the system at its working point with the use of Bode- and Nyquist plots. The main reason for this emphasis is the strong focus in other control related courses on (non-linear) time domain related methods while linearised frequency domain related methods are still dominantly applied in the industry. A mechatronic engineer should be able to work with both methods and use them where appropriate. Some supporting disciplines, like power-electronics and electromechanics, are not part of the BSc program of mechanical engineers. For this reason this course introduces these disciplines in connection with mechanical dynamics and PID-motion control principles to realise an optimally designed motion system. The target application for the lectures are motion systems that combine high speed movements with extreme precision. The course covers the following three main subjects: 1: Dynamics of motion systems in the time and frequency domain, including analytical frequency transfer functions that are represented in Bode and Nyquist plots. 2: Electromechanical actuators, mainly based on the electromagnetic Lorentz principle. Reluctance force and piezoelectric actuators will be shortly presented to complete the overview. 3: Motion control in the frequency domain with PID and model-based feedforward control-principles that effectively deal with the mechanical dynamic anomalies (resonances and eigenmodes) of the plant. The other relevant discipline, electronics and position measurement systems is dealt with in another course: ME46005, Physics and measurement. The most important educational element that will be addressed is the necessary knowledge of the physical phenomena that act on motion systems, to be able to critically judge results obtained with simulation software. The lectures challenge the capability of students to match simulation models with reality, to translate a real system into a sufficiently simplified dynamic model and use the derived dynamic properties to design a suitable, practically realisable controller. This course increases the understanding what a position control system does in reality in terms of virtual mechanical properties like stiffness and damping that are added to the mechanical plant by a closed loop feedback controller. It is shown how a motion system can be analysed and modelled top-down with approximating (scalar and linearised) calculations by hand, giving a sufficient feel of the problem to make valuable concept design decisions in an early stage. With this method students learn to work more efficiently by starting their design with a quick and dirty global analysis to prove feasibility or direct further detailed modelling in specific problem areas.	
Study Goals	Can analyse and derive improvements to the dynamic behaviour of an actuator-driven mechanical structure with maximum 6th order plant dynamics (incl actuator and amplifier) by means of Bode and Nyquist plots. Can select and calculate a single-axis functional electromagnetic actuator for a given specification, working according to the Lorentz and reluctance force generation principle. Can select a proper amplifier to drive the above mentioned electromagnetic actuator. Can understand the basic concept of piezoelectric actuators and their application in mechatronic systems Can identify and apply feedforward and feedback P-, PD- or PID-motion controller settings for a given plant, consisting of a dynamically realistic power amplifier, electromagnetic actuator and mechanical structure with an ideal sensor, to achieve a stable system targeting a specified maximum bandwidth or disturbance rejection.	
Education Method	Lectures are given in 12*2 lecture hours, with presentations on the theory and practice of active-controlled motion systems. In addition, students will work on assignments to implement the theory learned in the course.	
Literature and Study Materials	Only the book is allowed at the examination. Therefore, no printed copy of the first edition, no computers, e-readers, smartphones, or other items are permitted. However, small notes written on the pages of the book are allowed.	
Books	The Design of High Performance Mechatronics - 3rd Revised Edition by R. Munnig Schmidt, G. Schitter, A. Rankers and Jan Van Eijk	
Assessment	The Final Assignment will account for 50% of the final grade. The remaining 50% of the score will be based on an open-book written examination. The examination will consist of questions covering the above-mentioned study goals.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	

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Contact	s.h.hosseinniakani@tudelft.nl

ME46095	Multiphysics Modelling using COMSOL	4
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Contact Hours / Week x/x/x/x	2 hour lectures on a weekly basis	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In applied mechanics one is often confronted with a multi-physics or coupled problem: A problem that requires the (simultaneous) solution of more than one type of physical process or phenomenon in order to accurately describe the problem. Examples of multiphysics problems are fluid-structure interaction, thermal-structure interaction and electro-thermal-structure interaction, possibly combined with a control problem. Particularly in the field of Mechatronic design and MEMS multiphysics problems are frequently encountered. COMSOL MultiPhysics is a finite element code, which can be used both as a MATLAB toolbox and as a standalone program, which is particularly suited for the simulation of multi-physics systems. In this course the student will learn to recognize different types of multi-physics coupling and methods for their efficient numerical solution using COMSOL. Short homework assignments are used to practise the use of COMSOL on different types of problems and in a final assignment the student is asked to study a multi-physics problem using COMSOL.	
Study Goals	The student must be able to: 1. recognize multiphysics coupling in complex problems 2. distinguish between different types of coupling one-directional vs. bi- or multi-directional interface vs. field strong vs. weak 3. describe numerical solution techniques applicable to coupled problems 4. use COMSOL MultiPhysics on coupled problems	
Education Method	(Online) Lectures (2 hours per week) / Self study	
Assessment	Report	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: WB2454-07	
Department	ME Department Precision & Microsystems Engineering	

ME46300	Optics	4
Responsible Instructor	Dr.ir. S.F. Pereira	
Instructor	Dr. N. Bhattacharya	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to optics: light propagation, the vectorial nature of light, interference, diffraction, optical spectra, geometrical optics	
Study Goals	At the completion of this course the student will be able to - have a good understanding of the basics of optics (i.e. making new connections between learned concepts and applying what has been learned to new situations) - explain the basic laws of polarization, birefringence, interference and diffraction - apply the theory to solve simple problems - put the theory in practice by performing experiments.	
Education Method	Lectures, exercise sections and practicals	
Books	Textbook that will be used in this course: Introduction to Modern Optics, G.T. Fowles, second edition, ISBN 100486659577 plus lecture notes and slides (available on Brightspace)	
Assessment	Individual written exam (90% of the grade). If it is required that the exam will be online, we will have an oral check after the exam and similarity check. The grade of the practicals (10% of the final grade) will be the average of the written group reports.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Precision & Microsystems Engineering	

ME46310	Opto-Mechatronics	4
Responsible Instructor	Dr. N. Bhattacharya	
Instructor	Dr.ir. S. Iskander-Rizk	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	Optical sensors and actuators are being integrated at an accelerated rate into mechatronics systems making these systems more precise, autonomous and intelligent. In this course the principles of optics, optomechanics and optical detectors are taught and discussed from the perspective of integration in optomechatronics systems for increased performance and improvement/addition in functionality. Topics included in this course are: 1) The different models of optics 2) Optical measurement methods. 3) Tolerances and sensitivities of mechanical positioning which influence optical performance of a machine. 4) Total system design where optical requirements can be translated to mechanical and mechatronical requirements. 5) Sources of light and optical detectors which can influence system performance (SNR, reliability, accuracy).	
Study Goals	At the end of this course the student can LO 1: Understand the practical applications and limitations of the different models of optics (geometric optics, diffraction optics, electromagnetic waves). LO 2: Understand the basic principles of imaging, interferometry, and spectrometry. LO 3: Understand how aberrations are related to mechanical tolerances and imaging quality. LO 4: Evaluate the function and performance of an optical system including the electronic readout module LO 5: Design an optical system for given environmental and application requirements (stress, thermal, kinematic effects) LO 6: Justify the mechatronic system components choices which can improve the performance of a given optomechanical system LO 7: Understand the influence of material choices and types of mounting optics	
Education Method	Lectures and Quizzes	
Assessment	Written exam + Quiz grade + 2 Guest lecture summary.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Broad Challenging Industry Mechanics Mechatronics Physics Signals	
Department	ME Department Precision & Microsystems Engineering	

MS43310	Materials at High Temperature	4
Responsible Instructor	Dr. V.A. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on Brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Materials Science & Engineering

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. L. di DiCarlo	
Responsible Instructor	Dr. M. Blaauboer	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	The final grade for the final exam is composed of the homework assignment grades (total of 30%), online quiz grades (total of 10%) and the grade obtained for the final exam (60%). A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course. The final grade for the resit exam consists of the resit exam grade only (100%). In other words grades for homework and quizzes do not count towards the final grade of the resit exam.	

RO47001	Robot Dynamics & Control	5
Responsible Instructor	L. Peternel	
Instructor	B. Shyrokau	
Instructor	J.M. Prendergast	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides the basics for deriving kinematic/dynamical models of a robot (such as a factory robotic arm, mobile robot, etc.) and for designing corresponding robot controllers. Students will learn how to derive the equations of motion and basic control equations themselves, and they will also learn how to use existing frameworks to do the same. THEORY SUBJECTS Introduction & Fundamentals -Course layout, instructors, grade scheme, etc. -Importance of the course and the big picture -Connection to real-world -Basic concepts/terms in robotics -Methods to describe robot pose -Numerical integration Recap of Control Theory -State-space model -Root locus -Bode plot -PID control -Concept of LQR and MPC control -Reference to online materials Endpoint Control for Robotic Manipulators -Big picture + motivation -Difference between position and torque-controlled robots -Endpoint position control (with a PID controller) -Endpoint impedance control (for position tracking tasks) -Endpoint force control (for force tracking tasks) Robotic Arm Kinematics -Deriving forward kinematics and Jacobian (for a 2-DoF planar arm) -Transformations between endpoint and joint space using Jacobian -Kinematic singularities and their solution -Kinematic redundancy and task priority control (for a 4-DoF planar arm) -Kinematics of parallel robots and differences compared to serial robots Robotic Arm Dynamics -Rigid-body dynamics model -Deriving dynamics model with Lagrange method for 2-DoF planar arm -Using the dynamics model in the control -Dynamic task priority control 3D Robotic Arm -Deriving 3D kinematics using the Denavit-Hartenberg convention -3D trajectory generation Planning & Motion Control -Orientation control -Stability analysis of the controller Kinematics & Dynamics of Mobile Robot / Automated Vehicle -Forward kinematic models -Wheel kinematic constraints -Robot kinematic constraints -Non-holonomic system -Derivation of linear bicycle model -Steady-state analysis Dynamics & Control of Mobile Robot / Automated Vehicle -Transient analysis -Frequency response -Path-following control PRACTICAL ASSIGNMENTS (individual) Assignment 1: The goal of this assignment is to learn how to control a 2-DoF robotic arm in Python to follow a prescribed trajectory using kinematic and dynamic equations. Assignment 2: The goal of this assignment is to learn how to control a multi-DoF robotic manipulator (3D kinematics, 3D trajectory, orientation control). Assignment 3: The goal of this assignment is to learn how to implement a linear bicycle model, and calculate and discuss primary handling characteristics, poles, stability analysis, and feedback control (PID).	
Study Goals	After completing this course, the students are able to:	

Education Method	-Derive the equations of motion for different types of robots -Create joint-level and endpoint-level controllers (including trajectory following) -Implement a simulation model in Python or Matlab -Analyse the designed control systems and reflect on the design choices -Lectures & interactive sessions (4 hours per week) -Practical assignments (homework)
Literature and Study Materials	Note: While attending lectures & interactive sessions is not mandatory, it is highly recommended. Assignments are not strictly mandatory, however, if you do not submit them you will lose the corresponding percentage of the final grade. That might mean that even if you pass the exam, you might not have a high enough final grade to pass the course. Study materials -Lecture slides -Papers, books, videos, etc. (links provided in Brightspace) Robotics textbooks (either one) -Lynch, K. M., & Park, F. C. (2017). Modern robotics. Cambridge University Press. -Craig, J. J. (2017). Introduction to robotics: mechanics and control. Pearson Education. -Siciliano, B., Sciavicco, L., Villani, L., & Oriolo, G. (2010). Robotics: modelling, planning and control. Springer Science & Business Media. Control theory textbooks (either one) -Ogata, K. (2010). Modern control engineering. Prentice hall. -Dorf, R. C., & Bishop, R. H. (2001). Modern Control Systems: Solutions Manual (Vol. 12). Prentice Hall. Useful toolboxes -Corke, Peter. Robotics Toolbox for Python. -Foote, Tully. "tf: The transform library." 2013 IEEE Conference on Technologies for Practical Robot Applications (TePRA). IEEE, 2013.
Prerequisites	-Linear algebra (including matrix and vector operations) -Calculus -Linear system theory at the BSc level (e.g., simple differential equations, Laplace transforms) -Basic programming experience at the BSc level (Python/Matlab)
Assessment	Number of assessments: -Four (3 assignments, 1 exam), all based on individual work. -The exam will be a computer/digital exam in ANS on campus. Grade calculation: -If and only if the exam grade is 5.5 or higher, the grade will consist of 30% of the average score of the assignments and 70% of the exam grade. The final grade must be 6.0 or higher to pass the course. Assignment submission lateness criteria: <1 hour: -1 point <1 day: -2 points <2 days: -4 points <3 days: -6 points >3 days: no points given
Enrolment / Application	Note: Assignments and exam grades are only valid for the academic year they were obtained in. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department Contact	ME Department Cognitive Robotics For general course inquiries: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 1 and related content: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 2 and related content: Micah Prendergast (J.M.Prendergast@tudelft.nl) For Assignment 3 and related content: Barys Shyrokau (B.Shyrokau@tudelft.nl)

RO47002	Machine Learning for Robotics	5
Responsible Instructor	Dr.ir. J. Kober	
Instructor	Dr. H. Caesar	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides a broad overview of machine learning techniques and their practical application in robotics. Intro to Machine Learning and python - Machine Learning in Robotics - Understanding the goal of machine learning, fundamental problems, high-level overview. Visualizing your data - Supervised vs Unsupervised vs Reinforcement learning - Model-driven vs Data-driven: White box, Gray box, Black box - Prior knowledge vs Unstructured. Feature extraction, linear regression + Deep Learning. Interpretability Hands-on Machine Learning - Example machine learning project in Robotics - Binary classification, decision boundaries, using logistic regression/SVM (one parameter) - How to perform a classification experiment. Dataset splitting, learning curves, metrics, comparing results Regression & Data collection - Regression and Data Collection in Robotics - Regression methods, least squares fitting - Overfitting, cross validation, regularization - Collecting (noisy) data, labelling data, outliers - High-dimensional data, data augmentation - Hyper-parameter optimization (grid search vs random search) Classification - Classification in Robotics - Parametric vs Non-parametric classifiers - Logistic regression - Decision tree, forest - Bayesian classification: Bayes' rule, naive Bayes, Gaussian Mixture Model - k-nearest neighbour - SVM, kernel-SVM, dual problem - Multi-class classification, metrics (confusion matrix), class imbalance Unsupervised Learning - Unsupervised Learning in Robotics - Clustering: K-means, Gaussian Mixture Model, DBSCAN - Dimensionality reduction: PCA, Local Linear Embedding (LLE) Neural Networks - Neural Networks in Robotics - Multi-Layer Perceptron, gradient descent - Neural Networks, and deep learning - Vanishing gradient problem, DropOut, Optimizers Advanced machine learning - Outlook: Recurrent neural networks, convolutional neural networks - Outlook: Data augmentation, AutoEncoder - Outlook: Reinforcement learning	
Study Goals	Knowledge, insight, judgement and skills: After completing this course, students will be able to - explain the role of ML in robotics, and describe possible use cases, and when not to use ML - explain the main concepts in ML (overfitting, training and validation, classification vs regression, cross validation, regularization, parametrics vs. non-parametrics, supervised/unsupervised/reinforcement learning) - explain the principles of well-established ML methods (e.g. Logistic regression, SVM, RFs, Neural Networks) - collect and preprocess data (normalization, feature selection, augmentation, splitting) for ML - analyze a ML problem, consider available data and computational resources, and select the appropriate ML methods to apply - perform experiments, evaluate the results, and draw sound conclusions - present the findings of a ML experiment clearly in a report using graphs and scientific English - write python code to visualize data and implement ML methods using popular python frameworks, such as pandas, numpy, matplotlib, jupyter notebook, sklearn, keras Transferable and interpersonal skills: After completing this course, students will be able to - collaborate and coordinate with a lab partner on coding and reporting tasks	
Education Method	Contact hours - Flipped classroom: pre-recorded lectures (2 hours per week) + classroom discussion (2 hours per week) - Q&A on practica (2 hours per week) Formative feedback - Quizzes -- Individual -- On material of previous lecture(s) -- Via BrightSpace at set time slots - Practica -- Pairwise collaboration -- Submission via BrightSpace -- Counts for knock-out criteria -- No late submission -- No resit Deadlines - For each practicum, the deadline is approx. one week after the practicum is published.	
Literature and Study Materials	Study materials - Slides (released shortly before lectures) - Book: "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent	

	Systems" 3rd edition, Aurélien Geron
Prerequisites	Suggested literature - "Pattern Recognition and Machine Learning", C. Bishop - "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Hastie, Tibshirani, Friedman TU Delft BSc Mechanical Engineering curriculum, or equivalent. Especially familiarity with - Programming in python - Linear Algebra - Probability theory
Assessment	Final assignment (30%) - Pairwise collaboration - Late submission: -1 grade point per day - No resit - Deadline: Will be released after the 6th practicum, giving the students approx. two weeks to read the assignment, collect the necessary data, perform their experiments, and write their report Exam (70%) - Individual - Written closed-book exam - Exam in Q1, resit in Q2 Knock-out criteria - At least 4 practicum grades need to be a pass (of current or previous academic year) - Minimum grade for final assignment is a 5 (of current or previous academic year) - Minimum grade for the exam is a 5 - Final grade minimum is a 6
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics

RO47007	Multidisciplinary Project	5
Responsible Instructor	Dr. C. Pek	
Instructor	Dr.ir. G.N. Saunders	
Instructor	Prof.dr.ir. M. Wisse	
Instructor	C. Hernandez Corbato	
Instructor	Ir. C.C. Claij	
Contact Hours / Week x/x/x/x	0.0.0.2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In this capstone project of the first year of the MSc Robotics, you will be challenged to use all the knowledge and skills you have gathered so far. Within a multidisciplinary team of up to 5 people, you will develop and test software for use in a real robot whilst working together for a real client in an industrial setting. We expect all your robotics skills (including your programming skills) to be up to date! The estimated workload of this course is 18 hours per week over 8 weeks. In this project, you will not just produce a software solution for a robot. You will also implement and test your software in a real robot and report and present your results as well as document your code, mirroring industry practice. Just like you would in industry, you will likely spend only 50% of your time coding and testing and the other 50% on meetings, systems design, report writing, (preparing) presentations, and documenting your code. These are no longer afterthoughts. In this project, you work closely with an industry client, realistically a possible future employer. That means there are expectations regarding your commitment and presence during the project. We expect you to behave cordially and respectfully towards the client and be on time for scheduled sessions. You will be given the option to express a preference for a client when you enroll if multiple clients take part in the course. Each of you will sign up with a specialist role in mind within the project, in which you are responsible for a specific part of the software design and implementation. It is no longer possible for one person to do all the programming. The specializations for which you can express a preference when you enroll are: - human-robot interaction, - navigation, planning, control - perception, and relate to the different disciplines within robotics. Within your team, you are expected to make a translation of societal issues from different perspectives (human, sustainability, safety, ethics, economic, etc. into intelligent robot solutions. Practically, each student team develops a complete, integrated software package for a complex robot system. You develop and test in a simulated environment, and then implement it in a real robot during the course before presenting your results in the final presentation to your client at the end. This course consists of 5 iterative phases with continuous reporting varying in lengths: - Phase 0: Introduction, meet your team, and get to know your robot. - Phase 1: Initiation and Planning - Getting organized, defining and planning your project. - Phase 2: Design & Execution - Software architecture design, initial (simulation) implementations and demonstration of concept to the client. - Phase 3: Implementation - Further design and implementation of your solution on your robot. - Phase 4: Testing & Completion - Finalizing your project and final reporting and code documentation, with a presentation of your implementation for the client in week 4.8. A detailed schedule will be available on Brightspace after the introduction lecture, likely in week 3.8.	
Study Goals	The learning objectives for this course are divided over two domains: 1) Knowledge, insight, judgment and skills By the end of the course, you should be able to: LO 1: Define a robotics use case with its functional and non-functional requirements. LO 2: Design robotic solutions by integrating knowledge of system engineering, current opportunities for robots, trends in robotics, and societal aspects. LO 3: Produce a robotic solution that meets the design specifications and considers sustainable, safety, ethical, economical and other relevant societal aspects. LO 4: Use functional architecture for planning and communicating robot software. LO 5: Communicate the multidisciplinary robot solution orally, in writing, and in code documentation according to robot software development guidelines. 2) Transferable and interpersonal skills. By the end of the course, you should be able to: LO 1: Formulate learning goals for your own personal development. LO 2: Reflect on your competencies (e.g., Teamwork, Leadership, Entrepreneurial thinking, Strategic multidisciplinary problem-solving), their development, and your personal learning goals. LO 3: Apply structured feedback to improve your performance or the performance of others. LO 4: Apply structured multidisciplinary software project management, with the use of different team roles and responsibilities.	
Education Method	Project work in teams supported by (video) lectures and workshops.	
Computer Use	During the project, we will be using different types of software and collaboration platforms. To save time and frustration, please ensure you have a working laptop that has the following software installed and have gained access to the following repository before the project starts: ROS: We expect you to have a working ROS environment on your laptop. Please use your knowledge and experience from RO47003 Robot Software Practicals and make sure that you align with the Ubuntu and ROS versions used in your clients robot. We will be using native installations of Ubuntu and ROS, and do not provide support for any other installation. Instructions on this will be published on Brightspace. Gitlab: All code and documentation for this project will be stored on GitLab. To be added to your groups repository on the GitLab environment for this course, go to: https://gitlab.tudelft.nl and log in using your NetID. Please do this as soon as you have been admitted to the course and have been assigned a group. Otherwise, you will not be able to access the required code for your project.	
Literature and Study Materials	During the project we expect you to use and refer to the following books. Please avail yourself prior to the course starting of either a hardcopy or an electronic copy of the book. [1] A Gentle Introduction to ROS Jason OKane, E-Book: Accessible via the authors website: https://jokane.net/agitr/ [2] Practical Model-Based Systems Engineering Jose Luis Fernandez & Carlos Hernandez Corbato, E-book: Accessible with the TU Delft library web extension In addition, you may also want to refer to these books. These are, however, not mandatory for the course: [3] Beginning Software Engineering Rod Stephens, Hardcopy: Available via Leeghwater, E-book: available through the TU Delft library extension. [4] Information Technology Project Management K. Schwalbe, Hardcopy: any online bookshop, E-Book: no free legal website available	
Prerequisites	Starting the project is only allowed when a student has successfully completed the course RO47003 "Robot Software Practicals" and has completed RO47000 Vision & Reflection. Without these preliminaries, the student will not be allowed to participate. Exemptions require written approval of the teachers of RO47003 and RO47000 that the student has obtained equivalent	

Assessment	qualifications. It is expected that you can code when you start this course. At the end of the course, your project needs to comply with a few requirements before we will grade it. These will be clearly listed in the course manual on Brightspace. Moreover, you must attend all mandatory sessions of the course. Excused absences are only permitted by receiving prior approval of a study advisor and the course staff. Absences due to medical reasons require a confirmation, e.g., by a study advisor. For this project, an individual grade is given. The grade for this project is a combination of the group grades for your final project including your final presentation, the Sprint assignments, and an individual grade based on your personal reflection. An individual bonus-malus of max. 1 grade point can be applied by staff in rare cases of excellence or underperformance. For more details on grading, see the course manual on Brightspace.
Enrolment / Application	Enrollment for the course will open at the beginning of Q3. Further details on how to enroll will be announced on Brightspace. All students must have registered before the announced registration deadline. It is the students responsibility to acquire the required registration information and deadline. After the registration deadline, late participation in the project is not possible anymore. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Special Information	Note: The mandatory introduction lecture and the ROS software & robot programming sessions will take place at the end of Q3, likely in week 3.8.
Remarks	This is a very intensive course which requires you to work 18 hours per week over a period of 8 weeks regardless of Public Holidays. The course requires intensive teamwork and good time management. On average there is 1 mandatory session per week you must be available for which can vary over the weeks depending on the availability of robots and industry clients. Past experience has shown that combining this course with too many electives and/or other intensive activities during weekdays is incompatible. Be prepared you will likely spend 40h per week on this project and any electives combined. It is also not recommended to start with your literature research during MDP.
Department	ME Department Cognitive Robotics

RO47015	Applied Experimental Methods	5
Responsible Instructor	Dr. Y. Vardar	
Instructor	Prof.dr.ir. D.A. Abbink	
Contact Hours / Week x/x/x/x	0.0.0.x	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Study Goals	After completing this course, students will be able to: - independently design, execute and analyse a human factors experiment - formulate a clear hypothesis in the field of human factors that can be tested experimentally, in a realistic amount of time with a realistic amount of means - set up a methodologically sound experiment to test this hypothesis - o select, adapt or build an experimental setup - o gather reliable experimental data on test subjects - o choose appropriate metrics - o apply correct statistical techniques - o choose the most relevant results, and construct the most clear figures to show them - critically reflect on your own experimental findings - communicate clearly through presentations and writing	
Education Method	Students will perform a mini-graduation project in six weeks, in small groups (2-3 students). They will independently design, perform, analyze and present a human factors experiment, guided by four supervisors.	
Literature and Study Materials	Scientific papers	
Assessment	The final grade is composed of the following sub grades 10% Presentation 1: Hypothesis 10% Presentation 2: Methods 10% Presentation 3: Analyzing Results (two main results figures, explain) 20% First Draft for Paper 50% Final Paper In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment can take place in a similar way.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Broad Design Group Dynamics/Project Organisation Group work Integrated Intensive Involved Lab Research Project planning / management Small groups	
Department	ME Department Cognitive Robotics	

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory	
Course Contents	Fundamentals of Coding in Python and MatLab Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Symbolic regression * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control We will aim for a balance of practical intuition and theoretical understanding: we will discuss examples of applications and go deeper into the theory when feasible.	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures (recorded and in class) and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: (i) practical assignment and (ii) written exam, open book. You must score at least 5.5 on both assignments to get a pass. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity to resit if they fail the written examination. * If your practical assignment is graded between 5 and 5.75, you will have two weeks to integrate the assessors feedback. If this is done satisfactorily, you can be rewarded with a mark that is not higher than 6/10. The exact instructions for the addition will be announced after the grading of the PA is concluded.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding course registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) exam registration is obligatory. More information regarding exam registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Artificial intelligence Complex Mathematics Intensive Linear Algebra Mathematics Modelling Programming Project Proofs	
Department	ME Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42030	Control for High Resolution Imaging	3
Responsible Instructor	Dr.ing. C.S. Smith	
Responsible Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	sc42065	
Expected prior knowledge	Basic knowledge of physics and control theory is expected. The first lectures recap in a condensed form the necessary knowledge in optics. Understanding of signal processing (in particular, the Fourier analysis) is expected for successful work on the homework assignments. The course involves some basic numeric simulations, so practical knowledge of Matlab/python/... is required.	
Course Contents	High-resolution imaging is crucial in scientific breakthroughs, such as discovering exoplanets in the habitable zone or understanding the origin and progress of diseases at a molecular level. For that purpose, special optical instruments like Extreme Large Telescopes or STED microscopy are developed. Optical aberrations are the key obstacle that hampers a clear vision and invites control engineers to step in. These are the disturbances induced by the medium, like turbulence in the case of astronomy, or by the specimen under investigation, like the change in refraction index due to inhomogeneities in the biological tissue. Adaptive Optics is a modern way to remove the effect of aberrations. This fascinating and expanding field in science is providing an excellent challenge to control engineers to improve image quality significantly by active control. This course will review the hardware necessary to control light waves in contemporary optical instruments, their modelling from a control engineering perspective, and discuss model-based control methodologies to do disturbance rejection.	
Study Goals	Understand the propagation of light, imaging and aberrations in the imaging process. Understand the operation principle of pupil-plane and focal-plane sensors to estimate the wavefront aberrations. Understand the design principles of opto-mechatronic wavefront corrector devices to correct wavefront aberrations. Develop spatial and temporal models of complete imaging systems and use these models in the design of model-based controllers for aberration correction.	
Education Method	Oral Presentations (seven lectures), homework assignments (4 in total), and self-study in the form of reading and presenting a scientific paper. Seven seminars (interactive discussions related to the homework assignments) are interleaved with the lectures, using the same time slots. Some homework assignments are allowed to be done in small groups; paper presentations are done in groups.	
Literature and Study Materials	Course Notes; selected chapters from published books and relevant scientific papers are provided as complementary reading material.	
Assessment	SC42030: (homework and paper presentation) 80% + oral examination 20%.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC4045	
Elective	Yes	
Tags	Broad Mathematics Optimalisation Physics Signals and Systems	
Department	ME Department Delft Center for Systems and Control	

SC42065	Adaptive Optics Design Project	3
Responsible Instructor	Dr.ing. C.S. Smith	
Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of signal/Fourier analysis, linear algebra, and optimisation is required. Basic skills in Python are required.	
Course Contents	The course consists of the realization of laboratory experiments to design and operate Adaptive Optics equipment to realize high-resolution imaging systems. Crucial in the design is the alignment of active optics systems and in the operation the development of algorithms for acquiring accurate information about the wavefront aberrations from intensity-based imaging components (like Shack-Hartmann sensors or CCD cameras) and using this wavefront information in the tuning of multivariable dynamic controllers to compensate in real-time the wavefront aberrations. The design is conducted under close supervision by world-leading experts in the field and is performed in groups of students. The size of the groups depends on the number of participants in this course. The course requires hands-on experiments, Python coding, and presenting the results in a report and/or a joint final presentation.	
Study Goals	Building insights about the key components in Adaptive Optics such as the wavefront reconstruction and the deformable mirror. As well as building the controller methodology to obtain a smart optics system for high-resolution imaging.	
Education Method	Project Based The work is subdivided into sessions. Each session results in a deliverable (e.g. some measurements or a piece of code). The work is performed in groups, which should be well balanced in, for instance, writing, coding, and theoretical skills.	
Computer Use	Lab computers are available, own laptops can be used	
Course Relations	Participation in SC42030 is required	
Practical Guide	A practical guide subdivided into sessions is provided.	
Assessment	Oral Presentation and evaluation of the written report in combination with the course sc42030	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Remarks	Old course code: SC4115	
Department	ME Department Delft Center for Systems and Control	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. R. Kortlever	

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced, analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	The final grade of the course consists of the following components: - Take home exam (50%) - oral exam (50%) Final grade calculation: $(0.5 * \text{take home exam}) + (0.5 * \text{oral exam})$ The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 take home exam: resubmit deliverable oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. H.M. Schuttelaars	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	After this course you are able to: - recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - solve the shortest path problem and the transportation problem as well as some small flow shop problems. - solve the travelling salesman problem by the Branch and Bound algorithm. - apply several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - reproduce basic theorems concerning the mentioned problems and is able to prove some of these theorems. - explain how to solve large scale problems, especially shortest path and vehicle routing problems. - define scheduling problems and complexity theory. - implement the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam and a written resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Elective	Yes	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr.ir. J.T. van Essen	

WI4156(TU)	Game theory	6
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Open to all MSc students with a strong command of mathematics. Ideally suited for math students with an interest in finance.	
Expected prior knowledge	Standard BSc math courses: calculus, linear algebra, probability.	
Course Contents	This is a course in "mathematical" game theory, which is the counterpart of "algorithmic" game theory (agent based modelling, this part of game theory is taught in the computer science department). We cover all subjects from combinatorial games to cooperative games.	
Study Goals	Understanding the mathematical theory of rational behaviour/conflicts/negotiations/social choice. Game theory is one of the most prominent economic theories: three recent Nobel prizes have been awarded to game theorists. We will cover a very wide area, ranging from pure mathematics to economics, with an emphasis on the mathematical aspects. This is a course in the Finance Track of the Applied Maths programme, but it is open to all MSc students (math and non-math).	
Education Method	Lectures	
Literature and Study Materials	Game Theory - Thomas Ferguson - UCLA: find it on the net.	
Books	T.S. Ferguson, A course in game theory, World Scientific, 2020	
Assessment	The final grade of the course consists of the following components: written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students are expected to have: superior knowledge and skill in elementary probability (details below); experience working with Python, Matlab, or R; and a mathematical inclination. If this does not apply to you and you still want to enroll: contact the instructor. Good skills in elementary probability are indispensable as they are the foundation for the more advanced material of the course. We build on the concepts, tools, and methods of elementary probability, and students taking this course must be (or rapidly become) black-belt masters in these foundations, specifically: Chapters 2-5, 7, 9, and 10 of Dekking et al; prior knowledge of Chapters 13 and 14 is helpful, though we will revisit these subjects during the course. Testing yourself and doing a thorough review in the first few weeks is required; we will also spend some time on this in class to help you do this efficiently; after this a diagnostic test (carrying part of the course grade) will assess your mastership; sub par performance means you cannot follow the course.	
Course Contents	This course on Monte Carlo simulation and stochastic processes consists of a mixture of these components: modelling and models (and some of their pluses and minuses), learning a number of stochastic processes and some of their properties, doing Monte Carlo simulations, analyzing the generated data and drawing conclusions from them. Contents and format of this course are intended to combine the following: 1. real (possibly stylized) problems as starting points; 2. enough about modelling to make participants aware of the impact of modelling decisions; 3. identification of a number of stochastic processes present in/related to models and problems; 4. (limited) treatment of underlying theory, guided by: (a) what is necessary to ensure that participants know how to do things right and (b) what enables them to do things smartly; 5. facts, properties, and theorems: (a) sincere attempt to understand where possible; (b) focus on practical implications and uses; (c) illustrations rather than proofs or theory; 6. useful tools and techniques from statistics, probability, and stochastic processes; 7. simulation as aid to: obtaining answers, intuitive understanding, illustration of theory; 8. providing as much opportunity as feasible for participants to work on/with problems from their own field. Tentative topics/contents list: Review basic probability: most to be done as self-study, some in class, plus a mandatory test (see Assessment). Models, modelling. Simulation: Convert a model to a program and run it. How to expose and locate errors. Data, empirical distribution; estimation. Accuracy, efficiency. Sensitivity. Rare event simulation. Experimental design. Conditional distributions and expectations; conditioning as a tool. Multivariate distributions: general (some examples); multivariate normal. Stochastic processes: Random walk. Poisson process; renewal processes. Markov chains: discrete time; continuous time; long-term behaviour; embedded Markov chain; birth and death process; Brownian motion. A snippet of diffusions, (solutions to) stochastic differential equations (SDEs).	
Study Goals	Learning objectives: after following the course, participants will be able to 1. in an iid setting, carry out Monte Carlo simulations; and 2. extract and collect, summarize, and interpret the data thus generated; and 3. assess accuracy/uncertainty in estimates, and solve questions like How many runs for 1% accuracy?; 4. the above (three) also in some non-iid settings, e.g., for a stationary process, using the batch-means method; 5. for small-sized practical problems: build a suitable model and simulate it; 6. extract and collect required model parameters from a problem description or background; 7. recognize where modelling choices are made; 8. make elementary modelling decisions themselves; 9. recognize and describe in mathematical terms (the properties of) a number of stochastic processes, among others: Markov chains, random walks, Brownian motion; 10. recognize/identify the covered stochastic processes in practical settings/problems; 11. use the tools and concepts from probability theory to describe, discuss, and manipulate (the components of) stochastic models; 12. where applicable, solve problems by means other than Monte Carlo, i.e., analytically or numerically, procuring exact answers, by identifying a stochastic process and using its known properties; for example, concerning long-term system-behaviour.	
Education Method	Weekly class meeting of two hours. About two thirds spent on lecture; about one third on questions and exercises. A number of so-called "Hands-on" assignments: a collection of exercises that combine the analytical content with computer simulation; to be done individually or with a partner. The participants are expected to perform a significant amount of simulation in Python, Matlab, or R (or other suitable language). Spread over the semester, five homework assignments. I strive to make a classroom available during the (probably) Friday lunch break, in which the participants can work on their own and where the TA (if there is one) is available for questions. In addition, I will hold a weekly drop-in office hour, most likely at lunch time (details on Brightspace).	
Literature and Study Materials	Unfortunately, for this course there is no really appropriate textbook, one that combines modelling, Monte Carlo simulation, and stochastic processes. For the review of basic probability and statistics, you are referred to Dekking et al, "A modern introduction to probability and statistics," Springer, 2005; the text is very well suited for individual review and practice, as it contains many worked examples, (quick) exercises, answers and full solutions. To be successful in the course, participants must have developed superior skill wrt the tools and techniques in the first half of the book (see "Expected prior knowledge"). Most of the material on stochastic processes can be found in "Probability Models" by Sheldon M. Ross; the book contains many worked examples and exercises; its heuristic and intuitive approach fits that of the course; references will be to the 10th Edition. A slightly more mathematical, but still very accessible reference covering the same area is "An introduction to stochastic modeling" by Howard M. Taylor and Samuel Karlin. Lecture slides, notes, and additional references will be posted on Brightspace.	

	A laptop is required and it is recommended that you bring it to class. Many languages and/or software packages are suitable: I prefer you use the one you are most familiar with. I am migrating to Python, but in the material presented in class and distributed via Brightspace, there still is some Matlab code. You may use a language/tool other than those three, but talk to me first (because you should be able to use your software on the exam and it should be possible to arrange this). In all cases: you should have the software installed and be familiar with its use.
Assessment	1. Probability test on knowledge and skills needed in/for the course (15%); passing score must be obtained to take the course. 2. Assignments (five; each counts for 4%), which may be done with a (=one) partner. 3. Final exam (on-campus): closed-book computer exam (with cheatsheet allowed, specs on Brightspace); counts for 65%. To pass the course, the probability test must be passed and the final exam grade must be 6 or higher; course grade is weighted average.
Permitted Materials during Tests	To the final exam you may bring a self-compiled formula sheet: one A4 page with normal-sized text on about 25 lines (or any reasonable equivalent of this), made by you, of things you would find helpful.
Elective	Yes
Tags	Adventurous Diverse Mathematics Modelling Practicals Stochastics
Co-Instructor	Dr.ir. G.N.J.C. Bierkens

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: * Bachelor level knowledge and ability in calculus, probability, statistics, optimization, linear algebra. * Programming skills in Python (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Recommended: * Linear Algebra and Optimization for Machine Learning (WI4635) * Statistical Inference (WI4455) * Knowledge of measure theory (as, for example, in AM2090)	
Course Contents	Statistical learning provides a probabilistic and statistical understanding of topics in machine learning. Note: this year the course will be part of the MasterMath programme. This may lead to some last-minute deviations from the information in the study guide. The overarching goal of the course is to develop methods for estimating (or 'learning') an unknown function from data or making predictions for unseen function outputs. The course aims to empower the student to make a justified decision in adopting machine learning approaches for practical problems and even design their own machine learning methodology using sound mathematical principles. This is achieved by studying key ideas, models, algorithms and theories related to the subject of statistical learning. In the first half of the course a collection of essential models, algorithms and techniques is introduced. In the second half of the course we adopt a Bayesian perspective on machine learning and study associated methods for analysing models and developing computational methodology.	
Study Goals	By the end of the course, you are able to: 1. Explain the challenges of statistical learning (e.g., curse of dimensionality, overfitting, bias-variance trade-off). 2. Implement machine learning algorithms and computational methods on artificial and real-world data sets (e.g., linear regression, logistic regression, sparse regression, neural networks, decision trees). 3. Compare the theoretical properties and practical aspects of machine learning approaches. 4. Validate machine learning approaches on arbitrary data sets. 5. Design tailor-made computational approaches for new learning problems by adapting, combining and/or extending methods and/or by employing a Bayesian approach.	
Education Method	Lectures, exercises, practical assignments, practice exam	
Literature and Study Materials	Lecture notes and slides will be made available. You may wish to consult additional references for background reading. Some suggestions of publicly available textbooks are (listed in order of suitability for this course): * Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/ (very explicit in terms of derivations; focusses on a Bayesian approach) * Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ (more advanced reference; requires some mathematical and statistical maturity since it shows relatively few derivations; focusses on a non-Bayesian approach) * Murphy, Kevin P (2022). Probabilistic machine learning: an introduction. MIT press, https://probml.github.io/pml-book/book1.html (very recent textbook on machine learning which discusses most topics; however not a very well thought out as textbook)	
Assessment	The final grade of the course consists of the following components: * Practical assignments during the course (30%) * Final written exam (70%) Final grade calculation: $(0.3 * \text{practical assignments} + 0.7 * \text{final written exam})$ Note: For both components, a sufficient grade (5.8) is required. Repair possibilities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5. * Assignments: resubmit deliverable * Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course material and literature are mostly self-contained, but prior knowledge equivalent to contents of the courses AM2010 Linear Algebra 2 and AM2020 Optimization from the Bachelor of Applied Mathematics programme is strongly recommended. Programming experience in Python or Matlab is expected.	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will discuss algorithmic foundations from the areas of linear algebra and optimization. This also includes aspects of their theory and efficient implementation. Among others, we will study + matrix decomposition and factorization techniques, + regression and classification problems and regularization, + iterative and continuous optimization methods, + hyper parameter optimization, + graph-based algorithms and clustering, + basic introduction to neural networks and algorithmic aspects neural networks (e.g. computational graphs and back-propagation)	
Study Goals	Upon completion of the course, you should be able to: + Categorize typical data sets based on their structure + Derive suitable mathematical formulations for common supervised and unsupervised machine learning tasks + Choose suitable algorithmic building blocks for solving mathematical formulations of common machine learning tasks + Analyze machine learning algorithms with respect to their properties, such as efficiency and robustness + Implement machine learning algorithms in a computationally efficient way	
Education Method	The course comprises 14 lectures and 14 exercise classes.	
Literature and Study Materials	Lecture notes, slides, and the main literature: + Charu Aggarwal. Linear algebra and optimization for machine learning. Springer, 2020. + Gilbert Strang. Linear algebra and learning from data. Wellesley-Cambridge Press Cambridge, 2019. + Jeremy Watt, Reza Borhani, and Aggelos K Katsaggelos. Machine learning refined: Foundations, algorithms, and applications. Cambridge University Press, 2020.	
Assessment	The final grade of the course consists of the following components: - Two group projects (0.4) - Written exam (0.6) Final grade calculation: $(0.4 * \text{group projects} + 0.6 * \text{written exam})$ Note: For the written exam, a score of at least 5.0 must be obtained. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group projects: resubmit deliverable - Written exam: There will be a resit for the written exam at a later point in the same academic year Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. D. de Laat	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	The final grade of the course consists of the following components: - Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). - Group programming project (group work) starting mid Q2 with submission deadline end Q2 (2/3 of final grade). Final grade calculation: $(1/3 * \text{homework assignments}) + (2/3 * \text{group programming project})$ but it is only valid when all homework programming assignments have been completed with sufficient results. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework programming assignments: submit new set of homework - Group programming project: submit improved version Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. J. Thies	

Year

Organization

Education

2024/2025

Aerospace Engineering

Master Aerospace Engineering

Profile III: Planetary Exploration	
Program Coordinator	Dr. S.J. de Vet
Introduction 1	Profile III: "Planetary Exploration" The Planetary Exploration profile is focused on objects in the Solar System and beyond: planets, moons and asteroids, and missions targeting these objects. The profile addresses how science questions drive new planetary space missions and how new knowledge is generated from measurements of past and current space missions, specifically on the structure, environment, surface characteristics and dynamics of the planetary objects. Your studies can include simulation with analytical or numerical models, analyzing observations from satellites and telescopes, performing laboratory experiments or studying field-analogues. Therefore, the profile offers theoretical components as well as practical components in design and engineering.

AE4868-1	Numerical Astrodynamics	4
Responsible Instructor	D. Dirks	
Instructor	Ir. K.J. Cowan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4874 I (Astrodynamics I) is required. Basic programming knowledge is expected.	
Course Contents	Basic orbit propagation/integration methods and options; simulation options in Tudat. Defining input for orbit propagation Linking analytical and numerical design methods in astrodynamics Details of numerical integration methods: multi-stage, multi-step and extrapolation methods (fixed and variable step size) Selection of numerical integration methods; quantification of integration errors	
Study Goals	In principle, the course will only use Tudatpy (the Python interface for the Tudat C++ software). Depending on student interest, part of the C++ layer of Tudat may be covered. Set up and run a numerical orbit propagation using the Tudat software suite from a given set of simulation settings Critically evaluate and analyze numerical orbit propagation results Interpret numerical propagation results, and relate numerical results to theoretical predictions Select numerical integration scheme for a given astrodynamics problem	
Education Method	The course uses the following education methods: Weekly live lectures covering specifics of the assignments, examples, topic introduction, etc. The bulk of the material is covered as pre-recorded Lectures on Tudat, Dynamical Modelling, Analytical vs. Numerical Mission Design and Numerical integration Working Lectures are held twice per week, in weeks 1-8 of the quarter. The course requires significant self-study by the students, both to master the material, and complete the assignments. However, ample support is provided through Brightspace, forum, and the working lecture. Successfully doing the course within a reasonable time requires students to make good use of these options.	
Computer Use	Laptop computer with at least 4 GB of RAM	
Course Relations	AE4866 (Propagation and Optimization in Astrodynamics); AE4870A (Rocket Motion); AE4872 (Satellite Orbit Determination); AE4890-11 (Planetary Sciences I); AE4894 (Practical Astrodynamics)	
Assessment	Individual take-home assignments. Assignments will consist of setting up, running, analyzing, and documenting astrodynamics simulations in Tudat. Partially completed Python code will be provided for the assignments, which is to be completed by the students, and used to answer questions in the assignments. Students will use Python for generating results during the course, and the language/tool of their choice for visualizing results. It is assumed that students are sufficiently proficient in Python (or are willing to independently make themselves so) to finish and run partially finished Python scripts. It is assumed that students are proficient in a data visualization tool (e.g. Matlab, Python's matplotlib, etc.) The resit of the course will be done during the summer period. During the resit, there will be no opportunities to ask questions of any kind (installation, coding, assignment, theory, interpretation, etc.). It is not possible to resit single assignments, only a resit of the full course is offered.	
Remarks	In case of capacity issues, access to the course may be limited for students not enrolled in the Spaceflight track. In case you want to take this course, but are following a different track, please contact the course manager.	

AE4872	Satellite Orbit Determination	6
Responsible Instructor	Dr.ir. E.J.O. Schrama	
Instructor	Prof.dr.ir. P.N.A.M. Visser	
Instructor	Dr.ir. J.A.A. van den IJssel	
Instructor	Dr.ir. B.C. Root	
Instructor	C. Siemes	
Responsible for assignments	C. Siemes	
Contact Hours / Week x/x/x/x	0/4/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	We assume that the participants are familiar with mechanics, analysis, linear algebra, and differential equations. We make use of MATLAB and the algebraic manipulation tool MAPLE, there are excellent primers for both tools. Some of the exercises assigned as student projects we expect that MATLAB and MAPLE will be used. If participants make use of other programming environments than no support can be guaranteed.	
Course Contents	Course contents 1 Dynamics - Introduction to dynamics - o Planetary Gravity field - o Tides and the three-body problem - o Hill radius and Roche limit - o Relation to planetary sciences and astrodynamics - Solving Equations of motion - o reformulate orbit problem as a system of ordinary differential equations - o efficiency and accuracy of numerical integration methods - o implementation of numerical integration methods 2 Observations techniques and reference systems - Observation techniques - o Laser, Doppler, optical and VLBI observations - o Refraction, Electromagnetism, radio- and optical technology - o Tropospheric and ionospheric refraction - o Relativity and the definition of time, - o Classification of time systems (UTC, TAI, etc) - o Light-time effect - o Quality of clocks (Allan Variance behaviour of clocks) - Reference systems - o Local and global coordinate systems - o Definition of geoid and reference ellipsoid, height systems - o Precession and nutation, polar motion, polar wander. - o Newton or Einstein, consequences for reference systems 3 Statistics - Random variables, probability density functions, moments, hypothesis testing - Least squares minimisation - o unconstrained linear parameter estimation, - o data weighting - o nonlinear parameter estimation. - Rank deficient equation systems - o Singular value decomposition and Empirical Orthogonal functions - o Penrose Moore inverse - o compatibility conditions - o general and homogeneous solutions - o constrained linear parameter estimation - Mechanisation of parameter estimation algorithms - o Choice of algorithms 4 Orbit determination - Perturbation analysis and variational problems - o state transition matrix for initial state vector problems - o partial derivatives for dynamical parameters - Parameter estimation - o Identification of parameters - o batch least squares - o Kalman filter, theory and implementation 5 Applications - Global Navigation Satellite Systems: - o Technology and terminology, - o various data processing strategies and available software - o Modelling deformation of the solid Earth, - o the Earths gravity field and thermospheric density - Satellite laser ranging and Doppler tracking via DORIS - o technology and terminology, results and applications - Observing changes in the cryosphere with satellites - Hydrology and Oceanography observed with satellites 6 Individual assignments in Q2 - An exercise related to dynamics, observation systems or reference systems - An exercise related to GNSS applied to orbit determination - An exercise related to Kalman filtering	

	7 Numerical Practical in Q3 Exercises with the Ghost software, typical examples are to solve: - o initial value problems (state vector estimation) - o problems with parameters in a dynamic model (drag parameter estimation) - o problems with time bias parameters
Study Goals	The candidate should be able to: 1) Explain the physical and mathematical aspects of orbit determination (OD), the topics are 1.1) Solar system dynamics 1.2) Equations of motion and variational equations 1.3) Parameter estimation 2) Construct transformations between various coordinate and time systems that play a role in OD 3) Examine error sources in satellite tracking data and implement error mitigation strategies 4) Make use of parameter estimation methods in the context of tracking data for OD 5) Apply relevant statistical techniques within the framework of OD 6) Discuss details relevant to scientific applications of satellite missions that depend on precise OD 6.1) GNSS techniques to model the deformation of the Earth 6.2) Satellite gravimetry to model the gravity field of the Earth 6.3) Satellite altimetry to model the ocean topography 6.4) Satellite altimeter missions to measure variations of land and sea ice 6.5) Combinations of 6.1 to 6.4 7) Apply OD with state-of-the-art software
Education Method	We provide lectures and classroom instructions during quarter 2. In quarter 3 there is a numerical practical to introduce the topics mentioned under the course contents on an MSc level for aerospace engineering students. In quarter 2 all lecture material will be made available, instruction hours will be organised to complement the material presented during the lectures, also, the instruction hours will be used to introduce individual assignments which are part of this course. In quarter 3 we run a numerical practical. During this practical software tools will be used to compute precise satellite orbits and which comes with the estimation of parameters with a state-of-the-art software package (e.g. GHOST or TUDAT). In addition a quality assessment of the results will be part of the practical. There are two variants of the numerical practical, before we start with Q3 participants will be able to make a choice between the variants.
Computer Use	Use will be made of Python or MATLAB (preferred) during the assignments, for the numerical practical use will be made of Linux and Fortran, or Python and Jupyter notebooks
Literature and Study Materials	Lecture notes will be provided on BrightSpace and they containing most of the material used in this course. Optional reference material: Statistical Orbit Determination, Schutz, Tapley and Born, Academic Press, ISBN: 9780126836301 Montenbruck and Gill, Satellite Orbits, Springer Verlag, doi: doi: 10.1007/978-3-642-58351-3 Planetary Sciences, Imke de Pater and Jack Lissauer, Cambridge University Press Additional information could be provided via BrightSpace
Assessment	The final grade will come together from: 1) Written examination resulting in a grade 2) Nominally two individual assignments resulting in a weighted average grade 3) Numerical practical resulting in "pass" or "fail". The rules are as follows: 1) The numerical practical result should be: "pass" 2) The written examination result should be at least a 5,0 3) In order to participate in the numerical practical you must have passed the individual assignments with at least a weighted average of 5,0 4) No claims can be made with regard to individual assignment results or numerical practical results obtained before the previous academic year. For ae4-872 in 2024-2025 it means that results from 2023-2024 are still accepted, however, we do not accept results from 2022-2023 or earlier academic years. (Academic years run from September to August)
Exam Hours	Conform the regulations there are two written exam opportunities within an academic year. These are three hour written exams at the end of quarter 2 and 3.
Permitted Materials during Tests	We provide a closed book examination, required information for the exam (like physical constants or a selected number of equations) can be made available on the examination form.
Tags	Analysis Elektricity & Magnetism Lineair Algebra Matlab Numeric Methods Physics Signals and Systems Technology

AE4874 I	Fundamentals of Astrodynamics	4
Responsible Instructor	Ir. K.J. Cowan	
Instructor	Dr. S. Gehly	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	ae4866 ae4868 ae4870 ae4872 ae4889 ae4894	
Summary	Thorough treatment of the fundamentals of astrodynamics; topics are treated in a mathematically cohesive way. You will acquire fundamental insights into the theory and methods of astrodynamics as well as working knowledge to compute and analyze spacecraft trajectories.	
Course Contents	Introduction to astrodynamics; two-body and many-body problems; relative motion; coordinates, reference frames, orbital elements, and time; geocentric, cislunar, and interplanetary flight; introduction to orbital perturbations.	
Study Goals	You will be able to: 1. Provide full and accurate descriptions of the fundamental concepts and phenomena of astrodynamics; 2. Explain the relationships between astrodynamics concepts & phenomena and the mathematics & physics theory that describes them; 3. Apply the relevant relationships, models, and methods for astrodynamics analysis; 4. Derive and manipulate the equations of motion for a spacecraft subject to gravitational and other forces; 5. Apply the equations of motion, their solutions, and related equations to analyze the absolute and relative positions and velocities of spacecraft as a function of time. Additional detail will be available in the course syllabus.	
Education Method	Lectures, workshops, self study	
Literature and Study Materials	Fundamentals of Astrodynamics, Karel Wakker, 2015. ISBN: 978-946-186-4192 available for download from the TU Delft Library / Institutional Repository also available to order as university reader in hardcopy Astrodynamics course website Orbital Mechanics for Engineering Students, Howard Curtis, 2013, 3rd edition. ISBN: 978-008-097-7478 (print) 978-008-097-7485 (ebook) available for download from the TU Delft Library also available to order from booksellers as hardcopy	
Assessment	Written exam	

AE4876-24	Planetary Missions	5
Responsible Instructor	Dr.ir. W. van der Wal	
Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	Dr.ir. E. Mooij	
Instructor	Prof. Dr. L. Gurvits	
Instructor	Dr. A. Menicucci	
Instructor	Dr. S.M. Cazaux	
Instructor	Dr. M.R.N. Rovira Navarro	
Instructor	Ir. L.H. Svedhem	
Instructor	E.S. Steenstra	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Important: the course has a cap of 40 students because of the number of staff members available to supervise. It is therefore mandatory to register for this course through Osiris. Priority will be given to students from the Planetary Exploration profile. Lectures by topical experts give deeper insight and understanding in planetary sciences topics, with an emphasis on applied technologies, and previous, current and future space missions, as well as data analysis. Lecture topics include: Planetary missions, space missions at ESA, JUICE mission to the Jupiter system, Earth observation, Gravity missions, Harmony mission for Earth observation, space dust, radiation & space weather, re-entry. Students will work in groups of 5 to 6 students on the conceptual design of a planetary space mission, supervised by a staff member from the Planetary Exploration section. The focus is on coming up with the science questions (the "why" of the mission) and the instruments and operations needed to answer the science questions.	
Study Goals	After completion, the student should be able to: - Explain the phases of a planetary mission, from concept to end-of-life. - Deduce science questions from the literature that can form the basis for a competitive planetary mission proposal - Describe the road map from quantities measured by a space mission to answers of the science questions - Produce a conceptual design of a planetary mission in a team, taking into account the state-of-the-art in instrument development and current and past missions. - Present in writing an orally a competitive proposal for a planetary mission.	
Education Method	Oral lectures and group work on a planetary mission proposal. 4 lecture hours per week are reserved for lectures, 2 lecture hours are reserved for project time for group work or mid-term and presentations. The group will need to work outside of the scheduled time as well.	
Literature and Study Materials	The lecturers will provide slides and recommended literature on Brightspace. General recommended literature: Fundamental Planetary Science by J. J. Lissauer & I. de Pater, Cambridge University Press, ISBN: 978-0521-61855-7 (paperback) This book is the same as the one that is used in the core course Planetary Exploration.	
Assessment	The grade for the course is a group grade for the Planetary Mission Project. Two partial grades will be given for the following components: - A document with science questions based on a literature survey submitted in the first half of the course - Report and poster of the final conceptual design There is no written exam or individual homework assignment for this course.	

AE4890-11	Planetary Exploration	4
Responsible Instructor	Prof.dr. L.L.A. Vermeersen	
Instructor	Dr.ir. W. van der Wal	
Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J.M. Potin	
Instructor	Dr. S. Paardekooper	
Instructor	Dr. S.J. de Vet	
Responsible for assignments	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4876 Planetary Sciences II	
Parts	Planetary Sciences: - Origin and Evolution of our Solar System - Planetary Surfaces and Interiors - Planetary Atmospheres - Planetary Magnetism - Planetary Ring Systems - Asteroids and Comets - Extrasolar Planets - Recent Topics in Planetary Science Current topics in planetary science with a focus on a specific topic each time the course is offered, such as Mars, Venus, Galilean satellites, comets, recent discoveries, spacecraft observations, or other developments.	
Course Contents	Planetary science is a major interdisciplinary field, combining aspects of astrophysics with geology, geophysics, meteorology, atmospheric and space science. This close relationship to geophysics, atmospheric and space sciences implies that the study of the planetary bodies in our Solar System and beyond offers the unique opportunity for comparison available to Earth scientists. This course teaches the concepts in the planetary physical sciences and solar system properties. The learning process is greatly enhanced by involving students in solving related problems.	
Study Goals	After the course the student should be able to: 1. Describe the primary physical processes in solar and planetary systems and the properties involved 2. Apply modelling techniques that are commonly used to describe the primary physical processes in solar and planetary systems 3. Demonstrate the use of physical principles to derive knowledge on the state and evolution of (extra-)solar system bodies from observations 4. Assess the quality of evidence of current knowledge of (extra-)solar system bodies 5. Assess the habitability of a (extra-)solar system body based on inferences of its properties and orbit	
Education Method	Classical oral lectures and classroom/take home assignments.	
Literature and Study Materials	Jack J. Lissauer and Imke de Pater, Fundamental Planetary Science, Cambridge University Press, ISBN: 9780521618557, 2013. PowerPoint/PDF files of the lectures and lecture notes.	
Books	Jack J. Lissauer & Imke de Pater, Fundamental Planetary Science, Cambridge Univ. Press, 2013.	
Assessment	The final grade for the planetary sciences course is determined from: - mark of a written exam - mark for mandatory assignments The mark for the written exam needs to be 5.0 or higher in all cases, no exceptions. If the mark for the exam is 5.0 or higher, grading for this course follows from averaging two parts with equal (so 0.5) weight: the mark for the classroom/take home exercises and the mark for the written exam. No opportunity will be given to redo the classroom/take home exercises within the same academic year. The grade for the classroom/homework assignments will remain valid for the next consecutive academic year, and so will annul the year thereafter	
Exam Hours	3 hours written exam	
Permitted Materials during Tests	Closed-Book exam, Non-programmable calculator is allowed.	
Set-up	Oral lectures and classroom/take home assignments.	

AE4893-24	Physics of Planetary Interiors	5
Responsible Instructor	Dr.ir. B.C. Root	
Instructor	Dr. M.R.N. Rovira Navarro	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course focuses on the different aspects of numerical modelling of planetary interiors. The interior of a planet or moon can be studied via observations of its gravity field, shape, surface features, rotation, tidal deformations and orbital evolution. To interpret these observations, the response of the bodies to forcings (thermal and mechanical) need to be modelled. As a student you will get hands-on experience in modelling planetary bodies with various numerical code packages. Different problems will be discussed, including solving the Stokes equation for internal mantle convection, static and time-variable gravity forward modelling as well as orbital evolution and relation to interior dynamics. You will be able to study a range of internal solid and fluid processes, consider their role in planetary evolution and link them to existing and planned observations. Lecture topics include: 1. Observations related to planetary interiors: Gravity field, rotation, tides, shape (topography, faults) - Example bodies and learn about different internal processes. You will learn how to calculate the internal gravity, density, and pressure of these bodies. 2. How to model fluid-solid mechanics of a planet? - Stokes equations in planetary science in spherical coordinates, rheology - Heat-transport: state equation exercise with different heat regimes - Mantle convection applications: dynamics and ocean flows 3. How to perform gravity field modelling - non-uniqueness, advanced isostasy/flexure models, density anomalies - Forward modelling density anomalies: spectral vs. volumetric methods - Inversion of lithosphere structure - Effect of mantle convection on the gravity field - Lessons learned from seismology on Earth 4. Tidal and loading deformation - Tidal and surface loads - Forward modelling of time-variable gravity field and deformation - Love numbers and they connection with interior properties. - Use of observations to constrain interior properties (e.g., detection of subsurface oceans, tidal heating in Io, etc) 5. Rotational dynamics - Rotational dynamics basics. - Main drivers of rotational dynamics - Effect of surface loads in the spin of a planet (Polar Wander) - Effects of tides in the spin of a planet (tidal de-spinning, spin-orbital resonances, librations) - Using observations of the rotation of a body to constrain its interior (e.g., librations and subsurface oceans) 6. Interior and orbit evolution - Perturbations to Keplerian orbits - Effect of tides in the orbital evolution of planets and moon - Third body perturbation effects, including secular and mean motion resonances - Feedback between interior and orbital evolution - Use of orbital observations to constrain the interior of planets.	
Study Goals	After the course you will be able to: 1. Describe the physical processes shaping planetary interiors, the numerical and analytical methods employed to study them, and their connection to observations 2. Analyse the physical processes shaping planetary interiors at different stages of planetary evolution and understand how they are interconnected. 3. Apply fundamental physical laws (Navier-Stokes, Poisson equation, etc.) in numerical modelling setups and assess the applicability of numerical codes in the planetary exploration. 4. Assess common approximations and limitations used in numerical models and observations in relation to the scientific study. 5. Critically read and analyse literature in planetary sciences, recognise the novelty and implications of relevant publications and be able to formulate new research questions in this context. 6. Infer the properties of planetary interiors by constraining models with available observations and present new findings in front of a scientific audience	
Education Method	Two lectures and one work session per week During the work sessions you can work on the 3 assignments focussing on different numerical methodologies Numerical modelling exercises will help you to get familiar with different numerical codes in the planetary science field. The assignments will be a selection of different numerical exercises connected to the lecture material that you will apply to a solar system body. First numerical exercise: How to model the radial profile of a planetary body (learning goal 1 and 2) Propagate state equation of an icy moon for different heating conditions Apply the heat transport equation to understand the phase transition, investigate conduction vs. convection Validate your code with literature (learning goal 5) Second numerical exercise: Gravity field modelling assignment (learning goal 4) Study best way to compute Bouguer anomaly of the planet: spectral vs. volumetric Perform gravity inversion of for 3D crustal/lithosphere structure. Apply isostasy and flexure theory Third numerical exercise: Assignment using existing software to run simulations tidal dynamics. Computation of Love numbers and comparison with available or future observations, and role of tides in spin/orbital evolution. Deliverable will be 3 scientific reports on the research you perform in these assignments, this to better prepare you in writing scientifically papers for your career as planetary scientist. A final poster presentation during a symposium simulates a conference set-up and you can reflect on your work with your fellow peers.	
Course Relations	Complementary to other courses: Planetary exploration, Planetary Missions and Measurement Strategies CIE4610 Gravity, Geodynamics and Climate Change; Spherical harmonics and basics of potential theory are treated there, but not gravity forward modelling. Several courses at TU Delft discuss Navier-Stokes equation, FEM and inversion, however by focussing on spherical coordinates	

Literature and Study Materials	and spherical shell models we believe this course is much more suited for potential planetary scientists/engineers. Various book chapters and peer-review journal articles: Murray, Carl D., and S. F. Dermott. 1999. Solar System Dynamics. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9781139174817 (ebook available in library webpage) Sabadini, R., Bert Vermeersen, and Gabriele Cambiotti. 2016. Global Dynamics of the Earth : Applications of Viscoelastic Relaxation Theory to Solid-Earth and Planetary Geophysics. Second edition. Dordrecht: Springer. https://doi.org/10.1007/978-94-017-7552-6 (ebook available via library webpage) Lecture notes and slides Software and tutorial manual
Assessment	Numerical assignments: (grade assignment 1 + grade assignment 2 + grade assignment 3)/3 is 90 percent of the final grade. the final 10 percent of the grade is obtained during the poster presentation at the symposium.

AE4895-24	Measurement Strategies for Planetary Science Missions	4
Responsible Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	For planetary science the research question has a profound impact on the measurement strategy and instrument to be used. Studying planetary objects and materials in the Solar System thus hinges on observing and obtaining measurements from either nearby (in contact) and from afar (remote sensing). In this course you will follow lectures, learn more about planetary analogues and gain hands-on experience as you explore different measurement techniques in the lab. By applying measurement techniques to a specific rocky or ice sample, and to a planetary analogue material, you will try to find out how the data products and the scale at which it is applied influences the research questions that a science mission can address.	
Study Goals	After completion of the course, the student will have obtained practical experience and gained a deeper understanding of how measurement techniques contribute to characterizing a physical-chemical property of a planetary object. After this course the student will be able to: LO1: Explain key scientific questions that space exploration seeks to answer for rocky (Mars) and icy (icy moons) planetary bodies. LO2: Understand the underlying physical principles of a measurement technique. LO3: Demonstrate to have working knowledge of measurement techniques by participating in a structured laboratory exercise. LO4: Evaluate how measurement techniques result in an improved understanding of materials and processes in planetary sciences.	
Education Method	Lectures; structured laboratory exercises focussing on either a rocky or an icy planetary analogue material.	
Assessment	Laboratory activities and maintaining a lab journal (20%) Presentation lab results and planetary integration (20%) Final report, in form of a short communication in paper format (60%)	
Enrolment / Application	The course has a limited capacity of 20 students, which will be allocated to students from the profile planetary exploration first, and then on the basis of first come, first serve (potentially filled up with students from other profiles and/or programs). This limit of 20 could be higher depending on the actual capacity of the laboratory facilities	

AE4896	Space instrumentation	4
Responsible Instructor	Dr. J.J.D. Loicq	
Contact Hours / Week x/x/x/x	4/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	The aim of this course is to provide students with an understanding of instrumentation and its role in observational strategies for missions. The course will explore the main observables of the universe and how they are measured in space. For each of the observables, the course will provide: i) The physical context of the observables and how they are generated, ii) The physics of the measure, iii) A review and explanation of the existing and under development space instruments, iv) The missions related to these observables. The course will be divided into three main parts: - The Fundamentals of Instrumentation - Remote sensing space Instrumentation - In-situ space Instrumentation.	
Study Goals	This course aims to familiarize students with the wide range of space instrumentation and observables. By the end of the course, students will be able to: - Understand and describe the relationships between instruments, missions, and observation strategies - Identify and explain the physical observables that space instruments measure, including light, ions, magnetic and electric fields, etc and related phenomena - Distinguish between the different types of space instruments, and explain how they are designed and constructed - Derive instrument requirements from scientific questions - Perform preliminary analysis and use the key driver parameters to size space instruments. Overall, this course will equip students with a comprehensive understanding of space instrumentation, enabling them to dive into space observables diversity and the complex processes involved in their measurement.	
Education Method	Lectures	
Assessment	The exam will test all theories provided in the lectures and skills practised in the provided exercises. It will be a closed-book written exam (3h). Formula sheet will be provided. The resit will have the same settings.	

AE4S12-23	Space systems engineering	4
Responsible Instructor	Prof.dr. E.K.A. Gill	
Contact Hours / Week x/x/x/x	4/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	General engineering knowledge at completed BSc level. AE1110, AE1111, AE2111, AE3211, or similar; AE3524 (for students, not having AE BSc)	
Parts	1. Introducing Space Systems Engineering 2. Designing Systems 3. Integrating Systems 4. Managing the Systems Engineering Process 5. Managing the technical effort 6. Managing interfaces and configuration; Conclusions	
Course Contents	The course covers Space System Engineering exemplified by the design of a spacecraft subsystem for a satellite mission of high societal impact. It introduces a process which allows the creation of successful systems, such as space applications, space missions and satellite subsystems. Methods and tools are presented and exercised which will improve the depth and breadth of Space Systems Engineering graduate level education at the TU Delft by emphasizing the need for the end-to-end approach and life cycle of space systems, including cost and risk handling and using state-of-the-art methods, such as elements of Model-based Systems Engineering. Upon completion, the student will have a firm understanding of advanced Systems Engineering and be able to apply adequate methods and tools which help to create successful systems.	
Study Goals	The course mission is to enable students to realize a successful space system in an end-to-end Systems Engineering approach. There are three high-level learning objectives: 1. Participants shall be able to explain the objectives and tasks of Systems Engineering for realizing successful systems together with their needs, potentials, benefits and limitations in a context which comprises Business Engineering and Management. 2. Participants shall be able to design an end-to-end Systems Engineering process demonstrating a smart balance of risks of cost, schedule, and performance. 3. Participants shall be able to use Systems Engineering methods and tools to design a high-level satellite subsystem, such as the Attitude and Orbit Control System (AOCS).	
Education Method	Lectures and tutorial Individual homeworks from lectures to practice material Group Assignment including peer-reviews to design a satellite sub-system, done by students in small groups The graded group report and graded individual homeworks comprise the formal course close-out.	
Literature and Study Materials	Required: Lecture Slides AE4-S12 Space Systems Engineering. Optional: Larson W. J.; Kirkpatrick D.; Sellers J. J.; Thomas L.D.; Verma D.: Applied Space Systems Engineering. McGraw Hill (2009). Optional: Space Mission Analysis and Design, 3rd edition (Space Technology Library, Vol. 8) Eds.: Wiley J. Larson, James R. Wertz (1999)	
Assessment	4 graded individual homeworks + 1 Group Assignment. Acceptance of 6 homeworks is required to pass the course. Passing both assessment elements is required to pass course.	
Enrolment / Application	Students have to enroll via Brightspace. Professionals are welcome to the course. Conditions for their participation and registration are available with the Management Assistants of the Department of Space Engineering.	
Set-up	The course comprises lectures, guest lectures, tutorial, group work, peer-review, and individual homeworks.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project.

The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5822	Thesis Space	45
Responsible Instructor	Prof.dr. E.K.A. Gill	
Responsible Instructor	Prof.dr. L.L.A. Vermeersen	
Responsible Instructor	Prof.dr.ir. P.N.A.M. Visser	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	e MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal).	

If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go' (fail). Please refer to Brightspace for more information.

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
	MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Space

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4143	Hypersonic Aerodynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Instructor	Dr.ir. L.M.G.F.M. Walpot	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* AE2130-III Aerodynamics 2 (Compressible Aerodynamics) or equivalent. In case you did not have such a course, please study chapters 7 to 12 of Fundamentals of Aerodynamics by J.D. Anderson * It is very highly recommended to also follow AE4120 (Viscous flows) * AE4140 Gasdynamics is nice to follow as well but not absolutely necessary	
Parts	Lecture and study material 1. Introduction: characteristics of hypersonic flow; applications. 2. Inviscid flow: 2D shocks and expansions; Newtonian theory. 3. Tangent- wedge/tangent cone-, shock-expansion theories; similarity parameters. 4. Euler equations; small perturbation theory; hypersonic equivalence principle; blast-wave theory. 5. Thin shock layer theory; computational results. 6. Viscous flow: boundary layer equations; laminar boundary layers. 7. Transition; turbulent boundary layers. 8. Viscous interactions; flat plate application. 9. Heat transfer: convective heating; results for flat plate, cone, blunt body. 10. Laminar and turbulent stagnation region heating; radiation: effects on surface heating; radiation from the shock layer; nocturnal radiation. 11. High temperature effects; chemical reactions; inviscid equilibrium flow; difference between equilibrium flow and frozen flow. 12. Non-equilibrium flow; high temperature effects in flow through normal and oblique shocks. 13. High temperature effects in flow through nozzles. 14. Chemical reacting viscous flow	
Course Contents	Hypersonic Flow: Inviscid flow, compressions and expansions, surface inclination methods, similarity rules, hypersonic small perturbations, Euler equations. Viscous flow, boundary layers, viscous interactions, heat transfer, radiation effects. High temperature effects, inviscid and viscous equilibrium and nonequilibrium flow. Rarefied gasdynamics.	
Study Goals	This course is designed to provide the student with fundamental knowledge of hypersonic aerothermodynamics. The course is set up to deliver engineering tools for conceptual design problems together with a physical understanding of various aspects of the broad gamma of hypersonic flow.	
Education Method	Lecture	
Literature and Study Materials	J.D. Anderson, High temperature gas dynamics, , - ISBN 0070016712 Recommended literature J.J. Bertin, Hypersonic aerothermodynamics, , - ISBN 1563470365	
Assessment	Oral exam	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4204	Knowledge Based Engineering	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ing. A. Heidebrecht	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some familiarity with Object Oriented modelling and (Python) programming	
Course Contents	The course will provide a theoretical and practical introduction to Knowledge Based Engineering (KBE) technology, as a means to address complex engineering problems through design automation/digitalization principles. Other addressed topics, strictly related to KBE include - UML/SysML modelling language (class and activity diagrams) - object oriented modelling paradigm	
Study Goals	The main goals are the following: - understanding of Knowledge Based Engineering theoretical foundations and programming principles and their application to support complex engineering design problems - hands-on experience in developing simple but representative KBE applications by means of a provided KBE system - ability to model product and process knowledge by means of UML/SysML	
Education Method	The course is organized into a set of interactive lectures and tutorials to be performed part in class and part at independently at home. During the tutorials, students are introduced to the KBE programming approach. After a quick refresh of the python language, the main functionalities of a commercial KBE system will be illustrated, through interactive programming sessions. A significant amount of time is left for self-study, where students will have to develop a KBE application in teams of two, to address an engineering design problem of their choice. Attendance to lecture and tutorials is not compulsory but very highly recommended. Although the course is spread over two periods, the second one (Q4) is reserved for self-study (i.e. the development of the KBE application) and the oral examinations (See assessment section)	
Computer Use	The use of a personal computer is required, with the Windows OS installed A selected KBE system will be used for the practical sessions and assignment (temporary licenses for this course will be provided)	
Course Relations	This course has a logical connection to the MDO course AE4205, being KBE an enabling technology to multidisciplinary design, analysis and optimization. However, the MDO course is not propaedeutic to this course, thus students who did not followed MDO can take the KBE course without any problem	
Literature and Study Materials	- lecture sheets and some lecture notes (only available digitally on Brightspace) - a few scientific papers - several programming tutorials and code samples	
Books	Suggested (not compulsory) book: Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 Please note that only Chapter 9 is required for this course (single chapters can be purchased on line at Wiley). The rest of the book is suggested literature for the MDO course AE4205 The book is available as e-book through the TU Delft library	
Assessment	There is no conventional written exam for this course. The assessment is based on the quality of a KBE app to be developed at home by teams of two students and demonstrated live during an oral examination session. Two examination sessions are planned per academic year. The KBE app is supposed to address a relevant engineering design problem, which will be proposed by the student teams in the early phase of the course. The proposed KBE assignment descriptions will require lecturer's approval. If during the oral examination the quality of the developed KBE application and its demonstration are evaluated to be unsatisfactory, the team will have to use the retake session. For the retake session, two options are offered: 1) submit and present an improved version of the app proposed for the nominal session. One penalty point applies. 2) submit and present a totally different app than proposed for the nominal session. No penalty points. The new KBE assignment proposal will have to be delivered after the nominal session and will require lecture's approval. The oral examinations (nominal and retake sessions) are taking place around the beginning and the end of the second period of the course (i.e. 4th quarter of the academic year), not necessarily during the standard exam weeks. Delivery dates and examinations days are announced at the beginning of the course	
Special Information	Temporary student licenses of ParaPy, the KBE system used in this course, will be provided only to the students actively participating to the course. Thus, not to guests or other participants enrolled via Brightspace. The licenses shall be used exclusively for the course assignment.	
Tags	Artificial intelligence Computer Engineering Group work Integrated Modelling Programming Programming concepts Programming Software Software Engineering	

AE4205	MDO for Aerospace Applications	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ir. M.F.M. Hoogreef	
Instructor	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Familiarity with MATLAB and basic aircraft design knowledge. MATLAB is used for tutorials and home assignment, thus students without MATLAB experience must gain this through self-study Students with no experience in aircraft design should consult textbooks like Torenbeek, Raymer and Roskam to familiarize with the fundamentals of wing sizing process and general concepts such as wing loading, aerodynamic coefficients, fuel fraction method, drag and its components, etc.	
Course Contents	The course will introduce students to multidisciplinary design optimization (MDO), as a modern methodology to support complex product development. Main topics: - What and why MDO - Parameterization techniques - Numerical optimisation methods (focus on gradient based methods) - Decomposition and coordination of complex computational systems - the eXtended Design Structure Matrix (XDSM) - MDO architectures - Intro to surrogate modeling, discrete&mixed integer problems, genetic algorithms and Multi Objective Optimisation	
Study Goals	The main goals are the following: - understanding of Multidisciplinary Design Optimization (MDO) methods and architectures - hands-on experience in setting up and solving MDO problems	
Education Method	The course consists of classical lectures and computer tutorials using own laptop. During the tutorials, students are introduced to some of the tools discussed during lecture and can make hands-on experience during interactive sessions with the teachers. In particular, students will be introduced to the MATLAB optimization function fmincon and its use for gradient based optimization. A significant amount of time is left for self-study, where students will have to work in pairs on one take-home assignment.	
Computer Use	The use of a personal computer is required MATLAB (including its optimization toolbox) will be used for the MDO practical sessions and assignment. Student licenses are provided by the TUD	
Literature and Study Materials	- lecture sheets and some notes (only available digitally on Brightspace) - scientific papers - some software modules - Suggested (not compulsory) book: - Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 - The book is available in e-format from the TUD library (Chapter 9 is used in the course AE4204 on Knowledge Based Engineering) - Engineering Design Optimization, by J.R.R.A. Martins and A. Ning (free on line)	
Assessment	There is no conventional exam for this course. The assessment is based on the quality of one take-home assignment, which will be executed in teams of two students. The assignment is about setting up and solving an MDO problem using Matlab. The deliverables are the Matlab code and a report. Two different take-home assignments are defined for the two yearly examination sessions.	
Tags	Algorithmics Analysis Matlab Modelling Optimalisation Programming Project	

AE4238	Aero Engine Technology	4
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	Dr. F. Yin	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery", "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Inlets and Exhaust - o Turbomachinery - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls - o New Engine /propulsion Concepts	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an flipped course in which the lectures are pre-recorded and the students are supposed to watch the lectures before coming to the classroom. The classroom interaction is mainly to have discussions and for tutorials. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13. Evaluate the calorific values for various fuels 14. Understand the basics of combustion and emission formation in Aero engines 15. Understand the working of gas turbine combustor 16. Understand the basic principles of gas turbine controls and evaluate different strategies for control 17. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a flipped teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

Reader	1. Gas Turbine Theory by H.I.H.Saravanamuttoo, G.F.C. Rogers, and H. Cohen 2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Prerequisites	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Assessment	The assessment is based on 2-3 assignments given during the course and a final exam. All assignments together will form 30% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
Permitted Materials during Tests	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4302	Avionics and Operations	3
Responsible Instructor	Dr.ir. C. Borst	
Instructor	Ir. T.J. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4316 AE4316P AE4321-15 AE4322	
Expected prior knowledge	Flight Dynamics System and Control Theory	
Parts	Week arrangement Lecture and study material 1. Introduction to avionics systems. -. Air Data systems (home study). 2. Gyroscopes, attitude reference systems. 3. Compasses, heading reference systems. 5. Flight deck instruments and integrated systems. 6. Navigation equations. 7. The Flight Management System (FMS). 8. Inertial Navigation Systems. 9. Radio navigation systems (ADF, VOR, DME). 10. Landing guidance systems (ILS, GPS). 11. Communication, Navigation, Surveillance (CNS). 12. Satellite navigation systems (GPS). 13. Air Traffic Management (ATM). 14. The Future Air Navigation System (FANS). (see week arrangement)	
Course Contents	This course provides a comprehensive, unified coverage of the principles of modern navigation equipment and systems, both in the aircraft and on the ground, including the aircraft instrumentation and flight-deck systems, with a special emphasis on the important trends in the global air navigation and air traffic management system.	
Study Goals	1. The student can describe in detail the working principles of the avionics systems treated in the course. 2. The student can demonstrate the avionics systems' functionalities, identify their strong points and weaknesses, and make comparisons between the avionics systems. 3. The student can evaluate, criticize, and appraise their usage in the current and future operational context.	
Education Method	Lecture	
Literature and Study Materials	Lecture slides and other lecture material are put on the BrightSpace The books mentioned below (Pallett & Coombs, Kayton & Fried) can be hired at the Control & Simulation division. Recommended literature: E.H.J. Pallett, L.F.E.Coombs, Aircraft Instruments and Integrated Navigation Systems, Longman, 1997 ISBN 0-582-08627-2. M. Kayton, W.R.Fried, Avionics Navigation Systems, Wiley, 1997 ISBN 0471547956.	
Assessment	Digital, closed-book exam on campus	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical singularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments. The assignments will be handed out near the end of Q3 and the hand-in deadline will be near the end of Q4. Exact dates will be announced during the lectures.	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Physical intelligence for interaction - Insect intelligence and Swarming - Spiking neural networks	
Study Goals	After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.	
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4866-1	Propagation and Optimisation in Astrodynamics	5
Responsible Instructor	D. Dirksen	
Instructor	Dr. S. Gehly	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4874 I (Astrodynamics I) is required; AE4868-1 (Numerical Astrodynamics) is required; Basic programming knowledge is expected.	
Course Contents	1. Introduction. Explanation of course assignment. Introduction to software. 2. Numerical propagation. Formulations of equations of motion. Selection of propagation method. 3. Overview of formulation of acceleration models. Relation between acceleration models and environment models. Influence of acceleration model selection on simulation fidelity/runtime. 4. Introduction to local optimization methods. Constraints in optimization problems. Robustness of optimal solutions. Optimal control. Examples of local optimization in astrodynamics. 5. Global optimization methods (genetic algorithms, particle swarm, etc.). Multi-objective optimization. Examples of global optimization in astrodynamics 6. Pre/post-processing of results. Analysis of numerical propagation/optimization results. Use of numerical tools (e.g. interpolation).	
Study Goals	1. Select and define the settings of a numerical astrodynamics simulation problem, based on top-level description and requirements. 2. Describe the relation between models of the physical environment and the propagated dynamical behaviour. 3. Define and set up local and global optimization problems for typical astrodynamics problems 4. Assess the quality of numerical orbit propagation for typical astrodynamics problems 5. Assess the quality of optimization results for typical astrodynamics problems	
Education Method	The course uses the following education methods: Weekly live lectures covering specifics of the assignments, examples, topic introduction, etc. The bulk of the material is covered as pre-recorded lectures on propagators, environment, acceleration models, design space exploration, optimization Working Lectures are held twice per week, in weeks 1-10 of the quarter. The course requires significant self-study by the students, both to master the material, and complete the assignments. However, ample support is provided through Brightspace, forum, and the working lecture. Successfully doing the course within a reasonable time requires students to make good use of these options.	
Computer Use	Laptop computer with minimum 4 GB RAM	
Course Relations	AE4870A (Rocket Motion); AE4870B (Re-Entry Systems); AE4872 (Satellite Orbit Determination); AE4890-11 (Planetary Sciences I); AE4894 (Practical Astrodynamics)	
Literature and Study Materials	Selected articles; lecture notes; lecture slides (details to be announced and distributed during the course).	
Assessment	The assessment is a combination of two individual and one group take-home assignments. The individual assignments are thematically related to the group assignments, and are used as a starting point for the group work . Students will use Python for generating results during the course, and the language/tool of their choice for visualizing results. It is assumed that students are sufficiently proficient in Python (or are willing to independently make themselves so) to finish and run partially finished Python scripts. It is assumed that students are proficient in a data visualization tool (e.g. Matlab, Python's matplotlib, etc.) For the group work, individual contributions to the group work must be clearly identifiable. The total grade is composed of: Individual assignments (individual grade): 50% Project (individual grade): 25% Project (group grade): 25% Each of these three grades must be ≥ 5.0 to pass the course. The final grade for the course must be ≥ 6 to pass the course. In the assignments, students will apply the material covered in the course to the analysis of a real astrodynamics problem. Weekly practical sessions will be held to discuss questions and issues with homework, the practical assignment and the numerical implementation. The resit of the course will be done during the summer period and/or Q1 of the following academic year. During the resit, there will be no opportunities to ask questions of any kind (installation, coding, assignment, theory, interpretation, etc.). It is not possible to resit single assignments, only a resit of the full course is offered.	
Remarks	**NOTE:** This course will have limited capacity (maximum ~25-30 students). The procedure on student admittance (if the maximum is reached) will be communicated to the students at the beginning of the academic year through Brightspace. In case of oversubscription, preference will be given to students from the Spaceflight track. To take part in the course, students must submit their preferences for an assignment topic. These must be submitted through Brightspace on the first Wednesday of the course's first lecture week, at the latest. Only enrolling for the course on Brightspace is NOT sufficient to indicate you wish to take part in the course. Instructions for submitting a topic preference will be posted on Brightspace one week before the start of the course, at the latest. Not submitting your topic preference in a timely manner means that you will NOT be able to take part in the course. For this course, it is expected that students are on campus for the full duration of this course (weeks 1-8 of the quarter in which it is taught). Being away for one week or more may result in exclusion from the course. In case of doubts or concerns, please contact the course manager. Working lectures are held during week 1-8 of the period, where students are expected to be present. It is required that students have followed the course AE4868 (Numerical Astrodynamics), and have submitted all assignments, and can still reasonably pass that course, before taking part in this course.	

AE4889	Special Topics in Astrodynamics	2
Responsible Instructor	Dr.ir. M.J. Heiligers	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	A modular and research-focussed course on advanced topics in astrodynamics. You will extend and deepen your insights and analytical abilities based on Fundamentals of Astrodynamics and related coursework, devoting particular attention to their application to a selection of advanced astrodynamics topics (see below).	
Course Contents	The course treats a selected set of advanced topics in astrodynamics; the exact topics change from year to year depending on the latest research developments and the instructors involved. Strictly as examples, the following topics were covered in previous years: - Circular restricted three-body problem (equilibrium points, their stability and manifolds, and periodic orbits around the equilibrium points, their stability and manifolds) - Solar-sail orbital dynamics (acceleration models and artificial equilibrium points in the circular restricted three-body problem) - Machine learning applied to astrodynamics	
Study Goals	Within the context of the selected astrodynamics topics: - Extend your insight and analytical abilities beyond the Fundamentals of Astrodynamics and related coursework; - Actively apply your knowledge and skills accumulated through Fundamentals of Astrodynamics and related coursework; - Tackle "open-ended assignments with a hands-on, scientific approach as a first foray into astrodynamics research.	
Education Method	Workshops, lectures, self-study	
Computer Use	Yes	
Prerequisites	Fundamentals of Astrodynamics (AE4874 I)	
Assessment	Assignments Additional detail will be available on the course website (Brightspace)	

AE4895-24	Measurement Strategies for Planetary Science Missions	4
Responsible Instructor	Dr. S.M. Cazaux	
Instructor	Dr. S.J. de Vet	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	For planetary science the research question has a profound impact on the measurement strategy and instrument to be used. Studying planetary objects and materials in the Solar System thus hinges on observing and obtaining measurements from either nearby (in contact) and from afar (remote sensing). In this course you will follow lectures, learn more about planetary analogues and gain hands-on experience as you explore different measurement techniques in the lab. By applying measurement techniques to a specific rocky or ice sample, and to a planetary analogue material, you will try to find out how the data products and the scale at which it is applied influences the research questions that a science mission can address.	
Study Goals	After completion of the course, the student will have obtained practical experience and gained a deeper understanding of how measurement techniques contribute to characterizing a physical-chemical property of a planetary object. After this course the student will be able to: LO1: Explain key scientific questions that space exploration seeks to answer for rocky (Mars) and icy (icy moons) planetary bodies. LO2: Understand the underlying physical principles of a measurement technique. LO3: Demonstrate to have working knowledge of measurement techniques by participating in a structured laboratory exercise. LO4: Evaluate how measurement techniques result in an improved understanding of materials and processes in planetary sciences.	
Education Method	Lectures; structured laboratory exercises focussing on either a rocky or an icy planetary analogue material.	
Assessment	Laboratory activities and maintaining a lab journal (20%) Presentation lab results and planetary integration (20%) Final report, in form of a short communication in paper format (60%)	
Enrolment / Application	The course has a limited capacity of 20 students, which will be allocated to students from the profile planetary exploration first, and then on the basis of first come, first serve (potentially filled up with students from other profiles and/or programs). This limit of 20 could be higher depending on the actual capacity of the laboratory facilities	

AE4897	Space Engineering Practical	8
Responsible Instructor	I. Akay	
Responsible Instructor	Ir. B.T.C. Zandbergen	
Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr. J. Guo	
Instructor	Dr.ir. R. Saathof	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	M.S. Uludag	
Instructor	Dr. B.V.S. Jyoti	
Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you will be taught on and experience the basic steps needed to successfully conduct experiments/simulations to verify/characterize a product/system as part of a real space engineering research project brought forward by industry and/or academia. Everyone who ever designed an experiment or conducted simulations to verify product/system requirements has experienced issues with improperly formulated requirement on one hand and the preparation and execution of simulations and experiments on the other. Typical problems that occur are inadequate planning (not planning in advance, lack of time), system/object not being in agreement with the design specifications, improper test set-up, instrument failures, missing data, convergence issues, bugs, extremely long calculation times, and improper requirements verification. Learning about the basic steps will allow the students to conduct such projects successfully in the future. The course is made up of two complementary parts, which are described in more detail below. Part 1 provides in-class instructions on how to perform specific tasks needed for conducting a proper test/verification project. The subjects dealt with in class (lectures/workshops) include (not necessarily in the order indicated): 1. Why this course, course organization, project phases, timeline, deliverables 2. Generating requirements including also test requirements, like test ranges and test tolerances, 3. Hypothesis formulation and assessment, 4. Design of Experiments, Experimental Measurements, and Error Propagation, 5. Test planning and the development of the test set-up and the accompanying Assembly, Test and Safety Procedures. 6. Data gathering/acquisition (both qualitative and quantitative) & automation 7. Results analysis 8. Other, like Safety, Types of Testing, Use of Standards Part 2 focuses on teams stepping through the project cycle of a experimentally oriented project assigned to them. The steps include: - Orientation (on project assignment, object/system to be researched/tested/verified under what conditions, goals) - Estimation of the performances of the test object by design analysis/simulations - Development of test hypothesis, and test requirements and criteria - Generation of test specification with success criteria - Generating a test plan - Assembly/build of test object (may be a mock-up in case of use of hazardous materials) - Assembly of (mock-up) test set up. early testing (including calibrations) and data elaboration to develop/check procedures - Test readiness review (if review is passed go on to next step, if not revise test plan) - Test execution - Data elaboration (observations and experimental data) - Discussion of results (data review) - Communicating the results/outcomes + project close-out For each project, project supervisors may arrange for more specific training/instructions related to the specific test set-up to be used. Project supervisors are also responsible for students conducting safe testing.	
Study Goals	Upon completion of the project the student shall be able to: 1. Describe and motivate the project cycle of an experimental project. 2. Formulate a test specification and test plan. 3. Set up and assemble/develop an appropriate test set-up and test procedures. 4. Perform data acquisition (both qualitative and quantitative data) with some degree of automation. 5. Execute measurements in a reproducible way and determine measurement uncertainties. 6. Analyze experimental data taking into account measurement uncertainties. 7. Discuss the outcomes of the project including the statistical validity of the data. 8. Report on the outcomes of an experimental project.	
Education Method	Lectures/workshops and Project The course is a practical lab project combined with lectures/workshops. The lectures/workshops are given by a variety of instructors. All instructors are expected to link the material they are teaching to the various projects. Instructors shall also strive to let students experience the work through a number of workshops/assignments that are geared to the projects the various teams are conducting. During the projects, instructors can be approached in case teams need help on elements instructed upon. Most of the instructions are scheduled in Q2, but to allow for proper alignment with the projects, some may move to Q3. The size of this part of the course is 2 EC (roughly 56 working hours for each participating student). Each project on offer is of size 6 EC and is conducted by a team of 5-6 students. All teams are assigned a staff member, with expertise in the field, as project supervisor. The role of the project supervisors is to provide intellectual guidance and offer advice on the planning and progress of the work. Project supervisors also are to check and review the progress made by the team and determine if a team can move on to the next phase in the project, thereby ensuring project progress and that test preparation activities and testing are conducted in a safe way. To streamline this, two formative assessments will be performed, which help students identify their strengths and weaknesses and target areas that need work. The first such assessment relates to the project	

	specification and the second one is prior to getting approval to conduct full scale testing. Additionally, during their project, students have an open invitation to consult the lecturers/instructors of the lecture/workshops part of the course for advise/feedback on issues taught in class or exercised upon in the workshops that are relevant to their project. Projects can be conducted in the Q2 to Q4 time frame of the academic year. The exact time frame is subject to negotiation between the team and the respective project supervisor. For instance, teams can propose running the project in a single quarter or spread over multiple quarters.
Literature and Study Materials	Appropriate literature and study materials will be provided throughout the course. For the lectures/workshops, material is provided for by the instructors relating to the topic/subject they instruct upon. For the specific projects, a project assignment will be made available that describes the test object/system to a great detail, including the relevant characteristics of the object/system. The project assignment will also describe the test goals as well as the type of testing to be performed (functional testing, life testing, reliability testing, failure testing, etc.) and the test conditions under which the testing is to be performed.
Assessment	The result of this course will be entirely based upon the results of the experimentally oriented project and the associated deliverables. These deliverables include: - Test specification - Test plan - Results Summary document, test data files, and a visual demonstration (video/photos/poster) of the testing conducted and the equipment used. For each deliverable a template/content description (or example) will be provided. Assessment will take place according to the detailed rubrics available on Brightspace. Assessment is by the project supervisor of the project (40% of the final grade) and the course instructors involved in the instructions (60% of the final grade). Assessment is coordinated by the course responsible instructor. A minimum grade of 6 is needed after rounding to the nearest half to pass the project. Late submission of the work results in a maximum penalty of 1.5 point (on a 10-scale). Due dates for submission will be communicated in due time via Brightspace. In case of an insufficient grade, teams are allowed one opportunity to resubmit an improved version of their report.
Enrolment / Application	This course is only open for enrollment for students enrolled in the Space Engineering profile of the track Space. A maximum of 45 students will be accommodated on a first-come, first-served basis. Students will have the opportunity to choose from a list of available projects that encompass different knowledge fields in the field of Space Engineering. Some past projects include: - Battery project: Novel capacitor-based spacecraft energy storage system - Propulsion project: Solid rocket motor assembly and testing (in cooperation with DARE) - Attitude control project: Assembly and testing of a PocketQube attitude determination and control system - Payload controller project: Assembly and testing of a small sat onboard data handling system

AE4898	Space debris tracking and mitigation	4
Responsible Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	AE4874 I (Fundamentals of Astrodynamics) is required. AE4868-1 (Numerical Astrodynamics), AE4894 (Practical Astrodynamics) and AE4870B (Re-Entry Systems) are recommended.	
Course Contents	The course will introduce the problem of space debris and associated aspects, such as how to detect it, how to avoid it, and potentially how to remove it. The contents of the course will be presented in a hybrid format, i.e., a combination of classical lectures and working lectures, where the students can practise with the introduced material, tools and techniques. The course content is roughly split into two parts. The first half of the course will introduce the problem of space debris: how much is out there, how large and heavy is it, where does it come from, and what kind of problems does it pose. Having identified the problems for the safe operation of space, the focus shifts to how we can predict potential collisions and how we can avoid those. Finally, how can the larger chunks of space debris be removed (Active Debris Removal), and if the debris enters the atmosphere and does not fully burn up: where will it impact on earth. The second part focuses on the observation and estimation of space objects in general, and space debris in particular. The students will learn about different sensors that can be used, either ground-based or space-based, the accuracy of measurements, and how this will affect the estimation process. Linear and nonlinear estimation techniques will be discussed, with application to determine translational state vectors, characterise physical parameters, and detect manoeuvres of space objects. The course will further consider methods of sensor management and data fusion.	
Study Goals	After the course the student shall: 1. Assess the risks posed by orbital debris on current and future space missions, including the probability and consequences of on-orbit collisions 2. Analyze the mitigation of space debris hazards through the use of post-mission disposal and active debris removal 3. Estimate the orbits and physical characteristics of resident space objects using simulated space object tracking data, including measurements from optical and/or radar sensors 4. Design a collision detection and avoidance strategy for use with a simulated space object catalog	
Education Method	Per week, two lectures are scheduled. One is a regular ("classical") lecture where topics related to the field are introduced, clarified with examples. The second lecture is a working lecture, where tools are introduced by means of tutorials and where students can begin on their assignments/lab work.	
Computer Use	Personal laptop that can run Tudat. The choice of programming language is free (Python knowledge is required to work with Tudat). No coding support is provided.	
Literature and Study Materials	The main course material consists of the lecture slides. Recommended papers and books will be suggested during the course.	
Assessment	The assessment is a combination of one individual and two group take-home assignments. A tutorial assignment (not graded) will be handed out at the beginning of the course. For the group work, individual contributions to the group work must be clearly identifiable. The total grade is composed of: Individual assignments (individual grade): 30% Project (individual grade): 30% Project (group grade): 40% Each of these three grades must be ≥ 5.0 to pass the course (final grade is the weighted average, which should be 5.75 or higher). The resit of the course will be done based on individual assignments. Support during this will be limited. It is not possible to resit single assignments, only a resit of the full course is offered.	
Enrolment / Application	The course is open to students of the Spaceflight Dynamics profile only; the maximum number of students that can participate is limited to a maximum of 30. To enroll, the student should write a motivation (limited to 5 lines) why this course is important/relevant to them, and send this by email to both instructors: e.mooij@tudelft.nl (Dr Erwin Mooij) and s.gehly@tudelft.nl (Dr. Steve Gehly). Enrollment closes 2 weeks before the start of the course, and students will be notified about the status of their enrollment before the course starts.	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods.	
	Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to:	
	1. Explain the different steps in a finite element analysis and apply them to practical engineering problems	
	2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function)	
	3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up	
	4. Use and interpret results from 1D and 2D elements in commercial finite element software	
	5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace.	
	Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments (only assignment 3 and 4 since they have the highest weight; assignment 1 and 2 cannot be redone).	
	Results for individual assignments are only valid for one academic year: when re-taking the course all assignments have to be done in that academic year.	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	P.P. Sundaramoorthy	
Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of lectures, working sessions, design review sessions and a final symposium. The course structure will be explained in detail in the first weekly module. The time and date of the design review(s) and final symposium will also be communicated in the first weekly module. Students are requested to reserve all scheduled timeslots in their agendas. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. Note that, the course will require group work. Groups will be formed in the fourth week of the course. Only students who pass the individual test are allowed to participate in a group. Students who are not part of a group cannot participate in the remainder of the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. During each module it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Individual test on course content 2. Written feedback on design specification sheet of another group in the course Students with a failing grade for the individual test, i.e. a grade below 5.8, will have a resit opportunity of the individual test in week 4 of the course. This resit will be in the form of a short oral exam. This resit is only available for students who did not pass the individual test. On passing the oral resit exam, students will receive a 5.8 as grade for the individual test. Students who fail the resit opportunity will not be able to participate in the group work, and therefore cannot participate in the remainder of the course. It is mandatory that students are present at the design review(s) and the final symposium presentations of their own group and those of other groups. The individual grade of students who fail to be present will be withheld. These students will not receive a final grade for the course as a result. Students of whom the individual grade which has been withheld, may take a resit. The resit will be in the form of a short oral exam. It will be possible to apply for a resit up to 15 working days after the publication of the grades in Osiris, but no later. A resit exam will then be scheduled by the principle lecturer of the course in consultation with the student in question. Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, Introduction to materials science" or books of similar nature. Be sure to check the entry level quiz before making up your mind about this course.	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, ISS, Hubble telescope, James Webb telescope, Venus orbiters, Launchers) will be presented as a carrier for material selection analysis. A holistic view of "materials science - manufacturing - structural analysis - system engineering for missions" will be presented. While students come mostly from different backgrounds, the goal of this class is to help them identify the perspective they use in the holistic view and to learn about the materials science perspective. Fundamentals of materials classes and their typical properties will be introduced. Current material choice will be analyzed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. In sessions of "ESA forum + student poster seminar", students will be given opportunities to interact with ESA experts and present their own findings from analyzing given missions, by translating desired properties into material structures and microstructures. In class discussion is an important part of learning. The structure of the lectures will be tailored to maximize the student involvement. The course contains the following topics: week 1: Holistic view and materials science perspective; Typical materials - metals, ceramics and polymers week 2: Typical materials - metals, ceramics and polymers week 3: Material degradation in space - vacuum; thermal cycling; week 4: Material degradation in space - AO; charging week 5: Material degradation in space - space radiation week 6: Material degradation in space - meteoroids and orbital debris; week 7: "ESA forum + student poster seminar" Materials testing for space applications; Extreme missions; Materials challenge in future space age (ESA guest speakers); Student poster seminar (Jan 6, 2025) week 8: Study period (Q and A) week 9: exam week (Jan 20. 2025)	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 12 lectures of 2h per week for the first 6 weeks. The last week is a whole-morning session of "ESA guest speaker forum + student poster presentation".	
Literature and Study Materials	A compilation of book chapters, lecture notes and review papers.	
Assessment	Poster presentation (40%) + final exam (60%). The students are to work in groups and do research on prescribed question/topics and deliver a final poster presentation (15% individual score, 25% group score) in week 7 to show their discoveries throughout the course. The final grade will be the weighted average of poster presentation and final exam. Example questions/topics will be made available on Brightspace.	
Enrolment / Application	For students not from "structures for space" profile, there will be a cap of 20 students maximum. Please enroll in brightspace (enrollment period Oct 28 to Nov 10, 2024) as soon as you decide to take the course. Once the maximal number is reached, enrollment will be closed.	

AE4ASM526	Spacecraft Thermal Design	3
Responsible Instructor	I. Uriol Balbin	
Instructor	C. Falsetti	
Instructor	N. van der Pas	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of orbital mechanics Understanding of general design spacecraft systems	
Course Contents	General knowledge of space thermal control hardware, system design and analysis processes are presented. Using the principles of heat transfer, and more specific of radiation and conduction, the methods and tools for thermal design of a spacecraft are presented. Numerical analysis will also be introduced via the software ESATAN-TMS.	
Study Goals	The student is able to generate a thermal design of a satellite using heat transfer analysis and thermal engineering techniques. In more detail the student will be able to: - Understand the spacecraft thermal environment - Define the thermal subsystem and compare the different hardware options. - Understand the design cycle of the thermal subsystem. - Apply the concepts of heat transfer (radiation, conduction), heat balance and heat flows - Generate and use analytical thermal analysis models - Generate and use numerical thermal analysis models	
Education Method	Lectures + homework Students are given access to a discussion board provided with the course in Brightspace. The discussion board will be used as the main communication tool to ask questions on the material taught or in preparation of an assignment. Answers to the questions may be given by fellow students and/or the lecturer(s).	
Computer Use	Software licenses are available during the lecture period Software can be used in the computer room or in the personal computer (Windows/Linux)	
Literature and Study Materials	Lecture notes available on Brightspace.	
Assessment	The final grade of this course is based on 2 homework assignments (60%) and one final exam (40%). A minimum grade of 5 is required in each part.	
Enrolment / Application	This class has a limit of 80 students, due to the limit on the software licenses. Please enroll in Brightspace before 4th of March 2025 if you intend to take this class.	

AE4S52-21	Collaborative Space System Design Project	5
Responsible Instructor	Ir. B.T.C. Zandbergen	
Instructor	Dr. S.M. Cazaux	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The Collaborative Space (System) Design Project (CSDP) is a TU Delft/Faculty of Aerospace Engineering (AE) Master course in engineering design. The course focuses on the conceptualization and preliminary design phase of a space mission, spacecraft, or space instrument and starts on a design challenge co-created by the CSDP organization team and a company, research group or other entity. All projects on offer are multi-disciplinary projects, i.e. projects that require different types of design work and knowledge to be combined to provide a design solution. All projects are carried out by a team of students that have to organize and manage themselves. Each project is to be organized in 3 phases similar to NASA's System Design Process as defined in a.o. "NASA's System Engineering Handbook". Some adaptations have been made though to make it fit in the limited course time. The phases are: 1) Exploration phase; In this phase the team is to explore the problem, the needs, the competition and past missions and to develop a proposal for the next phase of the project. This includes the identification of a range of high-potential concepts for study in phase 2, the work distribution, etc. 2) Concepts design studies phase; In phase 2, the high-potential concepts defined in phase 1 are analyzed in detail for feasibility, traded and a best concept is selected. Additionally the plan for the next phase is to be generated. 3) Detailed design phase. In this final phase of the project the single concept selected in phase 2 is worked out in detail and a plan is developed for the further development of the design. Each project phase ends with a review (feedback moment) wherein other teams and expert staff reflect on the outcomes generated and the engineering design methods used. During the quarter, workshops/instructions will be held to provide knowledge and training on selected management and engineering design topics, including agile management, and the use of integrated design modelling, i.e. the integration of all the geometry, configuration, analysis, and requirements verification into a generative, parametric, unified computational model where data is shared seamlessly between the different disciplines. The course does not focus on teaching the required disciplinary knowledge and experience, but rather focuses on decision making, the collaborative integration of the knowledge and experience available in the team, and the iterative design method using different levels of model fidelity to create a feasible design solution in answer to the problem identified by the "customer". As projects vary from year to year and may encompass knowledge that is not available in the team, this may require that participants actively acquire the knowledge required.	
Study Goals	The course aims to develop student skills in multi-disciplinary team projects from a challenge-driven perspective. In more detail, students will advance their ability in ...: - ... disciplinary design including modelling, simulation, visualization, quantitative analysis of alternatives, design tool verification, calibration and validation, and design refinement. - ... the process of engineering design (ABET definition), including the steps in design, the (iterative) nature of the process, development of a Straw Man design, and development of process models. - ... multidisciplinary design, thereby taking into account differences between the different disciplines involved in terms of a.o. differences in fidelity level of disciplinary models, and dissimilar assumptions; - ... systems engineering, including the design phasing, work breakdown and work distribution, modeling and interfaces, and the role of specialty engineering (e.g. cost-, RAMS-, and mass modelling and configuration design). - ... concurrent engineering, as opposed to the more classical sequential engineering (the waterfall method). - ... project management and teamwork (assigning roles and responsibilities, setting goals and objectives, coordination and management of team process, decision making, handling conflicts, creativity, empowerment and motivation, communication, and reflection on own work and work of others).	
Education Method	The course consists of: - Group project (4 EC or 112 working hours) - Workshops/instructions (1 EC or 28 working hours) For the group work, students work in teams of 6-12 students. Prior to the start of the course, students will be informed on the available projects as to allow them to state their preference(s), and project teams are formed. During the project, each team is guided by a design coach, who mainly focuses on the problem solving/engineering design process. Additionally, each team is free to organize experts (from industry, academia or other) for the different disciplines involved that can advise the team during the design. During the quarter workshops/instructions are given to train teams' project management and design integration skills.	
Assessment	The assessment of each team consists of an assessment by the workshop instructors and project coaches (80% of the final grade), team of experts not involved in instruction and workshops (10%) and by the other student teams (10%). The assessment of the workshop instructors and project coaches is per design phase: - Exploration phase (10%) - Concepts design studies phase (20%) - Final design phase (70%) The assessment of the instructors and coaches is based on a design report, a (design) management log (record keeping) and one or more design presentations and reviews. Elements that will be assessed relate to design competencies (problem formulation, requirements generation, design option identification, design analysis, design trades and design evaluation), workplace competencies (e.g. teamwork, creative thinking, decision making, planning and organizing), academic and personal effectiveness competencies (e.g. communications, modelling/experimenting and critical and analytical thinking). Assessment by the team of experts and by peer teams will be based on the performance of the teams at a project presentation session at the end of quarter 3.	
Enrolment / Application	Only students enrolled in the Space track can participate. There is no free enrollment for this course. Still at the start of Q1, the students can preliminary enroll for the course as to learn more about the CSDP, and past projects conducted (no solutions on offer). In Q1, an information session will be held informing interested students of the track Space on the projects available and on how	

they can apply for a project. Students can then apply and only if they are accepted, their enrollment will be made definite. All non-accepted students will be dis-enrolled.

For the year 24/25, the maximum number of participants for this course has been set to 20. Priority will be given to students enrolled in the Space Engineering profile of the Space track. If places remain, also students from the other profiles in the Track Space can participate.

AESM2001	Physical Principles of Earth System Observation	5.5
Responsible Instructor	Prof.dr.ir. F. Lopez Dekker	
Module Manager	Prof.dr.ir. F. Lopez Dekker	
Instructor	Dr. P.G. Ditmar	
Instructor	Prof.dr.ir. S.C. Steele-Dunne	
Instructor	Prof.dr.ir. F. Lopez Dekker	
Instructor	K. L��er	
Contact Hours / Week x/x/x/x	4 hours lectures or practicals	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Measuring is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Electromagnetic, seismic and gravity potential-field observations inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. The measurements can be acquired from spaceborne, airborne, surface and sub-surface-based sensors. This programme core module aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain geophysical variables. For example, students will learn how electromagnetic theory allows to use the intensity of radar echoes to yield information about rain rate, soil moisture, ocean roughness, or the layering of the subsurface. Similarly, they will learn how potential field theory can be applied to quantify mass changes of, e.g., the ice sheets. Students will be able to weigh the advantages and disadvantages of, for example, microwave versus visible and near-infrared observations for monitoring the Earth surface, and seismic and electromagnetic imaging in mapping the subsurface. Review basics of electromagnetic (EM) waves (propagation, polarization, spectra, Doppler, etc.) and propagation effects (attenuation, refraction, dispersion, polarimetric effects) Optical-infrared scattering and propagation (different reflection types, albedo, effect of wavelength, BRDF, Radiative transfer modelling of optical EM waves). Physical principles and limitations of VNIR/SWIR sensing systems (photographic systems, LiDAR, spectral systems including TIR) in terms of spatial and spectral resolution, sensitivity, atmospheric disturbances and corrections. Microwave scattering and propagation (Rayleigh, Mie scattering, rough surface, volume scattering, clear air, radiative transfer modelling of microwave scattering in the atmosphere and at the land surface). Radiometry (Radiometric Quantities and Units, Blackbody Radiation, emissivity at different frequencies). Radiative transfer modelling in passive microwave remote sensing of the Earths atmosphere and surface. Physical principles and limitations of microwave sensing systems (radiometer, non-imaging radar, imaging radar, ground-penetrating radar) in terms of spatial and spectral resolution, sensitivity, atmospheric propagation effects and corrections, radar range equation Seismic (including acoustic) waves (Newtons second law, stress tensor, equation of motion, elasto-dynamic wave equation, Plane P- and S-waves, spatial and temporal resolution) Potential fields (gravity, magnetic and electric fields) and their relationship to physical properties of a medium (Gauss law and its applications, Newtonian/Magnetic Potential, Laplace equation and harmonic functions, representation of potential fields, spherical harmonics) Choose an approach for the observation and measurement of a geophysical variable.	
Study Goals	After completing this module, students will be able to: Characterize electromagnetic waves and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and & subsurface). [ILO A] Describe the interaction of optical-infrared waves with the Earths atmosphere and surface and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of optical/infrared remote sensing in determining geophysical variables. [ILO A, D] Describe microwave scattering in the Earths atmosphere and at the Earths surface, and analyse the fundamental sensitivity to geophysical variables of interest. [ILO A] Describe the emission of thermal waves and microwaves from the Earths atmosphere and surface, and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of microwave remote sensing in determining geophysical variables. [ILO A, D] Explain mathematically how the elastodynamic wave equation is derived, characterize elastic waves, and explain their sensitivity to contrast parameters. [ILO A] Establish the linearized relationship between an observable and a given representation of the potential field, and synthesize the potential from functionals given on the surface of a sphere. [ILO A] Critically discuss the potential value and limitations of different types of measurements (I.e., different bands of the EM spectrum and EMR/seismic/potential field) and propose and defend a type of measurement to observe certain geophysical variables. [ILO F, G]	
Education Method	Lectures and group practicals/exercises. Feedback via quizzes (formative) and intermediate assignment results (formative).	
Assessment	Module is summatively assessed with a project, using 1 rubric (similar as thesis project) at the end of the period based on: Assignment report (may comprise different parts, i.e., based on a several assignments, report includes reflection on LOs) Presentation Oral discussion (may include questions on LOs not directly covered in the assignment work).	

Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Some background in electromagnetic theory and fundamentals of wave propagation. Solving linear partial and ordinary differential equations.
Academic Skills	Earth Observation, data interpretation.
Literature & Study Materials	Provided course material, published online lectures and material, book/e-books
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N.A.
Collegerama	No

AP3141	Environmental Physics	6
Responsible Instructor	Dr. S.R. de Roode	
Instructor	Prof.dr.ir. J.L. Kloosterman	
Instructor	Dr.ir. D.J. Verschuur	
Instructor	Ir. R.A. Verzijlbergh	
Instructor	Prof.dr.ir. B.J.H. van de Wiel	
Instructor	Dr.ir. S.J.A. van der Linden	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Physics at BSc level, including basic Transport Phenomena and Fluid Dynamics, thermodynamics, calculus and differential equations; use of Python and/or Matlab.	
Course Contents	*) Fundamentals of the environmental physics, world climate and the greenhouse-model, durable energy-resources (solar-, wind-, wave-, nuclear-, biomass-energy), transport of pollution in ecosystems, Air Quality; *) Fundamentals of harnessing sustainable nuclear energy and physical models for nuclear reactors to do so; *) Land-atmosphere interactions and turbulent transport processes. Fundamentals of dimensional analysis & generalization in environmental physics; *) Radiative transfer & Solar cell physics, PV system design & Renewable energy integration; *) Environmental noise, basic theory, noise calculation schemes	
Study Goals	To provide physicists with the knowledge and tools so that they can contribute to a more sustainable society from their own specific field of expertise: 1. Acquiring advanced knowledge of the basic physics underlying environmental issues such as: the earth energy budget; air-, water- and ground pollution; solar, wind, tidal, wave, biomass and nuclear energy technologies and environmental noise. 2. Being able to translate problems in the above areas into appropriate physical models and their relevant equations. 3. Being able to translate physical models and model equations into (e.g. Matlab/Python) computer models. 4. Being able to find (quantitative) solutions to model equations, using either theoretical analytical methods, or numerical methods. 5. Being able to quantitatively and physically evaluate, analyse and critically acclaim proposed solutions and strategies for environmental issues. 6. Being able to contribute to solutions and strategies for environmental issues.	
Education Method	Lectures and mandatory homework assignments.	
Computer Use	Homework requires computer programming in Matlab or Python	
Literature and Study Materials	Egbert Boeker and Rienk van Grondelle, Environmental Physics, Third edition, Wiley, 2011, ISBN 978-0-470-66676-0 + lecture notes and handouts	
Assessment	There is a series of mandatory (homework) Assignments. Please note that the Assignments have to be done in the prescribed periods, and that grades cannot be improved later. Lecturers can invite students for an oral assessment of their Assignments. The grade for the Assignments is the mean over all of them. At the end of the course there will be a written Exam. This Exam will be of an open-book type (see below). The final grade of the course is the average of the Exam grade and the Assignment grade. A passing grade can only be obtained if a score of at least 5.0 is obtained for both the Exam and for the Assignments. The Assignments and Exam must be done in the same academic year. The validity of result of the Assignments is restricted to one academic year. Resit exam can be a written exam (if enough candidates) or via an oral assessment (by appointment). In case an on-campus exam cannot be provided, there will an online open-book exam followed by an oral fraud-assessment of a random subset of students. See announcements on Brightspace.	
Permitted Materials during Tests	The exam will be done in a computer room. The computer will contain the PDFs of the book and PDFs of all handouts of the lectures, without access to internet. Besides this, you can take one A4 double-sides printed or hand-written with anything you prefer. And a simple calculator (not programmable) for possible calculations is allowed.	
Studyload/Week	2 lecture hours per week (14 weeks), 5 series of homework of 20 hours each, 40 hours of exam preparation.	

AP3181	Applied Multiphase Flows	6
Responsible Instructor	Dr. L. Portela	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course assumes some elementary knowledge of Fluid Mechanics and/or Transport Phenomena. The level of prior knowledge required is typically acquired in a BSc course in Engineering or Applied Physics. Prior knowledge of multiphase flow is not required.	
Course Contents	1. Introduction. Overview of multiphase flow. Examples of industrial and environmental flows. Examples of flows with heat transfer and phase change. Classification of multiphase flows: separated vs. dispersed. Brief introduction to flow patterns. 2. Balance equations. Physical mechanisms involved and material behavior. Constitutive equations and mechanistic models of behavior. Interfacial interactions and forces. Analogies and differences with respect to single-phase flow. Flow parameters, non-dimensional numbers and scaling. 3. Introduction to turbulence, turbulence modeling and turbulence effects. Analogies and differences with respect to single-phase flow. 4. Separated flows and interfacial phenomena. Stability associated with interfacial flows. Interfacial waves. 5. Dispersed flows. Interaction between the dispersed and continuous phases. Momentum, heat and mass transfer. Differences and similarities between solid particles, droplets and bubbles. 6. Dynamics of single particles: solid particles, droplets and bubbles. Bubble and droplet dynamics. Bubble growth and collapse. Introduction to cavitation. Droplet dynamics and breakup. 7. Inter-particle interactions and collisions. Droplet and bubble coalescence and breakup. Collision kernel. Population balance models and simulations. 8. Quasi-1D flows, both internal and external (channel, pipe, jet, etc.). Balance equations and simplifying assumptions. Simple mechanistic models. Similarities and differences between: gas-liquid, liquid-gas, liquid-liquid, solid-liquid, and solid-gas. Flow classification and flow patterns, both separated and dispersed. Physical explanation and underlying mechanisms in the construction of a flow map. Steady and transient phenomena. 9. Stirred flows, both internal and external (bubble columns, fluidized beds, stirred tanks, ocean and atmospheric turbulence, etc.). Similarities and differences with respect to quasi-1D flows. Motion-driven, pressure-driven and gravity-driven flows. Dilute and dense flows. Flows dominated by inter-particle interactions (dense fluidized beds, bubble columns, etc.). 10. Multiphase flows with complex fluids (granular flow, slurries, agglomerates, emulsions, foams, etc.). Internal mesoscale structures. Thermodynamic and physicochemical interactions. Complex particles (with a complex structure and/or a complex particle-dynamics). 11. Heat transfer and phase change. Boiling and condensation. Radiation. 12. Industrial flow examples. Complex flow in pipelines in the oil-gas industry. Pneumatic conveying and solid-gas flows. Process equipment. Microfluidics applications. 13. Environmental flow examples. Sediment transport in rivers and the ocean. Aerosols and particulates dispersion in the atmosphere. Droplets in clouds.	
Study Goals	To give a general unified perspective on multiphase flow, with an emphasis on: (i) the understanding of the several essential physical mechanisms involved; and (ii) the formulation of complex engineering problems in terms of simple physically-based models. The students will develop the ability to: (i) formulate the problems in a structured way, starting from first principles; (ii) obtain (approximate) solutions using standard analytical and numerical techniques; (iii) interpret and critically analyse the solutions obtained; (iv) report the formulation of the problems and its solution in a clear well-structured form; and (v) sustain a critical discussion on the formulation of the problems and its solution. Also, during the course the students (i) will acquire a broad knowledge of multiphase flow; and (ii) will develop the ability to read and understand scientific literature on multiphase flow.	
Education Method	The education method is based on the combination of regular lectures, homework and interaction with the instructor outside the lectures (regular and impromptu office hours). The Learning by Doing is a key component of the education method: the homework involves both standard problems and open-ended problems, with the initiative and creativity of the students being strongly encouraged. The interaction with the instructor outside the lectures plays a key role, and it involves an iterative process, based on the Learning by Struggling philosophy.	
Literature and Study Materials	The study material consists mostly of lecture notes. Also, during the course, sections of books, on specific topics, and articles, from the scientific literature, are suggested for reading. Possible auxiliary general reference-books on Multiphase Flow are: 1. Brennen, C.E. (2005). Fundamentals of Multiphase Flow. Cambridge University Press. 2. Ghiaasiaan, S. M. (2007). Two-Phase Flow, Boiling and Condensation: in Conventional and Miniature Systems. Cambridge University Press. 3. Kleinstreuer, C. (2003). Two-Phase Flow: Theory and Applications. Taylor & Francis. 4. Crowe, C.T., Schwarzkopf, J.D., Sommerfeld, M. and Tsuji, Y. (2012). Multiphase Flows with Droplets and Particles. Second Edition. CRC Press. 5. Fan, L-S and Zhu, C. (1998). Principles of Gas-Solid Flows. Cambridge University Press. 6. Peker, S.M. and Helvacı, S.S. (2008). Solid-Liquid Two Phase Flow. Elsevier. Possible auxiliary general reference-books on Fluid Mechanics and Transport Phenomena are: 1. Kundu, P.K. and Cohen, I.M. (2008). Fluid Mechanics. Fourth Edition. Academic Press. 2. Ghiaasiaan, S.M. (2011). Convective Heat and Mass Transfer. Cambridge University Press.	
Assessment	The assessment is based on the homework proposed during the course (30%) and on the final exam (70%). The final exam consists of a written part followed by an oral discussion with the instructor. The written part is open book: the students can consult any reference they wish (including the internet) and use any tool they wish (including a laptop). The oral discussion will be arranged individually, shortly after the written exam, in consultation between the student and the instructor. It is possible to have a resit of the final exam; the format of the resit is the same as the format of the original final exam, consisting of a written exam followed by an oral discussion with the instructor.	
Permitted Materials during Tests	Any material (including laptop).	
Elective	Yes	
Tags	Fluid Mechanics Physics	

AP3222	Nanotechnology	6
Responsible Instructor	S. Goswami	
Instructor	R.A. Norte	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic physics and mathematics are required; solid state physics, materials science and physics of microelectronics are recommended. Good understanding of the typical physics approach to problems is helpful.	
Course Contents	The synthesis and study of objects with nanometer-size functional elements require dedicated fabrication methods. This course will provide the technological and scientific background needed to master the methods of nanofabrication. Subjects include: material properties, growth and etching methods; pattern definition and transfer (=lithography); self-assembly; molecular manipulation; methods of inspection, analysis and characterization. The subjects will be taught with a focus on practical applications.	
Study Goals	The student can evaluate the merits of the fabrication technologies for nanotechnological applications. The student can design the processing steps for a nanotechnological application. The student can find, understand and apply the relevant scientific literature. The student can evaluate, model and understand the functioning of nano-objects. The student can execute a simple but complete nano fabrication task.	
Education Method	Lectures, expert lectures, student presentation & discussion sessions, and lab work.	
Literature and Study Materials	1. Sami Franssila, Introduction to Microfabrication, 2nd edition, Wiley 2. Copies of the lectures 3. Scientific papers for the student presentations and discussions 4. Lab-work manual	
Assessment	Written exam (35%), home work (15%), lab work (30%), and student presentation (20%). The scores for the written exam and the lab work should be both at least 5.0.	
Permitted Materials during Tests	Ruler, pocket calculator	
Enrolment / Application	Because of limited space and support in the cleanroom, a maximum of 30 students can do the course. Enrollment will start on Monday 6 January 2025 at 8:00am via MyTUDelft. Enrollment requests done before January 6 at 8:00am will not be considered. If you encounter any problems with this registration procedure, please contact Arno Haket, A.J.W.Haket@tudelft.nl, from 6 January. Students who do not have a Physics background are asked to submit a motivation (on prior knowledge, current activities and interest in nanotechnology) by e-mail to Srijit Goswami S.Goswami@tudelft.nl. Students who do not attend the first lecture (without prior notice) will have lower priority than those present. Students who do not have their bachelor degree at the start of the second semester (10 February 2025), will have lower priority than those with the degree.	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/education/admission-and-application/elective-course#c1323630	
Elective	Yes	
Tags	Lab Research Physics	

AP3701	Submm and terahertz physics and applications	3
Responsible Instructor	Dr. J.R. Gao	
Responsible Instructor	Dr. A.J.L. Adam	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	The course is intended for MSc students from different disciplines, including those from Applied Physics, Electrical Engineering, and Space Engineering, but it is also open to interested PhD students.	
Expected prior knowledge	Electromagnetism, Any introduction of Optics, and Introduction of solid-state Physics. If you miss some of them, we will also add necessary background knowledge during the course	
Course Contents	The submm and terahertz region of the electromagnetic spectrum, which spans from 100 GHz to 10 THz, has attracted a lot of attention in recent years due to the developments in new detection and generation techniques and due to the applications for astronomic observations in space and the potential applications as an imaging technique for security, communications and biomedical applications. The Submm&THz frequency range is rapidly emerging which allows unearthing new science and technology. We wish to provide a course in which students can get an introduction and learn the basic physics & principle, and will also be updated with the latest developments. The course will focus on the topics like Photonics, Time-Domain Spectroscopy, Antenna/Waveguide, terahertz detectors and sources, THz Space application. Here the titles of all the different lectures given: 1. Introduction to THz. Light Interaction with Matter 2. THz photonics (THz fibers, Photonic crystals & Metamaterials. High THz fields. Nonlinear interaction of THz waves with matter) 3. THz Imaging (THz Time Domain spectroscopy, Transmission, Reflection, Time of Flight Tomography, Near field, Tomography) 4. THz Sensing (Application to Chemistry, Security, Biology & Medicine; Pump-probe experiment) 5. Submm & THz Coupling (Waveguide, Si Lens, Antennas) 6. Sources 7. Nano-detectors and cameras 8. Submm & THz Sensing instruments from space (as a few examples) Additionally: A case study and design of an space instrument by a subgroup of students (Adam/Gao) (To be announced)	
Study Goals	Understand submm & terahertz electromagnetic radiation and propagation fundamentals, and the basics of typical terahertz detectors and sources. Distinguish different terahertz antennas with regard to their advantages and disadvantages for terahertz radiations. Explain the working principle of THz Time-Domain Spectroscopy. Build practical experience on a THz imaging setup and a heterodyne receiver system. Highlight recent advances of THz research and development from the academic and astronomic sectors. Summarize state-of-the-art THz applications, including space applications.	
Education Method	The course will be given as following: 1) one 2x45 min lecture per week 2) Home work every week on the topics following the lecture. The homework counts 30% of the grade. 3) Design an instrument concept and address a specific topics with groups and present 4) a written exam on all topics discussed during the lectures and homework We based our lectures on current research and science on THz technologies	
Literature and Study Materials	The lectures will be based on a few selected chapters of different books. For Gao's lectures: Chapter 27 "FIR heterodyne receivers by J.R. Gao" of a new book "Observing Photons in Space" and the book of "Terahertz Astronomy" by Christopher K. Walker. For Aurele's lectures: Terahertz Technology, Volume 77, A. Rostami, H. Rasooli, and H. Baghban, An Overview of the Technological and Scientific Achievements of the Terahertz, pp. 189. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011. (http://link.springer.com/chapter/10.1007/978-3-642-15793-6_1) After each lecture, the teachers will provide also the powerpoint slides used during the lectures.	
Assessment	20% from the homework 30 % from case/project study report/presentation; 50 % from the written exam	
Permitted Materials during Tests		
Elective	Yes	
Tags	Building & Spatial Development Physics Projects Technology	

BM41155	3D Printing	4
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Responsible Instructor	Dr. J. Zhou	
Instructor	Ing. M.A. Leeftang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	in Track I (Musculoskeletal Biomechanics)and Track II (Medical Devices and Bioelectronics) of MSc Biomedical Engineering BME) and in the specialization of Bio-Inspired Technology (ME-BMD-BITE)	
Expected prior knowledge	BSc in mechanical engineering, materials engineering, industrial design engineering and aerospace engineering with a basic knowledge of materials (e.g. WB2330)	
Parts	- Overview of AM technologies, their principles and applications - Materials selection (polymers, ceramics, metals and composites) - Powder-bed fusion processes - Powdered materials and powder fabrication - Powder characteristics in relation to AM - Structural integration during melting and sintering - Extrusion-based systems - Design for functionality and 3D printability Software issues and AM limitations - Other powder-based manufacturing technologies as alternatives to AM - Post-processing of AM products - Mechanical testing of AM products - Case studies AM for engineering and medical applications - Assignment and presentations	
Course Contents	Additive manufacturing (AM), also known as 3D printing the process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to conventional subtractive manufacturing technologies, is set to revolutionize manufacturing across a wide range of industries. The course 3D printing - AM technologies is designed to equip students with the fundamental knowledge of AM and basic hands-on experience in AM. The topics include various AM processes, their principles, capabilities and limitations, materials used for AM processes, powdered material fabrication and characterization, CAD designs for AM, alternative powder-based manufacturing technologies, current and future AM applications, as well as the mechanical evaluation of AM parts. The primary focus is placed on powder-based AM for metallic parts. In addition to a series of lectures, project groups, each composed of 3-5 students, will create their own designs of the objects made of polymer PLA, formulate requirements in terms of geometry, dimensions, tolerances, mass, surface finish, functionality and mechanical properties of these objects, redesign them by making use of SolidWorks, print them, test and evaluate 3D printed products against the requirements, and document the projects in the form of reports and presentations. The new design is expected to reflect the distinctive features and freedom that AM has to offer (thin walls, lattice structures, form-free channels, integrated parts/functions, and customization), but within the capacity of the printing machines (bounding box size, minimum feature size or wall thickness). The report and presentation are expected to contain: reason for the selection of the object, weakness of the existing design, new design features, actual testing results and recommendations.	
Study Goals	Students will be familiar with the concept of AM, various AM processes, AM applications, materials for AM, product design for AM, post-processing techniques and evaluation methods. In addition, students will gain basic hands-on experience in product development via AM. Once successfully completing the course, the students will: - Have a fundamental knowledge of 3D printing/AM technologies for a wide range of applications - Be able to explain the benefits, advantages, challenges and limitations of individual AM technologies as well as alternative powder-based manufacturing technologies - Have an insight into selective laser sintering and melting (SLS/SLM) processes regarding process control in relation to the geometry and mechanical properties of the products - Become familiar with the common (metallic) materials for 3D printing and their characterization techniques - Understand printable geometry design rules and the common methods of digital file creation - Become familiar with the common techniques for post-processing - Become familiar with the common methods for the mechanical evaluation of 3D printed materials and functional parts.	
Education Method	Lectures and student projects	
Computer Use	Attending lectures and performing assignment are both compulsory	
Course Relations	- Object design with Software (SolidWorks Student Engineering Kit)	
Literature and Study Materials	Biomaterials, Tissue biomechanics and other engineering courses in MSc biomedical engineering, materials science and engineering, mechanical engineering, marine technology, aerospace engineering, design for interaction, integrated design and strategic design - Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21) - Lecture materials	
Books	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21)	
Reader	- Lecture materials	
Assessment	Written exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information) or CSE3130 (Introduction to Quantum Computer Science)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution.	
Study Goals	The student will acquire: - A good understanding of the fundamental concepts of quantum communication - The ability to understand, design and analyze quantum communication protocols - Insight into the differences between classical and quantum communication and cryptography	
Education Method	- Flipped classroom (Online videos and materials, prepared for flip classroom teaching) - Weekly Quiz and Discussion Session (In person) - Weekly Tutorials on Homework Problems (In person)	
Literature and Study Materials	Book on tools and quantum cryptography: Introduction to Quantum Cryptography, Cambridge University Press, 2023 Lecture Notes Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: - Homework (weighting 60%) - Final Project (weighting 30%) - Weekly Quiz (weighting 10%) Final grade calculation: $0.6 * \text{Homework} + 0.3 * \text{Final Project} + 0.1 * \text{Quiz}$ A passing final grade for the course can only be earned when for the homework and final project at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: - Homework: re-submit deliverable - Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4205	Evolutionary Algorithms	5
Responsible Instructor	Dr.ir. P.A. Bouter	
Responsible Instructor	Prof.dr. P.A.N. Bosman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks.	
	This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, (GPU) parallelization, and real-world applications.	
Study Goals	At the end of this course, the student is able to: 1. Explain the key concepts underlying the main streams in Evolutionary Algorithm (EA) research, with in particular genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2. Explain key ingredients underlying the rationale of when these algorithms work and when they do not work. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3. Explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4. Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5. Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	8-10 lectures 4 smaller (individual) projects 1 larger group project	
Literature and Study Materials	Various scientific papers, course-specific texts, and slides	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment (weighting 30%) - Individual Assignments: weekly assignments (weighting 10%) Final grade calculation = 0.6 * Written Exam + 0.3 * Group Assignments + 0.1 * Individual Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4685	Machine Learning, a Bayesian Perspective	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. It is assumed that the student has seen a basic course on machine learning that has presented an overview of the general concepts.	
Course Contents	This course provides a solid mathematical foundation for machine learning algorithms, presented from a statistical (Bayesian) and optimization point of view. It is considered to be a second (more advanced) course, after an introductory course that would have presented the general concepts. Machine Learning is an umbrella for a range of methods coming from different scientific communities, such as (EE) statistical learning, statistical signal processing, adaptive signal processing, image processing and analysis, system identification and control, and (CS) pattern recognition, data mining and information retrieval, computer vision, and computational learning. The name Machine Learning indicates what these disciplines have in common: to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions. To this end, a number of diverse approaches have been developed, ranging from optimization of cost functions, whose goal is to optimize the deviation between what one observes from data and what the model predicts, to probabilistic models that attempt to model the statistical properties of the observed data.	
Course Contents Continuation	The course covers 3 introductory chapters and 9 regular chapters of a recently published rather exhaustive book. The introductory chapters summarize the mathematical language used in the course: conditional probability, distributions, random processes (as also seen in EE4C03), and repeat basic tools in Estimation (regression) and Detection (classification), as seen in ET4386. The course itself then presents three groups of techniques: - Bayesian learning: inference; conjugate priors; the EM algorithm; Gaussian mixture models; non-parametric models; variational bound approximation technique; sparse Bayesian learning; relevance vector machine (RVM). - Graphical models: Bayesian networks, factor graphs, message passing algorithms. Hidden Markov models. Learning graphical models. - Neural networks and deep learning: (probabilistic) CNN, variational autoencoder, GAN, learning deep networks.	
Study Goals	The objectives of the course are as follows: - Present the fundamentals of Bayesian learning, probabilistic graphical models, neural networks and deep learning in a unifying context. - For each of these, present compelling examples that demonstrate how the theory is applied in practice. Very often, this centers on the translation of a problem into a suitable model description, such that it becomes apparent which tools can be applied.	
Study Goals continuation	After the course, the student is able to - Translate a data analysis problem as a Bayesian machine learning problem through a 4-step process (problem, probabilistic model, inference algorithm, and solution). - Reason about the most appropriate tools to solve the data analysis problem. - Implement the most appropriate tools to solve the data analysis problem, e.g., for ECG or EEG analysis, radar, image analysis/denoising - Derive proofs that underly these tools and techniques. - Explore recent literature that builds upon the presented tools and techniques. - Compare the pros and cons of a variety of machine learning solutions in terms of societal aspects such as (but not limited to) quality control, ethics, robustness, and privacy.	
Education Method	Lectures; assignments. The course consists of 11 lectures of 2h per week theory, and 4 lectures/working sessions for example discussions. Each 2 weeks about 2 chapters are covered, centered around a compelling example/exercise/ data set that students take home to work on; the exercise is later discussed in depth during class. The examples are taken e.g. from biomedical applications (ECG, EEG data), time-frequency (spectral) data, radar, computer vision. Data sets will be provided.	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex optimization.	
Literature and Study Materials	Machine Learning, A Bayesian and Optimization Perspective by Sergios Theodoridis, Elsevier Academic Press, 2nd Edition (Feb 2020), hardcover ISBN: 9780128188033	
Assessment	The course is closed by a written exam (40% of final grade). There is one resit per year for the written exam. Several homework assignments count together towards 60% of the grade; the assignments involve solving machine learning problems, writing a report and essay, and giving a presentation. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework. On the slides, references to articles (i.e. numbered paragraphs) or to pages in the book "Graph Spectra for Complex Networks" (see Literature and Study Material) are made. The information in the book thus complements the slides.	
Literature and Study Materials	Graph Spectra For Complex Networks, second edition (2023) Cambridge University Press ISBN: 978-009-36680-9 See: https://www.nas.ewi.tudelft.nl/people/Piet/bookGraphSpectraCN.html	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4C12	Machine Learning for Electrical Engineering	5
Responsible Instructor	Dr. J.H.G. Dauwels	
Responsible Instructor	Dr. P.P. Vergara Barrios	
Responsible Instructor	Dr. J.L. Cremer	
Instructor	Dr. S.H. Tindemans	
Instructor	Dr.ing. R.K. Bishnoi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming skills in any programming language, elementary knowledge of probability theory and statistics	
Summary	This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.	
Course Contents	1. Introduction to data analytics 2. Regression 3. Classification 4. Feature engineering, selection 5. Neural Networks 6. Develop an ML workflow 7. Geometric Deep Learning 8. Tree-based methods 9. Reinforcement learning 10. Hardware in ML	
Study Goals	After this course, students will be able to 1. Draw conclusions by analysing data in the context of electrical engineering 2. Compare machine learning concepts 3. Use scikit-learn package in Python 4. Apply training strategies for machine learning models to electrical engineering problems 5. Design a workflow on an electrical engineering example	
Education Method	Interactive lectures, intuition-based,	
Course Relations	This course is mutual exclusive with the course SET3125. Students can not take this course if they have taken the course SET3125.	
Literature and Study Materials	Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.	
Assessment	Written exam (70%), project (20%), homework (10%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET3604LR	Electronic Circuits	3
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	Major topics: Electronics in information processing (sensor readout) Design principles and trajectory Signal processing functions Electronic design analysis Negative feedback amplifiers Operational amplifiers Components and networks Semiconductor diodes Power supplies and voltage regulators Semiconductor transistors and IC technology (basic) Oscillators DC motor basics	
Course Contents	This course is intended to provide gain insight, in a structured way, in the function of electronics as information-processing technology. The way of modelling will be discussed in relation to the physical characteristics of electronic components, circuits and systems. The course will provide the students with the skills to interpret circuit diagrams and measurement results and formulate specifications for electronic circuits that enable expert electronics designers to take proper design decisions when requested to design a high-performance application specific circuit. The students will learn the terminology that enables them to understand the explanation for these decisions by the experts (and potentially critically review them with respect to system level decisions). During the course many lecture demonstrations will be given, where the students both can observe the practical behavior of the test object and see what type of measurement equipment is used and how it is operated.	
Study Goals	After this course, the student -Is able to interpret electronic circuit diagrams and draw them according to standards -Is able to recognize and select electronic components -Is able to interpret and create Bode plots that describe the frequency behavior of an electronic circuit -Is able to analyze and design basic frequency filters -Is able to explain the basic behavior of semiconductor devices, like bipolar transistors, field effect transistors, diodes and special devices like thyristors. -Is able to determine the proper readout conditions for a sensor -Is able to determine the proper drive conditions for an actuator -Is able to interpret basic diode circuits like rectifiers -Is able to explain the basics of negative feedback systems and able to calculate and interpret the essential parameters -Is able to design a negative feedback amplifier using an Operational Amplifier (OpAmp) as active circuit that optimally meets the requirements -Is able to explain the basics of power circuits -Is able to name the different oscillator classes and explain the working principle the oscillators in each class -Is able to determine the class of a given oscillator -Is able to name the specific good and bad properties of each oscillator class -Is able to communicate with experts in electronic design and translate system specifications to specifications for the electronics in such a way that the expert can create an optimal design	
Education Method	Lecture. There is a mandatory practical with its own course code: ET3604LRP	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276 (older versions can very well be used for this course)	
Assessment	The exam is multiple-choice. During the lecture period, there will be one bonus test on oscillators. A pass grade gives a bonus point for the exam. The bonus point remains valid for the re-sit exam. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Dr. M.R. Vogt	
Instructor	Prof.dr.ir. O. Isabella	
Instructor	Dr.ir. H. Ziar	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	By the end of this course, you will be able to: 1. Explain the purpose of the components in different PV Systems using trends in their physical quantities. 2. Examine different power tracking methods and the importance of bypass diodes. 3. Calculate the energy yield of a PV system in conjunction with the following steps: 3a. Employ the calculation for optimal PV panel orientation from geographical survey. 3b. Determine the advantages and disadvantages of different PV systems topologies. 4. Design PV systems for different applications based on your own cost estimation and technical evaluation justified in a report	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	For students of the 2024 cohort: Final project report (33% of the final mark) and written test (67% of the final mark). For students of previous cohorts: Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr. R.T. Rajan	
Instructor	Dr.ir. R.C. Hendriks	
Instructor	Dr. J.H.G. Dauwels	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Tags	Algebra Algorithmics Artificial intelligence Linear Algebra Optimisation Signals Software Stochastics	

GEO1016	Photogrammetry and 3D Computer Vision	5
Course Coordinator	Dr. L. Nan	
Contact Hours / Week x/x/x/x	6 hours/week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Photogrammetry and 3D computer vision aim at recovering the structure of real-world objects/scenes from images. This course is about the theories, methodologies, and techniques of 3D computer vision for the built environment. In the term of this course, students will learn the basic knowledge and algorithms in 3D computer vision through a series of lectures, reading materials, lab exercises, and group assignments. The topics cover the whole pipeline of reconstructing 3D models from images: - Cameras models: how a point from the real world gets projected onto the image plane and how to recover the camera parameters from a set of observations; - Epipolar geometry: the geometric relations between 3D points and their images points; the constraints between the image points; - Image matching: define and match image features (SIFT) to establish correspondences between images; - Structure from motion: recover/refine geometry and camera parameters from a set of images; - Multi-view stereo and learning-based approaches for recovering dense geometry (e.g., point clouds) from images; - Surface reconstruction: obtain 3D surface models of real-world objects from point clouds.	
Study Goals	After finishing this course, the students will be able to: - apply linear algebra knowledge to implement basic 3D computer vision algorithms; - explain the main concepts in 3D computer vision (i.e., camera models, epipolar constraints, fundamental matrix, image matching, triangulation, structure from motion, bundle adjustment, multi-view stereo, and surface reconstruction); - explain the principles of the state-of-the-art 3D computer vision pipelines for 3D dense reconstruction from images; - evaluate methods for reconstructing smooth surfaces and piecewise planar objects, and choose applicable methods to solve specific reconstruction problems; - propose and implement solutions for reconstructing real-world buildings from images.	
Education Method	Lectures, reading materials, group assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is related to all fields of Geomatics for the Built Environment, from data acquisition, data management, data visualization, data analysis, 3D modeling, to applications.	
Literature and Study Materials	In addition to the lectures and handouts (distributed after each lecture), you will find relevant topics discussed in great detail in the following materials. [1] Y. Ma, et al. An Invitation to 3-D Vision: From Images to Models. 2001 [2] R. Hartley and A. Zisserman. Multiple View Geometry in Computer Vision. Second Edition. Cambridge University Press, 2004. [3] D. Forsyth and J. Ponce. Computer Vision: A Modern Approach. Second Edition. Pearson, 2012. [4] R. Hartley and P. Strum. Triangulation. Computer vision and image understanding. 1997. [5] Berthold K.P. Horn. Recovering baseline and orientation from essential matrix. 1990. [6] Y. Furukawa and J. Ponce. Accurate, Dense, and Robust Multiview Stereopsis. PAMI, 2010 [7] H. Hoppe, et al. Surface reconstruction from unorganized points. SIGGRAPH, 1992 [8] M. Kazhdan, M. Bolitho, H. Hoppe. Poisson surface reconstruction. SPG, 2006, 61-70 [9] L. Nan and P. Wonka. PolyFit: Polygonal Surface Reconstruction from Point Clouds. ICCV 2017.	
Practical Guide	In the assignments and group work sessions, students will experiment with the 3D computer vision techniques introduced in the lectures. - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; - Discuss the effect of parameters (if possible); - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Basic linear algebra and calculus; - Basic knowledge and skills in C++ programming. Students who have no experience in the above topics can still participate but should be willing to make efforts and invest some more time.	
Assessment	The assessment of this course consists of three group assignments and the final written exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded lower than 5.75 is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.	
Period of Education	Quarter 3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ME44312	Machine Learning for Transport and Multi-Machine Systems	3
Responsible Instructor	F. Schulte	
Instructor	Dr. B. Atasoy	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic programming	
Course Contents	The course gives an introduction to the basic concepts of machine learning (ML) (supervised, unsupervised, reinforcement learning) and puts them into relation to artificial intelligence and data analytics. Important fundamental machine learning algorithms including regression, classification, clustering, and neural networks are introduced and implemented in Python. The course also gives a brief overview of more advanced ML techniques that are used in the transportation and multi-machine domains such as reinforcement learning, graph-neural networks, and recommender systems. On this basis, machine learning methods are applied in group projects to transport and multi-machine systems using different types of data from different domains (e.g., AIS, GPS, New York Taxi, or machine failure data).	
Study Goals	-Understand the basic concepts of machine learning -Learn to implement widely-used machine learning algorithms -Apply machine learning methods to transport and multi-machine systems -Evaluate different machine learning methods for a problem setting -Conduct small ML projects	
Education Method	-Lectures -Group Work -Individual exercises for each machine learning algorithm	
Computer Use	Python on laptop	
Assessment	Group report	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Abstract Algorithmics Artificial intelligence Broad Business Design Group work Industry Modelling Optimalisation Programming Research Methods Stochastics Sustainability Technology Transport & Logistics Transport phenomena	
Percentage of Design	40%	
Design Content	Designing the right machine learning methodologies for different problem settings in the context of transport and multi-machine systems	
Department	ME Department Maritime & Transport Technology	
Contact	f.schulte@tudelft.nl b.atasoy@tudelft.nl	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Instruction if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Process & Energy	

ME45165	Equipment for Heat & Mass Transfer	5
Responsible Instructor	Prof.dr. K. Hooman	
Instructor	Dr. H.B. Eral	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Advanced reaction & separation systems Process plant design	
Expected prior knowledge	ME4500 (Advanced heat transfer)	
Summary	The course covers basic principles and methods required for the design and selection of heat and mass transfer equipment (for instance heat exchangers and thermal separation processes). This allows for the integration of heat & mass transfer principles to design process equipment.	
Course Contents	Rating and sizing of heat and mass transfer equipment such as heat exchangers, distillation, absorption, tripping, crystallization, adsorption, extraction and membrane separations will be covered. In particular: classification of various separation and heat transfer processes, the separation efficiency in single and multi-stage processes (and the influence of the process conditions and contacting modes on the separation). design procedures of different separation processes based on limitations of the thermodynamic equilibria. quantification of the influence of process condition on the design procedures of different separation processes based on limitations of the thermodynamic equilibria. characteristics of equipment for various separation processes the role of mass transfer in the sizing and design of the separation equipment. methods for the design and dimensioning of separation equipment. the use of computer tools for the design of heat exchangers and separation processes	
Study Goals	At the end of the course, the student will be able to rate and size heat and mass transfer equipment. In particular the student: 1. Can calculate overall heat transfer coefficient for heat exchangers; 2. Can select heat transfer equipment for given application; 3. Can apply effectiveness and Number of Transfer Units method to design heat exchangers; 4. Can apply pinch technology and heat integration for heat exchanger network. 5. Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes, 6. Comprehends the role of mass transfer in the dimensioning of equipment for separation processes 7. Can determine the separation efficiency for single and multi-stage separations and can quantify the influence of the process conditions and the contacting modes on the separation. 8. Can use computer based process simulation tools to analyze and design separation equipment	
Education Method	Lectures, assignments, self study, team work, tutorials	
Computer Use	Simulation of performance of distillation trays Standard engineering packages including ASPEN	
Literature and Study Materials	Course material: 1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738 2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Wiley & Sons, 2011. 3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer. 4: Course sheets / relevant papers 5: R K Sinnott, Gavin P Towler: Chemical engineering design, Coulson & Richardson's Chemical Engineering Series; Butterworth-Heinemann, an imprint of Elsevier: Kidlington, Oxford, United Kingdom, 2020. 6: Kays, W. M.; London, A. L. Compact Heat Exchangers, 3rd ed.; McGraw-Hill: New York, 1984. 7: Kroger Detlev G. Air-Cooled Heat Exchangers and Cooling Towers; Penwell Corp: Tulsa, Okl., 2004. Further reading Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son Geoffrey F. Hewitt, "Heat Exchanger Design Handbook", ISBN Online: 978-1-56700-422-9, Imperial College, England, UK	
Assessment	Formative: In-class exercises/quizzes Summative: Group project report & presentation (50%) Individual exam (50%)	
Permitted Materials during Tests	Course books and lecture notes	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Percentage of Design	50	
Design Content	Thermohydraulic design of heat exchangers; heat exchanger network, dimensioning pieces of equipment used in the	

	abovementioned processes
Department	ME Department Process & Energy

ME45170	Turbomachinery	4
Responsible Instructor	Prof.dr.ir. S.A. Klein	
Responsible Instructor	Prof.dr. R. Pecnik	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Thermodynamics, Fluid dynamics	
Course Contents	The course consists of the following parts: - Introduction and recap of the required thermodynamics. This class will give an overall overview of the relevance of turbomachinery, the general basics and the history. Furthermore the basic thermodynamics will be repeated (1 class) - Turbomachinery: basic theory and the application to the design of turbomachinery (Conservation equations and velocity triangles, turbines, compressors and pumps, 3D flow, radial machinery) (5 classes) - Application of compressor, pump and turbine maps. Limitations in the operation (stall, surge, NPSH etc) (1 class) - Gas turbine: components (turbine, compressor, combustor), gas turbine cycles, performance calculation,	
Study Goals	The student is able to describe the fundamental aspects of the working principle of turbomachinery. More specifically, the student must be able to: 1. reproduce and apply thermodynamic principles of turbomachinery 2. describe the basic fluid dynamics of turbomachinery and the way work is transferred between the working fluid and the turbomachinery, including the impact on performance and limitations of operations 3. apply compressor, pump and turbine curves in practical applications 4. describe different gas turbine cycles 5. describe the influence of different losses on the performance (power and efficiency) of gas turbines 6. synthesize different gas turbine cycles with respect to their application. 7. describe the principles of combustion as applied in gas turbines with respect to functionality and emissions 8. describe the part load characteristics of gas turbines and its relation to the functional characteristics of gas turbine components	
Education Method	Lectures and homework assignments	
Assessment	Closed book exam (60%) and home work assignments (40%). All homework assignments must be submitted. The grade for the homework assignments will only be included in the final grade if the exam grade is above 5.5/10	
Enrolment / Application	Registration: Standard enrollment via Brightspace	
Tags	Energy Energy & Industry Fluid Mechanics Thermodynamics	
Department	ME Department Process & Energy	

ME45203	Electrolysers, Fuel Cells and Batteries	4
Responsible Instructor	J.W. Haverkort	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Familiarity with transport phenomena (e.g. a course on heat transport), basic fluid dynamics, elementary differential equations, and introductory electrochemistry. Ideal preparation include the courses ME45201, CH3513, or SET3110. A previous course on fuel cells, batteries, or similar, or additional self-study may also be sufficient.	
Course Contents	Electrochemistry plays a pivotal role in the transition towards a sustainable energy future. Renewable electrical energy can be stored in batteries or used to synthesise chemicals or fuels in electrochemical reactors, like an electrolyser producing hydrogen. This course will introduce to a student, the knowledge and skills required to understand, model and optimize such electrochemical devices. Content overview: Porous electrode theory, multiphase flow, and transport models will be used to derive analytical current-voltage relations of various electrochemical devices including fuel cells, batteries, redox flow batteries, and electrolysers. Some of the topics that will be discussed in this course include: - The Nernst-Planck equations for ionic transport in binary electrolytes. - Porous electrode theory using the concept of an electrochemical Thiele modulus. - The flooded agglomerate and relative permeability models for gas diffusion electrodes. - The single particle and Newman's P2D battery models. - A vertical current and gas distribution model for electrolysers.	
Study Goals	After successful completion of this course, the student is able to independently: 1. explain and apply porous electrode theory. 2. derive 0D, 1D, and 2D mathematical models to describe the current-voltage relationship of various electrochemical cells including flow, multiple phases, and porous electrodes. 3. optimize an electrochemical cell design with respect to, for example, energy efficiency.	
Education Method	Two lecture hours + two exercise hours per week.	
Course Relations	This course is a natural sequel to ME45201, CH3513, and SET3110 for students interested in analytical modelling of electrochemical systems.	
Books	Electrolysers, Fuel Cells and Batteries: Analytical Modelling. (J.W. Haverkort) https://doi.org/10.59490/tb.93 (open access)	
Assessment	Written exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	

ME46006	Physics for Mechanical Engineers	4
Responsible Instructor	Dr. G.J. Verbiest	
Instructor	Prof.dr. P.G. Steeneken	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have Physics knowledge at the level of the final exam of the physics exam at Dutch highschools (VWO) and to have Mathematics knowledge at the level of the BSc education in Mechanical Engineering (e.g. WBMT2048). If they have not, they can still do the course but it will require more self-study from the book of Tipler.	
Summary	The course ME46006 Physics for Mechanical Engineers, aims to provide mechanical engineering students with a broad background in physics. The course offers basic understanding and skills in the field of physics, that will provide a background that can be useful in many engineering situations. It can serve as a starting point for more advanced physics studies and can also help mechanical engineers to interact with electrical engineers and physicists in their future career. Since the course aims for a broad coverage of the field of physics, the course material includes many different subjects at a basic to intermediate level. In contrast to other courses, this course favours to cover a relatively large number of different topics in order to provide a broad background, instead of covering a small number of topics in-depth. This means that the amount of material that needs to be studied is larger than for other courses. A bachelor degree in mechanical, electrical, aerospace, industrial engineering or applied physics is considered prerequisite.	
Course Contents	- Waves - Electrostatics and electricity - Magnetism - Electrodynamics - Optics - Interference and diffraction - Outlook into quantum mechanics	
Study Goals	At the end of the course, students should have knowledge, and should be able to solve problems on the following topics: Waves (Chapters 15&16 of Tipler): - The wave equation - Propagation, reflection and transmission of waves - The Doppler effect Electrostatics (Chapters 21-24): - Coulomb's law and electric forces - Electric fields and Gauss' law - Voltage and capacitors Electricity (Chapters 25 & 29): - Ohm's law - DC circuits Magnetism (Chapters 27-28, 30): - Gauss' law for magnetism - Lorentz', Ampere's, Lenz' and Faraday's law - Electromagnetic motors and inductors - Maxwell's equations Optics (Chapter 31-34): - Propagation and properties of light - Light-matter interaction - Imaging systems with light - Interference and diffraction - Limits of optical systems and quantum mechanics	
Education Method	Learning methods include power point presentations, class demonstrations, and recommended homework. Lecture notes will be made available on the Brightspace. Recommended homework will be on the concepts discussed in the class and solutions will be made available on the Brightspace. The course ends with a written exam.	
Literature and Study Materials	The course will closely follow selected chapters of the book of Tipler. In addition lecture slides will be provided on Brightspace.	
Books	Physics for Engineers and Scientists, 6th edition by Tipler and Mosca.	
Assessment	Written exam.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Precision & Microsystems Engineering	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Prof.dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Dr. O. Nejadseyfi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be 5.8 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest decimal. The maximum bonus score is 0.8. The bonus only applies to reports with grade ≥ 5.5 and submitted before the first report deadline.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	ME Department Precision & Microsystems Engineering	
Contact	Prof.dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

ME46085-23	Mechatronic System Design	4
Responsible Instructor	Dr. S.H. Hossein Nia Kani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The following are strictly required to be able to follow the course: - Basics of Multi-body Dynamics, including the ability to form equations of motion. - Basics of Electromagnetism, particularly Maxwell's equations. - Fundamentals of signals and systems (Laplace transforms, transfer function, bode plot, Nyquist stability). It is expected that students are able to use MATLAB or are interested in spending time to learn it while passing the course. This is required to complete the assignments.	
Course Contents	Course Contents Mechatronic system design deals with the design of controlled motion systems by the integration of functional elements from a multitude of disciplines. It starts with thinking how the required function can be realised by the combination of different subsystems according to a Systems Engineering approach. It should be noted that the control principles used in this course place a strong emphasis on frequency domain methods by linearising the system at its working point with the use of Bode- and Nyquist plots. The main reason for this emphasis is the strong focus in other control related courses on (non-linear) time domain related methods while linearised frequency domain related methods are still dominantly applied in the industry. A mechatronic engineer should be able to work with both methods and use them where appropriate. Some supporting disciplines, like power-electronics and electromechanics, are not part of the BSc program of mechanical engineers. For this reason this course introduces these disciplines in connection with mechanical dynamics and PID-motion control principles to realise an optimally designed motion system. The target application for the lectures are motion systems that combine high speed movements with extreme precision. The course covers the following three main subjects: 1: Dynamics of motion systems in the time and frequency domain, including analytical frequency transfer functions that are represented in Bode and Nyquist plots. 2: Electromechanical actuators, mainly based on the electromagnetic Lorentz principle. Reluctance force and piezoelectric actuators will be shortly presented to complete the overview. 3: Motion control in the frequency domain with PID and model-based feedforward control-principles that effectively deal with the mechanical dynamic anomalies (resonances and eigenmodes) of the plant. The other relevant discipline, electronics and position measurement systems is dealt with in another course: ME46005, Physics and measurement. The most important educational element that will be addressed is the necessary knowledge of the physical phenomena that act on motion systems, to be able to critically judge results obtained with simulation software. The lectures challenge the capability of students to match simulation models with reality, to translate a real system into a sufficiently simplified dynamic model and use the derived dynamic properties to design a suitable, practically realisable controller. This course increases the understanding what a position control system does in reality in terms of virtual mechanical properties like stiffness and damping that are added to the mechanical plant by a closed loop feedback controller. It is shown how a motion system can be analysed and modelled top-down with approximating (scalar and linearised) calculations by hand, giving a sufficient feel of the problem to make valuable concept design decisions in an early stage. With this method students learn to work more efficiently by starting their design with a quick and dirty global analysis to prove feasibility or direct further detailed modelling in specific problem areas.	
Study Goals	Can analyse and derive improvements to the dynamic behaviour of an actuator-driven mechanical structure with maximum 6th order plant dynamics (incl actuator and amplifier) by means of Bode and Nyquist plots. Can select and calculate a single-axis functional electromagnetic actuator for a given specification, working according to the Lorentz and reluctance force generation principle. Can select a proper amplifier to drive the above mentioned electromagnetic actuator. Can understand the basic concept of piezoelectric actuators and their application in mechatronic systems Can identify and apply feedforward and feedback P-, PD- or PID-motion controller settings for a given plant, consisting of a dynamically realistic power amplifier, electromagnetic actuator and mechanical structure with an ideal sensor, to achieve a stable system targeting a specified maximum bandwidth or disturbance rejection.	
Education Method	Lectures are given in 12*2 lecture hours, with presentations on the theory and practice of active-controlled motion systems. In addition, students will work on assignments to implement the theory learned in the course.	
Literature and Study Materials	Only the book is allowed at the examination. Therefore, no printed copy of the first edition, no computers, e-readers, smartphones, or other items are permitted. However, small notes written on the pages of the book are allowed.	
Books	The Design of High Performance Mechatronics - 3rd Revised Edition by R. Munnig Schmidt, G. Schitter, A. Rankers and Jan Van Eijk	
Assessment	The Final Assignment will account for 50% of the final grade. The remaining 50% of the score will be based on an open-book written examination. The examination will consist of questions covering the above-mentioned study goals.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams.	

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Contact	s.h.hosseinniakani@tudelft.nl

ME46095	Multiphysics Modelling using COMSOL	4
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Contact Hours / Week x/x/x/x	2 hour lectures on a weekly basis	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In applied mechanics one is often confronted with a multi-physics or coupled problem: A problem that requires the (simultaneous) solution of more than one type of physical process or phenomenon in order to accurately describe the problem. Examples of multiphysics problems are fluid-structure interaction, thermal-structure interaction and electro-thermal-structure interaction, possibly combined with a control problem. Particularly in the field of Mechatronic design and MEMS multiphysics problems are frequently encountered. COMSOL MultiPhysics is a finite element code, which can be used both as a MATLAB toolbox and as a standalone program, which is particularly suited for the simulation of multi-physics systems. In this course the student will learn to recognize different types of multi-physics coupling and methods for their efficient numerical solution using COMSOL. Short homework assignments are used to practise the use of COMSOL on different types of problems and in a final assignment the student is asked to study a multi-physics problem using COMSOL.	
Study Goals	The student must be able to: 1. recognize multiphysics coupling in complex problems 2. distinguish between different types of coupling one-directional vs. bi- or multi-directional interface vs. field strong vs. weak 3. describe numerical solution techniques applicable to coupled problems 4. use COMSOL MultiPhysics on coupled problems	
Education Method	(Online) Lectures (2 hours per week) / Self study	
Assessment	Report	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: WB2454-07	
Department	ME Department Precision & Microsystems Engineering	

ME46300	Optics	4
Responsible Instructor	Dr.ir. S.F. Pereira	
Instructor	Dr. N. Bhattacharya	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to optics: light propagation, the vectorial nature of light, interference, diffraction, optical spectra, geometrical optics	
Study Goals	At the completion of this course the student will be able to - have a good understanding of the basics of optics (i.e. making new connections between learned concepts and applying what has been learned to new situations) - explain the basic laws of polarization, birefringence, interference and diffraction - apply the theory to solve simple problems - put the theory in practice by performing experiments.	
Education Method	Lectures, exercise sections and practicals	
Books	Textbook that will be used in this course: Introduction to Modern Optics, G.T. Fowles, second edition, ISBN 100486659577 plus lecture notes and slides (available on Brightspace)	
Assessment	Individual written exam (90% of the grade). If it is required that the exam will be online, we will have an oral check after the exam and similarity check. The grade of the practicals (10% of the final grade) will be the average of the written group reports.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Precision & Microsystems Engineering	

ME46310	Opto-Mechatronics	4
Responsible Instructor	Dr. N. Bhattacharya	
Instructor	Dr.ir. S. Iskander-Rizk	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	Optical sensors and actuators are being integrated at an accelerated rate into mechatronics systems making these systems more precise, autonomous and intelligent. In this course the principles of optics, optomechanics and optical detectors are taught and discussed from the perspective of integration in optomechatronics systems for increased performance and improvement/addition in functionality. Topics included in this course are: 1) The different models of optics 2) Optical measurement methods. 3) Tolerances and sensitivities of mechanical positioning which influence optical performance of a machine. 4) Total system design where optical requirements can be translated to mechanical and mechatronical requirements. 5) Sources of light and optical detectors which can influence system performance (SNR, reliability, accuracy).	
Study Goals	At the end of this course the student can LO 1: Understand the practical applications and limitations of the different models of optics (geometric optics, diffraction optics, electromagnetic waves). LO 2: Understand the basic principles of imaging, interferometry, and spectrometry. LO 3: Understand how aberrations are related to mechanical tolerances and imaging quality. LO 4: Evaluate the function and performance of an optical system including the electronic readout module LO 5: Design an optical system for given environmental and application requirements (stress, thermal, kinematic effects) LO 6: Justify the mechatronic system components choices which can improve the performance of a given optomechanical system LO 7: Understand the influence of material choices and types of mounting optics	
Education Method	Lectures and Quizzes	
Assessment	Written exam + Quiz grade + 2 Guest lecture summary.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Broad Challenging Industry Mechanics Mechatronics Physics Signals	
Department	ME Department Precision & Microsystems Engineering	

MS43310	Materials at High Temperature	4
Responsible Instructor	Dr. V.A. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on Brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Materials Science & Engineering

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. L. di DiCarlo	
Responsible Instructor	Dr. M. Blaauboer	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	The final grade for the final exam is composed of the homework assignment grades (total of 30%), online quiz grades (total of 10%) and the grade obtained for the final exam (60%). A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course. The final grade for the resit exam consists of the resit exam grade only (100%). In other words grades for homework and quizzes do not count towards the final grade of the resit exam.	

RO47001	Robot Dynamics & Control	5
Responsible Instructor	L. Peternel	
Instructor	B. Shyrokau	
Instructor	J.M. Prendergast	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides the basics for deriving kinematic/dynamical models of a robot (such as a factory robotic arm, mobile robot, etc.) and for designing corresponding robot controllers. Students will learn how to derive the equations of motion and basic control equations themselves, and they will also learn how to use existing frameworks to do the same. THEORY SUBJECTS Introduction & Fundamentals -Course layout, instructors, grade scheme, etc. -Importance of the course and the big picture -Connection to real-world -Basic concepts/terms in robotics -Methods to describe robot pose -Numerical integration Recap of Control Theory -State-space model -Root locus -Bode plot -PID control -Concept of LQR and MPC control -Reference to online materials Endpoint Control for Robotic Manipulators -Big picture + motivation -Difference between position and torque-controlled robots -Endpoint position control (with a PID controller) -Endpoint impedance control (for position tracking tasks) -Endpoint force control (for force tracking tasks) Robotic Arm Kinematics -Deriving forward kinematics and Jacobian (for a 2-DoF planar arm) -Transformations between endpoint and joint space using Jacobian -Kinematic singularities and their solution -Kinematic redundancy and task priority control (for a 4-DoF planar arm) -Kinematics of parallel robots and differences compared to serial robots Robotic Arm Dynamics -Rigid-body dynamics model -Deriving dynamics model with Lagrange method for 2-DoF planar arm -Using the dynamics model in the control -Dynamic task priority control 3D Robotic Arm -Deriving 3D kinematics using the Denavit-Hartenberg convention -3D trajectory generation Planning & Motion Control -Orientation control -Stability analysis of the controller Kinematics & Dynamics of Mobile Robot / Automated Vehicle -Forward kinematic models -Wheel kinematic constraints -Robot kinematic constraints -Non-holonomic system -Derivation of linear bicycle model -Steady-state analysis Dynamics & Control of Mobile Robot / Automated Vehicle -Transient analysis -Frequency response -Path-following control PRACTICAL ASSIGNMENTS (individual) Assignment 1: The goal of this assignment is to learn how to control a 2-DoF robotic arm in Python to follow a prescribed trajectory using kinematic and dynamic equations. Assignment 2: The goal of this assignment is to learn how to control a multi-DoF robotic manipulator (3D kinematics, 3D trajectory, orientation control). Assignment 3: The goal of this assignment is to learn how to implement a linear bicycle model, and calculate and discuss primary handling characteristics, poles, stability analysis, and feedback control (PID).	
Study Goals	After completing this course, the students are able to:	

Education Method	-Derive the equations of motion for different types of robots -Create joint-level and endpoint-level controllers (including trajectory following) -Implement a simulation model in Python or Matlab -Analyse the designed control systems and reflect on the design choices -Lectures & interactive sessions (4 hours per week) -Practical assignments (homework)
Literature and Study Materials	Note: While attending lectures & interactive sessions is not mandatory, it is highly recommended. Assignments are not strictly mandatory, however, if you do not submit them you will lose the corresponding percentage of the final grade. That might mean that even if you pass the exam, you might not have a high enough final grade to pass the course. Study materials -Lecture slides -Papers, books, videos, etc. (links provided in Brightspace) Robotics textbooks (either one) -Lynch, K. M., & Park, F. C. (2017). Modern robotics. Cambridge University Press. -Craig, J. J. (2017). Introduction to robotics: mechanics and control. Pearson Education. -Siciliano, B., Sciavicco, L., Villani, L., & Oriolo, G. (2010). Robotics: modelling, planning and control. Springer Science & Business Media. Control theory textbooks (either one) -Ogata, K. (2010). Modern control engineering. Prentice hall. -Dorf, R. C., & Bishop, R. H. (2001). Modern Control Systems: Solutions Manual (Vol. 12). Prentice Hall. Useful toolboxes -Corke, Peter. Robotics Toolbox for Python. -Foote, Tully. "tf: The transform library." 2013 IEEE Conference on Technologies for Practical Robot Applications (TePRA). IEEE, 2013.
Prerequisites	-Linear algebra (including matrix and vector operations) -Calculus -Linear system theory at the BSc level (e.g., simple differential equations, Laplace transforms) -Basic programming experience at the BSc level (Python/Matlab)
Assessment	Number of assessments: -Four (3 assignments, 1 exam), all based on individual work. -The exam will be a computer/digital exam in ANS on campus. Grade calculation: -If and only if the exam grade is 5.5 or higher, the grade will consist of 30% of the average score of the assignments and 70% of the exam grade. The final grade must be 6.0 or higher to pass the course. Assignment submission lateness criteria: <1 hour: -1 point <1 day: -2 points <2 days: -4 points <3 days: -6 points >3 days: no points given
Enrolment / Application	Note: Assignments and exam grades are only valid for the academic year they were obtained in. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department Contact	ME Department Cognitive Robotics For general course inquiries: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 1 and related content: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 2 and related content: Micah Prendergast (J.M.Prendergast@tudelft.nl) For Assignment 3 and related content: Barys Shyrokau (B.Shyrokau@tudelft.nl)

RO47002	Machine Learning for Robotics	5
Responsible Instructor	Dr.ir. J. Kober	
Instructor	Dr. H. Caesar	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides a broad overview of machine learning techniques and their practical application in robotics. Intro to Machine Learning and python - Machine Learning in Robotics - Understanding the goal of machine learning, fundamental problems, high-level overview. Visualizing your data - Supervised vs Unsupervised vs Reinforcement learning - Model-driven vs Data-driven: White box, Gray box, Black box - Prior knowledge vs Unstructured. Feature extraction, linear regression + Deep Learning. Interpretability Hands-on Machine Learning - Example machine learning project in Robotics - Binary classification, decision boundaries, using logistic regression/SVM (one parameter) - How to perform a classification experiment. Dataset splitting, learning curves, metrics, comparing results Regression & Data collection - Regression and Data Collection in Robotics - Regression methods, least squares fitting - Overfitting, cross validation, regularization - Collecting (noisy) data, labelling data, outliers - High-dimensional data, data augmentation - Hyper-parameter optimization (grid search vs random search) Classification - Classification in Robotics - Parametric vs Non-parametric classifiers - Logistic regression - Decision tree, forest - Bayesian classification: Bayes' rule, naive Bayes, Gaussian Mixture Model - k-nearest neighbour - SVM, kernel-SVM, dual problem - Multi-class classification, metrics (confusion matrix), class imbalance Unsupervised Learning - Unsupervised Learning in Robotics - Clustering: K-means, Gaussian Mixture Model, DBSCAN - Dimensionality reduction: PCA, Local Linear Embedding (LLE) Neural Networks - Neural Networks in Robotics - Multi-Layer Perceptron, gradient descent - Neural Networks, and deep learning - Vanishing gradient problem, DropOut, Optimizers Advanced machine learning - Outlook: Recurrent neural networks, convolutional neural networks - Outlook: Data augmentation, AutoEncoder - Outlook: Reinforcement learning	
Study Goals	Knowledge, insight, judgement and skills: After completing this course, students will be able to - explain the role of ML in robotics, and describe possible use cases, and when not to use ML - explain the main concepts in ML (overfitting, training and validation, classification vs regression, cross validation, regularization, parametrics vs. non-parametrics, supervised/unsupervised/reinforcement learning) - explain the principles of well-established ML methods (e.g. Logistic regression, SVM, RFs, Neural Networks) - collect and preprocess data (normalization, feature selection, augmentation, splitting) for ML - analyze a ML problem, consider available data and computational resources, and select the appropriate ML methods to apply - perform experiments, evaluate the results, and draw sound conclusions - present the findings of a ML experiment clearly in a report using graphs and scientific English - write python code to visualize data and implement ML methods using popular python frameworks, such as pandas, numpy, matplotlib, jupyter notebook, sklearn, keras Transferable and interpersonal skills: After completing this course, students will be able to - collaborate and coordinate with a lab partner on coding and reporting tasks	
Education Method	Contact hours - Flipped classroom: pre-recorded lectures (2 hours per week) + classroom discussion (2 hours per week) - Q&A on practica (2 hours per week) Formative feedback - Quizzes -- Individual -- On material of previous lecture(s) -- Via BrightSpace at set time slots - Practica -- Pairwise collaboration -- Submission via BrightSpace -- Counts for knock-out criteria -- No late submission -- No resit Deadlines - For each practicum, the deadline is approx. one week after the practicum is published.	
Literature and Study Materials	Study materials - Slides (released shortly before lectures) - Book: "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent	

	Systems" 3rd edition, Aurélien Geron
Prerequisites	Suggested literature - "Pattern Recognition and Machine Learning", C. Bishop - "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Hastie, Tibshirani, Friedman TU Delft BSc Mechanical Engineering curriculum, or equivalent. Especially familiarity with - Programming in python - Linear Algebra - Probability theory
Assessment	Final assignment (30%) - Pairwise collaboration - Late submission: -1 grade point per day - No resit - Deadline: Will be released after the 6th practicum, giving the students approx. two weeks to read the assignment, collect the necessary data, perform their experiments, and write their report Exam (70%) - Individual - Written closed-book exam - Exam in Q1, resit in Q2 Knock-out criteria - At least 4 practicum grades need to be a pass (of current or previous academic year) - Minimum grade for final assignment is a 5 (of current or previous academic year) - Minimum grade for the exam is a 5 - Final grade minimum is a 6
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics

RO47007	Multidisciplinary Project	5
Responsible Instructor	Dr. C. Pek	
Instructor	Dr.ir. G.N. Saunders	
Instructor	Prof.dr.ir. M. Wisse	
Instructor	C. Hernandez Corbato	
Instructor	Ir. C.C. Claij	
Contact Hours / Week x/x/x/x	0.0.0.2	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In this capstone project of the first year of the MSc Robotics, you will be challenged to use all the knowledge and skills you have gathered so far. Within a multidisciplinary team of up to 5 people, you will develop and test software for use in a real robot whilst working together for a real client in an industrial setting. We expect all your robotics skills (including your programming skills) to be up to date! The estimated workload of this course is 18 hours per week over 8 weeks. In this project, you will not just produce a software solution for a robot. You will also implement and test your software in a real robot and report and present your results as well as document your code, mirroring industry practice. Just like you would in industry, you will likely spend only 50% of your time coding and testing and the other 50% on meetings, systems design, report writing, (preparing) presentations, and documenting your code. These are no longer afterthoughts. In this project, you work closely with an industry client, realistically a possible future employer. That means there are expectations regarding your commitment and presence during the project. We expect you to behave cordially and respectfully towards the client and be on time for scheduled sessions. You will be given the option to express a preference for a client when you enroll if multiple clients take part in the course. Each of you will sign up with a specialist role in mind within the project, in which you are responsible for a specific part of the software design and implementation. It is no longer possible for one person to do all the programming. The specializations for which you can express a preference when you enroll are: - human-robot interaction, - navigation, planning, control - perception, and relate to the different disciplines within robotics. Within your team, you are expected to make a translation of societal issues from different perspectives (human, sustainability, safety, ethics, economic, etc. into intelligent robot solutions. Practically, each student team develops a complete, integrated software package for a complex robot system. You develop and test in a simulated environment, and then implement it in a real robot during the course before presenting your results in the final presentation to your client at the end. This course consists of 5 iterative phases with continuous reporting varying in lengths: - Phase 0: Introduction, meet your team, and get to know your robot. - Phase 1: Initiation and Planning - Getting organized, defining and planning your project. - Phase 2: Design & Execution - Software architecture design, initial (simulation) implementations and demonstration of concept to the client. - Phase 3: Implementation - Further design and implementation of your solution on your robot. - Phase 4: Testing & Completion - Finalizing your project and final reporting and code documentation, with a presentation of your implementation for the client in week 4.8. A detailed schedule will be available on Brightspace after the introduction lecture, likely in week 3.8.	
Study Goals	The learning objectives for this course are divided over two domains: 1) Knowledge, insight, judgment and skills By the end of the course, you should be able to: LO 1: Define a robotics use case with its functional and non-functional requirements. LO 2: Design robotic solutions by integrating knowledge of system engineering, current opportunities for robots, trends in robotics, and societal aspects. LO 3: Produce a robotic solution that meets the design specifications and considers sustainable, safety, ethical, economical and other relevant societal aspects. LO 4: Use functional architecture for planning and communicating robot software. LO 5: Communicate the multidisciplinary robot solution orally, in writing, and in code documentation according to robot software development guidelines. 2) Transferable and interpersonal skills. By the end of the course, you should be able to: LO 1: Formulate learning goals for your own personal development. LO 2: Reflect on your competencies (e.g., Teamwork, Leadership, Entrepreneurial thinking, Strategic multidisciplinary problem-solving), their development, and your personal learning goals. LO 3: Apply structured feedback to improve your performance or the performance of others. LO 4: Apply structured multidisciplinary software project management, with the use of different team roles and responsibilities.	
Education Method	Project work in teams supported by (video) lectures and workshops.	
Computer Use	During the project, we will be using different types of software and collaboration platforms. To save time and frustration, please ensure you have a working laptop that has the following software installed and have gained access to the following repository before the project starts: ROS: We expect you to have a working ROS environment on your laptop. Please use your knowledge and experience from RO47003 Robot Software Practicals and make sure that you align with the Ubuntu and ROS versions used in your clients robot. We will be using native installations of Ubuntu and ROS, and do not provide support for any other installation. Instructions on this will be published on Brightspace. Gitlab: All code and documentation for this project will be stored on GitLab. To be added to your groups repository on the GitLab environment for this course, go to: https://gitlab.tudelft.nl and log in using your NetID. Please do this as soon as you have been admitted to the course and have been assigned a group. Otherwise, you will not be able to access the required code for your project.	
Literature and Study Materials	During the project we expect you to use and refer to the following books. Please avail yourself prior to the course starting of either a hardcopy or an electronic copy of the book. [1] A Gentle Introduction to ROS Jason OKane, E-Book: Accessible via the authors website: https://jokane.net/agitr/ [2] Practical Model-Based Systems Engineering Jose Luis Fernandez & Carlos Hernandez Corbato, E-book: Accessible with the TU Delft library web extension In addition, you may also want to refer to these books. These are, however, not mandatory for the course: [3] Beginning Software Engineering Rod Stephens, Hardcopy: Available via Leeghwater, E-book: available through the TU Delft library extension. [4] Information Technology Project Management K. Schwalbe, Hardcopy: any online bookshop, E-Book: no free legal website available	
Prerequisites	Starting the project is only allowed when a student has successfully completed the course RO47003 "Robot Software Practicals" and has completed RO47000 Vision & Reflection. Without these preliminaries, the student will not be allowed to participate. Exemptions require written approval of the teachers of RO47003 and RO47000 that the student has obtained equivalent	

Assessment	qualifications. It is expected that you can code when you start this course. At the end of the course, your project needs to comply with a few requirements before we will grade it. These will be clearly listed in the course manual on Brightspace. Moreover, you must attend all mandatory sessions of the course. Excused absences are only permitted by receiving prior approval of a study advisor and the course staff. Absences due to medical reasons require a confirmation, e.g., by a study advisor. For this project, an individual grade is given. The grade for this project is a combination of the group grades for your final project including your final presentation, the Sprint assignments, and an individual grade based on your personal reflection. An individual bonus-malus of max. 1 grade point can be applied by staff in rare cases of excellence or underperformance. For more details on grading, see the course manual on Brightspace.
Enrolment / Application	Enrollment for the course will open at the beginning of Q3. Further details on how to enroll will be announced on Brightspace. All students must have registered before the announced registration deadline. It is the students responsibility to acquire the required registration information and deadline. After the registration deadline, late participation in the project is not possible anymore. Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Special Information	Note: The mandatory introduction lecture and the ROS software & robot programming sessions will take place at the end of Q3, likely in week 3.8.
Remarks	This is a very intensive course which requires you to work 18 hours per week over a period of 8 weeks regardless of Public Holidays. The course requires intensive teamwork and good time management. On average there is 1 mandatory session per week you must be available for which can vary over the weeks depending on the availability of robots and industry clients. Past experience has shown that combining this course with too many electives and/or other intensive activities during weekdays is incompatible. Be prepared you will likely spend 40h per week on this project and any electives combined. It is also not recommended to start with your literature research during MDP.
Department	ME Department Cognitive Robotics

RO47015	Applied Experimental Methods	5
Responsible Instructor	Dr. Y. Vardar	
Instructor	Prof.dr.ir. D.A. Abbink	
Contact Hours / Week x/x/x/x	0.0.0.x	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Study Goals	After completing this course, students will be able to: - o independently design, execute and analyse a human factors experiment - o formulate a clear hypothesis in the field of human factors that can be tested experimentally, in a realistic amount of time with a realistic amount of means - o set up a methodologically sound experiment to test this hypothesis - o select, adapt or build an experimental setup - o gather reliable experimental data on test subjects - o choose appropriate metrics - o apply correct statistical techniques - o choose the most relevant results, and construct the most clear figures to show them - o critically reflect on your own experimental findings - o communicate clearly through presentations and writing	
Education Method	Students will perform a mini-graduation project in six weeks, in small groups (2-3 students). They will independently design, perform, analyze and present a human factors experiment, guided by four supervisors.	
Literature and Study Materials	Scientific papers	
Assessment	The final grade is composed of the following sub grades 10% Presentation 1: Hypothesis 10% Presentation 2: Methods 10% Presentation 3: Analyzing Results (two main results figures, explain) 20% First Draft for Paper 50% Final Paper In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment can take place in a similar way.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Broad Design Group Dynamics/Project Organisation Group work Integrated Intensive Involved Lab Research Project planning / management Small groups	
Department	ME Department Cognitive Robotics	

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory	
Course Contents	Fundamentals of Coding in Python and MatLab Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Symbolic regression * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control We will aim for a balance of practical intuition and theoretical understanding: we will discuss examples of applications and go deeper into the theory when feasible.	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures (recorded and in class) and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: (i) practical assignment and (ii) written exam, open book. You must score at least 5.5 on both assignments to get a pass. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity to resit if they fail the written examination. * If your practical assignment is graded between 5 and 5.75, you will have two weeks to integrate the assessors feedback. If this is done satisfactorily, you can be rewarded with a mark that is not higher than 6/10. The exact instructions for the addition will be announced after the grading of the PA is concluded.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding course registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) exam registration is obligatory. More information regarding exam registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Artificial intelligence Complex Mathematics Intensive Linear Algebra Mathematics Modelling Programming Project Proofs	
Department	ME Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42030	Control for High Resolution Imaging	3
Responsible Instructor	Dr.ing. C.S. Smith	
Responsible Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	sc42065	
Expected prior knowledge	Basic knowledge of physics and control theory is expected. The first lectures recap in a condensed form the necessary knowledge in optics. Understanding of signal processing (in particular, the Fourier analysis) is expected for successful work on the homework assignments. The course involves some basic numeric simulations, so practical knowledge of Matlab/python/... is required.	
Course Contents	High-resolution imaging is crucial in scientific breakthroughs, such as discovering exoplanets in the habitable zone or understanding the origin and progress of diseases at a molecular level. For that purpose, special optical instruments like Extreme Large Telescopes or STED microscopy are developed. Optical aberrations are the key obstacle that hampers a clear vision and invites control engineers to step in. These are the disturbances induced by the medium, like turbulence in the case of astronomy, or by the specimen under investigation, like the change in refraction index due to inhomogeneities in the biological tissue. Adaptive Optics is a modern way to remove the effect of aberrations. This fascinating and expanding field in science is providing an excellent challenge to control engineers to improve image quality significantly by active control. This course will review the hardware necessary to control light waves in contemporary optical instruments, their modelling from a control engineering perspective, and discuss model-based control methodologies to do disturbance rejection.	
Study Goals	Understand the propagation of light, imaging and aberrations in the imaging process. Understand the operation principle of pupil-plane and focal-plane sensors to estimate the wavefront aberrations. Understand the design principles of opto-mechatronic wavefront corrector devices to correct wavefront aberrations. Develop spatial and temporal models of complete imaging systems and use these models in the design of model-based controllers for aberration correction.	
Education Method	Oral Presentations (seven lectures), homework assignments (4 in total), and self-study in the form of reading and presenting a scientific paper. Seven seminars (interactive discussions related to the homework assignments) are interleaved with the lectures, using the same time slots. Some homework assignments are allowed to be done in small groups; paper presentations are done in groups.	
Literature and Study Materials	Course Notes; selected chapters from published books and relevant scientific papers are provided as complementary reading material.	
Assessment	SC42030: (homework and paper presentation) 80% + oral examination 20%.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC4045	
Elective	Yes	
Tags	Broad Mathematics Optimalisation Physics Signals and Systems	
Department	ME Department Delft Center for Systems and Control	

SC42065	Adaptive Optics Design Project	3
Responsible Instructor	Dr.ing. C.S. Smith	
Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of signal/Fourier analysis, linear algebra, and optimisation is required. Basic skills in Python are required.	
Course Contents	The course consists of the realization of laboratory experiments to design and operate Adaptive Optics equipment to realize high-resolution imaging systems. Crucial in the design is the alignment of active optics systems and in the operation the development of algorithms for acquiring accurate information about the wavefront aberrations from intensity-based imaging components (like Shack-Hartmann sensors or CCD cameras) and using this wavefront information in the tuning of multivariable dynamic controllers to compensate in real-time the wavefront aberrations. The design is conducted under close supervision by world-leading experts in the field and is performed in groups of students. The size of the groups depends on the number of participants in this course. The course requires hands-on experiments, Python coding, and presenting the results in a report and/or a joint final presentation.	
Study Goals	Building insights about the key components in Adaptive Optics such as the wavefront reconstruction and the deformable mirror. As well as building the controller methodology to obtain a smart optics system for high-resolution imaging.	
Education Method	Project Based The work is subdivided into sessions. Each session results in a deliverable (e.g. some measurements or a piece of code). The work is performed in groups, which should be well balanced in, for instance, writing, coding, and theoretical skills.	
Computer Use	Lab computers are available, own laptops can be used	
Course Relations	Participation in SC42030 is required	
Practical Guide	A practical guide subdivided into sessions is provided.	
Assessment	Oral Presentation and evaluation of the written report in combination with the course sc42030	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses . To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams . *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations .	
Remarks	Old course code: SC4115	
Department	ME Department Delft Center for Systems and Control	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. R. Kortlever	

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced, analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	The final grade of the course consists of the following components: - Take home exam (50%) - oral exam (50%) Final grade calculation: $(0.5 * \text{take home exam}) + (0.5 * \text{oral exam})$ The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 take home exam: resubmit deliverable oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. H.M. Schuttelaars	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	After this course you are able to: - recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - solve the shortest path problem and the transportation problem as well as some small flow shop problems. - solve the travelling salesman problem by the Branch and Bound algorithm. - apply several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - reproduce basic theorems concerning the mentioned problems and is able to prove some of these theorems. - explain how to solve large scale problems, especially shortest path and vehicle routing problems. - define scheduling problems and complexity theory. - implement the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam and a written resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Elective	Yes	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr.ir. J.T. van Essen	

WI4156(TU)	Game theory	6
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Open to all MSc students with a strong command of mathematics. Ideally suited for math students with an interest in finance.	
Expected prior knowledge	Standard BSc math courses: calculus, linear algebra, probability.	
Course Contents	This is a course in "mathematical" game theory, which is the counterpart of "algorithmic" game theory (agent based modelling, this part of game theory is taught in the computer science department). We cover all subjects from combinatorial games to cooperative games.	
Study Goals	Understanding the mathematical theory of rational behaviour/conflicts/negotiations/social choice. Game theory is one of the most prominent economic theories: three recent Nobel prizes have been awarded to game theorists. We will cover a very wide area, ranging from pure mathematics to economics, with an emphasis on the mathematical aspects. This is a course in the Finance Track of the Applied Maths programme, but it is open to all MSc students (math and non-math).	
Education Method	Lectures	
Literature and Study Materials	Game Theory - Thomas Ferguson - UCLA: find it on the net.	
Books	T.S. Ferguson, A course in game theory, World Scientific, 2020	
Assessment	The final grade of the course consists of the following components: written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students are expected to have: superior knowledge and skill in elementary probability (details below); experience working with Python, Matlab, or R; and a mathematical inclination. If this does not apply to you and you still want to enroll: contact the instructor. Good skills in elementary probability are indispensable as they are the foundation for the more advanced material of the course. We build on the concepts, tools, and methods of elementary probability, and students taking this course must be (or rapidly become) black-belt masters in these foundations, specifically: Chapters 2-5, 7, 9, and 10 of Dekking et al; prior knowledge of Chapters 13 and 14 is helpful, though we will revisit these subjects during the course. Testing yourself and doing a thorough review in the first few weeks is required; we will also spend some time on this in class to help you do this efficiently; after this a diagnostic test (carrying part of the course grade) will assess your mastership; sub par performance means you cannot follow the course.	
Course Contents	This course on Monte Carlo simulation and stochastic processes consists of a mixture of these components: modelling and models (and some of their pluses and minuses), learning a number of stochastic processes and some of their properties, doing Monte Carlo simulations, analyzing the generated data and drawing conclusions from them. Contents and format of this course are intended to combine the following: 1. real (possibly stylized) problems as starting points; 2. enough about modelling to make participants aware of the impact of modelling decisions; 3. identification of a number of stochastic processes present in/related to models and problems; 4. (limited) treatment of underlying theory, guided by: (a) what is necessary to ensure that participants know how to do things right and (b) what enables them to do things smartly; 5. facts, properties, and theorems: (a) sincere attempt to understand where possible; (b) focus on practical implications and uses; (c) illustrations rather than proofs or theory; 6. useful tools and techniques from statistics, probability, and stochastic processes; 7. simulation as aid to: obtaining answers, intuitive understanding, illustration of theory; 8. providing as much opportunity as feasible for participants to work on/with problems from their own field. Tentative topics/contents list: Review basic probability: most to be done as self-study, some in class, plus a mandatory test (see Assessment). Models, modelling. Simulation: Convert a model to a program and run it. How to expose and locate errors. Data, empirical distribution; estimation. Accuracy, efficiency. Sensitivity. Rare event simulation. Experimental design. Conditional distributions and expectations; conditioning as a tool. Multivariate distributions: general (some examples); multivariate normal. Stochastic processes: Random walk. Poisson process; renewal processes. Markov chains: discrete time; continuous time; long-term behaviour; embedded Markov chain; birth and death process; Brownian motion. A snippet of diffusions, (solutions to) stochastic differential equations (SDEs).	
Study Goals	Learning objectives: after following the course, participants will be able to 1. in an iid setting, carry out Monte Carlo simulations; and 2. extract and collect, summarize, and interpret the data thus generated; and 3. assess accuracy/uncertainty in estimates, and solve questions like How many runs for 1% accuracy?; 4. the above (three) also in some non-iid settings, e.g., for a stationary process, using the batch-means method; 5. for small-sized practical problems: build a suitable model and simulate it; 6. extract and collect required model parameters from a problem description or background; 7. recognize where modelling choices are made; 8. make elementary modelling decisions themselves; 9. recognize and describe in mathematical terms (the properties of) a number of stochastic processes, among others: Markov chains, random walks, Brownian motion; 10. recognize/identify the covered stochastic processes in practical settings/problems; 11. use the tools and concepts from probability theory to describe, discuss, and manipulate (the components of) stochastic models; 12. where applicable, solve problems by means other than Monte Carlo, i.e., analytically or numerically, procuring exact answers, by identifying a stochastic process and using its known properties; for example, concerning long-term system-behaviour.	
Education Method	Weekly class meeting of two hours. About two thirds spent on lecture; about one third on questions and exercises. A number of so-called "Hands-on" assignments: a collection of exercises that combine the analytical content with computer simulation; to be done individually or with a partner. The participants are expected to perform a significant amount of simulation in Python, Matlab, or R (or other suitable language). Spread over the semester, five homework assignments. I strive to make a classroom available during the (probably) Friday lunch break, in which the participants can work on their own and where the TA (if there is one) is available for questions. In addition, I will hold a weekly drop-in office hour, most likely at lunch time (details on Brightspace).	
Literature and Study Materials	Unfortunately, for this course there is no really appropriate textbook, one that combines modelling, Monte Carlo simulation, and stochastic processes. For the review of basic probability and statistics, you are referred to Dekking et al, "A modern introduction to probability and statistics," Springer, 2005; the text is very well suited for individual review and practice, as it contains many worked examples, (quick) exercises, answers and full solutions. To be successful in the course, participants must have developed superior skill wrt the tools and techniques in the first half of the book (see "Expected prior knowledge"). Most of the material on stochastic processes can be found in "Probability Models" by Sheldon M. Ross; the book contains many worked examples and exercises; its heuristic and intuitive approach fits that of the course; references will be to the 10th Edition. A slightly more mathematical, but still very accessible reference covering the same area is "An introduction to stochastic modeling" by Howard M. Taylor and Samuel Karlin. Lecture slides, notes, and additional references will be posted on Brightspace.	

	A laptop is required and it is recommended that you bring it to class. Many languages and/or software packages are suitable: I prefer you use the one you are most familiar with. I am migrating to Python, but in the material presented in class and distributed via Brightspace, there still is some Matlab code. You may use a language/tool other than those three, but talk to me first (because you should be able to use your software on the exam and it should be possible to arrange this). In all cases: you should have the software installed and be familiar with its use.
Assessment	1. Probability test on knowledge and skills needed in/for the course (15%); passing score must be obtained to take the course. 2. Assignments (five; each counts for 4%), which may be done with a (=one) partner. 3. Final exam (on-campus): closed-book computer exam (with cheatsheet allowed, specs on Brightspace); counts for 65%. To pass the course, the probability test must be passed and the final exam grade must be 6 or higher; course grade is weighted average.
Permitted Materials during Tests	To the final exam you may bring a self-compiled formula sheet: one A4 page with normal-sized text on about 25 lines (or any reasonable equivalent of this), made by you, of things you would find helpful.
Elective	Yes
Tags	Adventurous Diverse Mathematics Modelling Practicals Stochastics
Co-Instructor	Dr.ir. G.N.J.C. Bierkens

WI4630	Statistical Learning	6
Responsible Instructor	Dr.ir. G.N.J.C. Bierkens	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: * Bachelor level knowledge and ability in calculus, probability, statistics, optimization, linear algebra. * Programming skills in Python (AM1090). If you do not have this background, consult the book Think Python, https://greenteapress.com/wp/think-python/, and familiarize yourself with the Numpy and Matplotlib packages. Recommended: * Linear Algebra and Optimization for Machine Learning (WI4635) * Statistical Inference (WI4455) * Knowledge of measure theory (as, for example, in AM2090)	
Course Contents	Statistical learning provides a probabilistic and statistical understanding of topics in machine learning. Note: this year the course will be part of the MasterMath programme. This may lead to some last-minute deviations from the information in the study guide. The overarching goal of the course is to develop methods for estimating (or 'learning') an unknown function from data or making predictions for unseen function outputs. The course aims to empower the student to make a justified decision in adopting machine learning approaches for practical problems and even design their own machine learning methodology using sound mathematical principles. This is achieved by studying key ideas, models, algorithms and theories related to the subject of statistical learning. In the first half of the course a collection of essential models, algorithms and techniques is introduced. In the second half of the course we adopt a Bayesian perspective on machine learning and study associated methods for analysing models and developing computational methodology.	
Study Goals	By the end of the course, you are able to: 1. Explain the challenges of statistical learning (e.g., curse of dimensionality, overfitting, bias-variance trade-off). 2. Implement machine learning algorithms and computational methods on artificial and real-world data sets (e.g., linear regression, logistic regression, sparse regression, neural networks, decision trees). 3. Compare the theoretical properties and practical aspects of machine learning approaches. 4. Validate machine learning approaches on arbitrary data sets. 5. Design tailor-made computational approaches for new learning problems by adapting, combining and/or extending methods and/or by employing a Bayesian approach.	
Education Method	Lectures, exercises, practical assignments, practice exam	
Literature and Study Materials	Lecture notes and slides will be made available. You may wish to consult additional references for background reading. Some suggestions of publicly available textbooks are (listed in order of suitability for this course): * Bishop, C. M. (2006). Pattern recognition and machine learning. Springer, New York. https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/ (very explicit in terms of derivations; focusses on a Bayesian approach) * Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning (Second). Springer, New York. https://web.stanford.edu/~hastie/ElemStatLearn/ (more advanced reference; requires some mathematical and statistical maturity since it shows relatively few derivations; focusses on a non-Bayesian approach) * Murphy, Kevin P (2022). Probabilistic machine learning: an introduction. MIT press, https://probml.github.io/pml-book/book1.html (very recent textbook on machine learning which discusses most topics; however not a very well thought out as textbook)	
Assessment	The final grade of the course consists of the following components: * Practical assignments during the course (30%) * Final written exam (70%) Final grade calculation: $(0.3 * \text{practical assignments} + 0.7 * \text{final written exam})$ Note: For both components, a sufficient grade (5.8) is required. Repair possibilities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5. * Assignments: resubmit deliverable * Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4635	Linear Algebra and Optimization for Machine Learning	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course material and literature are mostly self-contained, but prior knowledge equivalent to contents of the courses AM2010 Linear Algebra 2 and AM2020 Optimization from the Bachelor of Applied Mathematics programme is strongly recommended. Programming experience in Python or Matlab is expected.	
Course Contents	State-of-the-art machine learning algorithms combine techniques from many different areas of mathematics. In this course, we will discuss algorithmic foundations from the areas of linear algebra and optimization. This also includes aspects of their theory and efficient implementation. Among others, we will study + matrix decomposition and factorization techniques, + regression and classification problems and regularization, + iterative and continuous optimization methods, + hyper parameter optimization, + graph-based algorithms and clustering, + basic introduction to neural networks and algorithmic aspects neural networks (e.g. computational graphs and back-propagation)	
Study Goals	Upon completion of the course, you should be able to: + Categorize typical data sets based on their structure + Derive suitable mathematical formulations for common supervised and unsupervised machine learning tasks + Choose suitable algorithmic building blocks for solving mathematical formulations of common machine learning tasks + Analyze machine learning algorithms with respect to their properties, such as efficiency and robustness + Implement machine learning algorithms in a computationally efficient way	
Education Method	The course comprises 14 lectures and 14 exercise classes.	
Literature and Study Materials	Lecture notes, slides, and the main literature: + Charu Aggarwal. Linear algebra and optimization for machine learning. Springer, 2020. + Gilbert Strang. Linear algebra and learning from data. Wellesley-Cambridge Press Cambridge, 2019. + Jeremy Watt, Reza Borhani, and Aggelos K Katsaggelos. Machine learning refined: Foundations, algorithms, and applications. Cambridge University Press, 2020.	
Assessment	The final grade of the course consists of the following components: - Two group projects (0.4) - Written exam (0.6) Final grade calculation: $(0.4 * \text{group projects} + 0.6 * \text{written exam})$ Note: For the written exam, a score of at least 5.0 must be obtained. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group projects: resubmit deliverable - Written exam: There will be a resit for the written exam at a later point in the same academic year Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. D. de Laat	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	The final grade of the course consists of the following components: - Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). - Group programming project (group work) starting mid Q2 with submission deadline end Q2 (2/3 of final grade). Final grade calculation: $(1/3 * \text{homework assignments}) + (2/3 * \text{group programming project})$ but it is only valid when all homework programming assignments have been completed with sufficient results. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework programming assignments: submit new set of homework - Group programming project: submit improved version Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. J. Thies	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Track 4: Aerospace Structures and Materials

MSc Coordinator C.D. Rans

Contact for students Calvin Rans (ASM MSc Track Coordinator)

Please use the following email address: MScCoordinator-ASM@tudelft.nl

Introduction 1

Official announcements are posted on the Aerospace Structures & Materials Msc Track Brightspace page

No structure is indestructible. Time, environment, repeated use, and misuse all take their toll on everyday structures, taking advantage of their inherent weaknesses and bringing them closer to failure. This is not the most comforting thought when walking onto an airplane, or driving over a bridge, but it is a reality that structural designers must face; how do you design something that you don't want to fail while accepting that it eventually will? The answer is: very carefully!

The challenge structural engineers are faced with is as follows. First, they must understand the raw materials and the level of engineering already applied in their creation. Second, they need to synthesize material behavior and required structural function into a working design. Third, compromises in the working design need to be made to address manufacturability. Finally, all of this must be completed while continually assessing the impact on the durability and longevity of the final structure.

The Aerospace Structures and Materials (ASM) MSc track aims to equip students with the necessary knowledge and practical skills necessary to tackle this challenge in an industrial or research environment. From an educational standpoint, students will be exposed to a broad range of courses examining this entire process in the context of the design, manufacturing, and analysis of composite aerospace structures. This will provide the foundation for subsequent specialization in one of the thematic profiles mirroring the steps in the design challenge described above: Material, Manufacturing, Structural Analysis and Design. From a research standpoint, students will have the opportunity to carry out high-level research and development in fields directly relevant to aerospace industry and that improve our capability to design the next generation of aerospace structures and materials. Thesis projects can be carried out with any of the research groups within the ASM department, regardless of the thematic profile selected by the student. Additionally, numerous opportunities exist to carry out research within industry under the close supervision of an ASM staff member.

The ASM Track is committed to making your MSc experience a memorable one. Next to offering a world-class education we also offer many opportunities to expand your horizon through industry involvement, visits and guest lectures. Our students have organized themselves in their own ASM student society, Enlighthness which organizes lunch lectures, company visits, drinks and the annual ASM career event.

The track program is built up of a common core of 15 EC followed by a selection from one of our three thematic profiles, after which you can further specialize or widen your horizon by choosing from a large selection of electives. The first year is dedicated mostly to courses in preparation of your second year, in which you first go on your internship and then work on your thesis project.

Our three thematic profiles are:

Profile 1: Materials and manufacturing
Profile 2: Design and safety of structures
Profile 3: Structures for space

You can choose the materials and manufacturing profile if your intention is to develop new aerospace materials and/or work in a production surroundings and translate the needs of the designer to production and vice versa. The design and safety of structures is for you if your interest is on structural design & stress engineering. The structures for space profile is when you want a more in-depth knowledge of structures specifically for space applications, such as space vehicles and launcher structures.

Program Goals

After completing the MSc Track Aerospace Structures & Materials the student will be able to:

- Develop design requirements for materials, manufacturing and structures
- Design a lightweight structure and explain the reasoning and the physics behind the design
- Design a material suitable for aerospace application and explain the reasoning and the physics behind the design
- Carry out a structural analysis using Finite Element Methods and/or other analytical and numerical techniques
- Explain the manufacturing processes and their applications
- Select suitable manufacturing processes
- Manufacture a prototype
- Explain and predict how a design will perform over its life-time and explain how the performance can be monitored

This is executed in a hands-on way with both practical and theoretical learning activities and exciting multi-disciplinary projects within the research framework of the department but also allowing for students' own input and ideas.

Program Structure 1

The 1st period (Q1), core module consists of five themed courses of 3 EC focused on aerospace structures, such as an aircraft wing

AE4ASM001 - Design of lightweight structures I: Composites & Metals
AE4ASM002 - Designing Materials with Aerospace Specific Properties
AE4ASM003 - Linear Modeling (incl. F.E.M)
AE4ASM004 - Manufacturing of Aerospace Structures & Materials
AE4ASM005 - Fatigue of Structures & Materials

Towards the end of middle of Q1, students select their thematic profile and their initial study plan including electives.

Program Structure 2

Practical issues: selection and updates in study program and administrative systems

During the MSc the preferred elective courses and thesis direction may change as the interest of the student becomes more clear. This change is facilitated by altering your course selection on the course selection system:

My Study Planning, <https://mystudyplanning.tudelft.nl>

Please make sure to submit your change request for approval by the MSc Track Coordinator who will also consult the thesis supervisor involved. The submit button can be found all the way at the bottom of the page. You will receive an e-mail notification when your request has been processed by the master Track Coordinator. This information is also forwarded by the Track Coordinator to the system managing the administrative information related to your studies (OSIRIS). It is advised to check in OSIRIS whether the change request has been processed by the central administrative office of TUD according to the data from My Study Planning. Please note that this may take several working days since the course selection system (My Study Planning) is not meant to be an automated process for OSIRIS. The information in the OSIRIS system is always leading.

Keeping your list of electives up-to-date is imperative as it may delay your thesis defense date.

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile I: Materials and Manufacturing

Program Coordinator	Dr. K. Masania
Contact for students	To contact the ASM MSc track coordinator send an email to MScCoordinator-ASM@tudelft.nl. Please indicate the profile (Materials and Manufacturing) for specific questions about the profile itself.
Introduction 1	Profile I Materials and Manufacturing Why? New materials and the way we manufacture them has been linked with human development since pre-historical times (e.g. Stone age, Bronze age). Fast forward 5000 years: Today Climate change is threatening Human Civilization. Critical sectors of our society as mobility, energy and infrastructures must change. New material and manufacturing are a key part of the solution for a new sustainable era. What? In the profile Materials and Manufacturing you will be equipped with the knowledge and tools to create innovative and creative solutions to kick off a new era of sustainable materials. You will learn about polymer science, polymer composite manufacturing, additive manufacturing and characterization techniques of materials and components. Electives with a strong Materials and Manufacturing flavor within the ASM Department are: - Aircraft Manufacturing Laboratory - Trinity exercise - Materials for Space - Functional Coatings - Experimental Techniques & NDT - Design of Self-healing materials - Industrial Composite Manufacturing - Sheet Metal Forming
Program Structure 1	In the ASM Track you do not have to select a profile until October. Use the core module in period 1 to select your final preference. Electives are decided upon in the second period whilst you are taking the profile courses to allow you to orientate yourself on your thesis topic and pick the relevant deepening electives upon the advice of your preferred thesis supervisor. You then register your choice of electives with the MSc track coordinator in December. TU Delft students who intend to start in February are requested to see the Track Coordinator by the end of November or early December to discuss their planning. Be mindful that a February start assumes you will take your internship from April of that year, so plan accordingly and start preparing for your internship in time.

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
Responsible Instructor	Dr.ir. O.K. Bergsma	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The global course setup is such that there lectures contain the following topics: - Thoughts behind lightness - Design allowables - Interaction between materials processes - Materials and their properties - Essentials of manufacturing - Processing to final products and their applications. - The principles of stresses in laminated composites - Recycling and other environmental aspects. - The principles of repair of structures. Guest lectures will be illustrating the course content.	
Study Goals	After succeeding this course the student should be able to: - explain parameters and their relationships, which play a role in the development of lightweight structures and parts. Examples: - be able to judge a structural design on conditions required to call a design a lightweight design. - be able to identify carbon glass aramid and dyneema fibres - be able to identify metals from composites on micro and macro scale. - be able to relate lightweight materials to typical strong and weak points in their performance. - be able to give examples of fibre morphologies - be able to argue the correlation between fibre content, orientation control, fibre length, manufacturing process and application. - be able to recall & argue unwanted stress distributions in composite materials	
Education Method	Lectures. The lectures will most likely be given in complete day sections (2 times 2 hours). Old collegeera recordings of lectures are available. Online lecture videos are available and explain the content in an other way. Guest lectures are two hour lectures.	
Assessment	Written exam, with grades converted to whole or half numbers. In case of 100 points the following distribution is most often used <points>[grade]: <0-14>[1]; <15>[1.5]; <16-24>[2]; <25>[2.5]; <26-34>[3]; <35>[3.5]; <36-49>[4]; <50>[4.5]; <51-54>[5]; <55>[5.5]; <56-64>[6]; <65>[6.5]; <66-74>[7]; <75>[7.5]; <76-84>[8]; <85>[8.5]; <86-94>[9]; <95>[9.5]; <96-100>[10];	

AE4ASM002	Designing Materials with Aerospace Specific Properties	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Prof.dr.ir. R. Benedictus	
Instructor	B. Atli-Veltin	
Instructor	Dr. Y. Tang	
Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Several courses follow on the general material concepts introduced in this 002 course, amongst others, within the ASM MSC track: -AE4ASM101 Polymer Science; (Q2) -AE4ASM103 Functional Coatings. (Q3)	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Ashby, Materials". The students must have completed the materials science courses in the first 3 years of the Aerospace Engineering Curriculum, or any other Engineering curriculum	
Course Contents	In this course an unorthodox approach to materials will be presented. Rather than memorising known routes to reach certain materials properties, the students will be trained to translate these desired properties into material structures and microstructures and to design suitable material production processes to realise these properties. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The course is divided in 7 lectures. Three lectures cover basic design rules of different material classes and their behaviour: metals, polymers, ceramics. Three lectures cover material behavior in aerospace relevant environments: exposed to high temperatures, as building blocks for H2 storage, as structural materials sensitive to environmental degradation. A last lecture deals with strategies to extend material life through the use of self-healing material concepts. -polymers -metals -ceramics and 'smart' materials -materials and H2 (energy transition) -materials at high temperatures -materials and environmental damage -materials and lifetime The students will need to deliver a report (assignment). For this they will get a set of questions they need to solve by reading the existing literature.	
Study Goals	The objective of the course is to train the student in reverse material engineering. This skill enables students to initiate and guide new material developments to meet future targets in the industry. By the end of the course, you should be able to: - explain structure and property (inter)relationships of metals, polymers and ceramics - explain the functionalities of aerospace relevant material properties - breakdown and translate these functionalities into material structures and microstructures using reverse engineering methodologies - explain material behaviour and function loss and strategies to decrease the impact of damage in the material function - give examples of new material concepts and explain the underlying concepts.	
Education Method	Lectures	
Literature and Study Materials	A compilation of lecture notes, review papers and assignment.	
Assessment	Assignment report (40%) + exam (60%) (a minimum of a 5 over 10 points in the exam is needed to pass)	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods. Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments (only assignment 3 and 4 since they have the highest weight; assignment 1 and 2 cannot be redone). Results for individual assignments are only valid for one academic year: when re-taking the course all assignments have to be done in that academic year.	

AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
Responsible Instructor	Dr. K. Masania	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The red line of the course is aiming at knowledge and understanding of manufacturing processes in relation to material properties and feasible product designs. The course contents include manufacturing processes for metallic and composite parts, the assembly of parts into large (sub)structures, and related topics like Quality control, organisation principles, finances, etc.	
Study Goals	The student should have a good knowledge and understanding of the mainstream manufacturing processes of structural materials (lightweight alloys, composites, hybrids). The student should be able to describe and motivate the processing procedures like the processing steps, required tooling and equipment, of manufacturing processes. The student should be able to select adequate manufacturing processes for designs of lightweight structures and components, and should be able to motivate his/her choices. The student should be able to analyse and synthesize interactions between materials, design and manufacturing processes. The student should be able to identify and explain manufacturing related flaws and inaccuracies and advice on how to prevent/limit those.	
Education Method	- Interactive lecture where contents are presented and where the flipped classroom is often used. - Exercises in class with some additional working material	
Assessment	written exam. Closed book.	

AE4ASM005	Fatigue of Structures & Materials	3
Responsible Instructor	Prof.dr.ing. R.C. Alderliesten	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	- Introduction to Fatigue (fatigue as a phenomenon; stress concentrations; residual stresses; fatigue properties of metallic and composite materials; fatigue strength of notched specimens, residual strength). - Fatigue damage mechanisms (initiation, crack growth, delamination growth, transverse matrix cracking, fibre failure). - Analysis methods (stress concentration factors, stress intensity factors; energy balance approaches, strain energy release rates). - Fatigue loading (Load Spectra, Fatigue under Constant- & Variable-Amplitude Loading). - Special Fatigue Conditions (surface treatments; fretting corrosion; corrosion fatigue; high-temperature and low-temperature fatigue, moisture ingress). - Fatigue and Damage Tolerance of Aircraft Structures: Regulations, tests, scatter, application of fatigue and damage tolerance methods.	
Study Goals	This course provides the students with engineering knowledge and skills to recognize and to analyse fatigue and damage tolerance problems in aircraft structures and materials. After the course the student must be able 1. Interpret and discuss the fatigue fracture features with respect to the characteristics of each phase in fatigue life 2. Define and determine stress concentration factors for notched structures with or without residual stresses 3. Explain and discuss S-N curves with respect to mean stress, material surface effects, and scatter, and perform fatigue life analyses considering mean stress and notch root plasticity 4. Assess the fatigue life of tension and shear joints, and explain limitations to the similarity principles (K,I,T) 5. Explain Linear Elastic Fracture Mechanics concepts for damage growth, and perform crack growth analyses with these concepts 6. Explain the consequences of variable- and constant amplitude loading on fatigue life and damage growth, and perform fatigue life analyses for arbitrary load spectra 7. Explain the effect of environment on fatigue life and fatigue phenomena 8. Perform residual strength analyses.	
Education Method	Selfstudy using the Online Course Material on Brightspace together with weekly plenary instruction sessions and discussions with fellow (online and campus) students or e-moderator on a forum.	
Literature and Study Materials	J. Schijve, Fatigue of Structures and Materials, 2nd edition, 2009, Springer Science+Business Media, B.V., ISBN-13: 978-1-4020-6807-2 e-ISBN-13: 978-1-4020-6808-9.	
Assessment	Readers, formative tests, documentation, and other relevant material provided via Brightspace.	
	Written exam & assignments Written exam at the end of the lecture period, 80% of total grade. Assignments during the lecture period, altogether 20% of the total grade. Deadlines of individual assignments are communicated via Brightspace.	

AE4ASM101TU-22	Polymer Science	4
Responsible Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of a number of lectures covering the most relevant aspects of polymer science and technology distributed in 6 blocks, each covered in one up to three lectures. After this a session on 3 minutes pitches by the students will be organized: -Introduction to polymer science: daily life examples, nomenclature, concept of polymers, initiators, polymer architectures and polymerization media. -Polymerization: condensation and addition polymerization under step-growth; radical, ionic and ring opening polymerization under chain-growth mechanisms, concept of copolymerisation and crosslinking, basic characterization techniques (such as GPC). -Polymers in Aerospace: brief outlook on composites, chemistry of thermosets, high-performance thermosets, high performance thermoplastics, mechanical properties. -Processing: basics of polymer processing and polymer-based structures manufacturing, fiber processing and properties, living materials and their potential. -Sustainable Polymers: a glance on green chemistry, introduction to lignocellulosic biomass, how to develop biobased monomers, recyclability and dynamic materials, examples on sustainable polymers, impacts on sustainability, a perspective on how can you design sustainable polymers? -Polymers for Energy and Environment- demonstrating a concept of photovoltaic devices and batteries, polymers used in these applications, polymer membranes and polymer coagulants for water remediation, polymer ionotronics. -Pitch talks: students to give 3 minutes presentations about developments/applications in the polymer field, convince audience as in business structures.	
Study Goals	By the end of this course, you should be able to: LO1: Identify polymer basics and analysis LO2: Assess polymer structure-property relationship for thermomechanical performance estimation LO3: Judge parameters and requirements for polymers to be qualified for aerospace materials design LO4: Define sustainability concepts in polymer science LO5: Present about recent developments in polymers field	
Education Method	Lectures and lab activity	
Literature and Study Materials	Seymour/Carraher's Polymer Chemistry: 10th Edition. Contains comprehensive information on polymer materials from chemistry perspective Lecture slides will be made available to students as study material	
Assessment	Written final exam and pitch talk (3 minutes) presentation Lab activity attendance is mandatory Transitions AE4ASM101TU -> AE4ASM101TU-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM101TU, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM101TU. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM101TU is not completed in 2022 2023, students will need to complete AE4ASM101TU-22 instead.	

AE4ASM110-22	Polymer Composite Manufacturing	4
Responsible Instructor	Dr. J.J.E. Teuwen	
Instructor	Dr.ir. D.M.J. Peeters	
Instructor	Dr.ir. B.C.P. Jongbloed	
Instructor	Dr. B. Caglar	
Instructor	R. Tao	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course explores in depth the manufacturing of polymer composite structures and its underlying physics. By understanding the relationships among physics, part/material quality and the design of the manufacturing process, you will be able to critically assess any given manufacturing process. The course is built around a number of polymer composite manufacturing processes highly relevant to the current and future aerospace industry for the manufacturing of high-performance individual parts (autoclave processing of prepreg, liquid composite moulding, thermoplastic composite processing), for the assembling of complex structures (manufacturing of integrated structures, thermoplastic composite welding) and for end of life and recycling. The main basic physical phenomena governing polymer composites manufacturing, e.g. flow (of polymer and of fibres), void formation, curing, shrinkage and crystallization, are interwoven within the course.	
Study Goals	At the end of this course, the student should be able to: - Describe the common manufacturing processes used in aerospace industry in terms of processing steps, tooling and equipment as well as advantages, limitations and applicability. - Correlate and analyse the processes and their principles/underlying physics with manufacturability, part quality and manufacturing design. - Compare and evaluate the different processes for their suitability of manufacturing common components (aircraft wing, fuselage etc). - Recognise manufacturing defects, identify their potential sources and recommend strategies to improve the quality of common manufacturing processes for simple component geometries to reduce scrap rate or to increase yield.	
Education Method	Interactive lectures with numerous in-class exercises. Assignments are used throughout the course to allow the students apply, deepen and test their knowledge on the manufacturing processes discussed during the course.	
Assessment	Oral exam (50% of the final grade) - exam format subject to change depending on amount of students - and mandatory assignments (50% of the final grade). A minimum grade of 5.0 out of 10 on both the assignment and the oral exam need be obtained to be able to pass the course.	

AE4ASM515-22	Characterization of materials and components	4
Responsible Instructor	Dr. B. Kumru	
Co-responsible for assignments	Dr.ir. C.M. de Zeeuw	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	General knowledge of materials engineering and science (in particular polymers and metals) at the level expected for a BSc graduate in a (mechanical) engineering discipline.	
Course Contents	This course is an introduction to the most common characterization tools to gather crucial structural and property information of materials and components. The focus is on knowing about several possible characterisation methods and their underlying principles of measurement as well as the strategies to extract the most relevant information from the tests for the purpose in mind. The characterisation techniques will be taught by various colleagues in each field. The methods to covered in this course are classified as follows: 1. Imaging of material structures and surfaces (microscopy, confocal, SEM, EDS) 2. Chemical analysis of polymers (FTIR, Raman spectroscopy, contact angle) 3. Structural characterisation 4. Thermomechanical analysis of polymers (DSC, TGA, DMA/Rheology) 5. Mechanical characterization	
Study Goals	Aim of the course is to provide (aerospace, mechanical, maritime) engineering students with adequate skills and knowledge such that later in their technical careers they can handle materials performance issue in which it is important to clarify the material characteristics such as to link actual material performance to its microstructure. Such knowledge and skills are important for determination of material in certain applications and while choosing engineering materials with desired properties. To this aim: - Students will get familiar with the underlying principles of measurement of principal materials characterisation techniques. - Students will be able to analyse and interpret data from each characterisation technique. - Students will be able to select the right characterisation techniques to correlate structure and property relationships over a wide range of engineering materials.	
Education Method	Lectures and hands on practices	
Assessment	Oral Exam and assignment	
	Transition AE4ASM515 -> AE4ASM515-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM515, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM515. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM515 is not completed in 2022 2023, students will need to complete AE4ASM515-22 instead.	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr.ir. J.A. Pascoe	
Instructor	Prof. C.A. Dransfeld	
Instructor	Dr. J. Wu	
Instructor	V. Yaghoubi Nasrabadi	
Instructor	P.G. Marchese	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project.

The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5722	Thesis Aerospace Structures & Materials	45
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal). If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go'	

(fail). Please refer to Brightspace for more information.

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Materials and Manufacturing

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4ASM103		Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	3 4		
Course Language	English		
Expected prior knowledge	The students must have general knowledge about materials science.		
Course Contents	In this course we introduce the design fundamentals that control specific properties in coatings, surface treatments and functional coatings and the techniques used to apply them and test them. The goal is therefore to provide the fundamentals of chemistry and physics that explain structure-property relationships relevant to understand and design surface treatments/coatings with: (i) all-round relevant properties such as adhesion, mechanical response, active (inhibitors) and passive (barrier) corrosion protection; and (ii) increasingly relevant functional properties such as antiicing, self-cleaning, self-healing, drag, structural colours, storage/release of active chemicals with nano and microparticles, and antifouling. Since material design rules are directly related to application methods, different technologies for coatings application will be covered such as electrophoretic deposition, powder coatings, anodizing, sol-gel and plasma electrolytic oxidation. Characterization strategies to obtain information about key properties will also be discussed. The fundamentals and characterization strategies covered in the course are applicable to coatings, adhesives and composites of interest in aerospace applications and other engineering domains.		
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.		
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.		
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.		
Assessment	Written report (40% of final grade) and written exam (60% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.		

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr.ir. C.M. de Zeeuw	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Compared to previous years the course starts in Q3 (not Q2). There will be a kick-off lecture in week 3.1	
Course Contents	Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four to five students about 8 day sections with support of staff are needed during this course. These will be planned on fore hand, preferably on fixed time slots. Most likely either Friday morning or Friday afternoon. We will not use all of those slots, but it will provide the flexibility to schedule things in the lab and make it easier to block their schedules. Finishing period As the starting period changed, the finishing period changes according, at the end of the fourth period (Q4, June) it is. The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Estimation of the maximal bending-torsion coupling for a strip of UD composite 4. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 5. Performing mechanical tests 6. Analysing the test results of the mechanical test, including failure analyses 7. Adhesion of thermoset laminates 8. Possibly preparation of lap shear specimens 9. Performing lap shear tests 10. Estimation of the mechanical properties of a sandwich panel 11. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 12. Performing a bending test on a sandwich panel. 13. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	This course was renewed in 2023-2024 by fixing the time schedule for students on beforehand.	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on beforehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM106	Stability & Analysis of Structures I	3
Responsible Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	This course introduces analytical and numerical techniques for the analysis of representative aerospace structures under mechanical loading, with particular emphasis on determining their stability. The course covers the connection between the principle of minimum potential energy, the variational formulation and the classical (differential) formulation of quasi-static structural problems. Basic structural components are analyzed using beam and plate theory. The basic theory of structural stability is introduced through asymptotic analysis in order to derive the stability (buckling) equations. Buckling equations are formulated and solved for geometrically non-linear elastic beams and plates. The methodology is extended to formulate stability analysis of representative aerospace components. Emphasis is placed on the understanding of the models, their assumptions, and ranges of applicability. The topics covered include: - Introduction to structural analysis - Review of global and local balance principles - Constitutive relations, material symmetries and strain energy - Formulations of structural problems: differential, variational and minimum formulations - Introduction to calculus of variations and approximate solution methods (Ritz and Galerkin) - Applications to beams and plates: solution to basic structural problems - Introduction to buckling: Eigenvalue buckling problem (bifurcation buckling) - Imperfection analysis (snap buckling) and stability analysis - Buckling of non-linear beams/columns and plates - Design methodology for thin-walled aerospace structures	
Study Goals	1- Understand the various formulations of equilibrium and equations of motion 2- Model simple aerospace structural components using engineering theories 3- Apply energy and variational methods for the evaluation of structural responses 4- Understand and being able to obtain the buckling equations for beams and plates 5- Develop insights into buckling phenomena of aerospace thin-walled structures	
Education Method	Lectures.	
Literature and Study Materials	Primary source: course slides provided in Brightspace Recommended references: T.H.G. Megson, Aircraft Structures for Engineering Students, Elsevier, Any edition. C.L. Dym and I.H. Shames, Solid Mechanics: A Variational Approach, Augmented Edition, Springer, available online through the TU Delft Library.	
Assessment	Written (take-home) assignments and oral exam	

AE4ASM108	Experimental Techniques & NDT	3
Responsible Instructor	Dr. A. Anisimov	
Instructor	Dr. R.M. Groves	
Responsible for assignments	K.P.W. Dissanayake	
Responsible for assignments	Dr. N. Yue	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of electronics, elements of signal processing (analogue, digital), basics of measurements and metrology	
Course Contents	This is a lecture course on developing the critical thinking needed to perform measurements in the Aerospace Structures and Materials Laboratory. This includes an introduction to the most common sensors for experimental mechanics, non-destructive testing and structural health monitoring, evaluation of the performance of these sensors, selection of sensors for different applications and signal processing and control algorithms.	
Study Goals	By the end of the course, you should be able to: 1. Analyse a measurement system for the aerospace sector and to determine if it is fit for a specific application. This learning objective acts as an umbrella objective, covering the whole course. Lower level objectives include that by the end of the course, you should be able to: 2. Explain the principle of operation of the most common sensors and instrumentation in the aerospace structures and materials laboratory. 3. Justify the selection of a sensor for a specific measurement application, based on sensor specifications and measurement principle. 4. Analyse a measurement chain and determine signal accuracy and noise levels. 5. Assess the performance of signal processing and feedback algorithms for their influence on measurement noise and accuracy.	
Education Method	Lectures, obligatory laboratory sessions, reader with self-evaluation questions.	
Literature and Study Materials	Course reader, pre-recorded videos with slides available on Brightspace. Additional material as datasheets, book chapters and papers are uploaded to Brightspace during the course.	
Reader	Course reader available on Brightspace	
Assessment	Open book written exam (30% of grade). Oral exam (40% of grade). Laboratory group assignments (30% of grade).	
Set-up	12 active lectures with problem-solving and discussion (weeks 2.1-2.6). Students have to prepare for these workshops using the material available on Brightspace, including pre-recorded videos. The material becomes available as the course progresses. 2 exam review workshops (week 2.7). It is particularly important to attend these sessions as the course lecturers will explain the structure and what is required in the open-book exam 5 obligatory laboratory assignments done in groups (weeks 2.1-2.5). Each practical is graded with a group grade for the written report (maximum of 4% towards the final course grade) and personal grade for the oral defence of the report (maximum of 2% towards the final course grade).	

AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
Responsible Instructor Responsible Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language	C. Kassapoglou Dr. D. Zarouchas 0/0/4/0 3 3 none English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis 3. Reliability analysis & Health monitoring a. Probability of Failure & Uncertainty quantification b. Structural Health Monitoring 4. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Understand the philosophy of reliability analysis and calculate the probability of failure of composite structures by combining numerical methods and the mechanics of composite structures - Understand the importance of structural health monitoring (SHM) and design a SHM system for a given composite structure - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Classical Laminate Theory: 3 2. Progressive damage analysis: 3 3. Reliability analysis & Health Monitoring: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory, progressive damage analysis, reliability analysis and health monitoring. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 10 points A2. Progressive damage and damage tolerance analysis 20 points A3. Reliability analysis & Health Monitoring. 30 points	

Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method).

B1. Design problem 40 points

The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2,A3 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM503	Sheet Metal Forming	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The lecture consists of two main parts: a theoretical part and an application part The theoretical part includes: - Introduction, formability, deformations, formability testing. - Stress- and strain states, forming limit curves, yield criteria, anisotropy, heat treatments. - Theory of plasticity The application part includes a number of forming processes, like - Simple forming processes like bending and stretching. - Several forming processes for more complex parts like deepdrawing, rubber forming, hydroforming and matched die forming. - Non-conventional processes like explosive forming and superplastic forming - The use of numerical tools for simulation of forming processes, and some other related topics.	
Course Contents	The theoretical part of the lecture will focus on the descriptions of deformations, strains, stresses, etc. In addition, the interaction between stresses and strains and their limits are expressed in a.o. yield criteria and forming limit diagrams. Such concepts are also used in the second, application part to describe and explain the do's and don'ts of the forming processes.	
Study Goals	After attending this course the student should be able to: - describe and explain the basic phenomena in plastic deformation of sheet metal - understand the interactions between multiple phenomena and how to exploit these - be able to relate the formability of materials to forming processes - have understanding and ability to combine the features of material behaviour, the forming process and the shape of the component.	
Education Method	Lecture, including discussions about real parts, pictures and videos.	
Literature and Study Materials	Hand-outs are the core material A list of recommended literature will be given during the first lecture (background information unless specified otherwise)	
Assessment	Written exam. Closed book.	

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	N. Eleftheroglou	
Responsible Instructor	Dr.ir. J.A. Pascoe	
Instructor	Prof.dr.ing. R.C. Alderliesten	
Instructor	Ir. J. Sinke	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). Regulations applicable to maintenance. - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals	At the end of this course, you should be able to: - Describe different maintenance strategies and select the appropriate one for a given problem, taking into account different aspects (e.g. technical, regulations, planning); - Describe the working principles of different non-destructive inspection techniques, their pros and cons and their applicability to different materials; - Explain the relationship between certification and maintenance requirements and (documentation of) procedures; - Develop a maintenance programme for a simple case; - Select an appropriate design philosophy to ensure structural integrity of a given part throughout the aircraft lifetime.	
Education Method	Lectures.	
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam, closed book (60% of final grade) and group assignment (40% of final grade).	

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	lectures and practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Adisson-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of aerospace structures and structural components. The equation of motion for the dynamic response is formulated in matrix form using finite element discretization or direct modelling discretization (rigid bodies and flexible connectors). Various solution techniques are used for solving transient dynamic response and vibration problems, including modal analysis. The course also covers experimental techniques to obtain model parameters and test the response of a physical system. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE4ASM514TU	Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.	
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.	
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.	
Education Method	Lectures.	
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.	
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.	

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	The grade for this course will be a Pass/Fail grade. This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Alternatively, having earned a Bachelor degree in aerospace engineering will give access to the course as well. Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December - Week 4.8: 15 May - Week 1.6: 30 September	

Set-up	The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and 3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3
	There is an intentional overlap so that continuity and handover is guaranteed

AE4ASM520	Industrial Composite Manufacturing	3
Responsible Instructor	Prof. C.A. Dransfeld	
Responsible Instructor	Dr. D. Stefaniak	
Contact Hours / Week x/x/x/x	0/0/4/0 COURSE NOT OFFERED IN 2024-2025	
	Students that have participated in this course and have not yet passed the course, please contact C.A.Dransfeld@tudelft.nl	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Expected BSc diploma	
Course Contents	The course deepens the understanding on different composite manufacturing technologies for aerostructures used at the industrial scale and addresses on specific challenges evoked especially by automation and large part size. Special pitfalls with respect material selection, mould design, process induced part distortion and vacuum setup are discussed to apply the basic understanding of the manufacturing processes to several specific industrial challenges. Finally, these manufacturing processes are also evaluated with respect their sustainability and are investigated in depth towards their energy, materials and auxiliaries consumption in context of the product life cycle.	
Study Goals	At the end of this course, the student should be able to: - Transfer the basic principles of composite manufacturing processes to real and more complex industrial examples - To gain insight in material selection and its consequences in an industrial context - To recognize manufacturing processes and the linked pitfalls based on the resulting part or the utilized mold - Understand the challenges in sizing composite manufacturing systems for large size components - Understand the importance and dependencies of performance assessment indicators in composite manufacturing like robustness, energy consumption, occupation time, non-operating costs and flexibility.	
Education Method	Interactive lectures based on industrial case studies. Some exercises are accompanied by Python Scripts used in Jupyter Notebooks to deepen the understanding and facilitate interactivity. The required software is provided as a browser-based version and previous knowledge in programming is not compulsory. It is intended to visit the Center for Lightweight Production Technology at the German Aerospace Center (DLR) and Airbus site in Stade, Germany.	
Computer Use	yes	
Course Relations	The course is given by a guest lecturer from DLR (German Aerospace Center)	
Literature and Study Materials	will be defined at the beginning of the course	
Prerequisites	AE4ASM001, AE4ASM004, AE4ASM110	
Assessment	Oral exam (50% of the final grade) and mandatory assignment (50% of the final grade).	
Enrolment / Application		
Remarks	Students that need to retake AE4ASM520 can contact the responsible lecturer for a resit.	
Set-up	2 weeks with 2day-blocks and two weeks with one 1day-blocks (e.g. 3.1, 3.3, 3.6, 3.8); each day will consist of four lecture hours	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr.ir. J.A. Pascoe	
Instructor	Prof. C.A. Dransfeld	
Instructor	Dr. J. Wu	
Instructor	V. Yaghoubi Nasrabadi	
Instructor	P.G. Marchese	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE4ASM522	Applied Aircraft Aeroelasticity	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	X. Wang	
Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE4ASM506 (mandatory)	
Course Contents	This course is a follow-up course of Aeroelasticity AE4ASM506. Taking and passing the Aeroelasticity course AE4ASM506 is mandatory to pass this course. This course builds on the basics and regular aeroelastic phenomena that were introduced in AE4ASM506. The extension of the knowledge is twofold: 1. Introduction of nonlinear aeroelasticity 2. Introduction to control of flexible aircraft. 3. Introduction to potential flow aerodynamics and its coupling to structural FE models to form closely coupled aeroelastic models The course aims at placing the aeroelastic knowledge into a flexible aircraft context, and it will provide the student with new insights into the aircraft design process and aircraft design choices. Finally, understanding the contents of this course will prepare the student for the design of future aircraft configurations beyond the currently established aircraft concepts.	
Study Goals	At the end of the course, the student should be able to: 1. Know advanced nonlinear aeroelastic phenomena due to structural and aerodynamic nonlinearities 2. How to approach aeroelastic control of flexible aircraft 3. Know and apply the basics of potential flow theory as it is the prevalent aerodynamic theory used in aeroelastic modeling 4. Understand the difference between closely and loosely coupled aeroelastic systems. 5. Know how to couple panel method aerodynamics with FEM to form closely coupled aeroelastic system	
Education Method	The course will be taught in a classroom and hands-on sessions, circumstances permitting.	
Books	Recommended: 1. Introduction to nonlinear aeroelasticity. Dimitriadis, Grigorios. John Wiley & Sons, 2017. 2. Introduction to structural dynamics and aeroelasticity. Hodges, Dewey H., and G. Alvin Pierce. Cambridge University Press, 2011. 3. Aeroelasticity. Bisplinghoff, Raymond L., Holt Ashley, and Robert L. Halfman.&nbsp;Courier Corporation, 2013. 4. Etkin B. Dynamics of Atmospheric flight. Courier Corporation; 2012.&nbsp; 5. Introduction to aircraft aeroelasticity and loads. Wright, Jan Robert, and Jonathan Edward Cooper. John Wiley & Sons, 2008.	
Reader	Course notes	
Assessment	Take home assignments with oral close-out session.	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	P.P. Sundaramoorthy	
Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses:	
Course Contents	- AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of lectures, working sessions, design review sessions and a final symposium. The course structure will be explained in detail in the first weekly module. The time and date of the design review(s) and final symposium will also be communicated in the first weekly module. Students are requested to reserve all scheduled timeslots in their agendas. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. Note that, the course will require group work. Groups will be formed in the fourth week of the course. Only students who pass the individual test are allowed to participate in a group. Students who are not part of a group cannot participate in the remainder of the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. During each module it will be possible to discuss with the lecturers and fellow students.	
	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Individual test on course content 2. Written feedback on design specification sheet of another group in the course Students with a failing grade for the individual test, i.e. a grade below 5.8, will have a resit opportunity of the individual test in week 4 of the course. This resit will be in the form of a short oral exam. This resit is only available for students who did not pass the individual test. On passing the oral resit exam, students will receive a 5.8 as grade for the individual test. Students who fail the resit opportunity will not be able to participate in the group work, and therefore cannot participate in the remainder of the course. It is mandatory that students are present at the design review(s) and the final symposium presentations of their own group and those of other groups. The individual grade of students who fail to be present will be withheld. These students will not receive a final grade for the course as a result. Students of whom the individual grade which has been withheld, may take a resit. The resit will be in the form of a short oral exam. It will be possible to apply for a resit up to 15 working days after the publication of the grades in Osiris, but no later. A resit exam will then be scheduled by the principle lecturer of the course in consultation with the student in question. Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, Introduction to materials science" or books of similar nature. Be sure to check the entry level quiz before making up your mind about this course.	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, ISS, Hubble telescope, James Webb telescope, Venus orbiters, Launchers) will be presented as a carrier for material selection analysis. A holistic view of "materials science - manufacturing - structural analysis - system engineering for missions" will be presented. While students come mostly from different backgrounds, the goal of this class is to help them identify the perspective they use in the holistic view and to learn about the materials science perspective. Fundamentals of materials classes and their typical properties will be introduced. Current material choice will be analyzed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. In sessions of "ESA forum + student poster seminar", students will be given opportunities to interact with ESA experts and present their own findings from analyzing given missions, by translating desired properties into material structures and microstructures. In class discussion is an important part of learning. The structure of the lectures will be tailored to maximize the student involvement. The course contains the following topics: week 1: Holistic view and materials science perspective; Typical materials - metals, ceramics and polymers week 2: Typical materials - metals, ceramics and polymers week 3: Material degradation in space - vacuum; thermal cycling; week 4: Material degradation in space - AO; charging week 5: Material degradation in space - space radiation week 6: Material degradation in space - meteoroids and orbital debris; week 7: "ESA forum + student poster seminar" Materials testing for space applications; Extreme missions; Materials challenge in future space age (ESA guest speakers); Student poster seminar (Jan 6, 2025) week 8: Study period (Q and A) week 9: exam week (Jan 20. 2025)	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 12 lectures of 2h per week for the first 6 weeks. The last week is a whole-morning session of "ESA guest speaker forum + student poster presentation".	
Literature and Study Materials	A compilation of book chapters, lecture notes and review papers.	
Assessment	Poster presentation (40%) + final exam (60%). The students are to work in groups and do research on prescribed question/topics and deliver a final poster presentation (15% individual score, 25% group score) in week 7 to show their discoveries throughout the course. The final grade will be the weighted average of poster presentation and final exam. Example questions/topics will be made available on Brightspace.	
Enrolment / Application	For students not from "structures for space" profile, there will be a cap of 20 students maximum. Please enroll in brightspace (enrollment period Oct 28 to Nov 10, 2024) as soon as you decide to take the course. Once the maximal number is reached, enrollment will be closed.	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Design and Safety of Structures

Program Coordinator	Dr.ir. J.A. Pascoe
Contact for Students	To contact the ASM MSc track coordinator send an email to MScCoordinator-ASM@tudelft.nl. Please indicate the profile (DSS) for specific questions about the profile itself.
Introduction 1	Profile II Design and Safety of Structures The design and safety of structures (DSS) profile is for you if your intention is to become a structural designer or stress engineer. It focusses on accurately calculating and predicting stresses in structures and coming up with clever structural solutions through advanced design techniques and optimization. In the DSS profile you will learn more about bucking & structural analysis; the trinity in structures, materials and manufacturing; the design & optimization of composite reinforced structures, joining methods and experimental techniques. You will also learn about techniques for detecting structural damage, and how to analyse the damage tolerance of aerospace structures. This profile allows you to specialize in structures with an Aerospace focus but also allows you to go into neighbouring fields. Within the different research groups in the Aerospace Structures and Materials Department (ASM) there are plenty of thesis opportunities available. Electives with a strong Design and Safety of Structures flavour within the ASM Department are: - AE4ASM522 Applied Aircraft Aeroelasticity - AE4ASM505 Non-linear (structural) modelling (using F.E.M.) - AE4ASM511 Structural Analysis & Stability of Structures II - AE4ASM510 Design & Analysis of Composite Structures II - AE4ASM514TU Continuum Mechanics - AE4ASM517 Aircraft Manufacturing Laboratory - AE4ASM524 Spacecraft Structures Development - AE4ASM504 Structural Integrity & Maintenance - AE4ASM105 Trinity Exercise - AE4ASM110-22 Polymer Composite Manufacturing
Program Structure 1	In the ASM Track you do not have to select a profile until October. Use the core module in period 1 to select your final preference. Electives are decided upon in the second period whilst you are taking the profile courses to allow you to orientate yourself on your thesis topic and pick the relevant deepening electives upon the advice of your preferred thesis supervisor. You then register your choice of electives with the MSc track coordinator in December. TU Delft students who intend to start in February are requested to see the Track Coordinator by the end of November or early December to discuss their planning. Be mindful that a February start assumes you will take your internship from April of that year, so plan accordingly and start preparing for your internship in time.

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
Responsible Instructor	Dr.ir. O.K. Bergsma	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The global course setup is such that there lectures contain the following topics: - Thoughts behind lightness - Design allowables - Interaction between materials processes - Materials and their properties - Essentials of manufacturing - Processing to final products and their applications. - The principles of stresses in laminated composites - Recycling and other environmental aspects. - The principles of repair of structures. Guest lectures will be illustrating the course content.	
Study Goals	After succeeding this course the student should be able to: - explain parameters and their relationships, which play a role in the development of lightweight structures and parts. Examples: - be able to judge a structural design on conditions required to call a design a lightweight design. - be able to identify carbon glass aramid and dyneema fibres - be able to identify metals from composites on micro and macro scale. - be able to relate lightweight materials to typical strong and weak points in their performance. - be able to give examples of fibre morphologies - be able to argue the correlation between fibre content, orientation control, fibre length, manufacturing process and application. - be able to recall & argue unwanted stress distributions in composite materials	
Education Method	Lectures. The lectures will most likely be given in complete day sections (2 times 2 hours). Old collegerama recordings of lectures are available. Online lecture videos are available and explain the content in an other way. Guest lectures are two hour lectures.	
Assessment	Written exam, with grades converted to whole or half numbers. In case of 100 points the following distribution is most often used <points>[grade]: <0-14>[1]; <15>[1.5]; <16-24>[2]; <25>[2.5]; <26-34>[3]; <35>[3.5]; <36-49>[4]; <50>[4.5]; <51-54>[5]; <55>[5.5]; <56-64>[6]; <65>[6.5]; <66-74>[7]; <75>[7.5]; <76-84>[8]; <85>[8.5]; <86-94>[9]; <95>[9.5]; <96-100>[10];	

AE4ASM002	Designing Materials with Aerospace Specific Properties	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Prof.dr.ir. R. Benedictus	
Instructor	B. Atli-Veltin	
Instructor	Dr. Y. Tang	
Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Several courses follow on the general material concepts introduced in this 002 course, amongst others, within the ASM MSC track: -AE4ASM101 Polymer Science; (Q2) -AE4ASM103 Functional Coatings. (Q3)	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Ashby, Materials". The students must have completed the materials science courses in the first 3 years of the Aerospace Engineering Curriculum, or any other Engineering curriculum	
Course Contents	In this course an unorthodox approach to materials will be presented. Rather than memorising known routes to reach certain materials properties, the students will be trained to translate these desired properties into material structures and microstructures and to design suitable material production processes to realise these properties. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The course is divided in 7 lectures. Three lectures cover basic design rules of different material classes and their behaviour: metals, polymers, ceramics. Three lectures cover material behavior in aerospace relevant environments: exposed to high temperatures, as building blocks for H2 storage, as structural materials sensitive to environmental degradation. A last lecture deals with strategies to extend material life through the use of self-healing material concepts. -polymers -metals -ceramics and 'smart' materials -materials and H2 (energy transition) -materials at high temperatures -materials and environmental damage -materials and lifetime The students will need to deliver a report (assignment). For this they will get a set of questions they need to solve by reading the existing literature.	
Study Goals	The objective of the course is to train the student in reverse material engineering. This skill enables students to initiate and guide new material developments to meet future targets in the industry. By the end of the course, you should be able to: - explain structure and property (inter)relationships of metals, polymers and ceramics - explain the functionalities of aerospace relevant material properties - breakdown and translate these functionalities into material structures and microstructures using reverse engineering methodologies - explain material behaviour and function loss and strategies to decrease the impact of damage in the material function - give examples of new material concepts and explain the underlying concepts.	
Education Method	Lectures	
Literature and Study Materials	A compilation of lecture notes, review papers and assignment.	
Assessment	Assignment report (40%) + exam (60%) (a minimum of a 5 over 10 points in the exam is needed to pass)	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods. Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments (only assignment 3 and 4 since they have the highest weight; assignment 1 and 2 cannot be redone). Results for individual assignments are only valid for one academic year: when re-taking the course all assignments have to be done in that academic year.	

AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
Responsible Instructor	Dr. K. Masania	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The red line of the course is aiming at knowledge and understanding of manufacturing processes in relation to material properties and feasible product designs. The course contents include manufacturing processes for metallic and composite parts, the assembly of parts into large (sub)structures, and related topics like Quality control, organisation principles, finances, etc.	
Study Goals	The student should have a good knowledge and understanding of the mainstream manufacturing processes of structural materials (lightweight alloys, composites, hybrids). The student should be able to describe and motivate the processing procedures like the processing steps, required tooling and equipment, of manufacturing processes. The student should be able to select adequate manufacturing processes for designs of lightweight structures and components, and should be able to motivate his/her choices. The student should be able to analyse and synthesize interactions between materials, design and manufacturing processes. The student should be able to identify and explain manufacturing related flaws and inaccuracies and advice on how to prevent/limit those.	
Education Method	- Interactive lecture where contents are presented and where the flipped classroom is often used. - Exercises in class with some additional working material	
Assessment	written exam. Closed book.	

AE4ASM005	Fatigue of Structures & Materials	3
Responsible Instructor	Prof.dr.ing. R.C. Alderliesten	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	- Introduction to Fatigue (fatigue as a phenomenon; stress concentrations; residual stresses; fatigue properties of metallic and composite materials; fatigue strength of notched specimens, residual strength). - Fatigue damage mechanisms (initiation, crack growth, delamination growth, transverse matrix cracking, fibre failure). - Analysis methods (stress concentration factors, stress intensity factors; energy balance approaches, strain energy release rates). - Fatigue loading (Load Spectra, Fatigue under Constant- & Variable-Amplitude Loading). - Special Fatigue Conditions (surface treatments; fretting corrosion; corrosion fatigue; high-temperature and low-temperature fatigue, moisture ingress). - Fatigue and Damage Tolerance of Aircraft Structures: Regulations, tests, scatter, application of fatigue and damage tolerance methods.	
Study Goals	This course provides the students with engineering knowledge and skills to recognize and to analyse fatigue and damage tolerance problems in aircraft structures and materials. After the course the student must be able 1. Interpret and discuss the fatigue fracture features with respect to the characteristics of each phase in fatigue life 2. Define and determine stress concentration factors for notched structures with or without residual stresses 3. Explain and discuss S-N curves with respect to mean stress, material surface effects, and scatter, and perform fatigue life analyses considering mean stress and notch root plasticity 4. Assess the fatigue life of tension and shear joints, and explain limitations to the similarity principles (K,I,T) 5. Explain Linear Elastic Fracture Mechanics concepts for damage growth, and perform crack growth analyses with these concepts 6. Explain the consequences of variable- and constant amplitude loading on fatigue life and damage growth, and perform fatigue life analyses for arbitrary load spectra 7. Explain the effect of environment on fatigue life and fatigue phenomena 8. Perform residual strength analyses.	
Education Method	Selfstudy using the Online Course Material on Brightspace together with weekly plenary instruction sessions and discussions with fellow (online and campus) students or e-moderator on a forum.	
Literature and Study Materials	J. Schijve, Fatigue of Structures and Materials, 2nd edition, 2009, Springer Science+Business Media, B.V., ISBN-13: 978-1-4020-6807-2 e-ISBN-13: 978-1-4020-6808-9.	
Assessment	Readers, formative tests, documentation, and other relevant material provided via Brightspace.	
	Written exam & assignments Written exam at the end of the lecture period, 80% of total grade. Assignments during the lecture period, altogether 20% of the total grade. Deadlines of individual assignments are communicated via Brightspace.	

AE4ASM106	Stability & Analysis of Structures I	3
Responsible Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	This course introduces analytical and numerical techniques for the analysis of representative aerospace structures under mechanical loading, with particular emphasis on determining their stability. The course covers the connection between the principle of minimum potential energy, the variational formulation and the classical (differential) formulation of quasi-static structural problems. Basic structural components are analyzed using beam and plate theory. The basic theory of structural stability is introduced through asymptotic analysis in order to derive the stability (buckling) equations. Buckling equations are formulated and solved for geometrically non-linear elastic beams and plates. The methodology is extended to formulate stability analysis of representative aerospace components. Emphasis is placed on the understanding of the models, their assumptions, and ranges of applicability. The topics covered include: - Introduction to structural analysis - Review of global and local balance principles - Constitutive relations, material symmetries and strain energy - Formulations of structural problems: differential, variational and minimum formulations - Introduction to calculus of variations and approximate solution methods (Ritz and Galerkin) - Applications to beams and plates: solution to basic structural problems - Introduction to buckling: Eigenvalue buckling problem (bifurcation buckling) - Imperfection analysis (snap buckling) and stability analysis - Buckling of non-linear beams/columns and plates - Design methodology for thin-walled aerospace structures	
Study Goals	1- Understand the various formulations of equilibrium and equations of motion 2- Model simple aerospace structural components using engineering theories 3- Apply energy and variational methods for the evaluation of structural responses 4- Understand and being able to obtain the buckling equations for beams and plates 5- Develop insights into buckling phenomena of aerospace thin-walled structures	
Education Method	Lectures.	
Literature and Study Materials	Primary source: course slides provided in Brightspace Recommended references: T.H.G. Megson, Aircraft Structures for Engineering Students, Elsevier, Any edition. C.L. Dym and I.H. Shames, Solid Mechanics: A Variational Approach, Augmented Edition, Springer, available online through the TU Delft Library.	
Assessment	Written (take-home) assignments and oral exam	

AE4ASM108	Experimental Techniques & NDT	3
Responsible Instructor	Dr. A. Anisimov	
Instructor	Dr. R.M. Groves	
Responsible for assignments	K.P.W. Dissanayake	
Responsible for assignments	Dr. N. Yue	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of electronics, elements of signal processing (analogue, digital), basics of measurements and metrology	
Course Contents	This is a lecture course on developing the critical thinking needed to perform measurements in the Aerospace Structures and Materials Laboratory. This includes an introduction to the most common sensors for experimental mechanics, non-destructive testing and structural health monitoring, evaluation of the performance of these sensors, selection of sensors for different applications and signal processing and control algorithms.	
Study Goals	By the end of the course, you should be able to: 1. Analyse a measurement system for the aerospace sector and to determine if it is fit for a specific application. This learning objective acts as an umbrella objective, covering the whole course. Lower level objectives include that by the end of the course, you should be able to: 2. Explain the principle of operation of the most common sensors and instrumentation in the aerospace structures and materials laboratory. 3. Justify the selection of a sensor for a specific measurement application, based on sensor specifications and measurement principle. 4. Analyse a measurement chain and determine signal accuracy and noise levels. 5. Assess the performance of signal processing and feedback algorithms for their influence on measurement noise and accuracy.	
Education Method	Lectures, obligatory laboratory sessions, reader with self-evaluation questions.	
Literature and Study Materials	Course reader, pre-recorded videos with slides available on Brightspace. Additional material as datasheets, book chapters and papers are uploaded to Brightspace during the course.	
Reader	Course reader available on Brightspace	
Assessment	Open book written exam (30% of grade). Oral exam (40% of grade). Laboratory group assignments (30% of grade).	
Set-up	12 active lectures with problem-solving and discussion (weeks 2.1-2.6). Students have to prepare for these workshops using the material available on Brightspace, including pre-recorded videos. The material becomes available as the course progresses. 2 exam review workshops (week 2.7). It is particularly important to attend these sessions as the course lecturers will explain the structure and what is required in the open-book exam 5 obligatory laboratory assignments done in groups (weeks 2.1-2.5). Each practical is graded with a group grade for the written report (maximum of 4% towards the final course grade) and personal grade for the oral defence of the report (maximum of 2% towards the final course grade).	

AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis 3. Reliability analysis & Health monitoring a. Probability of Failure & Uncertainty quantification b. Structural Health Monitoring 4. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Understand the philosophy of reliability analysis and calculate the probability of failure of composite structures by combining numerical methods and the mechanics of composite structures - Understand the importance of structural health monitoring (SHM) and design a SHM system for a given composite structure - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Classical Laminate Theory: 3 2. Progressive damage analysis: 3 3. Reliability analysis & Health Monitoring: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory, progressive damage analysis, reliability analysis and health monitoring. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 10 points A2. Progressive damage and damage tolerance analysis 20 points A3. Reliability analysis & Health Monitoring. 30 points	

Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method).

B1. Design problem 40 points

The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2,A3 and part B1) should at least be rewarded with 55% of the attributed points ($55\% \cdot 60$ and $55\% \cdot 40$), otherwise the student will fail. In addition, the overall average grade should at least be a 5.8 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project.

The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5722	Thesis Aerospace Structures & Materials	45
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal). If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go'	

(fail). Please refer to Brightspace for more information.

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Design and Safety of Structures

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4ASM101TU-22	Polymer Science	4
Responsible Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of a number of lectures covering the most relevant aspects of polymer science and technology distributed in 6 blocks, each covered in one up to three lectures. After this a session on 3 minutes pitches by the students will be organized: -Introduction to polymer science: daily life examples, nomenclature, concept of polymers, initiators, polymer architectures and polymerization media. -Polymerization: condensation and addition polymerization under step-growth; radical, ionic and ring opening polymerization under chain-growth mechanisms, concept of copolymerisation and crosslinking, basic characterization techniques (such as GPC). -Polymers in Aerospace: brief outlook on composites, chemistry of thermosets, high-performance thermosets, high performance thermoplastics, mechanical properties. -Processing: basics of polymer processing and polymer-based structures manufacturing, fiber processing and properties, living materials and their potential. -Sustainable Polymers: a glance on green chemistry, introduction to lignocellulosic biomass, how to develop biobased monomers, recyclability and dynamic materials, examples on sustainable polymers, impacts on sustainability, a perspective on how can you design sustainable polymers? -Polymers for Energy and Environment- demonstrating a concept of photovoltaic devices and batteries, polymers used in these applications, polymer membranes and polymer coagulants for water remediation, polymer ionotronics. -Pitch talks: students to give 3 minutes presentations about developments/applications in the polymer field, convince audience as in business structures.	
Study Goals	By the end of this course, you should be able to: LO1: Identify polymer basics and analysis LO2: Assess polymer structure-property relationship for thermomechanical performance estimation LO3: Judge parameters and requirements for polymers to be qualified for aerospace materials design LO4: Define sustainability concepts in polymer science LO5: Present about recent developments in polymers field	
Education Method	Lectures and lab activity	
Literature and Study Materials	Seymour/Carraher's Polymer Chemistry: 10th Edition. Contains comprehensive information on polymer materials from chemistry perspective Lecture slides will be made available to students as study material	
Assessment	Written final exam and pitch talk (3 minutes) presentation Lab activity attendance is mandatory Transitions AE4ASM101TU -> AE4ASM101TU-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM101TU, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM101TU. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM101TU is not completed in 2022 2023, students will need to complete AE4ASM101TU-22 instead.	

AE4ASM103		Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	3 4		
Course Language	English		
Expected prior knowledge	The students must have general knowledge about materials science.		
Course Contents	In this course we introduce the design fundamentals that control specific properties in coatings, surface treatments and functional coatings and the techniques used to apply them and test them. The goal is therefore to provide the fundamentals of chemistry and physics that explain structure-property relationships relevant to understand and design surface treatments/coatings with: (i) all-round relevant properties such as adhesion, mechanical response, active (inhibitors) and passive (barrier) corrosion protection; and (ii) increasingly relevant functional properties such as antiicing, self-cleaning, self-healing, drag, structural colours, storage/release of active chemicals with nano and microparticles, and antifouling. Since material design rules are directly related to application methods, different technologies for coatings application will be covered such as electrophoretic deposition, powder coatings, anodizing, sol-gel and plasma electrolytic oxidation. Characterization strategies to obtain information about key properties will also be discussed. The fundamentals and characterization strategies covered in the course are applicable to coatings, adhesives and composites of interest in aerospace applications and other engineering domains.		
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.		
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.		
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.		
Assessment	Written report (40% of final grade) and written exam (60% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.		

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr.ir. C.M. de Zeeuw	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Compared to previous years the course starts in Q3 (not Q2). There will be a kick-off lecture in week 3.1	
Course Contents	Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four to five students about 8 day sections with support of staff are needed during this course. These will be planned on fore hand, preferably on fixed time slots. Most likely either Friday morning or Friday afternoon. We will not use all of those slots, but it will provide the flexibility to schedule things in the lab and make it easier to block their schedules. Finishing period As the starting period changed, the finishing period changes according, at the end of the fourth period (Q4, June) it is. The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Estimation of the maximal bending-torsion coupling for a strip of UD composite 4. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 5. Performing mechanical tests 6. Analysing the test results of the mechanical test, including failure analyses 7. Adhesion of thermoset laminates 8. Possibly preparation of lap shear specimens 9. Performing lap shear tests 10. Estimation of the mechanical properties of a sandwich panel 11. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 12. Performing a bending test on a sandwich panel. 13. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	This course was renewed in 2023-2024 by fixing the time schedule for students on beforehand.	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on beforehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM110-22	Polymer Composite Manufacturing	4
Responsible Instructor	Dr. J.J.E. Teuwen	
Instructor	Dr.ir. D.M.J. Peeters	
Instructor	Dr.ir. B.C.P. Jongbloed	
Instructor	Dr. B. Caglar	
Instructor	R. Tao	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course explores in depth the manufacturing of polymer composite structures and its underlying physics. By understanding the relationships among physics, part/material quality and the design of the manufacturing process, you will be able to critically assess any given manufacturing process. The course is built around a number of polymer composite manufacturing processes highly relevant to the current and future aerospace industry for the manufacturing of high-performance individual parts (autoclave processing of prepreg, liquid composite moulding, thermoplastic composite processing), for the assembling of complex structures (manufacturing of integrated structures, thermoplastic composite welding) and for end of life and recycling. The main basic physical phenomena governing polymer composites manufacturing, e.g. flow (of polymer and of fibres), void formation, curing, shrinkage and crystallization, are interwoven within the course.	
Study Goals	At the end of this course, the student should be able to: - Describe the common manufacturing processes used in aerospace industry in terms of processing steps, tooling and equipment as well as advantages, limitations and applicability. - Correlate and analyse the processes and their principles/underlying physics with manufacturability, part quality and manufacturing design. - Compare and evaluate the different processes for their suitability of manufacturing common components (aircraft wing, fuselage etc). - Recognise manufacturing defects, identify their potential sources and recommend strategies to improve the quality of common manufacturing processes for simple component geometries to reduce scrap rate or to increase yield.	
Education Method	Interactive lectures with numerous in-class exercises. Assignments are used throughout the course to allow the students apply, deepen and test their knowledge on the manufacturing processes discussed during the course.	
Assessment	Oral exam (50% of the final grade) - exam format subject to change depending on amount of students - and mandatory assignments (50% of the final grade). A minimum grade of 5.0 out of 10 on both the assignment and the oral exam need be obtained to be able to pass the course.	

AE4ASM503	Sheet Metal Forming	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The lecture consists of two main parts: a theoretical part and an application part The theoretical part includes: - Introduction, formability, deformations, formability testing. - Stress- and strain states, forming limit curves, yield criteria, anisotropy, heat treatments. - Theory of plasticity The application part includes a number of forming processes, like - Simple forming processes like bending and stretching. - Several forming processes for more complex parts like deepdrawing, rubber forming, hydroforming and matched die forming. - Non-conventional processes like explosive forming and superplastic forming - The use of numerical tools for simulation of forming processes, and some other related topics.	
Course Contents	The theoretical part of the lecture will focus on the descriptions of deformations, strains, stresses, etc. In addition, the interaction between stresses and strains and their limits are expressed in a.o. yield criteria and forming limit diagrams. Such concepts are also used in the second, application part to describe and explain the do's and don'ts of the forming processes.	
Study Goals	After attending this course the student should be able to: - describe and explain the basic phenomena in plastic deformation of sheet metal - understand the interactions between multiple phenomena and how to exploit these - be able to relate the formability of materials to forming processes - have understanding and ability to combine the features of material behaviour, the forming process and the shape of the component.	
Education Method	Lecture, including discussions about real parts, pictures and videos.	
Literature and Study Materials	Hand-outs are the core material A list of recommended literature will be given during the first lecture (background information unless specified otherwise)	
Assessment	Written exam. Closed book.	

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	N. Eleftheroglou	
Responsible Instructor	Dr.ir. J.A. Pascoe	
Instructor	Prof.dr.ing. R.C. Alderliesten	
Instructor	Ir. J. Sinke	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). Regulations applicable to maintenance. - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals	At the end of this course, you should be able to: - Describe different maintenance strategies and select the appropriate one for a given problem, taking into account different aspects (e.g. technical, regulations, planning); - Describe the working principles of different non-destructive inspection techniques, their pros and cons and their applicability to different materials; - Explain the relationship between certification and maintenance requirements and (documentation of) procedures; - Develop a maintenance programme for a simple case; - Select an appropriate design philosophy to ensure structural integrity of a given part throughout the aircraft lifetime.	
Education Method	Lectures.	
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam, closed book (60% of final grade) and group assignment (40% of final grade).	

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	lectures and practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of aerospace structures and structural components. The equation of motion for the dynamic response is formulated in matrix form using finite element discretization or direct modelling discretization (rigid bodies and flexible connectors). Various solution techniques are used for solving transient dynamic response and vibration problems, including modal analysis. The course also covers experimental techniques to obtain model parameters and test the response of a physical system. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE4ASM514TU	Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.	
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.	
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.	
Education Method	Lectures.	
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.	
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.	

AE4ASM515-22		Characterization of materials and components	4
Responsible Instructor	Dr. B. Kumru		
Co-responsible for assignments	Dr.ir. C.M. de Zeeuw		
Contact Hours / Week x/x/x/x	0/2/0/0		
Education Period	2		
Start Education	2		
Exam Period	2 3		
Course Language	English		
Expected prior knowledge	General knowledge of materials engineering and science (in particular polymers and metals) at the level expected for a BSc graduate in a (mechanical) engineering discipline.		
Course Contents	This course is an introduction to the most common characterization tools to gather crucial structural and property information of materials and components. The focus is on knowing about several possible characterisation methods and their underlying principles of measurement as well as the strategies to extract the most relevant information from the tests for the purpose in mind. The characterisation techniques will be taught by various colleagues in each field. The methods to covered in this course are classified as follows: 1. Imaging of material structures and surfaces (microscopy, confocal, SEM, EDS) 2. Chemical analysis of polymers (FTIR, Raman spectroscopy, contact angle) 3. Structural characterisation 4. Thermomechanical analysis of polymers (DSC, TGA, DMA/Rheology) 5. Mechanical characterization		
Study Goals	Aim of the course is to provide (aerospace, mechanical, maritime) engineering students with adequate skills and knowledge such that later in their technical careers they can handle materials performance issue in which it is important to clarify the material characteristics such as to link actual material performance to its microstructure. Such knowledge and skills are important for determination of material in certain applications and while choosing engineering materials with desired properties. To this aim: - Students will get familiar with the underlying principles of measurement of principal materials characterisation techniques. - Students will be able to analyse and interpret data from each characterisation technique. - Students will be able to select the right characterisation techniques to correlate structure and property relationships over a wide range of engineering materials.		
Education Method	Lectures and hands on practices		
Assessment	Oral Exam and assignment Transition AE4ASM515 -> AE4ASM515-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM515, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM515. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM515 is not completed in 2022 2023, students will need to complete AE4ASM515-22 instead.		

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	The grade for this course will be a Pass/Fail grade. This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Alternatively, having earned a Bachelor degree in aerospace engineering will give access to the course as well. Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December - Week 4.8: 15 May - Week 1.6: 30 September	

Set-up	The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and 3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3
	There is an intentional overlap so that continuity and handover is guaranteed

AE4ASM520	Industrial Composite Manufacturing	3
Responsible Instructor	Prof. C.A. Dransfeld	
Responsible Instructor	Dr. D. Stefaniak	
Contact Hours / Week x/x/x/x	0/0/4/0 COURSE NOT OFFERED IN 2024-2025	
	Students that have participated in this course and have not yet passed the course, please contact C.A.Dransfeld@tudelft.nl	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Expected BSc diploma	
Course Contents	The course deepens the understanding on different composite manufacturing technologies for aerostructures used at the industrial scale and addresses on specific challenges evoked especially by automation and large part size. Special pitfalls with respect material selection, mould design, process induced part distortion and vacuum setup are discussed to apply the basic understanding of the manufacturing processes to several specific industrial challenges. Finally, these manufacturing processes are also evaluated with respect their sustainability and are investigated in depth towards their energy, materials and auxiliaries consumption in context of the product life cycle.	
Study Goals	At the end of this course, the student should be able to: - Transfer the basic principles of composite manufacturing processes to real and more complex industrial examples - To gain insight in material selection and its consequences in an industrial context - To recognize manufacturing processes and the linked pitfalls based on the resulting part or the utilized mold - Understand the challenges in sizing composite manufacturing systems for large size components - Understand the importance and dependencies of performance assessment indicators in composite manufacturing like robustness, energy consumption, occupation time, non-operating costs and flexibility.	
Education Method	Interactive lectures based on industrial case studies. Some exercises are accompanied by Python Scripts used in Jupyter Notebooks to deepen the understanding and facilitate interactivity. The required software is provided as a browser-based version and previous knowledge in programming is not compulsory. It is intended to visit the Center for Lightweight Production Technology at the German Aerospace Center (DLR) and Airbus site in Stade, Germany.	
Computer Use	yes	
Course Relations	The course is given by a guest lecturer from DLR (German Aerospace Center)	
Literature and Study Materials	will be defined at the beginning of the course	
Prerequisites	AE4ASM001, AE4ASM004, AE4ASM110	
Assessment	Oral exam (50% of the final grade) and mandatory assignment (50% of the final grade).	
Enrolment / Application		
Remarks	Students that need to retake AE4ASM520 can contact the responsible lecturer for a resit.	
Set-up	2 weeks with 2day-blocks and two weeks with one 1day-blocks (e.g. 3.1, 3.3, 3.6, 3.8); each day will consist of four lecture hours	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr.ir. J.A. Pascoe	
Instructor	Prof. C.A. Dransfeld	
Instructor	Dr. J. Wu	
Instructor	V. Yaghoubi Nasrabadi	
Instructor	P.G. Marchese	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE4ASM522	Applied Aircraft Aeroelasticity	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	X. Wang	
Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE4ASM506 (mandatory)	
Course Contents	This course is a follow-up course of Aeroelasticity AE4ASM506. Taking and passing the Aeroelasticity course AE4ASM506 is mandatory to pass this course. This course builds on the basics and regular aeroelastic phenomena that were introduced in AE4ASM506. The extension of the knowledge is twofold: 1. Introduction of nonlinear aeroelasticity 2. Introduction to control of flexible aircraft. 3. Introduction to potential flow aerodynamics and its coupling to structural FE models to form closely coupled aeroelastic models The course aims at placing the aeroelastic knowledge into a flexible aircraft context, and it will provide the student with new insights into the aircraft design process and aircraft design choices. Finally, understanding the contents of this course will prepare the student for the design of future aircraft configurations beyond the currently established aircraft concepts.	
Study Goals	At the end of the course, the student should be able to: 1. Know advanced nonlinear aeroelastic phenomena due to structural and aerodynamic nonlinearities 2. How to approach aeroelastic control of flexible aircraft 3. Know and apply the basics of potential flow theory as it is the prevalent aerodynamic theory used in aeroelastic modeling 4. Understand the difference between closely and loosely coupled aeroelastic systems. 5. Know how to couple panel method aerodynamics with FEM to form closely coupled aeroelastic system	
Education Method	The course will be taught in a classroom and hands-on sessions, circumstances permitting.	
Books	Recommended: 1. Introduction to nonlinear aeroelasticity. Dimitriadis, Grigorios. John Wiley & Sons, 2017. 2. Introduction to structural dynamics and aeroelasticity. Hodges, Dewey H., and G. Alvin Pierce. Cambridge University Press, 2011. 3. Aeroelasticity. Bisplinghoff, Raymond L., Holt Ashley, and Robert L. Halfman.&nbsp;Courier Corporation, 2013. 4. Etkin B. Dynamics of Atmospheric flight. Courier Corporation; 2012.&nbsp; 5. Introduction to aircraft aeroelasticity and loads. Wright, Jan Robert, and Jonathan Edward Cooper. John Wiley & Sons, 2008.	
Reader	Course notes	
Assessment	Take home assignments with oral close-out session.	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	P.P. Sundaramoorthy	
Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of lectures, working sessions, design review sessions and a final symposium. The course structure will be explained in detail in the first weekly module. The time and date of the design review(s) and final symposium will also be communicated in the first weekly module. Students are requested to reserve all scheduled timeslots in their agendas. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. Note that, the course will require group work. Groups will be formed in the fourth week of the course. Only students who pass the individual test are allowed to participate in a group. Students who are not part of a group cannot participate in the remainder of the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. During each module it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Individual test on course content 2. Written feedback on design specification sheet of another group in the course Students with a failing grade for the individual test, i.e. a grade below 5.8, will have a resit opportunity of the individual test in week 4 of the course. This resit will be in the form of a short oral exam. This resit is only available for students who did not pass the individual test. On passing the oral resit exam, students will receive a 5.8 as grade for the individual test. Students who fail the resit opportunity will not be able to participate in the group work, and therefore cannot participate in the remainder of the course. It is mandatory that students are present at the design review(s) and the final symposium presentations of their own group and those of other groups. The individual grade of students who fail to be present will be withheld. These students will not receive a final grade for the course as a result. Students of whom the individual grade which has been withheld, may take a resit. The resit will be in the form of a short oral exam. It will be possible to apply for a resit up to 15 working days after the publication of the grades in Osiris, but no later. A resit exam will then be scheduled by the principle lecturer of the course in consultation with the student in question. Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, Introduction to materials science" or books of similar nature. Be sure to check the entry level quiz before making up your mind about this course.	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, ISS, Hubble telescope, James Webb telescope, Venus orbiters, Launchers) will be presented as a carrier for material selection analysis. A holistic view of "materials science - manufacturing - structural analysis - system engineering for missions" will be presented. While students come mostly from different backgrounds, the goal of this class is to help them identify the perspective they use in the holistic view and to learn about the materials science perspective. Fundamentals of materials classes and their typical properties will be introduced. Current material choice will be analyzed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. In sessions of "ESA forum + student poster seminar", students will be given opportunities to interact with ESA experts and present their own findings from analyzing given missions, by translating desired properties into material structures and microstructures. In class discussion is an important part of learning. The structure of the lectures will be tailored to maximize the student involvement. The course contains the following topics: week 1: Holistic view and materials science perspective; Typical materials - metals, ceramics and polymers week 2: Typical materials - metals, ceramics and polymers week 3: Material degradation in space - vacuum; thermal cycling; week 4: Material degradation in space - AO; charging week 5: Material degradation in space - space radiation week 6: Material degradation in space - meteoroids and orbital debris; week 7: "ESA forum + student poster seminar" Materials testing for space applications; Extreme missions; Materials challenge in future space age (ESA guest speakers); Student poster seminar (Jan 6, 2025) week 8: Study period (Q and A) week 9: exam week (Jan 20. 2025)	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 12 lectures of 2h per week for the first 6 weeks. The last week is a whole-morning session of "ESA guest speaker forum + student poster presentation".	
Literature and Study Materials	A compilation of book chapters, lecture notes and review papers.	
Assessment	Poster presentation (40%) + final exam (60%). The students are to work in groups and do research on prescribed question/topics and deliver a final poster presentation (15% individual score, 25% group score) in week 7 to show their discoveries throughout the course. The final grade will be the weighted average of poster presentation and final exam. Example questions/topics will be made available on Brightspace.	
Enrolment / Application	For students not from "structures for space" profile, there will be a cap of 20 students maximum. Please enroll in brightspace (enrollment period Oct 28 to Nov 10, 2024) as soon as you decide to take the course. Once the maximal number is reached, enrollment will be closed.	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile III: Structures for Space

Program Coordinator	I. Uriol Balbin
Contact for students	To contact the ASM MSc track coordinator send an email to MScCoordinator-ASM@tudelft.nl. Please indicate the profile (SfS) for specific questions about the profile itself.
Introduction 1	Profile III Structures for Space The Structures for Space (S4S) profile is for you if your intention is to work in the rapidly growing space sector as a structural engineer in the design, analysis and verification of spacecraft and launcher structures. In this profile you will learn the technologies and principles required to design, analyze and test the different types of space structures. You will also become familiar with spacecraft and launcher structures and will develop strategies necessary to transform the technical, economic and societal requirements into an integrated structural solution. Finally, you will be able to use these methods and tools to solve practical structural problems for present and future space exploration. Electives with a strong Structures for Space flavour within the ASM Department are: - Additive Manufacturing - Fundamentals of Aeroelasticity - Trinity Exercise - Stability & Analysis of Structures II - Spacecraft Thermal Design Electives with a strong Structures for Space flavour within the Aerospace Faculty are: - MicroSat Engineering - Spacecraft Attitude Dynamics and Control
Program Structure 1	TU Delft students who intend to start in February are requested to see the Track Coordinator by the end of November or early December to discuss their planning. Be mindful that a February start assumes you will take your internship from April of that year, so plan accordingly and start preparing for your internship in time.

AE4870A	Rocket Motion	3
Responsible Instructor	Ir. M.C. Naeije	
Instructor	Ir. W.J.F. Simons	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	Course Overview	
Study Goals	1. Introduction and recap previous courses on orbital mechanics 2. Fundamentals of rocket motion 3. Launch trajectories in a homogeneous gravity field, in vacuum and in an atmosphere (2D); vertical flight, constant pitch angle, gravity, and sounding rockets. 4. Theory of the multi-stage rocket; optimal mass distribution. 5. Ballistic flight over the Earth; 2D, and 3D over spherical (non-rotating and rotating Earth). 6. Launch systems design, unconventional launchers like air launch 7. Summary and question hour (workout exam question example) This course introduces the fundamentals of rocket motion by studying the motion of rockets under different circumstances. Emphasising on analytical solutions of the equations of motion, it will provide insight into the qualitative and quantitative aspects of launch trajectories of rockets in a homogeneous gravitational field, the performance of single and multi-stage rockets, and the exoatmospheric ballistic flight over the Earth. It discusses launcher design considerations and unconventional launch systems like air launch. At the end of the course you should be able to LO1: Evaluate the performance of existing space launchers that make use of single and multi-stage rockets. LO2: Analytically derive equations of rocket motion for specific launch and rocket applications in a homogeneous gravitational field. LO3: Simulate a range of rocket trajectory scenarios (e.g. in Matlab or python).	
Education Method	Lectures	
Literature and Study Materials	To be studied: Lecture slides AE4870A Rocket Motion Lecture notes AE4-870A Rocket Motion (selected chapters) Further reading: Turner, Martin J.L., Rocket and Spacecraft propulsion: principles, practice and new developments. 3rd Edition. ISBN 978-3-540-69202-7 (selected parts)	
Assessment	Closed-book written exam	

AE4870B	Re-Entry Systems	3
Responsible Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	IMPORTANT NOTICE: DUE TO THE LARGE INFLUX OF STUDENTS, THE COURSE IS ONLY OPEN TO STUDENTS FROM THE FOLLOWING PROFILES: SPACEFLIGHT DYNAMICS STRUCTURES FOR SPACE THIS CAP MAY BE LIFTED IN CASE CAPACITY IS AVAILABLE. ===== The course aims at giving a complete overview of re-entry systems, mainly from a mission point of view, but also from the system point of view. To begin with the mission, the fundamental principles of mechanics (Newton's Laws) are applied to derive the equations of motion for translational and rotational motion. Subsequently, simplifications are applied to give a description of planar ballistic, gliding and skipping flight, where the focus will be on analytical solutions for quantifying maximum mechanical and thermal loads. Next, typical mission applications, such as planetary entry and descent with parachutes and/or propulsion systems are discussed, as well as terminal-area flight of winged entry vehicles. Special topics (not covered each year): To extend the mission applications, attention is also paid to the atmospheric flight aspects during an aero-gravity assist, aerocapture and aerobraking. As many re-entry missions rely on the application of guidance, navigation and control systems, the fundamentals of each of these three sub-systems will be discussed and applied in simple examples, such that after the course the student has a starting point for further study. The relation between vehicle configurations (capsule, winged, lifting body, etc.) and mission aspects, such as flight time and (maximum) system loads is a typical system aspect that will pass review. Finally, all presented theory will be combined in a so-called development plan for building a simulator that can be used for performance analysis of both controlled and uncontrolled re-entry vehicles. How to use such a simulator efficiently is depending on what kind of simulation and analysis technique will be used. Therefore, the course is concluded with an overview of such techniques like Monte-Carlo Analysis, and how to extract the required information from the simulated data.	
Course Contents Continuation	Lecture Topics (the actual topics covered may vary per year): #01 Introduction #02 Entry Environment and Aeroheating #03 Fundamentals of Motion #04 Ballistic entry #05 Gliding Entry #06 Skip Entry #07 Guidance, Navigation and Control #08 Planetary Entry and Descent #09 Advanced Descent and Landing Systems #10 Terminal Area Energy Management	
Study Goals	At the end of this course, the student will be able to (depending on topics covered): 1. Identify the influence of the planetary environment on the motion of and loads on an entry vehicle 2. Derive the equations of planar motion, as well as key performance indicators 3. Understand the three unpowered entry mechanisms (ballistic, gliding and skip entry), and derive the corresponding equations of motion 4. Apply a sizing methodology to design an entry descent system (parachute and/or propulsion) 5. Understand the fundamental functions of a Guidance, Navigation and Control System and design a simple system 6. Name the characteristic phases of the Terminal Area flight, and derive the equations for maximum range and maximum dive (steady-state approximation) 7. Solve typical re-entry problems through a combination of physical insight, analytical skills, and numerical evaluation NOTE: particular emphasis is put on deriving the relevant equations, as this provides insight in the physics and underlying assumptions. These derivations are an integral part of the exam	
Education Method	Classical lectures (2 x 2 hrs per week). The course is structured in a number of blocks, which each deal with an aspect of re-entry missions and systems. Many real-life examples will be used to highlight certain theoretical aspects, and to show the applicability of the presented theory. Typical exam questions will be used to prepare the student for the exam by testing their insight in the presented material.	
Course Relations	AE3212-I - Aerospace Flight Dynamics and Simulation AE3212-II - Simulation, Verification and Simulation AE4870A - Rocket Motion	
Literature and Study Materials	1. Lecture Notes and Lecture Slides (Mandatory) The lecture notes are in the process of being published, so they may be replaced by the published book. To be announced through Brightspace and in class. Selected references for additional study: 1. Anderson, J. (1989). Hypersonic and High Temperature Gas Dynamics. McGraw-Hill. 2. Bertin, J. (1994). Hypersonic Aerothermodynamics. AIAA. 3. Cornelisse, J., Schoeyer, H., and Wakker, K.F. (1979). Rocket propulsion and spaceflight dynamics. Pitman, London. 4. Hankey, W. (1988). Re-Entry Aerodynamics. AIAA. 5. Mulder, J., van Staveren, W., van der Vaart, J., de Weerd, E., In 't Veld, A. and Mooij, E. (2013). Aerospace Flight Dynamics and Simulation. Delft University of Technology, Lecture Notes AE3-212-I. 6. Regan, F. and Anandakrishnan, S. (1993). Dynamics of Atmospheric Re-Entry. AIAA. 7. Vinh, N. (1981). Optimal trajectories in atmospheric flight. Elsevier. 8. Vinh, N., Busemann, A., and Culp, R. (1980). Hypersonic and planetary entry flight mechanics. The University of Michigan.	
Assessment	Closed-book, written examination.	

Note: in previous years the grade was composed of partial grades for homework and exam. Exam grades from previous years are not carried over to the current year

AE4ASM001	Design of lightweight structures I: Composites & Metals	3
Responsible Instructor	Dr.ir. O.K. Bergsma	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The global course setup is such that there lectures contain the following topics: - Thoughts behind lightness - Design allowables - Interaction between materials processes - Materials and their properties - Essentials of manufacturing - Processing to final products and their applications. - The principles of stresses in laminated composites - Recycling and other environmental aspects. - The principles of repair of structures.	
Study Goals	Guest lectures will be illustrating the course content. After succeeding this course the student should be able to: - explain parameters and their relationships, which play a role in the development of lightweight structures and parts. Examples: - be able to judge a structural design on conditions required to call a design a lightweight design. - be able to identify carbon glass aramid and dyneema fibres - be able to identify metals from composites on micro and macro scale. - be able to relate lightweight materials to typical strong and weak points in their performance. - be able to give examples of fibre morphologies - be able to argue the correlation between fibre content, orientation control, fibre length, manufacturing process and application. - be able to recall & argue unwanted stress distributions in composite materials	
Education Method	Lectures. The lectures will most likely be given in complete day sections (2 times 2 hours). Old collegerama recordings of lectures are available. Online lecture videos are available and explain the content in an other way. Guest lectures are two hour lectures.	
Assessment	Written exam, with grades converted to whole or half numbers. In case of 100 points the following distribution is most often used <points>[grade]: <0-14>[1]; <15>[1.5]; <16-24>[2]; <25>[2.5]; <26-34>[3]; <35>[3.5]; <36-49>[4]; <50>[4.5]; <51-54>[5]; <55>[5.5]; <56-64>[6]; <65>[6.5]; <66-74>[7]; <75>[7.5]; <76-84>[8]; <85>[8.5]; <86-94>[9]; <95>[9.5]; <96-100>[10];	

AE4ASM002	Designing Materials with Aerospace Specific Properties	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Prof.dr.ir. R. Benedictus	
Instructor	B. Atli-Veltin	
Instructor	Dr. Y. Tang	
Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Several courses follow on the general material concepts introduced in this 002 course, amongst others, within the ASM MSC track: -AE4ASM101 Polymer Science; (Q2) -AE4ASM103 Functional Coatings. (Q3)	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Ashby, Materials". The students must have completed the materials science courses in the first 3 years of the Aerospace Engineering Curriculum, or any other Engineering curriculum	
Course Contents	In this course an unorthodox approach to materials will be presented. Rather than memorising known routes to reach certain materials properties, the students will be trained to translate these desired properties into material structures and microstructures and to design suitable material production processes to realise these properties. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. The course is divided in 7 lectures. Three lectures cover basic design rules of different material classes and their behaviour: metals, polymers, ceramics. Three lectures cover material behavior in aerospace relevant environments: exposed to high temperatures, as building blocks for H2 storage, as structural materials sensitive to environmental degradation. A last lecture deals with strategies to extend material life through the use of self-healing material concepts. -polymers -metals -ceramics and 'smart' materials -materials and H2 (energy transition) -materials at high temperatures -materials and environmental damage -materials and lifetime The students will need to deliver a report (assignment). For this they will get a set of questions they need to solve by reading the existing literature.	
Study Goals	The objective of the course is to train the student in reverse material engineering. This skill enables students to initiate and guide new material developments to meet future targets in the industry. By the end of the course, you should be able to: - explain structure and property (inter)relationships of metals, polymers and ceramics - explain the functionalities of aerospace relevant material properties - breakdown and translate these functionalities into material structures and microstructures using reverse engineering methodologies - explain material behaviour and function loss and strategies to decrease the impact of damage in the material function - give examples of new material concepts and explain the underlying concepts.	
Education Method	Lectures	
Literature and Study Materials	A compilation of lecture notes, review papers and assignment.	
Assessment	Assignment report (40%) + exam (60%) (a minimum of a 5 over 10 points in the exam is needed to pass)	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods. Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments (only assignment 3 and 4 since they have the highest weight; assignment 1 and 2 cannot be redone). Results for individual assignments are only valid for one academic year: when re-taking the course all assignments have to be done in that academic year.	

AE4ASM004	Manufacturing of Aerospace Structures & Materials	3
Responsible Instructor	Dr. K. Masania	
Instructor	I. Fernandez Villegas	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The red line of the course is aiming at knowledge and understanding of manufacturing processes in relation to material properties and feasible product designs. The course contents include manufacturing processes for metallic and composite parts, the assembly of parts into large (sub)structures, and related topics like Quality control, organisation principles, finances, etc.	
Study Goals	The student should have a good knowledge and understanding of the mainstream manufacturing processes of structural materials (lightweight alloys, composites, hybrids). The student should be able to describe and motivate the processing procedures like the processing steps, required tooling and equipment, of manufacturing processes. The student should be able to select adequate manufacturing processes for designs of lightweight structures and components, and should be able to motivate his/her choices. The student should be able to analyse and synthesize interactions between materials, design and manufacturing processes. The student should be able to identify and explain manufacturing related flaws and inaccuracies and advice on how to prevent/limit those.	
Education Method	- Interactive lecture where contents are presented and where the flipped classroom is often used. - Exercises in class with some additional working material	
Assessment	written exam. Closed book.	

AE4ASM005	Fatigue of Structures & Materials	3
Responsible Instructor	Prof.dr.ing. R.C. Alderliesten	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	- Introduction to Fatigue (fatigue as a phenomenon; stress concentrations; residual stresses; fatigue properties of metallic and composite materials; fatigue strength of notched specimens, residual strength). - Fatigue damage mechanisms (initiation, crack growth, delamination growth, transverse matrix cracking, fibre failure). - Analysis methods (stress concentration factors, stress intensity factors; energy balance approaches, strain energy release rates). - Fatigue loading (Load Spectra, Fatigue under Constant- & Variable-Amplitude Loading). - Special Fatigue Conditions (surface treatments; fretting corrosion; corrosion fatigue; high-temperature and low-temperature fatigue, moisture ingress). - Fatigue and Damage Tolerance of Aircraft Structures: Regulations, tests, scatter, application of fatigue and damage tolerance methods.	
Study Goals	This course provides the students with engineering knowledge and skills to recognize and to analyse fatigue and damage tolerance problems in aircraft structures and materials. After the course the student must be able 1. Interpret and discuss the fatigue fracture features with respect to the characteristics of each phase in fatigue life 2. Define and determine stress concentration factors for notched structures with or without residual stresses 3. Explain and discuss S-N curves with respect to mean stress, material surface effects, and scatter, and perform fatigue life analyses considering mean stress and notch root plasticity 4. Assess the fatigue life of tension and shear joints, and explain limitations to the similarity principles (K,I,T) 5. Explain Linear Elastic Fracture Mechanics concepts for damage growth, and perform crack growth analyses with these concepts 6. Explain the consequences of variable- and constant amplitude loading on fatigue life and damage growth, and perform fatigue life analyses for arbitrary load spectra 7. Explain the effect of environment on fatigue life and fatigue phenomena 8. Perform residual strength analyses.	
Education Method	Selfstudy using the Online Course Material on Brightspace together with weekly plenary instruction sessions and discussions with fellow (online and campus) students or e-moderator on a forum.	
Literature and Study Materials	J. Schijve, Fatigue of Structures and Materials, 2nd edition, 2009, Springer Science+Business Media, B.V., ISBN-13: 978-1-4020-6807-2 e-ISBN-13: 978-1-4020-6808-9.	
Assessment	Readers, formative tests, documentation, and other relevant material provided via Brightspace.	
	Written exam & assignments Written exam at the end of the lecture period, 80% of total grade. Assignments during the lecture period, altogether 20% of the total grade. Deadlines of individual assignments are communicated via Brightspace.	

AE4ASM106	Stability & Analysis of Structures I	3
Responsible Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	This course introduces analytical and numerical techniques for the analysis of representative aerospace structures under mechanical loading, with particular emphasis on determining their stability. The course covers the connection between the principle of minimum potential energy, the variational formulation and the classical (differential) formulation of quasi-static structural problems. Basic structural components are analyzed using beam and plate theory. The basic theory of structural stability is introduced through asymptotic analysis in order to derive the stability (buckling) equations. Buckling equations are formulated and solved for geometrically non-linear elastic beams and plates. The methodology is extended to formulate stability analysis of representative aerospace components. Emphasis is placed on the understanding of the models, their assumptions, and ranges of applicability. The topics covered include: - Introduction to structural analysis - Review of global and local balance principles - Constitutive relations, material symmetries and strain energy - Formulations of structural problems: differential, variational and minimum formulations - Introduction to calculus of variations and approximate solution methods (Ritz and Galerkin) - Applications to beams and plates: solution to basic structural problems - Introduction to buckling: Eigenvalue buckling problem (bifurcation buckling) - Imperfection analysis (snap buckling) and stability analysis - Buckling of non-linear beams/columns and plates - Design methodology for thin-walled aerospace structures	
Study Goals	1- Understand the various formulations of equilibrium and equations of motion 2- Model simple aerospace structural components using engineering theories 3- Apply energy and variational methods for the evaluation of structural responses 4- Understand and being able to obtain the buckling equations for beams and plates 5- Develop insights into buckling phenomena of aerospace thin-walled structures	
Education Method	Lectures.	
Literature and Study Materials	Primary source: course slides provided in Brightspace Recommended references: T.H.G. Megson, Aircraft Structures for Engineering Students, Elsevier, Any edition. C.L. Dym and I.H. Shames, Solid Mechanics: A Variational Approach, Augmented Edition, Springer, available online through the TU Delft Library.	
Assessment	Written (take-home) assignments and oral exam	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	P.P. Sundaramoorthy	
Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of lectures, working sessions, design review sessions and a final symposium. The course structure will be explained in detail in the first weekly module. The time and date of the design review(s) and final symposium will also be communicated in the first weekly module. Students are requested to reserve all scheduled timeslots in their agendas. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. Note that, the course will require group work. Groups will be formed in the fourth week of the course. Only students who pass the individual test are allowed to participate in a group. Students who are not part of a group cannot participate in the remainder of the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. During each module it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Individual test on course content 2. Written feedback on design specification sheet of another group in the course Students with a failing grade for the individual test, i.e. a grade below 5.8, will have a resit opportunity of the individual test in week 4 of the course. This resit will be in the form of a short oral exam. This resit is only available for students who did not pass the individual test. On passing the oral resit exam, students will receive a 5.8 as grade for the individual test. Students who fail the resit opportunity will not be able to participate in the group work, and therefore cannot participate in the remainder of the course. It is mandatory that students are present at the design review(s) and the final symposium presentations of their own group and those of other groups. The individual grade of students who fail to be present will be withheld. These students will not receive a final grade for the course as a result. Students of whom the individual grade which has been withheld, may take a resit. The resit will be in the form of a short oral exam. It will be possible to apply for a resit up to 15 working days after the publication of the grades in Osiris, but no later. A resit exam will then be scheduled by the principle lecturer of the course in consultation with the student in question. Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, Introduction to materials science" or books of similar nature. Be sure to check the entry level quiz before making up your mind about this course.	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, ISS, Hubble telescope, James Webb telescope, Venus orbiters, Launchers) will be presented as a carrier for material selection analysis. A holistic view of "materials science - manufacturing - structural analysis - system engineering for missions" will be presented. While students come mostly from different backgrounds, the goal of this class is to help them identify the perspective they use in the holistic view and to learn about the materials science perspective. Fundamentals of materials classes and their typical properties will be introduced. Current material choice will be analyzed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. In sessions of "ESA forum + student poster seminar", students will be given opportunities to interact with ESA experts and present their own findings from analyzing given missions, by translating desired properties into material structures and microstructures. In class discussion is an important part of learning. The structure of the lectures will be tailored to maximize the student involvement. The course contains the following topics: week 1: Holistic view and materials science perspective; Typical materials - metals, ceramics and polymers week 2: Typical materials - metals, ceramics and polymers week 3: Material degradation in space - vacuum; thermal cycling; week 4: Material degradation in space - AO; charging week 5: Material degradation in space - space radiation week 6: Material degradation in space - meteoroids and orbital debris; week 7: "ESA forum + student poster seminar" Materials testing for space applications; Extreme missions; Materials challenge in future space age (ESA guest speakers); Student poster seminar (Jan 6, 2025) week 8: Study period (Q and A) week 9: exam week (Jan 20. 2025)	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 12 lectures of 2h per week for the first 6 weeks. The last week is a whole-morning session of "ESA guest speaker forum + student poster presentation".	
Literature and Study Materials	A compilation of book chapters, lecture notes and review papers.	
Assessment	Poster presentation (40%) + final exam (60%). The students are to work in groups and do research on prescribed question/topics and deliver a final poster presentation (15% individual score, 25% group score) in week 7 to show their discoveries throughout the course. The final grade will be the weighted average of poster presentation and final exam. Example questions/topics will be made available on Brightspace.	
Enrolment / Application	For students not from "structures for space" profile, there will be a cap of 20 students maximum. Please enroll in brightspace (enrollment period Oct 28 to Nov 10, 2024) as soon as you decide to take the course. Once the maximal number is reached, enrollment will be closed.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project.

The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5722	Thesis Aerospace Structures & Materials	45
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal). If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go'	

(fail). Please refer to Brightspace for more information.

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Structures for Space

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4ASM101TU-22	Polymer Science	4
Responsible Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of a number of lectures covering the most relevant aspects of polymer science and technology distributed in 6 blocks, each covered in one up to three lectures. After this a session on 3 minutes pitches by the students will be organized: -Introduction to polymer science: daily life examples, nomenclature, concept of polymers, initiators, polymer architectures and polymerization media. -Polymerization: condensation and addition polymerization under step-growth; radical, ionic and ring opening polymerization under chain-growth mechanisms, concept of copolymerisation and crosslinking, basic characterization techniques (such as GPC). -Polymers in Aerospace: brief outlook on composites, chemistry of thermosets, high-performance thermosets, high performance thermoplastics, mechanical properties. -Processing: basics of polymer processing and polymer-based structures manufacturing, fiber processing and properties, living materials and their potential. -Sustainable Polymers: a glance on green chemistry, introduction to lignocellulosic biomass, how to develop biobased monomers, recyclability and dynamic materials, examples on sustainable polymers, impacts on sustainability, a perspective on how can you design sustainable polymers? -Polymers for Energy and Environment- demonstrating a concept of photovoltaic devices and batteries, polymers used in these applications, polymer membranes and polymer coagulants for water remediation, polymer ionotronics. -Pitch talks: students to give 3 minutes presentations about developments/applications in the polymer field, convince audience as in business structures.	
Study Goals	By the end of this course, you should be able to: LO1: Identify polymer basics and analysis LO2: Assess polymer structure-property relationship for thermomechanical performance estimation LO3: Judge parameters and requirements for polymers to be qualified for aerospace materials design LO4: Define sustainability concepts in polymer science LO5: Present about recent developments in polymers field	
Education Method	Lectures and lab activity	
Literature and Study Materials	Seymour/Carraher's Polymer Chemistry: 10th Edition. Contains comprehensive information on polymer materials from chemistry perspective Lecture slides will be made available to students as study material	
Assessment	Written final exam and pitch talk (3 minutes) presentation Lab activity attendance is mandatory Transitions AE4ASM101TU -> AE4ASM101TU-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM101TU, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM101TU. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM101TU is not completed in 2022 2023, students will need to complete AE4ASM101TU-22 instead.	

AE4ASM103	Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	In this course we introduce the design fundamentals that control specific properties in coatings, surface treatments and functional coatings and the techniques used to apply them and test them. The goal is therefore to provide the fundamentals of chemistry and physics that explain structure-property relationships relevant to understand and design surface treatments/coatings with: (i) all-round relevant properties such as adhesion, mechanical response, active (inhibitors) and passive (barrier) corrosion protection; and (ii) increasingly relevant functional properties such as antiicing, self-cleaning, self-healing, drag, structural colours, storage/release of active chemicals with nano and microparticles, and antifouling. Since material design rules are directly related to application methods, different technologies for coatings application will be covered such as electrophoretic deposition, powder coatings, anodizing, sol-gel and plasma electrolytic oxidation. Characterization strategies to obtain information about key properties will also be discussed. The fundamentals and characterization strategies covered in the course are applicable to coatings, adhesives and composites of interest in aerospace applications and other engineering domains.	
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.	
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.	
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.	
Assessment	Written report (40% of final grade) and written exam (60% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.	

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr.ir. C.M. de Zeeuw	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Compared to previous years the course starts in Q3 (not Q2). There will be a kick-off lecture in week 3.1	
Course Contents	Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four to five students about 8 day sections with support of staff are needed during this course. These will be planned on fore hand, preferably on fixed time slots. Most likely either Friday morning or Friday afternoon. We will not use all of those slots, but it will provide the flexibility to schedule things in the lab and make it easier to block their schedules. Finishing period As the starting period changed, the finishing period changes according, at the end of the fourth period (Q4, June) it is. The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Estimation of the maximal bending-torsion coupling for a strip of UD composite 4. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 5. Performing mechanical tests 6. Analysing the test results of the mechanical test, including failure analyses 7. Adhesion of thermoset laminates 8. Possibly preparation of lap shear specimens 9. Performing lap shear tests 10. Estimation of the mechanical properties of a sandwich panel 11. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 12. Performing a bending test on a sandwich panel. 13. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	This course was renewed in 2023-2024 by fixing the time schedule for students on beforehand.	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on beforehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM108	Experimental Techniques & NDT	3
Responsible Instructor	Dr. A. Anisimov	
Instructor	Dr. R.M. Groves	
Responsible for assignments	K.P.W. Dissanayake	
Responsible for assignments	Dr. N. Yue	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of electronics, elements of signal processing (analogue, digital), basics of measurements and metrology	
Course Contents	This is a lecture course on developing the critical thinking needed to perform measurements in the Aerospace Structures and Materials Laboratory. This includes an introduction to the most common sensors for experimental mechanics, non-destructive testing and structural health monitoring, evaluation of the performance of these sensors, selection of sensors for different applications and signal processing and control algorithms.	
Study Goals	By the end of the course, you should be able to: 1. Analyse a measurement system for the aerospace sector and to determine if it is fit for a specific application. This learning objective acts as an umbrella objective, covering the whole course. Lower level objectives include that by the end of the course, you should be able to: 2. Explain the principle of operation of the most common sensors and instrumentation in the aerospace structures and materials laboratory. 3. Justify the selection of a sensor for a specific measurement application, based on sensor specifications and measurement principle. 4. Analyse a measurement chain and determine signal accuracy and noise levels. 5. Assess the performance of signal processing and feedback algorithms for their influence on measurement noise and accuracy.	
Education Method	Lectures, obligatory laboratory sessions, reader with self-evaluation questions.	
Literature and Study Materials	Course reader, pre-recorded videos with slides available on Brightspace. Additional material as datasheets, book chapters and papers are uploaded to Brightspace during the course.	
Reader	Course reader available on Brightspace	
Assessment	Open book written exam (30% of grade). Oral exam (40% of grade). Laboratory group assignments (30% of grade).	
Set-up	12 active lectures with problem-solving and discussion (weeks 2.1-2.6). Students have to prepare for these workshops using the material available on Brightspace, including pre-recorded videos. The material becomes available as the course progresses. 2 exam review workshops (week 2.7). It is particularly important to attend these sessions as the course lecturers will explain the structure and what is required in the open-book exam 5 obligatory laboratory assignments done in groups (weeks 2.1-2.5). Each practical is graded with a group grade for the written report (maximum of 4% towards the final course grade) and personal grade for the oral defence of the report (maximum of 2% towards the final course grade).	

AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
Responsible Instructor Responsible Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Expected prior knowledge	C. Kassapoglou Dr. D. Zarouchas 0/0/4/0 3 3 none English Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis 3. Reliability analysis & Health monitoring a. Probability of Failure & Uncertainty quantification b. Structural Health Monitoring 4. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Understand the philosophy of reliability analysis and calculate the probability of failure of composite structures by combining numerical methods and the mechanics of composite structures - Understand the importance of structural health monitoring (SHM) and design a SHM system for a given composite structure - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Classical Laminate Theory: 3 2. Progressive damage analysis: 3 3. Reliability analysis & Health Monitoring: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory, progressive damage analysis, reliability analysis and health monitoring. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013.	
Assessment	Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace. Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 10 points A2. Progressive damage and damage tolerance analysis 20 points A3. Reliability analysis & Health Monitoring. 30 points	

Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method).

B1. Design problem 40 points

The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2,A3 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM110-22	Polymer Composite Manufacturing	4
Responsible Instructor	Dr. J.J.E. Teuwen	
Instructor	Dr.ir. D.M.J. Peeters	
Instructor	Dr.ir. B.C.P. Jongbloed	
Instructor	Dr. B. Caglar	
Instructor	R. Tao	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course explores in depth the manufacturing of polymer composite structures and its underlying physics. By understanding the relationships among physics, part/material quality and the design of the manufacturing process, you will be able to critically assess any given manufacturing process. The course is built around a number of polymer composite manufacturing processes highly relevant to the current and future aerospace industry for the manufacturing of high-performance individual parts (autoclave processing of prepreg, liquid composite moulding, thermoplastic composite processing), for the assembling of complex structures (manufacturing of integrated structures, thermoplastic composite welding) and for end of life and recycling. The main basic physical phenomena governing polymer composites manufacturing, e.g. flow (of polymer and of fibres), void formation, curing, shrinkage and crystallization, are interwoven within the course.	
Study Goals	At the end of this course, the student should be able to: Describe the common manufacturing processes used in aerospace industry in terms of processing steps, tooling and equipment as well as advantages, limitations and applicability. Correlate and analyse the processes and their principles/underlying physics with manufacturability, part quality and manufacturing design. Compare and evaluate the different processes for their suitability of manufacturing common components (aircraft wing, fuselage etc). Recognise manufacturing defects, identify their potential sources and recommend strategies to improve the quality of common manufacturing processes for simple component geometries to reduce scrap rate or to increase yield.	
Education Method	Interactive lectures with numerous in-class exercises. Assignments are used throughout the course to allow the students apply, deepen and test their knowledge on the manufacturing processes discussed during the course.	
Assessment	Oral exam (50% of the final grade) - exam format subject to change depending on amount of students - and mandatory assignments (50% of the final grade). A minimum grade of 5.0 out of 10 on both the assignment and the oral exam need be obtained to be able to pass the course.	

AE4ASM503	Sheet Metal Forming	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The lecture consists of two main parts: a theoretical part and an application part The theoretical part includes: - Introduction, formability, deformations, formability testing. - Stress- and strain states, forming limit curves, yield criteria, anisotropy, heat treatments. - Theory of plasticity The application part includes a number of forming processes, like - Simple forming processes like bending and stretching. - Several forming processes for more complex parts like deepdrawing, rubber forming, hydroforming and matched die forming. - Non-conventional processes like explosive forming and superplastic forming - The use of numerical tools for simulation of forming processes, and some other related topics.	
Course Contents	The theoretical part of the lecture will focus on the descriptions of deformations, strains, stresses, etc. In addition, the interaction between stresses and strains and their limits are expressed in a.o. yield criteria and forming limit diagrams. Such concepts are also used in the second, application part to describe and explain the do's and don't's of the forming processes.	
Study Goals	After attending this course the student should be able to: - describe and explain the basic phenomena in plastic deformation of sheet metal - understand the interactions between multiple phenomena and how to exploit these - be able to relate the formability of materials to forming processes - have understanding and ability to combine the features of material behaviour, the forming process and the shape of the component.	
Education Method	Lecture, including discussions about real parts, pictures and videos.	
Literature and Study Materials	Hand-outs are the core material A list of recommended literature will be given during the first lecture (background information unless specified otherwise)	
Assessment	Written exam. Closed book.	

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	N. Eleftheroglou	
Responsible Instructor	Dr.ir. J.A. Pascoe	
Instructor	Prof.dr.ing. R.C. Alderliesten	
Instructor	Ir. J. Sinke	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). Regulations applicable to maintenance. - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals	At the end of this course, you should be able to: - Describe different maintenance strategies and select the appropriate one for a given problem, taking into account different aspects (e.g. technical, regulations, planning); - Describe the working principles of different non-destructive inspection techniques, their pros and cons and their applicability to different materials; - Explain the relationship between certification and maintenance requirements and (documentation of) procedures; - Develop a maintenance programme for a simple case; - Select an appropriate design philosophy to ensure structural integrity of a given part throughout the aircraft lifetime.	
Education Method	Lectures.	
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam, closed book (60% of final grade) and group assignment (40% of final grade).	

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	lectures and practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Adisson-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of aerospace structures and structural components. The equation of motion for the dynamic response is formulated in matrix form using finite element discretization or direct modelling discretization (rigid bodies and flexible connectors). Various solution techniques are used for solving transient dynamic response and vibration problems, including modal analysis. The course also covers experimental techniques to obtain model parameters and test the response of a physical system. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE4ASM514TU		Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	none		
Course Language	English		
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.		
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.		
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.		
Education Method	Lectures.		
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.		
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.		

AE4ASM515-22		Characterization of materials and components	4
Responsible Instructor	Dr. B. Kumru		
Co-responsible for assignments	Dr.ir. C.M. de Zeeuw		
Contact Hours / Week x/x/x/x	0/2/0/0		
Education Period	2		
Start Education	2		
Exam Period	2 3		
Course Language	English		
Expected prior knowledge	General knowledge of materials engineering and science (in particular polymers and metals) at the level expected for a BSc graduate in a (mechanical) engineering discipline.		
Course Contents	This course is an introduction to the most common characterization tools to gather crucial structural and property information of materials and components. The focus is on knowing about several possible characterisation methods and their underlying principles of measurement as well as the strategies to extract the most relevant information from the tests for the purpose in mind. The characterisation techniques will be taught by various colleagues in each field. The methods to covered in this course are classified as follows: 1. Imaging of material structures and surfaces (microscopy, confocal, SEM, EDS) 2. Chemical analysis of polymers (FTIR, Raman spectroscopy, contact angle) 3. Structural characterisation 4. Thermomechanical analysis of polymers (DSC, TGA, DMA/Rheology) 5. Mechanical characterization		
Study Goals	Aim of the course is to provide (aerospace, mechanical, maritime) engineering students with adequate skills and knowledge such that later in their technical careers they can handle materials performance issue in which it is important to clarify the material characteristics such as to link actual material performance to its microstructure. Such knowledge and skills are important for determination of material in certain applications and while choosing engineering materials with desired properties. To this aim: - Students will get familiar with the underlying principles of measurement of principal materials characterisation techniques. - Students will be able to analyse and interpret data from each characterisation technique. - Students will be able to select the right characterisation techniques to correlate structure and property relationships over a wide range of engineering materials.		
Education Method	Lectures and hands on practices		
Assessment	Oral Exam and assignment Transition AE4ASM515 -> AE4ASM515-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM515, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM515. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM515 is not completed in 2022 2023, students will need to complete AE4ASM515-22 instead.		

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	The grade for this course will be a Pass/Fail grade. This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Alternatively, having earned a Bachelor degree in aerospace engineering will give access to the course as well. Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December - Week 4.8: 15 May - Week 1.6: 30 September	

Set-up	The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and 3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3
	There is an intentional overlap so that continuity and handover is guaranteed

AE4ASM520	Industrial Composite Manufacturing	3
Responsible Instructor	Prof. C.A. Dransfeld	
Responsible Instructor	Dr. D. Stefaniak	
Contact Hours / Week x/x/x/x	0/0/4/0 COURSE NOT OFFERED IN 2024-2025	
	Students that have participated in this course and have not yet passed the course, please contact C.A.Dransfeld@tudelft.nl	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Expected BSc diploma	
Course Contents	The course deepens the understanding on different composite manufacturing technologies for aerostructures used at the industrial scale and addresses on specific challenges evoked especially by automation and large part size. Special pitfalls with respect material selection, mould design, process induced part distortion and vacuum setup are discussed to apply the basic understanding of the manufacturing processes to several specific industrial challenges. Finally, these manufacturing processes are also evaluated with respect their sustainability and are investigated in depth towards their energy, materials and auxiliaries consumption in context of the product life cycle.	
Study Goals	At the end of this course, the student should be able to: - Transfer the basic principles of composite manufacturing processes to real and more complex industrial examples - To gain insight in material selection and its consequences in an industrial context - To recognize manufacturing processes and the linked pitfalls based on the resulting part or the utilized mold - Understand the challenges in sizing composite manufacturing systems for large size components - Understand the importance and dependencies of performance assessment indicators in composite manufacturing like robustness, energy consumption, occupation time, non-operating costs and flexibility.	
Education Method	Interactive lectures based on industrial case studies. Some exercises are accompanied by Python Scripts used in Jupyter Notebooks to deepen the understanding and facilitate interactivity. The required software is provided as a browser-based version and previous knowledge in programming is not compulsory. It is intended to visit the Center for Lightweight Production Technology at the German Aerospace Center (DLR) and Airbus site in Stade, Germany.	
Computer Use	yes	
Course Relations	The course is given by a guest lecturer from DLR (German Aerospace Center)	
Literature and Study Materials	will be defined at the beginning of the course	
Prerequisites	AE4ASM001, AE4ASM004, AE4ASM110	
Assessment	Oral exam (50% of the final grade) and mandatory assignment (50% of the final grade).	
Enrolment / Application		
Remarks	Students that need to retake AE4ASM520 can contact the responsible lecturer for a resit.	
Set-up	2 weeks with 2day-blocks and two weeks with one 1day-blocks (e.g. 3.1, 3.3, 3.6, 3.8); each day will consist of four lecture hours	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr.ir. J.A. Pascoe	
Instructor	Prof. C.A. Dransfeld	
Instructor	Dr. J. Wu	
Instructor	V. Yaghoubi Nasrabadi	
Instructor	P.G. Marchese	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE4ASM522	Applied Aircraft Aeroelasticity	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	X. Wang	
Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE4ASM506 (mandatory)	
Course Contents	This course is a follow-up course of Aeroelasticity AE4ASM506. Taking and passing the Aeroelasticity course AE4ASM506 is mandatory to pass this course. This course builds on the basics and regular aeroelastic phenomena that were introduced in AE4ASM506. The extension of the knowledge is twofold: 1. Introduction of nonlinear aeroelasticity 2. Introduction to control of flexible aircraft 3. Introduction to potential flow aerodynamics and its coupling to structural FE models to form closely coupled aeroelastic models The course aims at placing the aeroelastic knowledge into a flexible aircraft context, and it will provide the student with new insights into the aircraft design process and aircraft design choices. Finally, understanding the contents of this course will prepare the student for the design of future aircraft configurations beyond the currently established aircraft concepts.	
Study Goals	At the end of the course, the student should be able to: 1. Know advanced nonlinear aeroelastic phenomena due to structural and aerodynamic nonlinearities 2. How to approach aeroelastic control of flexible aircraft 3. Know and apply the basics of potential flow theory as it is the prevalent aerodynamic theory used in aeroelastic modeling 4. Understand the difference between closely and loosely coupled aeroelastic systems. 5. Know how to couple panel method aerodynamics with FEM to form closely coupled aeroelastic system	
Education Method	The course will be taught in a classroom and hands-on sessions, circumstances permitting.	
Books	Recommended: 1. Introduction to nonlinear aeroelasticity. Dimitriadis, Grigorios. John Wiley & Sons, 2017. 2. Introduction to structural dynamics and aeroelasticity. Hodges, Dewey H., and G. Alvin Pierce. Cambridge University Press, 2011. 3. Aeroelasticity. Bisplinghoff, Raymond L., Holt Ashley, and Robert L. Halfman.&nbsp;Courier Corporation, 2013. 4. Etkin B. Dynamics of Atmospheric flight. Courier Corporation; 2012.&nbsp; 5. Introduction to aircraft aeroelasticity and loads. Wright, Jan Robert, and Jonathan Edward Cooper. John Wiley & Sons, 2008.	
Reader	Course notes	
Assessment	Take home assignments with oral close-out session.	

AE4ASM526	Spacecraft Thermal Design	3
Responsible Instructor	I. Uriol Balbin	
Instructor	C. Falsetti	
Instructor	N. van der Pas	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of orbital mechanics Understanding of general design spacecraft systems	
Course Contents	General knowledge of space thermal control hardware, system design and analysis processes are presented. Using the principles of heat transfer, and more specific of radiation and conduction, the methods and tools for thermal design of a spacecraft are presented. Numerical analysis will also be introduced via the software ESATAN-TMS.	
Study Goals	The student is able to generate a thermal design of a satellite using heat transfer analysis and thermal engineering techniques. In more detail the student will be able to: - Understand the spacecraft thermal environment - Define the thermal subsystem and compare the different hardware options. - Understand the design cycle of the thermal subsystem. - Apply the concepts of heat transfer (radiation, conduction), heat balance and heat flows - Generate and use analytical thermal analysis models - Generate and use numerical thermal analysis models	
Education Method	Lectures + homework Students are given access to a discussion board provided with the course in Brightspace. The discussion board will be used as the main communication tool to ask questions on the material taught or in preparation of an assignment. Answers to the questions may be given by fellow students and/or the lecturer(s).	
Computer Use	Software licenses are available during the lecture period Software can be used in the computer room or in the personal computer (Windows/Linux)	
Literature and Study Materials	Lecture notes available on Brightspace.	
Assessment	The final grade of this course is based on 2 homework assignments (60%) and one final exam (40%). A minimum grade of 5 is required in each part.	
Enrolment / Application	This class has a limit of 80 students, due to the limit on the software licenses. Please enroll in Brightspace before 4th of March 2025 if you intend to take this class.	

AE4S10	Microsat Engineering	4
Responsible Instructor	Dr. J. Guo	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Lectures from experts on various aspects of small satellites and group/individual projects for practicing.	
Course Contents	The course covers the engineering of small satellites. The concept and definition of micro-, nano-, pico-, and femto-satellites is introduced along with the past, present, and future of small satellites in general. The course comprises seven to eight lectures, partly from professionals in the space industry. The lectures are intended to familiarise the students with current industry trends and state-of the art research in small satellite engineering. The lectures vary each year and potential lecture topics include communications, AOCS, onboard computer, distributed space systems, miniaturised payloads, small satellite missions and applications, etc.	
Study Goals	The course will enable students to design and engineer small satellite missions. The high-level learning objectives of the course are: The students shall be able to - explain key aspects of small satellite missions - explain and describe elements in the architecture of a spacecraft mission - discuss in detail at least one element in the architecture of a spacecraft mission - present and defend their design of an element in the architecture of a spacecraft mission - show familiarity with current trends in space industry and small satellite research	
Education Method	Lectures (Provided partly by professionals from the space industry). Group work (Students work on assignment in groups; Coaching will be provided to guide the students). Individual work (Students work on small individual assignments).	
Literature and Study Materials	Optional: Wertz J.R., Larson W.J. (eds.); Space Mission Analysis and Design; 3rd edition Kluwer Academic Publishers (2003). Mandatory: Lecture Notes AE4-S10 MicroSatellite Engineering.	
Prerequisites	General mathematical, physical and engineering knowledge from completed BSc study. In particular, AE1101 + AE1102 + AE3201, or similar	
Assessment	There will be two assessment components A Group Assignment and an individual assignment. Group assignment The final demonstration or report will be graded on the following aspects: - Planning - Presentation, argumentation and defence of the design - Functionality and Usability of the deliverable - Context of Small satellite (How well has the student team incorporated current small satellite research and industry trends/needs in the design) - Independence - Creativity Individual assignment This assignment will be graded on the following aspects: - Understanding and summary - Critical review of articles in a specified technical field of small satellites Please note: there is no written exam and, therefore, also no re-sit exam for this course.	
Permitted Materials during Tests	N/A	
Enrolment / Application	Students have to enroll via the BrightSpace site of the course. Due to the limited capacity, this course is open to Space Engineering profile students only.	
Set-up	The course comprises lectures, guest lectures, team sessions, group work and individual work.	

Track 5: Flight Performance and Propulsion	
MSc Coordinator	Dr. F. Oliviero
Introduction 1	The main goal of the Flight Performance and Propulsion Master track is to contribute to the improvement of flight performance and propulsion. To achieve that we focus on innovative aircraft configurations and novel propulsion concepts. Therefore, aircraft -engine integration is also an important topic of research. These research areas are highly relevant if we want to contribute to the ambitious targets to reduce the environmental impact of aviation, as proposed by the Advisory Council for Aviation Research and Innovation Research in Europe (ACARE). Our mission is to advance the design of innovative aircraft configurations and propulsion concepts, by exploration of: 1. New technologies and unconventional airplane configurations, 2. Advanced flight physics modeling to improve the prediction and simulation of air-vehicle performance, 3. New methods and tools to improve the quality and effectiveness of the design process. The staff members of the Flight Performance and Propulsion Master track have a range of different expertise, varying from air-breathing propulsion systems, flight mechanics, aircraft design and integration, combustion, thermodynamics, systems engineering and multidisciplinary design optimization. The knowledge and competences within the Flight Performance and Propulsion section are therefore grouped in three key areas: 1. Propulsion & Power 2. Flight Mechanics 3. Aircraft Design & Design Methodologies The FPP Master Track offers an educational program consisting of two profiles and a shared set of core courses, relevant to both profiles. FPP students can choose one of the two profiles, according to their interests and development goals. Profile I (FP) is focused on flight performance and includes mandatory courses in the areas of aircraft aerodynamic design, experimental simulations, aeroelastics and digital design methodologies. Profile II (PP) offers a different set of compulsory courses dedicated to propulsion & power, such as turbomachinery, internal flows, combustion, heat transfer and modeling and simulation methods for propulsion and power. The common set of courses provides knowledge that is relevant to both profiles and includes subjects such as aircraft aerodynamics, CFD, aero engine technology and advanced design methods, such as multidisciplinary design optimization (MDO) Next to the track and profile courses, students can choose among a large number of elective courses, in consultation with the master track coordinator, such to develop in other relevant scientific and technological domains, as well as to acquire or reinforce skills for personal and professional development. The competence acquired during the first year of the Master program is put to fruition in the final year during the research period dedicated to the thesis. During the Master thesis, the student is engaged in a project most often within one of the active research programs. The project is carried out under the supervision of one of the academic staff members, but often the student also benefits from the closer supervision of one of the PhD candidates involved in the same project. Thesis projects in cooperation with industry and research institutes are also very common.

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile I: Flight Performance

Program Coordinator	Dr. F. Oliviero
Introduction 1	Education: The profile program consists of courses that deal with aerodynamic design of high subsonic transport aircraft, aeroelasticity, experimental simulations and digital design automation (KBE). Some of the courses have a more theoretical approach and aim to provide a fundamental understanding of the governing physics whereas other courses are more applied (e.g. include experimental sessions in the wind tunnel), others require good programming skills. Elective courses are grouped in three areas, Flight mechanics, Propulsion integration and Aircraft Design & Design Methodologies. A selection of elective courses is made in consultation with the profile coordinator. The competence acquired during the first year of the Master program is put to fruition in the final year during the research period dedicated to the thesis. During the Master thesis, the student is engaged in a project most often within one of the active research programs. The project is carried out under the supervision of one of the academic staff members, but often the student also benefits from the closer supervision of one of the PhD candidates involved in the same project. Below, an overview of the research activities is provided. Research: We focus our research on advanced and innovative aircraft configurations, novel propulsion concepts and aircraft engine integration. Our mission is to advance the design of innovative aircraft configurations and propulsion concepts, by exploration of new technologies to obtain novel or improved solutions, advances in flight physics to improve the prediction and simulation of air-vehicle performance, and new methods and tools to improve the quality and effectiveness of the design process. The knowledge and competences are therefore clustered in three key areas: aircraft design & design methodologies, flight mechanics, and propulsion integration. Aircraft design & design methodologies Major technological advances in flight performance can be realized by new unconventional airplane designs. To support the development of such novel configurations, advanced design methodologies (e.g. Multidisciplinary Design Optimization and Knowledge Based Engineering) are required that go beyond statistical and analytical methods. To that extent, design support tools are developed that are able to sustain the evolutionary improvement of current aircraft design, as well as to support the investigation of novel aircraft configurations and assess the impact of new technologies such as hybrid-electric propulsion, morphing structures or flow control. Flight mechanics It is our objective to focus on two specific aspects of flight mechanics and thereby to advance the current state of the art: - Investigation of the inherent flying qualities of novel fixed wing aircraft configurations and scaled models to enable scaled flight testing - Automatic generation of flight mechanics simulation models for application in multi-disciplinary design optimization frameworks Propulsion integration Apart from the optimisation of isolated aircraft engines, the efficient integration of engines to the airframe is also of crucial importance to maximise aircraft performance and minimise noise emissions. One major ongoing project within this domain focusses on the aerodynamic and acoustic performance of propellers integrated on the airframe. Apart from the analysis and optimization of the isolated propeller system, also the interactions between the airframe and the propeller need to be considered. The goal of the investigations is to reduce the performance and noise penalties associated with these airframe interaction effects. Students involved in the project perform experiments in the faculty's various wind tunnel facilities using powered propeller test rigs and advanced measurement techniques, while also CFD tools are applied for numerical evaluations.

AE4115-23	Experimental Simulations	4
Responsible Instructor	Dr.ir. T. Sinnige	
Instructor	Dr.ir. W.J. Baars	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling; fractional analysis; dimensional analysis (Buckingham-Pi theorem); rigorous methods of obtaining dimensionless parameters; 3 Fundamentals of aerodynamic similarity: fundamentals of aerodynamics necessary for the interpretation of similarity between aerodynamic test results and full-scale; 4 Mach number and Reynolds number effects: practical considerations with respect to implications of dissimilarities between model-scale testing and full-scale applications; 5 Wind tunnel facilities: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 6 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 7 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 8 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 9 Design of experiments: factorial designs, response-surface models; 10 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report; 11 Engine integration: engine simulators, performance of those and their similarity with full-scale engines; 12 Acoustic testing: noise generation mechanisms and their appropriate scaling; definition of sound pressure levels; measurement requirements and devices; far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report;	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stroemungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports (Q2 + Q3) and oral exam (wk 3.10 or 4.1).	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter.	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Prof.dr.ir. L.L.M. Veldhuis	
Instructor	L.T. Lima Pereira	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130-22, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during the first 3 weeks of Q1).	
Assessment	The assessment is done through 5 assignments that are performed on a personal basis. The assignments are to be combined in 1 report that should be delivered after the finalization of the lecture at the end of Q2. Additional information on the timing and contents of the assignments will be presented on BrightSpace.	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4204	Knowledge Based Engineering	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ing. A. Heidebrecht	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some familiarity with Object Oriented modelling and (Python) programming	
Course Contents	The course will provide a theoretical and practical introduction to Knowledge Based Engineering (KBE) technology, as a means to address complex engineering problems through design automation/digitalization principles. Other addressed topics, strictly related to KBE include - UML/SysML modelling language (class and activity diagrams) - object oriented modelling paradigm	
Study Goals	The main goals are the following: - understanding of Knowledge Based Engineering theoretical foundations and programming principles and their application to support complex engineering design problems - hands-on experience in developing simple but representative KBE applications by means of a provided KBE system - ability to model product and process knowledge by means of UML/SysML	
Education Method	The course is organized into a set of interactive lectures and tutorials to be performed part in class and part at independently at home. During the tutorials, students are introduced to the KBE programming approach. After a quick refresh of the python language, the main functionalities of a commercial KBE system will be illustrated, through interactive programming sessions. A significant amount of time is left for self-study, where students will have to develop a KBE application in teams of two, to address an engineering design problem of their choice. Attendance to lecture and tutorials is not compulsory but very highly recommended. Although the course is spread over two periods, the second one (Q4) is reserved for self-study (i.e. the development of the KBE application) and the oral examinations (See assessment section)	
Computer Use	The use of a personal computer is required, with the Windows OS installed A selected KBE system will be used for the practical sessions and assignment (temporary licenses for this course will be provided)	
Course Relations	This course has a logical connection to the MDO course AE4205, being KBE an enabling technology to multidisciplinary design, analysis and optimization. However, the MDO course is not propaedeutic to this course, thus students who did not followed MDO can take the KBE course without any problem	
Literature and Study Materials	- lecture sheets and some lecture notes (only available digitally on Brightspace) - a few scientific papers - several programming tutorials and code samples	
Books	Suggested (not compulsory) book: Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 Please note that only Chapter 9 is required for this course (single chapters can be purchased on line at Wiley). The rest of the book is suggested literature for the MDO course AE4205 The book is available as e-book through the TU Delft library	
Assessment	There is no conventional written exam for this course. The assessment is based on the quality of a KBE app to be developed at home by teams of two students and demonstrated live during an oral examination session. Two examination sessions are planned per academic year. The KBE app is supposed to address a relevant engineering design problem, which will be proposed by the student teams in the early phase of the course. The proposed KBE assignment descriptions will require lecturer's approval. If during the oral examination the quality of the developed KBE application and its demonstration are evaluated to be unsatisfactory, the team will have to use the retake session. For the retake session, two options are offered: 1) submit and present an improved version of the app proposed for the nominal session. One penalty point applies. 2) submit and present a totally different app than proposed for the nominal session. No penalty points. The new KBE assignment proposal will have to be delivered after the nominal session and will require lecture's approval. The oral examinations (nominal and retake sessions) are taking place around the beginning and the end of the second period of the course (i.e. 4th quarter of the academic year), not necessarily during the standard exam weeks. Delivery dates and examinations days are announced at the beginning of the course	
Special Information	Temporary student licenses of ParaPy, the KBE system used in this course, will be provided only to the students actively participating to the course. Thus, not to guests or other participants enrolled via Brightspace. The licenses shall be used exclusively for the course assignment.	
Tags	Artificial intelligence Computer Engineering Group work Integrated Modelling Programming Programming concepts Programming Software Software Engineering	

AE4205	MDO for Aerospace Applications	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ir. M.F.M. Hoogreef	
Instructor	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Familiarity with MATLAB and basic aircraft design knowledge. MATLAB is used for tutorials and home assignment, thus students without MATLAB experience must gain this through self-study Students with no experience in aircraft design should consult textbooks like Torenbeek, Raymer and Roskam to familiarize with the fundamentals of wing sizing process and general concepts such as wing loading, aerodynamic coefficients, fuel fraction method, drag and its components, etc.	
Course Contents	The course will introduce students to multidisciplinary design optimization (MDO), as a modern methodology to support complex product development. Main topics: - What and why MDO - Parameterization techniques - Numerical optimisation methods (focus on gradient based methods) - Decomposition and coordination of complex computational systems - the eXtended Design Structure Matrix (XDSM) - MDO architectures - Intro to surrogate modeling, discrete&mixed integer problems, genetic algorithms and Multi Objective Optimisation	
Study Goals	The main goals are the following: - understanding of Multidisciplinary Design Optimization (MDO) methods and architectures - hands-on experience in setting up and solving MDO problems	
Education Method	The course consists of classical lectures and computer tutorials using own laptop. During the tutorials, students are introduced to some of the tools discussed during lecture and can make hands-on experience during interactive sessions with the teachers. In particular, students will be introduced to the MATLAB optimization function fmincon and its use for gradient based optimization. A significant amount of time is left for self-study, where students will have to work in pairs on one take-home assignment.	
Computer Use	The use of a personal computer is required MATLAB (including its optimization toolbox) will be used for the MDO practical sessions and assignment. Student licenses are provided by the TUD	
Literature and Study Materials	- lecture sheets and some notes (only available digitally on Brightspace) - scientific papers - some software modules - Suggested (not compulsory) book: - Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 - The book is available in e-format from the TUD library (Chapter 9 is used in the course AE4204 on Knowledge Based Engineering) - Engineering Design Optimization, by J.R.R.A. Martins and A. Ning (free on line)	
Assessment	There is no conventional exam for this course. The assessment is based on the quality of one take-home assignment, which will be executed in teams of two students. The assignment is about setting up and solving an MDO problem using Matlab. The deliverables are the Matlab code and a report. Two different take-home assignments are defined for the two yearly examination sessions.	
Tags	Algorithmics Analysis Matlab Modelling Optimalisation Programming Project	

AE4238	Aero Engine Technology	4
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	Dr. F. Yin	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery", "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Inlets and Exhaust - o Turbomachinery - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls - o New Engine /propulsion Concepts	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an flipped course in which the lectures are pre-recorded and the students are supposed to watch the lectures before coming to the classroom. The classroom interaction is mainly to have discussions and for tutorials. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13. Evaluate the calorific values for various fuels 14. Understand the basics of combustion and emission formation in Aero engines 15. Understand the working of gas turbine combustor 16. Understand the basic principles of gas turbine controls and evaluate different strategies for control 17. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a flipped teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

Reader	1. Gas Turbine Theory by H.I.H.Saravnamuttoo, G.F.C. Rogers, and H. Cohen 2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Prerequisites	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Assessment	The assessment is based on 2-3 assignments given during the course and a final exam. All assignments together will form 30% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
Permitted Materials during Tests	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4240	Advanced Aircraft Design I	4
Responsible Instructor	Dr.ir. R. Vos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The design of high-subsonic transport aircraft is the topic of this course. The main goal is to demonstrate how the geometry of the aircraft can be shaped to ensure favorable flow conditions that satisfy requirements on performance, stability and controllability. The course is divided in the following lectures: 1. Design objectives 2. Boundary-layer aerodynamics and subsonic cruise drag. 3. Relation between supersonic velocity and pressure coefficient. Relation between geometry and pressure distribution. 4. Interference drag and area ruling. 5. Pressure distribution on airfoil sections at high and low speed. Development of the transonic airfoil. Design of supercritical and sonic airfoils. 6. Reynolds number effects. 7. Low-speed stalling characteristics of airfoil sections. Reynolds number and Mach number effects on maximum lift coefficient. High-speed stall (buffet) on airfoil sections. 8. Airfoils equipped with high lift devices: mutual interaction, types of flow separation. 9. Development of the swept-wing concept: historical background, principle of wing sweep, transonic flow characteristics, effect on boundary layer, effect on maximum lift coefficient, fences. 10. First generation of swept wing aircraft: tip-stall, aeroelastic deformation. 11. Pressure distribution on finite swept wings: spanwise drag distribution, forward sweep, fuselage interference, improving velocity distribution at root and tip, minimizing pitching moment. Design considerations for pressure distributions on finite swept wings. 12. Stalling characteristics of full scale aircraft: stall requirements, complexity vs. maximum lift, spanwise stall onset, deep stall. 13. Horizontal tail surface design: functions and requirements, planform shape, aeroelastic effects, elevator lock. 14. The buffet onset boundary: requirements, effect on flight envelope. Flight characteristics between MMO and MD: requirements, Mach tuck, Mach trim compensation. 15. Vertical tail surface design: functions and requirements, planform shape, effect of dorsal fin. Control surface design: functions and requirements, linearity, aeroelasticity, stall. 16. Wing control surfaces: spoilers and ailerons 17. Propeller slipstream effects: effect on spanwise lift distribution, effect on longitudinal balance, effect on directional balance in OEI condition. 18. Engine intake design: lip design for take-off and cruise conditions, drag in OEI condition. 19. Exhaust design: effect of ambient pressure, boat tail drag. Thrust reversers: types, interference with empennage. 20. Examples of modern wing design	
Course Contents	In this course it is demonstrated how the exterior design of high-subsonic transport aircraft can be manipulated in order to meet specified performance objectives and satisfy relevant certification constraints. It is shown what geometric characteristics of the airplane allow for a high operating Mach number while keeping fuel burn at acceptable levels. In particular, it is demonstrated how the exterior of all major airplane components can be shaped such as to adequately handle low-subsonic and transonic flow. It is demonstrated how the aerodynamic design and the operation at low speed and high speed affects the airplane in terms of balance, stability, aeroelastic behavior, weight, controllability, cruise performance, and field performance. Through many historic and contemporary examples the student learns to relate the functionality of high-subsonic transport airplanes to their external design.	
Study Goals	Students are able to: 1. Recognize the aerodynamic phenomena occurring around external shapes at a variety of conditions such as Mach number, Reynolds numbers or angles of attack. 2. Derive how these phenomena can be affected by the detailed external shape and interposition of various airplane components such as wing, fuselage, tail, nacelle, pylon, and fuselage. 3. Show how wing and tail movables can be applied to locally affect the flow at low subsonic and/or high-subsonic conditions. 4. Explain the effect of the aerodynamic characteristics on the airplanes cruise performance, field performance, stability, balance, and controllability. 5. Identify the advantages and disadvantages of various external design features on the airplanes aerodynamic performance and cross-disciplinary characteristics such as aeroelasticity, weight, or practical limitations resulting from the operational use of the aircraft.	
Education Method	Lectures	
Literature and Study Materials	Mandatory textbooks: - Aerodynamic Design of Transport Aircraft, Obert, E, IOS Press, 2009 - Introduction to Transonic Aerodynamics, Vos, R. and Farokhi, S, Springer, 2015	
Prerequisites	Aerospace Design and Systems Engineering Elements I Aerospace Design and Systems Engineering Elements II Aerodynamics I (subsonic aerodynamics) Aerodynamics II (supersonic aerodynamics) Systems Engineering and Aerospace Design Flight Dynamics and Simulation	
Assessment	Written exam	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Adisson-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project. The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5222	Thesis Flight Performance & Propulsion	45
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal). If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go'	

(fail). Please refer to Brightspace for more information.

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Flight Performance

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4120	Viscous Flows	3
Responsible Instructor	M. Kotsonis	
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Instructor	Dr.ir. T. Michelis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Responsible Instructor	W. Yu	
Gast Instructor	Dr.ir. T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic aerodynamics, level BSc Aerospace Engineering.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade elementmomentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsens Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4136-24	CFD 2: Discretisation Techniques	5
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results. After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignement and oral examination.	
Enrolment / Application	Brightspace	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.																																						

	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Responsible Instructor	Dr. S.J. Hulshoff	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4140	Gas Dynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 none	
Course Language	English	
Expected prior knowledge	AE2130-III Compressible aerodynamics (chapter 7 - 12 of Fundamentals of Aerodynamics by J.D. Anderson) WI3150TU Partial differential equations	
Course Contents	Knowledge of computer programming (Python, Matlab or similar) in order to successfully complete the tasks - How to apply the basic laws of mechanics and thermodynamics to describe compressible flows. - How can we properly describe a compressible flow field using the Euler equation + jump relations. - Understand the concept of characteristics and invariants in the context of linear and non-linear flows. - How to apply characteristics for non-linear flows for the computation of isentropic unsteady flows. - Understand the relation between shock waves and characteristics. - Application of Hugoniot and Poisson curves to solve a Riemann problem. - Definition of characteristics for 2D steady flows, and similarity with 1D unsteady flows (time-like and space-like). - How can use the method of characteristics and method of waves to compute a 2D compressible flow field. - Investigate the effect of viscosity and heat transfer in a 1D flow (Fanno and Rayleigh flow).	
Study Goals	At the end of this course, the student will be able to: - Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. - Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow and discuss the terms. - Derive the jump relations for the Euler equations and describe their relation to shock waves. - Discuss the role of entropy in combination with the jump relation for a correct description of a flow field. - Explain the concept of characteristics and invariants for 1D unsteady and 2D steady flows and how to use them for flow field computations (MOC). - Understand the role of characteristics in shock wave formation, elaborate on the theory of simple waves. - Derive the equations governing 1D flows through channels and nozzles in presence of viscosity and heat transfer. Explain the physical phenomena and processes that occur.	
Education Method	Lectures and self study	
Literature and Study Materials	Reader (can be downloaded from Brightspace) Further recommended literature M.J. Zucrow, J.D.Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. John D. Anderson jr., Modern Compressible Flow with Historical Perspective 2nd edition MacGraw-Hill 1990; H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001	
Prerequisites	Compressible aerodynamics Fundamentals of Aerodynamics by J.D. Anderson ch 7 - 10	
Assessment	Take home tasks + oral examination	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	M. Li	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Introduction to measurement techniques 2. Flow visualization (classroom assignment) 3. Pressure measurements 4. Hot Wire Anemometry (lab exercise) 5. Laser Doppler Velocimetry 6. Particle Image Velocimetry (classroom assignment+lab exercise) 7. InfraRed Thermography 8. Schlieren, Shadowgraphy and Background Oriented Schlieren 9. Fundamentals of data processing	

AE4206-24	Turbomachinery	4
Responsible Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The course covers fundamental concepts of internal flows in turbomachines and related design aspects. The typical course schedule consists of the following lectures: Lecture 0: Introduction course overview: topics, structure and assessment Lecture 1: Recall of Turbomachinery Fundamentals & Principles integral governing equations with applications, relative motion, thermodynamic transformations in turbomachinery stages, velocity triangles with practical examples; Lecture 2: Unsteady Flows in Turbomachinery concept of reduced frequency with applications; work exchange in turbomachinery rows; Lecture 3: Similarity in Turbomachinery similarity analysis for incompressible flows, Balje and Smith diagrams, accounting for compressibility, stage dimensionless performance parameters; Lecture 4: Fluid-dynamic Design of Turbomachinery design procedure, mean-line design model, radial equilibrium model, CFD-based models Lecture 5-6: Axial and Radial Turbines examples of axial and radial turbines, typical turbine stages, blade profiles for turbines, flow characteristics in subsonic, transonic, and supersonic cascades, basic concepts of diffusers, aerodynamics of radial impellers Lecture 7-8: Axial and Radial Compressors examples of axial and radial compressors, typical compressor stages, blade profiles for compressors, flow characteristics and unique incidence, stall and surge instability, slip factor, diffusers for radial compressors, design maps Seminar(s): topic(s) may vary but usually include turbomachinery aero-mechanics, shape optimisation with CFD methods, rocket engine turbo-pumps, and advanced turbomachinery components for aero-engines Tutorial: training session on the software tools provided for the team project	
Course Contents	This course educates aerospace students to apply theoretical knowledge in the field of fluid- mechanics and thermodynamics for analyzing and designing a thermal turbomachinery. Typical examples are: axial and radial compressors of gas turbine engines, industrial gas and steam turbines, radial turbines of automotive turbochargers, turbo-expanders and compressors used in refrigeration and liquefaction industry, micro-turbines for power generation. Since the turbomachinery field is intrinsically interdisciplinary, it is assumed that students already possess basic knowledge of thermodynamics, fluid-mechanics, applied mathematics, propulsion systems, internal flows, acquired in previous courses. Prerequisite of the course are therefore bachelor and master courses addressing fundamental (e.g. Thermodynamics, Internal Flows) and applied aspects (e.g. Aero-Engines) of air-breathing engines. Starting from basic concepts of gas-dynamics and fluid-machinery, the course brings the student to the level that he is able to select a turbomachinery for a specific application, perform the conceptual design of the stages, and eventually predict the flow characteristics within the blade passages by means of conceptual physical models and more sophisticated simulation tools.	
Study Goals	After the course the student will be able to: Classify and illustrate the main features of a turbomachine Select the most appropriate turbomachine for a given aerospace application Apply first principles to simplified turbomachinery configurations Identify the most prominent loss sources of a given turbomachinery stage Define the aerodynamic blade design of turbomachinery cascades Predict the performance of a turbomachine with analytical and numerical methods Devise solutions for performance improvement of turbomachinery cascades based on physical understanding Perform the conceptual and detailed fluid-dynamic design of a turbomachine for propulsion systems using analytical and numerical tools	
Education Method	Lectures, concept quizzes, concept exercises, and tutorials in preparation to the team assignment.	
Books	- S.L. Dixon, C. A. Hall &#147;Fluid-Mechanics and Thermodynamics of Turbomachinery&#148;; 7th edition, 2013 - M. Schobeiri, &#147;Turbomachinery Flow Physics and Dynamic Performance&#148;; 2nd edition, Springer, 2012 - E.A. Baskharone, "Principles of Turbomachinery in Air-Breathing Engine", Cambridge University Press	
Assessment	The final course grade results from the weighted average of: the assessment of the final project assignment, counting for 40% of the final grade (individual score 1-10, no threshold, 1 bonus point for best project); the assessment of the written exam, counting for 60% of the final grade (individual score 1-10, minimum threshold: 6); The written exam takes place at the end of course. The exam consists of concept exercises covering the material discussed during classes. The final assignment is meant to challenge students with the real engineering problem of designing a turbomachine for propulsion or power generation application. The concept exercises are numerical exercises that are formulated to stimulate critical thinking on turbomachinery problems of practical application. They also serve as diagnostic tool to help students self-assessing their level of knowledge and understanding.	
Permitted Materials during Tests	Lectures notes, slides, formula sheets and textbooks are allowed during the written exam. Laptops can be used for calculations.	
Set-up	The course is organized in 8 teaching sessions, 2/3 seminar(s), and a possible visit to a turbomachinery company. Lectures usually consist of a theoretical part integrated with in-class exercises and quizzes to stimulate critical thinking and learning-by-doing. During classes, the students, autonomously or in groups, are engaged in a series of learning activities, supervised by the team of instructors. Their solution is briefly discussed at the beginning of the next class. The seminar(s) are devoted to show students current research topics in turbomachinery and are given by recognised experts in the field.	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor	Dr. D. Ragni	
Instructor	Prof.dr.ir. M. Snellen	
Co-responsible for assignments	Dr. R. Zamponi	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources.	
Study Goals	The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4261	Internal Flows	3
Responsible Instructor	C. Falsetti	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In this course we address fluid dynamic phenomena of interest in internal flow situations for aerospace propulsion applications. The goal of the course is to develop physical insight into the phenomena that characterize internal flow in fluid machinery. As such we will discuss not just what happens, but why it happens and what are the consequences for propulsion systems. The course covers topic, which are not dealt with in courses or texts about external fluid dynamics. The flows described in Internal Flows are generally rotational, often three-dimensional, unsteady, and sometimes occurring in non-inertial (e.g., rotating) coordinate systems. The course can be viewed as the development of flow models and ideas to enable allow physical insight into the behavior of three-dimensional and unsteady flows. One of the objectives of the course is to provide an increased ability to interpret computational and experimental results and hence to effectively extract conclusions about the key features of complex internal flows. This course is inspired by 16.540 Internal Flow from the Department of Aeronautics and Astronautics at MIT. Much of the material is based on the lectures of Profesor E. M. Greitzer, with permission.	
Study Goals	By the end of the course, you should be able to: 1. Understand the underlying principles governing internal flow phenomena in fluid machinery by explaining not only "what" occurred but also "why" ; 2. Interpret temperature, pressure, velocity data to predict the behavior of internal flows. 3. Assess the effect of three-dimensional and unsteady flows on the design of a turbo-machine.	
Education Method	This course will use innovative didactic methods that emphasize active learning, as open questions, flipped classroom, and peer teaching. Students will be engaged through active feedback systems and online material. Given that a textbook exists for the course material, it is the intent of the lectures to discuss the material rather than to present it for the first time in class. As such the students will be expected to read the material before class and to engage in in-class discussions of the reading material by answering questions and elaborating concepts. Here we list a high-level description of the course contents. More detailed information are available in the complete syllabus on Brightspace. Lecture 1 Structure and Content of the Course, Introduction to Flow Regimes Lectures 1-2 Useful Basic Ideas Pressure fields and streamline curvature Upstream influence in turbomachines Control volume description of propulsion systems Asymmetry of real fluid motions Behavior of an inlet and turbomachinery system Lectures 3-6 Vorticity and Circulation Kinematics of vorticity Dynamics of vorticity Behavior of vortex lines Rotational flow description in terms of vorticity and virculation Mixing enhancement due to streamwise vorticity Lectures 7-9 Flow in Rotating Passages Non-dimensional parameters in a rotating flow Conserved quantities in a steady rotating flow Phenomena in flows where rotation dominates Rotating Channel Flow in Constant Area Straight Passages Boundary layers in rotating straight channels Rotating Flow in Turbomachinery Passages Secondary flow in rotating blade rows Lectures 10-12 Loss Sources and Loss Accounting Appropriate metrics for loss Boundary layer losses Mixing Losses Averaging of a non-uniform flow Effect of pressure level on average properties and mixing losses Loss characterization in turbomachinery cascades Lectures 13-14 Compressible Flows & Heat transfer in fluid machinery Lectures 13-14 Flows with heat transfer and phase change Heat transfer and temperature effects Flows undergoing a phase-change	
Literature and Study Materials	Internal Flow: Concept and Applications, E. M. Greitzer, C. S. Tan, M. B. Graf, Cambridge University Press http://www.cambridge.org/gb/academic/subjects/engineering/thermal-fluids-engineering/internal-flow-concepts-and-applications?format=PB#MciShGFvqgyx6G4p.97	
Assessment	The student will be asked to develop concept questions and corresponding answers which help to illustrate the physical principles and approximations that are made. This will be done in small teams (six or seven members) so you can prepare and discuss the questions among yourselves. The concept questions are to be developed each week by each team. Each week one	

team is selected to present and discuss their concept question to the class, in a flipped classroom mode.

Approximately five short (~20 minute) "concept quizzes" will be given during class, roughly equally spaced throughout the term. The total of these quizzes will count for 25% of the final course grade.

The final exam is an oral exam lasting approximately 30 minutes, and it will count for 75% of the final course grade.

A rubric will be provided at the beginning of the course.

AE4262	Combustion for propulsion and power technologies	4
Responsible Instructor	Dr. I. Langella	
Responsible Instructor	F. De Domenico	
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	G. Kelesidis	
Contact Hours / Week x/x/x/x	0/0/0/4+2lab	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology	
Summary	This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.	
Course Contents	1. Thermodynamics of combustion 2. Chemical kinetics 3. Transport processes 4. Premixed and diffusion flames 5. Laminar and turbulent reactive flows 6. Combustion experiments and diagnostics 7. Combustion modelling 8. Laboratory Tutorial & Exercise 9. Practical combustion systems for aerospace applications 10. Pollutants and emissions 11. Hydrogen combustion physics and technology	
Study Goals	Write chemical equations and perform stoichiometry calculations. Compute heating values of fuels, enthalpies of formation and adiabatic flame temperature. Apply principles of thermodynamics to analyse reacting mixtures at chemical equilibrium. Describe the fundamentals of the physics involved in premixed- and diffusion flames, laminar and turbulent. Discuss the physics related to turbulent reacting flows and chemistry interaction. Explain experimental measurement techniques and their role in understanding combustion. Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems. Provide phenomenological descriptions of pollutant formation of conventional combustion systems. Elaborate on the potential and challenges related to the use of alternative fuels in combustion systems.	
Education Method	The course consists of weekly lectures and exercise sessions (practicals). In addition, the students are required to complete a project assignment and perform a laboratory exercise. The project assignment is relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with the investigation of methane or hydrogen laminar flames. The students will gain hands-on experience in designing the experimental matrix, acquiring data, performing data analysis and writing a technical report. Both laboratory exercise and assignment are mandatory to be admitted to the oral exam.	
Course Relations	Aero Engine Technology Hydrogen Technology	
Literature and Study Materials	Slides, lecture notes and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION;	
Books	J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"	
Assessment	To pass the course, the students must Complete laboratory exercise and project assignment; and Pass the oral exam with a minimum grade of 5.5 out of 10 The project assignment consists of several parts, including analysis on the data collected during the laboratory exercise and further theoretical and numerical analyses of combustion phenomena. Note attendance to the laboratory exercise and submission of the assignment are both compulsory for the admission to the oral exam. The work on the assignment should be summarized in a report, to be written by a single student or in a group of maximum three. The final grade is the result of the oral examination (60%) and the grade obtained in the assignment (40%).	

AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5
Responsible Instructor	Prof.dr.ir. P. Colonna di Paliano	
Instructor	Dr.ir. W.P.J. Visser	
Instructor	Prof. F. Casella	
Instructor	C.M. De Servi	
Contact Hours / Week x/x/x/x	0/x/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	AE1240-I Thermodynamics AE2230-II Propulsion and Power AE1205 Programming and Scientific Computing in Python for Aerospace AE4238 Aero Engine Technology ME45000 Advanced Heat Transfer AE2235-I Aerospace Systems & Control Theory	
Course Contents	2nd EDUCATION PERIOD Part 1 (2 ECTS) Module 1 Introduction, Context, Foundations Module 2 Conservation equations Module 3 Modeling paradigms Module 4 Numerical methods and software Module 5 Modelica Module 6 Constitutive equations Module 7 Components and system modeling Module 8 Verification and validation 3rd EDUCATION PERIOD Part 1 (1 ECTS) Module 9 Model-based control Exam on Part 1 Part 2 (2 ECTS) Module 10 - Team Project to be chosen among a Aero engine with GSP or GTPsim b Power or propulsion system with Modelica	
Study Goals	After learning the content of the course, the student will be able to: Given an engineering problem related to propulsion and power systems, use the 9-steps method to create or select the appropriate model and run and interpret simulations in order to obtain a good solution of such problem, and communicate the results. In particular, this overarching objective can be obtained by developing the following theoretical capabilities: 1. Describe the role and types of models in Propulsion and Power Systems Engineering, and define which different modeling paradigms and numerical methods are most appropriate or needed to develop a system model given its purpose. 2. Formulate a mathematical model for a typical propulsion and power system or component, by first analyzing the functionality of a system by means of a process flow diagram; by defining the energy, mass, and momentum conservation balances for the system of interest, and choosing the most appropriate form of conservation equations; by selecting the constitutive equations required to close the mathematical model (thermophysical models of fluids, chemical reaction eqs, heat transfer correlations, etc.). 3. Choose and configure numerical techniques for the solution of non-linear algebraic and differential-algebraic equation systems, which result from the formulation of a mathematical model. 4. Implement and code a system model of a power and propulsion application by adopting an object-oriented modeling approach (use of modularity, hierarchy, predefined connectors and inter-module variables). 5. Evaluate the reliability and possibly the fidelity of a model in the light of its purpose (model validation) 6. Use system models to solve engineering problems such as system performance assessment, preliminary design of the system and its components, the design of control strategies and the tuning of controller parameters, as well as communicate the results of the engineering analysis both verbally, and by means of a technical report.	
Education Method	Self-study using the Online Course Material on Brightspace (video lectures, related slides and other downloadable documents) Weekly sessions with instructors either online or in classroom, to discuss students' questions and to practice by means of assignments. Some sessions also require the use of computers and software. Students are strongly advised to make use of the BrightSpace Forum (Q&A) to post questions and interact with instructors and other students. In the second part of the course, students learn how to apply the knowledge acquired during the first part of the course, by working on an exemplary modeling and simulation project, in teams. Instructors provide support and supervision to the project.	
Computer Use	Students learn the Modelica modeling language and optionally the Gas Turbine Simulation program (GSP) or GTPsim. More information on these programs is provided on BrightSpace: Course Information -> Resources and Tools -> Open Modelica and GSP	
Literature and Study Materials	Video lectures on BrightSpace. Other material (PDF's) is made available for download on BrightSpace, see Content -> Modules	
Books	Reference text for thermodynamics: W. C. Reynolds and P. Colonna. Thermodynamics: fundamentals and engineering applications. Cambridge University Press, 2018.	

**Reader
Assessment**

Reference text for numerical methods:

P. Moin. Fundamentals of engineering numerical analysis. Cambridge University Press, 2nd Edition, 2010. Chapter 4, Numerical Integration of Ordinary Differential Equations.

The reader is constituted by the slides of each module, which can be downloaded from BrightSpace (PDF's).

Part 1: Exercises

The first module is tested with a Quiz.

The other modules are tested with take home exercises which are explained and partially carried out during the weekly sessions. The deliverables must be uploaded on BrightSpace by the delivery dates that are posted there. The exercises are preparatory for the Exam on Part 1 and are graded.

Part 1: Exam

The knowledge of Part 1 is assessed with a combination of a take-home exam and an oral examination.

It covers the material of the 9 Modules of Part 1 of the course. Duration of the take-home exam: approximately 5 days. The date in which the exam text is made available and the delivery date are posted on BrightSpace.

The oral exam is held within three weeks from the delivery date of the Take-home exam and questions are related to the solution of the take home exam and the solutions of the Module exercises.

Part 2: Simulation project

Teams of students (2-3) are given an assignment that requires the development of a system model either in GSP/GTPSim or in OpenModelica/Dymola

Graded deliverables: technical report, model files.

A short oral examination is conducted to assess the knowledge and understanding about the project of each member of the team, separately. The grade of each student for this part of the course is determined by considering both the project report and its execution and the oral exam.

Grading

the final grade is calculated as

Final Grade = $0.2 \times \text{Exercises Grade} + 0.4 \times \text{Exam Part 1 Grade (written + oral)} + 0.4 \times (\text{Simulation Project} + \text{Oral Exam Grade})$.

A final grade can be obtained only if the grade of the Exam Part 1 and of Part 2 are sufficient (≥ 6).

Resit exams

One opportunity to resit the Exam Part 1 is provided during the second quarter the course is running in.

The resit exam of Part 2 consists in an additional oral exam in which the student is tested about the content of the Simulation Project. Additional work may be required to compensate for deficiencies assessed during the failed examination.

AE4301	Automatic Flight Control System Design	3
Responsible Instructor	Dr. A. Jamshidnejad	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4303	
Expected prior knowledge	The following prior knowledge is required: AE3302 AE2204 (until 2012/2013) AE2235-I (from 2013/2014)	
Parts	- Introduction & recapitulation of systems and control theory . Laplace transformation . Elementary closed loop systems . Transfer functions . Poles and zeros . First order systems . Second order systems . Pole placement for simple systems . Root locus method . Characteristic equation . Angle and magnitude conditions . State space formulation . Controllability, observability . Basic controllers: P,PI,PD,PID . Frequency response (Bode plots) - Introduction: Course use and arrangement a. Why automatic flight control systems? b. Function of the flight control system in civil aviation c. Recapitulation of theory on flight dynamics d. Review of the different frames of reference: wind, stability, and body. e. Non-linear equations of motion. f. Trim and linearization of the non-linear equations of motion. g. Linearized longitudinal dynamics using a statespace representation and the equivalent frequency domain form. - Performance and handling qualities a. Handling quality criteria b. The Control Anticipation Parameter (CAP) c. Gibsons Phase rate and Frequency criterion - Dynamic stability augmentation a. yaw dampers b. pitch dampers c. phugoid dampers - Static stability augmentation a. angle of attack feedback to improve static margin b. load factor feedback to improve manoeuvre margin c. sideslip feedback to improve directional static stability - Basic longitudinal autopilot modes a. pitch attitude hold mode b. altitude hold mode c. airspeed hold mode (using autothrottle) d. vertical speed - Basic lateral autopilot modes a. roll angle hold mode b. coordinated roll angle hold mode c. turn rate at constant altitude and speed d. heading angle hold mode - Longitudinal and lateral guidance modes a. glideslope hold mode b. automatic flare mode c. localizer hold mode d. VOR hold mode	
Course Contents	Classical control is still predominantly used in aerospace industry for the design and analysis of automatic flight control systems. Various existing control systems such as Stability Augmentation Systems (SAS), Control Augmentation Systems (CAS) and fly-bywire systems are reviewed in detail. The emphasis of the course lies in demonstrating, through application of classical frequency domain and state space techniques, how to design systems that fulfill the requirements imposed by the aviation authorities, with emphasis on understanding the benefits and limitations of such systems.	
Study Goals	After this course the student should be able to: - Understand dynamics of linear time invariant systems and design controllers for such system for the desired performance	

	- substantiate the function of a Flight Control System(FCS) in civil aviation. - apply the theory of flight dynamics and control to FCS design for civil aviation. - verify if a given FCS satisfies the handling qualities for civil aviation criteria. - design static and dynamic stability augmentation systems. - design all longitudinal and lateral autopilot modes for civil aviation.
Education Method	Lectures
Literature and Study Materials	Course material to support the exercises will be posted on the Brightspace. Recommended literature - Any standard textbook on systems and control theory (e.g., Modern Control Engineering by Ogata) - M.V. Cook, Principles in flight dynamics, Edward Arnold, London, 1997 ISBN 0-340-63200-3. - B.L. Stevens, F.L.Lewis, Aircraft control and simulation, Wiley, New York, 1992 ISBN 0471613975. - J. Roskam, Airplane flight dynamics and control Part II, , ISBN 1-8845885-18-7.
Assessment	Written closed-book examination
Set-up	At the end of each lecture, a simple take home assignment is given in order to gain experience in working with the course material. There will be a written examination at the end of the course. In the related practical AE4301P a control system must be designed that satisfies certain desired requirements.

AE4301P	Exercise Automatic Flight Control System Design	1
Responsible Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301 Automatic Flight Control System Design AE3212-I Flight Dynamics AE2235-I Systems and Control Theory	
Course Contents	The goal of this exercise is to design a flight control system using classical control theory, using a simulation model in Simulink. The exercise makes you in a practical way familiar with the implementation of inner control loop design, handling qualities satisfying CAP/Gibson/MIL-specifications, and the design of a higher level autopilot mode such as glideslope tracking or terrain following.	
Study Goals	To become familiar with classical flight controllers and their design, and to gain insight in handling qualities of open-loop and controlled aircraft.	
Education Method	General Practice Exercise.	
Literature and Study Materials	Course material to support the exercise will be posted on Brightspace.	
Assessment	Report	
Remarks	A short report about the design of this controller using classical control theory needs to be written. This report must be handed in during the second quarter. There is a resit-opportunity in the 3rd or 4th quarter.	
Set-up	The exercise is a homework assignment to be solved using Matlab and Simulink.	

AE4314-21	Helicopter Performance, Stability and Control	4
Responsible Instructor	Dr. M.D. Pavel	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Aircraft performance Aerodynamics Flight dynamics	
Parts	Week Arrangement Lecture and study material 1. Lay-out of the helicopter. Qualitative description of the rotor and control systems. Definition of the tip plane, control plane, hub plane and several co-ordinate systems. Blade element analysis of hover performance. 2. Performance calculations of vertical flight. Turbulent Wake State. Autorotation. Performance calculations in forward flight. Blade element analysis. Glauerts hypothesis. Comparison with performance of fixed wing aircraft. 3. Flapping dynamics of centrally hinged rotors. Equations of motion of helicopter with only pitch degree of freedom. Stability and control. 4. Flapping dynamics in forward flight. Symmetric equations of motion. Stability and control in forward flight. 5. Trim conditions, linearised equations of motions. Representation in the complex plane and comparison with fixed wing aircraft. 6. Influence of semi-rigid rotorsystems on the equations of motion and on the characteristic handling characteristics. 7. Flap/Torsion blade dynamics. Divergence. Classical blade flutter, Structural lay-up to avoid flutter. Introduction to ground resonance instability.	
Course Contents	General lay-out of helicopters, performance calculations during hover, vertical climb and descent and forward flight. Helicopter blade dynamics. Introduction to helicopter control characteristics: symmetrical equations of motion handling- and control properties. Introduction to aeroelasticity of rotorcraft.	
Study Goals	1. An example of the synthesis of a large diversity of subject areas, aimed at the analysis of a complicated technical object. 2. Introduction to aerodynamics, flight mechanics and aeroelasticity of helicopters, which enables the student to independently study more advanced literature.	
Education Method	A combination of Online and Classical lecturing courses. Articles Readings and discussions. Questions and Discussions fora	
Literature and Study Materials	"Helicopter performance, Stability and Control", reader AE4-214, 2002, Faculty of Aerospace Engineering "Basic Helicopter Aerodynamics" by John Seddon& Simon Newman, Blackwell Science, 2002 "Principle of Helicopter Aerodynamics", Gordon Leishman, Cambridge Aerospace Series, 2000	
Assessment	Multiple-choice tests during the course (optional). Presentation of a helicopter-related subject in a small Symposium (obligatory) One final take-home assignment at the end of the course (obligatory) +Oral Exam (obligatory)	
Special Information	A helicopter flight in SIMONA simulator may be given to the students following this course depending on the availability of SIMONA simulator.	
Set-up	Online education and Classical lecturing	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Physical intelligence for interaction - Insect intelligence and Swarming - Spiking neural networks	
Study Goals	After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.	
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4431-23	Aircraft noise	3
Responsible Instructor	Prof.dr. D.G. Simons	
Gast Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is required for AE4463P-23 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics (dBA meter) - Dutch aircraft noise model NRM or airframe noise model (ANOPP method) - digital signal analysis and acoustic imaging	
Expected prior knowledge	Basic knowledge on calculus and first year physics.	
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and tert's band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phone and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. Acoustic imaging: Principle of beamforming. Imaging aircraft noise data. (Note: this topic is only touched upon in this course, the details (including application) are treated in AE4463P-23) Aircraft noise sources: Mechanism of propeller noise and blade passage frequency. Mechanism of turbo engine noise. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise. ANOPP modelling method. Programming assignment about modelling airframe noise (landing gear, flaps, slats, etc.) using the ANOPP method.	
Study Goals	Understand the relationship between aviation and the resulting noise levels, especially around airports.	
Education Method	Lectures, exercises, programming assignment	
Literature and Study Materials	Lecture slides, exercises with worked-out solutions, programming assignment Reader: Course AE4431-23 Aircraft Noise Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2023	
Assessment	- Written exam (max 8 points) (a formula sheet will be provided) - programming assignment (max 2 points)	
Set-up	Lectures, exercises, programming assignment	

AE4462-17		Aircraft Emissions and Climate Effects	4
Responsible Instructor	Prof.dr. V. Grewe		
Instructor	Dr. I.C. Dedoussi		
Responsible for assignments	V.R. Meijer		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	3 4		
Course Language	English		
Expected prior knowledge	Basic knowledge on calculus, fluid mechanics and first year physics.		
Course Contents	The course introduces the students in the state-of-the-art capabilities, research issues and challenges in climate effects of aviation. Topics include: - Combustion and emissions - Overview of all environmental effects of aviation - Air pollution relevant atmospheric chemistry and physics - Local air pollution modelling - Aviation environmental policy, new technologies, and other mitigation options - General circulation of the atmosphere - Atmospheric chemistry and contrail formation - Climate change caused by air traffic emissions - Modelling of climate impact from aviation - Application and show cases for the assessment of mitigation options supporting decision making, e.g. for climate optimized trajectories		
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the climate impact of air traffic. These elements are required, for example, for the assessment of the air quality and climate impact of aviation and mitigation measures, including climate optimised trajectories/routing, low climate impact routing and consequences for aircraft design and supersonic transport.		
Education Method	Lectures and exercises		
Literature and Study Materials	Lecture slides and hand-outs		
Assessment	Written exam (Resits may be oral, depending on the situation) Assignments Both the written exam and the assignments need to be passed with a 6.0 or higher.		

AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course AE4463P-23 aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) must be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and also applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B, C and F are required. A choice can be made between D and E.	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods.	
Study Goals	Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models. At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace.	
	Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments (only assignment 3 and 4 since they have the highest weight; assignment 1 and 2 cannot be redone).	
	Results for individual assignments are only valid for one academic year: when re-taking the course all assignments have to be done in that academic year.	

AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
Responsible Instructor Responsible Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Expected prior knowledge	C. Kassapoglou Dr. D. Zarouchas 0/0/4/0 3 3 none English Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis 3. Reliability analysis & Health monitoring a. Probability of Failure & Uncertainty quantification b. Structural Health Monitoring 4. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Understand the philosophy of reliability analysis and calculate the probability of failure of composite structures by combining numerical methods and the mechanics of composite structures - Understand the importance of structural health monitoring (SHM) and design a SHM system for a given composite structure - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Classical Laminate Theory: 3 2. Progressive damage analysis: 3 3. Reliability analysis & Health Monitoring: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory, progressive damage analysis, reliability analysis and health monitoring. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 10 points A2. Progressive damage and damage tolerance analysis 20 points A3. Reliability analysis & Health Monitoring. 30 points	

Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method).

B1. Design problem 40 points

The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2,A3 and part B1) should at least be rewarded with 55% of the attributed points ($55\% \cdot 60$ and $55\% \cdot 40$), otherwise the student will fail. In addition, the overall average grade should at least be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1	
	2	
	3	
	4	
Start Education	1	
	3	
	4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work.	
	You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course.	
	The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses	
Assessment	All students will have a task in the production team as well as at least one additional organizational task. Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report	
	A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	The grade for this course will be a Pass/Fail grade.	
	This course is open to students from all Master tracks.	
	Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply.	
	As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses	
	Alternatively, having earned a Bachelor degree in aerospace engineering will give access to the course as well. Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December - Week 4.8: 15 May - Week 1.6: 30 September	

Set-up	The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and 3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3 There is an intentional overlap so that continuity and handover is guaranteed

ET4117	Electrical Machines and Drives	4
Responsible Instructor	Dr.ir. H. Vahedi	
Responsible Instructor	Dr. G.R. Chandra Mouli	
Instructor	W. Shi	
Responsible for assignments	H. Dialani	
Responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2014 (7th edition) ISBN 9781259254666	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Instruction if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Process & Energy	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Prof.dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Dr. O. Nejadseyfi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be 5.8 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest decimal. The maximum bonus score is 0.8. The bonus only applies to reports with grade ≥ 5.5 and submitted before the first report deadline.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	ME Department Precision & Microsystems Engineering	
Contact	Prof.dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. R. Kortlever	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Power and Propulsion

Program Coordinator	Dr. I. Langella
Contact for students	The profile coordinator is Dr. Ivan Langella (I.Langella@tudelft.nl)
Program Title	Master of Science in Propulsion and Power
Director of Education	Ir. Joris Melkert
Contact for Students	If you have specific question about the profile, you can send your queries to Dr. Ivan Langella (i.langella@tudelft.nl).
Introduction 1	Education. The MSc profile of propulsion and power is a part of the MSc track Flight Performance and Propulsion. In this profile the students will be able to specialize in courses related to aircraft propulsion, especially pertaining to increasing sustainability in propulsion systems. Several mandatory profile courses form the core of the knowledge that specializes our graduate students, covering relevant aspects in turbomachinery, combustion, propulsion system, component design and their integration. Elective courses and PPD (Professional and Personal Development) courses are chosen in consultation with the Profile Coordinator, and these are grouped in Advanced Topics, Methods, skills, etc. so that the student can choose subjects related to these aspects. The competence acquired during the first year of the Master program is put to fruition in during the 2nd year dedicated to internship and thesis during which the student is engaged in a research project. The thesis project is carried out under the supervision of one of the academic staff members, but often the student also benefits from the closer supervision of one of the PhD candidates involved in the same project. Below, an overview of the research activities is provided. Research: In our stride to make aviation sustainable, Aircraft propulsion is the best way to reduce fuel consumption and emissions. The use of alternative fuels and their usage in an aircraft is determined by the propulsion system. The propulsion system account for more than 90% of the life-cycle emissions and around 30% of the life-cycle cost. Education in topics related to propulsion is at the core of all aerospace Master programs in the world and TU Delft is no exception. The courses of our profile create the knowledge needed to effectively perform thesis work within one of the running research programs. They provide an exciting opportunity to learn about cutting-edge technologies that will significantly improve the efficiency and sustainability of aviation. NextGen Aircraft Propulsion Systems: We are currently pursuing studies of novel concepts with the potential to leapfrog propulsion systems to a much higher efficiency level and much lower emissions. For example, new engine architectures and new thermodynamic cycles like hybrid aircraft engines, multi-fuel engines, electrically assisted engines, etc. Novel Fuels: One of the enablers for sustainable aviation is alternative fuels like biofuels, hydrogen, synthetic fuels, etc. We are working on new engine configurations that have the ability to use novel fuels like hydrogen or hydrogen along with bio-fuels and synthetic fuels. These technologies allow for a leap forward in terms of environmental friendliness, thanks to the adoption of sustainable fuels. Investigations are theoretical, numerical and also experimental, thanks to the newly established Clean Combustion Lab. Novel Combustion Systems: In the Clean Combustion lab, we are investigating low NOx combustion technologies for novel energy carriers like Hydrogen, biofuels, methanol, or a mixture of different fuels. We have state of the art Laser diagnostic techniques like Laser Induced Fluorescence, Coherent Anti-stokes Raman Spectroscopy, etc. We are also using Large Eddy Simulations and Direct Numerical Simulations to understand and model combustion and combustion systems. Waste Heat Recovery: we are looking into recovering waste heat from engines to increase the thermodynamic efficiency and looking at configurations such as Combined-Cycle Engine (CC Engine) in which the exhaust from a gas turbine provides thermal energy to closed-cycle supercritical CO2 turbine, thereby increasing the efficiency.

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Prof.dr.ir. L.L.M. Veldhuis	
Instructor	L.T. Lima Pereira	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130-22, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during the first 3 weeks of Q1).	
Assessment	The assessment is done through 5 assignments that are performed on a personal basis. The assignments are to be combined in 1 report that should be delivered after the finalization of the lecture at the end of Q2. Additional information on the timing and contents of the assignments will be presented on BrightSpace.	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4205	MDO for Aerospace Applications	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ir. M.F.M. Hoogreef	
Instructor	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Familiarity with MATLAB and basic aircraft design knowledge. MATLAB is used for tutorials and home assignment, thus students without MATLAB experience must gain this through self-study Students with no experience in aircraft design should consult textbooks like Torenbeek, Raymer and Roskam to familiarize with the fundamentals of wing sizing process and general concepts such as wing loading, aerodynamic coefficients, fuel fraction method, drag and its components, etc.	
Course Contents	The course will introduce students to multidisciplinary design optimization (MDO), as a modern methodology to support complex product development. Main topics: - What and why MDO - Parameterization techniques - Numerical optimisation methods (focus on gradient based methods) - Decomposition and coordination of complex computational systems - the eXtended Design Structure Matrix (XDSM) - MDO architectures - Intro to surrogate modeling, discrete&mixed integer problems, genetic algorithms and Multi Objective Optimisation	
Study Goals	The main goals are the following: - understanding of Multidisciplinary Design Optimization (MDO) methods and architectures - hands-on experience in setting up and solving MDO problems	
Education Method	The course consists of classical lectures and computer tutorials using own laptop. During the tutorials, students are introduced to some of the tools discussed during lecture and can make hands-on experience during interactive sessions with the teachers. In particular, students will be introduced to the MATLAB optimization function fmincon and its use for gradient based optimization. A significant amount of time is left for self-study, where students will have to work in pairs on one take-home assignment.	
Computer Use	The use of a personal computer is required MATLAB (including its optimization toolbox) will be used for the MDO practical sessions and assignment. Student licenses are provided by the TUD	
Literature and Study Materials	- lecture sheets and some notes (only available digitally on Brightspace) - scientific papers - some software modules - Suggested (not compulsory) book: - Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszczanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 - The book is available in e-format from the TUD library (Chapter 9 is used in the course AE4204 on Knowledge Based Engineering) - Engineering Design Optimization, by J.R.R.A. Martins and A. Ning (free on line)	
Assessment	There is no conventional exam for this course. The assessment is based on the quality of one take-home assignment, which will be executed in teams of two students. The assignment is about setting up and solving an MDO problem using Matlab. The deliverables are the Matlab code and a report. Two different take-home assignments are defined for the two yearly examination sessions.	
Tags	Algorithmics Analysis Matlab Modelling Optimalisation Programming Project	

AE4206-24	Turbomachinery	4
Responsible Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The course covers fundamental concepts of internal flows in turbomachines and related design aspects. The typical course schedule consists of the following lectures: Lecture 0: Introduction course overview: topics, structure and assessment Lecture 1: Recall of Turbomachinery Fundamentals & Principles integral governing equations with applications, relative motion, thermodynamic transformations in turbomachinery stages, velocity triangles with practical examples; Lecture 2: Unsteady Flows in Turbomachinery concept of reduced frequency with applications; work exchange in turbomachinery rows; Lecture 3: Similarity in Turbomachinery similarity analysis for incompressible flows, Balje and Smith diagrams, accounting for compressibility, stage dimensionless performance parameters; Lecture 4: Fluid-dynamic Design of Turbomachinery design procedure, mean-line design model, radial equilibrium model, CFD-based models Lecture 5-6: Axial and Radial Turbines examples of axial and radial turbines, typical turbine stages, blade profiles for turbines, flow characteristics in subsonic, transonic, and supersonic cascades, basic concepts of diffusers, aerodynamics of radial impellers Lecture 7-8: Axial and Radial Compressors examples of axial and radial compressors, typical compressor stages, blade profiles for compressors, flow characteristics and unique incidence, stall and surge instability, slip factor, diffusers for radial compressors, design maps Seminar(s): topic(s) may vary but usually include turbomachinery aero-mechanics, shape optimisation with CFD methods, rocket engine turbo-pumps, and advanced turbomachinery components for aero-engines Tutorial: training session on the software tools provided for the team project	
Course Contents	This course educates aerospace students to apply theoretical knowledge in the field of fluid- mechanics and thermodynamics for analyzing and designing a thermal turbomachinery. Typical examples are: axial and radial compressors of gas turbine engines, industrial gas and steam turbines, radial turbines of automotive turbochargers, turbo-expanders and compressors used in refrigeration and liquefaction industry, micro-turbines for power generation. Since the turbomachinery field is intrinsically interdisciplinary, it is assumed that students already possess basic knowledge of thermodynamics, fluid-mechanics, applied mathematics, propulsion systems, internal flows, acquired in previous courses. Prerequisite of the course are therefore bachelor and master courses addressing fundamental (e.g. Thermodynamics, Internal Flows) and applied aspects (e.g. Aero-Engines) of air-breathing engines. Starting from basic concepts of gas-dynamics and fluid-machinery, the course brings the student to the level that he is able to select a turbomachinery for a specific application, perform the conceptual design of the stages, and eventually predict the flow characteristics within the blade passages by means of conceptual physical models and more sophisticated simulation tools.	
Study Goals	After the course the student will be able to: - Classify and illustrate the main features of a turbomachine - Select the most appropriate turbomachine for a given aerospace application - Apply first principles to simplified turbomachinery configurations - Identify the most prominent loss sources of a given turbomachinery stage - Define the aerodynamic blade design of turbomachinery cascades - Predict the performance of a turbomachine with analytical and numerical methods - Devise solutions for performance improvement of turbomachinery cascades based on physical understanding - Perform the conceptual and detailed fluid-dynamic design of a turbomachine for propulsion systems using analytical and numerical tools	
Education Method	Lectures, concept quizzes, concept exercises, and tutorials in preparation to the team assignment.	
Books	- S.L. Dixon, C. A. Hall &#147;Fluid-Mechanics and Thermodynamics of Turbomachinery&#148;; 7th edition, 2013 - M. Schobeiri, &#147;Turbomachinery Flow Physics and Dynamic Performance&#148;; 2nd edition, Springer, 2012 - E.A. Baskharone, "Principles of Turbomachinery in Air-Breathing Engine", Cambridge University Press	
Assessment	The final course grade results from the weighted average of: - the assessment of the final project assignment, counting for 40% of the final grade (individual score 1-10, no threshold, 1 bonus point for best project); - the assessment of the written exam, counting for 60% of the final grade (individual score 1-10, minimum threshold: 6); The written exam takes place at the end of course. The exam consists of concept exercises covering the material discussed during classes. The final assignment is meant to challenge students with the real engineering problem of designing a turbomachine for propulsion or power generation application. The concept exercises are numerical exercises that are formulated to stimulate critical thinking on turbomachinery problems of practical application. They also serve as diagnostic tool to help students self-assessing their level of knowledge and understanding.	
Permitted Materials during Tests	Lectures notes, slides, formula sheets and textbooks are allowed during the written exam. Laptops can be used for calculations.	
Set-up	The course is organized in 8 teaching sessions, 2/3 seminar(s), and a possible visit to a turbomachinery company. Lectures usually consist of a theoretical part integrated with in-class exercises and quizzes to stimulate critical thinking and learning-by-doing. During classes, the students, autonomously or in groups, are engaged in a series of learning activities, supervised by the team of instructors. Their solution is briefly discussed at the beginning of the next class. The seminar(s) are devoted to show students current research topics in turbomachinery and are given by recognised experts in the field.	

AE4238	Aero Engine Technology	4
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	Dr. F. Yin	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery", "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Inlets and Exhaust - o Turbomachinery - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls - o New Engine /propulsion Concepts	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an flipped course in which the lectures are pre-recorded and the students are supposed to watch the lectures before coming to the classroom. The classroom interaction is mainly to have discussions and for tutorials. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13. Evaluate the calorific values for various fuels 14. Understand the basics of combustion and emission formation in Aero engines 15. Understand the working of gas turbine combustor 16. Understand the basic principles of gas turbine controls and evaluate different strategies for control 17. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a flipped teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

	1. Gas Turbine Theory by H.I.H.Saravanamuttoo, G.F.C. Rogers, and H. Cohen 2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons
Reader	A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Prerequisites	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Assessment	The assessment is based on 2-3 assignments given during the course and a final exam. All assignments together will form 30% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
Permitted Materials during Tests	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4261	Internal Flows	3
Responsible Instructor	C. Falsetti	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In this course we address fluid dynamic phenomena of interest in internal flow situations for aerospace propulsion applications. The goal of the course is to develop physical insight into the phenomena that characterize internal flow in fluid machinery. As such we will discuss not just what happens, but why it happens and what are the consequences for propulsion systems. The course covers topic, which are not dealt with in courses or texts about external fluid dynamics. The flows described in Internal Flows are generally rotational, often three-dimensional, unsteady, and sometimes occurring in non-inertial (e.g., rotating) coordinate systems. The course can be viewed as the development of flow models and ideas to enable allow physical insight into the behavior of three-dimensional and unsteady flows. One of the objectives of the course is to provide an increased ability to interpret computational and experimental results and hence to effectively extract conclusions about the key features of complex internal flows. This course is inspired by 16.540 Internal Flow from the Department of Aeronautics and Astronautics at MIT. Much of the material is based on the lectures of Profesor E. M. Greitzer, with permission.	
Study Goals	By the end of the course, you should be able to: 1. Understand the underlying principles governing internal flow phenomena in fluid machinery by explaining not only "what" occurred but also "why" ; 2. Interpret temperature, pressure, velocity data to predict the behavior of internal flows. 3. Assess the effect of three-dimensional and unsteady flows on the design of a turbo-machine.	
Education Method	This course will use innovative didactic methods that emphasize active learning, as open questions, flipped classroom, and peer teaching. Students will be engaged through active feedback systems and online material. Given that a textbook exists for the course material, it is the intent of the lectures to discuss the material rather than to present it for the first time in class. As such the students will be expected to read the material before class and to engage in in-class discussions of the reading material by answering questions and elaborating concepts. Here we list a high-level description of the course contents. More detailed information are available in the complete syllabus on Brightspace. Lecture 1 Structure and Content of the Course, Introduction to Flow Regimes Lectures 1-2 Useful Basic Ideas Pressure fields and streamline curvature Upstream influence in turbomachines Control volume description of propulsion systems Asymmetry of real fluid motions Behavior of an inlet and turbomachinery system Lectures 3-6 Vorticity and Circulation Kinematics of vorticity Dynamics of vorticity Behavior of vortex lines Rotational flow description in terms of vorticity and virculation Mixing enhancement due to streamwise vorticity Lectures 7-9 Flow in Rotating Passages Non-dimensional parameters in a rotating flow Conserved quantities in a steady rotating flow Phenomena in flows where rotation dominates Rotating Channel Flow in Constant Area Straight Passages Boundary layers in rotating straight channels Rotating Flow in Turbomachinery Passages Secondary flow in rotating blade rows Lectures 10-12 Loss Sources and Loss Accounting Appropriate metrics for loss Boundary layer losses Mixing Losses Averaging of a non-uniform flow Effect of pressure level on average properties and mixing losses Loss characterization in turbomachinery cascades Lectures 13-14 Compressible Flows & Heat transfer in fluid machinery Lectures 13-14 Flows with heat transfer and phase change Heat transfer and temperature effects Flows undergoing a phase-change	
Literature and Study Materials	Internal Flow: Concept and Applications, E. M. Greitzer, C. S. Tan, M. B. Graf, Cambridge University Press http://www.cambridge.org/gb/academic/subjects/engineering/thermal-fluids-engineering/internal-flow-concepts-and-applications?format=PB#MciShGFvqgyx6G4p.97	
Assessment	The student will be asked to develop concept questions and corresponding answers which help to illustrate the physical principles and approximations that are made. This will be done in small teams (six or seven members) so you can prepare and discuss the questions among yourselves. The concept questions are to be developed each week by each team. Each week one	

team is selected to present and discuss their concept question to the class, in a flipped classroom mode.

Approximately five short (~20 minute) "concept quizzes" will be given during class, roughly equally spaced throughout the term. The total of these quizzes will count for 25% of the final course grade.

The final exam is an oral exam lasting approximately 30 minutes, and it will count for 75% of the final course grade.

A rubric will be provided at the beginning of the course.

AE4262	Combustion for propulsion and power technologies	4
Responsible Instructor	Dr. I. Langella	
Responsible Instructor	F. De Domenico	
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	G. Kelesidis	
Contact Hours / Week x/x/x/x	0/0/0/4+2lab	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology	
Summary	This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.	
Course Contents	1. Thermodynamics of combustion 2. Chemical kinetics 3. Transport processes 4. Premixed and diffusion flames 5. Laminar and turbulent reactive flows 6. Combustion experiments and diagnostics 7. Combustion modelling 8. Laboratory Tutorial & Exercise 9. Practical combustion systems for aerospace applications 10. Pollutants and emissions 11. Hydrogen combustion physics and technology	
Study Goals	Write chemical equations and perform stoichiometry calculations. Compute heating values of fuels, enthalpies of formation and adiabatic flame temperature. Apply principles of thermodynamics to analyse reacting mixtures at chemical equilibrium. Describe the fundamentals of the physics involved in premixed- and diffusion flames, laminar and turbulent. Discuss the physics related to turbulent reacting flows and chemistry interaction. Explain experimental measurement techniques and their role in understanding combustion. Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems. Provide phenomenological descriptions of pollutant formation of conventional combustion systems. Elaborate on the potential and challenges related to the use of alternative fuels in combustion systems.	
Education Method	The course consists of weekly lectures and exercise sessions (practicals). In addition, the students are required to complete a project assignment and perform a laboratory exercise. The project assignment is relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with the investigation of methane or hydrogen laminar flames. The students will gain hands-on experience in designing the experimental matrix, acquiring data, performing data analysis and writing a technical report. Both laboratory exercise and assignment are mandatory to be admitted to the oral exam.	
Course Relations	Aero Engine Technology Hydrogen Technology	
Literature and Study Materials	Slides, lecture notes and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION;	
Books	J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"	
Assessment	To pass the course, the students must Complete laboratory exercise and project assignment; and Pass the oral exam with a minimum grade of 5.5 out of 10 The project assignment consists of several parts, including analysis on the data collected during the laboratory exercise and further theoretical and numerical analyses of combustion phenomena. Note attendance to the laboratory exercise and submission of the assignment are both compulsory for the admission to the oral exam. The work on the assignment should be summarized in a report, to be written by a single student or in a group of maximum three. The final grade is the result of the oral examination (60%) and the grade obtained in the assignment (40%).	

AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5
Responsible Instructor	Prof.dr.ir. P. Colonna di Paliano	
Instructor	Dr.ir. W.P.J. Visser	
Instructor	Prof. F. Casella	
Instructor	C.M. De Servi	
Contact Hours / Week x/x/x/x	0/x/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	AE1240-I Thermodynamics AE2230-II Propulsion and Power AE1205 Programming and Scientific Computing in Python for Aerospace AE4238 Aero Engine Technology ME45000 Advanced Heat Transfer AE2235-I Aerospace Systems & Control Theory	
Course Contents	2nd EDUCATION PERIOD Part 1 (2 ECTS) Module 1 Introduction, Context, Foundations Module 2 Conservation equations Module 3 Modeling paradigms Module 4 Numerical methods and software Module 5 Modelica Module 6 Constitutive equations Module 7 Components and system modeling Module 8 Verification and validation 3rd EDUCATION PERIOD Part 1 (1 ECTS) Module 9 Model-based control Exam on Part 1 Part 2 (2 ECTS) Module 10 - Team Project to be chosen among a Aero engine with GSP or GTPsim b Power or propulsion system with Modelica	
Study Goals	After learning the content of the course, the student will be able to: Given an engineering problem related to propulsion and power systems, use the 9-steps method to create or select the appropriate model and run and interpret simulations in order to obtain a good solution of such problem, and communicate the results. In particular, this overarching objective can be obtained by developing the following theoretical capabilities: 1. Describe the role and types of models in Propulsion and Power Systems Engineering, and define which different modeling paradigms and numerical methods are most appropriate or needed to develop a system model given its purpose. 2. Formulate a mathematical model for a typical propulsion and power system or component, by first analyzing the functionality of a system by means of a process flow diagram; by defining the energy, mass, and momentum conservation balances for the system of interest, and choosing the most appropriate form of conservation equations; by selecting the constitutive equations required to close the mathematical model (thermophysical models of fluids, chemical reaction eqs, heat transfer correlations, etc.). 3. Choose and configure numerical techniques for the solution of non-linear algebraic and differential-algebraic equation systems, which result from the formulation of a mathematical model. 4. Implement and code a system model of a power and propulsion application by adopting an object-oriented modeling approach (use of modularity, hierarchy, predefined connectors and inter-module variables). 5. Evaluate the reliability and possibly the fidelity of a model in the light of its purpose (model validation) 6. Use system models to solve engineering problems such as system performance assessment, preliminary design of the system and its components, the design of control strategies and the tuning of controller parameters, as well as communicate the results of the engineering analysis both verbally, and by means of a technical report.	
Education Method	Self-study using the Online Course Material on Brightspace (video lectures, related slides and other downloadable documents) Weekly sessions with instructors either online or in classroom, to discuss students' questions and to practice by means of assignments. Some sessions also require the use of computers and software. Students are strongly advised to make use of the BrightSpace Forum (Q&A) to post questions and interact with instructors and other students. In the second part of the course, students learn how to apply the knowledge acquired during the first part of the course, by working on an exemplary modeling and simulation project, in teams. Instructors provide support and supervision to the project.	
Computer Use	Students learn the Modelica modeling language and optionally the Gas Turbine Simulation program (GSP) or GTPsim. More information on these programs is provided on BrightSpace: Course Information -> Resources and Tools -> Open Modelica and GSP	
Literature and Study Materials	Video lectures on BrightSpace. Other material (PDF's) is made available for download on BrightSpace, see Content -> Modules	
Books	Reference text for thermodynamics: W. C. Reynolds and P. Colonna. Thermodynamics: fundamentals and engineering applications. Cambridge University Press, 2018.	

**Reader
Assessment**

Reference text for numerical methods:

P. Moin. Fundamentals of engineering numerical analysis. Cambridge University Press, 2nd Edition, 2010. Chapter 4, Numerical Integration of Ordinary Differential Equations.

The reader is constituted by the slides of each module, which can be downloaded from BrightSpace (PDF's).

Part 1: Exercises

The first module is tested with a Quiz.

The other modules are tested with take home exercises which are explained and partially carried out during the weekly sessions. The deliverables must be uploaded on BrightSpace by the delivery dates that are posted there. The exercises are preparatory for the Exam on Part 1 and are graded.

Part 1: Exam

The knowledge of Part 1 is assessed with a combination of a take-home exam and an oral examination.

It covers the material of the 9 Modules of Part 1 of the course. Duration of the take-home exam: approximately 5 days. The date in which the exam text is made available and the delivery date are posted on BrightSpace.

The oral exam is held within three weeks from the delivery date of the Take-home exam and questions are related to the solution of the take home exam and the solutions of the Module exercises.

Part 2: Simulation project

Teams of students (2-3) are given an assignment that requires the development of a system model either in GSP/GTPSim or in OpenModelica/Dymola

Graded deliverables: technical report, model files.

A short oral examination is conducted to assess the knowledge and understanding about the project of each member of the team, separately. The grade of each student for this part of the course is determined by considering both the project report and its execution and the oral exam.

Grading

the final grade is calculated as

Final Grade = $0.2 \times \text{Exercises Grade} + 0.4 \times \text{Exam Part 1 Grade (written + oral)} + 0.4 \times (\text{Simulation Project} + \text{Oral Exam Grade})$.

A final grade can be obtained only if the grade of the Exam Part 1 and of Part 2 are sufficient (≥ 6).

Resit exams

One opportunity to resit the Exam Part 1 is provided during the second quarter the course is running in.

The resit exam of Part 2 consists in an additional oral exam in which the student is tested about the content of the Simulation Project. Additional work may be required to compensate for deficiencies assessed during the failed examination.

AE5051	Internship	15
Responsible Instructor	M.J. Oude Alink	
Instructor	Ir. J.A. Melkert	
Instructor	Ir. I.A. Houthuijsen	
Contact Hours / Week x/x/x/x	Within the MSc programme structure, the internship is aimed to start in Q1 of the second year	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success.	
Course Contents	The aim of the internship is to expose students to a real professional work environment different from that of TU Delft. The internship has a minimum duration of 10 consecutive weeks on a full-time basis. Full time means approx. 40 hrs p/wk and without interruption (vacation). To guarantee the real professional work experience outside the perimeter of TU Delft the internship has to take place at a company/research institute or at another university. The internship provides the opportunity to get a glance of the technical, social, economic and organizational aspects of Aerospace Engineering or related professional fields. It enables students to acquire professional skills different from those taught in the rest of the MSc program. During the internship the student will apply her/his academic skills and knowledge acquired during the MSc courses already taken so far. Therefore, earned a minimum of 30 EC before starting the internship is mandatory. The student has to aim for a project or tasks at academic master level that have well-defined outcomes and clear deliverables.	
Study Goals	Please find below the learning objectives of the internship course AE5051. By the end of the internship: 1. You can apply engineering knowledge and (technical) skills in a professional setting in an analytical academic way; 2. You can demonstrate this knowledge and these skills in a technical report; 3. You are able to demonstrate competences and (collaboration) skills in a professional environment; 4. You can write a self-reflection report about your personal functioning and the progress/improvement(s) that you made during your internship.	
Education Method	Practical professional work experience for a period of at least 10 consecutive weeks full time (approx.40 hrs p/wk without interruption) at the location at the company/research institute/other university, including desk research and writing of a technical internship report. (Partly) remote internships can be allowed, but always need to be arranged in consultation with the Internship Coordinator. The 10 week period may in exceptional cases, and solely in case of an internship abroad, be extended to a maximum of 6 months. This has to be discussed with and approved by the Internship Coordinator before the start of the internship. The internship placement should be in an environment different from that of TU Delft. Internship companies should have a clear hierarchical structure, e.g. employees, managers, higher management, CEO work on their own layers. internships at start-up companies have to meet additional criteria to regular internships (an internship at your own start-up is not allowed). Internships within the Netherlands have a maximum duration of 5 months. Internships abroad may have a maximum duration of 6 months due to the adaptation to intercultural differences (hierarchy, local languages, etc.) and the potential need to arrange practical matters on the spot (housing, insurance). This has to be discussed with and approved by the Internship Coordinator before the start of the internship and will not lead to issuing more credits.	
Literature and Study Materials	Please consult Brightspace AE5051.	
Prerequisites	A minimum of 30 EC obtained during the first MSc AE year before the start date of the internship is necessary to perform at the right level of knowledge and skills and to make the internship a success. A completed and approved internship approval form must be submitted via the OnStage system (tudelft-onstage.xebic.com), preferably 4 weeks in advance for checking. Students from cohort 2022-2023 and after who are admitted to the MSc AE are required to follow the procedure for the Internship AE5051 as described on Brightspace AE5051. Retroactive approval/exemption is NOT possible.	
Assessment	There are four milestones in the evaluation of the internship: 1. Internship Approval All assignments must be individually approved in advance by the Internship Office. Assignment: academic MSc level, analytical and related to your MSc AE study. No later than four weeks before students start their internship they must hand in the "Approval Form" to the Internship Office via the OnStage system (tudelft-onstage.xebic.com). In this form students have to describe in detail a clear assignment/project students will work on, including the planned deliverables and outcomes, their choice regarding the "Engineering Profession", and they also have to provide a "Self Evaluation" in which they define their personal learning goals for the internship period, and their plan how to fulfil them. Course list from Osiris with obtained EC points has to be uploaded in OnStage under study progress. 2. Midterm evaluation When their internship is half-way students have to fill in the Evaluation Form together with their internship supervisor. Once filled in and signed by the supervisor, students upload this evaluation form in the OnStage system. 3. Final evaluation At the end of the internship, the supervisor fills in the Evaluation form once again (same grading form as the half-way evaluation form). This final evaluation form will be included in the internship report and can be of help when writing the Self-Reflection chapter. Students have to make sure to have received the final evaluation form from their internship supervisor before leaving the company. 4. Internship report Once students have finished their internship, they must upload a digital version of their internship report in the OnStage system no later than 2 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. Students have to write their report according to the guidelines for internships reports that can be found on Brightspace AE5051. The digital Onstage system has been built to treat content strictly confidential, this means that all reports can be handed in online. Please consult Brightspace AE5051 for more information on confidentiality, guidelines, and forms to be included. The internship report is assessed at three criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form);	

2. Dedicated assignment about the Engineering Profession; topic to be determined before start of the internship;
3. Personal reflection on student's personal professional performance in the internship and the personal learning goals they defined in the Self-Evaluation prior to the start.

The internship report is first checked by the Internship Office, to see if it meets all the requirements and can then be forwarded to an assessor. It is then assessed by the assessor and can be rejected if the student does not meet the requirements as mentioned on Brightspace AE5051, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out.

Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is accepted for grading by the Internship Office and forwarded to an internship report assessor. The result of the report will be made public no later than at the last day of the period following the period in which the report was handed in.

Enrolment / Application

Officially registered and enrolled as a full-time Master student Aerospace Engineering.

Special Information

The Joint Interdisciplinary Project (JIP) is accepted as an alternative for the internship at Aerospace Engineering. It is a 10 week full time project (15EC) and the course will run between September and November with student teams of all disciplines (for AE students: an additional elective course of at least 3EC is required). The projects are co-created by students, academic staff and industrial companies and are scaffolded by workshops about professional tools and methods that will be used in the project.

The JIP is a good option for students who are eager to finish their MSc study within two years and want to learn how to work and think in an interdisciplinary and holistic industrial challenge. Note that there is a selection procedure for the JIP. You can only join the project if you have been selected.

More information can be found at <https://www.jointinterdisciplinaryproject.nl>

Set-up

1. Each internship has to be approved by the Internship Office prior to the start.

The responsibility of determining whether or not an internship opportunity meets the relevancy criteria rests with the Internship Coordinator. It is the students responsibility to receive approval from the Internship Coordinator no later than four weeks before the start of their internship. Handing in an approval form late might result in not getting your internship approved in time, and as a consequence not knowing if your internship assignment meets all the requirements. Consequently, no credits will be awarded for the internship. For the approval it is necessary to register your internship in the Onstage system (tudelft-onstage.xebic.com).

2. Students are free in their choice of internship. It is not necessary that the company or assignment is aligned with the specialization of the chosen track.

3. The internship can be held at the same company/research institute/other university as where the Final Thesis is done. The internship and thesis assignments must be two individual assignments. The internship cannot form the literature study for the thesis. The two can be somewhat related, but cannot be the same project, and require different deliverables. Students need to finish their internship before starting their thesis.

4. Students are responsible for arranging their own internship.

5. Internships at start-up companies have to meet additional criteria to regular internships. There is a start-up questionnaire available to be filled in by start-ups, through which the Internship Coordinator can check whether it meets all the requirements for students to do their internship at. Please email the filled-in questionnaire to the Internship Office to have it checked.

6. Doing an internship at your own start-up company is not allowed.

7. Activities in one of the TUD Student Teams can never count as an internship. For example: Dream Teams, IRP Delft, Science without Borders.

AE5222	Thesis Flight Performance & Propulsion	45
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The MSc programme in Aerospace Engineering is completed with a MSc thesis, the graduation project, which comprises 45 EC. The thesis is intended to provide students with the ability to do in-depth research in a particular area of their interest and make a considerable contribution to the knowledge of that area of expertise. Students have to address (a) specific research question(s) and are responsible for the planning and execution of the project. Before starting your thesis a few conditions have to be met: - You must have finished the first year of the Master's programme, with a maximum of 5 ECs outstanding. - You have informed your supervisor about your topic. - You have ensured your course list in MyTUDelft/Osiris is correct. If a student wants to do their thesis before their internship, please refer to the Implementation Regulations (available on the AE Student Portal), Paragraph 3, Article 9, Only for Masters Degree Programme, under 3: In exceptional circumstances, the student can request to complete the internship after the thesis via the Internship Office. Students can carry out their research within the facultys research groups, where the student normally becomes a member of a research team working within a particular area. Alternatively, students can also work outside the faculty for their thesis and do their thesis at a company or institute. At the start of the project, the student prepares a thesis plan describing research questions and defining a number of milestones including deliverables, which is then approved by the supervisor. During the course of the project, you are encouraged to maintain frequent contact with the supervisor, and possibly other members of the research staff as well. At each meeting, the supervisor provides feedback on your performance. At the end of the process, you submit a final report and present your work. During the thesis defence, students will defend their work in front of an Assessment committee. Students are expected to include a comprehensive and critical synthesis of the relevant literature and also embody a substantive contribution to theoretical knowledge, an empirical investigation or a design contribution in their thesis. More (detailed) information can be found on Brightspace: Master thesis Aerospace Engineering	
Study Goals	During the MSc thesis, the student will show their knowledge, understanding and skills at an academic master's level with respect to independently planning and executing a research or design project. On completing the thesis, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - Produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into - Scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	The format is self-study, but the student is supervised by one of our academic staff members, sometimes combined with one of their post-docs or PhDs, and/or a supervisor from a company or other (research) institute. The thesis supervisor(s) will provide assistance and direction to the student until the thesis has been successfully defended. Milestone meetings with the thesis supervisor(s) are mandatory and additional meetings to review your progress are encouraged. Available thesis topics can be found on the Brightspace community pages for each track and can also be discussed with the profile coordinator. Students are free to select a thesis topic from any member of staff within the faculty provided that the supervisor(s) of their choice feels that the student has obtained sufficient prior knowledge for the topic Please note that in some cases you will not be allowed to do your thesis in industry. The MSc Track Coordinator has the final say in this matter. Students are welcome to propose their own thesis topic of choice or acquire a topic outside the university on their own, but must be aware that the final approval of such a topic lies with the TU Delft supervisor(s) and the Track Coordinator. Topics deemed of insufficient academic level or too far outside the realm of expertise of the department will not be approved.	
Assessment	When you have started your thesis you must organize a Kick-Off meeting with your supervisor(s) to define the timeline of your project (Research Proposal review, Mid-Term review, Green-Light review and Thesis defence) and fill out the MSc Kick-off form (available on the AE Student Portal) together with your supervisor(s). Please submit all forms in accordance with the instructions as detailed on the appropriate form. At the end of the thesis, students are required to defend their thesis before an Assessment committee. The assessment is performed by a group of experts, based on the final thesis, a public presentation and a private defence session. The public presentation is held in front of a broad audience and should last about 30 minutes plus 15 minutes for discussion and is followed by a one-hour private defence with the Assessment committee, where the student answers questions from their supervisor(s) and external experts. For assessment of the thesis work, the following aspects are considered: 1. Content and scientific & engineering approach (choose design or scientific significance) 2. Literature study & research proposal 3. Report 4. Presentation & defence 5. Creativity & initiative 6. Project management. The assessment of the final thesis will be done based on the final objectives of the MSc curriculum. The faculty has introduced a new schematic using rubrics for this, for students who started the master's in 2022 and after. For students who started the master's in 2021 and before, the old rubrics will be used (both available on the AE Student Portal). If a student fails to meet the thesis requirements at any stage during the review milestone meetings, they may receive a 'No-Go'	

(fail). Please refer to Brightspace for more information.

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Instruction if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Process & Energy	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Power and Propulsion

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4115-23	Experimental Simulations	4
Responsible Instructor	Dr.ir. T. Sinnige	
Instructor	Dr.ir. W.J. Baars	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling; fractional analysis; dimensional analysis (Buckingham-Pi theorem); rigorous methods of obtaining dimensionless parameters; 3 Fundamentals of aerodynamic similarity: fundamentals of aerodynamics necessary for the interpretation of similarity between aerodynamic test results and full-scale; 4 Mach number and Reynolds number effects: practical considerations with respect to implications of dissimilarities between model-scale testing and full-scale applications; 5 Wind tunnel facilities: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 6 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 7 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 8 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 9 Design of experiments: factorial designs, response-surface models; 10 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report; 11 Engine integration: engine simulators, performance of those and their similarity with full-scale engines; 12 Acoustic testing: noise generation mechanisms and their appropriate scaling; definition of sound pressure levels; measurement requirements and devices; far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report;	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stromungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports (Q2 + Q3) and oral exam (wk 3.10 or 4.1).	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter.	

AE4120	Viscous Flows	3
Responsible Instructor	M. Kotsonis	
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Instructor	Dr.ir. T. Michelis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4136-24	CFD 2: Discretisation Techniques	5
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results. After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignement and oral examination.	
Enrolment / Application	Brightspace	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Responsible Instructor	Dr. S.J. Hulshoff	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4140	Gas Dynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 none	
Course Language	English	
Expected prior knowledge	AE2130-III Compressible aerodynamics (chapter 7 - 12 of Fundamentals of Aerodynamics by J.D. Anderson) WI3150TU Partial differential equations	
Course Contents	Knowledge of computer programming (Python, Matlab or similar) in order to successfully complete the tasks - How to apply the basic laws of mechanics and thermodynamics to describe compressible flows. - How can we properly describe a compressible flow field using the Euler equation + jump relations. - Understand the concept of characteristics and invariants in the context of linear and non-linear flows. - How to apply characteristics for non-linear flows for the computation of isentropic unsteady flows. - Understand the relation between shock waves and characteristics. - Application of Hugoniot and Poisson curves to solve a Riemann problem. - Definition of characteristics for 2D steady flows, and similarity with 1D unsteady flows (time-like and space-like). - How can use the method of characteristics and method of waves to compute a 2D compressible flow field. - Investigate the effect of viscosity and heat transfer in a 1D flow (Fanno and Rayleigh flow).	
Study Goals	At the end of this course, the student will be able to: - Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. - Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow and discuss the terms. - Derive the jump relations for the Euler equations and describe their relation to shock waves. - Discuss the role of entropy in combination with the jump relation for a correct description of a flow field. - Explain the concept of characteristics and invariants for 1D unsteady and 2D steady flows and how to use them for flow field computations (MOC). - Understand the role of characteristics in shock wave formation, elaborate on the theory of simple waves. - Derive the equations governing 1D flows through channels and nozzles in presence of viscosity and heat transfer. Explain the physical phenomena and processes that occur.	
Education Method	Lectures and self study	
Literature and Study Materials	Reader (can be downloaded from Brightspace) Further recommended literature M.J. Zucrow, J.D.Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. John D. Anderson jr., Modern Compressible Flow with Historical Perspective 2nd edition MacGraw-Hill 1990; H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001	
Prerequisites	Compressible aerodynamics Fundamentals of Aerodynamics by J.D. Anderson ch 7 - 10	
Assessment	Take home tasks + oral examination	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	M. Li	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Introduction to measurement techniques 2. Flow visualization (classroom assignment) 3. Pressure measurements 4. Hot Wire Anemometry (lab exercise) 5. Laser Doppler Velocimetry 6. Particle Image Velocimetry (classroom assignment+lab exercise) 7. InfraRed Thermography 8. Schlieren, Shadowgraphy and Background Oriented Schlieren 9. Fundamentals of data processing	

AE4204	Knowledge Based Engineering	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ing. A. Heidebrecht	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some familiarity with Object Oriented modelling and (Python) programming	
Course Contents	The course will provide a theoretical and practical introduction to Knowledge Based Engineering (KBE) technology, as a means to address complex engineering problems through design automation/digitalization principles. Other addressed topics, strictly related to KBE include - UML/SysML modelling language (class and activity diagrams) - object oriented modelling paradigm	
Study Goals	The main goals are the following: - understanding of Knowledge Based Engineering theoretical foundations and programming principles and their application to support complex engineering design problems - hands-on experience in developing simple but representative KBE applications by means of a provided KBE system - ability to model product and process knowledge by means of UML/SysML	
Education Method	The course is organized into a set of interactive lectures and tutorials to be performed part in class and part at independently at home. During the tutorials, students are introduced to the KBE programming approach. After a quick refresh of the python language, the main functionalities of a commercial KBE system will be illustrated, through interactive programming sessions. A significant amount of time is left for self-study, where students will have to develop a KBE application in teams of two, to address an engineering design problem of their choice. Attendance to lecture and tutorials is not compulsory but very highly recommended. Although the course is spread over two periods, the second one (Q4) is reserved for self-study (i.e. the development of the KBE application) and the oral examinations (See assessment section)	
Computer Use	The use of a personal computer is required, with the Windows OS installed A selected KBE system will be used for the practical sessions and assignment (temporary licenses for this course will be provided)	
Course Relations	This course has a logical connection to the MDO course AE4205, being KBE an enabling technology to multidisciplinary design, analysis and optimization. However, the MDO course is not propaedeutic to this course, thus students who did not followed MDO can take the KBE course without any problem	
Literature and Study Materials	- lecture sheets and some lecture notes (only available digitally on Brightspace) - a few scientific papers - several programming tutorials and code samples	
Books	Suggested (not compulsory) book: Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 Please note that only Chapter 9 is required for this course (single chapters can be purchased on line at Wiley). The rest of the book is suggested literature for the MDO course AE4205 The book is available as e-book through the TU Delft library	
Assessment	There is no conventional written exam for this course. The assessment is based on the quality of a KBE app to be developed at home by teams of two students and demonstrated live during an oral examination session. Two examination sessions are planned per academic year. The KBE app is supposed to address a relevant engineering design problem, which will be proposed by the student teams in the early phase of the course. The proposed KBE assignment descriptions will require lecturer's approval. If during the oral examination the quality of the developed KBE application and its demonstration are evaluated to be unsatisfactory, the team will have to use the retake session. For the retake session, two options are offered: 1) submit and present an improved version of the app proposed for the nominal session. One penalty point applies. 2) submit and present a totally different app than proposed for the nominal session. No penalty points. The new KBE assignment proposal will have to be delivered after the nominal session and will require lecture's approval. The oral examinations (nominal and retake sessions) are taking place around the beginning and the end of the second period of the course (i.e. 4th quarter of the academic year), not necessarily during the standard exam weeks. Delivery dates and examinations days are announced at the beginning of the course	
Special Information	Temporary student licenses of ParaPy, the KBE system used in this course, will be provided only to the students actively participating to the course. Thus, not to guests or other participants enrolled via Brightspace. The licenses shall be used exclusively for the course assignment.	
Tags	Artificial intelligence Computer Engineering Group work Integrated Modelling Programming Programming concepts Programming Software Software Engineering	

AE4240	Advanced Aircraft Design I	4
Responsible Instructor	Dr.ir. R. Vos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The design of high-subsonic transport aircraft is the topic of this course. The main goal is to demonstrate how the geometry of the aircraft can be shaped to ensure favorable flow conditions that satisfy requirements on performance, stability and controllability. The course is divided in the following lectures: 1. Design objectives 2. Boundary-layer aerodynamics and subsonic cruise drag. 3. Relation between supersonic velocity and pressure coefficient. Relation between geometry and pressure distribution. 4. Interference drag and area ruling. 5. Pressure distribution on airfoil sections at high and low speed. Development of the transonic airfoil. Design of supercritical and sonic airfoils. 6. Reynolds number effects. 7. Low-speed stalling characteristics of airfoil sections. Reynolds number and Mach number effects on maximum lift coefficient. High-speed stall (buffet) on airfoil sections. 8. Airfoils equipped with high lift devices: mutual interaction, types of flow separation. 9. Development of the swept-wing concept: historical background, principle of wing sweep, transonic flow characteristics, effect on boundary layer, effect on maximum lift coefficient, fences. 10. First generation of swept wing aircraft: tip-stall, aeroelastic deformation. 11. Pressure distribution on finite swept wings: spanwise drag distribution, forward sweep, fuselage interference, improving velocity distribution at root and tip, minimizing pitching moment. Design considerations for pressure distributions on finite swept wings. 12. Stalling characteristics of full scale aircraft: stall requirements, complexity vs. maximum lift, spanwise stall onset, deep stall. 13. Horizontal tail surface design: functions and requirements, planform shape, aeroelastic effects, elevator lock. 14. The buffet onset boundary: requirements, effect on flight envelope. Flight characteristics between MMO and MD: requirements, Mach tuck, Mach trim compensation. 15. Vertical tail surface design: functions and requirements, planform shape, effect of dorsal fin. Control surface design: functions and requirements, linearity, aeroelasticity, stall. 16. Wing control surfaces: spoilers and ailerons 17. Propeller slipstream effects: effect on spanwise lift distribution, effect on longitudinal balance, effect on directional balance in OEI condition. 18. Engine intake design: lip design for take-off and cruise conditions, drag in OEI condition. 19. Exhaust design: effect of ambient pressure, boat tail drag. Thrust reversers: types, interference with empennage. 20. Examples of modern wing design	
Course Contents	In this course it is demonstrated how the exterior design of high-subsonic transport aircraft can be manipulated in order to meet specified performance objectives and satisfy relevant certification constraints. It is shown what geometric characteristics of the airplane allow for a high operating Mach number while keeping fuel burn at acceptable levels. In particular, it is demonstrated how the exterior of all major airplane components can be shaped such as to adequately handle low-subsonic and transonic flow. It is demonstrated how the aerodynamic design and the operation at low speed and high speed affects the airplane in terms of balance, stability, aeroelastic behavior, weight, controllability, cruise performance, and field performance. Through many historic and contemporary examples the student learns to relate the functionality of high-subsonic transport airplanes to their external design.	
Study Goals	Students are able to: 1. Recognize the aerodynamic phenomena occurring around external shapes at a variety of conditions such as Mach number, Reynolds numbers or angles of attack. 2. Derive how these phenomena can be affected by the detailed external shape and interposition of various airplane components such as wing, fuselage, tail, nacelle, pylon, and fuselage. 3. Show how wing and tail movables can be applied to locally affect the flow at low subsonic and/or high-subsonic conditions. 4. Explain the effect of the aerodynamic characteristics on the airplanes cruise performance, field performance, stability, balance, and controllability. 5. Identify the advantages and disadvantages of various external design features on the airplanes aerodynamic performance and cross-disciplinary characteristics such as aeroelasticity, weight, or practical limitations resulting from the operational use of the aircraft.	
Education Method	Lectures	
Literature and Study Materials	Mandatory textbooks: - Aerodynamic Design of Transport Aircraft, Obert, E, IOS Press, 2009 - Introduction to Transonic Aerodynamics, Vos, R. and Farokhi, S, Springer, 2015	
Prerequisites	Aerospace Design and Systems Engineering Elements I Aerospace Design and Systems Engineering Elements II Aerodynamics I (subsonic aerodynamics) Aerodynamics II (supersonic aerodynamics) Systems Engineering and Aerospace Design Flight Dynamics and Simulation	
Assessment	Written exam	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor	Dr. D. Ragni	
Instructor	Prof.dr.ir. M. Snellen	
Co-responsible for assignments	Dr. R. Zamponi	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources.	
Study Goals	The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Physical intelligence for interaction - Insect intelligence and Swarming - Spiking neural networks	
Study Goals	After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.	
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4431-23	Aircraft noise	3
Responsible Instructor	Prof.dr. D.G. Simons	
Gast Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is required for AE4463P-23 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics (dBA meter) - Dutch aircraft noise model NRM or airframe noise model (ANOPP method) - digital signal analysis and acoustic imaging	
Expected prior knowledge	Basic knowledge on calculus and first year physics.	
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and tert's band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phone and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. Acoustic imaging: Principle of beamforming. Imaging aircraft noise data. (Note: this topic is only touched upon in this course, the details (including application) are treated in AE4463P-23) Aircraft noise sources: Mechanism of propeller noise and blade passage frequency. Mechanism of turbo engine noise. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise. ANOPP modelling method. Programming assignment about modelling airframe noise (landing gear, flaps, slats, etc.) using the ANOPP method.	
Study Goals	Understand the relationship between aviation and the resulting noise levels, especially around airports.	
Education Method	Lectures, exercises, programming assignment	
Literature and Study Materials	Lecture slides, exercises with worked-out solutions, programming assignment Reader: Course AE4431-23 Aircraft Noise Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2023	
Assessment	- Written exam (max 8 points) (a formula sheet will be provided) - programming assignment (max 2 points)	
Set-up	Lectures, exercises, programming assignment	

AE4462-17		Aircraft Emissions and Climate Effects	4
Responsible Instructor	Prof.dr. V. Grewe		
Instructor	Dr. I.C. Dedoussi		
Responsible for assignments	V.R. Meijer		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	3 4		
Course Language	English		
Expected prior knowledge	Basic knowledge on calculus, fluid mechanics and first year physics.		
Course Contents	The course introduces the students in the state-of-the-art capabilities, research issues and challenges in climate effects of aviation. Topics include: - Combustion and emissions - Overview of all environmental effects of aviation - Air pollution relevant atmospheric chemistry and physics - Local air pollution modelling - Aviation environmental policy, new technologies, and other mitigation options - General circulation of the atmosphere - Atmospheric chemistry and contrail formation - Climate change caused by air traffic emissions - Modelling of climate impact from aviation - Application and show cases for the assessment of mitigation options supporting decision making, e.g. for climate optimized trajectories		
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the climate impact of air traffic. These elements are required, for example, for the assessment of the air quality and climate impact of aviation and mitigation measures, including climate optimised trajectories/routing, low climate impact routing and consequences for aircraft design and supersonic transport.		
Education Method	Lectures and exercises		
Literature and Study Materials	Lecture slides and hand-outs		
Assessment	Written exam (Resits may be oral, depending on the situation) Assignments Both the written exam and the assignments need to be passed with a 6.0 or higher.		

AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course AE4463P-23 aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) must be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and also applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B, C and F are required. A choice can be made between D and E.	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods.	
Study Goals	Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Education Method	At the end of this course students are able to: 1. Explain the different steps in a finite element analysis and apply them to practical engineering problems 2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function) 3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up 4. Use and interpret results from 1D and 2D elements in commercial finite element software 5. Explain and perform verification and validation of results obtained using finite element principles	
Assessment	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace. Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments (only assignment 3 and 4 since they have the highest weight; assignment 1 and 2 cannot be redone). Results for individual assignments are only valid for one academic year: when re-taking the course all assignments have to be done in that academic year.	

AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
Responsible Instructor Responsible Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language	C. Kassapoglou Dr. D. Zarouchas 0/0/4/0 3 3 none English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis 3. Reliability analysis & Health monitoring a. Probability of Failure & Uncertainty quantification b. Structural Health Monitoring 4. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Understand the philosophy of reliability analysis and calculate the probability of failure of composite structures by combining numerical methods and the mechanics of composite structures - Understand the importance of structural health monitoring (SHM) and design a SHM system for a given composite structure - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Classical Laminate Theory: 3 2. Progressive damage analysis: 3 3. Reliability analysis & Health Monitoring: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory, progressive damage analysis, reliability analysis and health monitoring. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 10 points A2. Progressive damage and damage tolerance analysis 20 points A3. Reliability analysis & Health Monitoring. 30 points	

Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method).

B1. Design problem 40 points

The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2,A3 and part B1) should at least be rewarded with 55% of the attributed points ($55\% \cdot 60$ and $55\% \cdot 40$), otherwise the student will fail. In addition, the overall average grade should at least be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3 4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses All students will have a task in the production team as well as at least one additional organizational task.	
Assessment	Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	The grade for this course will be a Pass/Fail grade. This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Alternatively, having earned a Bachelor degree in aerospace engineering will give access to the course as well. Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December - Week 4.8: 15 May - Week 1.6: 30 September	

Set-up	The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and 3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3 There is an intentional overlap so that continuity and handover is guaranteed

ET4117	Electrical Machines and Drives	4
Responsible Instructor	Dr.ir. H. Vahedi	
Responsible Instructor	Dr. G.R. Chandra Mouli	
Instructor	W. Shi	
Responsible for assignments	H. Dialani	
Responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2014 (7th edition) ISBN 9781259254666	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Prof.dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Dr. O. Nejadseyfi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be 5.8 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest decimal. The maximum bonus score is 0.8. The bonus only applies to reports with grade ≥ 5.5 and submitted before the first report deadline.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	ME Department Precision & Microsystems Engineering	
Contact	Prof.dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

MS43310	Materials at High Temperature	4
Responsible Instructor	Dr. V.A. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour
Study Goals	Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on Brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Materials Science & Engineering

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: - Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. - Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. - Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. R. Kortlever	

Year

Organization

Education

2024/2025

Aerospace Engineering

Master Aerospace Engineering

Track 6: European Wind Energy Master Track Rotor Design	
Program Coordinator	W. Yu

Year

Organization

Education

2024/2025

Aerospace Engineering

Master Aerospace Engineering

Profile I: Aerodynamics	
Program Coordinator	W. Yu

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Responsible Instructor	W. Yu	
Gast Instructor	Dr.ir. T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic aerodynamics, level BSc Aerospace Engineering.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade elementmomentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsens Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	M. Li	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Introduction to measurement techniques 2. Flow visualization (classroom assignment) 3. Pressure measurements 4. Hot Wire Anemometry (lab exercise) 5. Laser Doppler Velocimetry 6. Particle Image Velocimetry (classroom assignment+lab exercise) 7. InfraRed Thermography 8. Schlieren, Shadowgraphy and Background Oriented Schlieren 9. Fundamentals of data processing	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Prof.dr. D.A. von Terzi	
Instructor	Dr.ir. M.B. Zaayer	
Instructor	Dr.ir. D.A.M. De Tavernier	
Instructor	Dr. D. Zappalá	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control The use of standards for load calculations Other optional material (tbd)	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): - AE4W13 Site conditions for wind turbine design - AE4135 Rotor / wake aerodynamics - AE4W21-14 Wind turbine aeroelasticity - AE4W31 Floating offshore wind energy - ET4117 Electrical machines and drives - OE44135 Offshore wind support structures	

AE4W13	Site Conditions for Wind Turbine Design	3
Responsible Instructor	Dr.ir. W.A.A.M. Bierbooms	
Instructor	Dr.ing. R. Schmehl	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 Different, to be announced	
Course Language	English	
Expected prior knowledge	It is assumed that the student has a basic understanding of fluid mechanics and stochastics (on Brightspace self tests are available). A basic skill of Matlab is assumed: you should be able to read from and write to a text file, write and/or modify simple functions (m-files). Furthermore, it is recommended that you already have a basic knowledge of wind turbines; e.g. AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Course Contents	Introduction Meteorology Atmospheric Boundary Layer Fluid Dynamics Turbulence Wind power meteorology Measurement instruments Statistics Methods in turbulence Offshore wind climate External conditions in standards Wave and current climate and loading Example analysis site data Turbulence spectra Satellite remote sensing techniques Simulation of turbulent wind fields and gusts	
Study Goals	After successfully completing the course, students will be able to: Discuss the relevance of particular site data for wind turbine design, operation and siting. Analyse simple problems related to wind power meteorology Interpret scientific dissemination by experts on main points Apply analysis methods to convert basic site data to wind turbine design parameters	
Education Method	Lecture + assignments + practical	
Course Relations	Follow up courses AE4W09, AE4135	
Literature and Study Materials	www.eos.ubc.ca/books/Practical_Meteorology Chapters 12 to 14; Pijush K. Kundu, Ira M. Cohen, Fluid Mechanics, Elsevier. Lecture notes	
Prerequisites	AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Assessment	Written / digital exam (scheduled in the exam period after Q4). The resit is an written exam on basis of pass/fail (i.e. in case of a pass the grade 6 will be put in Osiris).	
Set-up	The course combines lectures and student assignments/practical. For several topics guest speakers will be invited.	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	W. Yu	
Instructor	Dr.ir. C.J. Simao Ferreira	
Gast Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Preferred: AE4ASM506 (Aeroelasticity). It's preferred that the student has completed or is attending this or an equivalent course.	
Course Contents	The course will cover the main structural and aerodynamic phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. What are the key characterized parameters for an aeroelastic system? 2. How to construct a structural model of a wind turbine blade? 3. How to define if resonance occurs to a system and understand its relevance to wind turbines and how to mitigate it? 4. How to define if instability occurs to a system and its relevance to wind turbines and understand the role of aerodynamic damping? 5. Dynamic inflow and dynamic stall. 6. How to construct an aeroelastic model by coupling the structural and aerodynamic effects?	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand the dynamics and loads of a wind turbine due to the strong coupling of a) external loads(aerodynamics) b) internal structural loads c) inertia loads; 2. solve the dynamic structural response based on the equations of motions; 3. Calculate the unsteady aerodynamic loads of a wind turbine under dynamic inflow and dynamic stall; 4. apply the knowledge to program their own aeroelastic models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE5912	Thesis Wind Energy Rotor Design	45
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Course Coordinator	Dr. J.J.E. Teuwen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The thesis project engages the student in innovative research on a topic related to wind energy. The student addresses a specific research question, and is responsible for the planning and execution of the project. The project may be carried out in industry (preferably one of the EWEM partners), but most are carried out within the Wind Energy research group or Aerospace Structures and Materials department of TUDelft or at DTU, where the student is normally engaged as part of a research team working within a particular area. Please notice that DTU has a strict deadline for submitting your final thesis report (deadline for delivery).	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	Self-study. Frequent meetings with the graduation supervisors, where feedback is given on the students progress and performance, and also suggestions for improvement. More information can be found on Brightspace; "organizations" - Wind Energy: https://brightspace.tudelft.nl/d2l/home/43780 (Aerodynamics Profile) - Aerospace Structures & Materials MSc Track: https://brightspace.tudelft.nl/d2l/home/43786 (Structures and Composites Profile) See also: https://www.tudelft.nl/lr/organisatie/afdelingen/flow-physics-and-technology/wind-energy and https://www.tudelft.nl/lr/organisatie/afdelingen/aerospace-structures-and-materials	
Assessment	The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty has introduced a new schematic using a rubrics for this. This assessment matrix is the same for all students who are doing an MSc in Aerospace Engineering. Your assessment will be based upon the work you have carried out during your thesis as well as the way you carried it out; your written thesis report; your oral presentation of your thesis work; and your defense of your thesis work in front of a committee of experts. This committee includes your supervisor, other members of the academic staff of the research group and an external member from another research group as well as any external supervisors and industry guests as applicable. For the detailed rules on the composition of your MSc thesis committee, please see the MSc Regulations on the AE-Airport.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Aerodynamics

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub.,1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Prof.dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Instructor	Dr. F. Taruffi	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

AE5055	Professional Training	6
Responsible Instructor	W. Yu	
Contact Hours / Week x/x/x/x	N.A.	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The aim of the professional training is to expose students to a real work environment for a period of 4 weeks (consecutive) on a full-time basis.	
Study Goals	- To increase your understanding of employment options after graduation - To get a better perception of your tasks and responsibilities within a professional environment - To create an understanding of the context in which wind energy is practiced by industry, institutes and organizations.	
Education Method	This is an elective. You have to arrange the internship yourself. Before you start you should ask me for approval about the specific topic (specify it in a 1 pager describing what you are going to do in your internship and how that is going to help you meet the learning objectives of the course). Deadline for topic approval and internship arrangements: 15 April (afterwards you are not allowed to do an internship anymore). Here you can find contact persons: https://www.tudelft.nl/lr/organisatie/afdelingen/flow-physics-and-technology/wind-energy/education/master-projects/msc-possibilities-industry You can of course also contact one of the EWEM partners. Also keep in mind that DTU nowadays does not provide much room for delay in your MSc: you have to start your thesis project in November and finish it in August. This implies that you have to do your internship during the summer.	
Assessment	Once students have finished their internship, they must submit a report to the responsible instructor of this course and your internship supervisor at the company no later than 4 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. The internship report is assessed on two criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form); 2. Personal reflection on student's personal professional performance in the internship and the attainment of the study goals of the course as outlined above. The internship report is assessed and can be rejected if the student does not meet the requirements outlined above, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out. Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is assessed. The result of the report (pass/fail) will be made public no later than at the last day of the period following the period in which the report was handed in.	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Profile II: Structures and Composites	
Program Coordinator	N. Eleftheroglou

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr.ir. C.M. de Zeeuw	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Compared to previous years the course starts in Q3 (not Q2). There will be a kick-off lecture in week 3.1	
Course Contents	Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four to five students about 8 day sections with support of staff are needed during this course. These will be planned on fore hand, preferably on fixed time slots. Most likely either Friday morning or Friday afternoon. We will not use all of those slots, but it will provide the flexibility to schedule things in the lab and make it easier to block their schedules. Finishing period As the starting period changed, the finishing period changes according, at the end of the fourth period (Q4, June) it is. The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Estimation of the maximal bending-torsion coupling for a strip of UD composite 4. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 5. Performing mechanical tests 6. Analysing the test results of the mechanical test, including failure analyses 7. Adhesion of thermoset laminates 8. Possibly preparation of lap shear specimens 9. Performing lap shear tests 10. Estimation of the mechanical properties of a sandwich panel 11. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 12. Performing a bending test on a sandwich panel. 13. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	This course was renewed in 2023-2024 by fixing the time schedule for students on beforehand.	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on beforehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
Responsible Instructor	C. Kassapoglou	
Responsible Instructor	Dr. D. Zarouchas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis 3. Reliability analysis & Health monitoring a. Probability of Failure & Uncertainty quantification b. Structural Health Monitoring 4. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Understand the philosophy of reliability analysis and calculate the probability of failure of composite structures by combining numerical methods and the mechanics of composite structures - Understand the importance of structural health monitoring (SHM) and design a SHM system for a given composite structure - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Classical Laminate Theory: 3 2. Progressive damage analysis: 3 3. Reliability analysis & Health Monitoring: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory, progressive damage analysis, reliability analysis and health monitoring. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 10 points A2. Progressive damage and damage tolerance analysis 20 points A3. Reliability analysis & Health Monitoring. 30 points	

Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method).

B1. Design problem 40 points

The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2,A3 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	lectures and practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Adisson-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Prof.dr. D.A. von Terzi	
Instructor	Dr.ir. M.B. Zaayer	
Instructor	Dr.ir. D.A.M. De Tavernier	
Instructor	Dr. D. Zappalá	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control The use of standards for load calculations Other optional material (tbd)	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): - AE4W13 Site conditions for wind turbine design - AE4135 Rotor / wake aerodynamics - AE4W21-14 Wind turbine aeroelasticity - AE4W31 Floating offshore wind energy - ET4117 Electrical machines and drives - OE44135 Offshore wind support structures	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	W. Yu	
Instructor	Dr.ir. C.J. Simao Ferreira	
Gast Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Preferred: AE4ASM506 (Aeroelasticity). It's preferred that the student has completed or is attending this or an equivalent course.	
Course Contents	The course will cover the main structural and aerodynamic phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. What are the key characterized parameters for an aeroelastic system? 2. How to construct a structural model of a wind turbine blade? 3. How to define if resonance occurs to a system and understand its relevance to wind turbines and how to mitigate it? 4. How to define if instability occurs to a system and its relevance to wind turbines and understand the role of aerodynamic damping? 5. Dynamic inflow and dynamic stall. 6. How to construct an aeroelastic model by coupling the structural and aerodynamic effects?	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand the dynamics and loads of a wind turbine due to the strong coupling of a) external loads(aerodynamics) b) internal structural loads c) inertia loads; 2. solve the dynamic structural response based on the equations of motions; 3. Calculate the unsteady aerodynamic loads of a wind turbine under dynamic inflow and dynamic stall; 4. apply the knowledge to program their own aeroelastic models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE5912	Thesis Wind Energy Rotor Design	45
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Course Coordinator	Dr. J.J.E. Teuwen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The thesis project engages the student in innovative research on a topic related to wind energy. The student addresses a specific research question, and is responsible for the planning and execution of the project. The project may be carried out in industry (preferably one of the EWEM partners), but most are carried out within the Wind Energy research group or Aerospace Structures and Materials department of TUDelft or at DTU, where the student is normally engaged as part of a research team working within a particular area. Please notice that DTU has a strict deadline for submitting your final thesis report (deadline for delivery).	
Study Goals	Learning objectives MSc Thesis During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research or design project. On completing the Graduation Project, the student should be able to: - Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; - Formulate a theoretical and/or experimental framework and delineate a research and/or design problem such that it can be solved; - Interpret obtained results in a critical manner; - To produce results with scientific/technological relevance that can be published in scientific literature; - Critically reflect on the work performed at the level of their peers in their particular field; - Present the work performed in a well-structured report or a draft scientific paper that incorporates verification of methods and tools and is written in correct English; - Present work performed in a structured way through an oral presentation to their peers and wider audience; - Defend their work in presence of their peers where the work is placed into scientific context and explain the contribution to the field; - Plan the project efficiently considering resources and methodology; - Execute the project controlling the process, effectively using the plan.	
Education Method	Self-study. Frequent meetings with the graduation supervisors, where feedback is given on the students progress and performance, and also suggestions for improvement. More information can be found on Brightspace; "organizations" - Wind Energy: https://brightspace.tudelft.nl/d2l/home/43780 (Aerodynamics Profile) - Aerospace Structures & Materials MSc Track: https://brightspace.tudelft.nl/d2l/home/43786 (Structures and Composites Profile) See also: https://www.tudelft.nl/lr/organisatie/afdelingen/flow-physics-and-technology/wind-energy and https://www.tudelft.nl/lr/organisatie/afdelingen/aerospace-structures-and-materials	
Assessment	The assessment of the Master thesis will be done on the basis of the final objectives of the MSc curriculum. The faculty has introduced a new schematic using a rubrics for this. This assessment matrix is the same for all students who are doing an MSc in Aerospace Engineering. Your assessment will be based upon the work you have carried out during your thesis as well as the way you carried it out; your written thesis report; your oral presentation of your thesis work; and your defense of your thesis work in front of a committee of experts. This committee includes your supervisor, other members of the academic staff of the research group and an external member from another research group as well as any external supervisors and industry guests as applicable. For the detailed rules on the composition of your MSc thesis committee, please see the MSc Regulations on the AE-Airport.	

WM0324LR	Ethics and Engineering for Aerospace Engineering	3
Module Manager	Dr.ir. U. Pesch	
Responsible for assignments	Dr.ir. U. Pesch	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/4/4/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students should have completed a considerable part of the curriculum in Aerospace Engineering	
Parts	Week arrangement Lecture and study material 1. Introduction 2. Code of ethics 3. Argumentation and reasoning 4. Ethics. 5. Risks and hazards of technology 6. Responsibility in and of organizations	
Course Contents	This course describes and analyses the responsibility of engineers in the light of philosophical, historical and juridical backgrounds. Topics covered include: Description and analysis of the problems encountered by engineers who want to act responsible. Codes of ethics for engineers. Argumentation and reasoning. Uncertainty, ignorance, risks, and their implications for responsible behavior. (Philosophical) ethics, the foundation of (criteria) for good and bad, right and wrong, responsible and irresponsible behavior. Responsibility within and of organizations; the role of law.	
Study Goals	The course has two major study goals: 1) To get acquainted with the theoretical insights and relevant concepts in ethics of technology 2) To experience how this theoretical knowledge could be applied to an engineering case The first will be assessed via weekly assignments on brightspace. The second will be assessed with an essay that students write in trios or pairs. After the course students should: - be able to recognize and analyze the ethical aspects and problems of their future professional practice and to conduct a solution-oriented debate about such problems; - have knowledge of relevant backgrounds (ethics, law, responsibility in and of organizations, historical developments).	
Education Method	Lectures and tutorials There are three plenary lectures, that are given in week 1-3. From week 2-6, there are small-group tutorials (compulsory attendance). During the lecturing period, students will also make assignments.	
Literature and Study Materials	1) Book: Ethics, Technology, and Engineering: An Introduction, by Ibo van de Poel & Lambèr Royakkers - ISBN: 978-1-4443-3095-3 (approximately 30 euro) Available at Waltman book shop at the entrance of Industrial Design building 2) Course material available on Brightspace	
Assessment	The grade will be based on weekly assignments that have to be made before each tutorial session. Furthermore the grade will be based on a number of additional reflection assignments that involve writing and discussing.	
Enrolment / Application	TIMELY ENROLLMENT on brightspace is COMPULSORY; deadline is TWO WEEKS PRIOR to the START date of the COURSE. In addition to a general enrollment in brightspace you should also enroll in a specific tutorial group. We cannot guarantee placement if you don't enroll for tutorials in time. See brightspace for further instruction.	
Remarks	Q3: priority for EWEM students	
Category	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/ MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Preferred Electives Profile Structures and Composites

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub., 1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of aerospace structures and structural components. The equation of motion for the dynamic response is formulated in matrix form using finite element discretization or direct modelling discretization (rigid bodies and flexible connectors). Various solution techniques are used for solving transient dynamic response and vibration problems, including modal analysis. The course also covers experimental techniques to obtain model parameters and test the response of a physical system. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Prof.dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Instructor	Dr. F. Taruffi	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

AE5055	Professional Training	6
Responsible Instructor	W. Yu	
Contact Hours / Week x/x/x/x	N.A.	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The aim of the professional training is to expose students to a real work environment for a period of 4 weeks (consecutive) on a full-time basis.	
Study Goals	- To increase your understanding of employment options after graduation - To get a better perception of your tasks and responsibilities within a professional environment - To create an understanding of the context in which wind energy is practiced by industry, institutes and organizations.	
Education Method	This is an elective. You have to arrange the internship yourself. Before you start you should ask me for approval about the specific topic (specify it in a 1 pager describing what you are going to do in your internship and how that is going to help you meet the learning objectives of the course). Deadline for topic approval and internship arrangements: 15 April (afterwards you are not allowed to do an internship anymore). Here you can find contact persons: https://www.tudelft.nl/lr/organisatie/afdelingen/flow-physics-and-technology/wind-energy/education/master-projects/msc-possibilities-industry You can of course also contact one of the EWEM partners. Also keep in mind that DTU nowadays does not provide much room for delay in your MSc: you have to start your thesis project in November and finish it in August. This implies that you have to do your internship during the summer.	
Assessment	Once students have finished their internship, they must submit a report to the responsible instructor of this course and your internship supervisor at the company no later than 4 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. The internship report is assessed on two criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form); 2. Personal reflection on student's personal professional performance in the internship and the attainment of the study goals of the course as outlined above. The internship report is assessed and can be rejected if the student does not meet the requirements outlined above, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out. Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is assessed. The result of the report (pass/fail) will be made public no later than at the last day of the period following the period in which the report was handed in.	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Master LR Keuzevakken

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AE4115-23	Experimental Simulations	4
Responsible Instructor	Dr.ir. T. Sinnige	
Instructor	Dr.ir. W.J. Baars	
Contact Hours / Week x/x/x/x	0/4/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Topics per lecture: 1 Introduction to course and lab exercise 2 Theory of similarities and scaling; fractional analysis; dimensional analysis (Buckingham-Pi theorem); rigorous methods of obtaining dimensionless parameters; 3 Fundamentals of aerodynamic similarity: fundamentals of aerodynamics necessary for the interpretation of similarity between aerodynamic test results and full-scale; 4 Mach number and Reynolds number effects: practical considerations with respect to implications of dissimilarities between model-scale testing and full-scale applications; 5 Wind tunnel facilities: classification of wind tunnels according to mode of operation and/or domain of operation in the non-dimensional aerodynamic space, power requirements, techniques to obtain high quality flow (turbulence, acoustics, pressure distributions etc.); 6 Boundary corrections: scaling from wind-tunnel environment to free flight - theory; 7 Boundary corrections: scaling from wind-tunnel environment to free flight - application; 8 Design of experiments: statistical approach to test design, randomization vs. one factor at a time approach; 9 Design of experiments: factorial designs, response-surface models; 10 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report; 11 Engine integration: engine simulators, performance of those and their similarity with full-scale engines; 12 Acoustic testing: noise generation mechanisms and their appropriate scaling; definition of sound pressure levels; measurement requirements and devices; far field measurements and noise source localization/identification; 13 Balance measurements (guest lecture); 14 Pre-test support session: discussion of questions/doubts that have come up while preparing the pre-test report;	
Course Contents	Experimental simulation (in aeronautics) deals with the systematic approach required to design and perform good experiments simulating flight. It looks at the deficiencies of scaling real flight into a ground testing simulation and defines the validity of simplifications required for testing. Particular attention is given to the simulation of aircraft propulsion and noise generation in ground testing facilities.	
Study Goals	At the end of this course, the student will be able to: - Reduce a simulation challenge to its dominant nondimensional scaling parameters - Identify the necessary hardware required for ground based testing of the identified parameters - Create an effective test plan for a low-speed wind-tunnel test to satisfy predefined measurement objectives - Reflect on the possibilities of and limitations to obtaining data from wind-tunnel tests to describe aircraft behavior in free flight - Evaluate the power integration effects on aircraft performance and noise for a propeller-driven aircraft	
Education Method	Lectures and lab exercise	
Literature and Study Materials	G.I. Barenblatt: Scaling; G.I. Barenblatt: Scaling, self-similarity and intermediate asymptotics; J. Zierep: Aehnlichkeitsgesetze und Modellregeln der Stroemungslehre; J.B. Barlow, W.H. Rae, A. Pope: Low speed wind tunnel testing; Liepman and Roshko: Elements of gas dynamics.	
Assessment	Group test reports (Q2 + Q3) and oral exam (wk 3.10 or 4.1).	
Set-up	Two lectures per week plus preparation of lab exercise in the second quarter. Execution and analysis of lab exercise and oral exam in the third quarter.	

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoustics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagrangian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub., 1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	

AE4120	Viscous Flows	3
Responsible Instructor	M. Kotsonis	
Responsible Instructor	Dr.ir. B.W. van Oudheusden	
Instructor	Dr.ir. T. Michelis	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic inviscid incompressible and compressible fluid dynamics, thermodynamics, (partial) differential equations	
Parts	The course material is covered in the book 'Viscous Fluid Flow' by F.M. White (see below)	
Course Contents	The transport equations of mass, momentum and energy for flows with viscosity and heat conduction: the Navier-Stokes equations; molecular transport properties; boundary layer simplifications. Incompressible laminar flows: exact solutions, self-similar and non-similar boundary layers; approximate (integral) methods for boundary layer computations. Laminar flows with thermal and compressibility effects. Stability of laminar flows; transition. Turbulent flows: basic concepts, law of the wall and defect law, equilibrium boundary layers, introduction to turbulence modelling.	
Study Goals	Understanding and operational knowledge of viscous flow concepts and their relevance to applications in the aeronautical domain.	
Education Method	Oral lectures (further information will be communicated through BrightSpace)	
Books	F.M. White, Viscous fluid flow, McGraw-Hill, 2006, 3rd ed ISBN 0-07-124493-X (or 2nd or 4th edition, depending on availability)	
Assessment	Individual task exam by appointment	
Special Information	More specific information can be found on Brightspace.	

AE4130-22	Aircraft Aerodynamics	4
Responsible Instructor	Prof.dr.ir. L.L.M. Veldhuis	
Instructor	L.T. Lima Pereira	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	After a short recap on inviscid flow the first part of the course deals with boundary layer theory. Topics addressed are: the laminar boundary layer, the transition process, the turbulent boundary layer, laminar and turbulent flow separation, the separation bubble, lift and drag. The second part of the course starts with general information on drag, useful for the aerodynamic design of aircraft. The course continues with the analysis and design of single and multi-component airfoils, illustrated by examples of CFD analyses and windtunnel experiments. Special topics like winglets, high lift systems, flow control and propeller propulsion for sustainable aircraft applications will be treated as well. Some aerodynamic analysis and design codes will be demonstrated during the course.	
Study Goals	The course is designed to provide the student with the basic theoretical and experimental tools for the aerodynamic design of aircraft. At the end the student will be able to apply basic aerodynamics concepts as well as some usefull design codes.	
Education Method	The lecture is based on oral presentations distributed over 14 lectures. During the lectures some design tools will be demonstrated.	
Books	Recommended literature: 1. J.D. Anderson, Fundamentals of Aerodynamics. Mc Graw Hill, 4th edition 2. J.J. Bertin, Aerodynamics for Engineers, Prentice Hall, 4th edition 3. J. Katz and A. Plotkin, Low-speed Aerodynamics - From being theory to panel methods, McGraw-Hill 4. M. Drela, Flkight Vehicle Aerodynamics, MIT press 5. H. Schlichting, Boundary Layer Theory, McGraw-Hill	
Reader	The Reader "Aircraft Aerodynamics Primer", ae4130-22, will be made available to the students.	
Prerequisites	To successfully follow this course the participant is expected to have a basic knowledge on inviscid flows and viscous flows at BSc level. A good indicator of key knowledge that is expected to be present is the material discussed in the book of J.D. Anderson: "Fundamentals of Aerodynamics". Chapters 1-6, and preferably Ch. 15 should have been studied prior to starting this lecture (or be studied during the first 3 weeks of Q1).	
Assessment	The assessment is done through 5 assignments that are performed on a personal basis. The assignments are to be combined in 1 report that should be delivered after the finalization of the lecture at the end of Q2. Additional information on the timing and contents of the assignments will be presented on BrightSpace.	

AE4135	Rotor / wake Aerodynamics	4
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Responsible Instructor	W. Yu	
Gast Instructor	Dr.ir. T. Sinnige	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic aerodynamics, level BSc Aerospace Engineering.	
Parts	The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.	
Course Contents	Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices Blade elementmomentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors. Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design. Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling. Vortex panel methods. Advanced wake models. Acceleration potential method. Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure. 3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight. Unsteady aerodynamics and dynamic stall effects. Theodorsens Theory. Effects of tower shadow and wind shear. Aeroacoustics and rotor aeroacoustics. Vertical axis wind turbine rotor and Voight-Schneider propeller Effects of inflow turbulence intensity on blade loads. Near and far wake structure Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.	
Study Goals	Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail. "Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.	
Education Method	Lecture and exercises + 3 group assignment + final written (digital) or oral exam	
Literature and Study Materials	Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students Recommended literature The recommended literature for each chapter will be indicated in the course lectures.	
Assessment	Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.	
Remarks	Fundamental knowledge of Applied Aerodynamics is required.	
Set-up	There will be three major group assignments that must be completed and reported during the course. The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.	

AE4136-24	CFD 2: Discretisation Techniques	5
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Final thesis work in structure-preserving discretization techniques	
Expected prior knowledge	Basic calculus and physics	
Course Contents	In these lectures the fundamental principles underlying the approximation of partial differential equations will be explained. This unifying approach will be applied to finite difference methods, finite volume methods and finite elements. The main focus will be on the approximation of physical models. Extensions to high order methods and the application to curvilinear domains will be addressed	
Study Goals	Understanding the structure of physical models and the discrete representation. Ability to implement numerical schemes. Interpretation of numerical results. After this course the student is expected to be able to write a simple Navier-Stokes solver. To be able to verify and validate the code and to interpret results obtained by numerical means.	
Education Method	Lectures	
Computer Use	For the final assignment a computer and Matlab or Python is required	
Course Relations	This course is part of the sequence of CFD lectures in Aerodynamics, but can be followed as an independent course	
Literature and Study Materials	Lecture notes	
Assessment	Final assignement and oral examination.	
Enrolment / Application	Brightspace	
Tags	Abstract Algebra Calculus Challenging Computer Engineering Fluid Mechanics Mathematics Matlab Programming Software Software Engineering	

AE4138-18	CFD 4: Uncertainty Quantification	3																																								
Responsible Instructor	Dr. R.P. Dwight																																									
Contact Hours / Week x/x/x/x	0/0/0/2																																									
Education Period	4																																									
Start Education	4																																									
Exam Period	4																																									
Course Language	English																																									
Course Contents	This course is a general introduction to the numerical-statistical field of Uncertainty Quantification (UQ), with a particular focus on methods relevant for fluid dynamics (both CFD and to a lesser extent experimental methods), and machine learning. Prerequisites - Probability and statistics (fundamentals) - Numerical analysis (fundamentals) - CFD (familiarity and interest) - Scientific programming in Python (basics) The latter is for completing the programming homework assignments. Python is recommended and supported, but other languages may be used if preferred.																																									
Study Goals	We aim for a broad overview of the field of uncertainty quantification, touching on all sub-topics of recent interest: The role of uncertainty in science and engineering The place of probability and statistics in describing and quantifying uncertainty Numerical methods for pushing probabilistic quantities through complex models Monte-Carlo Linear models and Taylor series approximation of nonlinear models Stochastic spectral techniques - polynomial chaos Quadrature rules on stochastic spaces Sparse-grid quadrature and interpolation Numerical methods for Bayesian inversion (calibration) Linear-Gaussian analytic results - Kalman gain Markov-chain Monte-Carlo sampling methods (MCMC) Maximum-a-posteriori estimates (MAP) Variational inversion of complex models (adjoints) Hessian approximations of posterior covariance Gaussian process regression (Kriging) Gaussian processes Kriging Generalizations - co-kriging, gradient-enhanced Kriging, etc. At several places we will build a connection to machine learning, and demonstrate how the numerical-statistical approach informs those methods, and addresses issues of identifiability, overfitting, etc. In practical terms, at the end of the course you should be capable of, e.g.: Propagating (specified) uncertainty in input parameters through a simulation code. Estimating the effect of experimental uncertainty on measurements. Performing (non)linear regression with appropriate statistical models and quantified uncertainty. Calibrating turbulence closure models in RANS or LES against experimental data. Combining data and simulation into a coherent single prediction (data-assimilation). Regressing data with Kriging. Identifying Some important topics we will not have time to touch on are: Kalman filtering Classical stochastic PDEs (though all our PDEs will be random)																																									
Education Method	Lectures In each lecture we generally consider a different numerical technique - in a self-contained way. I.e. covering all of motivation, problem statement, solution methods, and analysis of results. Most weeks a lecture will be followed by an in-class programming assignment. The course will be structured as follows: <table><tr><th>Week</th><th>Lecture 1 (45mins)</th><th>Lecture 2 (45mins)</th><th>Homework (~3-6 hours)</th><th>Reading</th></tr><tr><td>1</td><td>UQ motivation</td><td>Probability refresher</td><td>Reading</td><td>Ch. 1/4</td></tr><tr><td>2</td><td>Polynomial chaos</td><td>[PC for XFOil]</td><td>Finalize []</td><td>Ch 10.</td></tr><tr><td>3</td><td>Sparse grids</td><td>[Sparse grids]</td><td>Finalize []</td><td>Ch 11.</td></tr><tr><td>4</td><td>Gaussian-processes</td><td>Kriging surrogates</td><td>[Kriging]</td><td>Ch 13.</td></tr><tr><td>5</td><td>Adjoints</td><td>[Adjoint Poisson]</td><td>Finalize []</td><td>--</td></tr><tr><td>6</td><td>Variational inverses</td><td>[Inversion]</td><td>Finalize []</td><td>--</td></tr><tr><td>7</td><td>Markov-chain MC</td><td>Turbulence UQ</td><td>RANS calibration</td><td>Ch 8.</td></tr></table> ----- [] - In class computer assignment (with laptops) Square brackets [] imply coding exercises, in which the theory just learnt is applied. The exercises will be made available as the course progresses, are in Python, and are due by the following week, and are pass/fail. Each should take ~3-6 hours depending strongly on your level of programming. Note: since the tutorials will be in the lecture room, please bring laptops to the lectures! I will bring power-strips! The final project "RANS model calibration" is a little larger, and should take ~1 week to complete. It will be due within 1-2 months of the end of the course (date to be announced). Book Throughout the course we will use Ralph C. Smith's "Uncertainty quantification: Theory, implementation and applications", SIAM Computational Science and Engineering, ISBN 978-1-611973-21-1, 2014 (though any edition is fine). Selected reading from this book will be provided most weeks, consisting typically of ~10-15 pages/week - except where the book does not cover the topics discussed. Chapters in the above table are an indication - we don't expect you to read the complete chapters in most cases.		Week	Lecture 1 (45mins)	Lecture 2 (45mins)	Homework (~3-6 hours)	Reading	1	UQ motivation	Probability refresher	Reading	Ch. 1/4	2	Polynomial chaos	[PC for XFOil]	Finalize []	Ch 10.	3	Sparse grids	[Sparse grids]	Finalize []	Ch 11.	4	Gaussian-processes	Kriging surrogates	[Kriging]	Ch 13.	5	Adjoints	[Adjoint Poisson]	Finalize []	--	6	Variational inverses	[Inversion]	Finalize []	--	7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.
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6	Variational inverses	[Inversion]	Finalize []	--																																						
7	Markov-chain MC	Turbulence UQ	RANS calibration	Ch 8.																																						

	Python programming tutorials The 5 tutorials are in the form of IPython notebooks. These can be worked through on your own laptops if you have a complete scientific Python install (numpy, scipy, ipython, etc.) - Anaconda with Python 3 is recommended, see https://www.continuum.io/downloads. For those with no programming experience, previous students have recommended Codecademy, see https://www.codecademy.com/learn/python. The notebooks require only relatively basic Python knowledge, and some experience of using numpy. In addition, to ensure easy access for everyone I've setup a server at: https://ipython.lr.tudelft.nl:8888. The port number is important!! Your username is your NetID, and your password will be assigned upon enrolling in the course. The Module notebooks will appear on this server, as well as here on Brightspace. Moreover, on this server you can run the notebooks from within any browser, make modifications, plots, etc. Please just bear in mind that your code is not running locally, but on the server, which has limited capacity. So please don't perform long-running computations. Final project An optional final project involves a deeper dive into RANS model calibration using Markov Chain Monte-Carlo.
Assessment	Final grade consists of a combination of: 4 Python projects (code only, no student-written report) An oral exam OR a written exam (to be decided) All projects may be done alone or in pairs - orals are individual.
Elective	Yes

AE4139	CFD 3: Large Eddy Simulation	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Responsible Instructor	Dr. S.J. Hulshoff	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction to turbulence; scale separation; spectral turbulence theory for homogeneous turbulence; Reynolds stress balance in free shear layers and wall bounded turbulence; functional and structural subgrid-scale models for large-eddy simulation; resolution requirements and cost of large-eddy simulations; numerical methods for large-eddy simulation; analysis and validation of large-eddy simulation data; case studies.	
Study Goals	Understand fundamentals of turbulence theory for canonical flows. Understand mathematical foundations of large-eddy simulation, and the relation between physical, mathematical and numerical modeling. Be familiar with different concepts of functional and structural subgrid-scale modeling and multi-scale modeling. Be able to analyze turbulence data. Be aware of uncertainties and errors and be able to critically evaluate simulation results.	
Education Method	Lecture, exercises, and take-home assignments	
Assessment	Take-home assignments and written exam	
Elective	Yes	

AE4140	Gas Dynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 none	
Course Language	English	
Expected prior knowledge	AE2130-III Compressible aerodynamics (chapter 7 - 12 of Fundamentals of Aerodynamics by J.D. Anderson) WI3150TU Partial differential equations	
Course Contents	Knowledge of computer programming (Python, Matlab or similar) in order to successfully complete the tasks - How to apply the basic laws of mechanics and thermodynamics to describe compressible flows. - How can we properly describe a compressible flow field using the Euler equation + jump relations. - Understand the concept of characteristics and invariants in the context of linear and non-linear flows. - How to apply characteristics for non-linear flows for the computation of isentropic unsteady flows. - Understand the relation between shock waves and characteristics. - Application of Hugoniot and Poisson curves to solve a Riemann problem. - Definition of characteristics for 2D steady flows, and similarity with 1D unsteady flows (time-like and space-like). - How can use the method of characteristics and method of waves to compute a 2D compressible flow field. - Investigate the effect of viscosity and heat transfer in a 1D flow (Fanno and Rayleigh flow).	
Study Goals	At the end of this course, the student will be able to: - Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. - Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow and discuss the terms. - Derive the jump relations for the Euler equations and describe their relation to shock waves. - Discuss the role of entropy in combination with the jump relation for a correct description of a flow field. - Explain the concept of characteristics and invariants for 1D unsteady and 2D steady flows and how to use them for flow field computations (MOC). - Understand the role of characteristics in shock wave formation, elaborate on the theory of simple waves. - Derive the equations governing 1D flows through channels and nozzles in presence of viscosity and heat transfer. Explain the physical phenomena and processes that occur.	
Education Method	Lectures and self study	
Literature and Study Materials	Reader (can be downloaded from Brightspace) Further recommended literature M.J. Zucrow, J.D.Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. John D. Anderson jr., Modern Compressible Flow with Historical Perspective 2nd edition MacGraw-Hill 1990; H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001	
Prerequisites	Compressible aerodynamics Fundamentals of Aerodynamics by J.D. Anderson ch 7 - 10	
Assessment	Take home tasks + oral examination	

AE4143	Hypersonic Aerodynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Instructor	Dr.ir. L.M.G.F.M. Walpot	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	* AE2130-III Aerodynamics 2 (Compressible Aerodynamics) or equivalent. In case you did not have such a course, please study chapters 7 to 12 of Fundamentals of Aerodynamics by J.D. Anderson * It is very highly recommended to also follow AE4120 (Viscous flows) * AE4140 Gasdynamics is nice to follow as well but not absolutely necessary	
Parts	Lecture and study material 1. Introduction: characteristics of hypersonic flow; applications. 2. Inviscid flow: 2D shocks and expansions; Newtonian theory. 3. Tangent- wedge/tangent cone-, shock-expansion theories; similarity parameters. 4. Euler equations; small perturbation theory; hypersonic equivalence principle; blast-wave theory. 5. Thin shock layer theory; computational results. 6. Viscous flow: boundary layer equations; laminar boundary layers. 7. Transition; turbulent boundary layers. 8. Viscous interactions; flat plate application. 9. Heat transfer: convective heating; results for flat plate, cone, blunt body. 10. Laminar and turbulent stagnation region heating; radiation: effects on surface heating; radiation from the shock layer; nocturnal radiation. 11. High temperature effects; chemical reactions; inviscid equilibrium flow; difference between equilibrium flow and frozen flow. 12. Non-equilibrium flow; high temperature effects in flow through normal and oblique shocks. 13. High temperature effects in flow through nozzles. 14. Chemical reacting viscous flow	
Course Contents	Hypersonic Flow: Inviscid flow, compressions and expansions, surface inclination methods, similarity rules, hypersonic small perturbations, Euler equations. Viscous flow, boundary layers, viscous interactions, heat transfer, radiation effects. High temperature effects, inviscid and viscous equilibrium and nonequilibrium flow. Rarefied gasdynamics.	
Study Goals	This course is designed to provide the student with fundamental knowledge of hypersonic aerothermodynamics. The course is set up to deliver engineering tools for conceptual design problems together with a physical understanding of various aspects of the broad gamma of hypersonic flow.	
Education Method	Lecture	
Literature and Study Materials	J.D. Anderson, High temperature gas dynamics, , - ISBN 0070016712 Recommended literature J.J. Bertin, Hypersonic aerothermodynamics, , - ISBN 1563470365	
Assessment	Oral exam	

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	M. Li	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Introduction to measurement techniques 2. Flow visualization (classroom assignment) 3. Pressure measurements 4. Hot Wire Anemometry (lab exercise) 5. Laser Doppler Velocimetry 6. Particle Image Velocimetry (classroom assignment+lab exercise) 7. InfraRed Thermography 8. Schlieren, Shadowgraphy and Background Oriented Schlieren 9. Fundamentals of data processing	

AE4185	Fluid Flow Data Processing and Visualization	3
Responsible Instructor	Dr. D. Modesti	
Responsible Instructor	Dr. N.A.K. Doan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is a general introduction to data processing techniques used in fluid mechanics to analyse unsteady flows. Specifically, techniques related to the flow visualization and the postprocessing of numerical data will be presented. The overall content of the course is: 1. Review of fundamentals of turbulent flows (energy cascade, Reynolds-averaged equations, Reynolds stress, turbulent kinetic energy, vorticity equation, turbulence spectrum, energy budget, ...) 2. Characteristics of flow datasets (Nyquist criterion, sampling and resolution, first principle of visualization, vortex/Q criteria, ...) 3. Statistical analysis in flows - Deterministic/random data - Averaging (ensemble, condition, phase) - Probability: PDFs, statistical moments, conditional PDFs - Correlation function - Ergodic hypothesis - Statistical convergence and uncertainty 4. Temporal and spectral data processing - Fourier transform - Physical/spectral space and multi-dimensional FFTs - Auto/cross spectra - Integral length/time scale - Convolution (filtering) and transfer function 5. Modal analysis and coherent structure eduction - Proper Orthogonal Decomposition - Dynamic Mode Decomposition - Basics of reduced-order modelling	
Study Goals	At the end of this course, the student should be able to: - Visualize fluid mechanics datasets using relevant quantities - Analyse fluid mechanics datasets using statistical methods - Apply temporal and spectral analysis methods to analyse flow datasets - Postprocess fluid mechanics datasets using modal-based methods	
Education Method	The course will be a combination of lectures, scheduled tutorials and homework assignments.	
Assessment	The course will be assessed with a final assignment (60% of the final grade) and an oral exam (40% of the final grade). To get a passing grade (6.0 or higher), the student must get at least a 5.0 on both the final assignment and the oral exam.	

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4204	Knowledge Based Engineering	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ing. A. Heidebrecht	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some familiarity with Object Oriented modelling and (Python) programming	
Course Contents	The course will provide a theoretical and practical introduction to Knowledge Based Engineering (KBE) technology, as a means to address complex engineering problems through design automation/digitalization principles. Other addressed topics, strictly related to KBE include - UML/SysML modelling language (class and activity diagrams) - object oriented modelling paradigm	
Study Goals	The main goals are the following: - understanding of Knowledge Based Engineering theoretical foundations and programming principles and their application to support complex engineering design problems - hands-on experience in developing simple but representative KBE applications by means of a provided KBE system - ability to model product and process knowledge by means of UML/SysML	
Education Method	The course is organized into a set of interactive lectures and tutorials to be performed part in class and part at independently at home. During the tutorials, students are introduced to the KBE programming approach. After a quick refresh of the python language, the main functionalities of a commercial KBE system will be illustrated, through interactive programming sessions. A significant amount of time is left for self-study, where students will have to develop a KBE application in teams of two, to address an engineering design problem of their choice. Attendance to lecture and tutorials is not compulsory but very highly recommended. Although the course is spread over two periods, the second one (Q4) is reserved for self-study (i.e. the development of the KBE application) and the oral examinations (See assessment section)	
Computer Use	The use of a personal computer is required, with the Windows OS installed A selected KBE system will be used for the practical sessions and assignment (temporary licenses for this course will be provided)	
Course Relations	This course has a logical connection to the MDO course AE4205, being KBE an enabling technology to multidisciplinary design, analysis and optimization. However, the MDO course is not propaedeutic to this course, thus students who did not followed MDO can take the KBE course without any problem	
Literature and Study Materials	- lecture sheets and some lecture notes (only available digitally on Brightspace) - a few scientific papers - several programming tutorials and code samples	
Books	Suggested (not compulsory) book: Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszcanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 Please note that only Chapter 9 is required for this course (single chapters can be purchased on line at Wiley). The rest of the book is suggested literature for the MDO course AE4205 The book is available as e-book through the TU Delft library	
Assessment	There is no conventional written exam for this course. The assessment is based on the quality of a KBE app to be developed at home by teams of two students and demonstrated live during an oral examination session. Two examination sessions are planned per academic year. The KBE app is supposed to address a relevant engineering design problem, which will be proposed by the student teams in the early phase of the course. The proposed KBE assignment descriptions will require lecturer's approval. If during the oral examination the quality of the developed KBE application and its demonstration are evaluated to be unsatisfactory, the team will have to use the retake session. For the retake session, two options are offered: 1) submit and present an improved version of the app proposed for the nominal session. One penalty point applies. 2) submit and present a totally different app than proposed for the nominal session. No penalty points. The new KBE assignment proposal will have to be delivered after the nominal session and will require lecture's approval. The oral examinations (nominal and retake sessions) are taking place around the beginning and the end of the second period of the course (i.e. 4th quarter of the academic year), not necessarily during the standard exam weeks. Delivery dates and examinations days are announced at the beginning of the course	
Special Information	Temporary student licenses of ParaPy, the KBE system used in this course, will be provided only to the students actively participating to the course. Thus, not to guests or other participants enrolled via Brightspace. The licenses shall be used exclusively for the course assignment.	
Tags	Artificial intelligence Computer Engineering Group work Integrated Modelling Programming Programming concepts Programming Software Software Engineering	

AE4206-24	Turbomachinery	4
Responsible Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The course covers fundamental concepts of internal flows in turbomachines and related design aspects. The typical course schedule consists of the following lectures: Lecture 0: Introduction course overview: topics, structure and assessment Lecture 1: Recall of Turbomachinery Fundamentals & Principles integral governing equations with applications, relative motion, thermodynamic transformations in turbomachinery stages, velocity triangles with practical examples; Lecture 2: Unsteady Flows in Turbomachinery concept of reduced frequency with applications; work exchange in turbomachinery rows; Lecture 3: Similarity in Turbomachinery similarity analysis for incompressible flows, Balje and Smith diagrams, accounting for compressibility, stage dimensionless performance parameters; Lecture 4: Fluid-dynamic Design of Turbomachinery design procedure, mean-line design model, radial equilibrium model, CFD-based models Lecture 5-6: Axial and Radial Turbines examples of axial and radial turbines, typical turbine stages, blade profiles for turbines, flow characteristics in subsonic, transonic, and supersonic cascades, basic concepts of diffusers, aerodynamics of radial impellers Lecture 7-8: Axial and Radial Compressors examples of axial and radial compressors, typical compressor stages, blade profiles for compressors, flow characteristics and unique incidence, stall and surge instability, slip factor, diffusers for radial compressors, design maps Seminar(s): topic(s) may vary but usually include turbomachinery aero-mechanics, shape optimisation with CFD methods, rocket engine turbo-pumps, and advanced turbomachinery components for aero-engines Tutorial: training session on the software tools provided for the team project	
Course Contents	This course educates aerospace students to apply theoretical knowledge in the field of fluid- mechanics and thermodynamics for analyzing and designing a thermal turbomachinery. Typical examples are: axial and radial compressors of gas turbine engines, industrial gas and steam turbines, radial turbines of automotive turbochargers, turbo-expanders and compressors used in refrigeration and liquefaction industry, micro-turbines for power generation. Since the turbomachinery field is intrinsically interdisciplinary, it is assumed that students already possess basic knowledge of thermodynamics, fluid-mechanics, applied mathematics, propulsion systems, internal flows, acquired in previous courses. Prerequisite of the course are therefore bachelor and master courses addressing fundamental (e.g. Thermodynamics, Internal Flows) and applied aspects (e.g. Aero-Engines) of air-breathing engines. Starting from basic concepts of gas-dynamics and fluid-machinery, the course brings the student to the level that he is able to select a turbomachinery for a specific application, perform the conceptual design of the stages, and eventually predict the flow characteristics within the blade passages by means of conceptual physical models and more sophisticated simulation tools.	
Study Goals	After the course the student will be able to: Classify and illustrate the main features of a turbomachine Select the most appropriate turbomachine for a given aerospace application Apply first principles to simplified turbomachinery configurations Identify the most prominent loss sources of a given turbomachinery stage Define the aerodynamic blade design of turbomachinery cascades Predict the performance of a turbomachine with analytical and numerical methods Devise solutions for performance improvement of turbomachinery cascades based on physical understanding Perform the conceptual and detailed fluid-dynamic design of a turbomachine for propulsion systems using analytical and numerical tools	
Education Method	Lectures, concept quizzes, concept exercises, and tutorials in preparation to the team assignment.	
Books	- S.L. Dixon, C. A. Hall &#147;Fluid-Mechanics and Thermodynamics of Turbomachinery&#148;; 7th edition, 2013 - M. Schobeiri, &#147;Turbomachinery Flow Physics and Dynamic Performance&#148;; 2nd edition, Springer, 2012 - E.A. Baskharone, "Principles of Turbomachinery in Air-Breathing Engine", Cambridge University Press	
Assessment	The final course grade results from the weighted average of: the assessment of the final project assignment, counting for 40% of the final grade (individual score 1-10, no threshold, 1 bonus point for best project); the assessment of the written exam, counting for 60% of the final grade (individual score 1-10, minimum threshold: 6); The written exam takes place at the end of course. The exam consists of concept exercises covering the material discussed during classes. The final assignment is meant to challenge students with the real engineering problem of designing a turbomachine for propulsion or power generation application. The concept exercises are numerical exercises that are formulated to stimulate critical thinking on turbomachinery problems of practical application. They also serve as diagnostic tool to help students self-assessing their level of knowledge and understanding.	
Permitted Materials during Tests	Lectures notes, slides, formula sheets and textbooks are allowed during the written exam. Laptops can be used for calculations.	
Set-up	The course is organized in 8 teaching sessions, 2/3 seminar(s), and a possible visit to a turbomachinery company. Lectures usually consist of a theoretical part integrated with in-class exercises and quizzes to stimulate critical thinking and learning-by-doing. During classes, the students, autonomously or in groups, are engaged in a series of learning activities, supervised by the team of instructors. Their solution is briefly discussed at the beginning of the next class. The seminar(s) are devoted to show students current research topics in turbomachinery and are given by recognised experts in the field.	

AE4238	Aero Engine Technology	4
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	Dr. F. Yin	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is mandatory course for the MSc program in Fight Performance and Propulsion. This course is beneficial for other courses related to gas turbines like "Turbomachinery", "Combustion for Propulsion and Power Technologies" and "Modelling, Simulation and Applications of Propulsion and Power Systems".	
Expected prior knowledge	It is important that the students following this course should have the basic understanding of thermodynamic systems and fundamentals of gas turbine engines. Students should have followed "Propulsion and Power (AE2230)" or any other similar course.	
Parts	The important modules of this course are - o Basics of Aero Engine - o Aero Engine performance - o Environmental Impact of Aviation - o Inlets and Exhaust - o Turbomachinery - o Gas Turbine Emissions - o Gas Turbine Combustion - o Gas turbine performance - o Gas Turbine Controls - o New Engine /propulsion Concepts	
Course Contents	The course presents advanced concepts in aircraft propulsion. This is an flipped course in which the lectures are pre-recorded and the students are supposed to watch the lectures before coming to the classroom. The classroom interaction is mainly to have discussions and for tutorials. The course is aimed at looking into the details of an aircraft engine, the various components of gas turbine and their interaction. The course is divided into various modules which deal with the various aspects / disciplines that are essential in an aero engine. The important modules of this course are engine performance, sustainability in aviation, new engine/propulsion concepts, engine inlets, turbo machinery, combustion, engine exhaust systems, gas turbine performance and engine controls. Throughout the course, practical examples of systems from aircraft engines and gas turbines will be used to demonstrate the various methods and techniques. The learning goals will be supported by lab visits and guest lectures by experts from aviation industry.	
Study Goals	After the course the students will have basic knowledge of the various modules and disciplines that play an important role in aircraft propulsion. The specific study goals of the course are 1. Calculate thermodynamic parameters of Brayton cycle and variations of the cycle. 2. Identify differences in cycle characteristics between Brayton cycle and other thermodynamic cycles. 3. Calculate the design point thermodynamic performance of an aircraft gas turbine 4. Evaluate the effect of engine design parameters on the engine performance 5. Evaluate the effect of engine architecture on its performance 6. Compare the efficacy of different engine architecture for different missions 7. understanding the environmental effects of different aircraft emissions 8. Evaluate various options of making aviation sustainable 9. Understand the basic fluid mechanics involved in turbomachinery 10. Perform the conceptual design of axial compressors and turbines 11. Compute the flow angles and flow properties within a axial and radial turbomachinery components 12. Determine matching conditions for compressors and turbines 13. Evaluate the calorific values for various fuels 14. Understand the basics of combustion and emission formation in Aero engines 15. Understand the working of gas turbine combustor 16. Understand the basic principles of gas turbine controls and evaluate different strategies for control 17. Enhance reporting and presentation skills	
Education Method	The medium of education is English. The course adapts a flipped teaching method. Students are supposed to watch the pre-recorded lectures online and then come to the classroom to discuss their doubts and to solve the tutorial problems. Throughout the course, practical examples of aircraft systems and aircraft engines will be used to demonstrate various methods and techniques. All lecture slides are uploaded on Brightspace. Video recordings of the lectures will be made available on the brightspace. Flipped Classroom technique will be used. The understanding of the students is reinforced during the assignments and tutorials.	
Course Relations	This course will be beneficial for other courses within the FPP Profile. - Advanced Aircraft Design I & II - Turbomachinery - Internal Flows - Combustion for Propulsion and Power Systems - Modelling for Propulsion and Power Systems - Experimental Simulations - Thermal Rocket Propulsion	
Books	There is no mandatory book for this course. however students can refer to the following book to get more information about the subject.	

Reader	1. Gas Turbine Theory by H.I.H.Saravnamuttoo, G.F.C. Rogers, and H. Cohen 2. "Aircraft Propulsion" by Farokhi, John Wiley and Sons A pdf of the reader will be provided which can be used for some of the modules covered in the course. All the lecture slides will be made available on Brightspace. Please note that the contents of the reader is slightly different than the lectures.
Prerequisites	The students are expected to know the thermodynamics and working principles of a gas turbine / aero engine. Students are expected to have undergone AE2203-Propulsion and Power (BSc course within the aerospace engineering at TU Delft) or a similar course.
Assessment	The assessment is based on 2-3 assignments given during the course and a final exam. All assignments together will form 30% of the final grade and the final exam will form rest of the grade. Students should note that they need to atleast get a grade of 6 in the written exam in order to pass the course. Students who fail the written exam also fail the course.
Permitted Materials during Tests	The students are permitted to use a non programmable calculator. A formula sheet will be provided during the exam. Students are not allowed to bring their own formula sheet, lecture notes, etc.

AE4240	Advanced Aircraft Design I	4
Responsible Instructor	Dr.ir. R. Vos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The design of high-subsonic transport aircraft is the topic of this course. The main goal is to demonstrate how the geometry of the aircraft can be shaped to ensure favorable flow conditions that satisfy requirements on performance, stability and controllability. The course is divided in the following lectures: 1. Design objectives 2. Boundary-layer aerodynamics and subsonic cruise drag. 3. Relation between supersonic velocity and pressure coefficient. Relation between geometry and pressure distribution. 4. Interference drag and area ruling. 5. Pressure distribution on airfoil sections at high and low speed. Development of the transonic airfoil. Design of supercritical and sonic airfoils. 6. Reynolds number effects. 7. Low-speed stalling characteristics of airfoil sections. Reynolds number and Mach number effects on maximum lift coefficient. High-speed stall (buffet) on airfoil sections. 8. Airfoils equipped with high lift devices: mutual interaction, types of flow separation. 9. Development of the swept-wing concept: historical background, principle of wing sweep, transonic flow characteristics, effect on boundary layer, effect on maximum lift coefficient, fences. 10. First generation of swept wing aircraft: tip-stall, aeroelastic deformation. 11. Pressure distribution on finite swept wings: spanwise drag distribution, forward sweep, fuselage interference, improving velocity distribution at root and tip, minimizing pitching moment. Design considerations for pressure distributions on finite swept wings. 12. Stalling characteristics of full scale aircraft: stall requirements, complexity vs. maximum lift, spanwise stall onset, deep stall. 13. Horizontal tail surface design: functions and requirements, planform shape, aeroelastic effects, elevator lock. 14. The buffet onset boundary: requirements, effect on flight envelope. Flight characteristics between MMO and MD: requirements, Mach tuck, Mach trim compensation. 15. Vertical tail surface design: functions and requirements, planform shape, effect of dorsal fin. Control surface design: functions and requirements, linearity, aeroelasticity, stall. 16. Wing control surfaces: spoilers and ailerons 17. Propeller slipstream effects: effect on spanwise lift distribution, effect on longitudinal balance, effect on directional balance in OEI condition. 18. Engine intake design: lip design for take-off and cruise conditions, drag in OEI condition. 19. Exhaust design: effect of ambient pressure, boat tail drag. Thrust reversers: types, interference with empennage. 20. Examples of modern wing design	
Course Contents	In this course it is demonstrated how the exterior design of high-subsonic transport aircraft can be manipulated in order to meet specified performance objectives and satisfy relevant certification constraints. It is shown what geometric characteristics of the airplane allow for a high operating Mach number while keeping fuel burn at acceptable levels. In particular, it is demonstrated how the exterior of all major airplane components can be shaped such as to adequately handle low-subsonic and transonic flow. It is demonstrated how the aerodynamic design and the operation at low speed and high speed affects the airplane in terms of balance, stability, aeroelastic behavior, weight, controllability, cruise performance, and field performance. Through many historic and contemporary examples the student learns to relate the functionality of high-subsonic transport airplanes to their external design.	
Study Goals	Students are able to: 1. Recognize the aerodynamic phenomena occurring around external shapes at a variety of conditions such as Mach number, Reynolds numbers or angles of attack. 2. Derive how these phenomena can be affected by the detailed external shape and interposition of various airplane components such as wing, fuselage, tail, nacelle, pylon, and fuselage. 3. Show how wing and tail movables can be applied to locally affect the flow at low subsonic and/or high-subsonic conditions. 4. Explain the effect of the aerodynamic characteristics on the airplanes cruise performance, field performance, stability, balance, and controllability. 5. Identify the advantages and disadvantages of various external design features on the airplanes aerodynamic performance and cross-disciplinary characteristics such as aeroelasticity, weight, or practical limitations resulting from the operational use of the aircraft.	
Education Method	Lectures	
Literature and Study Materials	Mandatory textbooks: - Aerodynamic Design of Transport Aircraft, Obert, E, IOS Press, 2009 - Introduction to Transonic Aerodynamics, Vos, R. and Farokhi, S, Springer, 2015	
Prerequisites	Aerospace Design and Systems Engineering Elements I Aerospace Design and Systems Engineering Elements II Aerodynamics I (subsonic aerodynamics) Aerodynamics II (supersonic aerodynamics) Systems Engineering and Aerospace Design Flight Dynamics and Simulation	
Assessment	Written exam	

AE4260A	Fundamental of Aeroacoustics	2
Responsible Instructor	D. Casalino	
Instructor	Dr. D. Ragni	
Instructor	Dr.ir. F. de Prenter	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Aeroacoustics is one of the most important challenges for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Fundamentals of Aeroacoustics is to provide the students with a fundamental knowledge about aerodynamic noise generation and propagation. The course is structured in two main areas: 1) Fundamentals of Aeroacoustics: This module provides the theoretical elements to model and predict sound generation from unsteady flows of practical relevance in the aerospace and wind-energy field. Students will learn how to: describe how aerodynamic noise is generated and radiated; identify the most important sound generation mechanisms in practical situations and select the suitable aeroacoustic model to describe those mechanisms; setup a numerical simulation to predict the noise generated by elementary unsteady flows. 2) Basics of Computational AeroAcoustics: This module provides an introduction to the usage of High-Fidelity Computational AeroAcoustics methods to address problems of industrial interest. Students will learn how to setup a numerical simulation, analyze and present the results of the simulation.	
Study Goals	The course is meant to give the students a basic knowledge of aeroacoustics with basics of numerical modelling for aerodynamics and sound Fundamental knowledge and industrial applications will be covered to help the students who either want to continue in an academic or industrial field. By the end of this course, you will be able to: - Be able to describe how sound and noise sources are generated - Identify the most important sources of noise and the most important aeroacoustic analogies - Be able to setup a simple numerical computation for the study of noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, reference books and reference downloadable material (videos, tools, etc.). Relevant books and reference material for the course: Aeroacoustics of Flight Vehicles: Theory and Practice. Volume 1: Noise Sources, Hubbard, 1991 https://ntrs.nasa.gov/search.jsp?R=19920001380 Additional books and reference material (by relevance): Analytical and Numerical Methods in Vortex-Body Aeroacoustics, Casalino, 2002: http://acoustique.ec-lyon.fr/publi/casalino_thesis.pdf An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf Aeroacoustics, Goldstein, 1976 Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 Theory of Vortex Sound, Howe, 2003	
Assessment	The assessment consists of two mandatory parts: an oral examination and a numerical assignment with a report of the activities performed in groups to be defined during the course. One report per group of students will be delivered and presented by each student during the interview. 1) Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the whole program of the course. Key element for a successful oral examination are: - conciseness of the answer to the question; - argumentation with formulas and demonstrations; - capability of the student to show understanding and not memorization of the material. 2) Assignment (20%) Groups of students will be formed during the course and a practical assignment will be carried out leading to a group report. During the examination, 1-2 additional questions about the report will be asked. Students should prove understanding of the entire activity performed by the group. ----- Transition regulation for AE4260 (4 EC) ----- The course: AE4260A Fundamentals of Aeroacoustics (2EC) AE4260B Experimental Applications of Aeroacoustics (2EC) replace the AE4260 Experimental Aeroacoustics (4 EC) from year 2018 on. Students still have 2X resits for the old course AE4260 (4EC) in Academic year 2018-19. You can discuss with the instructor Daniele Ragni when to schedule those. From 2019-20 onwards, AE4260A and AE4260B are the replacement courses.	

AE4260B	Experimental Applications of Aeroacoustics	2
Responsible Instructor	Dr. D. Ragni	
Instructor	Prof.dr.ir. M. Snellen	
Co-responsible for assignments	Dr. R. Zamponi	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Aeroacoustics is one of the most disciplines for engineering students approaching either an industrial or an academic career in propulsion, wind energy and aerodynamics. The aim of the course Experimental Aeroacoustics is to provide the students with knowledge on how to measure and design an experiment for the study of noise sources in many industrial applications. The course is structured in three main areas: 1) Introduction to Aeroacoustics: This area introduces the basic knowledge about sound, sound pressure levels, sound power level and wave equations. 2) Measurement Techniques for Noise: In this part of the course the students will cover the basic of signal theory, microphone measurements and beamforming. The most innovative techniques for simultaneous flow visualization and noise computations will be also covered in this phase. 3) Experimental Aeroacoustic Applications Many engineering systems are dealing today with the problem of noise emission. The students will explore in this module many practical applications, including wind-turbine, propeller/propfan and jet/cavity noise sources.	
Study Goals	The course is meant to give the students a basic knowledge of experimental aeroacoustics. Fundamental knowledge and industrial applications will be covered to help the students who either want to continue with an academic or industrial career. By the end of this course, you will be able to: - Describe how sound and noise sources are generated - Identify the most important sources of noise in industrial applications and give indications on how to accurately measure them - Predict and eventually propose your own ideas on possible mitigation techniques for noise sources in many engineering applications (including cars, rotors, aircraft)	
Education Method	All required course material will be given during the course. The material will consist of lecture slides, a reference book and extra downloadable material. Relevant books for the course material in order of importance: - Aeroacoustic Measurements, Muller, 2002 (Course Book) - Theory of Vortex Sound, Howe, 2003 - Aeroacoustic of Flying Vehicles, Hubbard, 1991 Other useful books: - Mechanics of Flow-Induced Sound and Vibrations, Blake, 1986 - Acoustics, Kuttruff, 2016 Free downloadable material: - An introduction to Acoustics, Rienstra & Hirschberg, 2001: http://www.win.tue.nl/~sjoerdr/papers/boek.pdf - Basics of Aeroacoustics, Delfs, 2014: http://www.dlr.de/as/en/Portaldata/5/Resources/dokumente/abteilungen/abt_ta/Notes_Basics_of_Aeroacoustics_Delfs.pdf	
Assessment	The assessment consists of two MANDATORY parts, an oral examination + an assignment with a report of the activities in the laboratory (experimental activities in groups to be defined during the course). One report per group of students will be delivered and presented by each student during the interview. Oral examination (80%) Consisting of a one-to-one interview with the students. 3-5 questions will be asked covering the program in the indicated modules (1-2-3) + 1-2 additional questions on the report from the group of the student. Relevant for a successful oral examinations are: 1) Conciseness of the answer to the question with practical argumentation in the indicate fields: aeroacoustics/aerodynamics/signal processing. 2) Argumentation with formulas and demonstrations is appreciated if backed-up with understanding of the mathematics behind it. 3) Capability of the student to show understanding and not memorization of the material. Assignment (20%) Groups of students (8-10) will be formed during the course and laboratory activities will be carried out. About 2-3 hours per group will be dedicated to the presentation of the experimental activities and the acquisition of the data (the students will not be responsible of the preparation of the setup). The groups will analyse the data and prepare a report of the activities. Each student will bring his group report to the oral examination.	

AE4261	Internal Flows	3
Responsible Instructor	C. Falsetti	
Instructor	Dr. M. Pini	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In this course we address fluid dynamic phenomena of interest in internal flow situations for aerospace propulsion applications. The goal of the course is to develop physical insight into the phenomena that characterize internal flow in fluid machinery. As such we will discuss not just what happens, but why it happens and what are the consequences for propulsion systems. The course covers topic, which are not dealt with in courses or texts about external fluid dynamics. The flows described in Internal Flows are generally rotational, often three-dimensional, unsteady, and sometimes occurring in non-inertial (e.g., rotating) coordinate systems. The course can be viewed as the development of flow models and ideas to enable allow physical insight into the behavior of three-dimensional and unsteady flows. One of the objectives of the course is to provide an increased ability to interpret computational and experimental results and hence to effectively extract conclusions about the key features of complex internal flows. This course is inspired by 16.540 Internal Flow from the Department of Aeronautics and Astronautics at MIT. Much of the material is based on the lectures of Profesor E. M. Greitzer, with permission.	
Study Goals	By the end of the course, you should be able to: 1. Understand the underlying principles governing internal flow phenomena in fluid machinery by explaining not only "what" occurred but also "why" ; 2. Interpret temperature, pressure, velocity data to predict the behavior of internal flows. 3. Assess the effect of three-dimensional and unsteady flows on the design of a turbo-machine.	
Education Method	This course will use innovative didactic methods that emphasize active learning, as open questions, flipped classroom, and peer teaching. Students will be engaged through active feedback systems and online material. Given that a textbook exists for the course material, it is the intent of the lectures to discuss the material rather than to present it for the first time in class. As such the students will be expected to read the material before class and to engage in in-class discussions of the reading material by answering questions and elaborating concepts. Here we list a high-level description of the course contents. More detailed information are available in the complete syllabus on Brightspace. Lecture 1 Structure and Content of the Course, Introduction to Flow Regimes Lectures 1-2 Useful Basic Ideas Pressure fields and streamline curvature Upstream influence in turbomachines Control volume description of propulsion systems Asymmetry of real fluid motions Behavior of an inlet and turbomachinery system Lectures 3-6 Vorticity and Circulation Kinematics of vorticity Dynamics of vorticity Behavior of vortex lines Rotational flow description in terms of vorticity and virculation Mixing enhancement due to streamwise vorticity Lectures 7-9 Flow in Rotating Passages Non-dimensional parameters in a rotating flow Conserved quantities in a steady rotating flow Phenomena in flows where rotation dominates Rotating Channel Flow in Constant Area Straight Passages Boundary layers in rotating straight channels Rotating Flow in Turbomachinery Passages Secondary flow in rotating blade rows Lectures 10-12 Loss Sources and Loss Accounting Appropriate metrics for loss Boundary layer losses Mixing Losses Averaging of a non-uniform flow Effect of pressure level on average properties and mixing losses Loss characterization in turbomachinery cascades Lectures 13-14 Compressible Flows & Heat transfer in fluid machinery Lectures 13-14 Flows with heat transfer and phase change Heat transfer and temperature effects Flows undergoing a phase-change	
Literature and Study Materials	Internal Flow: Concept and Applications, E. M. Greitzer, C. S. Tan, M. B. Graf, Cambridge University Press http://www.cambridge.org/gb/academic/subjects/engineering/thermal-fluids-engineering/internal-flow-concepts-and-applications?format=PB#MciShGFvqgyx6G4p.97	
Assessment	The student will be asked to develop concept questions and corresponding answers which help to illustrate the physical principles and approximations that are made. This will be done in small teams (six or seven members) so you can prepare and discuss the questions among yourselves. The concept questions are to be developed each week by each team. Each week one	

team is selected to present and discuss their concept question to the class, in a flipped classroom mode.

Approximately five short (~20 minute) "concept quizzes" will be given during class, roughly equally spaced throughout the term. The total of these quizzes will count for 25% of the final course grade.

The final exam is an oral exam lasting approximately 30 minutes, and it will count for 75% of the final course grade.

A rubric will be provided at the beginning of the course.

AE4262	Combustion for propulsion and power technologies	4
Responsible Instructor	Dr. I. Langella	
Responsible Instructor	F. De Domenico	
Responsible Instructor	Prof.dr. A. Gangoli Rao	
Instructor	G. Kelesidis	
Contact Hours / Week x/x/x/x	0/0/0/4+2lab	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology	
Summary	This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.	
Course Contents	1. Thermodynamics of combustion 2. Chemical kinetics 3. Transport processes 4. Premixed and diffusion flames 5. Laminar and turbulent reactive flows 6. Combustion experiments and diagnostics 7. Combustion modelling 8. Laboratory Tutorial & Exercise 9. Practical combustion systems for aerospace applications 10. Pollutants and emissions 11. Hydrogen combustion physics and technology	
Study Goals	Write chemical equations and perform stoichiometry calculations. Compute heating values of fuels, enthalpies of formation and adiabatic flame temperature. Apply principles of thermodynamics to analyse reacting mixtures at chemical equilibrium. Describe the fundamentals of the physics involved in premixed- and diffusion flames, laminar and turbulent. Discuss the physics related to turbulent reacting flows and chemistry interaction. Explain experimental measurement techniques and their role in understanding combustion. Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems. Provide phenomenological descriptions of pollutant formation of conventional combustion systems. Elaborate on the potential and challenges related to the use of alternative fuels in combustion systems.	
Education Method	The course consists of weekly lectures and exercise sessions (practicals). In addition, the students are required to complete a project assignment and perform a laboratory exercise. The project assignment is relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with the investigation of methane or hydrogen laminar flames. The students will gain hands-on experience in designing the experimental matrix, acquiring data, performing data analysis and writing a technical report. Both laboratory exercise and assignment are mandatory to be admitted to the oral exam.	
Course Relations	Aero Engine Technology Hydrogen Technology	
Literature and Study Materials	Slides, lecture notes and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION;	
Books	J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"	
Assessment	To pass the course, the students must Complete laboratory exercise and project assignment; and Pass the oral exam with a minimum grade of 5.5 out of 10 The project assignment consists of several parts, including analysis on the data collected during the laboratory exercise and further theoretical and numerical analyses of combustion phenomena. Note attendance to the laboratory exercise and submission of the assignment are both compulsory for the admission to the oral exam. The work on the assignment should be summarized in a report, to be written by a single student or in a group of maximum three. The final grade is the result of the oral examination (60%) and the grade obtained in the assignment (40%).	

AE4263	Modeling, Simulation and Application of Propulsion and Power Systems	5
Responsible Instructor	Prof.dr.ir. P. Colonna di Paliano	
Instructor	Dr.ir. W.P.J. Visser	
Instructor	Prof. F. Casella	
Instructor	C.M. De Servi	
Contact Hours / Week x/x/x/x	0/x/x/0	
Education Period	2 3	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	AE1240-I Thermodynamics AE2230-II Propulsion and Power AE1205 Programming and Scientific Computing in Python for Aerospace AE4238 Aero Engine Technology ME45000 Advanced Heat Transfer AE2235-I Aerospace Systems & Control Theory	
Course Contents	2nd EDUCATION PERIOD Part 1 (2 ECTS) Module 1 Introduction, Context, Foundations Module 2 Conservation equations Module 3 Modeling paradigms Module 4 Numerical methods and software Module 5 Modelica Module 6 Constitutive equations Module 7 Components and system modeling Module 8 Verification and validation 3rd EDUCATION PERIOD Part 1 (1 ECTS) Module 9 Model-based control Exam on Part 1 Part 2 (2 ECTS) Module 10 - Team Project to be chosen among a Aero engine with GSP or GTPsim b Power or propulsion system with Modelica	
Study Goals	After learning the content of the course, the student will be able to: Given an engineering problem related to propulsion and power systems, use the 9-steps method to create or select the appropriate model and run and interpret simulations in order to obtain a good solution of such problem, and communicate the results. In particular, this overarching objective can be obtained by developing the following theoretical capabilities: 1. Describe the role and types of models in Propulsion and Power Systems Engineering, and define which different modeling paradigms and numerical methods are most appropriate or needed to develop a system model given its purpose. 2. Formulate a mathematical model for a typical propulsion and power system or component, by first analyzing the functionality of a system by means of a process flow diagram; by defining the energy, mass, and momentum conservation balances for the system of interest, and choosing the most appropriate form of conservation equations; by selecting the constitutive equations required to close the mathematical model (thermophysical models of fluids, chemical reaction eqs, heat transfer correlations, etc.). 3. Choose and configure numerical techniques for the solution of non-linear algebraic and differential-algebraic equation systems, which result from the formulation of a mathematical model. 4. Implement and code a system model of a power and propulsion application by adopting an object-oriented modeling approach (use of modularity, hierarchy, predefined connectors and inter-module variables). 5. Evaluate the reliability and possibly the fidelity of a model in the light of its purpose (model validation) 6. Use system models to solve engineering problems such as system performance assessment, preliminary design of the system and its components, the design of control strategies and the tuning of controller parameters, as well as communicate the results of the engineering analysis both verbally, and by means of a technical report.	
Education Method	Self-study using the Online Course Material on Brightspace (video lectures, related slides and other downloadable documents) Weekly sessions with instructors either online or in classroom, to discuss students' questions and to practice by means of assignments. Some sessions also require the use of computers and software. Students are strongly advised to make use of the BrightSpace Forum (Q&A) to post questions and interact with instructors and other students. In the second part of the course, students learn how to apply the knowledge acquired during the first part of the course, by working on an exemplary modeling and simulation project, in teams. Instructors provide support and supervision to the project.	
Computer Use	Students learn the Modelica modeling language and optionally the Gas Turbine Simulation program (GSP) or GTPsim. More information on these programs is provided on BrightSpace: Course Information -> Resources and Tools -> Open Modelica and GSP	
Literature and Study Materials	Video lectures on BrightSpace. Other material (PDF's) is made available for download on BrightSpace, see Content -> Modules	
Books	Reference text for thermodynamics: W. C. Reynolds and P. Colonna. Thermodynamics: fundamentals and engineering applications. Cambridge University Press, 2018.	

Reader	Reference text for numerical methods:
	P. Moin. Fundamentals of engineering numerical analysis. Cambridge University Press, 2nd Edition, 2010. Chapter 4, Numerical Integration of Ordinary Differential Equations.
Assessment	The reader is constituted by the slides of each module, which can be downloaded from BrightSpace (PDF's).
	Part 1: Exercises The first module is tested with a Quiz. The other modules are tested with take home exercises which are explained and partially carried out during the weekly sessions. The deliverables must be uploaded on BrightSpace by the delivery dates that are posted there. The exercises are preparatory for the Exam on Part 1 and are graded. Part 1: Exam The knowledge of Part 1 is assessed with a combination of a take-home exam and an oral examination. It covers the material of the 9 Modules of Part 1 of the course. Duration of the take-home exam: approximately 5 days. The date in which the exam text is made available and the delivery date are posted on BrightSpace. The oral exam is held within three weeks from the delivery date of the Take-home exam and questions are related to the solution of the take home exam and the solutions of the Module exercises. Part 2: Simulation project Teams of students (2-3) are given an assignment that requires the development of a system model either in GSP/GTPSim or in OpenModelica/Dymola Graded deliverables: technical report, model files. A short oral examination is conducted to assess the knowledge and understanding about the project of each member of the team, separately. The grade of each student for this part of the course is determined by considering both the project report and its execution and the oral exam. Grading the final grade is calculated as Final Grade = 0.2 x Exercises Grade + 0.4 x Exam Part 1 Grade (written + oral) + 0.4 x (Simulation Project + Oral Exam Grade). A final grade can be obtained only if the grade of the Exam Part 1 and of Part 2 are sufficient (≥ 6). Resit exams One opportunity to resit the Exam Part 1 is provided during the second quarter the course is running in. The resit exam of Part 2 consists in an additional oral exam in which the student is tested about the content of the Simulation Project. Additional work may be required to compensate for deficiencies assessed during the failed examination.

AE4270	Control and Operations Project	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. J. Ellerbroek	
Instructor	Dr. G.C.H.E. de Croon	
Instructor	E.J.J. Smeur	
Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	///	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This course code is a placeholder for miscellaneous student assignments within the Control and Operations track. For example, a lecturer would like to have some data processed or have a nice simulation model that can be used for demonstration purposes in the lectures. Through this course code, we now have a way to compensate students for their efforts by considering it as an elective of 4 ECTS. If there are such assignments available, they will be announced.	
Study Goals	n/a, depends on the nature of the assignment	
Education Method	n/a	
Assessment	A (short) report needs to be written about the assignment. The content needs to be discussed with the supervisor of the assignment.	
Elective	Yes	

AE4301	Automatic Flight Control System Design	3
Responsible Instructor	Dr. A. Jamshidnejad	
Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	AE4303	
Expected prior knowledge	The following prior knowledge is required: AE3302 AE2204 (until 2012/2013) AE2235-I (from 2013/2014)	
Parts	1. Introduction & recapitulation of systems and control theory . Laplace transformation . Elementary closed loop systems . Transfer functions . Poles and zeros . First order systems . Second order systems . Pole placement for simple systems . Root locus method . Characteristic equation . Angle and magnitude conditions . State space formulation . Controllability, observability . Basic controllers: P,PI,PD,PID . Frequency response (Bode plots) 2. Introduction: Course use and arrangement a. Why automatic flight control systems? b. Function of the flight control system in civil aviation c. Recapitulation of theory on flight dynamics d. Review of the different frames of reference: wind, stability, and body. e. Non-linear equations of motion. f. Trim and linearization of the non-linear equations of motion. g. Linearized longitudinal dynamics using a statespace representation and the equivalent frequency domain form. 3. Performance and handling qualities a. Handling quality criteria b. The Control Anticipation Parameter (CAP) c. Gibsons Phase rate and Frequency criterion 4. Dynamic stability augmentation a. yaw dampers b. pitch dampers c. phugoid dampers 5. Static stability augmentation a. angle of attack feedback to improve static margin b. load factor feedback to improve manoeuvre margin c. sideslip feedback to improve directional static stability 6. Basic longitudinal autopilot modes a. pitch attitude hold mode b. altitude hold mode c. airspeed hold mode (using autothrottle) d. vertical speed 7. Basic lateral autopilot modes a. roll angle hold mode b. coordinated roll angle hold mode c. turn rate at constant altitude and speed d. heading angle hold mode 8. Longitudinal and lateral guidance modes a. glideslope hold mode b. automatic flare mode c. localizer hold mode d. VOR hold mode	
Course Contents	Classical control is still predominantly used in aerospace industry for the design and analysis of automatic flight control systems. Various existing control systems such as Stability Augmentation Systems (SAS), Control Augmentation Systems (CAS) and fly-bywire systems are reviewed in detail. The emphasis of the course lies in demonstrating, through application of classical frequency domain and state space techniques, how to design systems that fulfill the requirements imposed by the aviation authorities, with emphasis on understanding the benefits and limitations of such systems.	
Study Goals	After this course the student should be able to: - Understand dynamics of linear time invariant systems and design controllers for such system for the desired performance	

	- substantiate the function of a Flight Control System(FCS) in civil aviation. - apply the theory of flight dynamics and control to FCS design for civil aviation. - verify if a given FCS satisfies the handling qualities for civil aviation criteria. - design static and dynamic stability augmentation systems. - design all longitudinal and lateral autopilot modes for civil aviation.
Education Method	Lectures
Literature and Study Materials	Course material to support the exercises will be posted on the Brightspace. Recommended literature - Any standard textbook on systems and control theory (e.g., Modern Control Engineering by Ogata) - M.V. Cook, Principles in flight dynamics, Edward Arnold, London, 1997 ISBN 0-340-63200-3. - B.L. Stevens, F.L.Lewis, Aircraft control and simulation, Wiley, New York, 1992 ISBN 0471613975. - J. Roskam, Airplane flight dynamics and control Part II, , ISBN 1-8845885-18-7.
Assessment	Written closed-book examination
Set-up	At the end of each lecture, a simple take home assignment is given in order to gain experience in working with the course material. There will be a written examination at the end of the course. In the related practical AE4301P a control system must be designed that satisfies certain desired requirements.

AE4301P	Exercise Automatic Flight Control System Design	1
Responsible Instructor	E.J.J. Smeur	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301 Automatic Flight Control System Design AE3212-I Flight Dynamics AE2235-I Systems and Control Theory	
Course Contents	The goal of this exercise is to design a flight control system using classical control theory, using a simulation model in Simulink. The exercise makes you in a practical way familiar with the implementation of inner control loop design, handling qualities satisfying CAP/Gibson/MIL-specifications, and the design of a higher level autopilot mode such as glideslope tracking or terrain following.	
Study Goals	To become familiar with classical flight controllers and their design, and to gain insight in handling qualities of open-loop and controlled aircraft.	
Education Method	General Practice Exercise.	
Literature and Study Materials	Course material to support the exercise will be posted on Brightspace.	
Assessment	Report	
Remarks	A short report about the design of this controller using classical control theory needs to be written. This report must be handed in during the second quarter. There is a resit-opportunity in the 3rd or 4th quarter.	
Set-up	The exercise is a homework assignment to be solved using Matlab and Simulink.	

AE4311	Nonlinear and Adaptive Flight Control	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE2235-I AE3212-I AE4301 AE4301P AE4312/AE4320	
Course Contents	Safety and survivability of modern civil and military aircraft when airframe and structural failures have occurred in flight have become a key issue in the aerospace industry nowadays. This course introduces nonlinear and adaptive control techniques to enhance reconfigurabilities of aircraft flight control systems. Advanced nonlinear and adaptive control techniques such as Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion (INDI), Adaptive Backstepping (ABS) and Incremental Backstepping are described together with applications to fault tolerant and reconfigurable aircraft flight control system designs. Unconventional yet available control effectors are used through control allocation algorithms to achieve required performance.	
Study Goals	Students will be able to acquire the skills in designing fault tolerant and reconfigurable flight control systems using Adaptive Nonlinear Dynamic Inversion (ANDI), Incremental Nonlinear Dynamic Inversion, Adaptive Backstepping (ABS) and Incremental Backstepping control techniques.	
Education Method	Lectures altered with assisted computer exercises	
Literature and Study Materials	Handouts of lecture slides (available on Brightspace)	
	Recommended literature: J.-J.E. Slotine & Weiping Li, Applied non-linear control, Englewood Cliffs : Prentice-Hall, 1991, ISBN 0-13-040890-5 M. Krstic, et al, Nonlinear and adaptive control design, John Wiley & Sons, Inc., ISBN 0-471-12732-9	
Assessment	Individual Take-home assignment	
Remarks	Hand-in exercise and a report of the design of an advanced controller. The report must be handed in at the request date given by the course instructor.	
Set-up	Course contents Introduction to advanced flight control Nonlinear aircraft flight control based on Nonlinear Dynamic Inversion Adaptive Nonlinear Dynamic Inversion Incremental Nonlinear Dynamic Inversion Aircraft fault tolerant flight control based on Adaptive and Incremental Nonlinear Dynamic Inversion Nonlinear control based on backstepping Nonlinear aircraft flight control based on backstepping Adaptive Backstepping Incremental Backstepping Aircraft fault tolerant flight control based on Adaptive and Incremental Backstepping Control allocation	

AE4313-20	Spacecraft Attitude Dynamics and Control	4
Responsible Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Parts	Week arrangement Week 1. Introduction to spacecraft dynamics and control, Rotational kinematics with quaternions Week 2. Rigid body dynamics Week 3. Spacecraft attitude determination techniques and attitude estimation Week 4. Attitude control actuators Week 5. Spacecraft passive attitude control Week 6. Spacecraft active attitude control with momentum based actuators Week 7. Advanced control techniques for spacecraft attitude control	
Course Contents	Spacecraft attitude control is one of the most important subsystems for most space vehicles. It controls the orientation and rotational rate of the spacecraft to a required accuracy. The attitude control system uses attitude sensors and control actuators to determine and correct the attitude errors of the spacecraft. Unlike the flight dynamics in the third education year, the rotational kinematics of space vehicles in this course is thoroughly taught using quaternion algebra, in order to give a more general technique in modelling spacecraft rotational motion without having mathematical singularities. State estimation techniques such as Kalman filter and extended Kalman filter are introduced for estimating spacecraft attitude from attitude sensors. Several techniques in sensor integration and data fusion are also taught for enhancing the performance of the attitude determination system. Control actuators include reaction wheels, momentum biased wheels, thrusters, magnetic coils and Control Moment Gyros (CMGs). The single gimbal CMGs with their applications to spacecraft attitude control is specifically addressed. The course discusses not only the passive attitude control concepts but more dedicates to active attitude control system designs. Particularly the quaternion feedback control technique is enhanced. Conventional design of spacecraft attitude control systems is based on the linear control theory. In this lecture advanced control techniques such as Nonlinear Dynamic Inversion (NDI) is also introduced for highly maneuverable spacecraft.	
Study Goals	Students will be able to acquire classical and advanced techniques for spacecraft attitude control.	
Education Method	Lectures	
Literature and Study Materials	Recommended literature M.J. Sidi, Spacecraft dynamics and control, a practical engineering approach, Cambridge Univ. Press, Cambridge, 1997 ISBN 0521550726. B. Wie, Space vehicle dynamics and control, AIAA Education Series, AIAA Inc., 1998 .	
Assessment	Individual take-home assignments. The assignments will be handed out near the end of Q3 and the hand-in deadline will be near the end of Q4. Exact dates will be announced during the lectures.	

AE4314-21	Helicopter Performance, Stability and Control	4
Responsible Instructor	Dr. M.D. Pavel	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Aircraft performance Aerodynamics Flight dynamics	
Parts	Week Arrangement Lecture and study material 1. Lay-out of the helicopter. Qualitative description of the rotor and control systems. Definition of the tip plane, control plane, hub plane and several co-ordinate systems. Blade element analysis of hover performance. 2. Performance calculations of vertical flight. Turbulent Wake State. Autorotation. Performance calculations in forward flight. Blade element analysis. Glauerts hypothesis. Comparison with performance of fixed wing aircraft. 3. Flapping dynamics of centrally hinged rotors. Equations of motion of helicopter with only pitch degree of freedom. Stability and control. 4. Flapping dynamics in forward flight. Symmetric equations of motion. Stability and control in forward flight. 5. Trim conditions, linearised equations of motions. Representation in the complex plane and comparison with fixed wing aircraft. 6. Influence of semi-rigid rotorsystems on the equations of motion and on the characteristic handling characteristics. 7. Flap/Torsion blade dynamics. Divergence. Classical blade flutter, Structural lay-up to avoid flutter. Introduction to ground resonance instability.	
Course Contents	General lay-out of helicopters, performance calculations during hover, vertical climb and descent and forward flight. Helicopter blade dynamics. Introduction to helicopter control characteristics: symmetrical equations of motion handling- and control properties. Introduction to aeroelasticity of rotorcraft.	
Study Goals	1. An example of the synthesis of a large diversity of subject areas, aimed at the analysis of a complicated technical object. 2. Introduction to aerodynamics, flight mechanics and aeroelasticity of helicopters, which enables the student to independently study more advanced literature.	
Education Method	A combination of Online and Classical lecturing courses. Articles Readings and discussions. Questions and Discussions fora	
Literature and Study Materials	"Helicopter performance, Stability and Control", reader AE4-214, 2002, Faculty of Aerospace Engineering "Basic Helicopter Aerodynamics" by John Seddon& Simon Newman, Blackwell Science, 2002 "Principle of Helicopter Aerodynamics", Gordon Leishman, Cambridge Aerospace Series, 2000	
Assessment	Multiple-choice tests during the course (optional). Presentation of a helicopter-related subject in a small Symposium (obligatory) One final take-home assignment at the end of the course (obligatory) +Oral Exam (obligatory)	
Special Information	A helicopter flight in SIMONA simulator may be given to the students following this course depending on the availability of SIMONA simulator.	
Set-up	Online education and Classical lecturing	

AE4316P	Advanced Topics in Aerospace Human-Machine Systems	3
Responsible Instructor	Dr.ir. D.M. Pool	
Instructor	Dr.ir. M.M. van Paassen	
Instructor	Dr.ir. C. Borst	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	This course is an elective extension to the AE4316 (core) course for the C&S master profile. Prior knowledge from AE4316 is expected and mandatory.	
Course Contents	This course further explores the various aspects of supervisory control and manual control (cybernetics) and their applications in aerospace as introduced in the AE4316 lecture series. Through a number of lectures in the first weeks, this course provides detailed practical information on the current advanced state-of-the art methods used for measuring, analyzing, and modeling human performance and behavior in supervisory and manual control tasks and how such methods are applied in applications relevant to aerospace. Furthermore, through extensive practical work performed in small groups, students will gain practical experience with the advanced analysis of human-machine system performance through an experiment they need to plan and perform themselves. In this experiment, which will be performed in one of the simulator setups available at Control & Simulation (i.e., the Air Traffic Management Laboratory, the Human-Machine Laboratory or the SIMONA Research Simulator), students will collect data for an experiment of their choice, analyze the collected data, and reflect on the conclusions drawn from their experimental measurements.	
Study Goals	1. Formulate a good research question and hypotheses for a supervisory or manual control experiment. 2. Demonstrate understanding of the direct link between an experiments independent, dependent and control variables and the data analysis that can be performed on its data. 3. Analyze data from a supervisory or manual control experiment using state-of-the-art data analysis techniques and statistical tests. 4. Compare experimental findings with theoretical constructs and elementary models of human performance or control behavior. 5. Reflect on the scientific conclusions drawn from a set of human subject experiment data and their reliability. 6. Reflect on the consequences of experiment design choices on the scope and validity of human subject experiment results.	
Education Method	Classroom lectures, twice a week in the first three lecture weeks (3.1-3.3). After that, students will work in groups on the design and execution of an experiment they will perform in one of the simulation facilities at Control & Simulation.	
Literature and Study Materials	The presentation slides of all lectures will be made available through Brightspace Recommended reading for this course: 1. Jagacinski, Richard J., and John Flach. Control theory for humans: Quantitative approaches to modeling performance. Taylor & Francis, 2011. 2. Wickens, Christopher D., Lee, John D., Liu, Yili, and Gordon Becker, Sallie E. An Introduction to Human Factors Engineering, Second Edition, Pearson Education Inc., 2004. 3. Field, Andy. Discovering statistics using SPSS. Sage publications, 2009.	
Assessment	No written exam. Instead, grading for AE4316P will be based on the quality of two group deliverables: an Experiment plan (deadline end week 3.4, 20% of final grade) and a final Scientific Article (deadline end of Q3, 80% of final grade).	
Elective	Yes	
Set-up	Classroom lectures will be provided in the first three weeks. The following topics will be addressed in the lectures: 1. Introduction and experiment descriptions. 2. Human subject experiment design. 3. Experimental data analysis and statistics. 4. Cognitive Work Analysis. 5. Multi-channel manual control behavior identification. In addition to the lectures, students work in small groups on a practical assignment. This effort includes: developing an experimental plan, collecting and analyzing data, and writing a scientific article.	

AE4317	Autonomous Flight of Micro Air Vehicles	4
Responsible Instructor	Dr.ir. C. de Wagter	
Instructor	Dr. G.C.H.E. de Croon	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Programming experience (any language) If the student has no previous background in C / C++, we expect the student to do an online tutorial before/at the beginning of the course. Desired prior knowledge: Programming experience with C / C++ AE4393 (Avionics and Operations)	
Course Contents	This course covers the challenges and existing state-of-the-art methods for enabling autonomous flight of Micro Air Vehicles (MAVs), ranging from 20-gram flapping wings to 1 kg quad rotors. The emphasis is on computationally efficient, bio-inspired approaches to MAV autonomous flight.	
Study Goals	The theoretical knowledge will be applied by the students in the practical assignment, in which student groups program quad rotors in order to avoid obstacles in TU Delft's Cyberzoo. (1) Students can explain the differences between MAVs and larger aircraft in terms of control, design, aerodynamics, and electronics. (2) Students can explain how the above-mentioned differences affect the task of achieving autonomous flight with MAVs. (3) Students can list and explain Artificial Intelligence (AI) and control techniques useful for autonomous flight of Micro Air Vehicles (MAVs). a) Bio-inspired robotics b) Vision-based navigation (4) Students can apply the learned AI and control techniques to, and create new algorithms for an autonomous flight task. (5) Students can reproduce the current state of regulations for MAV operations. In addition, they can assess and discuss the safety and ethical aspects of MAV operations.	
Education Method	The course will consist of theory (7 lectures) and a practical assignment (6 practical sessions and a competition session). Brief description of practical assignment: Student teams will develop an efficient, purely vision-based approach to navigate through an obstacle field autonomously. The algorithm must run onboard the Parrot Bebop Drone, a commercially available quadrotor with a range of sensors. Importantly for the project, it has a single frontal camera. Hence, the developed algorithm concerns a monocular vision solution to obstacle avoidance.	
Assessment	The assessment of the course will consist of 50% of the theoretical exam and 50% of the practical assignment. Assessment of the practical assignment is based on the performance of the drone in the given task, the approach as explained in a presentation, and the code produced by the team.	

AE4320	System Identification of Aerospace Vehicles	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. D.M. Pool	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4301.	
Parts	Week arrangement 1. Introduction. A global overview of the complete system identification cycle applied to aerospace vehicles, from sensor data acquisition and state estimation to aircraft aerodynamic model identification and validation. Introduction to sensor data acquisition and management, state and parameter estimation, and the two-step method for aerodynamic model identification. Finally an overview is give of advanced modelling techniques. 2. State estimation. Introduction to the essential role of the Kalman Filter (KF) in aircraft state estimation. The extension of the standard KF to nonlinear systems will be introduced in the form of the Extended Kalman Filter/Smotherer (EKF/EKS), and iterated extended Kalman filter (IEKF). Applications of the Kalman filters in aerospace navigation, attitude determination, on-board sensor calibration, sensor integration, and data fusion will be discussed. 3. Parameter estimation. Introduction to the process of parameter estimation and its role in creating accurate aerodynamic models of aircraft. Introduction to four of the most widely used parameter estimation techniques; Least squares (LS), Weighted Least squares (WLS), Total Least Squares (TLS), and Maximum Likelihood (ML) parameter estimation. Additionally, recursive parameter estimation methods like Recursive Least Squares (RLS) will be introduced. 4. Advanced optimization techniques. Some problems cannot be solved adequately with standard linear optimization techniques. In that case, a nonlinear optimization method must be used. In this part of the course, a powerful global nonlinear optimization technique based on interval analysis will be introduced. 5. Advanced modelling methods: Neural Networks. Neural Networks are a class of black-box function approximators that can be used to model nonlinear systems like Micro Air Vehicles and high performance aircraft. The application of neural networks to aircraft system identification is discussed, and it will be shown how neural networks can be used inside a framework for parameter estimation. 6. Advanced modelling methods: Multivariate Splines. Multivariate Simplex B-Splines are a powerful type of white-box function approximators that can be used to create accurate models of aircraft with highly nonlinear aerodynamics such as MAVs and high performance aircraft. A number of unique methods for assessing the quality of the created models are discussed, and pointers towards future research are given. 7. Demonstration: Aircraft system identification with multivariate splines. An in-depth demonstration will be given of the process of Aircraft system identification with multivariate splines using real flight test data from the Cessna Citation II laboratory aircraft. The goal is not only to create an accurate flight dynamics model of the Citation II, but also to apply various model validation techniques to assess the quality of the created model.	
Course Contents	Accurate aerodynamic models play a crucial role in the design and operation of flight simulators and flight control systems. The creation of accurate aerodynamic models from CFD, wind tunnel, and flight test data has historically been a highly challenging task. This is a direct result of the nonlinear nature of aircraft (aero)dynamics and the fact that not all aircraft states can be measured directly. As a consequence, the aerospace vehicle parameter identification problem constitutes a joint parameter-state estimation problem. It is the aim of this course to provide the student with a complete overview of the system identification cycle as it is currently applied to aerospace systems, and to introduce the student to the current state of the art in the field of aerodynamic model identification. The course consists of 7 parts covering the entire system identification cycle. In the first part of the course, the process of data acquisition using on-board sensors including accelerometers, gyroscopes, GPS and various air data sensors is discussed. It will be demonstrated that measurements made by real-world sensors are contaminated with noise, and are sometimes biased. Additionally, some aircraft states, like the true angle of attack, cannot be measured directly and must be reconstructed by combining sensor measurements. The second part of the course introduces the concept of state estimation in which prior physical knowledge of the system is used in a Kalman filter (KF) to estimate the true aircraft states from the measured states. Next to the ordinary Kalman filter, the Extended Kalman filter (EKF) and the Iterated extended Kalman filter (IEKF) will be introduced. In the third part of the course various methods for the estimation of model parameters will be discussed. It will be shown how the results from the state estimation are used in combination with a parameter estimator. This part not only focuses on offline batch parameter estimation methods like ordinary least squares (OLS), weighted least squares (WLS), total least squares (TLS), and maximum likelihood (ML) estimators, but also on recursive parameter estimation methods like recursive least squares (RLS) that can be used online. Special attention will be paid on the process of choosing a parameter estimator that is right for the job. The fourth part of the course introduces an advanced global nonlinear optimization method based on interval analysis. In aerospace system identification, so-called non-convex and nonlinear optimization problems are often encountered. Such problems can be solved with global nonlinear optimization methods like interval analysis. The fifth and sixth part of the course focuses on two advanced model structures that can be used in combination with the earlier introduced parameter estimation methods. In the fifth part of the course the neural network black-box function approximator is introduced. It is shown how neural networks are used to approximate scattered multidimensional data. In the sixth part of the course a new method for aerodynamic model identification based on multivariate simplex B-splines is introduced. This method was recently developed at the Faculty of Aerospace Engineering of the TU-Delft, and has a number of advantages over existing methods. For example, the simplex B-splines have a transparent model structure, are general in any number of dimensions, and can be computed efficiently in real-time.	

Study Goals	In the final part of the course, all theory introduced in the first six parts is used in a demonstration that uses real flight data from a real-life system like the Cessna Citation II laboratory aircraft, or a Micro Air Vehicle to identify a multivariate spline based aerodynamic model. The created model is validated using various techniques. Finally, pointers are given towards further research in the field of aerodynamic model validation. After completing the course, students will have a thorough understanding of the theoretical elements of aircraft system identification. Students will be able to apply these elements to go from real CFD, wind tunnel, and flight test data to an accurate aerodynamic model that can be used in simulator or flight control system applications.
Education Method	Lectures.
Literature and Study Materials	Q.P. Chu, Modern flight test technologies and system identification, lecture notes, Faculty of Aerospace Engineering, Delft University of technology. C.C. de Visser, Global Nonlinear Model Identification with Multivariate Splines, PhD thesis, Faculty of Aerospace Engineering, Delft University of technology. Recommended literature V. Klein and E. A. Morelli, Aircraft System Identification, AIAA, 2006. J.A. Mulder, J.K.Sridhar, J.Breeman, Nonlinear Analysis and Manoeuvre Design, Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 2, , May 1994 R.E. Maine, K.W.Illiff, AGARD Flight Test Techniques Series on Identification of Dynamic Systems, Applications to Aircraft, AGARD AG-300, Vol. 3, Part. 1, , December 1986 . Eric Walter and Luc Pronzato, Identification of parametric models, Springer, ISBN 3-540-76119-5.
Assessment	Individual Take-home assignments

AE4321-15	Air Traffic Management	4
Responsible Instructor	Prof.dr.ir. J.M. Hoekstra	
Instructor	Dr.ir. J. Ellerbroek	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Course Contents	The course gives an Introduction in Air Traffic Management and ATM research topics. 1. Introduction and history of Air Traffic Management, organisational & regulatory aspects 2. Airspace set-up, flight planning and navigation 3. Communication: Information services, Voice R/T, Data Link 4. Navigation: Great Circle calculations, WGS'84, approximations, beacons/navaids, landing aids 5. Surveillance: Radar basics, primary-secondary radar, radar types, ADS-B 6. Separation: Separation Minima, Near Misses, Closest Point of Approach Calculations, wind vortex categories 7. New concepts: Airborne separation assurance, 4D contract TBO, paired approaches, etc. 8. ATC Simulations: fast-time, human in the loop 9. ATC Research organisations and large scale programs (SESAR, NextGen) 10. ATM research topics: interactively discussed in class based on presentations by students While the first part of the course is mainly lectures, the second part of the course is much more interactive. Here pairs of students get assigned a paper, project or case to study and form an opinion on. They should explain and review it in both a brief essay and during a short presentation in class, followed by a discussion with the class in which they need to defend their view on this study. In this way the students are familiarized with different aspects and opinions on the 'hot topics' in ATM research currently.	
Study Goals	This course is designed to give the students an appreciation of the current ATM systems, throughout the world, with emphasis on the core area of Europe, in preparing the existing ATC systems for the predicted increase of traffic. Emerging technologies will be described, with the associated implementation aspects like Air Traffic Controllers' systems, the history and future of ATC, airline perspective, European and US new ATM programmes SESAR and NEXTGEN. Emphasis is on both the current technology as well as future concepts and technology which are currently being researched.	
Education Method	Lectures will be given on-line via livestreaming. Assignments will be given to the students for the second period. Students need to present their understanding of and opinion on a case, project or paper on-line. Discussions on this with the other students are part of the interactive method used in the second period.	
Reader	Hand-outs of the presentations are available on Brightspace.	
Assessment	Written, on site exam (50%) and the assignment (essay+presentation, also 50%). Both parts need at least a 5.0.	

AE4323	Real-time Distributed Flight and Space Simulation	3
Responsible Instructor	Ir. O. Stroosma	
Instructor	Ing. E.H.H. Thung	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE1205 (Programming & Scientific Computing in Python, or equivalent introduction to programming) AE4322 (Piloted Flight Simulation)	
Course Contents	Lecture topics, not necessarily in chronological order 1) C++ introduction 2) Real-time software engineering 3) Delft University Environment for Communication and Activation (DUECA) 4) Modelling tools 5) Verification and Validation 6) Software project management Programming assignments, individually and in a team. The course is completed with a practical project in which students program, demonstrate and document a distributed simulation with an aerospace theme. Attendance in the project meetings is mandatory.	
Study Goals	1) The student can identify the software issues related to real-time, distributed simulation. 2) The student can actively apply the theory of rigid body dynamics in real-time simulation. 3) The student can combine the modelling of dynamical systems with numerical integration routines. 4) The student can use quaternions to represent rigid body attitudes. 5) The student can implement simulation models using different software languages and tools (C++, Simulink). 6) The student can create 3D models for computer visualisation and optimise these for real-time simulation. 7) The student can (in a team) implement a real-time simulation on a distributed platform, using advanced software tools.	
Education Method	Lectures, programming instruction, and programming assignments.	
Literature and Study Materials	Lecture slides and other materials are put on Brightspace.	
Assessment	Individual assignment during 2nd/3rd week and team project report. Attendance at the team project meetings in week 3 and later is mandatory.	

AE4324	Physical Interaction for Aerial and Space Robots	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	S. Hamaza	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course covers the fundamentals of Robotics and delves deeper into the mechanics of physical interaction for aerial and free-flying systems. Topics span from modelling, design and motion planning of fixed robots, to the latest developments in control of floating-base robots physically interacting with objects. Many of these concepts will then be reproduced on a real robotic system, as part of the assignment. The course is structured in the format of lectures, in-class exercises and home assignment. The final examination will be a written exam. TOPICS: Introduction & Applications Robot Morphologies Robot Kinematics The Jacobian Matrix Robot Dynamics Trajectory Planning Interaction Control for Fixed Robots Interaction Control for Floating-base Robots Contact-based Navigation ASSIGNMENT: The goal is to code the kinematics, workspace, trajectory planning and an arbitrary interaction task of a real robotic arm, e.g. grasping/contour following/writing/. The grade includes the simulated experiments in ROS/MATLAB/OPENAIGYM; the real-world demo; and the report. The grade of the assignment should be higher than 5.	
Study Goals	After completing this course, the students are able to: -Identify different robot morphologies and associate the correct modelling rules -Derive the kinematics of simple and complex morphologies -Implement contact-less control laws for robot arms -Implement contact-based control laws for robot arms -Explain and implement the different methods in interaction control theory -Implement a simulation model in Python or MATLAB -Integrate software & hardware, familiarise with robotic hardware and learn how to debug it.	
Education Method	-Lectures & in-class exercises (4 hours per week) -Home assignment	
Assessment	Individual assignment (weight 40%) - Simulated results of robot kinematics and planning task - Real-world implementation on a 5-DoFs arm - Report & Code Written Exam (weight 60%) To pass, the exam grade must be sufficient (≥ 6) and the assignment grade cannot be lower than 5.	

AE4350	Bio-inspired Intelligence and learning for Aerospace Applications	3
Responsible Instructor	Dr. G.C.H.E. de Croon	
Instructor	Dr.ir. E. van Kampen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course will cover the following subjects: - Introduction Bio-inspired Intelligence - Introduction to Artificial Neural Networks - Introduction to Reinforcement Learning - Reinforcement Learning for Aerospace Control - Evolutionary robotics - Physical intelligence for interaction - Insect intelligence and Swarming - Spiking neural networks	
Study Goals		After this course the student will: - Understand the history and relevance of bio-inspired intelligence in Aerospace Engineering. - be familiar with the basic theory behind the main bio-inspired intelligence methods. - be able to implement his/her own bio-inspired intelligence algorithm.
Education Method	The course consists of a series of lectures. Part of the lectures will be about the basic concepts in bio-inspired intelligence. The other lectures will focus on some state-of-the-art research, where one or more recent scientific publications will be discussed.	
Assessment	The assessment of this course will be through an individual take-home assignment. For this assignment, the student selects one of the bio-inspired algorithms that is treated during the course and implements this to a system of their choice. Deliverables are a scientific report and the code that is used to generate the results in the report.	

AE4351	Robust Flight Control	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	S. Theodoulis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE2235-I AE3212-I WI2056LR AE4301 AE4301P	
Course Contents	The vast majority of aerospace vehicles, spanning from UAVs and rotorcrafts to military or commercial aircraft and all the way up to space launchers, need to fly in a secure, predictable and repeatable way. To achieve this, the on-board flight control system needs to be designed to withstand (i.e. to be robust against) a whole array of uncertainties, disturbances, noise, rapidly-varying flight conditions and imperfect components. Furthermore, the designer needs to provide with solid guarantees and verifiable certificates that the flight control system must meet. This course is aimed at offering to the students hands-on theoretical and practical knowledge on the modelling, design and analysis of robust flight control laws, paying special attention to the industrial applicability of the tools albeit with a research twist.	
Study Goals	The learning objectives of this course are the following: LO1: Calculate multivariable mathematical models of aerospace systems that include system uncertainty LO2: Evaluate feedback loop robust stability and performance via the Structured Singular Value (SSV) and compare against more traditional methods LO3: Breakdown the fundamental control performance requirements and limitations intrinsic to the aerospace system models LO4: Identify and fix robustness deficiencies of linear-quadratic methods (LQR, LQG, H₂) for aerospace control system design (LQG/LTR) LO5: Compute and simulate robust control laws using modern H₂/H_∞ theory (mixed sensitivity, loop shaping) using the latest tools in MATLAB/Simulink To illustrate the above concepts the short period dynamics of a highly manoeuvrable aerospace vehicle are considered and various control and stability augmentation systems are designed. The results and simulations are done in MATLAB/Simulink.	
Education Method	Lectures together with assisted computer exercises in MATLAB/Simulink	
Books	Some books that may be used from the students for reference are: - Multivariable Feedback Control: Analysis and Design, Ian Postlethwaite and Sigurd Skogestad - Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems, Brian L. Stevens, Eric N. Johnson, and Frank L. Lewis - Robust Multivariable Control of Aerospace Systems, Declan Bates and Ian Postlethwaite	
Assessment	The assessment is a take-home assignment which involves the step-by-step design and comparison of various robust control laws taught in the course. The system used is the same one considered during the course, namely the longitudinal short period dynamics of a highly agile aerospace vehicle.	

AE4352	Mathematical and human-inspired decision making	3
Responsible Instructor	Dr. A. Jamshidnejad	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course you will learn about model predictive control (MPC) as well as human-inspired decision making (particularly, how fuzzy logic is used for decision making)	
Study Goals	After finishing this course successfully, the student should be able to: - Formulate model predictive control (MPC) problem for mathematical decision making - Develop fuzzy logic controllers for human-inspired decision making - Implement MPC controllers and fuzzy logic controllers for case study problems	
Education Method	Lectures	
Assessment	Group project (report + presentation of results)	

AE4422-20	Agent-based Modelling and Simulation in Air Transport	4
Responsible Instructor	Dr. O.A. Sharpans'kykh	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Introduction. Agents and Multiagent systems. Agent-based modelling architectures. Examples from air transportation. Emergence in Multiagent systems. Agent-based simulation. Agent-based modeling and simulation tools. Agent-based coordination, planning, and scheduling in air transportation. Nature-inspired approaches to solve optimization problems. Swarm intelligence. Adaptive behavior and learning in agent-based systems. Collaborative decision making in air transportation. Negotiation, auctions, game-theoretic approaches. Agent-based model analysis: sensitivity, uncertainty, robustness. Validation of agent-based models.	
Study Goals	The student has to be able: - to formulate a practical air transportation problem as an agent or a multiagent system model; - to identify appropriate agent-based methods and to apply them; - to implement agent-based models; - to perform agent-based simulation, interpret and analyze simulation results; - to be able to apply agent-based optimization techniques.	
Education Method	Lectures and practical sessions	
Assessment	65% of the final grade for the course is the grade for a group assignment; 35% is the grade for the individual defense.	
Tags	Adventurous Algorithmics Analysis Artificial intelligence Group work Intensive Modelling Programming Small groups Transport & Logistics Transport phenomena	

AE4431-23	Aircraft noise	3
Responsible Instructor	Prof.dr. D.G. Simons	
Gast Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is required for AE4463P-23 'Advanced aircraft noise modelling and measurements'. In this follow-up course you will practice with real aircraft noise data on the following topics: - noise metrics (dBA meter) - Dutch aircraft noise model NRM or airframe noise model (ANOPP method) - digital signal analysis and acoustic imaging	
Expected prior knowledge	Basic knowledge on calculus and first year physics.	
Course Contents	Aircraft noise Basics of acoustics: Physics of sound waves. Harmonic waves, sound speed, wave front and rays. Reflection, refraction and diffraction of sound waves. The dB scale for acoustic power, sound intensity and sound pressure level. Interference, adding sound pressure levels and the standing wave. Directionality of sound sources. Periodic and broadband noise. Doppler effect and shock waves. Wave equation and its basic solutions: Derivation of the wave equation from conservation of mass, momentum and energy. Plane waves, acoustic resistance. Harmonic solution to wave equation. spherical waves, characteristic acoustic impedance. Propagation of sound in the atmosphere: Geometrical spreading and sound attenuation due to friction. Sound pressure level calculations as a function of distance from the source. Derivation of reflection and transmission coefficient of sound when going from one medium to another medium. Critical angle. Effect of temperature gradient on sound propagation. Calculation of shadow zone distance. Ray tracing. Ground effect. Sound propagation - special situations Sound transmission through a layer, e.g. a wall. Mass law. Propagation in enclosures, room acoustics, diffuse sound field, reverberation time, Sabine's law. Acoustic filters. Attenuation of sound in ducts (with changing cross-sections), e.g. exhaust systems. Helmholtz resonator. Acoustic lining and its application in turbofan engines. Noise barriers. Acoustic signal analysis: Fourier transform (continuous and discrete), power spectral density and spectrum level. Octave band and tert's band analysis, pressure band level. Effect on bandwidth on measured aircraft spectra. Overall sound pressure level. Spectrogram. Examples of aircraft noise measurements, e.g. from flyovers. Properties of microphones. Noise metrics: Human perception of sound, loudness and the phone and sone scale. Equal loudness level contours. Overall loudness level for broadband noise. Equal noisiness curves, overall noy value and perceived noise level. A-weighting and overall A-weighted sound pressure level. Effect of the duration of sound on human perception, equivalent A-weighted sound pressure level and sound exposure level (SEL). Single flyover noise contours. Noise indices for total noise exposure (Lden). Noise certification. Dutch aircraft noise model NRM. Directional acoustic sources: Monopole, dipole and quadrupole source. Line array of monopoles. Rayleigh integral, loudspeakers. Acoustic imaging: Principle of beamforming. Imaging aircraft noise data. (Note: this topic is only touched upon in this course, the details (including application) are treated in AE4463P-23) Aircraft noise sources: Mechanism of propeller noise and blade passage frequency. Mechanism of turbo engine noise. Effect of bypassing on exhaust jet noise and effect of acoustic lining on fan noise. Properties of airframe (aerodynamic) noise. ANOPP modelling method. Programming assignment about modelling airframe noise (landing gear, flaps, slats, etc.) using the ANOPP method.	
Study Goals	Understand the relationship between aviation and the resulting noise levels, especially around airports.	
Education Method	Lectures, exercises, programming assignment	
Literature and Study Materials	Lecture slides, exercises with worked-out solutions, programming assignment Reader: Course AE4431-23 Aircraft Noise Introduction to general acoustics and aircraft noise Prof. dr. D.G. Simons and Prof. dr. ir. M. Snellen August 2023	
Assessment	- Written exam (max 8 points) (a formula sheet will be provided) - programming assignment (max 2 points)	
Set-up	Lectures, exercises, programming assignment	

AE4439	Sustainable Air Transport Modelling Project	2
Responsible Instructor	A. Bombelli	
Instructor	Ir. P.C. Roling	
Instructor	M.J. Ribeiro	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	All mandatory SAT courses.	
Course Contents	The course has a project format. It is meant to deepen knowledge and skills in mathematical and computational modelling in application to specific sustainable air transport problems. The course can be done under supervision of any SAT lecturer, who is willing to supervise such a project. Specific project themes can be very diverse, but should be in line with the research areas of the SAT profile. A specific theme can be defined in consultation with a supervisor. It is advisable to work on a project in Q3 or Q4, when the basic modelling techniques have been learned in Q1 and Q2.	
Study Goals	n/a, depend on the project	
Education Method	The course has a project format.	
Assessment	A report needs to be written about the project. The content needs to be discussed with the supervisor.	

AE4446	Airport and Cargo Operations	4
Responsible Instructor	A. Bombelli	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of the following lectures: Lecture 1 Introduction Lecture 2 Landside operations Lecture 3 Airside operations Lecture 4 The Gate assignment problem Lecture 5 The Aircraft taxiing problem Lecture 6 Air cargo supply chain Lecture 7 Bin packing problems in air cargo operations Lecture 8 Introduction to graph theory and scheduling problems Lecture 9 Introduction to stochastic programming Lecture 10-14 Guest lectures A short quiz is given after lectures 2 to 9. Bonus points for the test can be earned by participating in these quizzes. Note that the sequence of the lectures might change pending the availability of the invited speakers. In addition, if not enough invited speakers are found, such lectures will be replaced with lectures covering additional topics (e.g., uncertainty in airport operations) or a Q&A related to the assignment. Knowledge of linear optimization and familiarity with Mixed Integer Linear Programming (MILP) formulations is a strongly suggested pre-requisite because many models presented in the lecture and the assignment are MILPs that the students must understand and (potentially) reproduce. Additional resources, examples, and tutorials will be provided to help students.	
Study Goals	At the end of the course, students will be able to: - Understand and critically assess key airport characteristics - Understand the air cargo supply chain in terms of stakeholders and operations - Understand, reproduce, and modify mathematical models describing different operations occurring inside or around airports	
Education Method	The course consists of about 8/9 theoretical lectures and 5/6 guest lectures (based on availability of guests). Students work in a group of up to 4 students to assess and reproduce an airport/cargo related optimization model (MILP model) and write a report. As such, some knowledge about linear programming is a strongly suggested pre-requisite for the course (e.g., courses such as Operations Optimization or Quantitative Methods for Logistics are recommended courses). A closed-book computer test is used to test and grade the individual knowledge of the students.	
Literature and Study Materials	Slides and other information and supplementary material are posted on Brightspace. Some lectures and the assignment are based on academic papers describing MILP models. Such papers are also uploaded on Brightspace.	
Books	Airport Systems: Planning, Design, and Management, Richard de Neufville and Amadeo Odoni, McGraw-Hill	
Assessment	40% Individual computer test (must be 5.0 or higher) 50% Group report (must be 5.0 or higher) 10% Homework quizzes	
Permitted Materials during Tests	Non graphical calculator	

AE4462-17	Aircraft Emissions and Climate Effects	4
Responsible Instructor	Prof.dr. V. Grewe	
Instructor	Dr. I.C. Dedoussi	
Responsible for assignments	V.R. Meijer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on calculus, fluid mechanics and first year physics.	
Course Contents	The course introduces the students in the state-of-the-art capabilities, research issues and challenges in climate effects of aviation. Topics include: - Combustion and emissions - Overview of all environmental effects of aviation - Air pollution relevant atmospheric chemistry and physics - Local air pollution modelling - Aviation environmental policy, new technologies, and other mitigation options - General circulation of the atmosphere - Atmospheric chemistry and contrail formation - Climate change caused by air traffic emissions - Modelling of climate impact from aviation - Application and show cases for the assessment of mitigation options supporting decision making, e.g. for climate optimized trajectories	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the climate impact of air traffic. These elements are required, for example, for the assessment of the air quality and climate impact of aviation and mitigation measures, including climate optimised trajectories/routing, low climate impact routing and consequences for aircraft design and supersonic transport.	
Education Method	Lectures and exercises	
Literature and Study Materials	Lecture slides and hand-outs	
Assessment	Written exam (Resits may be oral, depending on the situation) Assignments Both the written exam and the assignments need to be passed with a 6.0 or higher.	

AE4463P-23	Advanced Aircraft Noise Modelling and Measurement	4
Responsible Instructor	Prof.dr.ir. M. Snellen	
Contact Hours / Week x/x/x/x	0/2/0/0 This course can also be followed in Q3, please contact the lecturer	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course AE4463P-23 aims at providing students with a solid background in the modelling and measurement of aircraft noise. The course consists of lectures and a number of assignments that focus on: A. Noise metrics Microphone data, consisting of measured pressures as a function of time will be provided. From these data both the overall sound pressure level (OSPL) and the A-weighted OSPL should be determined as a function of time. Based on this, the Sound Exposure Level (SEL) must be calculated. B. Digital Signal Processing (DSP) Lecture notes on this topic will be provided. The corresponding assignments focus on resolution in the frequency domain after Fourier Transforming the time signals, the effect of weighting the signals, sampling and cyclic convolution. C. Analog Signal Processing Lecture notes on this topic will be provided. For circuits related to acoustic measurements, an important element is the operational amplifier. The assignment focuses on a circuit containing this element and aims at providing an understanding of the properties of this amplifier circuit. The assignment is partly written, but also contains the actual building and testing of the circuit and also applying it for acoustic measurements. D. Aircraft noise source modelling For this assignment a program needs to be developed for calculating the noise levels for the various airframe components (clean wing, slat, flap, main landing gear). E. Calculating noise footprints from Noise-Power-Distance tables Current noise monitoring is based on models using the so-called Noise-Power-Distance (NPD) tables, specifying noise levels as a function of aircraft engine settings and distance. In the assignment such NPD table is used, together with an aircraft flight profile, to calculate the noise footprint for a certain aircraft. F. Acoustic imaging A series of assignments that focus on acoustic imaging, ranging from calculating the theoretical directivity of acoustic arrays, to imaging the noise from a real aircraft fly-over. Out of all above assignments, A, B, C and F are required. A choice can be made between D and E.	
Study Goals	The course aims at providing the students with a thorough understanding of the theory and modelling of the aircraft as a noise source and the state-of-the-art techniques for measuring aircraft noise. The classes are shaped around assignments to ensure application of the theory. The assignments address both the basics of modelling and signal analysis, but also more advanced techniques, with emphasis on the powerful technique of acoustic imaging with phased arrays. These elements are essential, for example, for identifying and localizing acoustic sources, accurate noise contour calculations and low noise aircraft design.	
Education Method	The material of this project will be provided to students that have enrolled in the course. Hours will be scheduled where the assignments can be discussed with and questions can be posed to the responsible lecturer. In addition a number of lectures will be given. The assignments can be carried out in groups, but this should be discussed with the lecturer in advance.	
Assessment	Assignments.	

AE4465	Maintenance Modeling & Analysis	4
Responsible Instructor	A. Bombelli	
Instructor	Dr. I.I. de Pater	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Maintenance is a critical operational activity in aviation, and is crucial to retain the aircrafts airworthiness. With the current developments in AI, the field of maintenance is quickly evolving as well, and new maintenance strategies involve methods from machine learning and data science. However, also classical reliability methods, probability and statistics remain very important in the field of maintenance. This course will cover all these topics. Specifically, the course consists of three parts where you will learn to: I Introduction to maintenance: 1. Describe the concepts of maintenance and reliability. 2. Apply the basic theories and models of probability and statistics to a variety of problems (recap). II Preventive maintenance (i.e., classical reliability methods): 1. Describe the concepts of preventive maintenance. 2. Fit an appropriate distribution to historically observed lifetimes of the component 3. Optimize a preventive maintenance strategy based on the usage of the component, using stochastic processes. If there is time left, you will also learn the basics of optimizing the inventory (spare parts) management. III Predictive maintenance: 1. Describe the concepts from predictive maintenance. 2. Estimate the current and future degradation and predict the failure time (remaining useful life) of aircraft components with various machine learning/data science techniques.	
Study Goals	By the end of the course, you should be able to 1. Optimize a preventive maintenance strategy for a component based on the usage of this component, including a. Fit an appropriate distribution to historically observed lifetimes of the component b. Identify the optimal maintenance time of the component based on a distribution of the historical lifetime, with various problem characteristics c. Explain the stochastic processes underlying this optimization procedure. 2. Apply the techniques from predictive maintenance to predict the future health of a component based on the sensor measurements of this component, including a. Estimate the current and future degradation in a component b. Predict the failure time of a component using various machine learning/data science techniques.	
Education Method	The content of the course will be given via on-campus lectures and tutorials. The students will also have access to the lecture slides and exercises on Brightspace. There will be assignment(s) to be solved either in groups or individually. The assignment(s) may involve coding problems, open questions, or mathematical exercises. The primary bibliography for this course will be: The presentation slides of all lectures will be made available through Brightspace. Non-mandatory reference literature: - Kumar, U. D., Crocker, J., Knezevic, J., & El-Haram, M. (2000). Reliability, Maintainability and Logistic Support - A Lifecycle Approach. Boston: Kluwer Academic Publishers. - Arts, J. J. (2017). Maintenance modeling and optimization. (BETA publicatie : working papers; Vol. 526). Technische Universiteit Eindhoven. - Axsäter, S. (2015). Inventory control (Vol. 225). Springer.	
Assessment	The course's final grade will comprise the individual exam grade (60%) and individual or group assignment(s) (40%). It is required that students have a grade equivalent or higher than 5.75 in the individual exam and an average grade equivalent or higher than 5.75 in the assignments to pass the course.	
Elective	Yes	

AE4889	Special Topics in Astrodynamics	2
Responsible Instructor	Dr.ir. M.J. Heiligers	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Summary	A modular and research-focussed course on advanced topics in astrodynamics. You will extend and deepen your insights and analytical abilities based on Fundamentals of Astrodynamics and related coursework, devoting particular attention to their application to a selection of advanced astrodynamics topics (see below).	
Course Contents	The course treats a selected set of advanced topics in astrodynamics; the exact topics change from year to year depending on the latest research developments and the instructors involved. Strictly as examples, the following topics were covered in previous years: - Circular restricted three-body problem (equilibrium points, their stability and manifolds, and periodic orbits around the equilibrium points, their stability and manifolds) - Solar-sail orbital dynamics (acceleration models and artificial equilibrium points in the circular restricted three-body problem) - Machine learning applied to astrodynamics	
Study Goals	Within the context of the selected astrodynamics topics: - Extend your insight and analytical abilities beyond the Fundamentals of Astrodynamics and related coursework; - Actively apply your knowledge and skills accumulated through Fundamentals of Astrodynamics and related coursework; - Tackle "open-ended assignments with a hands-on, scientific approach as a first foray into astrodynamics research.	
Education Method	Workshops, lectures, self-study	
Computer Use	Yes	
Prerequisites	Fundamentals of Astrodynamics (AE4874 I)	
Assessment	Assignments	
	Additional detail will be available on the course website (Brightspace)	

AE4897	Space Engineering Practical	8
Responsible Instructor	I. Akay	
Responsible Instructor	Ir. B.T.C. Zandbergen	
Instructor	Dr.ir. J. Bouwmeester	
Instructor	Dr. J. Guo	
Instructor	Dr.ir. R. Saathof	
Instructor	A. Cervone	
Instructor	Dr. A. Menicucci	
Instructor	S. Speretta	
Instructor	M.S. Uludag	
Instructor	Dr. B.V.S. Jyoti	
Instructor	Dr. J.A.M. Vanhamel	
Contact Hours / Week x/x/x/x	0/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you will be taught on and experience the basic steps needed to successfully conduct experiments/simulations to verify/characterize a product/system as part of a real space engineering research project brought forward by industry and/or academia. Everyone who ever designed an experiment or conducted simulations to verify product/system requirements has experienced issues with improperly formulated requirement on one hand and the preparation and execution of simulations and experiments on the other. Typical problems that occur are inadequate planning (not planning in advance, lack of time), system/object not being in agreement with the design specifications, improper test set-up, instrument failures, missing data, convergence issues, bugs, extremely long calculation times, and improper requirements verification. Learning about the basic steps will allow the students to conduct such projects successfully in the future. The course is made up of two complementary parts, which are described in more detail below. Part 1 provides in-class instructions on how to perform specific tasks needed for conducting a proper test/verification project. The subjects dealt with in class (lectures/workshops) include (not necessarily in the order indicated): 1. Why this course, course organization, project phases, timeline, deliverables 2. Generating requirements including also test requirements, like test ranges and test tolerances, 3. Hypothesis formulation and assessment, 4. Design of Experiments, Experimental Measurements, and Error Propagation, 5. Test planning and the development of the test set-up and the accompanying Assembly, Test and Safety Procedures. 6. Data gathering/acquisition (both qualitative and quantitative) & automation 7. Results analysis 8. Other, like Safety, Types of Testing, Use of Standards Part 2 focuses on teams stepping through the project cycle of a experimentally oriented project assigned to them. The steps include: - Orientation (on project assignment, object/system to be researched/tested/verified under what conditions, goals) - Estimation of the performances of the test object by design analysis/simulations - Development of test hypothesis, and test requirements and criteria - Generation of test specification with success criteria - Generating a test plan - Assembly/build of test object (may be a mock-up in case of use of hazardous materials) - Assembly of (mock-up) test set up. early testing (including calibrations) and data elaboration to develop/check procedures - Test readiness review (if review is passed go on to next step, if not revise test plan) - Test execution - Data elaboration (observations and experimental data) - Discussion of results (data review) - Communicating the results/outcomes + project close-out For each project, project supervisors may arrange for more specific training/instructions related to the specific test set-up to be used. Project supervisors are also responsible for students conducting safe testing.	
Study Goals	Upon completion of the project the student shall be able to: 1. Describe and motivate the project cycle of an experimental project. 2. Formulate a test specification and test plan. 3. Set up and assemble/develop an appropriate test set-up and test procedures. 4. Perform data acquisition (both qualitative and quantitative data) with some degree of automation. 5. Execute measurements in a reproducible way and determine measurement uncertainties. 6. Analyze experimental data taking into account measurement uncertainties. 7. Discuss the outcomes of the project including the statistical validity of the data. 8. Report on the outcomes of an experimental project.	
Education Method	Lectures/workshops and Project The course is a practical lab project combined with lectures/workshops. The lectures/workshops are given by a variety of instructors. All instructors are expected to link the material they are teaching to the various projects. Instructors shall also strive to let students experience the work through a number of workshops/assignments that are geared to the projects the various teams are conducting. During the projects, instructors can be approached in case teams need help on elements instructed upon. Most of the instructions are scheduled in Q2, but to allow for proper alignment with the projects, some may move to Q3. The size of this part of the course is 2 EC (roughly 56 working hours for each participating student). Each project on offer is of size 6 EC and is conducted by a team of 5-6 students. All teams are assigned a staff member, with expertise in the field, as project supervisor. The role of the project supervisors is to provide intellectual guidance and offer advice on the planning and progress of the work. Project supervisors also are to check and review the progress made by the team and determine if a team can move on to the next phase in the project, thereby ensuring project progress and that test preparation activities and testing are conducted in a safe way. To streamline this, two formative assessments will be performed, which help students identify their strengths and weaknesses and target areas that need work. The first such assessment relates to the project	

	specification and the second one is prior to getting approval to conduct full scale testing. Additionally, during their project, students have an open invitation to consult the lecturers/instructors of the lecture/workshops part of the course for advise/feedback on issues taught in class or exercised upon in the workshops that are relevant to their project. Projects can be conducted in the Q2 to Q4 time frame of the academic year. The exact time frame is subject to negotiation between the team and the respective project supervisor. For instance, teams can propose running the project in a single quarter or spread over multiple quarters.
Literature and Study Materials	Appropriate literature and study materials will be provided throughout the course. For the lectures/workshops, material is provided for by the instructors relating to the topic/subject they instruct upon. For the specific projects, a project assignment will be made available that describes the test object/system to a great detail, including the relevant characteristics of the object/system. The project assignment will also describe the test goals as well as the type of testing to be performed (functional testing, life testing, reliability testing, failure testing, etc.) and the test conditions under which the testing is to be performed.
Assessment	The result of this course will be entirely based upon the results of the experimentally oriented project and the associated deliverables. These deliverables include: - Test specification - Test plan - Results Summary document, test data files, and a visual demonstration (video/photos/poster) of the testing conducted and the equipment used. For each deliverable a template/content description (or example) will be provided. Assessment will take place according to the detailed rubrics available on Brightspace. Assessment is by the project supervisor of the project (40% of the final grade) and the course instructors involved in the instructions (60% of the final grade). Assessment is coordinated by the course responsible instructor. A minimum grade of 6 is needed after rounding to the nearest half to pass the project. Late submission of the work results in a maximum penalty of 1.5 point (on a 10-scale). Due dates for submission will be communicated in due time via Brightspace. In case of an insufficient grade, teams are allowed one opportunity to resubmit an improved version of their report.
Enrolment / Application	This course is only open for enrollment for students enrolled in the Space Engineering profile of the track Space. A maximum of 45 students will be accommodated on a first-come, first-served basis. Students will have the opportunity to choose from a list of available projects that encompass different knowledge fields in the field of Space Engineering. Some past projects include: - Battery project: Novel capacitor-based spacecraft energy storage system - Propulsion project: Solid rocket motor assembly and testing (in cooperation with DARE) - Attitude control project: Assembly and testing of a PocketQube attitude determination and control system - Payload controller project: Assembly and testing of a small sat onboard data handling system

AE4898	Space debris tracking and mitigation	4
Responsible Instructor	Dr.ir. E. Mooij	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	AE4874 I (Fundamentals of Astrodynamics) is required. AE4868-1 (Numerical Astrodynamics), AE4894 (Practical Astrodynamics) and AE4870B (Re-Entry Systems) are recommended.	
Course Contents	The course will introduce the problem of space debris and associated aspects, such as how to detect it, how to avoid it, and potentially how to remove it. The contents of the course will be presented in a hybrid format, i.e., a combination of classical lectures and working lectures, where the students can practise with the introduced material, tools and techniques. The course content is roughly split into two parts. The first half of the course will introduce the problem of space debris: how much is out there, how large and heavy is it, where does it come from, and what kind of problems does it pose. Having identified the problems for the safe operation of space, the focus shifts to how we can predict potential collisions and how we can avoid those. Finally, how can the larger chunks of space debris be removed (Active Debris Removal), and if the debris enters the atmosphere and does not fully burn up: where will it impact on earth. The second part focuses on the observation and estimation of space objects in general, and space debris in particular. The students will learn about different sensors that can be used, either ground-based or space-based, the accuracy of measurements, and how this will affect the estimation process. Linear and nonlinear estimation techniques will be discussed, with application to determine translational state vectors, characterise physical parameters, and detect manoeuvres of space objects. The course will further consider methods of sensor management and data fusion.	
Study Goals	After the course the student shall: 1. Assess the risks posed by orbital debris on current and future space missions, including the probability and consequences of on-orbit collisions 2. Analyze the mitigation of space debris hazards through the use of post-mission disposal and active debris removal 3. Estimate the orbits and physical characteristics of resident space objects using simulated space object tracking data, including measurements from optical and/or radar sensors 4. Design a collision detection and avoidance strategy for use with a simulated space object catalog	
Education Method	Per week, two lectures are scheduled. One is a regular ("classical") lecture where topics related to the field are introduced, clarified with examples. The second lecture is a working lecture, where tools are introduced by means of tutorials and where students can begin on their assignments/lab work.	
Computer Use	Personal laptop that can run Tudat. The choice of programming language is free (Python knowledge is required to work with Tudat). No coding support is provided.	
Literature and Study Materials	The main course material consists of the lecture slides. Recommended papers and books will be suggested during the course.	
Assessment	The assessment is a combination of one individual and two group take-home assignments. A tutorial assignment (not graded) will be handed out at the beginning of the course. For the group work, individual contributions to the group work must be clearly identifiable. The total grade is composed of: Individual assignments (individual grade): 30% Project (individual grade): 30% Project (group grade): 40% Each of these three grades must be ≥ 5.0 to pass the course (final grade is the weighted average, which should be 5.75 or higher). The resit of the course will be done based on individual assignments. Support during this will be limited. It is not possible to resit single assignments, only a resit of the full course is offered.	
Enrolment / Application	The course is open to students of the Spaceflight Dynamics profile only; the maximum number of students that can participate is limited to a maximum of 30. To enroll, the student should write a motivation (limited to 5 lines) why this course is important/relevant to them, and send this by email to both instructors: e.mooij@tudelft.nl (Dr Erwin Mooij) and s.gehly@tudelft.nl (Dr. Steve Gehly). Enrollment closes 2 weeks before the start of the course, and students will be notified about the status of their enrollment before the course starts.	

AE4ASM003	Linear Modeling (incl. F.E.M)	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mechanics (statics, mechanics of materials), mathematics (linear algebra) and programming (Python or Matlab) skills	
Course Contents	Learn how to model real life engineering problems using Finite Element Methods.	
	Computational methods in structural analysis are of prime importance in industry as tools to assess the efficiency and performance of structures in the field of aerospace, mechanical, civil and biomedical engineering. A combination of theoretical and practical knowledge in finite element analysis are valuable skills needed to address such problems in industry. To efficiently model a real life engineering problem using finite element analysis and predict its future behaviour, an engineer must possess a strong theoretical understanding of the finite element method (FEM) along with the understanding of the importance of verification and validation of such computational models.	
Study Goals	At the end of this course students are able to:	
	1. Explain the different steps in a finite element analysis and apply them to practical engineering problems	
	2. Explain and apply basic principles behind finite element analysis (i.e., minimum total potential energy and weighted residual function)	
	3. Develop and implement 1D (bar, truss, beam and frame) and 2D (triangular and rectangular) elements in a finite element set-up	
	4. Use and interpret results from 1D and 2D elements in commercial finite element software	
	5. Explain and perform verification and validation of results obtained using finite element principles	
Education Method	Weekly lecture and practical	
Assessment	4 homework assignments (not all with the same weight) throughout the course: for deadlines and weighing factors, see Brightspace.	
	Students who do not obtain a passing grade (i.e., average, rounded grade of 6/10) will have the opportunity to do a replacement assignment for up to 2 assignments (only assignment 3 and 4 since they have the highest weight; assignment 1 and 2 cannot be redone).	
	Results for individual assignments are only valid for one academic year: when re-taking the course all assignments have to be done in that academic year.	

AE4ASM101TU-22	Polymer Science	4
Responsible Instructor	Dr. B. Kumru	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course consists of a number of lectures covering the most relevant aspects of polymer science and technology distributed in 6 blocks, each covered in one up to three lectures. After this a session on 3 minutes pitches by the students will be organized: -Introduction to polymer science: daily life examples, nomenclature, concept of polymers, initiators, polymer architectures and polymerization media. -Polymerization: condensation and addition polymerization under step-growth; radical, ionic and ring opening polymerization under chain-growth mechanisms, concept of copolymerisation and crosslinking, basic characterization techniques (such as GPC). -Polymers in Aerospace: brief outlook on composites, chemistry of thermosets, high-performance thermosets, high performance thermoplastics, mechanical properties. -Processing: basics of polymer processing and polymer-based structures manufacturing, fiber processing and properties, living materials and their potential. -Sustainable Polymers: a glance on green chemistry, introduction to lignocellulosic biomass, how to develop biobased monomers, recyclability and dynamic materials, examples on sustainable polymers, impacts on sustainability, a perspective on how can you design sustainable polymers? -Polymers for Energy and Environment- demonstrating a concept of photovoltaic devices and batteries, polymers used in these applications, polymer membranes and polymer coagulants for water remediation, polymer ionotronics. -Pitch talks: students to give 3 minutes presentations about developments/applications in the polymer field, convince audience as in business structures.	
Study Goals	By the end of this course, you should be able to: LO1: Identify polymer basics and analysis LO2: Assess polymer structure-property relationship for thermomechanical performance estimation LO3: Judge parameters and requirements for polymers to be qualified for aerospace materials design LO4: Define sustainability concepts in polymer science LO5: Present about recent developments in polymers field	
Education Method	Lectures and lab activity	
Literature and Study Materials	Seymour/Carraher's Polymer Chemistry: 10th Edition. Contains comprehensive information on polymer materials from chemistry perspective Lecture slides will be made available to students as study material	
Assessment	Written final exam and pitch talk (3 minutes) presentation Lab activity attendance is mandatory Transitions AE4ASM101TU -> AE4ASM101TU-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM101TU, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM101TU. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM101TU is not completed in 2022 2023, students will need to complete AE4ASM101TU-22 instead.	

AE4ASM103	Functional Coatings	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The students must have general knowledge about materials science.	
Course Contents	In this course we introduce the design fundamentals that control specific properties in coatings, surface treatments and functional coatings and the techniques used to apply them and test them. The goal is therefore to provide the fundamentals of chemistry and physics that explain structure-property relationships relevant to understand and design surface treatments/coatings with: (i) all-round relevant properties such as adhesion, mechanical response, active (inhibitors) and passive (barrier) corrosion protection; and (ii) increasingly relevant functional properties such as antiicing, self-cleaning, self-healing, drag, structural colours, storage/release of active chemicals with nano and microparticles, and antifouling. Since material design rules are directly related to application methods, different technologies for coatings application will be covered such as electrophoretic deposition, powder coatings, anodizing, sol-gel and plasma electrolytic oxidation. Characterization strategies to obtain information about key properties will also be discussed. The fundamentals and characterization strategies covered in the course are applicable to coatings, adhesives and composites of interest in aerospace applications and other engineering domains.	
Study Goals	The students will be able to: explain the key mechanisms leading to relevant coating properties and functionalities; use a tool-box to develop new coating systems; relate formulation parameters (chemistry, particles), design and production parameters with final properties; identify suitable techniques to characterize coating properties (e.g. mechanics, viscoelastic properties, barrier, adhesion, crosslinking density); support sound solutions to known problems by analyzing literature.	
Education Method	The course consists of a set of lectures and one assignment followed by a presentation. The assignment is individual. The topic for the assignment will preferentially be unique for each student and related to the course content on coating functionalities. If the number of students registered exceeds a critical number the assignments will be performed in small groups.	
Literature and Study Materials	The lectures (and facilitated slides in pdf) will be supported by selected literature and book chapters that will be facilitated during the lectures. The students will also use specific literature for their assignments.	
Assessment	Written report (40% of final grade) and written exam (60% of final grade). A minimum of a 5 over 10 points will be required in the written exam to make an average with the report.	

AE4ASM105	Trinity Exercise	4
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr.ir. C.M. de Zeeuw	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	none	
Course Language	English	
Parts	Starting period Compared to previous years the course starts in Q3 (not Q2). There will be a kick-off lecture in week 3.1	
Course Contents	Participants Depending on the theme students choose in the track Aerospace Materials and Structures this course is either compulsory or an elective. When the course is compulsory students can always participate. When the course is elective students might be refused to participate for logistic reasons, such as availability of supporting staff and equipment. In case students need to be refused the responsible instructor decides on clear grounds who will be refused. Week arrangement Per team of four to five students about 8 day sections with support of staff are needed during this course. These will be planned on fore hand, preferably on fixed time slots. Most likely either Friday morning or Friday afternoon. We will not use all of those slots, but it will provide the flexibility to schedule things in the lab and make it easier to block their schedules. Finishing period As the starting period changed, the finishing period changes according, at the end of the fourth period (Q4, June) it is. The name "Composite Trinity Exercise" is based on the fact that designing composite structures requires knowledge of production methods, materials and geometrical design. The exercise is defined in following parts: 1. Manufacturing of thermoplastic and thermoset laminates 2. Determination of fibre volume fraction, void volume fraction and density 3. Estimation of the maximal bending-torsion coupling for a strip of UD composite 4. Possibly manufacturing of specimens for mechanical tests, including adhesive bonding of tabs 5. Performing mechanical tests 6. Analysing the test results of the mechanical test, including failure analyses 7. Adhesion of thermoset laminates 8. Possibly preparation of lap shear specimens 9. Performing lap shear tests 10. Estimation of the mechanical properties of a sandwich panel 11. Manufacturing of a sandwich panel by either vacuum infusion, pressing or vacuum bagging. 12. Performing a bending test on a sandwich panel. 13. Writing a test report.	
Study Goals	After succeeding this course the student should be able to: - produce vacuum infusion based laminates - produce thermoplastic based laminates - evaluate laminate quality - determine fibre volume fraction of laminates. - prepare test specimen according to standards - perform tests according to test standards - apply statistical methods for determination of design allowables. - evaluate the difference between modelling and reality.	
Education Method	Laboratory Exercise.	
Literature and Study Materials	The reader and schedule belonging to this course is handed out on at the kick off of the exercise. Other study materials, such as test standards, test planners, material specifications can be found on Brightspace. Some study materials need to be found via internet by the students themselves.	
Assessment	A written report by each group. This report is assessed and evaluated. Passing this course will result in a V.	
Remarks	This course was renewed in 2023-2024 by fixing the time schedule for students on beforehand.	
Set-up	The students work in teams of three to five persons (preferably four). The division of the students into the teams is done on beforehand (shortly before the start). The duration of the exercise is estimated at 20 day sections. Some of the day sections are planned by the supporting staff. The others are planned by the students.	

AE4ASM106	Stability & Analysis of Structures I	3
Responsible Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	This course introduces analytical and numerical techniques for the analysis of representative aerospace structures under mechanical loading, with particular emphasis on determining their stability. The course covers the connection between the principle of minimum potential energy, the variational formulation and the classical (differential) formulation of quasi-static structural problems. Basic structural components are analyzed using beam and plate theory. The basic theory of structural stability is introduced through asymptotic analysis in order to derive the stability (buckling) equations. Buckling equations are formulated and solved for geometrically non-linear elastic beams and plates. The methodology is extended to formulate stability analysis of representative aerospace components. Emphasis is placed on the understanding of the models, their assumptions, and ranges of applicability. The topics covered include: - Introduction to structural analysis - Review of global and local balance principles - Constitutive relations, material symmetries and strain energy - Formulations of structural problems: differential, variational and minimum formulations - Introduction to calculus of variations and approximate solution methods (Ritz and Galerkin) - Applications to beams and plates: solution to basic structural problems - Introduction to buckling: Eigenvalue buckling problem (bifurcation buckling) - Imperfection analysis (snap buckling) and stability analysis - Buckling of non-linear beams/columns and plates - Design methodology for thin-walled aerospace structures	
Study Goals	1- Understand the various formulations of equilibrium and equations of motion 2- Model simple aerospace structural components using engineering theories 3- Apply energy and variational methods for the evaluation of structural responses 4- Understand and being able to obtain the buckling equations for beams and plates 5- Develop insights into buckling phenomena of aerospace thin-walled structures	
Education Method	Lectures.	
Literature and Study Materials	Primary source: course slides provided in Brightspace Recommended references: T.H.G. Megson, Aircraft Structures for Engineering Students, Elsevier, Any edition. C.L. Dym and I.H. Shames, Solid Mechanics: A Variational Approach, Augmented Edition, Springer, available online through the TU Delft Library.	
Assessment	Written (take-home) assignments and oral exam	

AE4ASM108	Experimental Techniques & NDT	3
Responsible Instructor	Dr. A. Anisimov	
Instructor	Dr. R.M. Groves	
Responsible for assignments	K.P.W. Dissanayake	
Responsible for assignments	Dr. N. Yue	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of electronics, elements of signal processing (analogue, digital), basics of measurements and metrology	
Course Contents	This is a lecture course on developing the critical thinking needed to perform measurements in the Aerospace Structures and Materials Laboratory. This includes an introduction to the most common sensors for experimental mechanics, non-destructive testing and structural health monitoring, evaluation of the performance of these sensors, selection of sensors for different applications and signal processing and control algorithms.	
Study Goals	By the end of the course, you should be able to: 1. Analyse a measurement system for the aerospace sector and to determine if it is fit for a specific application. This learning objective acts as an umbrella objective, covering the whole course. Lower level objectives include that by the end of the course, you should be able to: 2. Explain the principle of operation of the most common sensors and instrumentation in the aerospace structures and materials laboratory. 3. Justify the selection of a sensor for a specific measurement application, based on sensor specifications and measurement principle. 4. Analyse a measurement chain and determine signal accuracy and noise levels. 5. Assess the performance of signal processing and feedback algorithms for their influence on measurement noise and accuracy.	
Education Method	Lectures, obligatory laboratory sessions, reader with self-evaluation questions.	
Literature and Study Materials	Course reader, pre-recorded videos with slides available on Brightspace. Additional material as datasheets, book chapters and papers are uploaded to Brightspace during the course.	
Reader	Course reader available on Brightspace	
Assessment	Open book written exam (30% of grade). Oral exam (40% of grade). Laboratory group assignments (30% of grade).	
Set-up	12 active lectures with problem-solving and discussion (weeks 2.1-2.6). Students have to prepare for these workshops using the material available on Brightspace, including pre-recorded videos. The material becomes available as the course progresses. 2 exam review workshops (week 2.7). It is particularly important to attend these sessions as the course lecturers will explain the structure and what is required in the open-book exam 5 obligatory laboratory assignments done in groups (weeks 2.1-2.5). Each practical is graded with a group grade for the written report (maximum of 4% towards the final course grade) and personal grade for the oral defence of the report (maximum of 2% towards the final course grade).	

AE4ASM109	Mechanics & Structural Health Monitoring of Composite Structures	5
Responsible Instructor Responsible Instructor Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Expected prior knowledge	C. Kassapoglou Dr. D. Zarouchas 0/0/4/0 3 3 none English Knowledge of Mechanics of Materials Knowledge of Linear Algebra Knowledge of Theory of Elasticity Affinity with infinite series Affinity with Partial Differential Equations Understanding of Experimental Mechanics Programming skills Minimum standards as taught in the BSc. programme of AE. See relevant course descriptions in the AE BSc. study guide	
Course Contents	1. Classical Lamination Theory a. Short Overview of Materials. Composites Design Philosophy b. Theory of Elasticity c. Engineering Constants d. Stress & Strain Transformations / Implications for Testing e. Thin laminates 2. Progressive damage analysis a. Failure criteria b. First ply failure & Last ply failure c. Damage tolerance analysis 3. Reliability analysis & Health monitoring a. Probability of Failure & Uncertainty quantification b. Structural Health Monitoring 4. Basic Stress Solutions and Buckling of Composite Plates a. Typical Airframe Elements b. Airframe Design Process, Materials & Damage c. Plate Governing Equations / Solution of the PDE d. Energy Minimization Methods e. Buckling of Composite Plates	
Study Goals	The students should develop in-depth understanding and insight with regard to the basic mechanics of composite materials and structures, and be able to apply the lectures theories and methods to tackle a variety of basic composite design problems. In addition, the students should become able to expand the provided analysis tools towards more advanced solutions for their graduation thesis work. Furthermore, the students should be able to understand and apply related scientific literature. At the end of this course the student will be able to: - Understand the basic mechanics of composite materials and structures as listed under "course content" - Show insight into the theory of progressive damage analysis of composite materials and structures by applying the theory and methods and tools listed under the course contents to solve a variety of basic composite structure design problems - Understand the philosophy of reliability analysis and calculate the probability of failure of composite structures by combining numerical methods and the mechanics of composite structures - Understand the importance of structural health monitoring (SHM) and design a SHM system for a given composite structure - Demonstrate the ability to expand the theory, methods and tools towards more advanced solutions in real practice such as students may encounter during their thesis - Understand, select on relevance, and apply additional theory, tools and methods on composite materials and structures found in relevant scientific literature to (design) problems based on the material taught in class	
Education Method	Lectures in sessions of 2 X 45 minutes. The distribution in sessions is: 1. Classical Laminate Theory: 3 2. Progressive damage analysis: 3 3. Reliability analysis & Health Monitoring: 4 4. Basic Stress Solutions and Buckling of Comp. Plates: 4	
Literature and Study Materials	Daniel IM, Ishai O. Engineering Mechanics of Composite Materials 2nd Edition, Oxford University Press, 2006. Zarouchas D. Lecture slides on the Classical Lamination Theory, progressive damage analysis, reliability analysis and health monitoring. Available on Brightspace. Kassapoglou C. "Design and Analysis of Composite Structures", 2nd Edition, Wiley, 2013. Kassapoglou C. Lecture slides on Basic Stress Solutions and Buckling of Composite Plates. Available on Brightspace.	
Assessment	Parts A1 and A2 will be examined by means of a take-home assignment. The students should demonstrate correct understanding of the conditions and limitations of the applied methodologies and should prove their ability to solve design problems by the utilization of basic programming techniques and proper results interpretation. They can earn a maximum of 60 points according to the following breakdown: A1. ABD and stress analysis of composite plates 10 points A2. Progressive damage and damage tolerance analysis 20 points A3. Reliability analysis & Health Monitoring. 30 points	

Part B1 will be assessed on the basis a design problem. The design problem will address a realistic composite structure and will require use of all aspects of part (4) as well as use of the Classical Laminated Plate Theory (2) and Failure criteria (3) (see education method).

B1. Design problem 40 points

The maximum score a student can get is thus 100 points. There is no possibility to omit point-related exam parts, for example handing-in a report regarding only parts A1+B1. Every part (Part A1,A2,A3 and part B1) should at least be rewarded with 55% of the attributed points (55%*60 and 55%*40), otherwise the student will fail. In addition, the overall average grade should at least be a 58 on the basis of 100. Grades will be rounded to the nearest half as per regulations.

Students who have handed-in the initial assignments for both parts and have received an overall grade of 5.5 or lower are entitled to a resit assignment in the same academic year. In that case, the student should resit the failed part only. If the student grades higher than 5.5 (pass) and wants to amend the grade, should resit both parts. The date from which the resit assignments are available is announced on Brightspace but will always be during the July-August period.

Each assignment will have a deadline published on Brightspace. Deadlines must be met, no exceptions possible. Assignments are individual and if students work together on a design problem, they must state this in the report naming the collaborators and hand-in individual reports that are 100% original work. (no commonality!). Failure to do so will be seen as plagiarism.

AE4ASM110-22	Polymer Composite Manufacturing	4
Responsible Instructor	Dr. J.J.E. Teuwen	
Instructor	Dr.ir. D.M.J. Peeters	
Instructor	Dr.ir. B.C.P. Jongbloed	
Instructor	Dr. B. Caglar	
Instructor	R. Tao	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The course explores in depth the manufacturing of polymer composite structures and its underlying physics. By understanding the relationships among physics, part/material quality and the design of the manufacturing process, you will be able to critically assess any given manufacturing process. The course is built around a number of polymer composite manufacturing processes highly relevant to the current and future aerospace industry for the manufacturing of high-performance individual parts (autoclave processing of prepreg, liquid composite moulding, thermoplastic composite processing), for the assembling of complex structures (manufacturing of integrated structures, thermoplastic composite welding) and for end of life and recycling. The main basic physical phenomena governing polymer composites manufacturing, e.g. flow (of polymer and of fibres), void formation, curing, shrinkage and crystallization, are interwoven within the course.	
Study Goals	At the end of this course, the student should be able to: Describe the common manufacturing processes used in aerospace industry in terms of processing steps, tooling and equipment as well as advantages, limitations and applicability. Correlate and analyse the processes and their principles/underlying physics with manufacturability, part quality and manufacturing design. Compare and evaluate the different processes for their suitability of manufacturing common components (aircraft wing, fuselage etc). Recognise manufacturing defects, identify their potential sources and recommend strategies to improve the quality of common manufacturing processes for simple component geometries to reduce scrap rate or to increase yield.	
Education Method	Interactive lectures with numerous in-class exercises. Assignments are used throughout the course to allow the students apply, deepen and test their knowledge on the manufacturing processes discussed during the course.	
Assessment	Oral exam (50% of the final grade) - exam format subject to change depending on amount of students - and mandatory assignments (50% of the final grade). A minimum grade of 5.0 out of 10 on both the assignment and the oral exam need be obtained to be able to pass the course.	

AE4ASM503	Sheet Metal Forming	3
Responsible Instructor	Ir. J. Sinke	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The lecture consists of two main parts: a theoretical part and an application part The theoretical part includes: - Introduction, formability, deformations, formability testing. - Stress- and strain states, forming limit curves, yield criteria, anisotropy, heat treatments. - Theory of plasticity The application part includes a number of forming processes, like - Simple forming processes like bending and stretching. - Several forming processes for more complex parts like deepdrawing, rubber forming, hydroforming and matched die forming. - Non-conventional processes like explosive forming and superplastic forming - The use of numerical tools for simulation of forming processes, and some other related topics.	
Course Contents	The theoretical part of the lecture will focus on the descriptions of deformations, strains, stresses, etc. In addition, the interaction between stresses and strains and their limits are expressed in a.o. yield criteria and forming limit diagrams. Such concepts are also used in the second, application part to describe and explain the do's and don't's of the forming processes.	
Study Goals	After attending this course the student should be able to: - describe and explain the basic phenomena in plastic deformation of sheet metal - understand the interactions between multiple phenomena and how to exploit these - be able to relate the formability of materials to forming processes - have understanding and ability to combine the features of material behaviour, the forming process and the shape of the component.	
Education Method	Lecture, including discussions about real parts, pictures and videos.	
Literature and Study Materials	Hand-outs are the core material A list of recommended literature will be given during the first lecture (background information unless specified otherwise)	
Assessment	Written exam. Closed book.	

AE4ASM504	Structural Integrity and Maintenance	3
Responsible Instructor	N. Eleftheroglou	
Responsible Instructor	Dr.ir. J.A. Pascoe	
Instructor	Prof.dr.ing. R.C. Alderliesten	
Instructor	Ir. J. Sinke	
Instructor	Dr. R.M. Groves	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on aircraft structures and materials, and on deterioration of aircraft during operation due to fatigue, corrosion, impact, etc.	
Course Contents	The course is split in a generic part and a part with special topics. For a number of topics, guest lecturers are invited. Generic part - Introduction: Developments in aviation, aircraft technology and safety, effects on Maintenance. Design & safety concepts (safe life, fail safe, damage tolerance). - Maintenance Fundamentals: Definition of maintenance, Maintenance concepts and policies, Failures of airframe structures (Failure definition, Failure types and classes, Degradation types, Reliability, Probability of failures, Relation failure classes and actual failure types, such as fatigue, corrosion, etc.). - Airworthiness Requirements: Main aspects of DT requirements, Fatigue & Damage Tolerance Evaluation (Selection of Critical Locations, Classification of Structure). Regulations applicable to maintenance. - Maintenance of Airframe Structures (maintaining structural integrity): ATA-100 system hierarchy, MSG-3 methods (Structural Maintenance Program), Relation between design concepts (e.g. Damage Tolerance), actual design, maintenance and airworthiness requirements. How to translate design results in to maintenance, Structural Repair Manual, Health/condition monitoring (Load/Usage Monitoring & Fleet Life Management). Special Topics (guest lecturers) - Non Destructive Inspections of Airframe Structures: Developments in NDI methods, NDI methods in practical situations. - Load/Usage Monitoring & Fleet Life Management. - Repair of Airframe Structures: Developments in repair methods. - Structural Integrity and Maintenance of Gas Turbines.	
Study Goals	At the end of this course, you should be able to: - Describe different maintenance strategies and select the appropriate one for a given problem, taking into account different aspects (e.g. technical, regulations, planning); - Describe the working principles of different non-destructive inspection techniques, their pros and cons and their applicability to different materials; - Explain the relationship between certification and maintenance requirements and (documentation of) procedures; - Develop a maintenance programme for a simple case; - Select an appropriate design philosophy to ensure structural integrity of a given part throughout the aircraft lifetime.	
Education Method	Lectures.	
Literature and Study Materials	Lecture notes (hand-outs).	
Assessment	Written exam, closed book (60% of final grade) and group assignment (40% of final grade).	

AE4ASM505	Non-Linear Modeling (using F.E.M.)	3
Responsible Instructor	Dr. B. Chen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Learn how to model non-linear structural/solid mechanics problems using Finite Element Method (FEM). Computational methods, particularly FEM, are important tools to assess the efficiency and performance of materials & structures in the field of aerospace, mechanical, civil and biomedical engineering. Reduced performance and failure of materials & structures are mostly due to the effects of different nonlinearities such as buckling, yielding, damage and fracture. A combination of theoretical and practical knowledge in non-linear FEM are valuable skills needed to address such problems in industry and academia. Upon finishing this course, you will have the skill set needed to solve various non-linear structural/solid mechanics problems using FEM. Both the theoretical and the practical aspects will be covered. A free FEM package will be used for practical applications.	
Study Goals	- be able to explain the theories of non-linear FEM and use them to perform analytical work - be able to apply non-linear FEM to solve practical engineering problems - be able to identify and employ efficient modelling techniques	
Education Method	lectures and practical sessions	
Prerequisites	Mechanics of materials Structural Mechanics Buckling and Vibrations Linear Modelling including relevant linear FEM experience	
Assessment	2 homework assignments	

AE4ASM506	Fundamentals of Aeroelasticity	3
Responsible Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to the physical and analytical aspects of aeroelasticity. The breakdown of the course is: 1. Introduction to aeroelasticity and aeroelastic phenomena 2. Illustration of aeroelastic phenomena using simplified aerodynamic and structural models 3. Aerodynamic models for aeroelastic analysis 4. Structural models for aeroelastic analysis 5. Aeroelastic response to gust excitation 6. Aeroelastic models in state-space format 7. Aeroelastic aspects in the design of aircraft 8. Numerical aeroelastic calculations using custom-programmed software.	
Study Goals	At the end of the course the student should: 1. understand the physical processes which drive aeroelastic phenomena. 2. be able to formulate and solve aeroelastic response and instability problems. 3. be able to identify strengths and weaknesses of different aerodynamic and structural models for the analysis of a given aeroelastic condition. 4. be familiar with the role of aeroelasticity in aircraft design. 5. be able to program and solve simplified aeroelastic problems.	
Education Method	The lectures will be given in a lecture room, circumstances permitting. You will be asked to do some preparatory work at home prior to the lectures, and there are voluntary homeworks to process the material. Parallel to that, an equivalent online version of the course on Brightspace also exists. The course can be followed in the classroom, but also entirely online. A hybrid format is also an option, all depending on the students' preferences.	
Literature and Study Materials	Course notes with practice problems Additional references (optional): 1. J.R. Wright and J.E. Cooper, "Introduction to Aircraft Aeroelasticity and Loads", Wiley, 2007 2. E.H. Dowell (Editor), "A Modern Course in Aeroelasticity", 4th edition, Kluwer Academic Publishers, 2004 3. C.S. Bisplinghoff, H.Ashley, R.L.Halfman, "Aeroelasticity", Adisson-Wesley Publishing Company, 1955 4. Y.C. Fung, "An introduction to the theory of aeroelasticity", Dover Publications Inc., New-York, 1955.	
Assessment	Oral exam about course material and take-home assignment	
Special Information	The grades of the previous academic years for either the oral exam or the assignment cannot be transferred to the next academic year.	

AE4ASM510	Design & Analysis of Composite Structures II	3
Responsible Instructor	C. Kassapoglou	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Skin-stiffened structure: Analysis and Design. Failure modes with emphasis on skin-stiffener separation Sandwich structure: Analysis of different failure modes (buckling including transverse shear, wrinkling, crimping, dimpling). Issues with sandwich rampdown Design guidelines: Robust, manufacturable, and damage tolerant design Effects of damage: Holes, cracks, delaminations, impact. Analysis methods and design guidelines.	
Study Goals	Establish an understanding of how applied loads relate to local stresses and strains in composite structures. Use stresses and strains to predict failure. Combine failure methods into design approaches accounting for different components (skin and stiffeners), design approach (sandwich versus stiffened) and type of damage (hole, crack, delamination, impact)	
Education Method	Lectures	
Books	C. Kassapoglou "Modeling the effect of damage in composite structures", Wiley, 2015	
Prerequisites	AE4ASM109	
Assessment	Homework problems and a final project	

AE4ASM511	Stability & Analysis of Structures II	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	S.R. Turteltaub	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course covers numerical and experimental methods to evaluate the dynamic response of aerospace structures and structural components. The equation of motion for the dynamic response is formulated in matrix form using finite element discretization or direct modelling discretization (rigid bodies and flexible connectors). Various solution techniques are used for solving transient dynamic response and vibration problems, including modal analysis. The course also covers experimental techniques to obtain model parameters and test the response of a physical system. The topics covered include 1. Formulation of structural dynamics problems 2. Time-domain methods 3. Frequency-domain methods 4. Direct numerical integration 5. Experimental modal analysis	
Study Goals	At the end of the course, students should be able to: 1. Represent a structure using discrete finite element models derived from continuous formulation of beams and plates 2. Formulate and solve linear dynamic problems using modal reduction both in time and frequency domains 3. Formulate and solve general dynamic problems using direct time integration 4. Perform structural analysis based on dynamic responses 5. Identify main model parameters and understand how they quantitatively affect the structural dynamic response 6. Experimentally obtain natural frequencies, mode shapes and modal damping using different techniques 7. Validate finite element models using data from experimental modal analysis	
Education Method	Lectures, assignments, experiments, slides available on Brightspace.	
Literature and Study Materials	1. Slides available on website. 2. Recommended textbooks (not mandatory): Engineering Vibration Daniel J. Inman ISBN 0-13-228172-2 Pearson Prentice Hall Dynamics of Structures Ray W. Clough and Joseph Penzien ISBN-13: 978-0070113923 ISBN-10: 0070113920 (out of print / second-hand) Computers & Structures, Inc. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis Thomas J. R. Hughes ISBN-13: 978-0486411811 ISBN-10: 0486411818 Dover	
Assessment	Take-home assignments and oral exam.	

AE4ASM514TU	Continuum Mechanics	4
Responsible Instructor	Dr. B. Giovanardi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students taking this course are expected to be familiar with linear algebra, vector calculus and should have taken introductory courses in solid and fluid mechanics.	
Course Contents	In mechanics of continuous media, the theories of solid and fluid mechanics are presented in a unified framework. Basic notions of tensor algebra and analysis are covered in the first part of the course. Subsequently, a general description of deformations is introduced, from which the distinct measures of deformation and rates of deformation used in geometrically-nonlinear theories are obtained. Special emphasis is given to indicate the differences between the Eulerian and Lagrangian approaches as well as their connections, which are relevant for the proper description of fluid-structure interaction problems in the presence of large structural deformations. Fundamental principles (balance equations) are developed in both Lagrangian and Eulerian formulations, which naturally leads to the definition of different stress measures. The last part of the course is devoted to the constitutive theory where typical models for fluids and solids are presented. Topics include: 1. Introduction to tensor algebra and tensor analysis. 2. Kinematics: bodies and configurations. Description of deformations and motions. Lagrangian and Eulerian descriptions of a continuum. Material and spatial time derivatives. Theory of admissible deformations. Deformation and strain tensors. Polar decomposition (stretch and rotation tensors). Rates of deformation (stretching and spin tensor). 3. Kinetics: Mass and mass balance. Reynolds transport theorem. Body and contact forces. Balance of linear and angular momentum in Eulerian and Lagrangian formulations. Cauchy and Piola-Kirchhoff stresses. 4. Introduction to constitutive relations. Constitutive relations in elasticity and Newtonian fluid mechanics.	
Study Goals	General course aim: The purpose of this course is to provide a unified framework and a thorough understanding of the theories of solid and fluid mechanics. General principles, used in advanced theoretical and numerical work in mechanics, are emphasized. At the end of the course, students should be able to formulate structural and fluid mechanics problems, understand the origin of nonlinearities in each case, and be able to correctly interpret the results of analytical and/or numerical solutions. Learning outcomes: Students that successfully complete the course should be able to 1) Apply fundamental principles (balance equations) and constitutive equations to describe problems in solid mechanics and fluid mechanics. 2) Explain the difference between Lagrangian and Eulerian formulations, specifically addressing: a) the natural stress measures resulting in the two formulations, b) the translation of Eulerian measures (force, deformation, mass density) in their Lagrangian counterparts, and vice-versa, c) the meaning of the cartesian components of Eulerian and Lagrangian measures. 3) Formulate quasi-static boundary value problems and dynamic initial-boundary value problems in computational solid and fluid mechanics. 4) Solve simple solid and fluid mechanics problems analytically, either by hand or with software tools for symbolic computation. 5) Compute stress and strain measures in the context of geometric nonlinearities (large deformations) and constitutively nonlinear materials.	
Education Method	Lectures.	
Literature and Study Materials	1. Lecture notes, slides and videos available on website 2. Recommended literature: Books: Morton E. Gurtin, An Introduction to Continuum Mechanics, Academic Press, 1981 I-Shih Liu, Continuum Mechanics, Springer, 2002 Lawrence E. Malvern, Introduction to the Mechanics of a Continuous Medium, Prentice Hall, 1969 Peter Chadwick, Continuum Mechanics: Concise Theory and Problems, Dover, 1999 Raymond W. Ogden, Non-Linear Elastic Deformations, Dover, 1997. Additional resources: Rohan Abeyaratne, Lecture Notes on The Mechanics of Elastic Solids. Use a search engine to download latest version. For personal use only.	
Assessment	Take-home assignments (individual) + oral examination. Oral examination consists of two parts: (i) questions related to the take-home assignments and (ii) general theory questions.	

AE4ASM515-22		Characterization of materials and components	4
Responsible Instructor	Dr. B. Kumru		
Co-responsible for assignments	Dr.ir. C.M. de Zeeuw		
Contact Hours / Week x/x/x/x	0/2/0/0		
Education Period	2		
Start Education	2		
Exam Period	2 3		
Course Language	English		
Expected prior knowledge	General knowledge of materials engineering and science (in particular polymers and metals) at the level expected for a BSc graduate in a (mechanical) engineering discipline.		
Course Contents	This course is an introduction to the most common characterization tools to gather crucial structural and property information of materials and components. The focus is on knowing about several possible characterisation methods and their underlying principles of measurement as well as the strategies to extract the most relevant information from the tests for the purpose in mind. The characterisation techniques will be taught by various colleagues in each field. The methods to covered in this course are classified as follows: 1. Imaging of material structures and surfaces (microscopy, confocal, SEM, EDS) 2. Chemical analysis of polymers (FTIR, Raman spectroscopy, contact angle) 3. Structural characterisation 4. Thermomechanical analysis of polymers (DSC, TGA, DMA/Rheology) 5. Mechanical characterization		
Study Goals	Aim of the course is to provide (aerospace, mechanical, maritime) engineering students with adequate skills and knowledge such that later in their technical careers they can handle materials performance issue in which it is important to clarify the material characteristics such as to link actual material performance to its microstructure. Such knowledge and skills are important for determination of material in certain applications and while choosing engineering materials with desired properties. To this aim: - Students will get familiar with the underlying principles of measurement of principal materials characterisation techniques. - Students will be able to analyse and interpret data from each characterisation technique. - Students will be able to select the right characterisation techniques to correlate structure and property relationships over a wide range of engineering materials.		
Education Method	Lectures and hands on practices		
Assessment	Oral Exam and assignment Transition AE4ASM515 -> AE4ASM515-22 1. For students from the 2021 2022 cohort and earlier who have not yet completed AE4ASM515, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE4ASM515. b. Students can also follow the new course and incorporate this in their programme via a change request in MyStudyPlanning. c. If AE4ASM515 is not completed in 2022 2023, students will need to complete AE4ASM515-22 instead.		

AE4ASM517	Aircraft Manufacturing Laboratory	6
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Dr.ir. J.A. Pascoe	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1	
	2	
	3	
	4	
Start Education	1	
	3	
	4	
Exam Period	none	
Course Language	English	
Course Contents	When designing an aircraft (part) during your degree program, you have only had to manufacture simple test articles, but were not involved in manufacturing aircraft parts, let alone a part of an aircraft that will actually fly. In this course you will work with a team of 15-20 students and contribute to build a real (to be certified) aircraft based on a commercially available kit. The aircraft will be built in the Delft Aerospace Structures and Materials Laboratory. It is intended to make its first flight in 2020 or before. The total manufacturing will be done with successive groups of students. In addition to the actual construction work, also the whole documentation and certification process need to be handled in a professional way. Furthermore accompanying work, like the construction of a trailer, PR, etc. will part of the job.	
Study Goals	After completing the course, a student will be able to: - Operate within an aircraft manufacturing setting - Be able to keep a project and certification administration for aircraft production according to industry standards and understand its importance - Have insight in and be part of the safety culture and safety chain surrounding the manufacturing and certification of aircraft - Be able to organize and plan ones own work and that of the entire team and to manage information transfer to next production teams - Demonstrate understanding of an aerospace quality assurance system and has shown to be able to work in an organization exercising such a system - be able to carry out simple metal machining and manufacturing skills such as (pop) riveting to an acceptable standard for aircraft production - be able to understand and overview large sections of the assembly and their sequences in building an aircraft	
Education Method	Based on a commercially available kit-aircraft you will be involved in the entire manufacturing, production and certification process of the aircraft learning about all features of these processes with a combination of practical work, lectures and team work. You are expected to put in 8h per week (on average) over a 20-week period (totaling to 168 hrs (= 6 EC workload)), attend the mandatory lectures and safety and equipment training sessions and follow all relevant safety and production guidelines. The team can plan their own working hours taking into account exam scheduling. This is a self-paced course. The 20 weeks are roughly scheduled as: - Week 1&2: work on production skills, lab safety course, team building, planning and project handover with the previous building team - Week: 3 18: Production and related tasks* - Week 19 - 20: Handover, break in new team and complete all required paperwork that is still pending *related tasks are: - team management - quality assurance - health and safety monitoring - maintenance of workshop and tooling - cleaning of the workshop - documentation - PR - construction of tools and equipment needed for the project - taking relevant additional courses	
Assessment	All students will have a task in the production team as well as at least one additional organizational task. Students will receive a grade based on the following requirements; - Minimum attendance of 95% - Quality of work as judged by the staff - Operating in accordance with aerospace standards - Peer Evaluation on team work - Self Reflection report A fail for the first 2 requirements will always result in a fail for the course. There is no resit opportunity for this course. This is a one-time opportunity only. Once a student has started this course and drops out in due course, there will be no possibility to continue in a later team.	
Enrolment / Application	The grade for this course will be a Pass/Fail grade. This course is open to students from all Master tracks. Admission takes place based on an application letter or video, CV and grade list. An interview and/or skills assessment may be part of the selection process. Only students that have not taken part in the course before and have not taken part in the selection process before are allowed to apply. As the course will require sufficient advance knowledge students must have sufficient study progress to take part in this course. This implies that if you start in: - Week 3.1: you must have completed all your period 1 courses - Week 4.8: you must have completed all your period 1 and 2 courses - Week 1.6: you must have completed all your period 1 and 2 courses Alternatively, having earned a Bachelor degree in aerospace engineering will give access to the course as well. Applications may be send to Mrs. Shanta Visser, S.P.Visser@tudelft.nl by the following deadlines: - Week 3.1: 15 December - Week 4.8: 15 May - Week 1.6: 30 September	

Set-up	The application package should consist of: 1. an application letter in the form of a letter or a video with a maximum duration of 1.5 minutes, 2. a c.v. and 3. a grade list. Incomplete application packages will not be considered.
	Selection of team members will done such that the team composition is optimized. In addition to a strong academic performance we are looking for students with a strong affinity with aviation and with a team players attitude.
	This course can run continuously and students can start 3 times per year with the project kicking off in February 2017.
	The course periods are: - Week 3.1 Week 4.10 - Week 4.8 - week 1.6 (including the summer vacation) - Week 1.6 3.3
	There is an intentional overlap so that continuity and handover is guaranteed

AE4ASM520	Industrial Composite Manufacturing	3
Responsible Instructor	Prof. C.A. Dransfeld	
Responsible Instructor	Dr. D. Stefaniak	
Contact Hours / Week x/x/x/x	0/0/4/0 COURSE NOT OFFERED IN 2024-2025	
	Students that have participated in this course and have not yet passed the course, please contact C.A.Dransfeld@tudelft.nl	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Expected BSc diploma	
Course Contents	The course deepens the understanding on different composite manufacturing technologies for aerostructures used at the industrial scale and addresses on specific challenges evoked especially by automation and large part size. Special pitfalls with respect material selection, mould design, process induced part distortion and vacuum setup are discussed to apply the basic understanding of the manufacturing processes to several specific industrial challenges. Finally, these manufacturing processes are also evaluated with respect their sustainability and are investigated in depth towards their energy, materials and auxiliaries consumption in context of the product life cycle.	
Study Goals	At the end of this course, the student should be able to: - Transfer the basic principles of composite manufacturing processes to real and more complex industrial examples - To gain insight in material selection and its consequences in an industrial context - To recognize manufacturing processes and the linked pitfalls based on the resulting part or the utilized mold - Understand the challenges in sizing composite manufacturing systems for large size components - Understand the importance and dependencies of performance assessment indicators in composite manufacturing like robustness, energy consumption, occupation time, non-operating costs and flexibility.	
Education Method	Interactive lectures based on industrial case studies. Some exercises are accompanied by Python Scripts used in Jupyter Notebooks to deepen the understanding and facilitate interactivity. The required software is provided as a browser-based version and previous knowledge in programming is not compulsory. It is intended to visit the Center for Lightweight Production Technology at the German Aerospace Center (DLR) and Airbus site in Stade, Germany.	
Computer Use	yes	
Course Relations	The course is given by a guest lecturer from DLR (German Aerospace Center)	
Literature and Study Materials	will be defined at the beginning of the course	
Prerequisites	AE4ASM001, AE4ASM004, AE4ASM110	
Assessment	Oral exam (50% of the final grade) and mandatory assignment (50% of the final grade).	
Enrolment / Application		
Remarks	Students that need to retake AE4ASM520 can contact the responsible lecturer for a resit.	
Set-up	2 weeks with 2day-blocks and two weeks with one 1day-blocks (e.g. 3.1, 3.3, 3.6, 3.8); each day will consist of four lecture hours	

AE4ASM521	Additive Manufacturing	3
Responsible Instructor	Dr. K. Masania	
Instructor	Dr.ir. J.A. Pascoe	
Instructor	Prof. C.A. Dransfeld	
Instructor	Dr. J. Wu	
Instructor	V. Yaghoubi Nasrabadi	
Instructor	P.G. Marchese	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc in any disciplines of mechanical/civil/aerospace engineering, materials science, chemical engineering, chemistry, bionanoscience, applied sciences and industrial design engineering.	
Course Contents	To achieve ambitious sustainability goals that will become necessary in the future, we need to utilise lightweight materials with reduced environmental footprint more efficiently. Their structuring to prescribes load paths and manufacturing with minimal material waste will become key. This course focuses on studying the basics physics of additive manufacturing processes. The course covers the areas of biologically-inspired materials and processing, topology optimisation for design and explores emerging topics in additive manufacturing. During the course, students will gain an overview of existing additive manufacturing technologies of metals, ceramics, polymers, living and composite materials. Working in small groups, students will study self-assigned research topics and apply the acquired knowledge for a design assignment of a structural component. The understanding of the manufacturing process and the design knowledge, applying numerical optimisation methods, will lead to the creation of an own design which will be manufactured. All parts will be tested and evaluated under different criteria and critically assessed with their future perspective in mind.	
Study Goals	The course consists on the one part of lectures to allow the student to understand the fundamentals of materials processing as well as the design and optimisation methods relating to additive manufacturing. Selected guest lecture(s) will highlight how this knowledge is applied in specific fields or applications. In parallel to the lectures, a design assignment for a lightweight structure will help the students to explore processing knowledge and freedom associated to the design and optimisation methods to create an own structural design which will be tested in a final design competition.	
Education Method	Lectures (initially 4h per week, decreasing in second half) and project based learning with design assignment in small groups (initially 2h per week, increasing in second half).	
Computer Use	Own laptop, access to parametric design and optimisation software will be provided.	
Assessment	Based on assignment and group project with oral exam if necessary	

AE4ASM522	Applied Aircraft Aeroelasticity	3
Responsible Instructor	Dr. J. Sodja	
Responsible Instructor	X. Wang	
Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	AE4ASM506 (mandatory)	
Course Contents	This course is a follow-up course of Aeroelasticity AE4ASM506. Taking and passing the Aeroelasticity course AE4ASM506 is mandatory to pass this course. This course builds on the basics and regular aeroelastic phenomena that were introduced in AE4ASM506. The extension of the knowledge is twofold: 1. Introduction of nonlinear aeroelasticity 2. Introduction to control of flexible aircraft 3. Introduction to potential flow aerodynamics and its coupling to structural FE models to form closely coupled aeroelastic models The course aims at placing the aeroelastic knowledge into a flexible aircraft context, and it will provide the student with new insights into the aircraft design process and aircraft design choices. Finally, understanding the contents of this course will prepare the student for the design of future aircraft configurations beyond the currently established aircraft concepts.	
Study Goals	At the end of the course, the student should be able to: 1. Know advanced nonlinear aeroelastic phenomena due to structural and aerodynamic nonlinearities 2. How to approach aeroelastic control of flexible aircraft 3. Know and apply the basics of potential flow theory as it is the prevalent aerodynamic theory used in aeroelastic modeling 4. Understand the difference between closely and loosely coupled aeroelastic systems. 5. Know how to couple panel method aerodynamics with FEM to form closely coupled aeroelastic system	
Education Method	The course will be taught in a classroom and hands-on sessions, circumstances permitting.	
Books	Recommended: 1. Introduction to nonlinear aeroelasticity. Dimitriadis, Grigorios. John Wiley & Sons, 2017. 2. Introduction to structural dynamics and aeroelasticity. Hodges, Dewey H., and G. Alvin Pierce. Cambridge University Press, 2011. 3. Aeroelasticity. Bisplinghoff, Raymond L., Holt Ashley, and Robert L. Halfman.&nbsp;Courier Corporation, 2013. 4. Etkin B. Dynamics of Atmospheric flight. Courier Corporation; 2012.&nbsp; 5. Introduction to aircraft aeroelasticity and loads. Wright, Jan Robert, and Jonathan Edward Cooper. John Wiley & Sons, 2008.	
Reader	Course notes	
Assessment	Take home assignments with oral close-out session.	

AE4ASM524	Spacecraft Structures Development	3
Responsible Instructor	P.P. Sundaramoorthy	
Instructor	Dr.ir. J.M.J.F. van Campen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is strongly recommended that students planning to take this course will have followed multiple of the following courses: - AE4ASM106 Stability & Analysis of Structures I - AE4ASM505 Non-Linear Modeling (using F.E.M.) - AE4ASM511 Stability & Analysis of Structures II - AE4S12 Space Systems Engineering - AE4S20 Satellite Thermal Control	
Course Contents	Spacecraft and launcher structures are subjected to extreme environments, making their design and development and the verification thereof (DVV) very complex. In this course the design of the main spacecraft structure, the structural behaviour of non-structural spacecraft components and their interaction will be illustrated and addressed through the design of the structure of a spaceborne instrument. Thermal elastics and dynamics of the instrument structure will be considered. In small groups, participating students will have to design their own instrument structure, addressing both thermo-elastics and vibrations. Verification and validation of the designs will be addressed as well. The course will consist of lectures, working sessions, design review sessions and a final symposium. The course structure will be explained in detail in the first weekly module. The time and date of the design review(s) and final symposium will also be communicated in the first weekly module. Students are requested to reserve all scheduled timeslots in their agendas. Each module will consist of a combination of learning activities, e.g. reading suggested papers, watching videos and small assignments. Note that, the course will require group work. Groups will be formed in the fourth week of the course. Only students who pass the individual test are allowed to participate in a group. Students who are not part of a group cannot participate in the remainder of the course. For the purpose of the course, each group will be considered to be a company that tries to win the bid of a large space organization to engineer a space instrument. At the end of the course, a mini-symposium will be organized in which each group presents their design for the space instrument structure.	
Study Goals	The overall educational goal of the course is to provide students with an advanced understanding of the design, development and verification of spacecraft. Successful students will be able to: - Interpret and analyze a requirement specification for a spacecraft payload or instrument - Assess what the design drivers for the structural subsystem of a spacecraft payload or instrument are. - Describe how vibration and thermo-elastic considerations affect the design choices and interact with each other. - Develop a conceptual design and perform initial sizing by analysis of the structural (and to some extent, thermal) subsystem of a spacecraft payload or instrument. - Describe in detail how the design of a structural subsystem of a spacecraft payload or instrument can be verified and validated.	
Education Method	The course will be offered as a flipped classroom, group-based, course. The aim is to offer the course as a fully on-campus course, however the lecturers reserve the right to switch to an online course format if necessary. The course consists of a series of seven weekly modules set up by faculty staff and industry experts. During each module it will be possible to discuss with the lecturers and fellow students.	
Assessment	The assessment consists of two parts: group assessment and individual assessment, where the group assessment has a weight of 2/3 and the individual assessment a weight of 1/3. The group assessment consists of two parts: 1. Design specification sheet of group design (template will be provided) 2. Design presentation at final symposium The design specification sheet will address the following points: - Thermo-elastics design - Vibration design - Design integration - Systems Engineering - Verification & Validation The individual assessment is based on two parts: 1. Individual test on course content 2. Written feedback on design specification sheet of another group in the course Students with a failing grade for the individual test, i.e. a grade below 5.8, will have a resit opportunity of the individual test in week 4 of the course. This resit will be in the form of a short oral exam. This resit is only available for students who did not pass the individual test. On passing the oral resit exam, students will receive a 5.8 as grade for the individual test. Students who fail the resit opportunity will not be able to participate in the group work, and therefore cannot participate in the remainder of the course. It is mandatory that students are present at the design review(s) and the final symposium presentations of their own group and those of other groups. The individual grade of students who fail to be present will be withheld. These students will not receive a final grade for the course as a result. Students of whom the individual grade which has been withheld, may take a resit. The resit will be in the form of a short oral exam. It will be possible to apply for a resit up to 15 working days after the publication of the grades in Osiris, but no later. A resit exam will then be scheduled by the principle lecturer of the course in consultation with the student in question. Grading criteria will be defined in the first lecture.	

AE4ASM525	Materials for Space	3
Responsible Instructor	Dr. Y. Tang	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic materials science and engineering concepts as presented in Callister, Introduction to materials science" or books of similar nature. Be sure to check the entry level quiz before making up your mind about this course.	
Course Contents	In this course a number of space missions (for example Vikings, Opportunity, Perseverance, ISS, Hubble telescope, James Webb telescope, Venus orbiters, Launchers) will be presented as a carrier for material selection analysis. A holistic view of "materials science - manufacturing - structural analysis - system engineering for missions" will be presented. While students come mostly from different backgrounds, the goal of this class is to help them identify the perspective they use in the holistic view and to learn about the materials science perspective. Fundamentals of materials classes and their typical properties will be introduced. Current material choice will be analyzed and material designing principles will be explained. The concept of reverse material engineering for metals, polymers and inorganic materials will be demonstrated in a series of lectures. In sessions of "ESA forum + student poster seminar", students will be given opportunities to interact with ESA experts and present their own findings from analyzing given missions, by translating desired properties into material structures and microstructures. In class discussion is an important part of learning. The structure of the lectures will be tailored to maximize the student involvement. The course contains the following topics: week 1: Holistic view and materials science perspective; Typical materials - metals, ceramics and polymers week 2: Typical materials - metals, ceramics and polymers week 3: Material degradation in space - vacuum; thermal cycling; week 4: Material degradation in space - AO; charging week 5: Material degradation in space - space radiation week 6: Material degradation in space - meteoroids and orbital debris; week 7: "ESA forum + student poster seminar" Materials testing for space applications; Extreme missions; Materials challenge in future space age (ESA guest speakers); Student poster seminar (Jan 6, 2025) week 8: Study period (Q and A) week 9: exam week (Jan 20. 2025)	
Study Goals	By the end of the course, you should be able to: °LO1: Identify space environmental conditions and common materials used for space. °LO2: Explain material degradation mechanisms in materials under space conditions such as radiation, vacuum, thermal fluctuation etc. °LO3: Evaluate or analyse material selection in space-related situations through reverse materials engineering	
Education Method	The course consists of 12 lectures of 2h per week for the first 6 weeks. The last week is a whole-morning session of "ESA guest speaker forum + student poster presentation".	
Literature and Study Materials	A compilation of book chapters, lecture notes and review papers.	
Assessment	Poster presentation (40%) + final exam (60%). The students are to work in groups and do research on prescribed question/topics and deliver a final poster presentation (15% individual score, 25% group score) in week 7 to show their discoveries throughout the course. The final grade will be the weighted average of poster presentation and final exam. Example questions/topics will be made available on Brightspace.	
Enrolment / Application	For students not from "structures for space" profile, there will be a cap of 20 students maximum. Please enroll in brightspace (enrollment period Oct 28 to Nov 10, 2024) as soon as you decide to take the course. Once the maximal number is reached, enrollment will be closed.	

AE4ASM526	Spacecraft Thermal Design	3
Responsible Instructor	I. Uriol Balbin	
Instructor	C. Falsetti	
Instructor	N. van der Pas	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of orbital mechanics Understanding of general design spacecraft systems	
Course Contents	General knowledge of space thermal control hardware, system design and analysis processes are presented. Using the principles of heat transfer, and more specific of radiation and conduction, the methods and tools for thermal design of a spacecraft are presented. Numerical analysis will also be introduced via the software ESATAN-TMS.	
Study Goals	The student is able to generate a thermal design of a satellite using heat transfer analysis and thermal engineering techniques. In more detail the student will be able to: - Understand the spacecraft thermal environment - Define the thermal subsystem and compare the different hardware options. - Understand the design cycle of the thermal subsystem. - Apply the concepts of heat transfer (radiation, conduction), heat balance and heat flows - Generate and use analytical thermal analysis models - Generate and use numerical thermal analysis models	
Education Method	Lectures + homework Students are given access to a discussion board provided with the course in Brightspace. The discussion board will be used as the main communication tool to ask questions on the material taught or in preparation of an assignment. Answers to the questions may be given by fellow students and/or the lecturer(s).	
Computer Use	Software licenses are available during the lecture period Software can be used in the computer room or in the personal computer (Windows/Linux)	
Literature and Study Materials	Lecture notes available on Brightspace.	
Assessment	The final grade of this course is based on 2 homework assignments (60%) and one final exam (40%). A minimum grade of 5 is required in each part.	
Enrolment / Application	This class has a limit of 80 students, due to the limit on the software licenses. Please enroll in Brightspace before 4th of March 2025 if you intend to take this class.	

AE4S10	Microsat Engineering	4
Responsible Instructor	Dr. J. Guo	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Lectures from experts on various aspects of small satellites and group/individual projects for practicing.	
Course Contents	The course covers the engineering of small satellites. The concept and definition of micro-, nano-, pico-, and femto-satellites is introduced along with the past, present, and future of small satellites in general. The course comprises seven to eight lectures, partly from professionals in the space industry. The lectures are intended to familiarise the students with current industry trends and state-of the art research in small satellite engineering. The lectures vary each year and potential lecture topics include communications, AOCS, onboard computer, distributed space systems, miniaturised payloads, small satellite missions and applications, etc.	
Study Goals	The course will enable students to design and engineer small satellite missions. The high-level learning objectives of the course are: The students shall be able to - explain key aspects of small satellite missions - explain and describe elements in the architecture of a spacecraft mission - discuss in detail at least one element in the architecture of a spacecraft mission - present and defend their design of an element in the architecture of a spacecraft mission - show familiarity with current trends in space industry and small satellite research	
Education Method	Lectures (Provided partly by professionals from the space industry). Group work (Students work on assignment in groups; Coaching will be provided to guide the students). Individual work (Students work on small individual assignments).	
Literature and Study Materials	Optional: Wertz J.R., Larson W.J. (eds.); Space Mission Analysis and Design; 3rd edition Kluwer Academic Publishers (2003). Mandatory: Lecture Notes AE4-S10 MicroSatellite Engineering.	
Prerequisites	General mathematical, physical and engineering knowledge from completed BSc study. In particular, AE1101 + AE1102 + AE3201, or similar	
Assessment	There will be two assessment components A Group Assignment and an individual assignment. Group assignment The final demonstration or report will be graded on the following aspects: - Planning - Presentation, argumentation and defence of the design - Functionality and Usability of the deliverable - Context of Small satellite (How well has the student team incorporated current small satellite research and industry trends/needs in the design) - Independence - Creativity Individual assignment This assignment will be graded on the following aspects: - Understanding and summary - Critical review of articles in a specified technical field of small satellites Please note: there is no written exam and, therefore, also no re-sit exam for this course.	
Permitted Materials during Tests	N/A	
Enrolment / Application	Students have to enroll via the BrightSpace site of the course. Due to the limited capacity, this course is open to Space Engineering profile students only.	
Set-up	The course comprises lectures, guest lectures, team sessions, group work and individual work.	

AE4S52-21	Collaborative Space System Design Project	5
Responsible Instructor	Ir. B.T.C. Zandbergen	
Instructor	Dr. S.M. Cazaux	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The Collaborative Space (System) Design Project (CSDP) is a TU Delft/Faculty of Aerospace Engineering (AE) Master course in engineering design. The course focuses on the conceptualization and preliminary design phase of a space mission, spacecraft, or space instrument and starts on a design challenge co-created by the CSDP organization team and a company, research group or other entity. All projects on offer are multi-disciplinary projects, i.e. projects that require different types of design work and knowledge to be combined to provide a design solution. All projects are carried out by a team of students that have to organize and manage themselves. Each project is to be organized in 3 phases similar to NASA's System Design Process as defined in a.o. "NASA's System Engineering Handbook". Some adaptations have been made though to make it fit in the limited course time. The phases are: 1) Exploration phase; In this phase the team is to explore the problem, the needs, the competition and past missions and to develop a proposal for the next phase of the project. This includes the identification of a range of high-potential concepts for study in phase 2, the work distribution, etc. 2) Concepts design studies phase; In phase 2, the high-potential concepts defined in phase 1 are analyzed in detail for feasibility, traded and a best concept is selected. Additionally the plan for the next phase is to be generated. 3) Detailed design phase. In this final phase of the project the single concept selected in phase 2 is worked out in detail and a plan is developed for the further development of the design. Each project phase ends with a review (feedback moment) wherein other teams and expert staff reflect on the outcomes generated and the engineering design methods used. During the quarter, workshops/instructions will be held to provide knowledge and training on selected management and engineering design topics, including agile management, and the use of integrated design modelling, i.e. the integration of all the geometry, configuration, analysis, and requirements verification into a generative, parametric, unified computational model where data is shared seamlessly between the different disciplines. The course does not focus on teaching the required disciplinary knowledge and experience, but rather focuses on decision making, the collaborative integration of the knowledge and experience available in the team, and the iterative design method using different levels of model fidelity to create a feasible design solution in answer to the problem identified by the "customer". As projects vary from year to year and may encompass knowledge that is not available in the team, this may require that participants actively acquire the knowledge required.	
Study Goals	The course aims to develop student skills in multi-disciplinary team projects from a challenge-driven perspective. In more detail, students will advance their ability in ...: - ... disciplinary design including modelling, simulation, visualization, quantitative analysis of alternatives, design tool verification, calibration and validation, and design refinement. - ... the process of engineering design (ABET definition), including the steps in design, the (iterative) nature of the process, development of a Straw Man design, and development of process models. - ... multidisciplinary design, thereby taking into account differences between the different disciplines involved in terms of a.o. differences in fidelity level of disciplinary models, and dissimilar assumptions; - ... systems engineering, including the design phasing, work breakdown and work distribution, modeling and interfaces, and the role of specialty engineering (e.g. cost-, RAMS-, and mass modelling and configuration design). - ... concurrent engineering, as opposed to the more classical sequential engineering (the waterfall method). - ... project management and teamwork (assigning roles and responsibilities, setting goals and objectives, coordination and management of team process, decision making, handling conflicts, creativity, empowerment and motivation, communication, and reflection on own work and work of others).	
Education Method	The course consists of: - Group project (4 EC or 112 working hours) - Workshops/instructions (1 EC or 28 working hours) For the group work, students work in teams of 6-12 students. Prior to the start of the course, students will be informed on the available projects as to allow them to state their preference(s), and project teams are formed. During the project, each team is guided by a design coach, who mainly focuses on the problem solving/engineering design process. Additionally, each team is free to organize experts (from industry, academia or other) for the different disciplines involved that can advise the team during the design. During the quarter workshops/instructions are given to train teams' project management and design integration skills.	
Assessment	The assessment of each team consists of an assessment by the workshop instructors and project coaches (80% of the final grade), team of experts not involved in instruction and workshops (10%) and by the other student teams (10%). The assessment of the workshop instructors and project coaches is per design phase: - Exploration phase (10%) - Concepts design studies phase (20%) - Final design phase (70%) The assessment of the instructors and coaches is based on a design report, a (design) management log (record keeping) and one or more design presentations and reviews. Elements that will be assessed relate to design competencies (problem formulation, requirements generation, design option identification, design analysis, design trades and design evaluation), workplace competencies (e.g. teamwork, creative thinking, decision making, planning and organizing), academic and personal effectiveness competencies (e.g. communications, modelling/experimenting and critical and analytical thinking). Assessment by the team of experts and by peer teams will be based on the performance of the teams at a project presentation session at the end of quarter 3.	
Enrolment / Application	Only students enrolled in the Space track can participate. There is no free enrollment for this course. Still at the start of Q1, the students can preliminary enroll for the course as to learn more about the CSDP, and past projects conducted (no solutions on offer). In Q1, an information session will be held informing interested students of the track Space on the projects available and on how	

they can apply for a project. Students can then apply and only if they are accepted, their enrollment will be made definite. All non-accepted students will be dis-enrolled.

For the year 24/25, the maximum number of participants for this course has been set to 20. Priority will be given to students enrolled in the Space Engineering profile of the Space track. If places remain, also students from the other profiles in the Track Space can participate.

AE4T40	Airborne Wind Energy	3
Responsible Instructor	Dr.ing. R. Schmehl	
Instructor	Dr. D. Ragni	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	BSc. completed	
Parts	Lectures include latest material on the developments in the field of airborne wind energy, with a thematical focus on kite power generation.	
Course Contents	Lecture 1: Introduction to airborne wind energy Lecture 2: History of kite applications Lecture 3: Implemented AWE concepts Lecture 4: Physics of tethered flight Lecture 5: Theory of tethered flight in three dimensions Lecture 6: Flight navigation and maneuvering Lecture 7: Operation of AWE systems Lecture 8: Practical use of AWE systems Lecture 9: AWE in the future energy system Lecture 10: Design and flight dynamics of kites Lecture 11: Aerodynamics of kites Lecture 12: Aeroelasticity of kites Practical: AWE workshop with Jupyter Notebook	
Study Goals	Give an overview of developments in airborne wind energy, developng mathematical models for kites and designing kite power systems for industrial scale generation of energy.	
Education Method	Lecture & workshop	
Prerequisites	Any MSc student can take this course, given a good understanding of vector analysis and basic trigonometry.	
Assessment	Written exam	
Set-up	12 lectures of two lecture hours each in the first semester followed by a written examination.	

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr.ir. M.B. Zaayer	
Instructor	Prof.dr. D.A. von Terzi	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: - describe the components and configurations of wind turbines - describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine - describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design - explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. - identify the drivers of dynamic behaviour and calculate their typical values - describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure - describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Remarks	This course is an elective course for students from various faculties (AE, AS, ME, EEMCS, CEG, TPM). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W21-14	Wind Turbine Aeroelasticity	2
Responsible Instructor	W. Yu	
Instructor	Dr.ir. C.J. Simao Ferreira	
Gast Instructor	Prof.dr.ing. R. De Breuker	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Preferred: AE4ASM506 (Aeroelasticity). It's preferred that the student has completed or is attending this or an equivalent course.	
Course Contents	The course will cover the main structural and aerodynamic phenomena of wind turbine aeroelasticity. The students will program models to evaluate these phenomena and deepen their understanding. We will explore the following questions: 1. What are the key characterized parameters for an aeroelastic system? 2. How to construct a structural model of a wind turbine blade? 3. How to define if resonance occurs to a system and understand its relevance to wind turbines and how to mitigate it? 4. How to define if instability occurs to a system and its relevance to wind turbines and understand the role of aerodynamic damping? 5. Dynamic inflow and dynamic stall. 6. How to construct an aeroelastic model by coupling the structural and aerodynamic effects?	
Study Goals	General course objectives: The course is intended for students that already have some background in wind turbine design. The course will provide the students with an introduction to general and wind turbine specific aeroelastic phenomena and how to couple various structural and aerodynamic models in an aeroelastic setup. The course participants will be able to calculate the aeroelastic response of a wind turbine construction on time varying loads originating from atmospheric turbulence, wind gusts, wind shear, yaw, tower shadow as well as gravity and inertial loads. Learning objectives: A student who has met the objectives of the course will be able to: 1. understand the dynamics and loads of a wind turbine due to the strong coupling of a) external loads(aerodynamics) b) internal structural loads c) inertia loads; 2. solve the dynamic structural response based on the equations of motions; 3. Calculate the unsteady aerodynamic loads of a wind turbine under dynamic inflow and dynamic stall; 4. apply the knowledge to program their own aeroelastic models.	
Education Method	The lectures will be taught in a lecture room.	
Assessment	Take-home assignment.	

AE4W30	Wind Resource and Wind Farm Yield	4
Responsible Instructor	Prof.dr. S.J. Watson	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	This course will cover how the output of wind farm arrays is predicted based on modelling of the wind resource and combined wake interactions. The first part of the course will cover the different wind resource models which are used ranging from simple mass consistent models, to linearised solvers, computational fluid dynamics models and mesoscale models. The use of models in both onshore and offshore environments will be covered including the importance of atmospheric stability. The second part of the course will consider the application of models to assess the yield from wind farms considering wakes effects. Wake models will be reviewed including simple semi-empirical models to kinematic models and the modelling of wakes in CFD models. Once again, the importance of atmospheric stratification will be considered and the potential impact on wind farms of certain mesoscale phenomena, e.g. gravity waves offshore, will be assessed and quantified. Large wind farm phenomena will be studied including the so-called deep array effect and wind farm blockage.	
Study Goals	Course Aim: to give students a firm foundation in how to predict wind farm yield from prevailing wind conditions. Course Objectives: 1) To understand what tools are available to predict wind resource over a wide area 2) To apply to such models to potential wind farm sites onshore and offshore 3) To quantify the importance of terrain effects and atmospheric stability on the wind resource 4) To understand what tools are available to predict wake effects in a large wind farm 5) To apply wake models in a realistic wind farm situation 6) To appreciate the importance to wind farm yield of phenomena such as terrain complexity (onshore), proximity to the coast (offshore), atmospheric stability, mesoscale effects (e.g. gravity waves), wind farm layout and wind farm scale	
Education Method	Students will learn through a combination of guided reading, group work, online resources (e.g. lectures, videos, quizzes), face-to-face lectures, coursework and exams.	
Assessment	The students will undertake two course assignments (resource assessment and wind farm yield prediction) worth 30% each, spread out during the course and an oral exam worth 40% at the end of the course. Assignment 1: application of a model to assess and optimise wind resource and energy yield for an onshore site with a report to be assessed. Assignment 2: application of a model to assess and optimise energy yield for an offshore site incorporating different wake models with a report to be assessed. Exam: oral test to assess understanding of the principles of wind resource assessment and wind farm yield prediction.	

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Prof.dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Instructor	Dr. F. Taruffi	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

AE5055	Professional Training	6
Responsible Instructor	W. Yu	
Contact Hours / Week x/x/x/x	N.A.	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The aim of the professional training is to expose students to a real work environment for a period of 4 weeks (consecutive) on a full-time basis.	
Study Goals	- To increase your understanding of employment options after graduation - To get a better perception of your tasks and responsibilities within a professional environment - To create an understanding of the context in which wind energy is practiced by industry, institutes and organizations.	
Education Method	This is an elective. You have to arrange the internship yourself. Before you start you should ask me for approval about the specific topic (specify it in a 1 pager describing what you are going to do in your internship and how that is going to help you meet the learning objectives of the course). Deadline for topic approval and internship arrangements: 15 April (afterwards you are not allowed to do an internship anymore). Here you can find contact persons: https://www.tudelft.nl/lr/organisatie/afdelingen/flow-physics-and-technology/wind-energy/education/master-projects/msc-possibilities-industry You can of course also contact one of the EWEM partners. Also keep in mind that DTU nowadays does not provide much room for delay in your MSc: you have to start your thesis project in November and finish it in August. This implies that you have to do your internship during the summer.	
Assessment	Once students have finished their internship, they must submit a report to the responsible instructor of this course and your internship supervisor at the company no later than 4 weeks after completing the internship. The report has to comply with the rules of Technical Writing and thus should also be structured as such. The internship report is assessed on two criteria: 1. Quality of the work that student has done as assignment/project during the internship and the evaluation from their supervisor (Final Evaluation Form); 2. Personal reflection on student's personal professional performance in the internship and the attainment of the study goals of the course as outlined above. The internship report is assessed and can be rejected if the student does not meet the requirements outlined above, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for revising the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out. Grades will be registered in Osiris using the date that the student has handed in the version of the internship report that is assessed. The result of the report (pass/fail) will be made public no later than at the last day of the period following the period in which the report was handed in.	

AESM2001	Physical Principles of Earth System Observation	5.5
Responsible Instructor	Prof.dr.ir. F. Lopez Dekker	
Module Manager	Prof.dr.ir. F. Lopez Dekker	
Instructor	Dr. P.G. Ditmar	
Instructor	Prof.dr.ir. S.C. Steele-Dunne	
Instructor	Prof.dr.ir. F. Lopez Dekker	
Instructor	K. Löer	
Contact Hours / Week x/x/x/x	4 hours lectures or practicals	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Measuring is essential to characterize and explain processes in the Earth system, and a first step to assess, model and predict natural processes and human activities in and their impact on the Earth system. Electromagnetic, seismic and gravity potential-field observations inform us about a wide range of phenomena in the ocean, atmosphere, land surface, cryosphere and sub-surface. The measurements can be acquired from spaceborne, airborne, surface and sub-surface-based sensors. This programme core module aims to enable students i) to explain and apply the physical principles underlying the measurements, and ii) to assess what type of measurement could be used best to determine certain geophysical variables. For example, students will learn how electromagnetic theory allows to use the intensity of radar echoes to yield information about rain rate, soil moisture, ocean roughness, or the layering of the subsurface. Similarly, they will learn how potential field theory can be applied to quantify mass changes of, e.g., the ice sheets. Students will be able to weigh the advantages and disadvantages of, for example, microwave versus visible and near-infrared observations for monitoring the Earth surface, and seismic and electromagnetic imaging in mapping the subsurface. Review basics of electromagnetic (EM) waves (propagation, polarization, spectra, Doppler, etc.) and propagation effects (attenuation, refraction, dispersion, polarimetric effects) Optical-infrared scattering and propagation (different reflection types, albedo, effect of wavelength, BRDF, Radiative transfer modelling of optical EM waves). Physical principles and limitations of VNIR/SWIR sensing systems (photographic systems, LiDAR, spectral systems including TIR) in terms of spatial and spectral resolution, sensitivity, atmospheric disturbances and corrections. Microwave scattering and propagation (Rayleigh, Mie scattering, rough surface, volume scattering, clear air, radiative transfer modelling of microwave scattering in the atmosphere and at the land surface). Radiometry (Radiometric Quantities and Units, Blackbody Radiation, emissivity at different frequencies). Radiative transfer modelling in passive microwave remote sensing of the Earths atmosphere and surface. Physical principles and limitations of microwave sensing systems (radiometer, non-imaging radar, imaging radar, ground-penetrating radar) in terms of spatial and spectral resolution, sensitivity, atmospheric propagation effects and corrections, radar range equation Seismic (including acoustic) waves (Newtons second law, stress tensor, equation of motion, elasto-dynamic wave equation, Plane P- and S-waves, spatial and temporal resolution) Potential fields (gravity, magnetic and electric fields) and their relationship to physical properties of a medium (Gauss law and its applications, Newtonian/Magnetic Potential, Laplace equation and harmonic functions, representation of potential fields, spherical harmonics) Choose an approach for the observation and measurement of a geophysical variable.	
Study Goals	After completing this module, students will be able to: Characterize electromagnetic waves and describe mathematically the effects associated to their propagation through the different components of the Earth system (atmosphere, surface and & subsurface). [ILO A] Describe the interaction of optical-infrared waves with the Earths atmosphere and surface and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of optical/infrared remote sensing in determining geophysical variables. [ILO A, D] Describe microwave scattering in the Earths atmosphere and at the Earths surface, and analyse the fundamental sensitivity to geophysical variables of interest. [ILO A] Describe the emission of thermal waves and microwaves from the Earths atmosphere and surface, and apply radiative transfer models to simulate their propagation and interactions. [ILO A] Assess the potential value and limitations of microwave remote sensing in determining geophysical variables. [ILO A, D] Explain mathematically how the elastodynamic wave equation is derived, characterize elastic waves, and explain their sensitivity to contrast parameters. [ILO A] Establish the linearized relationship between an observable and a given representation of the potential field, and synthesize the potential from functionals given on the surface of a sphere. [ILO A] Critically discuss the potential value and limitations of different types of measurements (I.e., different bands of the EM spectrum and EMR/seismic/potential field) and propose and defend a type of measurement to observe certain geophysical variables. [ILO F, G]	
Education Method	Lectures and group practicals/exercises. Feedback via quizzes (formative) and intermediate assignment results (formative).	
Assessment	Module is summatively assessed with a project, using 1 rubric (similar as thesis project) at the end of the period based on: Assignment report (may comprise different parts, i.e., based on a several assignments, report includes reflection on LOs) Presentation Oral discussion (may include questions on LOs not directly covered in the assignment work).	

Expected prior Knowledge	A successful completion of the module requires a minimum grade of 5,8 per Exam/Assessment. Some background in electromagnetic theory and fundamentals of wave propagation. Solving linear partial and ordinary differential equations.
Academic Skills	Earth Observation, data interpretation.
Literature & Study Materials	Provided course material, published online lectures and material, book/e-books
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N.A.
Collegerama	No

ET3604LR	Electronic Circuits	3
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	Major topics: Electronics in information processing (sensor readout) Design principles and trajectory Signal processing functions Electronic design analysis Negative feedback amplifiers Operational amplifiers Components and networks Semiconductor diodes Power supplies and voltage regulators Semiconductor transistors and IC technology (basic) Oscillators DC motor basics	
Course Contents	This course is intended to provide gain insight, in a structured way, in the function of electronics as information-processing technology. The way of modelling will be discussed in relation to the physical characteristics of electronic components, circuits and systems. The course will provide the students with the skills to interpret circuit diagrams and measurement results and formulate specifications for electronic circuits that enable expert electronics designers to take proper design decisions when requested to design a high-performance application specific circuit. The students will learn the terminology that enables them to understand the explanation for these decisions by the experts (and potentially critically review them with respect to system level decisions). During the course many lecture demonstrations will be given, where the students both can observe the practical behavior of the test object and see what type of measurement equipment is used and how it is operated.	
Study Goals	After this course, the student -Is able to interpret electronic circuit diagrams and draw them according to standards -Is able to recognize and select electronic components -Is able to interpret and create Bode plots that describe the frequency behavior of an electronic circuit -Is able to analyze and design basic frequency filters -Is able to explain the basic behavior of semiconductor devices, like bipolar transistors, field effect transistors, diodes and special devices like thyristors. -Is able to determine the proper readout conditions for a sensor -Is able to determine the proper drive conditions for an actuator -Is able to interpret basic diode circuits like rectifiers -Is able to explain the basics of negative feedback systems and able to calculate and interpret the essential parameters -Is able to design a negative feedback amplifier using an Operational Amplifier (OpAmp) as active circuit that optimally meets the requirements -Is able to explain the basics of power circuits -Is able to name the different oscillator classes and explain the working principle the oscillators in each class -Is able to determine the class of a given oscillator -Is able to name the specific good and bad properties of each oscillator class -Is able to communicate with experts in electronic design and translate system specifications to specifications for the electronics in such a way that the expert can create an optimal design	
Education Method	Lecture. There is a mandatory practical with its own course code: ET3604LRP	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276 (older versions can very well be used for this course)	
Assessment	The exam is multiple-choice. During the lecture period, there will be one bonus test on oscillators. A pass grade gives a bonus point for the exam. The bonus point remains valid for the re-sit exam. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	Dr.ir. H. Vahedi	
Responsible Instructor	Dr. G.R. Chandra Mouli	
Instructor	W. Shi	
Responsible for assignments	H. Dialani	
Responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2014 (7th edition) ISBN 9781259254666	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'). The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Instruction if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Process & Energy	

ME45030	Turbulence	5
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Prof.dr.ir. J. Westerweel	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Advanced Fluid Dynamics (or equivalent)	
Course Contents	The course provides an introduction to the physical aspects of turbulent flows.	
Study Goals	(1) basic equations that describe turbulence; (2) transition to turbulence; (3) turbulence closure models; (4) energy cascade; (5) turbulent dispersion; (6) selected topics in turbulence	
Education Method	lectures and instructions	
Literature and Study Materials	Turbulence (2016) by Nieuwstadt, Boersma & Westerweel	
Assessment	Written exam	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Tags	Fluid Mechanics Mechanics Numeric Methods Physics	
Department	ME Department Process & Energy	

ME46060	Engineering Optimization: Concepts and Applications	3
Responsible Instructor	Prof.dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Instructor	Dr. O. Nejadseyfi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be 5.8 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest decimal. The maximum bonus score is 0.8. The bonus only applies to reports with grade ≥ 5.5 and submitted before the first report deadline.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	ME Department Precision & Microsystems Engineering	
Contact	Prof.dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

MS43310	Materials at High Temperature	4
Responsible Instructor	Dr. V.A. Popovich	
Instructor	P. Taheri	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students should already have knowledge on: a) Materials Science b) Mechanical Behaviour c) Basics of Fracture Mechanics d) Fatigue MS43005 - Structure and Properties of Materials covers aspect a). MS43100 - Science of Failure covers aspects b), c) and d).	
Summary	This course relates to materials exposed to high temperatures. It can be divided into two parts: Part A - Mechanical behaviour Part B - High temperature materials, processing and oxidation In the treatment hereof a number of aspects will be addressed: High temperature applications: - combustion engines and gas turbines - boiler materials (heater and steam pipes) - waste and bio mass incinerators - solid oxygen fuel cells Materials for high temperature applications: - Fe and Ni based alloys (Laves phases) - superalloys - intermetallics (TiAl, NiAl etc) - ceramics - refractory materials and coatings - properties-temperature relations Design & processing: - coating deposition - sintering - alloying	
Course Contents	A total of 14 two hour lectures will be taught (number of two hour lectures between brackets): Part A: High Temperature Mechanical Behaviour Creep in crystalline solids (metals and ceramics) (3) - creep testing - creep mechanisms - deformation mechanism maps - extrapolation procedures - creep resistant materials - creep fracture High temperature fatigue (metal alloys) (1) - testing - creep-fatigue interactions - effect of environment Thermomechanical fatigue (metal alloys) (2) - definitions - role of mechanical constraint - elastic / plastic shakedown - thermal ratcheting - testing - TMF in super alloys: effects of phasing, crystal orientation, time, environment - lifetime prediction models Part B: High Temperature oxidation and processing Introduction of high temperature oxidation (1) - cases, applications - methods of investigation - accelerated test - lifetime prediction High temperature processing of thin film materials (1½) - atomic layer deposition (ALD) - chemical vapor deposition (CVD) - physical vapor deposition (PVD) High-temperature surface engineering using (1½) - monolayer deposition (MLD) - high temperature doping - rapid thermal annealing Part C: High Temperature Materials (4) - Ceramics and composites - Thermal barrier coatings - Self-healing in Coatings at High Temperatures - Additive manufacturing and design of high temperature materials In addition to the lectures, exercises will be made and assignments will be given, for example: - prediction of creep lifetime	

Study Goals	- calculation of creep strain rate - assessing thermomechanical fatigue lifetime - assessment of thermal shock behaviour Students are able to: - identify the different mechanisms by which creep can occur in crystalline solids and explain the principle and usage of deformation mechanism maps - explain the principle of superplastic deformation - perform extrapolations of creep rupture data - identify effects of temperature and environment for high-temperature fatigue - describe the conditions for which thermal/thermomechanical fatigue play a role - describe and calculate conditions for which elastic or plastic shakedown occurs - illustrate the conditions for which thermal ratcheting occurs - explain how isothermal and thermomechanical fatigue are tested and how lifetime is predicted - describe possible mechanisms of thermal/thermomechanical fatigue in high-temperature materials and identify the major affecting parameters - describe the phenomenon of thermal shock and explain the major parameters affecting this - understand gas-solid interactions; thermodynamics and reaction kinetics - understand methods of investigation for materials applied at high temperatures - give materials solutions for high temperature applications
Education Method	Lectures, exercises and assignments
Literature and Study Materials	Slides, including notes (on brightspace) Reader (on Brightspace) on the basis of - The ASM Handbook - Deformation and Fracture Mechanics of Engineering Materials, R.W. Hertzberg - Papers and notes on specific applications and cases.
Assessment	Written closed-book exam
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Materials Science & Engineering

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory	
Course Contents	Fundamentals of Coding in Python and MatLab Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Symbolic regression * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control We will aim for a balance of practical intuition and theoretical understanding: we will discuss examples of applications and go deeper into the theory when feasible.	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures (recorded and in class) and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: (i) practical assignment and (ii) written exam, open book. You must score at least 5.5 on both assignments to get a pass. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity to resit if they fail the written examination. * If your practical assignment is graded between 5 and 5.75, you will have two weeks to integrate the assessors feedback. If this is done satisfactorily, you can be rewarded with a mark that is not higher than 6/10. The exact instructions for the addition will be announced after the grading of the PA is concluded.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding course registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) exam registration is obligatory. More information regarding exam registration, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Elective	Yes	
Tags	Artificial intelligence Complex Mathematics Intensive Linear Algebra Mathematics Modelling Programming Project Proofs	
Department	ME Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42035	Integration Project Systems and Control	5
Responsible Instructor	D. Boskos	
Instructor	Ir. S.C. Bregman	
Instructor	Dr. L. Laurenti	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	SC42015 Control Theory SC42155 Modeling of Dynamical Systems SC42095 Control Engineering SC42025 Filtering and Identification	
Course Contents	The course is based on practical laboratory sessions, in which students will gain hands-on experience with the application of control theory to real-world systems. Matlab and Simulink are used as the primary software environment for the design, analysis, and real-time implementation of the algorithms. Students work in groups of two in the lab, with a setup of their choice. The setups include for instance the rotary pendulum, helicopter model, inverted pendulum on a cart, rotational pendulum, reaction wheel pendulum, and magnetic levitation system.	
Study Goals	The goal of this course is to integrate and apply the theoretical knowledge gained in the compulsory courses of the Systems and Control curriculum. These courses cover the subjects of Control Theory, Physical Modeling, Filtering and Identification. The concepts and tools to be used include mechanistic modeling (based on principles like mass balances, Lagrange equations, etc.), filtering and estimation (e.g., Kalman filtering), linear control design and performance analysis, and system identification in open and closed loop. It is assumed that the students already know these concepts or are able to look them up in the literature. At the end of the course, the students should be able to: - Develop a mathematical model of a real-world dynamical system - Build a simulation model of the system in Matlab/Simulink - Identify the model parameters - Design a state estimator to recover unmeasured system states - Design controllers that regulate the derived system model - Integrate their acquired control engineering knowledge in laboratory setups	
Education Method	Project, demonstration	
Assessment	There is no written exam. The final grade is determined on the basis of a written report (50% of the grade), the final discussion of the results with the lecturers (50%) and the performance during the entire course period (rounding off the grade).	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. Due to the lab work that's needed to complete the project, it is required that you are enrolled on the Brightspace page and take part in the group/setup assignment procedure during the first week of the period. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Remarks	Old course code: SC52035	
Department	ME Department Delft Center for Systems and Control	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. R. Kortlever	

WI3150TU	Partial Differential Equations A	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, ordinary differential equations	
Course Contents	I: (WI3150TU) Introduction. Types of second order equations. Initial and initial boundary value problems. Fourier series. Quasi-linear, first order partial differential equations. Waves and reflections of waves. Separation of variables. Sturm-Liouville problems. Parabolic, elliptic and hyperbolic equations. Maximum principle. Diffusion and heat transport problems. Lectures (3 ECTS). II: (WI3151TU) Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972.	
Assessment	Written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. The resit consists of a written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. W.T. van Horssen	
Co-Instructor	Dr. H.M. Schuttelaars	

WI3151TU	Partial Differential Equations B	3
Responsible Instructor	T. Akkaya	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	WI3150TU	
Course Contents	Boundary value problems. Delta functions and distributions. Greens function for heat, wave and Laplace equations. Fourier and Laplace transform methods. Waves in R2 and in R3. Vibrations of membranes. Bessel functions. Shock waves. Lectures and a feedback assignment (3 ECTS).	
Study Goals	Many mathematical--physical problems can be formulated using partial differential equations. Therefore it is important to be able to both interpret and solve this type of equations. At the end of the course the student 1- is able to formulate various physical problems (wave--equation, heat--equation, transport--equations) in terms of partial differential equations. 2- has knowledge and understanding of various mathematical techniques which are necessary to solve these problems (Fourier--series, method of separation of variables, Sturm-Liouville problems, Greens' functions, Fourier- and Laplace transformations) and is able to apply these techniques to (simple) problems. 3- is able to interpret the solutions obtained and is able to place them in (a physical) context.	
Education Method	Lectures and a feedback assignment.	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	The feedback assignment and a written exam. The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The resit consists of a written exam (100%). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Judgement	The feedback assignment is mandatory, and as long as the assignment is not completed no final grading will be given.	
Co-Instructor	Dr. H.M. Schuttelaars	

WI4014TU	Numerical Analysis	6
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	Classification of partial differential equations and boundary and initial conditions. Maximum principle, uniqueness and non-negativity of the solution, elements of variational calculus. Finite- Difference, Finite-Volume and Finite-Element discretization methods. Solution methods for time-dependent and nonlinear problems. Direct and iterative linear solvers.	
Study Goals	Students will learn: the principles of finite-difference, finite-volume, and finite-element discretizations; the accuracy, consistency, and convergence of numerical solutions; the basic ideas of numerical linear algebra. They will be able to implement a simple PDE solver and to work with existing sophisticated numerical software.	
Education Method	Lectures (possibly, online), homework assignments, programming assignments, written report.	
Literature and Study Materials	[1] J.van Kan, A.Segal, and F.Vermolen, Numerical methods in scientific computing. VSSD, Delft, 2005, improved 2008, ISBN-13 978-90-71301-50-6 [2] Python programming: - Python Help: https://www.python.org/about/help/ - Numpy for Matlab users: https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html [3] FENICS Project: https://fenicsproject.org/ - Installation: https://fenicsproject.org/download/ - Tutorial: https://fenicsproject.org/tutorial/	
Prerequisites	Calculus of functions of several variables, vector calculus, introductory course on numerical analysis (numerical integration, error analysis), introductory course on partial differential equations, some experience with Python or Matlab.	
Assessment	Theoretical and computational assignments, graded reports. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Mathematics Modelling Numeric Methods	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced, analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	The final grade of the course consists of the following components: - Take home exam (50%) - oral exam (50%) Final grade calculation: $(0.5 * \text{take home exam}) + (0.5 * \text{oral exam})$ The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 take home exam: resubmit deliverable oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. H.M. Schuttelaars	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	After this course you are able to: - recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - solve the shortest path problem and the transportation problem as well as some small flow shop problems. - solve the travelling salesman problem by the Branch and Bound algorithm. - apply several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - reproduce basic theorems concerning the mentioned problems and is able to prove some of these theorems. - explain how to solve large scale problems, especially shortest path and vehicle routing problems. - define scheduling problems and complexity theory. - implement the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam and a written resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Elective	Yes	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr.ir. J.T. van Essen	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The final grade of the course consists of the following components: - homework exercises deadline start of Q2 leads to grade G1, - take home exam deadline half of January grade G2 - written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework exercises: resubmit deliverable - take home exam: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	mr.ir.drs. V.N.S.R. Dwarka	

WI4212	Advanced Numerical Methods	6
Responsible Instructor	Dr.ir. S. Jain	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Introductory numerical analysis, Introductory partial differential equations, Introductory continuum mechanics.	
Course Contents	This course is an introduction to hyperbolic partial differential equations and a powerful class of numerical methods for approximating their solution, including both linear problems and nonlinear conservation laws. These equations describe a wide range of wave propagation and transport phenomena arising in nearly every scientific and engineering discipline. Several applications are described in a self-contained manner, along with much of the mathematical theory of hyperbolic problems. High-resolution versions of Godunov's method are developed, in which Riemann problems are solved to determine the local wave structure and limiters are then applied to eliminate numerical oscillations. These methods were originally designed to capture shock waves accurately, but are also useful tools for studying linear wave-propagation problems, particularly in heterogeneous material.	
Study Goals	apply numerical methods to hyperbolic systems, study convergence, stability, monotonicity, know methods for 2 dimensional hyperbolic systems	
Education Method	Lectures	
Literature and Study Materials	Finite volume methods for hyperbolic problems R.J. LeVeque Cambridge, UK: Cambridge University Press, 2002. # ISBN-10: 0521009243 # ISBN-13: 978-0521009249	
Assessment	The final grade of the course consists of the following components: - Homework assignments - Group assignment - oral exam Final grade calculation: Provided that G_3 is larger than or equal to 6, the final grade is computed by the formula: $(G_1 + G_2 + 2 \cdot G_3)/4$. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework assignments: resubmit deliverable - Group assignment: resubmit deliverable - Oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. C.A. Urzúa Torres	

WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students are expected to have: superior knowledge and skill in elementary probability (details below); experience working with Python, Matlab, or R; and a mathematical inclination. If this does not apply to you and you still want to enroll: contact the instructor. Good skills in elementary probability are indispensable as they are the foundation for the more advanced material of the course. We build on the concepts, tools, and methods of elementary probability, and students taking this course must be (or rapidly become) black-belt masters in these foundations, specifically: Chapters 2-5, 7, 9, and 10 of Dekking et al; prior knowledge of Chapters 13 and 14 is helpful, though we will revisit these subjects during the course. Testing yourself and doing a thorough review in the first few weeks is required; we will also spend some time on this in class to help you do this efficiently; after this a diagnostic test (carrying part of the course grade) will assess your mastership; sub par performance means you cannot follow the course.	
Course Contents	This course on Monte Carlo simulation and stochastic processes consists of a mixture of these components: modelling and models (and some of their pluses and minuses), learning a number of stochastic processes and some of their properties, doing Monte Carlo simulations, analyzing the generated data and drawing conclusions from them. Contents and format of this course are intended to combine the following: 1. real (possibly stylized) problems as starting points; 2. enough about modelling to make participants aware of the impact of modelling decisions; 3. identification of a number of stochastic processes present in/related to models and problems; 4. (limited) treatment of underlying theory, guided by: (a) what is necessary to ensure that participants know how to do things right and (b) what enables them to do things smartly; 5. facts, properties, and theorems: (a) sincere attempt to understand where possible; (b) focus on practical implications and uses; (c) illustrations rather than proofs or theory; 6. useful tools and techniques from statistics, probability, and stochastic processes; 7. simulation as aid to: obtaining answers, intuitive understanding, illustration of theory; 8. providing as much opportunity as feasible for participants to work on/with problems from their own field. Tentative topics/contents list: Review basic probability: most to be done as self-study, some in class, plus a mandatory test (see Assessment). Models, modelling. Simulation: Convert a model to a program and run it. How to expose and locate errors. Data, empirical distribution; estimation. Accuracy, efficiency. Sensitivity. Rare event simulation. Experimental design. Conditional distributions and expectations; conditioning as a tool. Multivariate distributions: general (some examples); multivariate normal. Stochastic processes: Random walk. Poisson process; renewal processes. Markov chains: discrete time; continuous time; long-term behaviour; embedded Markov chain; birth and death process; Brownian motion. A snippet of diffusions, (solutions to) stochastic differential equations (SDEs).	
Study Goals	Learning objectives: after following the course, participants will be able to 1. in an iid setting, carry out Monte Carlo simulations; and 2. extract and collect, summarize, and interpret the data thus generated; and 3. assess accuracy/uncertainty in estimates, and solve questions like How many runs for 1% accuracy?; 4. the above (three) also in some non-iid settings, e.g., for a stationary process, using the batch-means method; 5. for small-sized practical problems: build a suitable model and simulate it; 6. extract and collect required model parameters from a problem description or background; 7. recognize where modelling choices are made; 8. make elementary modelling decisions themselves; 9. recognize and describe in mathematical terms (the properties of) a number of stochastic processes, among others: Markov chains, random walks, Brownian motion; 10. recognize/identify the covered stochastic processes in practical settings/problems; 11. use the tools and concepts from probability theory to describe, discuss, and manipulate (the components of) stochastic models; 12. where applicable, solve problems by means other than Monte Carlo, i.e., analytically or numerically, procuring exact answers, by identifying a stochastic process and using its known properties; for example, concerning long-term system-behaviour.	
Education Method	Weekly class meeting of two hours. About two thirds spent on lecture; about one third on questions and exercises. A number of so-called "Hands-on" assignments: a collection of exercises that combine the analytical content with computer simulation; to be done individually or with a partner. The participants are expected to perform a significant amount of simulation in Python, Matlab, or R (or other suitable language). Spread over the semester, five homework assignments. I strive to make a classroom available during the (probably) Friday lunch break, in which the participants can work on their own and where the TA (if there is one) is available for questions. In addition, I will hold a weekly drop-in office hour, most likely at lunch time (details on Brightspace).	
Literature and Study Materials	Unfortunately, for this course there is no really appropriate textbook, one that combines modelling, Monte Carlo simulation, and stochastic processes. For the review of basic probability and statistics, you are referred to Dekking et al, "A modern introduction to probability and statistics," Springer, 2005; the text is very well suited for individual review and practice, as it contains many worked examples, (quick) exercises, answers and full solutions. To be successful in the course, participants must have developed superior skill wrt the tools and techniques in the first half of the book (see "Expected prior knowledge"). Most of the material on stochastic processes can be found in "Probability Models" by Sheldon M. Ross; the book contains many worked examples and exercises; its heuristic and intuitive approach fits that of the course; references will be to the 10th Edition. A slightly more mathematical, but still very accessible reference covering the same area is "An introduction to stochastic modeling" by Howard M. Taylor and Samuel Karlin. Lecture slides, notes, and additional references will be posted on Brightspace.	

Assessment	A laptop is required and it is recommended that you bring it to class. Many languages and/or software packages are suitable: I prefer you use the one you are most familiar with. I am migrating to Python, but in the material presented in class and distributed via Brightspace, there still is some Matlab code. You may use a language/tool other than those three, but talk to me first (because you should be able to use your software on the exam and it should be possible to arrange this). In all cases: you should have the software installed and be familiar with its use. 1. Probability test on knowledge and skills needed in/for the course (15%); passing score must be obtained to take the course. 2. Assignments (five; each counts for 4%), which may be done with a (=one) partner. 3. Final exam (on-campus): closed-book computer exam (with cheatsheet allowed, specs on Brightspace); counts for 65%. To pass the course, the probability test must be passed and the final exam grade must be 6 or higher; course grade is weighted average.
Permitted Materials during Tests	To the final exam you may bring a self-compiled formula sheet: one A4 page with normal-sized text on about 25 lines (or any reasonable equivalent of this), made by you, of things you would find helpful.
Elective	Yes
Tags	Adventurous Diverse Mathematics Modelling Practical Stochastics
Co-Instructor	Dr.ir. G.N.J.C. Bierkens

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience.	
Literature and Study Materials	Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3. Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	The final grade of the course consists of the following components: - Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). - Group programming project (group work) starting mid Q2 with submission deadline end Q2 (2/3 of final grade). Final grade calculation: $(1/3 * \text{homework assignments}) + (2/3 * \text{group programming project})$ but it is only valid when all homework programming assignments have been completed with sufficient results. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework programming assignments: submit new set of homework - Group programming project: submit improved version Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. J. Thies	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will self-study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final exam + a speaking test. Contact the secretaries office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

Year	2024/2025
Organization	Aerospace Engineering
Education	Master Aerospace Engineering

Master AE PPD Courses

AE4006E	Sustainability in Engineering	5
Responsible Instructor	I. Fernandez Villegas	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/6/0/6-offered in both Q2 and Q4	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor in a relevant field	
Course Contents	As future engineers, students of TU Delft need to be aware of the social and ecological responsibilities that come with the engineering profession. This course is based on the principles of Blue Engineering, an innovative student-led workshop-style course developed at Technische Universität Berlin and currently running at more than 10 other German universities as well as TU Delft. The course facilitates creative, interdisciplinary and sometimes heated discussions on the issues posed by technology in society and in nature. It creates an inclusive and open learning environment which offers a refreshing alternative to teacher-centered instruction and offers a view beyond one's own horizon and an (inter-)active scrutiny of one's social and ecological responsibility. In this way, students add to their specialized knowledge an orientation knowledge and acquire competencies that help them to know their role in relation to technology and society and to shape it together with others. The course focuses on sustainability, topics such as technology assessment, engineers' responsibilities, neutrality of technology, the role of engineering in society, diversity in engineering as well as self-chosen topics. The following topics will be covered: - Week 1: Criteria for Technology Development / Role of Space Exploration / Decision Making (group decisions and introduction to TINS-D); - Week 2: Responsibility & Ethical Codes (Engineers- Responsibility and guiding legal frameworks); The Productivity Worldview (expert TINS-D analysis) - Week 3: The Value of Open Questions (connection to project section, effective teamwork for diverse groups) / Diversity & Technology (who do we design for?) - Week 4 - 5: Phase II - students facilitate existing learning modules; for example: UN Sustainable Development Goals, The Dilemma of Environmental Democracy, Life with Artificial Intelligence, Regulations for Today & Tomorrow, Planned Obsolescence, Circular Economy, Prenatal Transhumanism, and The Happiness Project. -; Week 6 - 7: Phase III - students facilitate newly developed learning modules on a topic of their choice; final session includes reflection on the course. - Week 8 - 10: Exam period free.	
Study Goals	At the end of the course, students will be able to: - Analyse and evaluate the present reciprocal relations of technology, individuals, nature, society and democracy by taking different perspectives. Students are able to state their personal perspective on them and act accordingly. - Cooperate with others to analyse and evaluate those relations. - 12 key competences of an Education for Sustainable Development: 1) perspective-taking; 2) anticipation; 3) gaining interdisciplinary knowledge; 4) dealing with incomplete and complex information; 5) cooperation; 6) coping with dilemmas of individual decision-making; 7) participation; 8) motivation; 9) reflection on own principles; 10) acting morally; 11) acting independently; 12) support of others - Design and co-conduct their own course material	
Education Method	The idea behind Blue Engineering is to enable students to shape their own learning process. Therefore, instead of teacher-centered education, student tutors facilitate and guide the weekly sessions, and the participants themselves get involved by developing their own teaching units as a semester project. Stimulating guest lectures on current topics are integrated throughout the course. Each week will consist of one 4-hour and one 2-hour seminar, with brief preparation and follow-up activities. Students complete the main part of preparation and follow-up material on a given session within class time, and write a weekly learning journal entry outside of class.	
Assessment	Presence in class is mandatory, only two sessions can be missed. If necessary due to COVID-19 related reasons online participation will be made possible for individual cases. The summative (graded) coursework consists of two parts: a personal learning journal (50%) and the semester project, which is to develop, perform and document a teaching unit (building block) in a team (50%).	
Enrolment / Application	Given the nature of the AE4006E "Sustainability in Engineering"; course and to ensure its education quality, the intake of students in the academic year 2024-2025 will be limited to 60 students in Q2 and 60 students in Q4. Also because this course thrives on multidisciplinary interactions, students from faculties other than the Aerospace Engineering Faculty will be given priority access to 20 of the above mentioned 60 spots (for each quarter in which the course is offered). The selection procedure put in place for this course and instructions to apply is described below. For all students, presence in the first session is a prerequisite to take the course. The selection procedure for this course will be as follows: - The students will need to submit their application by email to 4006E-AE@tudelft.nl within a prescribed application slot. The application slot for the course offering is as follows: - Q2 between September 30th at 08:00 and October 20th at 23:59. - Q4 between February 3rd at 08:00 and February 16th at 23:59. - Applications submitted outside of these slots will not be considered. - In their application, the students will need to provide answers within a total 100 word limit to the following three questions: (1) Which faculty are you in?; (2) What is your motivation to take the AE4006E course?; and (3) Which self-chosen topic would you like to explore further during the course? - The answers to the questions will be graded as pass or fail. The selection among all the applications that are given a pass will be done on a first-come, first-served basis based on the date and time in which the application was received. - The results of the selection process will be communicated via email within ten working days after the closing of the application period. - The above-described procedure will apply to the 40 spots in the course reserved for the students from the AE faculty as well as the 20 spots reserved for students from other faculties. In the case some of the 20 spots reserved for students from other faculties do not get filled out, they will be made available for students from the AE faculty.	

AE4007	AI for Mindfulness in Engineering	5
Responsible Instructor	Dr.ir. C.J. Simao Ferreira	
Contact Hours / Week x/x/x/x	0/1/1/1 COURSE NOT OFFERED IN 2024-2025	
Education Period	2 3 4	
Start Education	2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is an experiential community designed to empower both students and lecturers with the skills to integrate Artificial Intelligence (AI) into mindful graduation projects. It aims to enhance understanding and practical application of AI in personal and professional roles as researchers and designers while addressing the evolving role of AI in academic and professional environments. The aim is to facilitate the effective use of AI in academic research, fostering a deeper understanding of its impact on traditional academic values, creativity, skill requirements, and the evolving role of AI in education and engineering. This course explores the intersection of artificial intelligence (AI) and mindfulness practices within the engineering field, combined with the study of research methodologies. Students will learn how to leverage AI tools to foster mindful awareness in their work environment, enhance personal well-being, and improve team dynamics. The course combines theoretical foundations with practical applications, integrating mindfulness techniques and AI technologies to address stress, improve focus, and promote a balanced professional life. Additionally, students will explore the historical development and future trends of their respective engineering disciplines. The course incorporates the content of AE4010 Research Methodologies. The coursework builds upon the development of the research review and definition of an MSc thesis project or PhD research project. Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor. Important information: in the academic year of 2024-2025, we will be limiting the number of students that can join the course at a given quarter. Only 30 students can start the course at a given quarter. We will aim to distribute students throughout the year. If necessary, a selection system will be used.	
Study Goals	Learning Objectives: - Understand the basics of AI and mindfulness and their applications in engineering. - Learn to use AI tools to support mindfulness practices. - Apply mindfulness techniques to enhance personal and professional well-being. - Develop strategies to integrate AI-driven mindfulness practices into engineering workflows. - Analyze the impact of mindfulness on team dynamics and project outcomes. - Research and articulate the historical development of their specific engineering discipline. - Explore and predict future trends in their engineering discipline. - Facilitate the effective use of AI in academic research, understanding its impact on traditional academic values and skill requirements. - Develop a comprehensive research review and define an MSc thesis project or PhD research project. - Create and refine a portfolio that is peer-reviewed and supervised by an advisor.	
Education Method	The course employs a blend of lectures, practical workshops, and project-based learning. The methodology encourages collaborative learning between students and lecturers, fostering a mutual understanding of AIs potential and challenges in academic research. This approach facilitates collaborative learning, critical thinking, and the practical application of AI. The course also features guest lectures and interactions to provide real-world perspectives.	
Assessment	Students develop a portfolio that is peer-reviewed and reviewed by a supervisor/advisor.	

AE4010	Research Methodologies	2
Responsible Instructor	I. Uriol Balbin	
Contact Hours / Week x/x/x/x	Online	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The aim of the course is to be a research-driven preparation for the aerospace engineering MSc thesis in the final year of the MSc. It will help you prepare for the challenges of your thesis work. The course will consist of 6 units and will be taught as a fully online course using video lectures and other online contents. The set up is as follows: 1. Research Design in MSc - Introduction to research, research framework 2. Research Methods - Stages of a project, Research objective, research questions, research strategy, research methods 3. Data Analysis - Quantitative & Qualitative methods 4. Validation & Sustainable Research - How to validate & verify your work? What does it mean to conduct sustainable research? 5. Project Management - How to manage your project and your thesis progress. How to plan, expectations, Gantt Charts 6. Academic Writing - Learn the peculiarities of academic writing. The course has been designed to be suitable for distant learning but the exam will be on campus.	
Study Goals	At the end of the course you will be able to: 1. formulate a scientific research question(s). 2. correctly cite the relevant literature. 3. understand the appropriate methodologies and tools in research. 4. learn how to set-up a clear research plan.	
Education Method	(Online) Lectures and exercises.	
Literature and Study Materials	Chapters on Interdisciplinary Research and Research Skills - with contributions from van der Gaast, Koenders, Post, Menken en Keestra, Amsterdam University Press, ISBN: 9789463728256. Available online from the TU Delft Library	
Assessment	On-campus exam. In order to obtain a (passing) grade, a short project plan must be written, and a self-assessment and two peer reviews must be performed. There will be two exam opportunities. Please check Brightspace and mytimetable for further details. Only EWEM students who started their master in 2022 or 2023 will use the old assessment method: a (draft) Research Plan according to a prescribed format based on the topics of their MSc thesis.	
Permitted Materials during Tests	Students are permitted to use any handwritten/printed out notes. Access to digital and internet sources is prohibited.	
Enrolment / Application	Please enroll via BrightSpace. The course contents are always open.	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BM51007	Professional Development Journey	1
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	A.J. Loeve	
Instructor	Ir. P.H. de Jong	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamentals of Biomedical Engineering (BM41141)	
Course Contents	Professional development journey ===== In the studentcentered approach of the course Professional development journey (PDJ) students are motivated to explore their own interests, motivators and strength. Students investigate with reflection which moments they felt inspired and what they think is important to make their own choices and create their own learning path. The purpose of the mentorship meetings ===== The mentorship meetings will provide students of BME a space to share ideas and experiences. The sessions will be important to create a safe space for students to share and explore. It is a low stake space where they can talk and think freely. The sessions will be (peer)feedback and reflective oriented meetings to support the reflective assignments for their professional development. In the meetings students help and guide each other to full fill their personal goals. Students can share what they want to get better at and ask for feedback. During the meetings students also listen to their peers and have reflective conversations. Students will also reflect on their own personal goals and development. How they can make the decisions to support their personal growth according to their own goals for the future. Growth mindset (Q3) ===== Most people like the things they are good at. When you get rewarded by good grades or compliments you feel confident and are willing to work harder to get the next reward. In Q3 students explore their growth mindset and reflect on moments of happiness. This to enhance areas of fixed mindset. Reflective conversation (Q4) ===== For this meeting students gather feedback from different people around them (teachers, friends family) and in conversation with a peer about impression students made and how they come across. In the meeting mentors guide a reflective conversation where students evaluate and adjust chosen engineering roles and learning objectives for the decision for an internship/Jip/ or courses. Social context (Q4) ===== In Q4 students are placed in a new social context when joining a Jip or internship. Students make a reflective assignment which focuses on A new social context. Students who are following electives reflect on another moment where they joined a new social context.	
Study Goals	The student is able to 1. Reflect on decision making and problem solving: Analyze problems and decisions with reflection methods. Evaluate situations to create different options and select the best. 2. Create a growth mindset: Investigate and explore their own strengths to move from a fixed to a growth mindset. 3. Give and receive feedback: With feedback-seeking behavior and goal orientation stimulate professional development of peers and themselves. 4. Handle contextual and cultural differences: Discover (cultural) differences, cope with differences in team work and is able to choose another perspective.	
Education Method	Each quarter, interactive on-campus mentor sessions are organized. On Brightspace, the topics and assignments will be provided. During the sessions, students can discuss and work on their exercises and ask their peers and mentors for help.	
Assessment	The main assessment of this course is a portfolio assessment which contains three reflective assignments that are graded with Pass/Fail. If an assessment is judged as insufficient, the student can write an extended essay on the topic to achieve a passing grade.	
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.	
Department	ME Department Biomechanical Engineering	

CESE4020	Effective and Responsible Engineering	5
Responsible Instructor	Drs. J.A. Weitenberg	
Instructor	Drs. P.C. Post	
Instructor	O.E. Popa	
Contact Hours / Week x/x/x/x	0/0/0/5 Lectures (2x2h) Tutorial (1x2h)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Academic Writing: The Academic Writing course teaches students to write about relevant scientific research on an academic level. They will write a literature review on a subject connected to their specialisation, in which they will review relevant scientific sources and learn to compare and contrast these. Students will learn to reflect on their work and the work of their peers by doing peer reviews and processing feedback in revision rounds. Ethics: Students will familiarise themselves with the method of value-sensitive design as a method of approaching ethical problems in technological innovations. They will master this method and apply it in a case study of their choosing. Oral Presentations: The Oral Presentations course teaches students to prepare and deliver a presentation on a professional level, on a case study concerning value-sensitive design. Students will practice presenting during all lectures and will build their skills in a step-by-step approach, in which there is room for differentiation between those with less and those with further developed starting levels. Students will reflect on their progress in Oral Presentations and Ethics, incorporating the feedback from both lecturer and peers. All activities in this course have compulsory attendance.	
Study Goals	At the end of this course, students will: Master basic skills in academic writing. They will be able to write a literature review on a subject connected to their specialisation. Master basic principles of ethics. They will be able to recognize social and ethical issues in engineering and technology and critically evaluate possible solutions for the social and ethical issues coming up in engineering, by taking into account multiple perspectives and stakeholders. Master a professional level of presenting. They will be able to present an ethical issue connected to CESE to a mixed academic audience.	
Study Goals continuation	Academic Writing: At the end of this course, students will master basic skills in academic writing. They will be able to write a literature review on a subject connected to their specialisation. To achieve this goal they will be able to: 1. formulate a main question (or aim/purpose statement) with subquestions 2. support writing sufficiently and correctly with literature 3. support writing with argumentation 4. structure a text according to the scientific conventions 5. implement knowledge about the specific elements of a scientific text and their generic requirements in their own text. 6. use visualisations, such as tables and figures, appropriately. 7. apply a correct formal scientific style 8. apply the general writing strategies: genre analysis, outlining and revising. 9. determine further learning goals for their thesis project.. Ethics: Students will be able to: recognize social and ethical issues in engineering and technology and conceptualise these issues in terms of value conflicts critically evaluate possible solutions for the social and ethical issues coming up in engineering, by taking into account multiple perspectives and stakeholders. Oral Presentations: At the end of this course, students will master a professional level of presenting. They will be able to present an ethical issue connected to CESE. To achieve this goal they will be able to: 1. adjust academic and technical information to a target audience; 2. use content and rhetorical techniques to make presentations relevant and attractive to the audience; 3. use voice and body language to support the presentation; 4. apply a clear structure to a presentation; 5. use visualisation tools supporting the content of the presentation and operate these; 6. deal professionally with questions from the audience; 7. reflect on their presenting skills and formulate further personal learning goals.	
Education Method	Academic writing: 6 small-scale workshops in which students discuss and practice aspects of writing. Students will write a literature review on a subject connected to their specialization. Students will receive feedback from the lecturer as well as from their fellow students (peer feedback) to enhance their writing skills. Ethics: 1 lectures and 2 workshop on ethics. Students will then work on a case study for their final presentation. Oral Presentations: 7 small-scale workshops in which students discuss and practice aspects of presenting. Students will receive feedback on several presentation exercises from their lecturer as well as their fellow students (peer feedback) to enhance their presentation skills and prepare their final presentation on an ethical case study. Students make two reflection assignments. Attendance is mandatory for all activities.	
Literature and Study Materials	Academic writing: Brightspace course TU Write Ethics: Materials will be made available on Brightspace Oral presentations: Brightspace course TU Present	
Assessment	The final grade of CESE4020 consists of the following components: - Academic Writing: Written paper (pass/fail); (2,5 EC). - Ethics: Written assignment (pass/fail); (1 EC). - Oral Presentations: Final presentation: 50% group presentation and 50% individual presentation techniques (pass/fail) 1,5 EC). Two reflection assignments count towards the final OP grade. Repair possibilities: - Academic Writing: Reparation written paper	

Enrolment / Application	- Ethics: reparation written assignment - Oral Presentations: reparation for group presentation, individual presentation techniques, and/or reflection assignment(s).
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).
	This course is only open to students with an active and valid registration for the M-CESE, M-CE and M-ES programme

CME4200	Intercultural Relations and Diversity Competence	2
Responsible Instructor	Dr.ir. V.J. Cortes Arevalo	
Responsible for assignments	Dr. C.A. Benitez Avila	
Co-responsible for assignments	Dr.ir. V.J. Cortes Arevalo	
Contact Hours / Week x/x/x/x	Two workshops and four interactive lectures.	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Construction, Management, and Engineering projects are complex, innovative, multi-disciplinary, and intercultural projects developed in a dynamic environment and require engineers to competently combine engineering and organisation skills to achieve better the project expected results. This course helps you to look at intercultural relations as a communication process between people that work together and naturally belong to multiple social groups. The course helps to develop a diversity competence to address communication differences before and after they occur. Through role-playing games, class exercises, and a group essay, the students engage with the theory, reflect on their experience, and explore the role of inter and intra-cultural relations to develop a diversity competence that they can later apply in their professional practice.	
Study Goals	Departing from sociological and psychological theories on communication, this course covers themes about (cultural) diversity and inclusiveness in professional collaboration through the following learning objectives. The course takes a dynamic and multifaceted approach to culture as its starting point. At the end of this course, you will be able to... Discover and discuss the cultural and diversity dimensions in social/organisational phenomena, events, and activities related to the construction, management, and engineering practice. Apply the TOPOI model to assess and address the impact of these differences on interpersonal and intergroup interactions. Demonstrate cultural and diversity awareness by developing an appropriate and ethical attitude toward inter and intra-cultural collaboration and communication.	
Education Method	Two intensive half-day small-group training sessions of role-play and/or negotiation games (if Covid conditions allow). The aim is to reflect in the role you played. Blended classroom, that requires and expect you to prepare the book chapters, video clips, literature, and popular articles offered in Brightspace as a weekly learning unit. Interactive lectures with exercises and discussions in small groups that guide and provide input to writing the group essay for the test at the end of the course. Self-study, search and reference to relevant literature will support you in the execution and reporting of TOPOI analysis into a reflection report that students will submit in pairs, following the format we will introduce in week 2. See also 'special information' below.	
Course Relations	This course lays a foundation for the CME1201 Collaborative Design and engineering course; to make students aware of the inter and intra-cultural differences to improve their collaboration practice.	
Books	Edwin Hoffman and Arjan Verdooren. 2018. Diversity Competence. Cultures dont meet; people do. Uitgeverij Coutinho. Isbn 978 90 469 0598 2	
Assessment	We expect students' presence and active participation in the two workshops and the interactive lectures, as these provide input for the group essay. Grading is based on a written test in the form of a group reflective essay that applies the TOPOI analysis introduced in the course book by Hoffman and Verdooren (2018). The individual contribution to the group essay and course activities will also be part of the assessment. In case of failure (grade below 5.8), students can hand in a group addition to the project to be awarded a maximum grade of 6.0. The deadline for the addition is in the retake exam week immediately after the course ends. We will announce in Brightspace the exact instruction for the addition.	
Special Information	If circumstances prevent the game-play, some additional online materials may be offered. The course materials are organised in Brightspace in a blended learning fashion, requiring and expecting the students to prepare for the interactive lectures, which include small group discussions of the study material.	
Expected prior Knowledge	No prior knowledge is expected	
Academic Skills	Students are expected to write a short reflective essay in small groups, the format of which is provided.	
Literature & Study Materials	Video clips and a limited number of scientific articles available through BrightSpace. Course book: Edwin Hoffman and Arjan Verdooren. 2018. Diversity Competence. Cultures dont meet; people do. Uitgeverij Coutinho.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	not relevant	
Collegerama	No	

CME4300	Engineering Asset Management Systems	5
Responsible Instructor	Dr.ir. G.A. van Nederveen	
Instructor	Dr.ir. R. Binnekamp	
Instructor	A. Jagadeesh	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Engineering Asset Management (EAM) is a holistic approach to manage a system/portfolio of engineering assets (i.e., physical objects) that enables a certain quality of services (QoS), overarching and/or encompassing the worlds of Project & construction management and Operations & maintenance management (DBMO). EAM strategically balances and designs these QoS performance levels between: (a) economic user demands; (b) ecologic capital supply (c) isonomic organizational and financial means/support; embedded within a multi-stakeholders / multi-dimensional tripartism system context (open-source). EAM is integratively concerned with efficient and effective control and care of both one-off new engineering asset deliveries (projects management) and continuously safeguarding the desired quality of service levels of existing in-service assets (operations management) over the entire service life of the engineering assets base. These project and operations activities will be integrated into a strategic asset management planning system using multi-criteria decision/design optimization modelling (MCDO, open glass-box). Finally, this course will introduce some of the following important EAM topics (1) systems safety (2) resilience (3)outsourcing: a game of cooperation and (4) BIMSI asset management information. The first alternative view within this course is to provide a integrative viewpoint on EAM where it is viewed as a cyclic socio-eco purpose balancing act (economical, ecological, social). The second alternative viewpoint is the use of a service provider for reflectively practice the integrative projects and service management concepts into a multi-level strategic asset management plan. The third alternative viewpoint, is that the SPoIs EAM systems will be improved and or appraised using the Open Design principles: multi-criteria Open glass-box decision analysis/optimization , Open source systems theory, Open ended systems thinking combined with Open loops management. In this course the following concepts are addressed: 1. Your Service Provider of Interest (SPoI) and the Quality of Service (QoS) systems concept (introduction to strategic engineering asset -, project- and operations management and open design systems theory) 2. Zooming OUT the SPoIs embedding systems context and open-sources stakeholder constraints/ regulations 3. Zooming IN the open socio-eco purpose EAM system, tripartism system modelling and Open loops management (ODL, MI,MII) 4. Multi-criteria Quality of Service Decision Evaluation and Design optimization (MCDA/ MCDO) - The Projects Management and/or Operations Management solution space 5. Strategic Asset Management Planning (SAMP) integrating the Project & Construction Management and the Operations & Maintenance Management (the Project Delivery and the Service Operation Plan,: PDP and SOP) 6. Keeping the SPoI engineering assets Safe - Integral System Safety 7. Special EAM topics (1) Systems Resilience engineering (2)Outsourcing: a Game of Cooperation (3) the BIMSI EAM information concept For more course info see : https://www.open-design.school/ This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen Service Provider of Interest (SPoI) as a connective and practicing vehicle enabling the Open Design Learning (ODL) response. The SPoI can a transportation-, water management-, energy infrastructure and/or a real-estate system service provider.	
Study Goals	Students should be able: - To understand and be familiarized with the EAM concepts, principles and practices, by dialoguing these with the constructors, by navigating through the EAM reference books and by engaging to the SPoI. - To apply and relate these modellingEAM concepts, by dialoguing and experiencing these with the SPoI and the reflective practice. - To (re)work this EAM knowledge, by transforming and linking the SPoI-dialogues into new openings/ learning insights. - To demonstrate and develop the personal outcome of the aforementioned learning goals, by internalizing the openings/ learning achievements in an original ODL response integrating all (computer) models.	
Education Method	The course is based on the Open Design Learning concept (ODLc): every week students are asked to study specific concepts and apply these to a self-chosen Service Provider of Interest (SPoI) by means of a self-created ODL response and related open-glass-box computer models. The teachers incite the EAM concepts as a reflective practitioner using both reference books and dialogue questions from the students. The students have 2 hours of these concept and dialogue sessions and 4 hours of (computer) practical work sessions per week. During the work sessions, students can work on their ODL response under supervision of the teacher/constructor. During these practicals also Master class sessions will be organized.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a group deliverable based on the self-chosen SPoI (one group delivers one ODL response). The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen SPoI using a: 1) logical review and/or diagramming 2) computer model(s). --	

	The Open Design Learning commendation principle will be applied as a grading rubric for the ODL response. Both the SPoI content characteristics, and the students learning process are integrated within these commendation principles. We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude &Conspect /Convey /Convince
Expected prior Knowledge	CE, TPM, ABE or other engineering design related BSc.
Academic Skills	Skills to: - curiously connect to a committed system of interest; - critically convince others with a compelling response; - clearly convey a communicative response; - creatively convert concepts into a comparative response; - construct a properly compiled open glass box model; - contextualize in a combined and open space; - comprehensively conclude and cogently conspect your open ended solutions.
Literature & Study Materials	Reference books Hastings (2015) Physical Asset Management Graham M. Winch (2009). Managing Construction Projects. Wiley-Blackwell. Nigel Slack (2006). Operations Management (5th edition). Prentice-Hall. Crespo Márquez (2007). The Maintenance Management Framework, Springer. Reference reader (content) Binnekamp; Heukelum; Wolfert ;Zhilyaev (2022) Reader - Open Design Systems Reference scientific articles Kammouh, O ; Nogal, M ; Binnekamp, R ; Wolfert, A.R.M. / Multi-system intervention optimization for interdependent infrastructure. In: Automation in Construction. 2021 ; Vol. 127. pp. 1-11. Zhilyaev, D. ; Binnekamp, R. ; Wolfert, A.R.M. / Best Fit for Common Purpose : A Multi-Stakeholder Design Optimization Methodology for Construction Management . In: Buildings. 2022 ; Vol. 12, No. 5.
Judgement	After handing in the ODL response you will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

CME4400	Entrepreneurial Engineering	5
Responsible Instructor	Dr. D.F.J. Schraven	
Responsible for assignments	Ir. L.P.I.M. Hombergen	
Co-responsible for assignments	Prof.dr.ir. F.E.H.M. Smulders	
Co-responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	2 x 2hrs lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	CME students, the specialisation Design & Integration.	
Course Contents	The course deals with the entrepreneurial traits in the civil engineering domain. Students are introduced with innovation and sustainability transition theories, new economy concepts and business concepts. They are challenged by 2 assignments how to implement a technological innovation in the civil engineering market structure and processes. The course consists of two cumulative modules: Analysis module, in which you analyse the success of a retrospective innovation case as a group, informed by innovation and transition theories and tools as your basis and inspired by relevant guest lectures from innovation experts from the civil engineering practice. Development module, in which you create a business plan for a real-time technological innovation as a group, informed by business model innovation processes, transition and adaptation methods.	
Study Goals	Finishing the course, that students are able to: - Understand and analyse an innovation case, with innovation theory to identify the successes and failures on its adoption; - Understand specific market structures of the civil engineering domain, the relations between public and private roles in this domain, and the multifaceted reality thereof; - Understand the market processes, specifically the public tendering processes for innovation-oriented projects; - Analyse the technical aspects of a technology / innovation to identify the robustness and the alternative implementations thereof; - Analyse the operational context of the projected implementations of a technology; - Create the role of your business in the pursuit of implementing the technology / innovation; - Create an adaptive and embedding business plan (flexible and future oriented) for the case. - From the perspective of a client or an established market party: be able to identify, stimulate, guide and embrace innovative innovation concepts.	
Education Method	The course is an activity-based learning experience, which is accompanied by lectures with the intent to inform and inspire the student with theoretical and practical insights.	
Assessment	Both in the Analysis and Development module, the assignment will be prepared in groups in the form of a report.	
Expected prior Knowledge	No prior knowledge is required.	
Academic Skills	To critically analyse the success of a previous innovation and extract lessons from it, and to be able to develop a business plan and model for an innovation in a civil engineering context.	
Literature & Study Materials	A selection of Book Chapters, academic articles and case-based data.	
Judgement	The assignments make up 40% and 60% respectively for the final grade. The combined grade of the 2 group assignments should be a 6.0 or higher in order to pass. Also both assignments should minimally be a 5.0 to count.	
Permitted Materials during Exam	N/A	
Collegerama	No	

CME4601	Leadership and Strategic Management	5
Responsible Instructor	Prof.dr. P.W.C. Chan	
Instructor	Dr. M. Leijten	
Instructor	M.G.C. Bosch-Rekvelde	
Contact Hours / Week x/x/x/x	4hrs lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The climate emergency, rapid urbanisation and resultant need to provide more affordable housing quickly, along with the urgency to transform our food, water, energy and waste infrastructure are key societal and sustainability transitions of our time. Furthermore, technological advancements in the digital space and the changing demographics are also demanding a rethink in the ways organisations function. Moreover, successive crises such as the Global Financial Crisis in the late 2000s and the more recent Covid-19 Global Pandemic have triggered calls for radical change. Put another way, if we keep doing what we have always done, then we will always get the (unsustainable) results that we have always gotten. Thus, there is a need to do things differently, and this is where leadership and strategic management can play a significant role. In tackling the societal and sustainability transitions highlighted above, simply relying on technical knowledge is not enough. These transition challenges are wicked problems, where simple solutions and incremental change are not sufficient. To address these transitions, there is also a need to inspire people from different backgrounds to activate radical change in our actions. Leadership skills are therefore critical to influence others to take the path less travelled. In addition, projects are increasingly considered and commissioned as drivers of change. Therefore, leadership is deeply connected to the delivery of projects. The key topics of this course include: a. Context: Understanding the context of leadership and strategy in transitions; b. Leadership theories: Complementing management and leadership, and shifting the mindset from planning and control (outputs) to influencing change and transformation (outcomes); c. Strategic management: Shifting the framing from strategy formulation and (long-term) planning to purpose-driven implementation and strategy-as-practice; d. Creativity and innovation: Shifting towards more open forms of innovation, and the role leadership and organisational strategy can play; e. Communication and negotiation: Understanding the power of communication and negotiation practices to influence outcomes; f. Values and impacts: Critically reflecting on the value, values and impacts of leadership and strategic management.	
Study Goals	The aim of this course is to enable students to develop their leadership skills and knowledge about strategic management of engineering projects. The intended learning outcomes include: (a) to develop critical awareness of leadership theories and apply these appropriately in analysing their personal leadership preferences; (b) to explain the significance and process of leadership in influencing others to achieve beneficial outcomes in the management of projects; (c) to develop critical awareness and knowledge of the strategic role that project managers and project management play in supporting wider organisational change outside of the boundaries of the project organisation, and; (d) to develop knowledge of the management of projects as a value creating process that translates policies and strategies into practice.	
Education Method	Lectures, Discussions, Group and Individual Assignments	
Assessment	It is important that the teaching, learning and assessment strategy takes into account that leadership development requires the individual to critically reflect in practice. The students will be required to submit two written assignments, including (1) a group assignment that combines leadership and strategic management in a change context; (2) a critical review on leadership and strategic management, and; (3) an individual reflection of personal leadership as applied to the management of projects.	
Expected prior Knowledge	Not applicable	
Academic Skills	Critical Thinking, Reflective Practice, Communication (Written and Oral Presentation), and Working in Teams.	
Literature & Study Materials	There will be directed reading provided each week to cover different aspects of leadership and strategy management in relation to managing projects. The course is also connected with the urgent need to drive societal and sustainability transitions. The following report commissioned by the Association for Project Management provides the contextual background for this course. Morris, P. W. G. (2017) Climate Change And What The Project Management Profession Should Be Doing About It: A UK Perspective, Princes Risborough: Association for Project Management, https://www.apm.org.uk/media/7496/climate-change-report.pdf.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

CS4225	Educational Technologies	5
Responsible Instructor	Prof. M.M. Specht	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	* Theories of Human Information Processing and Learning * Learning Analytics * Personalisation and Adaptive Educational Systems * Multimodal Feedback Systems * Artificial Intelligence in Education * Realtime Learning Technologies	
Study Goals	The course will enable you to classify, understand, design and implement the core functionalities and systems for supporting human learning processes. As well current practices implemented as also approaches for technology enhanced learning currently researched will be presented. You will learn how educational technologies provide human learning process support, implement guidance and recommendation, create personalised learning support, as also give real-time feedback and support reflection of learners. In the final project you will identify a problem, design a solution based on the presented approaches and implement your own educational technology solution.	
Education Method	Lectures, Peer Reviews, Presentation and Final Report.	
Assessment	The final grade of the course consists of the following components: * Peer Reviews and midterm presentation (weighting 30%) * Final project (weighting 70%) Final grade calculation: $0.3 * \text{weekly assignments} + 0.7 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: * Weekly assignments: re-submit deliverable * Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	M.A. Migut	

CT3101	Project Management Basics	5
Responsible Instructor	Dr. Ir. R. Kuttantherappel Soman	
Instructor	M.G.C. Bosch-Rekvelde	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is an important cornerstone of the minor Project Management: from Nano to Mega. In this course the general PM principles will be taught such as: iron triangle, trade-offs, scoping, scheduling, risk management, monitoring and control, project start-up, project close-out. Also the more strategic context of a project in an organization is discussed including portfolio management and the business case of a project. Throughout, the project life-cycle acts as a basis: A. Conception B. Definition C. Execution, contracting, implementation D. Handover Trade-offs between time, cost, quality, and scope are extensively discussed, after discussing the different axes in detail: -SCOPE axis - setting the project boundaries, WBS; -TIME axis - planning & scheduling, network scheduling, critical path method (CPM), Program Evaluation and Review Technique (PERT), resource constraints; -COST axis - estimating, project cost accounting, scheduling and forecasting for costs; -QUALITY axis - risk, TQM, six sigma, other techniques. Attention is paid to both front-end development and project execution. During the project execution the execution team utilizes all the schedules, procedures and templates that were prepared and anticipated during prior phases. Unanticipated events and situations will inevitably be encountered, and the Project Manager and Project Team will have to deal with it (change management). Overall: People are key in successful project management!	
Study Goals	The course is aimed at getting to know the basics of project management and being able to apply the obtained knowledge on real life cases. More specifically, after following this course the student is able to 1. Explain the difference between projects, programs and portfolios. 2. Explain how value is developed in projects throughout the project life cycle (PLC). 3. Select and reflect upon an appropriate organisation structure for a project. 4. Construct a project planning using MS Project /PertMaster. 5. Create a risk register based upon a qualitative risk analysis. 6. Explain the role of change management techniques in project management.	
Education Method	This course will be mainly based on lectures (/guest lectures) with smaller assessed weekly assignments for elaborating on the various topics. We will also use study materials from the MOOC Management of Engineering Projects: Preparing for success!	
Course Relations	This course is part of the minor Project Management: from Nano to Mega (CT-MI-174). Also students of the minor Offshore Wind Energy follow this course. Non-minor students might participate as well, but always should contact the lecturer upfront.	
Assessment	The assessment consists of - Digital exam (half knowledge questions, half questions related to a casus) - Assignments (obligatory), further specified during lecture 1, which all should be completed before the final grade of the exam can be obtained. The course grade is based on the exam grade (but only when all assignments have been completed sufficiently).	
Tags	Business Project planning / management	
Expected prior Knowledge	No specific prior knowledge required.	
Academic Skills	This course will develop your writing and reflecting skills. And of course your project management skills!	
Literature & Study Materials	Management of Engineering Projects - People Are key. Hans L.M. Bakker and Jaap P. de Kleijn, editors. NAP - The Process Industry Competence Network. ISBN/EAN: 978-90-812162-0-3 This book can be ordered (student-price) via NAP or via the minor-staff (will be communicated in lecture 1).	
Judgement	The weekly assignments will be graded with fail - sufficient - good. All weekly assignments need to be at least sufficient: for a "fail", rework has to be done. The course grade is based on the exam grade (but only when all assignments have been completed sufficiently). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	No materials are permitted during the Exam, except Dutch-English (or vice versa) dictionaries. Also a simple calculator can be used.	
Collegerama	Yes	

EPA114A	Oral Presentation	0
Module Manager	Drs. A.E. Kam	
Instructor	Drs. P.C. Post	
Instructor	Drs. R.J. Hoffmann	
Co-responsible for assignments	Drs. P.C. Post	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	None	
Course Contents	In job advertisements for engineers applicants are always required to possess excellent communication skills. This means that they should at least be able to present their designs and research results fluently, for example during meetings with clients or with managers in their own company. A good presentation does not only require thorough preparation of content, but also good style. It takes quite a bit of skill to come across understandable for any particular audience and to stay in control of the situation. For this purpose EPA114A Oral Presentation offers students the possibility to gain experience in giving presentations under professional guidance in a small group.	
Study Goals	At the end of the course you should be able to: - Speak in public clearly and attractively without making beginner's errors; - Present complex content clearly, geared to both expert and non-expert audiences; - Create clearly structured presentations; - Make helpful visuals to support your presentation; - Deal with questions and reactions from the audience; - Evaluate your own and other people's presentations; - Show insight into theory and practice of speaking in public.	
Education Method	This course is a supportive course, connected to the EPA121A course Actor and Strategy Models. During this course we differentiate between students who have already taken a course in oral presentations during their bachelor program at Delft University of Technology and students who have not yet done so. The first three weeks of the course consist of a crash course in presentation skills for those students who have not done a course in presentation skills during their bachelor program at Delft University of Technology. This course is mandatory for those students. In the fourth week of the program a mandatory lecture will be given in which an EPA alumnus will tell about their experiences with giving presentations for consultancy clients. In week 5 all students will give a presentation about the model they have developed in the course EPA 121A. This presentation will be discussed by both the EPA121A lecturers as the presentation skills lecturers. In weeks 6 and 7 there is no oral skills presentation lecture. In week 8, all students will optionally, by subscription, have the opportunity to rehearse their final presentation for the course EPA 121A and receive formative feedback from the oral presentation skills lecturers. In week 9, all students present their final model for the EPA 121A course, which will be graded by the EPA 121A lecturers.	
Literature and Study Materials	TU Present (available on Brightspace) will be used as theoretical background for this course.	
Assessment	This course is a supportive course and therefore assessment in this course consists of formative feedback only. The presentations are not graded by the lecturers. Students will give a final presentation about the model they designed in the EPA121A course, that will be graded by the lecturers of that course.	

EPA142A	Ethics of Large System Interventions	5
Module Manager	S. Steinert	
Instructor	Dr. J.A. Annema	
Instructor	K.L.L. van Nunen	
Instructor	S. Steinert	
Responsible for assignments	S. Steinert	
Co-responsible for assignments	K.L.L. van Nunen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Large scale technical projects have potentially major impacts. They will have positive impacts; e.g., a new hydropower dam may provide affordable electricity, a new copper mine is supposed to provide profits for the owner and employment for the region, an airport extension may bring improved accessibility to a certain region, and so forth. However, in nearly all large scale technical projects will have more significant and adverse negative impacts, especially at the community level. They may form health and safety risks, potentially pollute the environment or may lead to negative social impacts such as a forced move of communities. The goal of this course is to teach the students how to identify, analyse and classifying the societal ethical and environmental impacts of large-scale projects. To this end, students will focus on the evaluation of the impacts of major technological projects while considering the ethical aspects as well as the risk analysis aspects both in evaluation and mitigation of those impacts. Risks can be quantified, qualified and/or monetarized. Four major assessment and evaluation methods will be discussed, i.e. Environmental Impact Assessment (EIA), Social Impact Assessment (SIA), Risk Analysis (RA) and Societal Cost Benefit Analysis (SCBA). A key component of the course is to scrutinise the ethical underpinnings of methods and outcomes that are associated with the assessment of large technological projects as well as with the four discussed assessment/evaluation methods.	
Study Goals	After taking the module the student will be able to: - Describe comprehensively the global technology development problems. - Distinguish and analyse first- and second-order impacts of technological initiatives, and relevant facts and perceptions. - Explain and argue the conceptual difference between goals and effects, and between scoping and screening in impact assessment. - Explain the main characteristics, requirements and structure of a number of impact assessment and evaluation methods, especially the following: - o Environmental Impact Assessment - o Social Impact Assessment - o Risk Analysis - o Social Cost Benefit Analysis - Explain and use ethical theories and tools for reflecting on the gathered impact assessment data, and for judging and making choices between alternative possibilities. - Describe and examine the (ethical) assumptions underlying the abovementioned methods and their inherent limitations. - Evaluate the scope and quality of existing impact studies and reports, identify quality gaps and poor assumptions of an existing Impact Assessment, and formulate improvements towards the permitting authorities	
Education Method	lectures, feedback sessions, group work	
Assessment	Assessment will be based on two parts that carry equal weight: 1. Group work (4 to maximum 5 students in a group) about a comprehensive impact assessment plan (consisting of writing a report). In case of an insufficient grade (below 5) for the groups work, the group will have the possibility to repair the groups work based on one feedback round with a supervisor. If sufficiently repaired, the repaired grade can become maximum a 6. Groups that have failed and want a higher grade than a 6 have to do the groups work again next academic year with a new topic. 2. An individual exam. Students must have a 5.0 or higher for both the group work grade and the exam grade and the average must be a rounded up 6.0 (at least 5.8) or higher to pass the course.	

EPA143A	Macro-economics for Policy Analysis	5
Module Manager	Dr. S.T.H. Storm	
Co-responsible for assignments	Dr. C.W.M. Naastepad	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	None	
Summary	Economics is about how we work, what we produce, how we interact, cooperate and compete, and how we distribute and ultimately use what we have produced. Because the economy is too important to be left to the economists, this course provides an introduction to economics that explains how a market economy works and develops, why it often does not work as well as we would like, how its macro-economic performance can be improved by regulation and (fiscal, monetary or other public) policy. Particular attention is given to conflicts between achieving different goals of macroeconomic policy at the same time and to different theoretical (modelling) approaches. The course reviews: the neoclassical macro model; the (post-)Keynesian macro model; the IS-LM model; the multisector input-output model; the model of a balance-of-payments constrained developing economy; and a model of a technologically dual economy. The course does not assume that the student has any prior knowledge of economics.	
Course Contents	To assess the macroeconomic impacts of policy economists use models of how the economy is supposed to work and how major actors (consumers; producers; employees; government) are supposed to take decisions. Such models are always based on prior theoretical assumptions. This course reviews the major macro-economic theories as well as their operationalization in terms of (applied) macro and CGE models. Theoretical approaches reviewed include: Neoclassical; Keynesian; input-output modelling; IS-LM models. We assess the model assumptions, mechanisms and the policy conclusions. Keywords: Macroeconomics: aggregate demand and aggregate supply; circular flow of incomes; national accounting; neo-classical and Keynesian economics; unemployment; fiscal and monetary policy; money creation; crowding out; stabilization policies; labour market regulation; financial markets; balance-of-payments constrained growth; input-output analysis.	
Study Goals	At the end of this course, students will be able to: 1. analyze and weigh up the short-term and long-term impacts of macroeconomic policy on income, employment, productivity and competitiveness. 2. critically examine and reflect on the existing alternative macroeconomic theories, models and policy debates. 3. understand how money is created and how the financial sector works, and how this affects the real economy. 4. assess the strengths and limitations of an applied macroeconomic model analysis of a real-life policy issue. 5. productively work with an input-output model (of the world economy).	
Education Method	Lectures. Students are expected to prepare for each class by reading/studying the indicated literature.	
Literature and Study Materials	To be announced.	
Assessment	The exam is a closed-book electronic (in-person, on campus) examination of 3 hours. The on-campus exam will be made in Ans. The passing grade for the exam is 5.8. Students will be offered two opportunities to do the exam in an academic year. In case of unforeseen circumstances (such as due to COVID-19), the exam will be given in the form of an on-line (remote) open-book & timed multiple-choice examination of 3 hours. Students may review their exam answers. The exam review will be held physically on campus (and on-paper) according to the time scheduled by the lecturer(s).	
Special Information	None	
Elective	Yes	

EPA223A	Academic Writing	0
Module Manager	Drs. A.E. Kam	
Instructor	Drs. A.E. Kam	
Instructor	P. van Delft	
Instructor	M. Looij	
Co-responsible for assignments	Drs. P.C. Post	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	This course is required for EPA students.	
Course Contents	This course is integrated with EPA113A, Political Decision-making. This means that the writing assignment that you do for EPA will be supported in the EPA223A Academic Writing course. However, where EPA113A focuses more on the content of the paper in relation to political decision-making, EPA223A Academic Writing focuses on the (structure of the) argumentation, the quality and readability of the text and on the writing process itself.	
Study Goals	At the end of the course you are able to: identify and apply the relevant author guidelines for the text genre you have to produce; ask critical questions about others arguments; give constructive feedback about the text of other writers; use different argumentation types to support claims; respond to others arguments effectively; structure arguments strategically; phrase arguments convincingly and adequately; apply an appropriate writing style for the intended audience; use generative AI for text production responsibly in accordance with the TU Delft guidelines.	
Education Method	The course consists of lectures and tutorials, supporting the writing process for both EPA113A and EPA223A. Students are expected to hand in several mandatory practice assignments.	
Literature and Study Materials	Online materials about argumentation and on writing for science and engineering will be made available for the participants through Brightspace.	
Assessment	Students will receive a final mark (pass/fail) when the final assignment meets the requirements. Handing in all practice assignments is mandatory in order to obtain a final grade. Please note that the grading for EPA223A is independent from the grading for EPA113A. A pass in one course does not guarantee a pass in the other course (and the other way around).	

IDEM102	Build Your Start-up 1: Idea Validation	5
Course Coordinator	Ir. M.J.J. Buijs	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	The BYS 1 + 2 elective teaches how to build and validate the fundamental building blocks of a startup. Students start with a venture idea that slowly gets morphed into an actual startup with paying customers. Its one of the few, if not the only, course where students actually get to implement their own solution in the market. BYS 1 focusses on idea development and validation with actual customers, industry experts and other external stakeholders by applying methods such as problem interviewing, agile design & lean startup principles to assess desirability and first hints of feasibility and viability. Key question: is the venture idea promising enough to operationalize it in Q2?	
Study Goals	Upon finishing this course, students are able to: 1. Justify their value proposition and initial customer segment using generated evidence 2. Generate evidence by means of experiments and interviews with potential customers 3. Identify and make explicit the risky assumptions and weaknesses in their startup in a validation board 4. Reflect on their actions as an entrepreneur by writing an individual reflection paper 5. Explain the potential value of their startup to potential investors by crafting and doing an investor pitch	
Education Method	In BYS courses, the emphasis is put on individual team coaching sessions, peer feedback and events where experienced entrepreneurs and experts (often course alumni) guide students pro-bono to validate their assumptions and build their idea. A mix of workshops and lectures is aimed to inspire and guide the students and forms a string of building blocks towards validating (BYS1) and operationalizing (BYS2) the venture idea. BYS courses are not ABOUT entrepreneurship. BYS courses are about DOING entrepreneurship. At BYS we tailor the journey around what your team needs at the very moment. We'll help students to Build Your Startup, one step at a time.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5659-1 Build Your Startup 1: Idea Validation.	
Assessment	The study goals of this course will be assessed by: - A Pitch of the startup and validation evidence (team assessment) (pass/fail) - A personal reflection using the EnJournal method (individual assessment) (pass/fail)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: bys-ide@tudelft.nl	

IDEM103	Build Your Start-up 2: Prototyping	5
Course Coordinator	Ir. M.J.J. Buijs	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	The BYS 1 + 2 elective teaches how to build and validate the fundamental building blocks of a startup. Students start with a venture idea that slowly gets morphed into an actual startup with paying customers. Its one of the few, if not the only, course where students actually get to implement their own solution in the market. BYS 2 focusses on operationalizing the venture idea by forecasting and experimenting to validate: Market need. Getting actually first paying customers, Go2Market strategy, marketing campaigns Technology. Creating MVPs Operations. Start operating as a real business, with actual roles and goals, setting up the supply chain Setting up prices. Key question: is there initial product-market fit that proves the venture could become a success?	
Study Goals	Upon finishing this course, students are able to: 1. Justify their value proposition and initial customer segment by (mockup or real) sales. 2. Define a project plan for further operationalisation of the startup after graduation. 3. Design & conduct experiments to research desirability, feasibility and viability of the venture. 4. Reflect on their progress as an entrepreneur & willingness to continue the journey & personal plan of action for the future by writing an individual reflection paper. 5. Show the progress of the startup & prove you are the team to do it to potential investors by crafting and doing an investor pitch.	
Education Method	In BYS courses, the emphasis is put on individual team coaching sessions, peer feedback and events where experienced entrepreneurs and experts (often course alumni) guide students pro-bono to validate their assumptions and build their idea. A mix of workshops and lectures is aimed to inspire and guide the students and forms a string of building blocks towards validating (BYS1) and operationalizing (BYS2) the venture idea. BYS courses are not ABOUT entrepreneurship. BYS courses are about DOING entrepreneurship. At BYS we tailor the journey around what your team needs at the very moment. We'll help students to Build Your Startup, one step at a time.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5659-2 Build Your Startup 2: Prototyping.	
Assessment	The study goals of this course will be assessed by: - A Pitch of the startup, plan for further operationalisation and validation evidence (team assessment) (pass/fail) - A personal reflection using the EnJournal method (pass/fail)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: bys-ide@tudelft.nl	

MOT9611	Project Entrepreneurship Thesis related	5
Module Manager	A. Boru	
Instructor	Dr. J.R. Ortt	
Instructor	Dr. L. Hartmann	
Instructor	Dr.ing. V.E. Scholten	
Instructor	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0,5/0,5/0,5/0,5	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	Students enrolled in the Master Annotation Entrepreneurship program.	
Expected prior knowledge	BEfore entering the entrepreneurship related project thesis, we expect you have passed the following courses: - MoT9610 - Entrepreneurship Basics Course - MoT9612 - Business Development Lab, short	
Course Contents	The specific content of the Project Entrepreneurship Thesis Related is to be defined together with the student and his/her supervisors in advance off the thesis period. However, a few guidelines for the structure of the work to be undertaken are: 1. The thesis theme should be directly related to the subject of the students Master Thesis. The relevance of the Project Entrepreneurship Thesis Related to the main Master Thesis should be clear; 2. The Project Entrepreneurship Thesis Related is a separate report that reflects a) the commercial and/or societal value of the technology, product or service which is subject of study in the master thesis, and b) a thorough investigation of the business model and organization by which the technology, product or service is brought to the market. The main goal is to articulate the identification and appropriation of the commercial and societal value of the research topic central to the master thesis project. This report can address several issues. Some examples: An exploration of the business opportunities of a technology or product/ service innovation and the possible partners that need to be included to bring it to commercial exploitation; An exploration of the possible market applications and product/service forms of a technology including a business model analyses of how to exploit it; An exploration of introduction strategies for a new high-tech product and the role of various business partners and stakeholders in the business model; A business model specification for a new high-tech product; A business plan for a new high-tech product; An investigation of the funding opportunities for the business model to improve and develop the technology further; 3. The project should state clearly whether it focusses on 'Starting your own Company' or 'new entrepreneurial venture projects within established organisations'; 4. The subject of the Project Entrepreneurship Thesis Related should include a reflection on the ambition of the student related to entrepreneurship; 5. In the case of a focus on 'Starting your own Company', thesis themes should be directly linked to the context of starting a new company. Ideally, the student should place himself as the entrepreneur willing to establish a start-up with the technology at hand either right after graduation or later; 6. In the case of a focus on 'new entrepreneurial venture projects within established organisations', the thesis should be directly linked to the context of an existing company. Ideally, the entrepreneurship thesis should be performed within the company; 7. Before commencing the Project Entrepreneurship Thesis Related project, students need to prepare a research proposal, including the activities and planning of work to be developed for the phases of literature review, field research, analysis and reporting. This plan should be reviewed by the Entrepreneurship supervisor. Students will be assessed on basis of this planning and its execution. A template for this plan is available from the DCE supervisors.	
Study Goals	The Project Entrepreneurship Thesis Related aims to develop and apply the skills and knowledge on entrepreneurship of students enrolled in the Master Annotation Entrepreneurship Program. The student applies all the knowledge and skills that he/she learned during the Annotation Entrepreneurship to a business problem related to the overarching Master Thesis, and shows an entrepreneurial attitude in tackling this problem. The Project Entrepreneurship Thesis Related should be seen as a complement to the overarching Master Thesis. The underlying idea is to make a bridge between the technical/ design aspects, examined in the students Master Thesis, and the business aspects of implementing the technology/product/service into the market/ society. The ability to make this connection is at the heart of technology-based entrepreneurship. The thesis, like the whole annotation program, can be focused on two main areas: Starting your own Company and new entrepreneurial venture projects within established organisations. On completing the Project Entrepreneurship Thesis Related, students should be able to: - Understand the practical business aspects of bringing a technology/ product/ service to the market (within the setting of a start-up or established firm); - Estimate a (potential) market and derive pricing strategies from this analysis. - Draw up the cost structure of a product or service and translate this into the accompanying financial statements. - Further develop their insights into what their motivations, ambitions and objectives are as entrepreneurs (in either a corporate setting or in a new start-up); - Be able to report their findings in a well-readable and attractive format, which should be easy to understand for decision makers; - Present their findings in the form of a short 'elevator pitch', so that decision makers understand the value of the concept and are motivated to know more about it.	
Education Method	The following learning methods can be applied: - Literature review associated to the thesis theme; - Field research and analysis (both on the technical and business aspects of the technology);	

Assessment	- The application of skills and knowledge acquired in the electives related to entrepreneurship, in a real context; - Critical dialogues and discussions on the findings and analysis done by the student with both the Master Thesis supervisor (for the technical aspects) and the Project Entrepreneurship Thesis Related supervisor (for the business aspects). The assessment is composed of four main elements: 1. Planning and execution of the thesis; including the necessary phases for literature review, field research, analysis and reporting. The timing for the delivery of the plan should be agreed upon with the supervisor for the Entrepreneurship Annotation Final Thesis, who will also evaluate it. The evaluation follows the following: (i) quality of the work plan (ii) ability to live up to the work plan and (iii) ability to be autonomous and having own initiative. The student should prove what he has learned about his own insights into what his motivations, ambitions and objectives with respect to entrepreneurship are. 2. Research quality of the Entrepreneurship Thesis Related; this is provided by a report and evaluated by the dedicated supervisor on the Project Entrepreneurship Thesis Related. The evaluation should follow these criteria: (i) ambition level, (ii) structure and boundaries, (iii) academic level with respect to the entrepreneurship skills taught at Delft University of Technology, (iv) originality, (v) usability, (vi) relevance and (vii) reflection on the product. Additional criteria, include: (i) the line of argument, (ii) use of English and (iii) design of the report. Criteria can vary by faculty and master program. 3. Personal reflection; the student will be judged on his ability to understand the practical business aspects of bringing a technology to the market, and to analyze real-life business problems in the field of entrepreneurship and innovation. 4 Elevator pitch and/or PowerPoint presentation given by the student at the end of the final thesis, evaluated by the supervisor. The evaluation should follow these criteria: (i) performance, (ii) ability to clearly answer to questions and (iii) usage of English language.
Elective	Yes
Category	MSc level

MS3320	The Biography of Art Objects: Understanding the Material Life Cycle of Artwork through Physical and	5
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Instructor	Dr. L. van Eijck	
Instructor	Dr.ir. D.J.M. Ngan-Tillard	
Instructor	Prof.dr. J. Dik	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	MS3350: Authenticity and Art Crime: A Challenging Case Study	
Expected prior knowledge	The basics of chemistry and physics will be briefly repeated in the class, so that the willingness to learn is required but no formal training in the fields of science or engineering.	
Course Contents	An artwork is often valued primarily on account of its attribution to a specific creator or location of origin, date or period, and provenance, i.e. not only on the basis of its physical appearance. While an attribution may provide an historical context to the work, establish quality and uniqueness, all these subjectively meaningful parameters are undermined should the work turn out to be a forgery. When authenticating art, experts eventually decide what is of cultural significance and what is not. Authenticity has concrete repercussions on the market, in the form of its influence on an artworks economic value, and in law, given the liability of experts and sellers for misattributions and the sale of forgeries. For the past two decades the art world has increasingly relied on forensics, i.e. the dedicated chemical and physical investigation of artwork to help answer questions of authenticity. Nowadays, the public museum field, but also the commercial markets such as auction houses and international art fairs involve forensic analysis in their vetting procedure. The expert opinion of the art expert (connoisseurship), the line of ownership of an artwork (the provenance), and materials analysis (forensics) are currently the three most important pillars to establish the authenticity of an artwork. The materials of artworks offer a unique insight into the making of art, covering the entire lifecycle of an object. Raw materials, their geographic and chronological origin, artist's working techniques, material decay, restoration treatments and re-use and recycling of art objects are all part of the artwork lifecycle. Understanding these aspects is key to understanding the artwork's biography, its provenance and, ultimately, the artwork's authenticity. The fundamentals of this life cycle will be presented in an introductory lecture and later illustrated throughout this course in case studies. The main focus of the introduction will be on paint and painted objects and the methods used for the investigation of historical paintings (UV, IRR, XRR). The tremendous methodological progress in non-destructive analysis and chemical and physical imaging diagnostics achieved in the past decades will be illustrated in a series of lectures, which take the variable entry level of the students into account. The lectures will start from the nature of matter (atoms, their cores and electron shells) and radiation, especially X-rays. This basic instruction will be consolidated by peer instruction in the project groups. The production and detection of X-rays and their interaction with matter will be presented and it will be explained how these are used to acquire computed tomography data. The value this non-invasive 3D technique has to gain insight into an objects lifecycle will be illustrated on pottery/ clay cuneiform tablets, stone (beads/ flints), glass (flasks and beads) and corroded metal (coins/ filigree sphere) objects. X-ray spectroscopy and diffraction will be discussed at a basic level and the use of these basic principles in MA-XRF and MA-XRD imaging will be illustrated in studies of pigments of 17th century paintings and archaeological objects from classical antiquity (Etruscan and Greek sites). The individual techniques will be concluded with an introduction to neutrons and their use in neutron radiography, tomography, gamma spectroscopy and neutron activation analysis with application examples. This series of lectures will be closed by two events: the students will visit the TU Delft campus and travel in groups to see instruments that are used for the techniques described above. Finally, in a formative test the student groups will be prepared for the final exam.	
Study Goals	After this course the student is capable to: Analyze the challenges presented by potential forgeries by interpreting different knowledge sources. Discuss the physical and chemical principles behind common analysis techniques and the build-up of artworks. Formulate the materials science questions for an art object. Propose suitable methods for investigating these questions. Interpret the acquired data to contribute to an answer in the discussion of a potential forgery.	
Education Method	Lectures Site and Laboratory visits	
Assessment	The grades of the students will be derived from a closed book, written exam.	
Department	ME Department Materials Science & Engineering	

SEN1131	Institutional Economics for Designing in Socio-technical Systems	3
Module Manager	Dr.ir. R. van Bergem	
Instructor	G. Tran	
Instructor	Dr.ir. R. van Bergem	
Responsible for assignments	Dr.ir. R. van Bergem	
Co-responsible for assignments	S. Renes	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides fundamental concepts from the field of Institutional Economics that are instrumental for the design of institutions in socio-technical systems. Relevant frameworks and theories will be elaborated in relation to the design of socio-technical systems. The fundamentals of institutional economics will be applied to concrete design challenges, like the transition to sustainable infrastructures. It will be demonstrated how to relate Institutional Economics to interdisciplinary research challenges, including engineering, political science, law and philosophy. This provides interesting perspectives to other courses in the curriculum. Typical questions that are addressed in this module include: What is the specific contribution of Institutional Economics to the design of socio-technical systems? How to frame and interpret design challenges from an Institutional Economic perspective? What is an institutional design? How can institutional designs be related to the technical design of complex systems? How to operationalize concepts from institutional economics and apply them to empirical design challenges?	
Study Goals	After the completion of the course, the student is able to: - to understand and analyse complex institutional arrangements from an Institutional Economic perspective - reflect on the possible contribution of Institutional Economics in addressing different design challenges - apply approaches from Institutional Economics to concrete design challenges - position and apply different approaches from Institutional Economics to interdisciplinary design challenges. - establish a coherent institutional design given the features of the technical system(s) embedded in their socio-political environment	
Education Method	Lectures and workshops	
Literature and Study Materials	Reader of relevant study material, including cases to be discussed in the workshops	
Assessment	On-campus: Digital exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: online: timed take-home exam (open book)	

SEN1141	Managing Multi-actor Decision-making	5
Module Manager	Prof.dr. T.A.P. Metze	
Instructor	Prof.mr.dr. J.A. de Bruijn	
Instructor	Dr. M. Leijten	
Instructor	F. Hirschhorn Zonana	
Instructor	Dr. F.S. Gürses	
Instructor	B. Wagner	
Co-responsible for assignments	B. Wagner	
Co-responsible for assignments	Dipl.-Math. B. Kasperek	
Co-responsible for assignments	Prof.dr. W.W. Veeneman	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Previous courses have focused on, first, diagnosing the complexity of systems and problems and, then, designing for systems and their institutions in the midst of this complexity. This course on multi-actor decision-making goes one step further: (reflecting on) realizing change. We focus on the actors in the system. How do they behave, individually and collectively? What does the system look like from an actor perspective? Why does intentional change often appear so hard? What limits the designability of systems? By studying characteristics of complex engineering systems (e.g. the increased importance of IT for design, the increased interdependencies between systems) as well as the specific actor characteristics (e.g. the global character of the manufacturers and supply chains, issues of plurality and inclusiveness we identify key factors that affect intentional change (i.e. the wickedness of problems, public values, strategic behaviour of actors and dynamics of the system). How do these factors affect decision-making and design and change of complex engineering systems? How can we assess the consequences of this decision-making complexity and how does this affect the potential for interventions for (re)design in these systems in deliberate and responsible ways? In this course, we identify key characteristics which affect multi-actor decision making in complex engineering designs. Furthermore, we assess how and under which conditions process managerial tactics may help to manage multi-actor decision-making with its plural rationalities and dynamics of negotiation and design.	
Study Goals	At the end of this course, students will be able to - explain why actors in networks behave as they do, and how their behaviour evolves - describe different network structures and their practical implications - assess the designability of institutions, systems, policies and technologies - argue why certain process managerial tactics might work under which conditions - create case-specific process managerial tactics to anticipate complex decision-making. - assess the quality of process designs - reflect on multi-actor decision-making in light of broader scientific and public policy trends and debates	
Education Method	Lectures Assignments Readings	
Assessment	Assessments group assignment analysing a multi-actor decision making process (1/2) individual assignments which enable students to reflect on prescribed literature (1/2)	

SEN1211	Agent-based Modelling	5
Module Manager	Dr. C.N. van der Wal	
Instructor	Dr.ir. I. Nikolic	
Instructor	Dr. O. Kammouh	
Instructor	Dr. C.N. van der Wal	
Co-responsible for assignments	Dr. O. Kammouh	
Co-responsible for assignments	Dr.ir. I. Nikolic	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course has two focus points: 1. agent-based modelling and 2. complex adaptive systems theory for ABM. Firstly, you will develop agent-based modelling skills by creating an agent-based model in a small group. This course is not meant as an introduction to basic programming, or a thorough introduction to Python/Mesa, but will rather focus on understanding the concepts of agent-based modelling and how model design choices affect the model created and how it can be used. Additional reading materials are provided for self-study in getting up to speed in basic programming, Python, and Mesa, as well as refresher exercises in the practical sessions. There will be a weekly lecture focused on agent-based modelling theory. In addition, there will be a weekly practical session focused on further developing your agent-based modelling skills. Secondly, there will be weekly lectures focusing on complex adaptive systems theory for agent-based models. Different characteristics of these systems, theoretical concepts and their applications will be discussed. Further reading will be offered to broaden your view and deepen your understanding.	
Study Goals	After following this course, the student can: 1. understand, modify and create agent-based simulations in Python with the Mesa library. 2. identify relevant properties of a system and determine whether the system can be classified as a complex adaptive system or not. 3. apply the definitions of complex adaptive system properties, as listed in the course topics, to different types of socio-technical systems and agent-based models 4. reflect on the traditional engineering systems thinking from the complex adaptive systems-perspective and understand the implications of changing the traditional perspective.	
Education Method	This course consists of a series of lectures and a modeling project. Lectures Lectures introduce the theoretical concepts and their application for agent-based models and complex adaptive systems. All the lectures form the basis for the practical sessions, as well as the knowledge for the theoretical exam. Modelling project You will be creating an agent-based model in a small group. The project deliverable consists of the simulation code and a report documenting the problem statement, model design, performed experiments and data analysis.	
Computer Use	The course involves developing and experimenting with simulations in Python with the Mesa library.	
Assessment	The final grade of the course is based 50% on the written theory exam and 50% on the modelling project (simulation + report). Both parts must be completed with a passing grade (at least 5.75) in order to pass the course.	

SEN124A	Design in Networked Systems	5
Module Manager	N.Y. Aydin	
Instructor	Dr.ir. P.W. Heijnen	
Instructor	Prof.dr. M.E. Warnier	
Responsible for assignments	N.Y. Aydin	
Co-responsible for assignments	Dr.ir. P.W. Heijnen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Non-CoSEM students who wish to follow the course as an elective are expected to have basic knowledge of mathematical subjects such as calculus, linear algebra, probability theory, and discrete maths, corresponding roughly to the level of a Bachelor engineering degree. In addition, some prior experience with programming, e.g. in Python or MatLab will be useful.	
Course Contents	In our highly connected world, networks exist in a multitude of contexts and domains, for example, in infrastructure, supply chains, transport, information, and social networks. These are socio-technical systems that consist of a large number of components and actors organized into a network. Analysing and managing such systems is a complex endeavour. This course provides an introduction to the analysis of networked systems and the design of interventions that aim to improve their efficiency. The analysis is based on graph theory, which studies structural network properties. Topics covered include, amongst other material, the following: - Network metrics - Social networks - Formation and evolution of networks - Complex networks - Optimal trees - Community detection - Epidemics	
Study Goals	Networked systems often have a high degree of complexity, both in terms of structure and in terms of their social and institutional environment. In this course, we aim to provide you with theoretical knowledge and practical skills that enable you to reason about networked systems and design mechanisms for improving their efficiency. After taking the module the student will be able to: Analyze structural properties of networked systems using graph theory. Apply a range of tools (Gephi, various Python libraries) for analysis of networked systems. Design a new mechanism that can be used as a basis for interventions in networked systems.	
Education Method	Mix of lectures, which discuss relevant examples within a CoSEM context, and labs, which provide the opportunity to apply theory discussed in lectures and the book. 4 contact hours per week.	
Course Relations	Book: Easley and Kleinberg (2010) Networks, Crowds and Markets Reasoning about a Highly connected World see: https://www.cs.cornell.edu/home/kleinber/networks-book/ for a (free) pdf of the book. The course also provides various research papers that are used to illustrate interventions. All covered material will be posted on BrightSpace.	
Assessment	The assessment consists of an exam of the covered material and a project about a relevant case. Both need to be passed, i.e. have a grade > 5.75 in order to pass the course. Both parts count for 50% towards the end grade. If you pass an individual part of the course (either exam or project) the grade stays valid for next year(s). In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: - the exam will be a timed take home exam (open book) - the assessment for the project will stay the same	

SEN1522	Electricity and Gas: Market Design and Policy Issues	5
Module Manager	Prof.dr.ir. L.J. de Vries	
Instructor	Dr. A.F. Correljé	
Responsible for assignments	Prof.dr.ir. L.J. de Vries	
Co-responsible for assignments	Dr. A.F. Correljé	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Economics of Infrastructures SPM2230 or EPA1233) or an equivalent introduction to micro economics is required.	
Course Contents	Introduction, models of liberalization, network regulation Conceptual framework of market design Gas system & actor description Electricity system & actor description Long-term development of electricity markets Long-term development of gas markets Policy options for intervention in electricity and gas markets Perspectives on long-term, sustainable energy market design Energy market modeling Electricity market simulation game.	
Study Goals	At the end of this course, students should be able to analyze energy policy issues from a technical, economic, and institutional perspective, to make informed choices regarding market design and policy options and to develop improvements to these options. In this course, the students technical, economic, legal and policy making skills are applied to the complex reality of regulation and market design in electricity and gas sectors.	
Education Method	The course makes use of interactive lectures. The reading materials are partly developed by ourselves and consist partly of relevant, recent policy-related documents. Quantitative modeling exercises are used to learn how electricity prices come about, the influence of policy instruments and renewable energy on these prices and to learn about investment decisions. An elaborate simulation game is used to apply this knowledge and gain insights in the dynamic, long-term development of an electricity market. Finally, an excursion provides in inside view in an energy company.	
Computer Use	An exercise requires the development of a simulation model. Excel may be used for this purpose. In addition, an on-line market simulation game is part of this course.	
Literature and Study Materials	Readers and other reading materials such as scientific papers will be provided through the online learning environment.	
Assessment	Digital exam (70%), modeling exercise (20%) and simulation game (10%). Each of these elements must be passed with a grade of 5,75 or higher.	
Permitted Materials during Tests	None.	
Remarks	This course overlaps in part with SET3055 and is therefore not available to M.Sc. SET students.	
Elective	Yes	
Tags	Economics Energy Policy Analysis	
Targetgroup	CoSEM Energy Track Students	

SEN1541	Sociotechnology of Future Energy Systems	5
Module Manager	Dr.ir. I. Bouwmans	
Instructor	Dr.ir. I. Bouwmans	
Instructor	Dr. A.F. Correljé	
Responsible for assignments	Dr.ir. I. Bouwmans	
Co-responsible for assignments	Dr. A.F. Correljé	
Contact Hours / Week x/x/x/x	0/0/0/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	PLEASE NOTE: This course is only open to CoSEM students. See under Target Group.	
Course Contents	This module addresses the changing technological and societal characteristics of future energy systems. Contemporary energy systems are confronted with technical and socio-economic developments, such as the climate crisis, renewable technologies, ICT-based control innovations, energy scarcity and environmental concerns. These changes will affect operational roles and responsibilities, market structures and actors interests, and business models for energy services. Moreover, they may require new forms of governance to ensure reliable operations that meet socio-economic performance criteria. The course discusses the technical and socio-economic aspects of future energy systems, dealing with issues like the integration of new forms of distributive solar PV or large-scale off-shore wind energy into existing networks, various forms of sustainable energy sources, alternative energy carriers, storage methods, and the dynamics of transition processes.	
Study Goals	At the end of this module the students know: - Which technical and societal developments can be expected in the energy sector. - How new energy technologies and societal developments may reshape operational roles and responsibilities, energy markets and business models, and governance arrangements. - How to compare and analyze future energy systems, develop plausible transition paths towards them, and estimate the potential of new energy technologies within various scenarios. At the end of this module the students can: - Formulate descriptions of the technical and societal characteristics of future energy systems. - Make a conceptual design of an energy system, given certain demands. - Formulate a balanced advice on issues in the energy sector.	
Education Method	Interactive sessions, self-study, online & class exercises, class discussion. Reading material: scientific papers.	
Assessment	Individual written exam (on the computer, on campus) consisting of essay questions. Any reference material on paper and/or in pdf format on memory stick is allowed; no internet access other than to the examination site is allowed. Sufficient participation in the online exercises (one exercise may be missed) is a prerequisite for taking the exam, but not for the resit.	
Targetgroup	This course is only for students from the MSc programme CoSEM. Students from other studies may follow the elective TPM001B (Q2).	

SEN1611	I&C Architecture Design	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	Dr. B.D. Rukanova	
Instructor	Dr. F. Kleiman	
Responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Co-responsible for assignments	G.A. De Reuver	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	· Basic knowledge of databases, including Entity-Relationship Diagram (ERD) · Basic programming skills (such as JavaScript) · Basic knowledge of Unified Modelling Language (UML) and software engineering (agile development) · Basic understanding of multi-actor system and stakeholder analyses methods This prior knowledge are part of the linkage program.	
Course Contents	More and more data is available using various sources like databases, social media or the Internet of Things (IoT). Information sharing requires the integration of these systems. The ability to execute and process data is a key capability required by public and private organizations. The large variety of heterogeneous data, applications and business processes results in a complex and fragmented landscape that need to be dealt with when architecting and engineering new systems. Path dependencies and legacy block the easy integration, whereas at the same time new technologies appear that need to be integrated in the complex systems landscape. The purpose of this course is to teach the architectural design of innovative and large-scale ICT infrastructures and services in the light of the challenges imposed by the requirements from the systems physical, economic and social environment. Emphasis will be put on the concepts and role of ICT-architecture and modelling in order to properly design ICT solutions within a multi-actor context. For this purpose application and data integration technologies and information and system quality theories will be addressed.	
Study Goals	-The student is familiar with the state-of-the-art knowledge of ICT-architecting, design and governance within the field of large-scale ICT-systems within a multi-actor context -The student is able to describe basic concepts related to architecting and designing large and complex ICT-infrastructure and service systems within a multi-actor context. -The student should master architecture theories, methods and tools with the ability to combine and switch between them when dealing with complex ICT-problems. -The student is able to structure and analyze problems and identify dilemmas arising during the design process with regard to designing large ICT-systems within a multi-actor context. -The student is able to apply system engineering and architecture-based approaches, methods and tools to deal with problems with regard to designing large ICT-systems. -The student is able to report about the use of architectural concepts, methods and tools for translating business needs into ICT-designs within constellation of public and private actors.	
Education Method	· Lectures. The lectures are aimed at providing an overview of the knowledge of this course. During the lectures exercises are given to practice to internalize the knowledge. · Guest lectures (obliged). The guest lectures are aimed at giving an overview of the state-of-the art in practice and provide insight into practical challenges when using ICT in a multi-actor domain. · Exercises During some of the lectures assignments are given to practice what is learned. · Assignments. Various assignments are given which together create the final report. This includes developing various architectural models. · Presentation. Students should present their results during a lectures. · Report. The results should be reported in a concise report including the argumentation of design choices, the resulting architecture and an evaluation	
Computer Use	Architecture modelling in Archimate tool (http://www.archimatetool.com/) BPMN modelling using Eunomia process builder (http://www.eunomia-process.com/), Bizagi (http://www.bizagi.com/) or LucidChart (https://www.lucidchart.com/)	
Literature and Study Materials	slides book (open access e-book) Nitesh Bharosa, Remco van Wijk, Niels de Winne, Marijn F.W.H.A. Janssen (2015). Challenging the chain. Governing the Automated Exchange and Processing of Business Information. ISBN: 978-1-61499-497-8 (open access). This book covers a range from governance to technical issues and integrates them. Open access book: http://ebooks.iospress.nl/book/challenging-the-chain-governing-the-automated-exchange-and-processing-of-business-information Reader providing an outline of the area and focus on some aspects which is complemented by papers which goes in more depth.	
Assessment	Group assignment is 60% of the grade. The students work together in groups of 4-6 persons. The assignment is assessed by a report which should adhere to the APA reference standards. The report cannot be updated after handing it in. Groups that will not pass will be given the opportunity to write a new report. The exam is 40% of the grade. The exam will be digital and held in a computer room at the university. Each grade should be 5.75 or higher If one part is not passed and the other part is then the other part will remain valid for the next year.	
Elective	Yes	
Tags	Design Information & Communication Modelling Prototyping Technology	
Targetgroup	SEN I&C track Information Architecture (IA) students Electives for MOT, EPA, Computer science students, and non-TBM students having the required knowledge in this field are welcome	

SEN1622	I&C Service Design	5
Module Manager	Dr. Y. Ding	
Instructor	Prof.dr.ir. G.A. de Reuver	
Instructor	Dr. Y. Ding	
Responsible for assignments	Dr. Y. Ding	
Co-responsible for assignments	Prof.dr.ir. G.A. de Reuver	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Mobile apps, AI, Internet-of-Things, Cloud Edge computing are enabling a range of new ICT services. Since these services are typically offered in ecosystems of interdependent actors, designing services that add value for users as well as ecosystem stakeholders is challenging. In this course, you learn how to design practical ICT services and develop a valorization plan. The course contains two main parts. In the first part, it is about designing innovative services that meet the needs of users. You will design service mockups and interview users to test your ideas. In the second part, it is about bringing that service idea to reality. You will design a valorization plan that specifies how to deal with external technologies, stakeholders and revenues. The course is practical and hands-on: you will create, test and plan your service ideas. At the same time, you will learn how to support design choices using relevant kernel and design theories.	
Study Goals	After the course, you are able to: - Analyze practical ICT services by applying concepts of design science research, action design research, agile development and service engineering - Evaluate practical requirements and make informed choices on supporting technologies and platforms, with specific attention to autonomous driving, healthcare IoT, AI, XR, and edge computing technologies - Evaluate an ICT-enabled service concept through semi-structured interviews - Design and illustrate a value-added service concept driven by ICT - Design a valorization plan that explicitly covers how and when to involve external stakeholders in a design process and integrate the ICT service with value network	
Education Method	Lectures, design project, peer review, presentations, guest lecture.	
Assessment	Group essay = 3 EC; Group mockup = 1.25 EC, Individual reflection document = 0.75 EC All three marks should be >5.75. Participation in peer reviewing, guest lecture and presentations is mandatory.	

SEN171A	Advanced Evaluation Methods for Transport Policy Decision-making	5
Module Manager	N. Mouter	
Instructor	Dr. J.A. Annema	
Instructor	N. Mouter	
Instructor	A.M. de Ruijter	
Instructor	M.O. de Vries	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Transport policies are almost always controversial. Several methods are applied worldwide to support decision-making in transport. Nowadays, societal cost-benefit analysis (CBA) is the most prominent and well-known transport policy appraisal methods. In this course you will learn about the newest scientific developments in these appraisal methods. Additionally, you will learn about new appraisal methods such as Participatory Value Evaluation.	
Study Goals	Describe and explain traditional evaluation methods (e.g. CBA) and non-traditional evaluation methods such as PVE. Apply a CBA and a PVE Explain and understand the pros, cons and ethical critiques of different evaluation methods for different evaluation circumstances Analyse the pros and cons of applying various evaluation methods for the assessment of a specific case study. Argue why an (combination of) evaluation methods is most applicable for a policy maker to deploy. Provide recommendations to a policy maker about further research and how to use results of the evaluation method that you suggested. Write a coherent academic essay in which arguments are underpinned with academic literature.	
Education Method	In formal lectures the methods and theories will be explained. By two exercises students will apply the methods and theories. Finally, by analysing a pre-defined real-world controversial transport policy case and at least ten scientific papers (which they have selected themselves) students will learn to critically reflect on appraisal outcomes. They have to write a concise essay to report their findings and reflections.	
Assessment	The exam is an individual essay of 3500 words (excluding appendices and list of references). (70%) A CBA assignment (15%) A PVE assignment (15%) The minimum grade for the assignments and the essay should be at least 5.0. The aggregate grade should be at least 5.75 Students can do a resit CBA assignment and PVE assignment. The essay can be repaired with 6.0 as a maximum grade.	
Elective	Yes	
Tags	Policy Analysis Transport & Logistics	

SEN1721	Travel Behaviour Research	5
Module Manager	M. Kroesen	
Instructor	Dr. E.J.E. Molin	
Instructor	M. Kroesen	
Instructor	B. Pudane	
Responsible for assignments	M. Kroesen	
Co-responsible for assignments	Dr. E.J.E. Molin	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Knowledge about basic statistics and data-analysis (probability density functions, hypothesis testing, regression) is absolutely required as prior knowledge. Without this knowledge it will be very difficult to follow this course.	
Course Contents	In this course students learn how to perform travel behaviour research and how to translate the findings from such research into recommendations for transport policy design. The students will acquire knowledge about six modelling paradigms in travel behaviour research, namely the econometric, psychological, geographical, biographical, marketing and sociological paradigm. The strengths and weaknesses of each paradigm will be discussed. In addition, three state-of-the-art methods to analyse travel behaviour (structural equation models, latent class choice models and latent class clusters models) will be treated. These methods figure prominently within the six modelling paradigms. For each method there will be a group assignment (so 3 in total). These group assignments will count for 50% of the grade.	
Study Goals	After this course the students will be able: (1) to carry out travel behavior research using three state-of-the-art statistical methods, namely structural equation modeling, latent class cluster analysis and latent class choice models. (2) to translate the study findings to recommendations for transport policy design.	
Education Method	Lectures Computer practicals Group project (4 persons)	
Computer Use	In the course the following statistical softwarepackages will be used: SPSS, Latent Gold, Apollo and AMOS	
Assessment	Knowledge about the modelling paradigms and three methods will be tested in an individual written exam in week 10, which accounts for 50% of the final grade. The group project will consist of three assignments which together account for 50% of the final grade. Both grades should be >5.75 to pass the course. Both parts can be separately passed.	

SEN1741	Innovations in Transport and Logistics	5
Module Manager	Prof. M. Comes	
Instructor	Dr. S. Balakrishnan	
Responsible for assignments	Prof. M. Comes	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The transport and logistics systems are continuously changing due to the introduction innovative concepts in vehicles, infrastructure, and services. Examples range from the introduction of the car, to e-bikes or fat bikes, drones, or high speed train (vehicles), Uber, GPS tracking, or satellite navigation (services), or charging facilities for electric driving (infrastructure). But many innovations fail. Others have undesired consequences that may undermine the resilience of the (transport) system or create unwanted cascading effects into other systems or services. Therefore, policy-makers, investors, start-ups and transport providers need to understand the chances for success and the implications of the innovations they pursue.	
Study Goals	By the end of the course, you should be able to: LO1: explain the main frameworks that describe the impact of innovations. LO2: analyse the main pathways that link transport and logistics innovations, and their societal impacts within the system (e.g., accessibility, safety, and environment). LO3: analyse the success and failure factors of candidate transport and logistic innovations using a combination of frameworks from the literature. LO4: examine the role of resilience capabilities of candidate transport and logistics innovations in reducing effects of unanticipated events. LO5: develop a conceptual model that illustrates the impacts of a candidate transport and logistics innovation.	
Education Method	Interactive teaching (frameworks, societal impacts, theories around innovation), based on material that you will receive beforehand Group projects, resulting in two presentations and a short essay Group debates	
Assessment	1. Group essay and group presentation 2. Individual exam (closed book) Each part has to be graded with at least 5.75. The final mark is composed as follows: part 1 40%; part 2 60%. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed individual assessment will be a timed open-book assignment, and it will be written at home.	
Elective	Yes	
Tags	Group work Involved Sustainability Technology Transport & Logistics	

SEN9110	Simulation Masterclass	5
Module Manager	Prof.dr.ir. A. Verbraeck	
Co-responsible for assignments	Y. Huang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Experience with a simulation environment such as Arena, Simio, SimPy, Salabim, Matlab Simulink, NetLogo, GAMA or MESA. Basic knowledge about probability theory and statistics.	
Course Contents	System Theory, Object Orientation, Discrete Event System Specification, Multi-Formalism Simulation, Distributed Simulation, Real-Time Simulation, Multi-Paradigm Simulation, Multi-Resolution Simulation, and Simulation Language selection will be the core topics of the course. After an introduction to system theory, the inner working of simulation environments will be illustrated on the basis of the DEVS, DESS, and DTSS formalisms. Then, possible integrating of the different formalisms will be shown. Several special topics will be taught, based on the latest research in simulation. This material will be illustrated in intensive and interactive courses. In addition to the lecture topics, several other simulation topics will be studied by groups of students, who will write a scientific paper, and present their findings in class. These topics can be focused on the MSc program that the students participate in; special topics to study are available for TIL, CoSEM, EPA, Computer Science, and other students. Finally, groups of students will study a simulation package in-depth and discuss the commonalities and differences with other packages. Again, the package chosen can be targeted at the MSc program of the students. TIL students can, e.g. study a package that is more aimed at logistics and transport, CoSEM students can focus on a package that is used in systems design, whereas EPA students can focus on a package used in policy analysis. Most packages can be made available for use from home.	
Study Goals	This course requires active participation and contains group work. After taking this course the student will have knowledge about: - systemic principles behind simulation, and clear understanding of key terms, such as state, time, transition, event, process, and activity; - internal working of different kinds of discrete event simulation languages and environments, with a focus on state transitions over time; - underlying theories and formalisms of discrete event simulation, such as DEVS, DTSS and DESS; - the DEVS paradigm, including state trajectories and transition functions; quantized DEVS (QDEVS); discretized time; Dynamic Structure DEVS (DSDEVS), and other DEVS variants; - transformation between simulation formalisms; the formalism-transformation graph; - object-oriented simulation environments; service-based simulation; - structure and abilities of distributed simulation; the concept of HLA to manage time and state in distributed simulations; classes of functions in a distributed simulation environment; - latest research activities in the field of simulation, with research topics like web-based simulation, real-time control using simulation, agent based simulation, interactive simulation and gaming, and simulation in special domains; - examples of successful and less successful simulation studies and the learning experiences of those studies; - important similarities and differences between simulation languages, and selection methods for simulation packages. Practical skills will have been developed with respect to: - usage of at least one simulation package that was previously unknown; - writing of a scientific paper on the topic of simulation. Insight will have been gained on: - comparing and contrasting simulation languages; - the importance of time and state relationships for any type of simulation modeling; - the formal foundations of the simulation domain.	
Education Method	Lectures about state-of-the-art simulation theory Computer use for studying one discrete-event or agent-based simulation package in-depth per group Paper writing by student groups on a special topic Lectures by student groups on a special topic.	
Computer Use	Each group of students will study and evaluate a discrete-event or agent-based simulation package. Examples are Simio, Arena, Enterprise Dynamics, Plant Simulation, AnyLogic, Jaamsim, DSOL, Salabim, MESA, GAMA, Netlogo, SimPy, and Repast Symphony. Also here, the simulation package to study can be specifically adapted to the group of students. Most packages can be made available for use from home.	
Course Relations	The course contains very little overlap with other simulation courses such as ME44305, EPA132A, EPA133A, EPA141A, TB231C / TB233B, etc.	
Literature and Study Materials	Set of scientific papers and book chapters that will be made available through Brightspace. Typically, every class will have on average 2 papers to study for the exam. The set of papers is also available as a Reader that can be obtained on paper.	
Reader	A paper-based or PDF version of the Reader is available. The content of the Reader is also electronically available in Brightspace.	
Assessment	The mark for this course will be based on the result of group assignments (paper, presentation, lab study of simulation package) and on an individual written exam. The exam will count for 50%, the group paper (including the class presentation) for 30% and the study of the simulation package for 20%. Each of the marks has to be 5.75 or higher to be included in the final mark.	
Permitted Materials during Tests	All written materials are allowed during the exam: papers, class notes, slides. Course materials are also available electronically during the exam.	
Elective	Yes	
Tags	Challenging Intensive Modelling Programming concepts Software Stochastics	
Targetgroup	Students in EPA, CoSEM, TIL, TEL, PEL, Computer Science, and any other MSc student with some prior knowledge about computer simulation.	
Extra Skills Trained	Writing a high-quality scientific paper; exploring a new simulation language / package.	

SEN9235	Game Design Project	5
Module Manager	Dr.ir. G. Bekebrede	
Instructor	Prof.dr.ir. A. Verbraeck	
Instructor	Dr.ir. G. Bekebrede	
Responsible for assignments	Dr.ir. G. Bekebrede	
Co-responsible for assignments	Prof.dr.ir. A. Verbraeck	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	The project is mandatory for the Advanced Modeling, Gaming and Design specialization, and open to students from all other faculties inside and outside the university.	
Expected prior knowledge	Bachelor degree	
Summary	The simulation game design project offers conceptual insights as well as hands on experience with simulation-gaming (SG). SG is an established field of practice with proven value for instance in the field of spatial and urban planning, ecology, engineering and design, public administration, business management, learning, research and consultancy. The staggering growth and success of the video gaming industry has triggered the interest in simulation games from paper based games to video games for learning and policy making - even more. Besides creativity and enthusiasm, there are no entry requirements. In this course, students acquire advanced knowledge in, and experience with, gaming-simulation for serious use, i.e. for policy- and decision support, organization and management and professional learning.	
Course Contents	Experienced speakers, from consultancy organizations (s.a. Accenture), game designers and universities, will give lectures on history, future, design, typology and facilitation of SG. (We will discuss the role of computers and video games, but emphasis is NOT on technology!) We will have ample opportunity to try out a number of (role-playing, board etc.) games that hold powerful messages about decision-making and management. Most of the work will be done in small teams, in which students will develop their own (non-digital) game to support a real-life or mock up case of policy/decision-making or organization and management. In a small group, you will also work on a scientific question related to game design.	
Study Goals	After completion of this course the student will have acquired knowledge and insights about: The history, backgrounds, key-concepts, formats and applications of simulation games. After completion of this course the student will be able to: 1. Design a (prototype) of a simulation game to be used for learning, research or intervention 2. Apply a game design cycle (or systematic design steps) 3. Define, conceptualize and construct the various game components 4. Facilitate simple simulation games	
Education Method	(Guest) lectures, workshops, game-play, game design, writing an essay Intensive workshops and group meetings Student group work	
Computer Use	Not likely; dependent on selected case. Possibility playing a computer game	
Course Relations	SEN9110 Simulation Masterclass (5 EC) Advanced Agent Based Modeling (5 ECTS) in4302 Building Serious Games (5 EC)	
Literature and Study Materials	To be announced (books and articles on gaming/simulation and game design). Reading materials will be made available via Brightspace.	
Assessment	Your final grade will be based on the following points: - Attendance and active participation during classes and workshops (required, no grade) - Grade for the game design assignment (group work 4-6 persons) counts for 80% of the final grade - Grade for the essay assignment (group work 2 persons) counts for 20% of the final grade - Facilitation of the warm-up game (same group as game design assignment) have to be done (required, no grade) Final grade of the course is $0,8 \times \text{grade game design} + 0,2 \times \text{grade essay assignment}$, and student have to active present and facilitated a warm-up game. To sufficiently finish the course the weighted average have to be 5,75 or higher. Retake: - For practical reasons, there is no retake opportunity for the game design assignment - The essay can be partly repaired, the essay can be improved, maximum grade of the essay is a 6 - If a student can not join the assigned date for the facilitation of the warm-up game, we will look for an additional opportunity during the course weeks.	
Enrolment / Application	Registration through Brightspace enrollment	
Special Information	Geertje Bekebrede (g.bekebrede@tudelft.nl)	
Remarks	Attendance during the class meetings is required.	
Elective	Yes	
Tags	Challenging Design Group work Intensive Project	
Targetgroup	Master students of the modelling, simulation and gaming specialization(MSG) Elective course for students CoSEM, EPA, MOT, other faculties such as Architecture, Industrial Design, EWI, Civil Engineering etc.	
Category	MSc level	

SEN9720	Logistics and Supply Chain Innovations	5
Module Manager	Ir. M.W. Ludema	
Instructor	Ir. M.W. Ludema	
Responsible for assignments	Ir. M.W. Ludema	
Co-responsible for assignments	Dr. J.H.R. van Duin	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	Supply chain management is about the exchange of product, information and money between companies that are part of a supply chains or demand/supply-network. Supply chain management contributes to the competitive strength of companies and supply chains in the market place. The supply chain management co-ordinates and controls operational business value adding processes in and between companies. For a large extent supply chain management determines the position of companies in the supply chain and marketplace and the flexibility of companies to respond to market changes and determines the efficiency of the internal processes within companies, as well as the inter-company processes. Logistical and supply chain decision-making frequently relates to situations in which conflicting wishes and desires from various parts of organizations have to be brought together cohesively. Gaining insight into the performance indicators of a company, its logistical costs, customer expectations and the opportunities to improve these aspects is of vital importance for the survivability of companies. All engineers will encounter logistics and supply chain management to a greater or lesser extent in the business and institutional environment in which they will be working or in which they will be carrying out their assignments. This course covers the supply chain management function in and between companies. This course will provide the student insight in the theoretical and practical background of supply chains, its current use and expected developments and a wide set of innovations that companies look into currently. The student learns and puts into practice the basic theoretical skills of a supply chain analyst, engineer and manager and can show the contribution of relevant innovations for companies in a supply chain from a competitive perspective. Theoretical and practical knowledge will be explained and exercised by means of a set of relevant cases and a large running group assignment.	
Course Contents	This course contains logistical, supply chain and innovations in logistics and supply chain topics. Logistics Topics: - The significance of the logistics system of companies in supply chains; - Methods for analyzing logistics systems of companies; - The role of inventories in logistics systems of companies; - The role of transport between logistics systems of companies; - Connecting purchasing and customer service within companies; - Value added logistics, outsourcing and third party logistics; - Performance measurement in logistics systems; - Logistics control and information systems; - Design of logistics networks and locations for distribution centers. Supply Chain Topics: - Global trade lanes, industrial clusters and supply chains; - Strategy issues and the supply chain; - Organizational issues and supply chains; - Supply Chain Mapping, Benchmarking and the SCOR-reference model; - Physical and Virtual Value Systems (MRP-I, MRP-II, ERP and APS); - Supply Chain Value Adding Processes; - Sustainability (triple P) within the supply chain; - Lean and Agile Supply Chains; - International Collaboration (Supply Chain Partnerships & Alliances); - Supply Chain Analysis; - Supply Chain Design. Innovations in Logistics and Supply Chain Management: - The Last-Mile of Supply Chains; - Synchromodal Logistics, Synchro Supply Chains and the Physical Internet; - Customs and Trade Compliance; - Orchestration and the application of Cross Chain Control Centers (4C); - Service Logistics (Asset Management, Maintenance Management and spare parts logistics); - Greening the Supply Chain, Circularity and Closed-Loop Supply Chains; - Smart Supply Chains (Automation, the Internet of Things and AI-applications); - Supply Chain Finance (Strategic Profit Model, Product Portfolio's and Resilience); - Smart Contracts, Block Chains Technology within the Supply Chain; - Alternative Organizing, Distributed Autonomous Organization as part of Supply Chains.	
Study Goals	This course contains logistical, supply chain and innovations in logistics and supply chain topics. Logistics (Management) Topics: - The significance of the logistics system of companies in supply chains; - Methods for analyzing logistics systems of companies; - The role of inventories in logistics systems of companies; - The role of transport between logistics systems of companies; - Connecting purchasing to customer service; - Value added logistics, outsourcing and third party logistics; - Performance Measurement in logistics systems of company reporting; - Logistics control and information systems; - Design of logistics networks and locations for distribution centers. Supply Chain (Management) Topics: - Global trade lanes, industrial clusters and supply chains; - Strategy issues and the supply chain; - Organizational issues and supply chains; - Supply Chain Mapping, Benchmarking and the SCOR-reference model; - Physical and Virtual Value Systems (MRP-I, MRP-II, ERP and APS); - Supply Chain Value Adding Processes;	

- Sustainable Supply Chains;
- Lean and Agile Supply Chains;
- International Collaboration (Supply Chain Partnerships & Alliances);
- Supply Chain Analysis;
- Supply Chain Design.

Innovations in Logistics and Supply Chain Management:

- The Last-Mile of Supply Chains;
- Synchromodal Logistics and Supply Chains;
- Customs and Trade Compliance;
- Cross Chain Control Centers (4C);
- Through-Life Logistics of Durable Systems;
- Service Logistics (Asset Management, Maintenance Management and spare parts logistics);
- Circularity, Reverse and Closed-Loop Supply Chains,
- Smart (ICT, Automated and AI-powered) Supply Chains
- Internet of Things (IoT) for the Supply Chain;
- Supply Chain Finance and Resilience;
- Smart Contracts & Block Chains Technology and the Supply Chain.
- Alternative Organized Supply Chain and the application of Distributed Autonomous Organizations.

Education Method

The course contains 17-24 lectures (2 or 3 time 2 hours each week), in-class discussions about presented topics and distributed articles and cases. During the course assignment groups of 4-5 students will work on a supply chain analysis and engineering assignment. During the course student work on a draft part (A, B & C) of a final report. The final report of the course assignment groups will be finalized and feedbacked. Furthermore the course contains self-study and some group assignment feedback sessions. The draft report of the course assignment groups will be discussed with the groups prior to final hand-in. Participation will add to a successful completion of the course. Group attendance during the (intermediary) presentations and feedback sessions are mandatory.

Literature and Study Materials

Suggested Course Materials will be published at the beginning of the course, though is continuously under construction (see manual on Bright Space), relevant articles will be related to weekly lectures. Students are stimulated to add to the suggested materials for their assignment relevant additional literature.

Practical Guide

See Bright Space for the youngest course manual, literature list and assignment manual (including assessment criteria).

Prerequisites

This module builds on the knowledge gained during courses like operations management, production logistics, logistics management or equivalent courses. Students without this knowledge are advised to spend additional study time during the first 2-3 weeks of the course.

Please consult the module manager / lecturer during the first lecture or earlier.

During the first two lectures an overview of relevant operational and logistics management issues will be provided by means of intensive lectures and a few in-class assignments.

Assessment

Final deliverable and group assessment:

Students will work in groups (4-5 students) towards a large assignment (in the form of a report) in one of the following example domains:

- Transportation Equipment Supply Chain;
- Construction Supply Chain;
- After-sales service /spare parts supply chain for the maintenance of durable goods;
- Supply chain of conditioned goods for the retail market;
- Supply chain of the inter city last mile;
- Care supply chain of the Medical Equipment;
- Other issues in respect to the interest of the student.

The focus is B2B (Business to Business), rather than B2C (Business to Consumer).

A final report including analysis of the chosen developed supply chain, impact assessment of a chosen logistics and supply chain innovation(s) and a design of a feasible future supply chain. The percentages of the Part A, B, C and final results are described in the Assessment Manual.

SEN9725	Supply Chain Gaming	5
Module Manager Instructor Instructor Responsible for assignments Co-responsible for assignments Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Required for	Ir. M.W. Ludema Ir. M.W. Ludema Dr. F. Kleiman Ir. M.W. Ludema Dr. J.H.R. van Duin 0/6/0/0 2 2 2 English Required for: MOT/EPA Specialization Supply Chain Management SEN9720 (Q1) or (strict) equivalent is obligatory Free elective: The course is offered as a free elective for students of other programs that finalized SEN9720.	
Expected prior knowledge	This course builds on the knowledge gained during courses like operations management, production logistics, logistics management, supply chain management or equivalent courses. The focus will be a supply chain orientation, with the focus on City Logistics. Students have had followed the course SEN9720 Logistics and Supply Chain Innovations (first quarter). Understanding of logistic and supply chain management trade-offs is necessary to be able to understand, analyze and develop supply chain game sessions. Students without this prior knowledge have to spend additional study time prior to this course. Please consult the module manager if this is your case. During the first two lectures an overview of relevant operational, logistics and supply chain management issues and trade-offs will be discussed and to the basics of game development. By means of small in-class assignments students discuss possible serious (board/card/roll-playing/computer) game development projects and themes that can be applied in their group assignment.	
Summary	Supply chain management considers product and information exchange between companies in supply chains. It determines a company's flexibility to respond to market changes and determines the efficiency of the internal processes within that company, as well as the inter-company processes. Logistical and supply chain decision-making frequently relates to situations in which conflicting wishes and desires from various parts of supply chain organizations have to be brought together cohesively. By means of analyzing existing games and developing supply chain game sessions more insight in these conflicts can be gained. If supply chain stakeholders play games they will improve their awareness about supply chain perspectives and mechanisms and possible means of actions to improve the alignment of supply chain processes. A serious game is a suitable tool to create awareness about relevant supply chain issues. It levels the experience of participants (either novice or experienced from the field). After playing the game it contributes to discussions and facilitates the exchange of knowledge and personal experience. Short introduction to some relevant supply chain concepts, playing a game and participating in relevant discussions together are part of what is called a game session. The supply chain domain chosen for the group assignment has been studied in depth and the relevant parts of this domain analyzed in detail before the game development process. Playing games is a good learning exercise, but designing and developing helps the game developer to understand the essence of the trade-off from the real world. One of the approaches that can be applied for design and development of a serious design session is the Triadic Game Design approach of Harteveld (2011). Games generally include three worlds, namely reality, meaning and play. These worlds must be in balance to provide a serious game which will be fun to play and also to learn from. The world reality is needed so the players of the game can relate to elements of the real world, this will make experience in the game intuitive and understandable. Meaning is needed, because the game should have a learning purpose. At last the world of play is needed to make a game playable. The game should have clear goals and rules. The play factor will guide players in the right direction of the learning goals of the game. The three worlds are equally important, therefore they should be balanced. Balancing of these three worlds, is what makes designing a game session complex. During the design, development and execution of a game session the players have to gain knowledge about a certain logistics or supply chain issues and/or trade-offs or the effect of changes in their part of the supply chain relative to changes in other parts of the supply chain. This could for example be the implementation of technological innovations to improve traceability in the logistic supply chain, the optimization of warehouses, the best collaboration approach, the understanding of the benefits of synchronicity. In general playing and designing games helps to create a better awareness about logistics and the supply chain in general and certain addressed problem areas in particular. In some cases games can help to gather data for analysis and research purposes.	
Course Contents	Common trade-offs in Logistics and Supply Chain Systems: 1) Fundamentals of logistics and supply chain systems; 2) Methods for analyzing logistics and supply chain systems on multiple levels; 3) Strategic, Tactical and Operational Trade-Offs in logistics and supply chain systems; 4) Game Concepts to facilitate improvement of the awareness of trade-offs in logistics and supply chain systems. 5) Discussing trade-offs and ideas of game concepts. Game Essentials: - Sharing (board and computer) game experiences and discussing fundamentals; - Flow experiences (Mihaly Csikszentmihalyi); - Structured Domain System Analysis Game Concept Decompositions and Construction; - Triadic Game Design approach; - Design Methods (from established needs to workable results); - Playing and Learning; - Lemonade Tycoon; - Rise of Industry; - Playing MSc-Students developed games. Game Design Assignment: - Towards understanding supply chain problem situations; - Developing Game Session Objectives; - Setting Boundaries; - In Depth Analysis of the problem situation and value of possible solutions;	

Study Goals	- Setting Requirements; - Developing the Game (Session) Concept; - Generating Solutions; - Evaluating Solutions; - Game Delivery and Reporting. This course gives students knowledge and understanding in theoretical and practical issues concerning the design, development and execution of supply chain gaming sessions. The fundamentals of game design and its application area supply chain analysis and management will be discussed and practiced. Important and problematic logistics and supply chain issues and opportunities for innovations to improve the supply chain from a multi-stakeholder and multi-perspective will be explained and related to game sessions design alternatives.
	Upon completion of this course the student should be able to: - Elaborate on how companies can develop and maintain their position as an adequate supply chain member. - Elaborate on the inter-relationship between processes in the silos of business logistics systems and the need to align them. - Understand important logistics and supply chain trade-offs that are suitable to incorporate in supply chain games. - Suggested and develop supply chain (re)design directions for the future of the companies in a supply chain that need awareness between supply chain partners that can be facilitated by means of game sessions. - Demonstrate that they can decompose logistics and supply chain activities in meaningful game elements. - Understand the fundamentals of strategic game design approaches and elaborate on reality, meaning and play aspects and their relationships. - Develop a practical descriptive and prescriptive design method for game session design. - Design, development a functional supply chain game session and be able to demonstrate and assess its play-ability.
Education Method	The course contains 14 lectures (7 times, i.e.2-4 hours each week), in-class discussions about presented topics and of distributed articles, cases and reviewing and playing games. The first two course weeks will be class room lectures, the third, fourth and fifth course weeks will be a combination of class room lectures and working on the group assignment, and the sixth and seventh week will be working on the group assignment and feedback. The last lecture will be playing each others game session. At the the end of the educational period students have to hand in the developed game and accompanying report and game session manual.
Assessment	During the course assignment groups will work on the design, the development and the execution of a supply chain game sessions on a relevant supply chain issue. Assessment: 100% game design consisting of the design project progress and the final delivered supply chain game and accompanying report (including a manual). An assessment manual/sheet will be shared during the group assignment choice and agreement.

TPM001B	Sociotechnology of Future Energy Systems	5
Module Manager	Dr.ir. I. Bouwmans	
Instructor	Dr. A.F. Correljé	
Co-responsible for assignments	Dr. A.F. Correljé	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	PLEASE NOTE: As of academic year 2025-2026 this course will be DISCONTINUED. The (identical) course SEN154a in Q4 will be open as elective. Students need to have sufficient prior knowledge in the fields of energy technology, economics and policy. They should be familiar with concepts such as intermittency, Carnot efficiency, BEVs, energy vs power, network externalities, sunk costs, fixed and variable costs, ROI and NPV.	
Course Contents	This module addresses the changing technological and societal characteristics of future energy systems. Contemporary energy systems are confronted with technical and socio-economic developments, such as the climate crisis, renewable technologies, ICT-based control innovations, energy scarcity and environmental concerns. These changes will affect operational roles and responsibilities, market structures and actors interests, and business models for energy services. Moreover, they may require new forms of governance to ensure reliable operations that meet socio-economic performance criteria. The course discusses the technical and socio-economic aspects of future energy systems, dealing with issues like the integration of new forms of distributive solar PV or large-scale off-shore wind energy into existing networks, various forms of sustainable energy sources, alternative energy carriers, storage methods, and the dynamics of transition processes.	
Study Goals	At the end of this module the students know: - Which technical and societal developments can be expected in the energy sector. - How new energy technologies and societal developments may reshape operational roles and responsibilities, energy markets and business models, and governance arrangements. - How to compare and analyze future energy systems, develop plausible transition paths towards them, and estimate the potential of new energy technologies within various scenarios. At the end of this module the students can: - Formulate descriptions of the technical and societal characteristics of future energy systems. - Make a conceptual design of an energy system, given certain demands. - Formulate a balanced advice on issues in the energy sector.	
Education Method	Interactive sessions, self-study, online & class exercises, class discussion. Reading material: scientific papers.	
Assessment	Individual written exam (on the computer, on campus) consisting of essay questions. Any reference material on paper and/or in pdf format on memory stick is allowed; no internet access other than to the examination site is allowed. Sufficient participation in the online exercises (one exercise may be missed) is a prerequisite for taking the exam, but not for the resit.	
Elective	Yes	
Targetgroup	Master students	

TPM007A	Talk Like TED	3
Module Manager	Drs. W.J. Blokzijl	
Instructor	Drs. W.J. Blokzijl	
Instructor	M.J.Y. Wackers	
Responsible for assignments	Drs. W.J. Blokzijl	
Co-responsible for assignments	M.J.Y. Wackers	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	A TEDx Talk is a showcase for speakers presenting great, well-formed ideas in under 18 minutes (ted.com). To give a TEDx Talk, a speaker needs more than just a great idea; he or she needs a variety of tools. An important one is knowledge of rhetorical theories and the ability to apply them in oral presentations. In this course students learn how to design and deliver a TEDx Talk, using rhetorical strategies and other techniques needed to captivate an audience.	
Study Goals	After successfully completing the course, students will have knowledge of rhetorical theories; have insight into features of rhetorical-communicative situations (mainly presentations); have knowledge of framing techniques and to be able to apply them in presentations; be able to analyse presentations based on rhetorical theories; be able to apply rhetorical theories in presentations; be able to deliver a public TED-like talk about a scientific and/or societally relevant topic; be able to reflect on their role as a rhetorical communicator.	
Education Method	Classes have a workshop approach, in which the relevant literature is discussed and presentation skills are practised. Students will read theory and literature on various aspects of rhetorical theory. Each class, a new aspect of rhetorical theory will be addressed and applied to examples from (everyday) speech practice. Besides, students work on and practise the TED-like talk they will deliver at the end of the course.	
Literature and Study Materials	Literature will be available on Brightspace	
Prerequisites	Students can only participate after successfully completing a course in Oral Presentations, offered by the Centre for Languages and Academic Skills (Dutch: Instituut voor Talen en Academische Vaardigheden). If you have completed a course elsewhere, check with the coordinator whether you can participate (Wim Blokzijl, w.j.blokzijl@tudelft.nl).	
Assessment	The final grade will be a weighted average of grades for the final presentation and for the final reflection report. Students performance and efforts during classes (e.g. contributions during discussions and prepared presentation assignments) may be taken into account to round off the final grade.	
Elective	Yes	
Category	MSc elective	

TPM010A	Cyber Crime Science	5
Module Manager	Dr. R.S. van Wegberg	
Instructor	Dr. R.S. van Wegberg	
Instructor	F.E.G. Miedema	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Introduction to crime science (i.e., how do scientists study crime), - Socio-technical research skills (i.e., research design, data collection, data analysis), - Studying cybercrime from a socio-technical perspective (e.g., evaluate the effectiveness of phishing detection).	
Study Goals	The student will acquire: A good understanding of the theoretical principles of crime science, An appreciation of the whole spectrum of different cybercrimes, Skills necessary to empirically research cybercrimes.	
Assessment	1. Research proposal (groups of 3 students) pass/no-pass; 2. Final conference grade - Research paper - Presentation	

TPM018A	English Grammar for the University	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/2/0/2	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The course is designed for students who want to improve their English grammar with the aim of enhancing their writing skills. We use a university grammar that addresses the limitations many students face when interacting in English. The focus of this course is on improving your grammar range and accuracy to help you express complex thoughts and ideas more clearly.	
Study Goals	During this course, students will: 1. expand their basic understanding of sentence structure in academic discourse 2. learn how to use punctuation in academic writing 2. demonstrate an understanding of more complex grammatical structures 3. write a variety of sentence types needed to write an academic text 4. write paragraphs to practise targeted grammar structures 5. demonstrate an understanding of grammar through text analysis of academic articles	
Education Method	This is a 7-week course with a weekly 90-minute class. Homework typically involves grammar theory as well as text and written assignments. In-class activities tend to focus on text analysis and writing assignments.	
Literature and Study Materials	Grammar for Academic Purposes 2 - Steve Marshall.	
Practical Guide		
Prerequisites		
Assessment	Assignments and a Final Test in week 7. Students have to complete weekly assignments with set deadlines. Students who submit their weekly assignments late or not at all may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). In the event of a missed class, the assignments for that particular week still need to be completed. Students receive their course credits when they complete four out of five weekly assignments at pass level and pass the Final Test in week 7.	
Enrolment / Application	Enrolment is through Brightspace.	
Remarks	The course will be taught if there is a minimum enrolment of 12 students.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	Bachelor's and Master's students. No Placement Test is required to take the course.	
Category	BSc and MSc level	

TPM021A	Economics and Finance	6
Course Coordinator	Dr. C.W.M. Naastepad	
Module Manager	Dr. S.T.H. Storm	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The courses Intermediate Economics and Cost-Benefit Analysis: Theory and Applications.	
Summary	The two decades preceding the recent global financial and economic crisis saw an exceptionally strong growth of (global) financial markets, including a large (unregulated) shadow banking system. While the stock of global financial assets increased manifold, steeply raising the ratio of financial assets to world output, household and private corporate debt also showed unprecedented increases. The profitability of financial-sector firms (banks, investment firms) increased to a multiple of that of firms producing goods and services. What does this process of financialisation mean for the economy? What is the shadow-banking system and which purposes does it serve? Are high rates of financial return (for shareholders) conducive to economic growth and technological progress? Does the short-run (profit) orientation of shareholders square with the longer-term strategic interests of firms? What are the economic consequences of firms' increased reliance on stock market finance and volatile global financial markets? What is securitization? What are derivatives markets and how do they work? The recent economic crisis raises important questions concerning the principles on which our financial system is founded. From various sides proposals have been made for a restructuring of our financial system to reduce its instability. Is financial market regulation desirable? Which principles govern the origination and destination of financial capital in our current financial-economic setting? Are other principles of financial-economic organization available? These are the central questions addressed in this course.	
Course Contents	financialisation and shareholder value maximization; asset markets; private and public debts; liquidity and the liquidity trap; credit risk transfer instruments (securitization and structured finance); derivatives markets; options; interest rate swaps; credit default swaps; mortgage-backed securities (including cdos); asset price inflation; debt deflation; the global financial crisis; the crisis in the European Union; debt crisis; crisis economics; monetary financing; the future of finance; financial regulation.	
Study Goals	At the end of this course, students will be able to: analyse and interpret developments in financial markets from the point of view of financial investors, firms, consumers, governments and (central) banks understand the structural causes of the recent financial crisis including its persistence and evolution into secular stagnation interpret and assess proposals to arrest the spread of the crisis contribute to discussions about a new financial architecture that will bring transparency and stability to the financial-economic system and its composing elements.	
Education Method	Lectures and assignments.	
Literature and Study Materials	To be announced.	
Assessment	The exam is an individual closed-book electronic (in-person, on campus) examination of 3 hours. The exam is a summative test and consists of multiple-choice questions and open (essay) questions. The multiple-choice questions count for 30% of the exam grade and the open questions for 70%. The individual on-campus exam will be made in Ans. The passing grade for the exam is 5.8. Students will be offered two opportunities to do the exam in an academic year. Students may review their exam answers. The exam review will be held physically on campus according to the time scheduled by the lecturer(s). In addition to the individual exam, the student will have to do a group assignment (writing a critical book review) in a group of 4 students. The group assignment is a bonus assessment in which the student can earn a maximum of 0.5 bonus-points. The final grade is equal to the exam grade (which must be 5.8 or higher) + the (possibly obtained) bonus-points; the final grade is a maximum of 10.	
Permitted Materials during Tests	None.	
Elective	Yes	
Category	MSc level	

TPM022A	Intermediate Economics	5
Module Manager	Dr. C.W.M. Naastepad	
Module Manager	Dr. S.T.H. Storm	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Specialisation "Economics & Finance"	
Expected prior knowledge	A completed introductory course in economics (e.g. MOT1421, EPA1223)	
Summary	This course introduces students to the basic theories and models underlying macroeconomic policymaking including New-Classical and New-Keynesian economics and Keynesian economics. Special attention will be given to central bank policy, based on NAIRU economics. We will look into causes and consequences of the rapid growth of the financial sector (financialisation) in the U.S. and Europe. Special attention will be given to the causes and consequences of the Global Financial Crisis of 2008 and its aftermath. Using this theoretical understanding we turn to an analysis of the origins of the Eurozone crisis and an evaluation of the costs and benefits of expansionary fiscal consolidation (or austerity economics). What role did the financial sector play in the Eurozone crisis? Why did Germany manage to recover from the financial crisis so quickly? Why did the crisis hit the Eurozone periphery (Greece, Italy, Portugal and Spain) so hard? How can growth in Europe be restored? To understand the role of finance in all this, we need to understand what money is, what it does and where it comes from. Therefore, the course concludes with a thorough analysis of the money creation process, and the roles played by the central bank, the state and the commercial banks in this process.	
Course Contents	NAIRU; money creation; central bank behaviour; lender-of-last-resort; fiscal and monetary policy; austerity; output gap; unemployment; regulation; labour market regulation; stabilization policy; financial markets; financialisation; global financial imbalances; Eurozone crisis; competitiveness; relative unit labour cost; liberal market economies; co-ordinated market economies; secular stagnation.	
Study Goals	At the end of this course, students will: understand how money is created and what this means for central bank policy. be able to appraise crucial differences between the main economic theories and what these mean for macroeconomic policy-making. be capable of diagnosing the origins of the Global Financial Crisis of 2008 and of evaluating policy responses to this crisis. have a keen understanding of the origins of the Eurozone crisis, as well as of how difficult it is to resolve this crisis. be able to identify key determinants of (technological) competitiveness.	
Education Method	Lectures and weekly exercise sets.	
Literature and Study Materials	to be announced	
Assessment	The exam is an individual closed-book electronic (in-person, on campus) examination of 3 hours. The exam is a summative test and consists of multiple-choice questions and open (essay) questions. The multiple-choice questions count for 80% of the exam grade and the open questions for 20%. The individual on-campus exam will be made in Ans. The passing grade for the exam is 5.8. Students will be offered two opportunities to do the exam in an academic year. Students may review their exam answers. The exam review will be held physically on campus according to the time scheduled by the lecturer(s).	
Elective	Yes	
Category	MSc level	

TPM025B	User-Centred Security	5
Module Manager	S.E. Parkin	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Cybersecurity attacks on organizations and services increasingly target the people who are involved, such as employees in companies or home computer users. Solutions to address this problem need to be feasible for individuals, alongside other user priorities (such as completing paid work). Without consideration of user skills or needs, solutions such as security training or browser warning pop-ups add effort and burden to the user, and encourage less secure behaviours as workarounds (such as writing down difficult-to-remember passwords to ensure system access). In this course, students will learn about the user perspective of security technologies. This will leverage key human factors concepts, around security usability and its connections to decisions in policy, planning, and technology investment. This will include how to assess a security solution from the perspective of different kinds of technology users and their tasks. By assessing the strengths and weaknesses of particular security mechanisms for users in practice (policies, training, monitoring, etc.), security implementation and management decisions can be made which better fit the context in which mechanisms are used. This can ensure long-term security which better matches the requirements of a particular user organisation or community.	
Study Goals	The student will: -Gain a sound understanding of security usability as a discipline, from assessing the context of use alongside primary tasks, to identifying the time and effort costs to users. -Obtain foundational skills in matching security technologies and processes to user abilities, motivations, and perceptions of security-related technologies. This will include examination of authentication and identity technologies, employee security training, trust in online contexts, privacy-related evaluations of personal data disclosure, etc. -Gain insights into the design of effective security technologies and their deployment from a human-centred perspective, to inform interface and policy design and ensure compliance with security expectations. This will include management of security behaviour change activities, and identifying how to position expert support to aid users, e.g., effective communication of policy, use of persuasive design in security.	
Education Method	Structured lectures, background reading, problem-driven group discussions	
Assessment	Group assignment on assigned reading material (20%); a final individual assignment is an essay with a feedback moment (80%). The group assignment can be repaired (only after a group report was first submitted) but cannot get more than 6.0 for a repair; the resit option for the individual assignment is an oral exam. To students that need to retake parts of TPM025A: you can still pass the course TPM025A with accompanying individual assignments in its old form of: Individual assignment on assigned reading material (20%); a final individual assignment is an essay with feedback moment (80%). Or, you can choose to partake in the new course TPM025B, with all new assessment; Written group assignment 20% minimum grade of 5.0 + Individual assignment 80% minimum grade of 5.0.	
Elective	Yes	
Category	MSc level	

TPM027A	Cyber Risk Management	5
Module Manager	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Dr. A.P. Afghari	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	MOTIVATION: The challenge of selecting the optimal technical, organisational, legal, and other preventive and repressive measures to reduce cyber risks to acceptable levels can only be understood in the context of the application of Cyber Risk Management. Risk Management is about analysing the relationships between threats, incidents and risks (here in the complex world of cyberspace), based on which an adequate set of countermeasures can be designed. SYNOPSIS: Risk (= the potential to losing something of value) can manifest itself in cyberspace in all kinds of ways: values at stake are financial wealth, health, physical condition (of people, materials, goods, infrastructures, etc.), well-being, reputation, privacy, trust, etc. Based on a conceptualisation of cyberspace and its various sub-domains (discussed in the project week of year 1), we introduce risk assessment approaches, both of qualitative and quantitative manner, illustrated with case studies. CONTENT: Cyberspace Risk concepts Probabilistic and adversarial risk System and attacker models Requirements and policies Risk assessment methods Qualitative and quantitative methods Measuring security and effectiveness	
Study Goals	After this course, students are able to: Understand and apply risk concepts to (possible) incidents in cyberspace Understand the theoretical principles of cyber risk management Explain the role of adversarial risk models, system models, and attacker profiles in cyber risk assessment Analyze and apply state-of-the-art cyber risk assessment methodologies to (complex, multi-step) cyber incidents Analyze the (expected) effect of measures that help to prevent the occurrence of cyber incidents Justify investments in cyber security	
Education Method	LECTURES: There are two lecture slots per week EXAM: There is a written exam at the end of the course. LANGUAGE: The course is taught in English. LECTURERS: Prof. dr. ir. Pieter v. Gelder (TUD/TPM) and guest lecturers.	
Assessment	#### Assessment Grading will be based on the individual written exam at the end of the course. Final mark of this course is determined by the grade of the individual exam (100%). The minimum grade for the individual exam to pass this course is 6.0. The grade for the individual exam is 2 years valid. The individual exam is summative. Detailed feedback on the individual exam can be reviewed by students by contacting the teaching staff of this course. #### Resit The students are allowed one re-sit per examination. It is not allowed to re-sit an examination or assignment for which they have received a pass (6,0 or higher). It is allowed to re-sit an examination which they haven't done during the first occasion. The re-sit format needs to be discussed with the teacher of the course in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit has to take place within the study year 2023-2024.	
Elective	Yes	
Targetgroup	MSc	
Category	MSc level	

TPM028B	Decision Making for Logistics and Transport Systems	5
Module Manager	M.Y. Maknoon	
Responsible for assignments	M.Y. Maknoon	
Co-responsible for assignments	Prof.dr. J. Rezaei	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students planning to enroll in this course are expected to have the following competencies: Operations Research Knowledge: A robust understanding of mathematical modeling and integer programming is required, along with familiarity with advanced problem-solving and theoretical concepts. Technical Skills: Proficiency in using Gurobi optimization software and Python programming is crucial, as these tools are extensively utilized throughout the course. Additional Guidance Students who do not meet these prerequisites are strongly encouraged to undertake preparatory measures. Recommended resources for self-improvement include online tutorials for Python and introductory courses on integer programming and optimization software. For more tailored advice or specific resource recommendations, students should contact the course coordinator.	
Course Contents	Transport and logistics systems are pivotal to economic health and societal well-being but face challenges from sustainability, economic efficiency, and environmental impacts. These systems are complex socio-technical entities where technology intersects with various social aspects: policy, organizational structures, and cultural factors, and where diverse stakeholders often have conflicting interests. This course highlights the importance of sophisticated decision-making models that incorporate advanced data analytics and operations research. These models are crucial for understanding and managing the intricate dynamics among stakeholders and system components. Course Objectives. By the end of this course, students will: Understand and frame transport and logistics challenges as decision-making problems. Utilize mathematical modeling to formulate these challenges. Validate and apply models to address real-world scenarios, enhanced through detailed case studies. Course Content: The curriculum is structured around decision-making at various planning levels, essential for effective transport and logistics management: Strategic Planning: Focuses on long-term network design and policy development with an emphasis on sustainability and macro-level efficiency. Tactical Decisions: Bridges strategic objectives and operational needs through medium-term planning, including resource allocation and service planning. Operational Management: Manages daily operations to ensure efficient, reliable service delivery, focusing on resource deployment and protocol adjustments. Real-Time Decisions: Employs quick, responsive decision-making to address immediate changes and unexpected challenges, ensuring system integrity and continuity.	
Study Goals	Upon completion, students will: Understand and apply decision-making frameworks to effectively manage transport and logistics challenges. Formulate and solve complex logistical problems using advanced analytical and modeling skills. Develop comprehensive solutions that integrate multiple perspectives, ensuring technical soundness, sustainability, and stakeholder acceptance.	
Education Method	This course employs an interactive teaching approach designed to engage students actively and deepen their understanding of decision-making models in transport and logistics. Students are expected to fully engage with all aspects of the course, a practice essential for a comprehensive learning experience and success in the field. The key educational components include: Lectures: Deliver foundational knowledge and theoretical insights, crucial for understanding complex decision-making in the field. Practice Questions: Reinforce learning through integrated questions that challenge students to apply concepts in varied contexts. Case Studies: Enable practical application of theoretical knowledge by tackling real-life scenarios, which enhances problem-solving skills. Tutorials: Facilitate the solving of problems and explanation of situations, showcasing the practical application of theoretical models to clarify complex concepts and boost problem-solving abilities.	
Assessment	This course's assessment consists of the following components: (1) Assignments	

	Students shall form groups of up to three members to complete two assignments. These assignments are designed to reinforce foundational concepts introduced in the course. Each assignment constitutes 15% of the final grade. The average of both assignments must be a minimum of 5.8 out of 10 to meet passing requirements. Should the average score of the group assignments fall below 5.8, an additional assignment will be offered in the fourth quarter (Q4). Successfully completing this assignment ensures a passing mark of 5.8 for the assignments component. (2) Written Examination The course includes a closed-book written examination designed to evaluate the students comprehensive understanding and analytical skills. To pass the written examination, a minimum score of 5.8 out of 10 is required. Reviews of the examination will be made available in the third quarter (Q3) following the announcement of final grades, offering students an opportunity to understand their performance. Should students not achieve a passing score on the written examination, a retake opportunity will be provided in Q3. Final Grade Calculation and Passing Criteria The final grade is determined as follows: Assignments collectively contribute 30% (15% per assignment). The written examination contributes the remaining 70%. The formula for calculating the final grade is: Final Grade=(0.15×Grade of Assignment 1)+(0.15×Grade of Assignment 2)+(0.70×Grade of Written Examination) Required minimum scores are as follows: Assignments: A combined average of at least 5.8 out of 10. Written Examination: A minimum score of 5.8 out of 10. Important Note on Assessment Validity Please be aware that the results from the assignments and the written examination are valid only for the current academic year. If students do not pass the assessments after all provided opportunities two attempts for the assignments and two for the written examination these results cannot be carried over to the next academic year. Failure to meet the course requirements within this timeframe will require students to re-enroll and complete all assessments again in a subsequent academic term. Elective Tags Yes Algorithmics Artificial intelligence Business Complex Mathematics Economics Mathematics Optimalisation Programming Stochastics Transport & Logistics Category MSc level
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TPM029A	Responsible Innovation	3
Module Manager	A.R. Gammon	
Responsible for assignments	C. Clare Shelley-Egan	
Responsible for assignments	A.R. Gammon	
Co-responsible for assignments	C. Clare Shelley-Egan	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	This course is part of the master specialisation Emerging Technology-based Innovation and Entrepreneurship	
Course Contents	Technology has become a main determinant of the quality of life of individuals and the quality of society. New technologies contribute to human well-being, but they may also introduce considerable risks to humans, the environment and future generations. We therefore have every reason to ascertain that the new technologies we develop as a society respect the values we hold dear. Various governments, companies and research funding agencies have now recognized this need for responsible innovation. Responsible innovation can, as a concept, be understood in a more substantive and in a more procedural sense. As a procedural notion, responsible innovation refers to a process of innovation that meets certain procedural norms, like anticipation, reflexivity, responsiveness and inclusiveness. As a substantive notion, responsible innovation refers to a process of innovation which results in certain products, i.e. innovative technologies, which reflect important moral values. This includes values like, health, safety, human welfare, sustainability, justice, inclusiveness, democracy, privacy, trust, and autonomy. In this course we will discuss the concept of responsible innovation and how it might be understood. We will discuss tools the approach of Value Sensitive Design (VSD) as a way to put in practice the more substantive notion of Responsible Innovation. We discuss each of the four procedural dimensions: anticipation, reflexivity, responsiveness and inclusiveness. For each, we will look at possible approaches, possibilities, and pitfalls. Lectures The following topics will be covered in the lectures: - The notion of Responsible Innovation (RI) - Notions of moral responsibility and for individuals and corporations/organizations - The approach of value Sensitive Design (VSD) - The four procedural dimensions of RI: anticipation, reflexivity, responsiveness and inclusiveness - How stakeholder publics are formed and how to engage publics in participatory processes for RI	
Study Goals	After this course, students should be able to: Provide and justify specific definitions of the notion of responsible innovation Explain the approach of Value Sensitive Design (VSD) and apply it to concrete cases and examples Relate responsible innovation to moral notions of responsibility Identify opportunities and challenges for responsible innovation in concrete cases with respect to the four dimensions of responsible innovation: anticipation, reflexivity, responsiveness and inclusiveness Develop strategies for implementing responsible innovation Work in groups to develop, over the duration of the course, a group project/case on responsible innovation and reflect on this process of group working and learning	
Education Method	Weekly class meetings, preparation of mini-lectures and required readings, and group work on projects, including in-class presentations.	
Literature and Study Materials	All readings are posted on Brightspace.	
Assessment	There is no exam for this course. Instead, assessment takes the form of several required assignments throughout the course. These are described and the weights of each assignment are indicated below. (A) 25% - weekly quizzes (in class, no option to retake missed quizzes. Best 5 out of 6, graded individually) (B) 15% - participation (engagement in class meetings, feedback on other groups' presentations, graded individually) (C) (graded per small group: 4-5 students) 20% - small group presentation 30% - final presentation (group poster) & writeup 10% - stakeholder feedback Final Grades The final grade is calculated: $((A) \times .25) + ((B) \times .2) + (\text{small group presentation} \times .2) + (\text{final group presentation \& writeup} \times .3) + (\text{stakeholder feedback} \times .1)$ To pass the course, the final grade has to be 5.75 or higher. All partial grades have to be 5.0 or higher. There is no resit because there is no exam. However, partial grades are valid for A and B and can be carried from one year to the next if students do not pass or complete all parts of the course. (Parts of C cannot be carried over: this means that students need to complete all parts of C in one year.) Carrying grades into future years is only allowed for insufficient or incomplete grades, not to improve passing grades.	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Philosophy Sustainability	
Targetgroup	Master students EPA, CoSEM and MOT.	

Category	MSc level
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TPM030A	Introduction to Cloud as Infrastructure: The effects of the new business of computing on practice	5
Module Manager	Dr. F.S. Gürses	
Instructor	Dr. F.S. Gürses	
Co-responsible for assignments	Dipl.-Math. B. Kasperek	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	0/0/0/2x2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Some understanding of software engineering or computer science is desirable.	
Course Contents	Why is there a rush to AI? What is the role of cloud computing in the multi-trillion dollar valuation of tech companies? Are mobile phones really personal devices? How has the way software is produced today changed the theory and practice of computing? And, what are the implications of the technical breakthroughs pushed by big tech companies (Google, Apple, Facebook, Amazon and Microsoft) for our societies? These are some of the questions we will ask during this course. The thesis of this course is that contemporary computational infrastructures (CI [0]) present an environment for software production that has become a key driver of political and economic developments globally. With computational infrastructures we refer to cloud + mobile devices as their accessories and by software production, we refer to the creation of economic value through the engineering of software based services. In this course, we will study how CI, developed and controlled by a handful of corporations in the business of computing, is vying to become the default environment for a movement that will transform relationships of production across the globe. To make things concrete, we will look at how digital policy on privacy and data protection attempts to but falls short of responding to the societal impact of computational infrastructures. The pivotal role that these computational infrastructures aim to play in production has great societal consequences, for example, in Europe. In an opinion piece published in the Financial Times on May 12, 2023, the president of France, Emmanuel Macron underlined the importance of re-industrialization of Europe to ensure its "strategic autonomy" while transitioning to a green and digital economy [1]. By that he refers to "taking back control of our supply chains, energy and innovation". However, a commitment to bringing back industrial production to mainland Europe, when digitalization requires outsourcing key aspects of production to US or Chinese based technology companies poses a fundamental challenge to this industrial strategy. The ambition to create a "European cloud", or replicating the cloud model in any other region of the world, lacks an understanding of the historical and economic underpinnings of current day computational infrastructures. In order to equip you with the ability to assess their global impact, the course will give you insights into how these computational infrastructures came to be and provide you with knowledge to better understand where their future political, economic and technical challenges may lie. [0] A mainframe, server rack, or even a personal computer can serve as any entity's computational infrastructure. In this course, however, we will use the term CI to refer to the dominant and corporate form of materially organizing computing for the development and global delivery of digital services, for example AWS by Amazon, Azure by Microsoft, or mobile devices offered by Apple or Google. Using popular terms, CI is cloud infrastructures plus end- or mobile devices as their accessories. [1] https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/european-industrial-strategy_en	
Study Goals	In this course you will discover how the way we produce software impacts how society is organized by reaching the following learning-objectives: Demonstrate a basic understanding of the history of computing through exemplary shifts in software production Develop an understanding of how corporate computational infrastructures serve as a powerful environment for software production Analyze how privacy or data protection by design are weakened in their protective role due to the rise of computational infrastructures Express strengths and weaknesses of current EU digital policy vis a vis computational infrastructures Speculate and predict possible futures of software production in CI and its ambition to transform global economic production	
Education Method	Lectures, work groups, case studies, presentations, self-study, writing.	
Assessment	10% engagement 40% 5 assignments 50% oral exam The total of the assignments, engagement as well as the oral exam must be graded with at least 5.0. The final grade consists of the average of the three parts (engagement, assignments and exam), and the rounded grade must be at least 6.0 (5.75 rounded up).	
Elective	Yes	
Tags	Artificial intelligence Business Computer Engineering Economics Energy & Industry Finance Group work Industry Information & Communication Software Software Engineering Sustainability Technology Telecommunication Transport & Logistics Water management	
Category	MSc level	

TPM033A	Sustainable Innovation and Transitions	3
Module Manager	Dr.ir. J.N. Quist	
Instructor	Dr. L.M. Kamp	
Co-responsible for assignments	Dr. L.M. Kamp	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	TPM specialisation on Emerging Technology-based Innovation and Entrepreneurship	
Course Contents	The course Sustainable Innovation and Transitions takes as starting points: (1) that technology and innovation on the one hand and society and users on the other hand mutually influence one another and evolve in a co-evolutionary way; (2) that sustainable innovations, as well as system innovations and transitions towards sustainability, are strongly needed to bring about sustainability, and; (3) that stakeholder involvement and participatory intervention instruments are needed and required to enable and realise implementation in a socially responsible way. The course evolves around the concepts of Functions of Innovation Systems, Strategic Niche Management, Socio-Technical Transitions, social innovation and sustainable business models. These frameworks and instruments can be applied to sustainable or social innovations, which can be a technical innovation (e.g. recycling, EVs, biogas), an organisational innovation (e.g. car sharing), or a bottom-up grassroots innovation (e.g. energy cooperatives, urban agriculture) in both industrialised and developing countries. In addition, intervention instruments like transition management and backcasting are part of the course. The course consists of two parts: (1) A theoretical part consisting of lectures, readings and discussions, which is assessed by a written exam. (2) A group assignment focusing on a self-chosen topic. Examples of systems that can be studied include regional eco-industrial parks, organic agriculture, circular economy systems and renewable energy innovation systems in both industrialised and developing countries.	
Study Goals	By the end of the course students have: A. Knowledge of (i) major system innovation theories & transition theory; (ii) frameworks of Functions of Innovation Systems, Social Innovation, Strategic Niche Management; (iii) main concepts for sustainable business models; (iv) Major multi-actor intervention instruments at the level of socio-technical systems, such as transition management, and backcasting. B. Understanding of: strengths and limitations of the above mentioned theories, frameworks, tools and instruments. C. Experience & skills in the application of the frameworks to a self-chosen topic in a step-by-step group assignment. Academic skills that are trained in the course include (i) development of research questions & proposal; (ii) writing coherent, well-structured and well argued academic reports; (iii) dealing with user and social aspects of innovation, and (iv) stakeholder analysis and plurality in stakeholder perspectives.	
Education Method	Lectures, tutorials, group work on innovation analysis, presentations, individual assignment,	
Literature and Study Materials	Readings consist of selected journal papers and book chapters on the topics relevant to the course; a reading list of articles and chapters will be provided.	
Assessment	Summative assessment consists of: (1) an individual open book written exam using open questions (that counts for 50%), (2) a group project on a chosen sustainable or social innovation (also counting for 50%). (3) A bonus assignment of 0.8 point on the exam grade is possible (till a maximum exam grade of 10.0), consisting of weekly reading assignments and the peer review of draft work of another group. Formative assessment takes place through peer review of draft work of another group as part of the group assignment. The final grade is calculated as follows: (1*written exam + 1*bonus assignment+1*group project)/2. Final grades are expressed by means of a figure between 1 and 10, rounded to the nearest half. Both exam (complemented with the bonus assignment point) and group project must be completed with at least 5.0 and the final grade must be at least or higher than 5.75. Only the final grade will be entered in OSIRIS. Please register for the exam in OSIRIS on time (see: https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams).	

TPM041A	Diversity and inclusion in Science, Technology, Engineering and Mathematics	3
Module Manager	Dr. L. Asveld	
Instructor	K. Maheshwari	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In this course, you'll delve into the vital topics of diversity and inclusion that are highly relevant to your future career, whether you're heading into scientific research or the development of technologies that have the potential to shape and impact society. Throughout the course, you'll gain insights into diversity and inclusion issues specifically pertaining to the fields of Science and Engineering, along with their real-world implications. You'll also have the opportunity to engage in critical discussions and reflections, drawing from examples or your own personal experiences. W In our classroom setting, you'll be encouraged to actively participate and bring your unique perspective to the discussions. Furthermore, you'll acquire knowledge about cutting-edge approaches and methods for fostering inclusivity within the realms of science and technology. As a group, we'll explore how inclusion-related matters can influence all of us involved in Science and Engineering, ultimately aiming to create a more inclusive research, innovation, and learning environment. This course combines traditional lectures with interactive seminar sessions, fostering an environment for thoughtful reflection on assignments and topics of personal interest.	
Study Goals	- (LO1) reflect on the societal impact of diversity and inclusion (D&I) in science and engineering a) explain issues of D&I in science and engineering, b) reflect on the role of D&I on scientific research and technology development, c) and the academic environment - (LO2) apply methods to pro-actively do inclusive science and engineering	
Education Method	The course is highly interactive, combining lectures (with many guest speakers), seminar discussions, and student presentations.	
Assessment	Graded, individually, pass or fail o Active participation during the seminar sessions (e.g. participation in group discussions, presentations, moderation of discussions, etc.). Graded, group or individually: Inclusive Science and Engineering Assignment o First a pass/fail student presentation as preliminary assessment o A final written report that will be graded A minimum of 5.5 and a pass has to be obtained in the graded assignments to pass the course	

TPM043A	Resilience of socio-technical-environmental systems	5
Module Manager	N.Y. Aydin	
Co-responsible for assignments	Dr. S.M. Copeland	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Research skills including conducting a literature review, formulating research questions, writing research reports, and peer reviewing research reports. Students have basic knowledge of quantitative research methods, including data analysis, basic programming skills and socio-technical-environmental systems.	
Course Contents	As unexpected disruptions due to climate change, societal fragmentation, geopolitical conflict, and infectious diseases are becoming more prominent, ensuring the resilience of socio-technical-environmental systems is more significant than ever. This course will provide students with the knowledge and skills to understand, assess, and enhance the resilience of various interconnected systems in the face of these challenges. The course overviews the key components of contemporary resilience thinking, which is important to assess and enhance the resilience of socio-technical-environmental systems in various domains such as water, transportation, energy, ethics, and urban. Lectures are tailored for students from various disciplines and programs to understand and apply resilience principles to their own fields: the course is designed to provide opportunities for participants to identify and address resilience challenges within their own domain. This course will incorporate guest lectures from experts and academics in various fields who have successfully applied resilience principles in their work to provide valuable insights for students. Thus, the course is structured as a blended course, consisting of in-class lectures with participants actively involved in discussions and individual and group assignments. For the course's final assessment, the participants will work in groups to develop a mini literature review on the resilience of socio-technical-environmental systems and develop an individual project report for evaluating the resilience of socio-technical-environmental systems. Students will develop case studies from their selected domain perspective. The participants will get opportunities to present and receive feedback on their assignments in class and via peer reviews.	
Study Goals	After successfully completing the course, the participants will be able to: - Describe the concepts and frameworks involved in resilience thinking and how they are applied to socio-technical-environmental systems - Discuss the strengths and limitations of resilience principles and concepts in the context of socio-technical-environmental systems and compare them and identify connections with other related concepts such as risk, robustness, vulnerability, and sustainability - Develop a real-world case study to apply the resilience metrics and assessment methodologies to selected domains - Evaluate the performance of socio-technical-environmental systems from various resilience dimensions and develop future alternatives to enhance resilience	
Education Method	The course will follow a blended education technique such as mix of lectures, guest lectures, discussions, and pre-class reading assignments about methods and techniques to evaluate resilience of socio technical and environmental systems.	
Assessment	Group assignment (20%): A report on a literature review (a mini literature review/min. 15 papers) on the resilience of socio-technical-environmental systems (Passing grade is 5.8 or higher). Group assignment: Presentation of the literature review report (10%) (Passing grade is 5.8 or higher). Individual assessment: Project Report (70%) where students will develop a case study to evaluate the resilience of socio-technical environmental systems and report their findings (Passing grade is 5.8 or higher). All three assignments must be completed with a passing grade (at least 5.8) in order to pass the course.	
Elective	Yes	
Tags	Analysis Building & Spatial Development Design Energy Ethics Group work Modelling Personal Policy Analysis Project Projects Research Methods Sustainability Technology Transport & Logistics Water management	
Targetgroup	Faculty of Technology, Policy, and Management (e.g., MSc Complex Systems Engineering and Management, MSc Engineering and Policy Analysis, MSc Industrial Ecology MSc Transport, Infrastructure and Logistics), Faculty of Architecture and the Built Environment (e.g., AMS MADE students), Civil Engineering and Geosciences (e.g., MSc Transport, Infrastructure and Logistics), Industrial Design Engineering (e.g., MSc Design for Interaction).	

TPM0637SET	Economic Policy for Sustainable Energy	4
Module Manager	Dr. E. Schröder	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge of algebra and calculus is required. In general, the typical master student in a STEM subject has mathematical competencies that exceed those required for this course. The course builds on some of the ideas covered in the SET course Economics and Regulations of Sustainable Energy Systems in Q1, especially the parts taught by Gideon Ndubuisi and Servaas Storm. We will review some of these ideas, but in general non-SET students will have to catch up.	
Course Contents	Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis? This is a course in environmental and ecological economics designed for master students with little background in economics. It introduces students to key economic concepts and theories with respect to: - consumption and production - efficiency and welfare - economic growth and sustainability - the choice of environmental policy targets and instruments - integrated assessment models	
Study Goals	After completing the course, students should be able to: - explain the main theoretical concepts used in environmental economics and how they relate to one another - explain how a market economy undervalues the environment - explain how environmental economists value the environment - apply economic theory to environmental problems such as climate change - evaluate economic policies for solving environmental problems such as climate change	
Education Method	Classes, quizzes, and exercises.	
Literature and Study Materials	The main text is: - Harris & Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor & Francis. The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.	
Assessment	The final grade is based exclusively on the performance in the final exam. In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.	

TPM0931SET	Sustainable Energy Innovation and Transitions	4
Module Manager	Dr. L.M. Kamp	
Instructor	Dr. L.M. Kamp	
Co-responsible for assignments	Dr.ir. J.N. Quist	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition. This course is the first course in the SET Economics & Society profile. It can also be followed by other MSc students as an elective. Due to overlap, is not allowed to follow this course if you have already followed course TPM033A or 4413TRIG.	
Study Goals	After this course, the student: 1. has knowledge of: * the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective. * the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability. 2. has insight in: * the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts. 3. has experience and basic skills in: * exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition. * formulating research questions. * writing a coherent, well-structured and well argued academic report. * critically reflecting on work of others and on their own work.	
Education Method	This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.	
Literature and Study Materials	Scientific articles which are made available via the TU library or via Brightspace.	
Assessment	Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Both reports must be completed with at least grade 5.0. Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.	
Elective	Yes	
Tags	Energy Energy & Industry Group work Sustainability	

TPM301C	Spoken English for International Communication (for Master students)	2
Module Manager	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/3/0/3	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	This speaking course focuses on the language and skills students need to participate successfully at the university. Students will engage in a number of activities both inside and out of the classroom - to practise and improve their English. The vocabulary focus will be on general academic language, so that after the course students can interact with more confidence in an academic setting.	
Study Goals	The aim of the course is to develop language skills needed for the university environment. You will learn phrases and words for a range of skills needed here, for example participating in discussions; rephrasing and explaining; giving an opinion, (dis)agreeing, making suggestions; describing and presenting a concept.	
Education Method	Group and pair work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	You can only take this course when you have taken our Placement Test, and received a Course Recommendation for this course.	
Assessment	We offer this Placement Test regularly. The Course Recommendation that you receive is valid for a year. This means, for example, that you can take the test in October, and sign up for a course that starts in April. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: https://www.tudelft.nl/tbm/itav/onderwijs/english-unit/placement-test Enrolment for the course (and the Placement Test) is via Brightspace. Type 'TPM301C'(or 'Placement Test') in Course search (Catalog). Make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule. You do not need to register for this course in Osiris. Please note that you must also enrol in a group to reserve a place.	
Remarks	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. The course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	MSc level	

TPM302B	Spoken English for Academic Purposes Advanced	2
Module Manager	Drs. M.A. Swennen	
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/0/3/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context. A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles. The sessions revolve around student interaction.	
Study Goals	The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information. Another study course goal is to expand and improve the students' general vocabulary.	
Education Method	Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English language skills.	
Assessment	Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. When you have taken our Placement Test you will receive a course recommendation. With this recommendation you can sign up for this course. You can only register for courses that are recommended to you, or do not require a Placement Test. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	The course will only be offered if there is a sufficient number of participants. Please note that it is not a presentation course.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc level	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure your thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English. You may not submit texts that were previously submitted for any other course. Students who submit their weekly assignments late, repeatedly miss deadlines or lack commitment may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). Students will receive their course credits when their work shows sufficient progress and weekly deadlines and all other conditions are met.	
Special Information	Enrolment is through Brightspace.	
Remarks	The course will only be offered when there is sufficient demand. If there are not enough enrolments, the course may be cancelled.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

TPM306A	Writing Academic English using AI	2
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/x/0/x	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	The Writing Academic English using AI course is designed to elevate students' writing proficiency through the strategic use of AI tools. Students learn the conventions of academic writing, such as text structure, paragraph structure, sentence structure, grammatical variation, and formal-vs-informal words. They will practise editing and proofreading with AI, focusing on improving grammar, style, and coherence. They also learn to critically evaluate the AI-generated responses. Additionally, the course will cover ethical considerations in the use of AI in writing, ensuring students understand issues related to plagiarism and data privacy.	
Study Goals	This course aims to familiarise students with academic writing conventions and show them how to use large language models to improve their writing.	
Education Method	This is a 7-week course with one lesson a week.	
Literature and Study Materials	Course material will be announced in Brightspace.	
Books	Course material will be announced in Brightspace.	
Prerequisites	Before enrolling in this course you must first take our Placement Test. When you have taken our Placement Test you will receive a course recommendation. With this recommendation you can sign up for this course. You can only register for courses that are recommended to you, or do not require a Placement Test. Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM306A' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Assessment	Continuous assessment. Participants submit various writing assignments in which they demonstrate their writing skills and their ability to critically evaluate the output given by AI tools.	
Remarks	The course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

TPM308A	Higher Advanced English for the University (for Master students)	3
Module Manager	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/0/x/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	The 14-week Higher Advanced English course is designed to enhance students' proficiency in English through an integrated approach, focusing on advanced linguistic skills and academic speaking and writing. The curriculum includes an in-depth study of complex grammatical structures and vocabulary common in academic contexts. Students will engage with a variety of texts to develop advanced reading comprehension and analytical skills. Emphasis will be placed on producing high-quality written work, such as a research proposal and article critique, and practising oral communication skills through presentations and discussions.	
Study Goals	This course aims to improve students' command of complex grammatical structures and sophisticated vocabulary, enabling precise and nuanced communication. It also looks to enhance students' speaking skills, as well as their ability to engage in discussions with clarity and confidence.	
Education Method	This is a 14-week course. We meet once a week.	
Assessment	Students should submit 6 writing assignments as well as do a group/pair presentation and student-led discussion. An 80% attendance rate is obligatory.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

TPM401A	Technology Entrepreneurship and Innovation	5
Module Manager	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Instructor	P.B.M. Vandekerckhove	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	This course offers theoretical fundamentals of entrepreneurship and technology innovation. You create a case study of an existing start-up analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an high-tech start-up to academic research on the topics of technology entrepreneurship and innovation.	
Course Contents	Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, dynamics and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to:	
	LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Describe and apply the high-tech start-up growth process. LO 4 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 5 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 6 Analyse an existing high-tech start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course.	
	Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ort, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	Enrollment via Brightspace	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft is needed.	
Category	MSc level	

TPM402A	Technology Entrepreneurship and Health	5
Module Manager	P.B.M. Vandekerckhove	
Instructor	H. Khodaei	
Responsible for assignments	P.B.M. Vandekerckhove	
Co-responsible for assignments	H. Khodaei	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	----- Due to some hick-ups, probable related to course code change, you cannot find the course in Brightspace or Timetables. This course used to carry the name MOT9614 - Basic Health Entrepreneurship. We would like you to enroll in the course TPM401A for Quarter 1. We will follow the schedule of this course, but acknowledge your interest in Health. -----	
Course Contents	This course has a focus on innovation and entrepreneurship in health care systems. Using methods and theories to analyse startups, you will gain a deep understanding of their early growth and success. You will analyse a case study of an existing health start-up and analyze its entrepreneurial journey, how the company has found an opportunity to improve healthcare and how it navigated the entrepreneurial ecosystem in healthcare. This course helps you prepare for a wide array of career in health entrepreneurship and the implementation of new solutions for health care. Throughout the course you will develop the skills to develop and act on innovative business opportunities in health systems. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies in healthcare systems and analyse the value of new technology for the various stakeholders in healthcare systems. The course is organized into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, discussions of cases, literature and assignments, and presentations.	
Study Goals	Upon finishing the course, you will be able to: 1. analyse healthcare challenges and reflect on the entrepreneurial opportunities for startups and established firms. 2. Describe and apply opportunity identification in healthcare by making use of the opportunity analysis canvas. 3. Analyse the growth process for academic spin-offs in healthcare from idea to entry in healthcare. 4. Recognize the importance of the entrepreneurial ecosystem and the main stakeholders in healthcare and how they influence and contribute to the start-up process. 5. Explain and apply methods for developing business models for new healthcare startup ventures and analyse how they can change over time. 6. Analyse an existing healthcare startup case and present your findings about the dynamics of its early growth. 7. Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing	
Education Method	The course is organized in 8 online sessions (8 weeks) following 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course.	
Assessment	The assessment is as follows: - Team: three case analyses of a startup company (30%) and a final report combining the individual cases analyses plus reflection and recommendation and presentation of the findings (20%) - Individual: Oral questions (30%) and theory quizzes (20%)	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed. All students across the various faculties of the Delft University of Technology can take this course. ==> You can directly enroll in Brightspace. ==> If you need to use the mytudelft app, you may see that you cannot register to this course. BUT that information is for the exam, which does not apply to this course, since there is no exam. Hence, you can register and enroll in Brightspace. ==> You may need approval from your exam committee, when you take this course within your elective space. But please register also in Brightspace and we can plan the course. ==> You can always take this course as an extra course, not part of your elective list, and it will be included in your grade list.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM403SET	Technology Entrepreneurship and Sustainability	4
Module Manager	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Instructor	H. Khodaei	
Instructor	M. Kolagar Daronkola	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	No specific knowledge is required prior to entering the course.	
Course Contents	This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course. This course is the third course in the SET Economics & Society profile.	
Study Goals	Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to: 1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing; 2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems; 3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products; 4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones; 5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular; 6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.	
Literature and Study Materials	Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted on Brightspace-site. Slides presented in class: Will be posted on Brightspace-site.	
Assessment	The assessment is composed of several elements: 1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade. 2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade. 3. Individual: Theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Targetgroup	Master students from all master programs can apply.	

TPM404B	Technology Entrepreneurship and Internationalization	5
Module Manager	H. Khodaei	
Instructor	J. Schmutzler	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship. Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to: LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Identify and analysis international opportunities across different markets. LO 4 Describe and apply the born-global start-up growth process. LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies. LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Elective	Yes	

TPM405A	Patent Law and Patent Policy	5
Module Manager	Dr. L. Hartmann	
Instructor	mr.dr. E. Visscher	
Instructor	M. Westerik	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Course Contents	Intellectual Property (IP) Rights are increasingly important in today's fast-moving, hyper-connected world. Awareness of the different types of IP is of vital importance, not only for large companies and startups, but also for university researchers and even individuals. Knowing what Intellectual Property is, who owns it, how to protect it, how to trade it, and how to enforce an IP Rights is essential to ensure freedom of operation and keep imitators at bay. In addition to the fundamentals of IP Rights, IP Law and IP Policy are central topics of this course. Whenever you create something, whether it is an invention, a book, a song, a painting or even a website, chances are that you will have to deal with Intellectual Property and the Rights that are associated with it. There are two sides of the medal: the IP Rights that you own and may have to enforce, and the IP Rights that are not yours and that you will have to deal with. This course is intended for entrepreneurial engineers in every discipline, whether it be the hard sciences (quantum computing or nanobiology) or the more creative sciences (architecture of industrial design), especially when it concerns innovative products, processes, services and design. Given the increasing demands placed by society on companies and universities to innovate and improve life and society at large, IP rights have become synonymous with successful innovation and startups. Intellectual property rights also play a key role in today's scientific community. Understanding how the legal framework affects both the science and the companies in the business of creating high-tech and innovative solutions for today's problems, is of paramount importance. In this course you will become familiar with the basics of European IP law and the most common types of IP law (including the requirements to obtain these rights). You will explore the process by which patents are granted, how patent rights can be exercised in court and how they can be used as a business tool. Modern-day IPR issues such as the patentability of computer-implemented inventions and biotech inventions will be addressed. Additionally, software protection (including open source) works and the related issues will be discussed. The topics will be presented on the basis of real-life cases. During the course, participants will analyze and present a case in which a university spin-off is confronted with a multitude of IP issues.	
Study Goals	Get to know the basics of European IP Law Understand the common types of IP law Understand the requirements to obtain IPR Understand the process of granting patents Understand how to exercise patent rights in court Understand how to use patent rights as a business modeling tool Understand patent rights in different industries	
Education Method	Lectures in class Group assignments	
Assessment	Exam Assignments	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply. PDEng students and PhD students are also welcome.	
Category	MSc level	

TPM406A	Corporate Entrepreneurship and Start-ups	6
Module Manager	Dr.ing. V.E. Scholten	
Responsible for assignments	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	This course examines how large corporations engage startups to tap into new markets, technologies and business models. You will develop knowledge and skills to analyse strategic situations of existing corporates and how they innovate through engaging startups. The course provides an eye on corporate startup engagement from both the corporate and startup perspective.	
Course Contents	Large international organizations often find themselves in a luxurious position compared to small start-ups. They have more financial capital, access to markets, and more technological knowledge. However, large international organizations are confronted with increasing external turbulence, technological advancement and globalization, which puts their luxurious position in jeopardy. At the same time the luxurious position constrains them to depart from present activity and focus on radical new innovations. Hence, corporates try to renew themselves through internal entrepreneurial behaviour, but also by initiating start-ups and acquiring start-ups to experiment with new technology. Many times, these start-ups originate in academic environments. Therefore we pay specific attention to the emergence of academic start-ups and how corporate organizations provide incubation facilities and venture capital support to engage startups to renew themselves. This course examines the various forms of corporate entrepreneurship and the relationship with corporate strategy in particular. You will develop knowledge and skills to analyze strategic situations and design the most appropriate solutions in terms of corporate entrepreneurship strategy and organization.	
Study Goals	Understand the concept of corporate entrepreneurship; the contexts, the forms, and the relationship with strategic management Analyze strategic situations and design appropriate corporate entrepreneurship strategies and organizations Analyse the contribution of internal and external ventures in the renewal of large corporates Understand the emergence of academic spin-offs and how they relate to the technology development of large established organisations Identify the various key actors that facilitate and support the emergence of new business ventures and academic spin-offs	
Education Method	Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Each student will be expected to be present during lectures (two times a week), actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.	
Literature and Study Materials	Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Practical Guide	Given the interactive nature of this course, attendance at classes is mandatory. Only very serious grounds for absence (such as illness accompanied with doctors note) are acceptable. Holidays and (paid or voluntary) work obligations do not count as acceptable reasons for absence. This is a full-time course with scheduled holidays. Unjustified non-attendance will result in expulsion from this course.	
Assessment	The assessment includes three components. 1) The first component is a team level online presentation of a case analyses of an existing company. Each team will analyse a practical company case provided to you and based on literature of a specific lecture session you will present your findings online in class (40%) 2) The second component is a written report of a large firm in which you will analyse with your team the startup engagement activities this firm has put in place to renew itself. The report is an analyses of the company activities w.r.t. corporate venturing and an reflection of these activities by making use of literature. The report accounts for 40% of the final grade. 3) The third component is an individual analyses of a startup that is involved in the corporate startup engagement activities of your team case company. It reflects on the motivation for the startup engagement from both startup and corporate perspective. It accounts for 20% of the final grade.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course. ==> You can directly enroll in Brightspace. ==> If you need to use the mytudelft app, you may see that you cannot register to this course. BUT that information is for the exam, which does not apply to this course, since there is no exam. Hence, you can register and enroll in Brightspace. ==> You may need approval from your exam committee, when you take this course within you elective space. But please register also in Brightspace and we can plan the course. ==> You can always take this course as an extra course, not part of your elective list, and it will be included in grade list.	
Special Information	Each student team will present during the lectures, actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM407B	Technology based marketing and sales	5
Module Manager	Dr.ing. V.E. Scholten	
Responsible for assignments	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	To take this course, no prerequisite or pre-work is required. If you have questions, please contact the course coordinator, Dr. Milad Kolagar: m.kolagar@tudelft.nl Attendance: As part of our attendance policy, we may ask at least one or two students from each group of four students to be present at the sessions, depending on the total number of students. In other words, individual attendance is not mandatory, but group attendance is. Note: It is certainly possible to discuss special circumstances with the main instructor (Milad) if necessary.	
Course Contents	The Technology-Based Marketing and Sales course will help you acquire an in-depth knowledge of a specialized aspect of technology-based business development that revolves around marketing. During this rapid technological evolution, businesses face both opportunities and challenges in creating value through digital and sustainable transitions. As markets increasingly emphasize sustainability, digital technologies, and service-driven business models, marketing strategies must evolve to address these transitions effectively. This course will support and provide you with a comprehensive understanding and application of marketing principles in today's ever-changing business environments. The course focuses on developing customer-centered value propositions, crafting adaptive marketing strategies, and leveraging ecosystem-driven innovation to gain competitive advantage. Further, it explores how businesses can harness the power of digital technologies, green practices, and service-orientation in order to unlock new sources of value while addressing the growing demand for responsible and sustainable products and services. In fact, you will learn how to design and implement a marketing strategy that incorporates the latest cutting-edge trends in the industry, including digitalization, green transition, service orientation, and ecosystem-driven innovation and entrepreneurship. After this course, you will be able to : - Align your marketing efforts with evolving customer expectations - Conduct hands-on market research and identify market opportunities - Craft marketing strategies by considering long-term customer value - Develop and apply transition-oriented (e.g., digital, green) business strategies - Understand and implement service- and solution-oriented business strategies - Build ecosystem-oriented capabilities to enable value co-creation with stakeholders - Design adaptive revenue models and pricing strategies - Develop sales processes and communication plans - Create short-term and long-term plans for your business	
Course Contents Continuation	Resources: Articles and lecture sheets (which will be uploaded to Brightspace)	
Study Goals	By the end of this course, you will be capable of acting competently as a 'techmarketer' within an entrepreneurial team and improving their entrepreneurial activity system and firm's competitive advantage.	
Education Method	Lectures and workshop settings, twice a week during 8 weeks.	
Assessment	Team assessment: a marketing plan submitted as report (100%) and, which has to be defended orally. An individual portfolio to give feedback to your peer teams is part of the final report.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course. ==> You can directly enroll in Brightspace. ==> If you need to use the mytudelft app, you may see that you cannot register to this course. BUT that information is for the exam, which does not apply to this course, since there is no exam. Hence, you can register and enroll in Brightspace. ==> You may need approval from your exam committee, when you take this course within your elective space. But please register also in Brightspace and we can plan the course. ==> You can always take this course as an extra course, not part of your elective list, and it will be included in grade list.	
Elective	Yes	
Category	MSc level	

TPM408A	Financing Technology Ventures	5
Module Manager	Dr. A. Giga	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Course Contents	This course focuses on financing of technology ventures and innovation from three different points of view that of an entrepreneur, a professional investor, and a policy-maker. From the entrepreneurs point of view, we will discuss financing strategies for different business models at the different stages of startup formation and examine the entrepreneurial finance industry: the sources of startup funding (from crowdfunding and venture capital to governments and corporations), motivations of entrepreneurs and investors, and the contracts between them (e.g. convertible note, equity deals, etc.). We also explore latest developments in entrepreneurial finance in more depth (e.g. crowdfunding, venture studios). Also, we will explore the basics of corporate governance important for startups (i.e. ownership, vesting schedule, control, and legal structure). From the professional investors angle, we will be discussing tools and concepts as well as evidence and metrics that are essential when assessing startups viability and profitability at different stages. We will go in more depth with techniques used to evaluate startups (i.e. discounted cash flow, real options, comparables, etc.). Lastly, we view the startup financing from the point of view of public and social impact investors and policy-makers that want to spur entrepreneurship and innovation through financing deep technology startups, and discuss major challenges in financing ventures with long and uncertain R&D, and the macroenvironment's effect on startup financing. The class lies at the intersection of academic research and insights from practice. The course is especially suitable for master students that are interested in (or in the process of) starting or joining a technology startup, as well as for those who are curious about the process and challenges of technology venture and innovation financing. The course is highly recommended for students who are considering developing a master thesis in this area.	
Study Goals	At the end of the course, students are expected to: (1) learn how to critically assess risks, financial viability, and profitability of a technology venture; (2) understand start-up financing ecosystem, investor incentives, the trade-offs with each funding source; (3) master fundamental concepts in finance, startup valuation techniques, and key financial contracts relevant in entrepreneurial finance; (4) apply the lessons to analyze various innovation investment programs and their structures.	
Education Method	The material is illustrated mostly through lectures and in-class discussions. Depending on the availability and scheduling, we may host guest speakers to share their experience related to the course material.	
Assessment	The assessment is based on the participation and assignments, some of which are in-class, and a final report. Final report is a great first step to exploring a master thesis topics of financing innovation and technology ventures. It will let you explore, in more depth, one of the topics discussed in class. The assessment of this course is Covid-19 proof.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Tags	Business Economics Finance	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM409A	Digital technology entrepreneurship and management	5
Module Manager	J. Gartner	
Instructor	J. Gartner	
Co-responsible for assignments	H. Khodaei	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Parts	The expected workload is 5 ECTS.	
Summary	This course offers the fundamentals of digital entrepreneurship and the management of emerging (digital) technologies. In this course, participants choose an upcoming digital technology of interest and assess it with different methods from multiple angles to evaluate its ability to create a competitive advantage or disrupt an industry or market. This course helps participants prepare for various career paths, from launching their product/service as an entrepreneur to making informed technology adoption and rejection decisions as an intrapreneur.	
Course Contents	Throughout the course, participants will learn to distinguish between emerging and disruptive technologies, understand influential dynamics such as innovation diffusion, hypes and emotions, and develop the methodical competencies to assess and evaluate an emerging technology in an environment of high information asymmetry. The goal is to learn to leverage the potential of emerging and digital innovations within a specific industry.	
Study Goals	Upon finishing the course, participants will be able to: LO 1 Describe the difference between emerging and disruptive technologies. LO 2 Understand current social and market dynamics that influence the diffusion and success of emerging technologies. LO 3 Develop a critical perspective on conflicting stakeholder interests. LO 4 Analyse an emerging technology's potential for a specific market/industry/company. LO 5 Adapt a business plan to the realities of digital technology. LO 6 Make informed decisions in the context of high information asymmetries.	
Education Method	The education includes: - Lectures - Self-learning study material - Formative Group assignments and hands-on exercises to apply the understanding of relevant material (contributing to 20% of the final grade). - One individual summative assignment to learn how to analyse and create an informed recommendation for action (contributing to 80% of the final grade).	
Assessment	Group work assessments Students will participate in formative group activities throughout the lectures contributing to 20% of the final grade. These activities may include minor in-class and home assignments, presentations, and collaborative projects. Depending on enrollment, group sizes will range from 2 to 6 members. To be considered successful, each assignment requires a grade of 5.0 or above. Individual assessment Students are expected to submit an individual research report at the course's conclusion, comprising 80% of the final grade. This report will be assessed based on the following criteria: - Depth and quality of analysis - Proper utilisation of theoretical concepts - Scope and quality of research and data collection - Strength and coherence of arguments and conclusions - Accuracy and comprehensiveness of the bibliography To pass, the final report must be submitted by November 10th, 2023 latest and achieve a grade of 6.0 or above. It is not possible to repeat or resubmit the report.	
Enrolment / Application	This course is part of the MOT specialisation ICT Design and Management and is also open for students from across the Delft University of Technology faculties. You can directly enrol into Brightspace. Brightspace enrollment will be activated by education support a couple of weeks before the course starts. No other means of registration, such as the app MyTuDelft, are needed.	
Special Information	Students are asked to register/enroll on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	

TPM411A	Idea to Start-up IT & AI	5
Module Manager	E. Çelik	
Instructor	E. Çelik	
Instructor	Dr. A. Giga	
Responsible for assignments	E. Çelik	
Co-responsible for assignments	Dr. A. Giga	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity using emerging digital technologies. Applications can range from mobility, cyber security to predictive maintenance etc. You will test its validity, and build a viable business model a roadmap from an idea to an innovation-driven company. On this journey, you will be coached by lecturers and learn practical insights from guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one.	
Course Contents	You will focus on information and computing technologies (e.g AI, big data, blockchain) that are transforming traditional industries and societies at large. To analyse the opportunities for new startups we focus on entrepreneurial tools such as the opportunity canvas; value proposition development and customer empathy mapping, market segmentation and value chain analyses and the initial operations to develop a minimum viable product.	
Study Goals	Develop a business Idea by identifying a relevant economic problem or unmet need that is addressed by digital solutions, and advanced information and computing technologies. Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment Analyse the competitive landscape and devise your products competitive advantage, given specific challenges of protecting IP in the digital world. Create a blueprint of your product. Identify relevant stakeholders in your startups ecosystem, outline the go-to-market strategy, and assess profitability. Understand how these strategies adjust when trying to implement your solution in well-established systems of traditional industries. Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by improving the business model. Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing	
Education Method	You will work in teams which are formed in the first sessions of the course. One aspect of a startup will be discussed each week and you will apply into your own projece immediately. You will present your work, provide peer-feedback as well and receive advise from coaches. The course ends with a jury pitch and a 4-pager of the business proposition. Team coaching will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Books	The primary textbook for this course is an organic handbook by Bill Aulet: Disciplined Entrepreneurship: 24 Steps to a Successful Startup. Wiley, 2023.	
Assessment	The assessment is based on your project. It includes class assignments and participation (40%), final business model report (30%), online pitch presentation (15%), and team evaluation (15%). Participation and Peer Evaluation grades pertain to each student. Class assignments, final business report, and pitch presentations are evaluated on a group level.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM412A	Idea to Start-up Health & Life Sciences	5
Module Manager	A. Boru	
Instructor	A. Boru	
Instructor	E. Çelik	
Instructor	P. Harish	
Responsible for assignments	A. Boru	
Co-responsible for assignments	P. Harish	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity, validate it, and build a viable business model a roadmap from an idea to a sustainable innovation-driven company. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one. During the eight teaching sessions we investigate societal challenges, how technology can address these and the way in which we can create viable entrepreneurial business opportunities. As a framework for this, we use the business planning process and students develop the essential parts of a business plan and investor pitch as to achieve this goal. Students will work in teams to develop a minimum viable business proposition. Central are the methodologies to conduct opportunity identification, market analysis, technology viability assessment, competitive positioning, team-building, product life-cycle planning, marketing strategy, sales channel analysis. Personal development is important and in the course of the program students will learn to think as an entrepreneur and act as a salesperson; a champion that has a specific eye for longevity of the business, sustaining growth and responsible innovation. Enrollment is through Brightspace (Osiris is not needed).	
Course Contents	We will make use of various methods, tools and techniques to develop a sound business plan: - Opportunity canvas - Value proposition development and customer empathy - Market segmentation - Value chain analyses - Stakeholder mapping - Competitive advantage analysis - Minimum viable product development - Key resource and capability analysis	
Study Goals	After completing this course you have a firm understanding of 1. Define the potential market segments and develop market research and segmentation for start-ups, through interaction with potential clients. 2. Develop a stakeholder map and recognise potential supporters and opponents and turn them into your favour (Include Health organizations, Insurance companies and other important players in the health care) 3. Reflect on leading a medical product design process how to manage actors and stakeholders to achieve their business goals 4. Identify the potential characteristics of your end user by developing persona through empathy map. 5. Ethics of collaboration with Doctors and Patients. 6. Design a medical product launch strategy and define steps for action for a specific National Systems 7. Define several potential launch strategies country depending on the needs of your product. (FDA, Insurance,) 8. Identify and develop your ideas into start-ups and analyze different factors to identify opportunities based on opportunity analysis canvas. 9. Identify and describe the main challenges or critical junctures that your company might face in the initial growth stages and describe the main activities that your company should follow to mitigate them. 10. Describe and apply the generic pattern of development and diffusions for your radically new high-tech products and identify the position of your innovative product in the pattern. 11. Identify your beach head market and calculate the TAM, SAM, SOM size for beachhead market. 12. Develop a stakeholder map and recognise potential supporters and opponents and turn them into your favour. 13. Conduct a competitor analysis and identify the main competitors that offer the alternatives and compare and analysis these alternatives against your proposition. 14. Describe the business model canvas that outlines the business case for your new product/service. 15. Calculate the Lifetime value of an Acquired Customer and calculate the Cost of Customer Acquisition 16. Define your initial Minimum Viable Business Product (MVP) 17. Make a first overview of your cash flow projections for the first 2-3 years based on the costs of product development and production, costs of marketing and sales effort and expected income generated	
Education Method	We will work in teams which are formed in the first sessions of the course. Topics are discussed using case studies. Students present their work, provide peer-feedback as well as receive advice from various coaches. The course ends with a jury pitch and a two-pager of the business proposition. During the course various assignments are given that will be discussed and presented in class.	
Assessment	The course has 3 grading components: - Active participation - In-class assignments - A business plan report In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed final assessment will be modified as follows: - Active participation in online lectures will be noted - the final (group) presentations will be online	

Enrolment / Application	the other submissions (in-class assignments, final business plan etc. are already online so no changes will be made. All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as osiris are needed.
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .
Elective	Yes
Targetgroup	Master students from all master programs can apply.
Category	MSc level

TPM413A	Idea to Start-up Energy & Sustainability	5
Module Manager	Dr. A. Giga	
Instructor	E. Çelik	
Instructor	Dr. A. Giga	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	E. Çelik	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	----- This course is NOT connected to the CleanTech Challenge. (A previous course Sustainable Business Study was linked to that competition, but this course is not in any way connected.) -----	
Course Contents	A sustainable business idea is a product, a process or a service that enables sustainability, where sustainability is defined as the capacity for the biosphere and human civilization to co-exist indefinitely. The course stimulates and assists students in creating a sustainable business idea and assessing the feasibility and the market potential thereof by testing it against prospective customers and key players in the specific industry of application. Students work together in teams which will be self-formed or if needed formed by the instructor with the aim to create optimal diversity. Each team needs to come up with an original sustainable startup idea, or a sustainable business idea connected to pressing sustainability challenges of our day. Each sustainable business idea must be technology-based we do not accept apps and definitely no consulting of any kind. The course consists of two interconnected parts: 1. Students attend lectures and workshops. 2. Teams carry out a feasibility study for their sustainable business idea.	
Study Goals	- Students learn how to assess the market potential of a sustainable product, process, or service. - Students gain basic knowledge on the commercialization of a sustainable business idea. - Students learn about different business models and gain insight in how to choose a market opportunity in the value chain. - Students gain practical knowledge about starting up a business. - Students experience real business practice by feedback from both lecturers, external advisers and experienced entrepreneurs in particular. - Students learn how to give a convincing business presentation. - Students learn to be critical about their own business case and those of their fellow students. - Students learn how to make a proper evaluation of the feasibility of their idea and business plan.	
Education Method	We offer a mix of lectures, case studies, and coaching sessions. The class is fast-paced and interactive. Many of the classes are in the form of coaching sessions and team exercises, therefore class attendance is mandatory. Lectures and online content will provide students with general guidelines for each business model milestone. Depending on availability, we will bring guest speakers, usually entrepreneurs, with experience in the energy and sustainability industry. Team coaching will help students to validate their ideas and develop their entrepreneurial skills.	
Literature and Study Materials	Mandatory books: ----- Aulet Disciplined Entrepreneurship: 24 Steps to a Successful Startup MacKay Sustainable Energy - Without the Hot Air Recommended books: ----- Scott Berkun The Myths of Innovation Goldstein Planetary Improvement More literature and reading suggestions will be provided during the course.	
Prerequisites	A BSc is required. Admission to Master students only.	
Assessment	The grade is based on: final group report, final presentation, midterm presentation and short progress report, peer evaluation and active participation in the course.	
Enrolment / Application	Enrol for the course on Brightspace	
Elective	Yes	
Tags	Business Economics Energy Project Sustainability	
Category	MSc level	

TPM414A	Idea to Start-up Deep Tech	5
Module Manager	Dr. A. Giga	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	E. Çelik	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Course Contents	In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity, test its validity, and build a viable business model a roadmap from an idea to a sustainable innovation-driven company. On this journey, you will be helped by guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology venture, whether you want to build or join one. While the starting point is tackling the societal and market problems and unmet needs, the solution ideas must be based on substantial scientific advances and cutting-edge engineering that can be used to build competitive advantage in the market (via patent for example). Feel free to bring your own ideas and technologies, but you do not need one to enjoy this course. All you need is curiosity about building a business on solid technological foundations. Fast-paced and interactive, the class consists of mostly coaching sessions and team exercises, therefore class attendance is very important. Online content and short in-class lectures provide you with general guidelines for each business model milestone. There will be weekly assignments, building blocks to a full-fledged business model. Final deliverables include: a logbook of primary market research (yes, youll do interviews!), business model report, and a pitch presentation in front of a jury of startup professionals. You start the course with forming teams around a market problem or unmet need. Then, you gradually build a business model, mainly by answering questions such as: Who are my customers and what value do I create for them? How big is this market? How should my minimum viable product look like? How do I build a competitive advantage in the market (e.g. by patenting)? How will I market and deliver this product to the customer? Can this be a profitable business? You seek answers through customer, stakeholder, and expert interviews, by developing step-by-step interview protocols and executing them.	
Study Goals	- Develop a business idea based on a relevant economic problem or unmet need. - Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment - Develop a blueprint of your product and map its position in a competitive landscape. - Identify relevant stakeholders in your startups ecosystem. - Formulate a business model and a go-to-market strategy. - Perform initial assessment of the profitability of the business idea. - Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. - Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by modifying the business model.	
Education Method	The course offers a mix of online lectures, case studies, and coaching sessions. Depending on availability, we will bring guest speakers with relevant industry and entrepreneurial experience. Team coaching with industry experts will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Assessment	The grade is based on: final group report, final presentation, midterm presentation and short progress report, peer evaluation and active participation in the course. Also, to pass the course, participation and regular submission of class assignments is necessary. Active participation and peer evaluation are assessed for each student, the rest is evaluated on a group level.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Tags	Business Economics Project Technology	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM415B	Idea to Start-up - Climate Action	5
Module Manager	F. Delgado Medina	
Instructor	F. Delgado Medina	
Instructor	Dr. C.A. Benitez Avila	
Responsible for assignments	F. Delgado Medina	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In the Global Climate Action course, you will work in a team to identify a start-up/challenge opportunity within the context of regeneration by challenging the as-usual buzzword narrative of sustainable entrepreneurship. You will analyze a global/regional/national/local context and look at the social, economic, environmental, political, cultural, and entrepreneurial ecosystem synergies and tradeoffs, identify a challenge, and co-develop an innovative startup idea or solution concept for tackling the framed challenge in the Global South given by our challenge owners located in regions of the world such Latin America, Africa and South-east Asia-keep in mind the challenge locations can change every year-. You will go from testing the ideas validity to building a viable business model with local partners using decolonial participatory design methods and alternative systems knowledge. In this course, you will go from an idea to a regenerative innovation-driven solution and learn the benefits and barriers of developing a business in the global/regional/national/local sustainability-related system. This course focuses on diving deep into sustainability and regenerative entrepreneurship and developing sustainable/circular business models that generate positive social and environmental impacts. You will focus on creating innovative solutions to some of todays urgencies. Transforming the value chain for unique marine biodiversity hotspots, thinking along on bio-inspired solutions, and working in rural agro-forestry farming with rural, local, and marginalized communities are some topics you will dive deep into during the course. The course narrative is built around entrepreneurial tools and alternative systems knowledge such as regenerative leadership, storytelling, skills for a new wave of change-makers, regenerative value propositions and customer and market segmentation, value chain and stakeholder analyses, and regenerative inclusive business modeling to analyze opportunities for new development-oriented start-ups and regenerative and emancipatory entrepreneurial solutions for and with communities.	
Study Goals	This course teaches students how to develop and co-design their own start-up or work on the Global South challenges related to Climate Action. Students will learn about essential entrepreneurship and apply it to the global/regional/national/local context, which requires understanding a different cultural context and designing for sustainable change. By the end of this course, you should be able to: Macro Level: Learning block 1 - Regenerative Shared Value Creation & Introducing the product to the market. LO1: Understand the concept and principles of regeneration and its importance in becoming the new transition of entrepreneurship. LO2: Define the shared value creation from an idea to a startup to boost the societal transition to regenerative systems for climate action. LO3: Analyze the competitive landscape (entrepreneurial ecosystems), stakeholders, and local communities roles and devise your products user advantage for different geographical and cultural contexts. LO4: Identify your potential market, beachhead market, and the size of the market. LO5: Identify and apply market research techniques to rigorously test and validate your business idea. Meso-level -Learning block 2 - Ideation & Air-tight Concept LO6: Deliver ideas, products, and services in response to regenerative challenges. LO7: Acquire skills and tools that can be used to facilitate a regenerative entrepreneurial process. Micro-level - Learning block 3 - The regenerative business model LO8: Explain and apply methods for creating regenerative business models for your startup idea/challenge. LO9: Analyze and identify the factors for adopting regenerative technology innovation for the specific customer segment of your startup idea/challenge.	
Education Method	A merger of lectures and guest lectures with serial and inspiring entrepreneurs and practitioners, fieldwork visits, workshop activities, group pitch deck presentations, and in-class discussions. Coaching sessions with local challenge owners-partners along the ideation process are highly encouraged and must be coordinated in advance with the partners in the Global South. The course is organized into weekly interactive lectures wherein theories of that week will be explained. Moreover, guest lectures will be given by speakers such as co-founders, experts, and nature and conservation practitioners, sharing their experiences, barriers, and learnings about starting a company. Lastly, workshops will be given during the lectures, wherein the students can apply the content to their specific start-up and work on their ideas.	
Assessment	In this course, attendance is mandatory. For excused absences, inform the lecturing team beforehand if possible. The final grade for this course consists of Final Report: 20% Story map: 40% Individual Journal: 30% Active participation (Lectures & coaching sessions): 10%	
Enrolment / Application	A minimum grade of 5.75 is required for each deliverable to pass the course. All students across the various faculties of the Delft University of Technology can take this course. => You can directly enroll in Brightspace. => If you need to use the mytudelft app, you may see that you cannot register to this course. BUT that information is for the exam, which does not apply to this course, since there is no exam. Hence, you can register and enroll in Brightspace. => You may need approval from your exam committee, when you take this course within your elective space. But please register also in Brightspace and we can plan the course. => You can always take this course as an extra course, not part of your elective list, and it will be included in grade list.	

TPM416A	Turning Technology into Business	6
Module Manager	Dr. L. Hartmann	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Turning Technology into Business (TTiB) aims to equip participants with a strong conceptual foundation to actively understand the dynamic process of technology-based entrepreneurship. Participants learn how business strategies are best formulated and how (through entrepreneurship) technology can create value. TTiB consists of seven lectures, introducing the theoretical backgrounds of technological, market and business analyses. A unique aspect of TTiB is that existing technologies (developed and patented by the TU Delft) are used as case subjects. The patent project is the focal point of the curriculum. Each group of 4-5 students will be assigned an original patent, and is expected to evaluate the commercial potential of this technology. This includes choosing promising applications (products) for the technology, and making recommendations for the most suitable business model to commercialize the technology. The patent project should provide a coherent and structured answer to the central question: which strategy is most likely to generate business from this patent? In addition to equipping participants with a thorough knowledge of the course subject, the section Technology, Strategy & Entrepreneurship (TSE) hopes that this course will be the starting point for a variety of university spin-offs in which the students will participate. To accommodate this, TSE offers the course TPM420A- Ready to Startup!	
Study Goals	This course aims to equip students with a strong conceptual foundation to an active understanding of two domains: (1) The dynamic process of technological innovation through concepts such as technology life-cycles, dominant design, disruptive technologies, Schumpeterian competition and the diffusion of innovations (2) How business strategies are formulated and, through entrepreneurship, technology can create value. This multi-faceted process of technology commercialization process is addressed in terms of assessing technology position, discovering market opportunities, competitive analysis, appropriability and the various modes of entrepreneurship.	
Education Method	7 interactive lectures, participant-centered case studies, homework assignments, classroom assignments, individual group coaching. Final Presentations in February.	
Literature and Study Materials	Will be provided during the course	
Assessment	Each group (of 4 or 5 students) writes a report on their findings, analysis and recommendations for the technology (patent) they used as a case for this course. On the Final Day, the group has to give a 10 minute oral presentation. The report makes up 80% of the final grade. The final grade is summative. This team grade will be modified according to the individual student's class contributions and his/her performance within the group (peer review). There is no resit possibility.	
Enrolment / Application	MAXIMUM CAPACITY is limited to 75 participants. Register on BrightSpace from October 1. Because of the huge popularity of this course, there is a selection procedure which consists of pre-course assignments. One of these assignments is a letter of motivation: why do you want to participate in this course? Based on the motivation and the other pre-course assignments, the final selection of 75 participants will be made.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Targetgroup	Advanced Master students, PDEng and PhD students, and staff researchers from all faculties within the TU Delft.	
Category	MSc level	

TPM420A	Ready to Start-up	6
Module Manager	P. Harish	
Instructor	Dr.ing. V.E. Scholten	
Instructor	P. Harish	
Responsible for assignments	P. Harish	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	This course is offered twice in a year; First starting week 3/4 of Q1 and ending in Q2 and second starting week 3/4 of Q3 until end Q4 Ready to Startup (RtS) is an elective master course that has been the crown jewel of entrepreneurial education at TU Delft for over two decades. Bringing together business experts and students eager to start companies, it has one simple goal: turn engineers into successful founders. RtS focuses predominantly on Business Strategy: gaining market traction, prototyping, growing the team, devising a financial plan and fundraising strategy, protecting IP, and foundational activities (setting up a company, sharing ownership, etc.) The course has the prerequisite that teams should be formed and have a business ideas that they want to further analyse and validate. It is meant for students who plan to start a company. After RtS, teams must be at the level required for YES!Delft incubation and Delft Enterprises investments.	
Course Contents		
Study Goals	At the end of the course, you will be able to - Reflect on your current business model and identify further necessary validation activities. Analyse market entry strategies and consider alternative models. - Learn how to negotiate, attract first users, and seek endorsement of stakeholders (such as investors and advisors). - Plan your core operational activities such as company foundation, IP protection, product development strategy and team development - Devise financial plans and appropriate fundraising strategies - Convey a convincing narrative of your business and be able to provide constructive feedback to other startup plans. - Demonstrate effective entrepreneurial behaviour by making swift decisions based on newly gathered information and adjusting your model accordingly.	
Education Method	The course is spread out over 12 in-person sessions on Wednesday afternoon (13h30-17h30), where the core elements of a startup plan are presented. You work on assignments that are aimed at specific elements of the core startup plan, and you will by apply them to your startup project and present your findings during the course. In this way you make evident progress with your startup. The course combines state of the art business methods, translated to the needs of starting entrepreneurs, with consults from practitioners. Topics are discussed using case studies. You will present your progress, provide peer-feedback as well as receive advice from mentor experts. The course ends with a Final Day, consisting of a jury pitch and handing in your business plan.	
Literature and Study Materials	Handouts and literature are available trough in Brightspace.	
Prerequisites	Enthusiasm, entrepreneurial spirit and a good business idea.	
Assessment	The course has 4 grading components. - Business plan: final report (40%) - Financial plan (10%) - Final Day performance (30%) - Overall progress: homework and attendance (20%) In general, you are only allowed to miss one lecture (with a valid reason submitted beforehand). Furthermore, we will monitor your progress during the course, with an official assessment halfway through the course.	
Enrolment / Application	Application process you need to register in advance to the course. Teams of students with clear ideas about a business they want to start supported with evidence that it addresses a need, will have priority to enter. ==> You can directly enroll in Brightspace. ==> If you need to use the mytudelft app, you may see that you cannot register to this course. BUT that information is for the exam, which does not apply to this course, since there is no exam. Hence, you can register and enroll in Brightspace. ==> You may need approval from your exam committee, when you take this course within your elective space. But please register also in Brightspace and we can plan the course. ==> You can always take this course as an extra course, not part of your elective list, and it will be included in grade list.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM423A	Technology Venture Development	15
Module Manager	Dr. A. Giga	
Instructor	J. Schmutzler	
Instructor	Dr. A. Giga	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	Join us in an interdisciplinary exploration of technology startups business model and strategy. This pioneering course offers you an opportunity to work with a technology-based startup from the local ecosystem or a corporate venture in an internship-like setting and help them progress. In a group of three to four students from different faculties and study programs, you will work on two projects: 1) Validate and challenge startups business model and strategy 2) Provide consulting on a specific business challenge provided by the startup. In addition, you will be coached and taught solid business theoretical foundation and practical lessons. You will apply your learnings directly, solving pressing business challenges and driving innovation. This hands-on experience allows to you communicate directly with startups key decision-makers and see the effects of your work making a difference. This course is a real-world expedition into business model innovation, designed to inspire, challenge, and prepare you for a future where you drive change. We expect you to spend around three days a week with the startup and about one day a week in the classroom setting. Given its 15 ECTS, we expect your full weekly commitment, as in any joint interdisciplinary project. The applications opens on April 30th and the application deadline will be on June 15th. Here is the link to the application form: https://forms.office.com/e/tX9hXqNM7u	
Course Contents	What to Expect Internship-like Experience: Step into the fast-paced world of startup or corporate venture, through our internship-like experience. Here, you'll gain invaluable hands-on exposure to real-world business practices, navigating challenges, and seizing opportunities alongside industry professionals. Lectures: Immerse yourself in captivating lectures that serve as the cornerstone of your theoretical understanding. Delivered by experts in the field, these sessions will unravel the intricacies of business model innovation, offering profound insights and cutting-edge perspectives. Each week delves into critical topics such as market dynamics, design thinking, team development, financial strategies, risk management, and growth mechanisms, ensuring you're equipped with a comprehensive toolkit for the entrepreneurial world. Sessions illuminate the intersection between technical innovation and business strategy, enriching your learning experience and equipping you with versatile skills for multifaceted business environments. Coaching Sessions: These sessions help you apply your newfound knowledge and skills to your cases and tackle pressing business challenges head-on, devising innovative solutions that drive tangible impact and transformation. What You'll Experience and Learn Interdisciplinary Exploration: Navigate the complex nexus of diverse disciplinary backgrounds in engineering, basic science and architecture and design, entrepreneurship, and management, transcending traditional disciplinary boundaries to craft innovative business models that redefine the status quo. Practical Application: Move beyond theoretical concepts and immerse yourself in practical, hands-on experiences. From market analysis to financial modelling, you'll apply cutting-edge techniques and frameworks to real-world scenarios, honing your problem-solving skills and business acumen. Innovation Mindset: Cultivate an entrepreneurial mindset rooted in creativity, resilience, and adaptability. Through design thinking workshops, risk management exercises, and growth strategy simulations, you'll embrace uncertainty as an opportunity for growth and innovation. Networking Opportunities: Forge meaningful connections with industry professionals, guest lecturers, and fellow classmates, expanding your professional network and gaining valuable insights into the startup ecosystem. Personal and Professional Growth: Reflect on your journey, identify your strengths, and chart a course for future success. Whether it's sharpening your communication skills or refining your leadership abilities, this course offers a transformative experience that extends far beyond the classroom. Why Choose This Elective Cutting-Edge Curriculum: Stay ahead of the curve with a curriculum that reflects the latest trends, technologies, and best practices in business model innovation. Hands-On Experience: Gain practical, real-world experience that sets you apart in today's competitive job market. Our internship-like experience and consulting project offer unparalleled opportunities for skill development and professional growth. Inspiring Faculty: Learn from esteemed faculty members who are experts in their fields and passionate about empowering the next generation of innovators and entrepreneurs. Impactful Learning: Make a meaningful impact with a course that challenges you to think critically, solve complex problems, and drive positive change in the world.	
Study Goals	After completing this course you will have learned to: - Identify and innovatively transform a business idea for a technology venture. - Conduct rigorous market analysis and validation using primary and secondary research methods. - Construct a detailed and financially viable business proposition. - Apply niche strategies to commercialize radical high-tech products. - Come up with innovative, viable, high quality recommendations that will solve real-life challenges for an existing venture, under	

	considerable time constraints. Employ relevant academic theories to support your recommendations; Effectively communicate your product and business proposition to diverse stakeholders. Emulate effective entrepreneurial behaviour in decision-making under uncertainty. Reflect on the evolution of entrepreneurial growth and business model adaptation.
Education Method	This course will entail a mix of lectures, and guest lectures. You will be provided with course materials. You will go through an onboarding process emulating a traineeship experience with the startup or corporate venture. There will be coaching sessions with experienced mentors while will offer guidelines, feedback, and tools to help you develop a solution to the consulting project.
Prerequisites	There are no prerequisites for this course.
Assessment	Your grade is determined as follows: 80% is group work and based on the quality of your group project. Quality is assessed based on your problem statement & presentation, midterm report & presentation and final report and presentation. Startup team will have a strong say on your performance. Adjustment is made through peer evaluation of your individual contribution to the team work. 20% is an individual grade based your personal learning and team reflections during the course.
Enrolment / Application	This course is open to master students across all TUD faculties starting their second year. MAXIMUM CAPACITY is contingent upon the capacity of our partners (startups and corporates) but initially limited to 30 participants. Please apply for the course no later than June 15th. If you would like to apply , fill out the application form: https://forms.office.com/e/tX9hXqNM7u If you are having trouble accessing the form, need help approving this course as your master programs elective, or have any other questions, please write to DCE@tudelft.nl If more than 30 participants apply for the course, we will select based on your application. If you apply after June 15th, you may still join if there is space available.
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.
Elective	Yes
Tags	Business Economics Finance
Targetgroup	MSc students from all faculties can apply.

TPM425B	Start-up Challenge	3
Module Manager	Dr. A. Giga	
Instructor	Dr. A. Giga	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	This course is a part of a TU Delft startup competition. If you would like to participate in this course, you have to be a part of the contest. This means you have to have an idea for a business that you would like to develop, and apply for the competition (either individually or as a team). Also, note that the competition (and course) runs through periods 3 and 4. It usually starts around the beginning of March and ends in the first week of June. The timeline for the competition is published well in advance.	
Course Contents	During the course/competition, you will focus on identifying a customer for your product or service and building a case for how your product addresses a customer problem or an unmet need. You will also perform a market analysis for this business idea. More specifically, you will assess the competitive advantage of your team and your product and the market size and segmentation, among other things. The focus of the course is on customer discovery and business model development. You should be talking to a lot of customers and stakeholders, thereby aggressively learning about the market you plan to enter. Throughout the course you will have an opportunity to present your project idea to industry professionals.	
Study Goals	After this course, you will have learned how to: - assess the problem-solution fit of your idea; - test, modify, and develop your idea by talking to customers and learning about the marketplace; - present your project idea; - emulate effective entrepreneurial behavior.	
Education Method	The course offers a mix of lectures, recommended online materials and books, coaching sessions, and meetings with professionals with industry and entrepreneurial experience. Meetings with industry experts will help you validate the idea and occasional individual coaching will aid your personal entrepreneurial development.	
Assessment	This is a pass/fail class. To pass the course, you must: - Submit intermediate and final deliverables of decent quality. - Show evidence of customer discovery and market learning via interviews - Attend lectures and meetings with professionals.	
Enrolment / Application	You need to apply for the competition to be eligible to participate in this course. The application deadline for the competition is in mid-February and the competition starts in March. E-mail dce@tudelft.nl to get the application form and all the other necessary information.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Tags	Adventurous Business Economics Project	
Targetgroup	This is a MSc elective. Master students from all master programs can apply.	

TPM426B	Start-up Challenge - Advanced	5
Module Manager	Dr. A. Giga	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	This course is a part of a TU Delft startup competition. If you would like to participate in this course, you have to be admitted to the advanced track of the of the contest. This means that you have started developing a business idea and already made some progress (e.g. prototype, customer discovery, market analysis and the business model). Also, note that the competition (and course) runs through periods 3 and 4. It usually starts around the beginning of March and ends in the first week of June. The timeline for the competition is published well in advance.	
Course Contents	During the course/competition, you will focus on testing and validating your customers need for your product or service and estimating the value it brings to your customer. You will also perform a market analysis. More specifically, you will assess the competitive advantage of your team and your product and the market size and segmentation, among other things. Additionally, you will build a detailed business model around your idea using the business model canvas and also create a detailed near-term execution plan with timeline and milestones. You will also start assessing the profitability of your business model and developing a financing strategy for it. The focus of the course is on customer discovery and business model development. You should be talking to a lot of customers and stakeholders, thereby aggressively learning about the market you plan to enter. Throughout the course you will have an opportunity to present your project idea to industry professionals.	
Study Goals	After completing this course, you will have learned to: - validate your value proposition and product-market fit; - build a viable and defensible business model; - test and modify your value proposition and the business model by talking to customers and relevant practitioners and gathering and analyzing market information; - present your project idea; - emulate effective entrepreneurial behavior.	
Education Method	The course offers a mix of lectures, recommended online materials and books, coaching sessions, and meetings with professionals with industry and entrepreneurial experience. Meetings with industry experts will help you validate the idea and occasional individual coaching will aid your personal entrepreneurial development.	
Assessment	This is a pass/fail class. To pass the course, you must: - submit workshop, intermediate and final deliverables of decent quality; - show evidence of customer discovery and market learning via interviews; - attend lectures, workshops, and meetings with professionals.	
Enrolment / Application	You need to apply and be admitted for the advanced track of the competition to be eligible to participate in this course. This requires that you show evidence of progress of your business idea. The application deadline for the competition is in mid-February and the competition starts in March. E-mail dce@tudelft.nl to get the application form and all the other necessary information.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Tags	Adventurous Business Economics Project	
Targetgroup	This is a MSc elective. Master students from all master programs can apply.	

TPM959A	Technology Battles	3
Module Manager	Dr. G. van de Kaa	
Instructor	Dr. G. van de Kaa	
Responsible for assignments	Dr. G. van de Kaa	
Co-responsible for assignments	Dr. A.C. Smit	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is organized into regular sessions. Each of the regular sessions consists of a mixture of lecturing and discussions. For each session you study the literature. Through class discussion we will further deepen our understanding of the topics and themes of the class. Every student is expected to participate in a class-wide discussion.	
Study Goals	After the course students are able to understand the theoretical background of standardization.	
Education Method	1 group assignment	
Assessment	Summative group assignment, with (normally) a group size of 3 students. Only a final grade, which counts for 100%, is registered into OSIRIS. The minimum grade to pass the course is 5.75 When the group assignment is not completed with a pass, a resit is possible.	
Elective	Yes	

WM-ITAV-4010	Scientific Writing	2
Module Manager	L. Meester	
Instructor	Drs. W.J. Blokzijl	
Instructor	Drs. B.M.D. van der Laaken	
Instructor	Drs. P.C. Post	
Instructor	Drs. A.E. Kam	
Instructor	M.J.Y. Wackers	
Instructor	S. Baars	
Instructor	M. Blikendaal	
Instructor	L.C. Schroten	
Instructor	A. Glasbergen-Plas	
Instructor	M. Looij	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1 2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you learn to write a scientific article, either a research article based on your own research data or a literature review about a subject of your own choice. This is a necessary skill for anyone who wants to pursue an academic career after their graduation, but it can also be used immediately for all academic texts you will write during your Master program, such as your Master thesis. In seven weeks, we will go through all steps of the writing process, from formulating a good main question and finding relevant literature, to the actual writing, re-writing and final editing. You will exercise with finding, reading and managing relevant academic literature, writing in an academic style, building a comprehensive argumentation, reviewing fellow students' articles and using other students and the instructor's comments to improve your own work.	
Study Goals	The purpose of this course is to learn how to write a scientific text. To achieve this, at the end of this course you will: - know what the main characteristics are of a scientific text - be able to formulate a main question - be able to find, critically read and manage scientific literature - be able to use literature properly and avoid plagiarism - be able to build up your argumentation - be able to structure an article according to the conventions in your field of study - be able to use scientific English style - be able to use tables and figures to support and communicate your results - be able to give feedback on somebody else's article - be able to use feedback for improving your work.	
Education Method	On campus practical, in 6 sessions (attendance mandatory). Every week you have to read some background information about scientific writing and hand in a part of your text. Participants must attend all sessions - one missed session is allowed only - and hand in all assignments in time. Students who receive a pass for this course are rewarded with 2 ects. This equals 56 hours of study. A total of 12 hours is spent on attending classes, in which you can ask questions, discuss the feedback you received on your work and discuss aspects of scientific writing with your fellow students; the remaining 44 hours is for self study, writing and revising. In seven weeks, from preparing lecture 2 up to handing in your final article in week 8, you will have to spend at least 6 hours of self study on this course, every week. It is important that you make sure you have this time in your personal schedule. At the beginning of the course you will mainly be reading up about scientific writing and the subject of your text. As the course proceeds, you will be spending more of your time on writing, giving feedback and revising your own text.	
Books	Theory about academic writing will be made available through Brightspace.	
Assessment	You write a scientific article of 2000-2400 words (excluding the list of references and the abstract). You have to hand in parts of your article every week. Your final grade is based on the final article. An evaluation form for the grading of the article is available throughout the course.	
Elective	Yes	
Category	MSc level	

WM0201TU	Technical Writing	2
Module Manager	Drs. P. van den Hoven	
Responsible for assignments	Drs. P. van den Hoven	
Co-responsible for assignments	Drs. A.E. Kam	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	Dutch	
Elective	Yes	
Category	BSc niveau	

WM0203TU	Oral Presentations	2
Module Manager	I.M. Schouten	
Instructor	L. Meester	
Instructor	I.M. Schouten	
Co-responsible for assignments	M. Wackers	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	Dutch	
Course Contents	In job advertisements for engineers applicants are always required to possess excellent communication skills. This means among other things that they should be able to present their designs and research results fluently, for example during meetings with clients or with managers in their own company. A persuasive presentation does not only require thorough preparation of content, but also good style. It takes quite a bit of skill to come across understandable for any particular audience and to stay in control of the situation. For this purpose the course Oral Presentations offers students the possibility to gain experience in giving presentations under professional guidance in a small group.	
Study Goals	At the end of the course you should be able to: Present complex material clearly to both expert and non-expert audiences; Apply structural elements that are offered in the course; Use presentation techniques such as: - Using various hand gestures - Using vocal variety - Keeping a good posture; Create attractive slides that support your presentation; Interact with your audience and respond to their reactions and questions professionally.	
Education Method	The first meeting will be an introduction to several presentation principles. The second, third, fourth, fifth and sixth meeting will be dedicated to practicing presentation skills. Students will be asked to prepare short presentations for this purpose, focusing on individual skills. These presentations will be evaluated by a panel of students and the teacher. For the seventh meeting students will be asked to give a final presentation about a subject related to their study. All presentations will be recorded on video. This will offer participants the possibility to evaluate their own presentations. Attendance is compulsory; no more than one lesson may be missed.	
Literature and Study Materials	Recommended: Van der Laaken, M. & van der Laaken, B. (2013). Presentation Techniques (Rev. ed.). Coutinho. Digital: online module in Brightspace.	
Assessment	A final mark will be based on the final presentation. Students who fail, will be given the opportunity to do a repair assignment.	
Enrolment / Application	Please enroll for a group in Brightspace: "wm0203tu Mondeling Presenteren". After enrollment you can subscribe in a group. Note: before subscribing check if you get ECTS for this course or if Oral Presentations (eng) is needed.	
Elective	Yes	
Category	BSc niveau	

WM0204TU	Meetings and Negotiations	2
Module Manager	Drs. P.C. Post	
Responsible for assignments	Drs. P.C. Post	
Co-responsible for assignments	Drs. A.E. Kam	
Contact Hours / Week x/x/x/x	0/0/0/2,5	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	Dutch	
Course Contents	Whenever you work in teams, you will have to have meetings, discuss proposals or defend your ideas. How to do that in an efficient and effective way, which is most likely to succeed? How to chair a meeting, what should you do as a chairperson? And how to contribute effectively to a meeting when you are a participant? Which techniques help you to explore, to clarify or to decide? You will get answers to these questions and gain practical knowledge and experience of how make the most of meetings.	
Study Goals	After this course, participants will have: - Insight into theory and techniques of discussions and meetings - Better skills in chairing meetings - Insight into how to apply meeting skills to different situations	
Education Method	3,5 hours per week course (5 lectures in total) with practical assignments, role games and theoretical insights.	
Literature and Study Materials	A course reader will be provided	
Assessment	Grading is based on attendance and quality of the participation	
Enrolment / Application	Please enroll via Brightspace into the course group - see instructions on Brightspace	
Elective	Yes	
Category	BSc niveau	

WM0312CIE	Philosophy, Technology Assessment and Ethics for CIE	4
Module Manager	A.R. Gammon	
Instructor	C. Clare Shelley-Egan	
Responsible for assignments	A.R. Gammon	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module explores philosophical and ethical questions in civil engineering, as well as providing a basic introduction into Technology Assessment.	
Study Goals	After having completed the course you: know the basic philosophical issues in engineering knowledge; can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	Virtual learning environment, introductory lecture and small number of question hours; mandatory assignment (more information will be provided during the introductory lecture and/or one of the question hours)	
Literature and Study Materials	- Ibo van de Poel and Lambèr Royakkers (2011). Ethics, Technology, and Engineering: An Introduction. Wiley Blackwell, West Sussex (ISBN 978-1-4443-3094-6 (hardcover) / 978-1-4443-3095-3 (paperback)). - Additional reading material - Virtual learning environment (Brightspace).	
Assessment	Exam (80%) + mandatory assignment (20%) To pass the course, students must earn a 6.0 or higher. Students can carry partial grades over into future years. Students can resit the exam for a higher grade even if their initial grade is a passing one. Students can re-submit the mandatory assignment only in the case of an incomplete or a failing grade, not to improve a passing grade.	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Tags	Ethics Philosophy	
Category	MSc niveau	

WM0318TN	Argumentation and Philosophy of Science	3
Module Manager	Dr. M. Astola	
Instructor	Dr. M. Astola	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	see Dutch version	
Study Goals	see Dutch version	
Education Method	see Dutch version	
Assessment	see Dutch version	

WM0320TU	Ethics and Engineering	3
Module Manager	M. Sand	
Instructor	M. Sand	
Instructor	A.R. Gammon	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and 5 interactive work sessions with partial assessment, concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	Written exam (60%, individual grade), presentation and active participation during the working group sessions (40%, same grade for all members of a presentation group). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade. These partial grades are only valid for 1 academic year, meaning that if one or both are not passed during resit, both assessments need to be retaken the following academic year.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enroll as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline. The Brightspace page will be activated a couple of weeks before the start of the course.	
Remarks	The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course WM0329TU (6 ects).	
Elective	Yes	
Category	MSc niveau	

WM0329TU	Ethics and Engineering	6
Module Manager	M. Sand	
Instructor	A.R. Gammon	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	4/0/0/0 or 0/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	The first part of this course completely coincides with the WM320TU course. The WM329TU course is the WM320TU course plus the writing of an essay. Please check the WM320TU Brightspace for all relevant information about lectures, work sessions, exam, and the essay. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. - are able to write an opinion article on ethical or societal aspects of technology, according to academic standards of writing - are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and 5 interactive work sessions concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	50%: grade for the participation in the course WM0320TU, which consists of a written exam (60%, individual grade), presentation, and active participation during the working group sessions (40%, same grade for all members of a presentation group). In order to pass the WM0320 part of the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. These partial grades are only valid for 1 academic year, meaning that if one or both are not passed during resit, both assessments need to be retaken the following academic year. 50%: grade of the essay, which is written individually (all steps in the essay writing procedure need to be followed). The grade of the essay (50%) conjoined with the grade of the WM0320TU part (50%) must be higher than 5.75. Only if the threshold is not met will it be possible to revise and resubmit the essay. It is not possible to retake the essay in another semester.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the number of workgroups. Please enrol as soon as possible and in any case before the start of the course via Brightspace.	
Remarks	The course is run twice each year, starting in quarters 1 and 3. The first part of this course (concluded with the written test) coincides with the course WM0320TU.	
Elective	Yes	
Category	MSc niveau	

WM0348TU	Christian Philosophy: Recent Developments	3
Module Manager	Prof.dr. M.J. de Vries	
Co-responsible for assignments	Prof.dr.ir. I.R. van de Poel	
Contact Hours / Week x/x/x/x	2/2/0/0/	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is not necessary to have finished wm0361tu and wmo371tu first, as this course can be done parallel. This is taught for the last time in 2024/2025.	
Course Contents	Students read a recent book on philosophy of science or philosophy of technology and make an analysis using the ideas and concepts that have been dealt with in courses wm0361tu and wm0371tu.	
Course Contents Continuation	This course is meant to offer a possibility to do the courses wm0361tu and wm0371tu at master level.	
Study Goals	Students learn to read a philosophical text and analyse it by using a framework from christian philosophy	
Education Method	Self study or lectures	
Assessment	Oral exam or essay	
Elective	Yes	
Category	MSc level	

WM0349WB	Philosophy of Engineering Science and Design	3
Module Manager	Dr. J.M. Duran	
Instructor	Dr. J.M. Duran	
Responsible for assignments	Dr. J.M. Duran	
Co-responsible for assignments	Ir. S.J. Zwart	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Course contents: (1) The goals of science; the character and scope of scientific claims. (2) The goals of engineering design; the nature of technical artefacts; the value-neutrality of technology. (3) The scientific method and the validation of scientific claims. (4) Methods of engineering design; the character and scope of design claims; the decision-making aspect of design. (5) The development of science; the objectivity of science; the notion of scientific progress. (6) The development of technology; social determinism and technical determinism. (7) The place and role of values in science and in engineering design.	
Study Goals	This course aims first of all to support students in developing a reflective and critical attitude with regard to empirical research underlying engineering science and engineering design at an academic level. Additionally, and in support of this primary goal, it aims to make students acquainted with views on the nature and goals of science and engineering design and how in these activities facts and values both have their role to play and how they interact.	
Education Method	The course is taught in the form of seven weekly sessions which each consist of a plenary lecture of two hours and a seminar of two hours during which students articulate and discuss their answers to a number of questions in smaller groups. Each seminar will be prepared and chaired by a team of students. For this task general and specific instructions, recommendations and suggestions will be available.	
Assessment	Assessment is through (1) a final individual exam, (2) participation in the seminars and (3) performance in preparing and chairing a seminar session. The exam result forms 50 percent of the final grade, assessment of general seminar participation 25 percent and assessment of the preparing and chairing role also 25 percent. In the expectation that no COVID restrictions will be in place while the course is being offered, the exam will be a closed-book on-campus exam, which may be either hand-written or digital (ANS). Which of the two will be announced at the start of the course.	
Category	MSc level	

WM0353TU	Climate Ethics	3
Module Manager	Prof.dr.ir. B. Taebi	
Instructor	Prof.dr.ir. B. Taebi	
Instructor	A. Melnyk	
Co-responsible for assignments	Prof.dr.ir. B. Taebi	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Climate change is considered to be one of the most urgent problems the world is currently facing. Two types of solutions are proposed namely mitigation and adaptation. Both climate change mitigation and adaptation are heavily relying on the progress of technology. Addressing the climate problem requires, however, more than just developing and applying a certain technology. It demands considerations of whose interests and whose rights are at stake. It also requires reflection on how and "under which condition" technology can change and shape the world for the better. In this course we will focus on the following ethical aspects of climate change: - Past emissions and responsibility to deal with climate change - Implications of global warming on human safety and security - The distribution of burdens and benefits, emission rights and international justice - Future generations and intergenerational justice - New developments after the Paris Agreement The course will be offered in English and it is primarily for MSc students. Enrollment on Bright Space is sufficient.	
Study Goals	- Creating awareness about climate change induced by human activities; - Gaining insights into ethical aspects of climate change and implications for law and policy-making; - Understanding and reflecting on the ethical and social conditions under which technological solutions considered to be ethically plausible; - Analysing and critically engaging with the international negotiations and agreements for dealing with climate change.	
Education Method	Lectures and discussion sessions. Students are expected to actively participate in the class, which is why attendance is mandatory.	
Literature and Study Materials	We will study different papers and book chapters on the ethics of climate change. The required reading material along with instruction on how to acquire this material will be announced on BS.	
Assessment	Class evaluation (1/5) and written exam (4/5). More instruction regarding the exam will be provided in class.	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/	
Elective	Yes	
Category	MSc level	

WM0361TU	Filosofie I: techniek- en wetenschapsfilosofie	3
Module Manager	Prof.dr. M.J. de Vries	
Co-responsible for assignments	Prof.dr.ir. I.R. van de Poel	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Course Contents	In this course important concepts in technology and engineering are introduced by means of elaborated case studies. This course is taught in 2024/2025 for the last time.	
Study Goals	Students learn to analyse technological developments from a philosophical perspective.	
Education Method	Lectures	
Assessment	Oral exam or small thesis	
Elective	Yes	
Category	BSc niveau	

WM0371TU	Today's Philosophy of Technology and MSc Ethics	3
Module Manager	Prof.dr. M.J. de Vries	
Co-responsible for assignments	Prof.dr.ir. I.R. van de Poel	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	Dutch	
Course Contents	In this course we read texts written by contemporary philosophers of technology. This course will be taught last in 2024/2025.	
Study Goals	Students learn about the current situation in philosophy of technology	
Education Method	Working lectures	
Assessment	Oral exam or small thesis	
Elective	Yes	
Category	BSc niveau	

WM0375TU	Ethics of Technological Risks	3
Module Manager	Prof.dr. S. Roeser	
Instructor	Prof.dr. S. Roeser	
Instructor	K. Maheshwari	
Responsible for assignments	Prof.dr. S. Roeser	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course investigates ethical aspects of technological risks. Debates about technological risks related to, for example, energy technologies, robotics and biotechnology frequently culminate in stalemates. This is due to the complexities and intricacies inherent to such debates as they involve scientific information and uncertainties, as well as ethical and emotional considerations. Conventional, quantitative approaches focus primarily on statistical information to risk. They do not explicitly incorporate important ethical considerations such as justice, fairness and autonomy. Emotions such as compassion, care and feelings of responsibility can draw attention to such ethical aspects of risky technologies. Taking emotions and ethical considerations seriously can lead to more fruitful deliberations between different stakeholders in which relevant concerns are taken seriously and are explicitly reflected upon. This course will study how approaches to ethical aspects of risk can lead to more morally responsible decision making and design of technological innovations.	
Study Goals	This course will provide students with an understanding of approaches to risk ethics and students will get experience with discussing and evaluating ethical aspects of risk	
Education Method	Self-study, lectures and tutorials (in smaller groups, ca 25-35 students, depending on the total number of students). All students will hold one group presentation (ca 5 students) at one of the lectures. Attending lectures and tutorials and giving a presentation is obligatory.	
Assessment	On campus written exam (individual assignment), this will count for 100% of the grade for this course. Students need to have a pass (5.75 or higher) in order to pass the course. There is a resit of the written exam in the 4th quarter (date to be announced). In case of unforeseen circumstances, eg related to COVID-19 measures, the on campus exam may be replaced with an online (eg take home) exam. [There is also a 5 ECTS version of this course, i.e. WM0376TU; the only difference is that for the 5 ECTS version students also write an essay in addition to the exam.]	
Special Information	Contact person for additional questions: S.Roeser@tudelft.nl	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Philosophy	
Category	MSc level	

WM0376TU	Ethics of Technological Risks	5
Module Manager	Prof.dr. S. Roeser	
Instructor	K. Maheshwari	
Responsible for assignments	Prof.dr. S. Roeser	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course investigates ethical aspects of technological risks. Debates about technological risks related to, for example, energy technologies, robotics and biotechnology frequently culminate in stalemates. This is due to the complexities and intricacies inherent to such debates as they involve scientific information and uncertainties, as well as ethical and emotional considerations. Conventional, quantitative approaches focus primarily on statistical information to risk. They do not explicitly incorporate important ethical considerations such as justice, fairness and autonomy. Emotions such as compassion, care and feelings of responsibility can draw attention to such ethical aspects of risky technologies. Taking emotions and ethical considerations seriously can lead to more fruitful deliberations between different stakeholders in which relevant concerns are taken seriously and are explicitly reflected upon. This course will study how approaches to ethical aspects of risk can lead to more morally responsible decision making and design of technological innovations.	
Study Goals	This course will provide students with an understanding of approaches to risk ethics and students will get experience with discussing and evaluating ethical aspects of risk.	
Education Method	Self-study, lectures and tutorials (in smaller groups, ca 25-35 students, depending on the total number of students). All students will hold one group presentation (ca 5 students) at one of the lectures. Attending lectures and tutorials and giving a presentation is obligatory.	
Assessment	On campus written exam (60% of the final grade) and essay (40% of the final grade). The exam is an individual assignment, the essay is written in pairs of students. Students need to have a pass (5.75 or higher) for each individual grade in order to pass the course. There is a resit of the written exam in the 4th quarter (date to be announced). In case students did not pass the essay initially, they have the opportunity to make further revisions within ca 2 weeks, but then the maximum grade is a 6, given the extra time. Partial grades are valid for the current and the following academic year. In case of unforeseen circumstances, eg related to COVID-19 measures, the on campus exam may be replaced with an online (eg take home) exam [There is also a 3 ECTS version of this course, i.e. WM0375TU; the only difference is that for the 3 ECTS version students do not write an essay.]	
Special Information	Contact person for additional questions: S.Roeser@tudelft.nl	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Philosophy	
Category	MSc level	

WM0801TU	Introduction to Safety Science	3
Module Manager	Dr. A.P. Afghari	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Dr. A.P. Afghari	
Co-responsible for assignments	Dr. E. Papadimitriou	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Students wishing to graduate with a final project in the area of safety supervised by the Safety Science and Security Group should have followed this course or one of the other introductory courses run by the group (WM0808TU, WM0821TU or WM0822TU)	
Expected prior knowledge	Technical academic BSc and common (scientific) sense	
Summary	System Safety Engineering focuses on development of a safety oriented pattern of thinking and a holistic approach. The tools that will be gained in this course will be helpful in recognizing, understanding, and analyzing hazards; and assessing risks in contemporary complex systems. Areas included are (1) Hazard analysis and safety assessment, (2) Human reliability assessment, and (3) Safety management.	
Course Contents	Introduction to system safety engineering Safety performance measurement Hazard analysis Failure modes and effects analysis Preliminary hazard analysis System safety assessment Reliability block diagram Fault & event trees Markov chain analysis Bayesian network Consequence assessment Risk ranking Human reliability assessment Decision making under uncertainty	
Study Goals	The students will learn how to perform safety assessment, reduce risk within acceptable levels, manage risk, improve system safety, and make risk-informed decisions to benefit the organization and the community.	
Education Method	Plenary lectures (mandatory) Exercises (carried out in small groups of ~2 students)	
Literature and Study Materials	Clifton A. Ericson, Hazard Analysis Techniques for System Safety, John Wiley and Sons; ISBN: 978-0471720195. Harold E. Ronald and Brian Moriarty. (1990). System Safety Engineering and Management, John Wiley and Sons, 2nd Edition.	
Prerequisites	Technical background preferred but certainly not mandatory	
Assessment	Summative assessment: A final exam consisting of open-ended and multiple-choice questions; The exam is based on book, papers, exercises and lecture slides; The final exam is individual and accounts for 100% of the final grade for the course; A re-sit exam is possible if needed. Formative assessment: In-class quizzes, discussions and games; Sample questions for final exam.	
Enrolment / Application	Enrol through Brightspace. Minor students are automatically enrolled into the course.	
Remarks	This course can be expanded to a System Safety/Reliability project	
Elective	Yes	
Category	MSc level	

WM1028AM	Ethics for Applied Mathematics	3
Module Manager	A.R. Gammon	
Responsible for assignments	Dr. S.N.R. Buijsman	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course is a study of broadly normative issues that arise in relation to applied mathematics. The following kinds of questions are examined: How can applied systems be built to promote, protect, preserve, and foster the pursuit of various values related to human wellbeing? What risks and uncertainties need to be accounted for in such systems? To what extent is it possible to empower individuals with applied systems while limiting the extent to which they can harm each other and themselves?	
Study Goals	1. Recognize and discuss normative and societal implications of mathematical applications and cases. 2. Explain key course concepts (e.g., bias, normativity, modeling assumptions) and their ethical dimensions. 3. Apply basic ethical frameworks and concepts to mathematical applications in society. 4. Think critically about the roles and responsibilities of mathematics professionals.	
Education Method	1 lecture and 1 tutorial per week	
Literature and Study Materials	Readings will be provided on Brightspace.	
Assessment	Final Assignment (Exam): 50% (individual) Tutorial Presentation: 25% (small groups) Ethical Assessment paper: 25% (small groups)	
Remarks	Contact the course instructor for disability accommodations: S.N.R.Buijsman@tudelft.nl	

WM1301TU	Ethics of Transportation	3
Module Manager	Dr. E. Ziliotti	
Responsible for assignments	Dr. E. Ziliotti	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems can see these problems within the broader background of moral, legal and political philosophy have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain. are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way	
Education Method	A series of 6 lectures and some working group sessions, focused on the discussion and application of the theories presented in the course	
Assessment	Written exam and assessment participation in working groups	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

WM1302TU	Ethics of Transportation	5
Module Manager	Dr. E. Ziliotti	
Responsible for assignments	Dr. E. Ziliotti	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course is identical to the WM1301 course, with the addition of a final essay. In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems - can see these problems within the broader background of moral and political philosophy - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way - can write a short academic essay of critical analysis of an ethical or societal issue related to transportation technology	
Education Method	A series of 6 lectures, active participation in some working groups, and the writing of an essay on ethical, social or legal aspects of technology to be written under the supervision of an instructor from the section Ethics of Technology at TBM	
Assessment	Written exam, assessment participation in working groups and the writing of an essay the ethical, social or legal aspect of technology to be written under the supervision of the course manager or an instructor from the section Ethics of Technology at TBM. In order to pass the course both the exam and the essay should be graded with at least 5.5 (and the average should be sufficient, that is higher than 5.75)	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

WM1401TU	Ethics of Healthcare Technologies	3
Module Manager	Prof.dr. S. Roeser	
Instructor	Dr. S. Rainey	
Co-responsible for assignments	Dr. S.M. Copeland	
Contact Hours / Week x/x/x/x	2/0/0/0 contact for students: Stephen Rainey via S.Rainey@tudelft.nl	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1401 (3ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

WM1402TU	Ethics of Healthcare Technologies	5
Module Manager	Prof.dr. S. Roeser	
Instructor	Dr. S. Rainey	
Instructor	Dr. J.J. Casey	
Assistant	Dr. G. Pozzi	
Co-responsible for assignments	Dr. S.M. Copeland	
Contact Hours / Week x/x/x/x	2/0/0/0 contact for students: Stephen Rainey via S.Rainey@tudelft.nl	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design. Regarding skills, students will have: developed skills in finding and interpreting relevant policy, regulations and codes of ethics for the field; formulated, debated and defended a substantiated position on the ethical issues at stake in current applications of technology in health care.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1402 (SECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions), Final paper or project (group).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

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